



DSD2

Technical documentation

Three-phase synchronous motors DSD2 28-132

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Technical data, motor components,
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Therefore, we cannot accept liability for the correctness of the information in this document.

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1. Three-phase synchronous motors DSD2-028-132



The motors of the DSD2 - servo series are non-cooled, surface or water-cooled three-phase synchronous motors. Low rotor inertia and optimized performance provides the 8-pole motors with an excellent accelerating power.

This motor series is covered to a speed range up to 6000 rpm. These motors are highly dynamic, excellent in terms of speed and position control and thus ideally suited for applications in machine tools, production machines (e.g., printing, packaging, textiles, plastics), in handling equipment and in the medical field.

1.1. General technical data

Version	IM B5	Horizontal installation position according to EN 60034-7
	IM V1	Vertical installation position, shaft end down, according to EN 60034-7
	IM V3	Vertical installation position, shaft end up, according to EN 60034-7 The shaft opening must be protected against water and dirt at protection class IP64
	IMB35	Size 132 Note: Sizes V1 and V3 are not available for DSD2-132.
Protection type	IP44	Size 028-036; Standard: without shaft seal ring
	IP54	Size 132
	IP64	Size 045-100; Standard: without shaft seal ring, with opposing plugs fitted and fully enclosed terminal boxes
	IP65	Size 045-100, Option: with shaft seal ring, with opposing plugs fitted and fully enclosed terminal boxes. Example configuration: DSD2-071SO65W-20-54-AOB-BPP-K-AN-O-DA1
	IP65	Size 045-100; Without consideration of shaft bushing with opposing plugs fitted and fully enclosed terminal boxes
	IP67	Size 045-100; Without consideration of the shaft bushing for IC410 and IC 3W7, fitted with mating connectors, not for motors with terminal box
Connection	Main connection	Refer to chapter 3.8.2, 3.5.10 and 3.5.11
	Encoder connection	Refer to chapter 3.5
		The main, encoder and fan connector can optionally be obtained in the Speedtec version
	Brake	Connection in the main connection, separate connection at the DSD2-132
	Temperature sensor	Standard in the main connection and optional in the encoder socket
Temperature sensor	PT1000.	linear temperature sensor for the analysis in the controller
Cooling type	IC 410	Size 028-100 surface-cooled without fan
	IC 416	Size 056-132 surface-cooled with fan
	IC 3W7	Size 045-132 water-cooled machine
Heating	$\Delta\theta = 105\text{ K}$	Insulation material class F according to EN 60034

Environmental conditions for running	Class 3K3/3Z12 as per DIN EN 60721-3-3, however: temperature range 0-40 °C	Represents 0 to 40 °C at 5 % to 85 % rel. humidity and an absolute humidity of 1 g/m³ to 25 g/m³ and an installation height up to approx. 1,400 m.
Environmental conditions for long-term storage	Class 1K2/1M1 DIN EN 60721-3-1, however: temperature range -15-60 °C	Represents -15 to 60 °C at 5 % to 85 % rel. humidity and an absolute humidity of 1 g/m³ to 25 g/m³; at temperatures below 3 °C you should drain the cooling water
Environmental conditions for transport	Class 2K2/2M1 DIN EN 60721-3-2, however: temperature range -15-60 °C	Represents -15 to 60 °C at 5 % to 85 % rel. humidity and an absolute humidity of 1 g/m³ to 25 g/m³; at temperatures below 3 °C you should drain the cooling water
Surface	untreated	Size 028-036
	Black matt	Size 045-132; RAL 9005
Bearing	D-side	Standard: Ball bearing, option: Roller bearing (size 056-100)
	N-side	Ball bearing, fixed bearing
Service life of bearing	L _{10h} 20.000 h	Standard value, anti-friction bearing with permanent grease lubrication
Vibration levels	A	according to DIN EN60034-14 (0530-Part 14) :2004-09
	B	Option: (only for ball bearings)
Concentricity	N	Standard: Normal DIN SPEC 42955 issued 1981*
	R	Option: Reduced acc to DIN SPEC 42955 issued 1981* (at the ball bearing only)
Vibration resistant up to	radial 3 g	10 Hz to 100 Hz according to EN 60068-2-6; DSD2-028-100
	axial 1 g	10 Hz to 100 Hz as per EN 60068-2-6 (Size 028-036 without brake)
	axial 0.5 g	10 Hz to 100 Hz as per EN 60068-2-6; DSD2-028-100
Flange	according to IEC standard	Centralization diameter: Clearance j6 at DSD2-028-100; h6 at DSD2-132
Shaft end	Cylindrical	Smooth as per DIN 748; also available with feather key DIN 6885
		Centralization with female thread as per DIN 332 Form D
Holding brake	Option	Permanent magnet brake at zero clearance at DSD2-028-100
		Spring brake with clearance at DSD2-132
Speed encoder	Resolver	Standard – see chapter 3.3
	SinCos encoder	Options – see chapter 3.3
Approvals	CE	Size 028-132
		Size 028-132
	UKCA	Size 028-132
	CEL	

*) DIN EN 50347:2003-09 non applicable, as valid for AC standard motors, only

**) If vibration loads are present, then measurings are required.

Based on these constructive reviews and evaluations with the company Baumüller will be carried out.

1.2. General safety instructions

The standard versions of the motors are unsuitable for operation in salty or aggressive atmospheres and are not suitable for erection outdoors. If the air is contaminated with dust particles or similar substances in the surrounding air when using air-cooled motors, which cannot be kept out efficiently by the filter elements in use then the manufacturer must be contacted to find a solution to the problem.

Suitable steps to reduce bearing currents are to be taken before commissioning the motor, depending on the application and system. The motor manufacturer must be consulted in this regard.

Note:

With allocation of the motor in a specific protection class, it is a standardized brief test procedure. This can vary considerably depending on the actual environmental conditions at the site of installation. Depending on the environmental conditions, such as the chemical consistency of the dust materials or the cooling media being used at the site of installation, evaluation of the suitability of the motor based on the type of protection is only possible to a limited extent (e.g. electrically conducting dust materials or aggressive coolant vapors or coolant fluids). In these cases, the motor must additionally be protected by appropriate measures on the machine side.

1.3. Definitions of power ratings

1.3.1. Definitions of power ratings for air-cooled machines

The power ratings (torques) listed in the table apply to continuous operation (S1) at the rated speed and a maximum ambient temperature of 40 °C, for machines installed below 1,000 m a.m.s.l.

If motors are to be operated at an ambient temperature of more than 40 °C, or altitudes above 1,000 m a.m.s.l., the required list power rating P_L (list torque M_n) is calculated from the product of factors k_1 and k_2 (specified in the table below) and the required power rating P (torque M).

Ambient temperature	40 °C	45 °C	50 °C	55 °C	60 °C
Correction factor k_1	1	1.06	1.13	1.22	1.34
Altitude a.m.s.l. up to	1,000 m	2,000 m	3,000 m	4,000 m	5,000 m
Correction factor k_2	1	1.07	1.16	1.27	1.55

Design changes may be necessary in the case of ambient temperatures above 40 °C and installation of motors in an enclosure: For this reason, it is imperative that the manufacturer is contacted.

If, in the case of an increasing site altitude above 1,000 m, the ambient temperature decreases by approx. 10 °C per 1,000 m increase, no power correction is necessary (note the minimum operating temperature).

1.3.2. Definitions of power ratings for water-cooled machines

The power ratings (torques) that appear in the list apply to permanent operation S1 at nominal speed, provided the cooling circuit requirements for water-cooled motors are met!

The reduction factors included in the table below must be considered when operating DSD2 motors with higher coolant inlet temperatures:

Coolant inlet temperature	25 °C	30 °C	35 °C	40 °C	45 °C
Percentage of list performance (torque)	100 %	97 %	95 %	92 %	89 %

1.4. Water cooling

1.4.1. Coolant consistency

The coolant must satisfy the following specifications:

Conditions	Unit	Value
Maximum permitted system pressure	bar	6
Temperature of coolant - for motor	° C	10 to 25
pH value (at 20° C)	---	6.5 to 9
Overall hardness	mmol/l	1.43 to 2.5
Chloride - Cl^-	mg/l	< 200
Sulphate - SO_4^{2-}	mg/l	< 200
Oil	mg/l	< 1
Permitted particle size of solid foreign objects, particles (e.g. sand)	mm	< 0.1

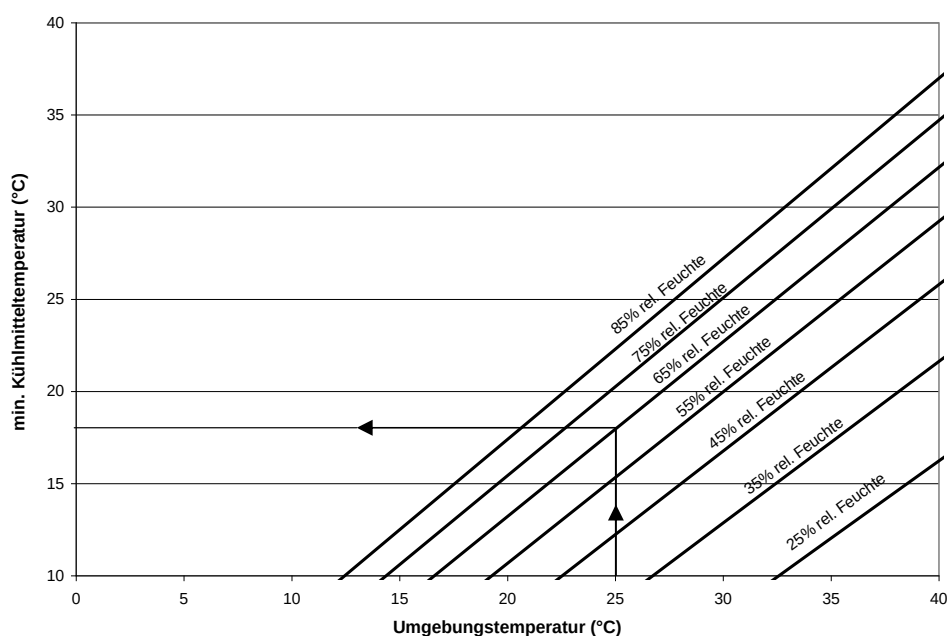
Clean water that is free of dirt and suspended matter must be used as a coolant.

Note:

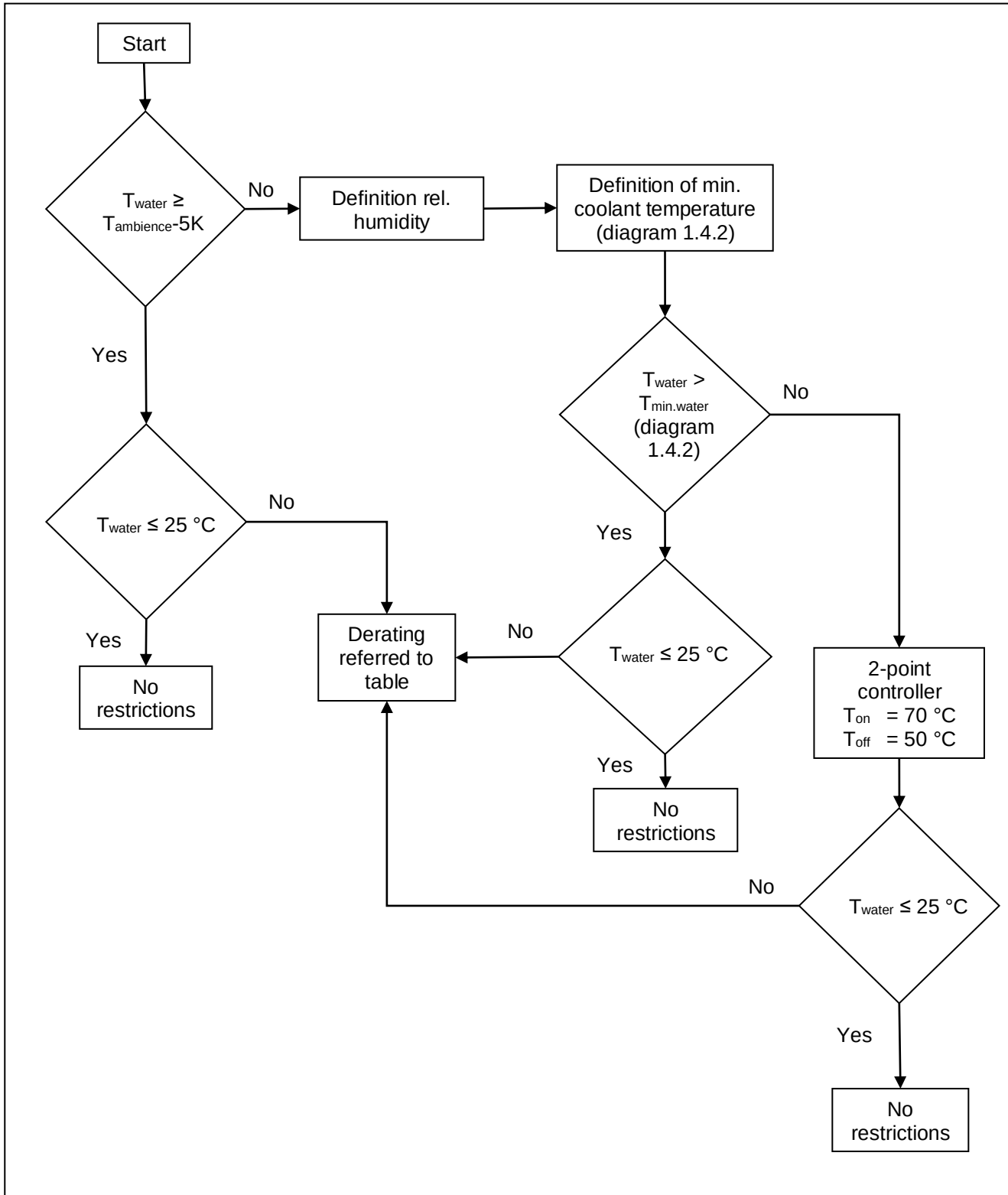
If the specific heat capacity is reduced by adding glycol for example, in dependence of the mixing ratio is a power reduction in the consequence, which is to be asked for at the manufacturer.

Compared to water cooling in the event of the use of hydraulic oil (HLP 46) a power rating reduction according to the overall length and speed of 20 to 25% arises for sizes 56-100 and 10% for size 132. The base is an inlet temperature of 35°C at both cooling mediums and an identical volume flow rate. The decrease of pressure is higher if using the hydraulic oil. Specific power ratings are available on request.

1.4.2. Min. coolant temperature against ambient temperature and humidity



The allowed coolant temperature depends on relative humidity and ambient temperature. For example with an ambient temperature of 25 °C and a relative humidity of 65% the minimum coolant temperature is 18 °C. Because these are limiting values on practical side a coolant temperature greater than 18 °C should be used. If this minimum coolant temperature will be under run the two- point controller of Baumüller drive must be used to avoid condensation.

**Note:**

The supply of cooling fluid must be interrupted to prevent condensation when storing for an extended period. In addition, at ambient temperatures $< 3^{\circ}\text{C}$ and if the motor has not run for an extended period, drain the cooling fluid to prevent damage caused by frost. When using anti-freeze you need to consult the manufacturer.

1.4.3. Specifications for required coolant volume flow

Motor type	Volume flow [l/min]	Pressure decrease ± 15 % [bar]	Heating [K]	Max. coolant pressure [bar]	Connection (2x) [mm]
DSD2-045SO..W	5	1,0	2	6	Quick coupling for the hose connection (Ø8 mm)
DSD2-045MO..W	5	1,2	2,5	6	Quick coupling for the hose connection (Ø8 mm)
DSD2-045LO..W	5	1,3	3	6	Quick coupling for the hose connection (Ø8 mm)

Motor type	Volume flow [l/min]	Pressure decrease ± 15 % [bar]	Heating [K]	Max. coolant pressure [bar]	Connection (2x) [mm]
DSD2-056SO..W	5	0.94	3	6	Quick coupling for the hose connection (Ø8 mm)
DSD2-056MO..W	5	0.96	4	6	Quick coupling for the hose connection (Ø8 mm)
DSD2-056LO..W	5	0.98	5	6	Quick coupling for the hose connection (Ø8 mm)

Motor type	Volume flow [l/min]	Pressure decrease ± 15 % [bar]	Heating [K]	Max. coolant pressure [bar]	Connection (2x) [mm]
DSD2-071SO..W	5	0.5	6	6	stainless steel tube Ø8x1
DSD2-071MO..W	5	0.65	7	6	stainless steel tube Ø8x1
DSD2-071LO..W	5	0.8	8	6	stainless steel tube Ø8x1

Motor type	Volume flow [l/min]	Pressure decrease ± 15 % [bar]	Heating [K]	Max. coolant pressure [bar]	Connection [mm]
DSD2-100SO..W	5	0.55	8	6	stainless steel tube Ø8x1
DSD2-100MO..W	5	0.65	9	6	stainless steel tube Ø8x1
DSD2-100LO..W	5	0.85	10	6	stainless steel tube Ø8x1
DSD2-100BO..W	5	1.0	12	6	stainless steel tube Ø8x1
DSD2-100XO..W	5	1.55	15	6	stainless steel tube Ø8x1

Motor type	Volume flow [l/min]	Pressure decrease ± 15 % [bar]	Heating [K]	Max. coolant pressure [bar]	Connection
DSD2-132KO..W	15	0.35	3	6	Female thread G1/2"
DSD2-132MO..W	15	0.35	4	6	Female thread G1/2"
DSD2-132LO..W	15	0.35	4	6	Female thread G1/2"
DSD2-132BO..W	15	0.35	5	6	Female thread G1/2"
DSD2-132XO..W	15	0.35	5	6	Female thread G1/2"
DSD2-132YO..W	15	0.35	6	6	Female thread G1/2"
DSD2-132YZ..W	15	0.35	6	6	Female thread G1/2"

Controlling the feed valve individually is possible, depending on the motor temperature measured by the temperature sensor.

Note:

The given cooling volume flows relate to the highest rotary speed of the relevant motor lengths.
It is possible to make an individual cooling unit evaluation on the basis of the motors power loss ($P_V = P_N / \eta_N - P_N$). The cooling unit should be scaled so that its cooling performance matches the motor power loss and so that 100% of the waste heat is diffused by the unit.

Enough additives for corrosion and germ protection must be mixed in. The additive type and dosage are based on recommendations from the additive manufacturer and the prevailing ambient conditions.

A lowering of the specific heating capacity leads to an output reduction in relation to the mixing ratio which should be enquired at the manufacturer.

1.4.4. Materials in the motor that contact the medium

The following materials that contact the medium are used in the motor:

Sizes 45 und 56:

Cooling system: Aluminum KTL coated.

Connections: In general, the motors are equipped with a quick coupling for a hose connection (Ø8 mm) in brass.

Seals: FPM

Sizes 71 and 100::

Cooling system: stainless steel

Water connections: According to standard, the motors are supplied with a stainless-steel tube Ø8x1 mm without additional connection technology. The water connection with the John Guest - quick connector SM 040 808 S can be optionally provided (dia 8 by dia 8). The water connection with the John Guest - quick connector SM 040808 S can be optionally provided (Ø8 mm by Ø8 mm) or equipped by a compression fitting Ø8L x G1/4". Please include this option including the order code when ordering.

Size 132:

Cooling system: Aluminum KTL coated.

Connections: Galvanized steel; Internal thread G1/2"

Seals: NBR

1.5. Winding insulation

DSD2-028-036:

The motors are designed for an operation on converters with intermediate link voltages of up to 540V + 10%.

DSD2-045-132:

The motors are designed for an operation on converters with intermediate link voltages of up to 640V. Higher intermediate link voltages of up to $\leq 800V$ are possible, if voltage spikes on the motor terminals are limited to $<1200V$ by suitable filters in the motor supply line.

1.6. Explanation of motor data

n_N	Rated speed [rpm]
M_0	Nominal torque [Nm] with speeds ≥ 1 [rpm] without time limit
I_0	nominal current [A] at M_0
$M_{0,max}$	Maximum static torque [Nm] with maximum current [A] and speed = 0. momentarily
$I_{0,max}$	Static current [A] at $M_{0,max}$; $I_{0,max}$ is the effective value
P_N	Rated output [kW] with M_N and n_N (see Performance definition)
M_N	Rated torque [Nm]
I_N	Rated effective current [A]
k_E / cold	Voltage constant (EMF) to [V per 1000 rpm]
f_N	Rated frequency [Hz]
J	Rotor inertia incl. resolver without holding brake [kgm ²]
m	Motor mass [kg]

When the converter is operating, the specified rated outputs and torques at the rated speed are achieved with a clocking frequency of ≥ 4 kHz in the power divider. A clocking frequency of > 6 kHz is recommended. All converters scheduled for use must have the option of field weakening as a mandatory requirement.

The **sizemaXX** drive configurator is available at www.baumueller.de for designing the motors and the overall drive system.

1.7. Type code

DSD2 -XXXXXXXX-XX-XX-XXX-XXX-X-XX-X-XXX	Type
DSD2- <u>XXX</u> XXXXXX-XX-XX-XXX-XXX-X-XX-X-XXX	Overall size 028 036 045 056 071 100 132
DSD2-XXXX <u>XX</u> XXX-XX-XX-XXX-XXX-X-XX-X-XXX	Overall length KO SO MO LO BO XO YO YZ
DSD2-XXXXXX <u>XX</u> -XX-XX-XXX-XXX-X-XX-X-XXX	Degree of protection 44 - Degree of protection IP44 54 - Degree of protection IP54 64 - Degree of protection IP64 65 - Degree of protection IP65
DSD2-XXXXXXX <u>X</u> -XX-XX-XXX-XXX-X-XX-X-XXX	Cooling type U - Without fan O - With fan, 230 VAC supply voltage for DSD2-056-100 A – axially fitted external fan R - radially fitted external fan W - Water cooling
DSD2-XXXXXXX- <u>XX</u> -XX-XXX-XXX-X-XX-X-XXX	Nominal speed class 10 – 1000 rpm 12 - 1200 rpm 15 - 1500 rpm 20 - 2000 rpm 25 - 2500 rpm 30 - 3000 rpm 40 - 4000 rpm 45 - 4500 rpm 60 - 6000 rpm
DSD2-XXXXXXX-XX- <u>XXI</u> -XXX-XXX-X-XX-X-XXX	Uzk_ DC 31 - 310 V 54 - 540 V 5 - 540 V; at DSD2-132

DSD2-XXXXXXXX-XX-XX- <u>X</u> XX-XXX-X-XX-X-XXX	Encoder type O - No encoder A - Resolver B - SEK52 C - SEL52 D - SRS50 E - SRM50 F - ECN1313 G - EQN1325 H - ECN1325 I - EQN1337 J - SEK37 (Size 028 / 036) K - SEL37 (Size 028 / 036) M - Resolver (Safety) N - SRS50-S (Safety) Q - SRM50-S (Safety) R - SKS36 S - SKS36-S (Safety) U - SKM36-S (Safety) X - EQI1331 Y - ECI1319 5 - ECN1325-S (Safety) 6 - EQN1337-S (Safety) a - EKS36 Hiperface DSL b - EKM36 Hiperface DSL g - EFS50 Hiperface DSL h - EFM50 Hiperface DSL p - ECI1319-S (Safety) q - EQI1331-S (Safety) r – EES37 Hiperface DSL s – EEM37 Hiperface DSL t - SEK37 u - SEL37 *On request
DSD2-XXXXXXXX-XX-XX- <u>X</u> XX-XXX-X-XX-X-XXX	Brake O - Without brake B - With PE-brake
DSD2-XXXXXXXX-XX-XX- <u>X</u> XX-XXX-X-XX-X-XXX	Shaft options A - Smooth shaft B - With parallel key
DSD2-132XXXXX-XX-X-XXX- <u>X</u> -XXX-XXX-XXX-XXX	Type of design* 1 - IM B3 2 - IM B5 7 - IM B35 8 - IM B35 flange 400 mm (size 132) * Coding at DSD2-132 only

DSD2-XXXXXXXX-XX-XX-XXX- <u>X</u> XX-X-XX-X-XXX	Type main connection M – Terminal box (PT1000 on main connection) N – Terminal box (PT1000 on encoder socket) B - Speedtec (PT1000 on main connection) D - Speedtec (PT1000 on encoder socket) U – Combi angle socket (PT1000 on main connection) X - Combi angle socket (PT1000 on encoder socket)
DSD2-XXXXXXXX-XX-XX-XXX- <u>XX</u> X-X-XX-X-XXX	Main outlet port T - Top L - Left with a view toward D-side on shaft end R - Right with a view toward D-side on shaft end D - DE (D-side) N - NDE (N-side) P - Pivoted Position main connection* * Coding at DSD2-132, only
DSD2-XXXXXXXX-XX-XX-XXX- <u>XX</u> X-X-XX-X-XXX	Encoder connection outlet O – without encoder socket T - Top L - Left with a view toward D-side on shaft end R - Right with a view toward D-side on shaft end D - DE (D-side) N - NDE (N-side) P – Pivoted Outlet main connection* * Coding at DSD2-132, only
DSD2-XXXXXXXX-XX-XX-XXX-XXX- <u>X</u> []XX-X-XXX	Bearing K - Ball bearing D-side R - Roller bearing D-side * No hyphen at DSD2-132
DSD2-XXXXXXXX-XX-XX-XXX-XXX-X- <u>X</u> X-X-XXX	Vibration level A - Vibration level A B - Vibration level B
DSD2-XXXXXXXX-XX-XX-XXX-XXX-X- <u>X</u> X-X-XXX	True running N - Normal R - Reduced
DSD2-132XXXXX-XX-X-XXX-XXX-X-XX- <u>[X]</u> XX-XXX	Position cooling* T - Top B - Bottom L - Left with viewing direction D-side R - Right with viewing direction D-side A - Axial fan * Coding at DSD2-132, only

DSD2-132XXXXX-XX-X-XXX-XXX-X-XX-X <u>X</u> X-XXX	Option cooling* O - without cooling option (for water cooling) J - Flat filter with filter mat rough M – Rectangular filter with filter mat rough Q - Without filter (for Δ/Y 265-345V / 460-600V) S - Flat filter with filter mat rough Version Δ/Y 265-345V / 460-600V U - Rectangular filter with filter mat rough Version Δ/Y 265-345V / 460-600V * Coding at DSD2-132, only
DSD2-XXXXXXXX-XX-XX-XXX-XXX-X-XX- <u>X</u> -XXX	Gear box / pump mounting O - Without transmission mount and without pump mounting A - BPE - Gear box B - BPEF - Gear box C - BPEA - Gear box D - BPN - Gear box E - BPNA - Gear box F - BPNF - Gear box G - BPV - Gear box H - BPVF - Gear box
DSD2-XXXXXXXX-XX-XX-XXX-XXX-X-XX-X- <u>XXX</u>	Further types 000 - No special version AJ1 - Water connection with a John-Guest-connector AP1 – Water connection with cutting ring fitting DA1 – Standard shaft sealing ring OS1 – Fan with 115 VAC supply voltage OS2 - Fan with 24 VDC supply voltage XXX - Special version (internal coding) Special coding is made alphanumeric

Example configuration:

DSD2-132LO54A-15-5-AOA-8-KTR-KAN-AAO-000

DSD2-132LO64W-15-54-AOB-BPP-K-AN-O-000

Motor with several expanded design attributes, such as shaft seal and fan with 24 VDC supply voltage:

DSD2-071SO64O-10-54-AOA-BPP-K-AN-O+DA1

2. Technical data

2.1. DSD2-028

DSD2-028..44U-.. (without fan)

1 AC 230 V mains voltage for converters with unregulated supply

Rated speed	Motor type	Stand-still torque ¹⁾	Stand-still current ¹⁾	max. static torque	Max. static current	Rated out-put ¹⁾	Rated torque ¹⁾	Rated current ¹⁾	Voltage constant	Rated frequency	Rotor inertia (motor) ²⁾	Weight ³⁾
n _N rpm		M ₀ Nm	I ₀ A	M _{0,max} Nm	I _{0,max} A	P _N kW	M _N Nm	I _N A	k _{E/cold} V/10001/min	f _N Hz	J kgcm ²	m kg
4500	DSD2-028SO44U-45-31	0.7	1.55	2	6.1	0.28	0.6	1.4	31	300	0.13	1.5
	DSD2-028MO44U-45-31	1.2	2.6	3.9	11.9	0.47	1	2.3	28.5	300	0.2	2
6000	DSD2-028SO44U-60-31	0.7	1.9	2	7.5	0.35	0.55	1.65	22.5	400	0.13	1.5
	DSD2-028MO44U-60-31	1.2	3	3.9	13.7	0.57	0.9	2.4	24.7	400	0.2	2

3 AC 400 V mains voltage for converters with unregulated supply

Rated speed	Motor type	Stand-still torque ¹⁾	Stand-still current ¹⁾	max. static torque	Max. static current	Rated out-put ¹⁾	Rated torque ¹⁾	Rated current ¹⁾	Voltage constant	Rated frequency	Rotor inertia (motor) ²⁾	Weight ³⁾
n _N rpm		M ₀ Nm	I ₀ A	M _{0,max} Nm	I _{0,max} A	P _N kW	M _N Nm	I _N A	k _{E/cold} V/10001/min	f _N Hz	J kgcm ²	m kg
4500	DSD2-028SO44U-45-54	0.7	1	2	4	0.28	0.6	0.9	42.4	300	kgcm ²	1.5
	DSD2-028MO44U-45-54	1.2	1.7	3.9	7.8	0.47	1	1.5	43.5	300	0.13	2
6000	DSD2-028SO44U-60-54	0.7	1	2	4	0.35	0.55	0.85	42.4	400	0.2	1.5
	DSD2-028MO44U-60-54	1.2	1.7	3.9	7.8	0.57	0.9	1.35	43.5	400	0.13	2

¹⁾ Coil overtemperature $\Delta T < 105K$; direct flange mounting (mounting plate 250 mm x 250 mm x 10 mm)

²⁾ Rotor moment of inertia with PE brake: +0.068 kgcm²

³⁾ Weight with PE brake: +0.7 kg

2.2. DSD2-036

DSD2-036..44U-.. (without fan)

1 AC 230 V mains voltage for converters with unregulated supply

Rated speed	Motor type	Stand-still torque ¹⁾	Stand-still current ¹⁾	max. static torque	Max. static current	Rated output ¹⁾	Rated torque ¹⁾	Rated current ¹⁾	Voltage constant	Rated frequency	Rotor inertia (motor) ²⁾	Weight ³⁾
n _N rpm		M ₀ Nm	I ₀ A	M _{0,max} Nm	I _{0,max} A	P _N kW	M _N Nm	I _N A	k _{E/cold} V/10001/min	f _N Hz	J kgcm ²	m kg
4000	DSD2-036SO44U-40-31	1.2	2.4	2.8	7.9	0.44	1.05	2.2	31.3	267	0.18	2.5
	DSD2-036MO44U-40-31	2	3.6	5.7	15.1	0.67	1.6	2.9	34.3	267	0.3	3.3
4500	DSD2-036LO44U-45-31	2.8	5	8.4	22	0.85	1.8	3.45	33.6	300	0.42	4.1
6000	DSD2-036SO44U-60-31	1.2	3.55	2.8	12	0.57	0.9	2.7	21.3	400	0.18	2.5
	DSD2-036MO44U-60-31	2	5	5.7	21	0.82	1.3	3.35	24.9	400	0.3	3.3
	DSD2-036LO44U-60-31	2.8	6.15	8.4	27.2	0.9	1.45	3.45	27.3	400	0.42	4.1

3 AC 400 V mains voltage for converters with unregulated supply

Rated speed	Motor type	Stand-still torque ¹⁾	Stand-still current ¹⁾	max. static torque	Max. static current	Rated output ¹⁾	Rated torque ¹⁾	Rated current ¹⁾	Voltage constant	Rated frequency	Rotor inertia (motor) ²⁾	Weight ³⁾
n _N rpm		M ₀ Nm	I ₀ A	M _{0,max} Nm	I _{0,max} A	P _N kW	M _N Nm	I _N A	k _{E/cold} V/10001/min	f _N Hz	J kgcm ²	m kg
4000	DSD2-036SO44U-40-54	1.2	1.45	2.8	4.7	0.44	1.05	1.35	52.7	267	0.18	2.5
4500	DSD2-036MO44U-45-54	2	2.55	5.7	10.7	0.71	1.5	1.95	48.6	300	0.3	3.3
	DSD2-036LO44U-45-54	2.8	3.45	8.4	15.3	0.85	1.8	2.3	49.3	300	0.42	4.1
6000	DSD2-036SO44U-60-54	1.2	2.05	2.8	7	0.57	0.9	1.5	37.4	400	0.18	2.5
	DSD2-036MO44U-60-54	2	2.95	5.7	12	0.82	1.3	2	42.1	400	0.3	3.3
	DSD2-036LO44U-60-54	2.8	3.85	8.4	17	0.9	1.45	2.15	43.8	400	0.42	4.1

¹⁾ Coil overtemperature $\Delta T < 105K$; direct flange mounting (mounting plate 250 mm x 250 mm x 10 mm)

²⁾ Rotor moment of inertia with PE brake: +0.18 kgcm²

³⁾ Weight with PE brake: +0.9 kg

2.3. DSD2-045

DSD2-045..64U-.. (without fan)

3 AC 400 V mains voltage for converters with unregulated supply

Rated speed	Motor type	Stand-still torque ¹⁾	Stand-still current ¹⁾	max. static torque	Max. static current	Rated output ¹⁾	Rated torque ¹⁾	Rated current ¹⁾	Voltage constant	Rated frequency	Rotor inertia (motor) ²⁾	Weight ³⁾
n_N rpm		M_0 Nm	I_0 A	$M_{0,max}$ Nm	$I_{0,max}$ A	P_N kW	M_N Nm	I_N A	$k_E/cold$ V/10001/min	f_N Hz	J kgcm ²	m kg
3000	DSD2-045SO64U-30-54	2.7	1.75	12	8.9	0.73	2.3	1.56	102	200	1,0	4.2
	DSD2-045MO64U-30-54	4.3	2.7	20	14.1	1.2	3.8	2.5	105	200	1,5	5.3
	DSD2-045LO64U-30-54	5.8	3.7	28	19.6	1.65	5.2	3.3	105	200	1,9	6.4
4500	DSD2-045SO64U-45-54	2.7	2.5	12	12.7	0.98	2.1	2	71.3	300	1,0	4.2
	DSD2-045MO64U-45-54	4.3	3.9	20	20.4	1.6	3.4	3.2	73	300	1,5	5.3
	DSD2-045LO64U-45-54	5.8	5.3	28	28.3	2.2	4.6	4.3	73	300	1,9	6.4
6000 ⁴⁾	DSD2-045SO64U-60-54	2.7	3.3	12	16.7	1.13	1.8	2.3	54.3	400	1,0	4.2
	DSD2-045MO64U-60-54	4.3	5.2	20	26.8	1.8	2.9	3.6	55.5	400	1,5	5.3
	DSD2-045LO64U-60-54	5.8	6.9	28	36.4	2.4	3.8	4.7	56.8	400	1,9	6.4

DSD2-045..64W-.. (water-cooled)

3 AC 400 V mains voltage for converters with uncontrolled supply

Rated speed	Motor type	Stand-still torque ¹⁾	Stand-still current ¹⁾	max. static torque	Max. static current	Rated output ¹⁾	Rated torque ¹⁾	Rated current ¹⁾	Voltage constant	Rated frequency	Rotor inertia (motor) ²⁾	Weight ³⁾
n_N min ⁻¹		M_0 Nm	I_0 A	$M_{0,max}$ Nm	$I_{0,max}$ A	P_N kW	M_N Nm	I_N A	$K_E/cold$ V/1000 min ⁻¹	f_N Hz	J kgcm ²	m kg
3000	DSD2-045SO64W-30-54	5,6	4,4	12,2	10,7	1,58	5	4,3	84,6	200	1,0	5
	DSD2-045MO64W-30-54	9,5	7	20	16,7	2,9	9,3	6,9	89	200	1,5	6,3
	DSD2-045LO64W-30-54	13	9,4	26	22,1	4	13,3	9,3	89,5	200	1,9	7,5
4500	DSD2-045SO64W-45-54	5,6	6,2	12,1	15,3	2,2	4,8	6,1	59,2	300	1,0	5
	DSD2-045MO64W-45-54	9,5	10	20	24	4,3	9,3	9,8	62	300	1,5	6,3
	DSD2-045LO64W-45-54	13	13,5	26	31,8	5,8	13	13,4	62,2	300	1,9	7,5
6000 ⁴⁾	DSD2-045SO64W-60-54	5,6	8,3	12	20,4	2,8	4,4	8	44,5	400	1,0	5
	DSD2-045MO64W-60-54	9,5	13	20	31,3	5,6	8,9	12,7	47,4	400	1,5	6,3
	DSD2-045LO64W-60-54	13	18	26	42,4	7,6	12,8	17,7	46,7	400	1,9	7,5

¹⁾ Coil overtemperature $\Delta T < 105K$; direct flange mounting (mounting plate 250 mm x 250 mm x 10 mm)

²⁾ Rotor moment of inertia with PE brake: +0.6 kgcm²

³⁾ Weight with PE brake: +1.0 kg

⁴⁾ In combination with Heidenhain absolute signal encoders available on request

2.4. DSD2-056

DSD2-056..64U-.. (without fan)

3 AC 400 V mains voltage for converters with unregulated supply

Rated speed	Motor type	Stand-still torque ¹⁾	Stand-still current ¹⁾	max. static torque	Max. static current	Rated out-put ¹⁾	Rated torque ¹⁾	Rated current ¹⁾	Voltage constant	Rated frequency	Rotor inertia (motor) ²⁾	Weight ³⁾
n _N rpm		M ₀ Nm	I ₀ A	M _{0,max} Nm	I _{0,max} A	P _N kW	M _N Nm	I _N A	k _{E/cold} V/10001/min	f _N Hz	J kgcm ²	m kg
2000	DSD2-056SO64U-20-54	7	3.5	25	13.7	1.28	6.1	3.1	134	133.3	3,6	8.3
	DSD2-056MO64U-20-54	11	5	41	21	1.95	9.3	4.3	144	133.3	5,1	10.6
	DSD2-056LO64U-20-54	14	6.4	57	28.9	2.3	11	5.2	146	133.3	6,6	13.0
3000	DSD2-056SO64U-30-54	7	4.9	25	19.1	1.77	5.6	4	96.2	200	3,6	8.3
	DSD2-056MO64U-30-54	11	7.2	41	30.3	2.7	8.4	5.7	100	200	5,1	10.6
	DSD2-056LO64U-30-54	14	9.3	57	42	3	9.6	6.5	100	200	6,6	13.0
4500	DSD2-056SO64U-45-54	7	7.2	25	28.3	2.3	5	5.2	65	300	3,6	8.3
	DSD2-056MO64U-45-54	11	10.6	41	44.9	3.3	7	7	67.4	300	5,1	10.6
	DSD2-056LO64U-45-54	14	13.2	56	59	3.3	6.9	6.9	71.2	300	6,6	13.0
6000 ⁴⁾	DSD2-056SO64U-60-54	7	9.2	25	36.1	2.7	4.3	5.9	50.9	400	3,6	8.3
	DSD2-056MO64U-60-54	11	13.7	41	58	3.4	5.4	7	52.3	400	5,1	10.6
	DSD2-056LO64U-60-54	14	17	56	77	2.5	4	5.6	55	400	6,6	13.0

DSD2-056..64O-.. (with fan)

3 AC 400 V mains voltage for converters with unregulated supply

Rated speed	Motor type	Stand-still torque ¹⁾	Stand-still current ¹⁾	max. static torque	Max. static current	Rated out-put ¹⁾	Rated torque ¹⁾	Rated current ¹⁾	Voltage constant	Rated frequency	Rotor inertia (motor) ²⁾	Weight ³⁾
n _N rpm		M ₀ Nm	I ₀ A	M _{0,max} Nm	I _{0,max} A	P _N kW	M _N Nm	I _N A	k _{E/cold} V/10001/min	f _N Hz	J kgcm ²	m kg
2000	DSD2-056SO64O-20-54	10	5	25	13.7	1.97	9.4	4.7	134	133.3	3,6	10.9
	DSD2-056MO64O-20-54	15	7.1	41	21	3	14	6.5	144	133.3	5,1	13.3
	DSD2-056LO64O-20-54	20	9.1	57	28.9	3.8	18	8.3	146	133.3	6,6	15.8
3000	DSD2-056SO64O-30-54	10	6.9	25	19.1	2.9	9.1	6.3	96.2	200	3,6	10.9
	DSD2-056MO64O-30-54	15	10.2	41	30.3	4.3	14	9	100	200	5,1	13.3
	DSD2-056LO64O-30-54	20	13.2	57	42	5.3	17	11.3	100	200	6,6	15.8
4500	DSD2-056SO64O-45-54	10	10.2	25	28.3	4	8.5	8.8	65	300	3,6	10.9
	DSD2-056MO64O-45-54	15	15.2	41	44.9	5.9	13	12.4	67.4	300	5,1	13.3
	DSD2-056LO64O-45-54	20	18.6	56	59	7	15	14.1	71.2	300	6,6	15.8
6000 ⁴⁾	DSD2-056SO64O-60-54	10	13.1	25	36.1	5	8	10.6	50.9	400	3,6	10.9
	DSD2-056MO64O-60-54	15	19.5	41	58	7.2	11	14.7	52.3	400	5,1	13.3
	DSD2-056LO64O-60-54	20	24	56	77	7.8	12	15.5	55	400	6,6	15.8

¹⁾ Coil overtemperature $\Delta T < 105K$; direct flange mounting (mounting plate 450 mm x 400 mm x 30 mm)

²⁾ Rotor moment of inertia with PE brake: +2.9 kgcm²

³⁾ Weight with PE brake: +2.0 kg

⁴⁾ in combination with Heidenhain absolute signal encoders available on request

DSD2-056..64W-.. (water cooled)**3 AC 400 V mains voltage for converters with unregulated supply**

Rated speed	Motor type	Stand-still torque ¹⁾	Stand-still current ¹⁾	max. static torque	Max. static current	Rated out-put ¹⁾	Rated torque ¹⁾	Rated current ¹⁾	Voltage constant	Rated frequency	Rotor inertia (motor) ²⁾	Weight ³⁾
n_N		M_0	I_0	$M_{0,max}$	$I_{0,max}$	P_N	M_N	I_N	$k_E/cold$	f_N	J	m
rpm		Nm	A	Nm	A	kW	Nm	A	V/10001/min	Hz	kgcm ²	kg
2000	DSD2-056SO64W-20-54	14	7.9	25	15.8	2.8	13	7.6	117	133.3	3,6	9.9
	DSD2-056MO64W-20-54	22	12.1	41	24.8	4.4	21	11.4	122	133.3	5,1	13.1
	DSD2-056LO64W-20-54	30	15.9	57	33.4	5.9	28	14.8	126	133.3	6,6	16.3
3000	DSD2-056SO64W-30-54	14	11.2	25	22.2	4	13	10.4	82.7	200.0	3,6	9.9
	DSD2-056MO64W-30-54	22	17.4	41	35.6	6.4	20	15.8	84.9	200.0	5,1	13.1
	DSD2-056LO64W-30-54	30	22.9	57	48.2	8.3	26	20.2	87.4	200.0	6,6	16.3
4500	DSD2-056SO64W-45-54	14	15.9	25	31.7	5.8	12	14.4	58	300.0	3,6	9.9
	DSD2-056MO64W-45-54	22	24.4	41	50	8.8	19	20.5	60.4	300.0	5,1	13.1
	DSD2-056LO64W-45-54	30	32.6	56	68	11	23	25.2	61.5	300.0	6,6	16.3
6000	DSD2-056SO64W-60-54	14	21.1	25	42	7.5	12	18.3	43.8	400.0	3,6	9.9
	DSD2-056MO64W-60-54	22	31.8	41	65	11	17	24.1	46.5	400.0	5,1	13.1
	DSD2-056LO64W-60-54	30	39.9	56	84	12	19	25.5	50.2	400.0	6,6	16.3

¹⁾ Coil overtemperature $\Delta T < 105K$; direct flange mounting (mounting plate 450 mm x 400 mm x 30 mm)

²⁾ Rotor moment of inertia with PE brake: +2.9 kgcm²

³⁾ Weight with PE brake: +2.0 kg

2.5. DSD2-071

DSD2-071..64U-.. (without fan)

3 AC 400 V mains voltage for converters with unregulated supply

Rated speed	Motor type	Stand-still torque ¹⁾	Stand-still current ¹⁾	max. static torque	Max. static current	Rated out-put ¹⁾	Rated torque ¹⁾	Rated current ¹⁾	Voltage constant	Rated frequency	Rotor inertia (motor) ²⁾	Weight ³⁾
n_N rpm		M_0 Nm	I_0 A	$M_{0,max}$ Nm	$I_{0,max}$ A	P_N kW	M_N Nm	I_N A	$k_E/cold$ V/10001/min	f_N Hz	J kgcm ²	m kg
2000	DSD2-071SO64U-20-54	17	7.8	53	28.3	3	14	6.7	143	133.3	11,7	15.8
	DSD2-071MO64U-20-54	22	10.2	79	41.3	3.8	18	8.4	146	133.3	15,3	20.0
	DSD2-071LO64U-20-54	26	11.7	105	54	4.2	20	9.1	149	133.3	18,9	24.1
3000	DSD2-071SO64U-30-54	17	11.4	53	41.3	4.1	13	9	97.8	200	11,7	15.8
	DSD2-071MO64U-30-54	22	14.7	79	60	5.1	16	10.9	101	200	15,3	20.0
	DSD2-071LO64U-30-54	26	17.1	105	79	5.4	17	11.3	102	200	18,9	24.1
4500	DSD2-071SO64U-45-54	17	17.1	53	62	5.3	11	11.6	65.2	300	11,7	15.8
	DSD2-071MO64U-45-54	22	21.4	79	87	6.2	13	13	69	300	15,3	20.0
	DSD2-071LO64U-45-54	26	25.1	105	115	5.9	13	12.4	69.6	300	18,9	24.1
6000 ⁴⁾	DSD2-071SO64U-60-54	17	22.3	53	81	5.8	9.2	12.6	50.2	400	11,7	15.8
	DSD2-071MO64U-60-54	22	28.3	79	115	6.3	10	13.3	52.2	400	15,3	20.0
	DSD2-071LO64U-60-54	26	31.9	105	147	5.1	8.1	10.5	54.7	400	18,9	24.1

DSD2-071..64O-.. (with fan)

3 AC 400 V mains voltage for converters with unregulated supply

Rated speed	Motor type	Stand-still torque ¹⁾	Stand-still current ¹⁾	max. static torque	Max. static current	Rated out-put ¹⁾	Rated torque ¹⁾	Rated current ¹⁾	Voltage constant	Rated frequency	Rotor inertia (motor) ²⁾	Weight ³⁾
n_N rpm		M_0 Nm	I_0 A	$M_{0,max}$ Nm	$I_{0,max}$ A	P_N kW	M_N Nm	I_N A	$k_E/cold$ V/10001/min	f_N Hz	J kgcm ²	m kg
2000	DSD2-071SO64O-20-54	24	11.2	53	28.3	4.7	22	10.5	143	133.3	11,7	18.6
	DSD2-071MO64O-20-54	34	15.7	79	41.3	6.6	31	14.3	146	133.3	15,3	22.6
	DSD2-071LO64O-20-54	44	19.6	105	54	8.1	39	17.5	149	133.3	18,9	26.9
3000	DSD2-071SO64O-30-54	24	16.4	53	41.3	6.8	22	14.8	97.8	200	11,7	18.6
	DSD2-071MO64O-30-54	34	22.6	79	60	9.4	30	19.7	101	200	15,3	22.6
	DSD2-071LO64O-30-54	44	28.7	105	79	11	36	24	102	200	18,9	26.9
4500	DSD2-071SO64O-45-54	24	24.6	53	62	9.7	21	21	65.2	300	11,7	18.6
	DSD2-071MO64O-45-54	34	33	79	87	13	28	26.7	69	300	15,3	22.6
	DSD2-071LO64O-45-54	44	42	105	115	15	32	31.3	69.6	300	18,9	26.9
6000 ⁴⁾	DSD2-071SO64O-60-54	24	32	53	81	12	19	26	50.2	400	11,7	18.6
	DSD2-071MO64O-60-54	34	43.6	79	115	16	25	32.6	52.2	400	15,3	22.6
	DSD2-071LO64O-60-54	44	53	105	147	18	28	34.6	54.7	400	18,9	26.9

¹⁾ Coil overtemperature $\Delta T < 105K$; direct flange mounting (mounting plate 450 mm x 400 mm x 30 mm)

²⁾ Rotor moment of inertia with PE brake: +7.9 kgcm²

³⁾ Weight with PE brake: +3.0 kg

⁴⁾ in combination with Heidenhain absolute signal encoders available on request

DSD2-071..64W-.. (water cooled)

3 AC 400 V mains voltage for converters with unregulated supply

Rated speed	Motor type	Stand-still torque ¹⁾	Stand-still current ¹⁾	max. static torque	Max. static current	Rated out-put ¹⁾	Rated torque ¹⁾	Rated current ¹⁾	Voltage constant	Rated frequency	Rotor inertia (motor) ²⁾	Weight ³⁾
n_N		M_0	I_0	$M_{0,max}$	$I_{0,max}$	P_N	M_N	I_N	$k_E/cold$	f_N	J	m
rpm		Nm	A	Nm	A	kW	Nm	A	V/10001/min	Hz	kgcm ²	kg
2000	DSD2-071SO64W-20-54	35	19.8	53	33.6	6.8	33	18.3	121	133.3	11,7	19
	DSD2-071MO64W-20-54	54	29.6	78	48.9	10	49	26.7	124	133.3	15,3	23
	DSD2-071LO64W-20-54	73	40.2	105	64	14	65	35.5	124	133.3	18,9	27
3000	DSD2-071SO64W-30-54	35	28.4	53	48.1	9.7	31	24.9	84.3	200	11,7	19
	DSD2-071MO64W-30-54	54	42.5	78	70	14	46	35.9	86.1	200	15,3	23
	DSD2-071LO64W-30-54	73	57	105	92	19	60	46.7	87	200	18,9	27
4500	DSD2-071SO64W-45-54	35	41.4	53	70	13	28	33	57.9	300	11,7	19
	DSD2-071MO64W-45-54	54	61	78	101	19	40	45.6	59.9	300	15,3	23
	DSD2-071LO64W-45-54	73	80	105	129	24	52	56	62.2	300	18,9	27
6000	DSD2-071SO64W-60-54	35	53	53	90	16	25	37.3	45.3	400	11,7	19
	DSD2-071MO64W-60-54	54	78	78	129	22	34	49.5	46.8	400	15,3	23
	DSD2-071LO64W-60-54	69	95	105	160	25	40	55	49.7	400	18,9	27

¹⁾ Coil overtemperature $\Delta T < 105K$; direct flange mounting (mounting plate 450 mm x 400 mm x 30 mm)

²⁾ Rotor moment of inertia with PE brake: +7.9 kgcm²

³⁾ Weight with PE brake: +3.0 kg

2.6. DSD2-100

DSD2-100..64U-.. (without fan)

3 AC 400 V mains voltage for converters with unregulated supply

Rated speed	Motor type	Stand-still torque ¹⁾	Stand-still current ¹⁾	max. static torque	Max. static current	Rated out-put ¹⁾	Rated torque ¹⁾	Rated current ¹⁾	Voltage constant	Rated fre-quency	Rotor inertia (motor) ²⁾	Weight ³⁾
n_N rpm		M_0 Nm	I_0 A	$M_{0,max}$ Nm	$I_{0,max}$ A	P_N kW	M_N Nm	I_N A	$k_E/cold$ V/10001/min	f_N Hz	J kgcm ²	m kg
1200	DSD2-100SO64U-12-54	42	11.8	105	35	4.2	33	9.4	240	80	52	36
	DSD2-100MO64U-12-54	60	16.5	160	51	5.8	46	12.8	245	80	70	46
	DSD2-100LO64U-12-54	76	20.5	210	66	7.1	57	15.3	251	80	88	56
	DSD2-100BO64U-12-54	90	24.4	265	83	8.1	65	17.6	250	80	105	66
	DSD2-100XO64U-12-54	95	25.3	380	143	8.7	69	18.7	258	80	139	73.2
2000	DSD2-100SO64U-20-54	42	19.1	105	57	5.9	28	13	148	133.3	52	36
	DSD2-100MO64U-20-54	60	26.2	160	81	7.9	38	16.7	154	133.3	70	46
	DSD2-100LO64U-20-54	76	33.2	210	108	9.4	45	19.8	155	133.3	88	56
	DSD2-100BO64U-20-54	90	39.1	265	133	10	49	21.5	156	133.3	105	66
	DSD2-100XO64U-20-54	95	38	380	215	10	48	19.7	172	133.3	139	73.2
3000	DSD2-100SO64U-30-54	42	26.8	105	80	6.9	22	14.3	105	200	52	36
	DSD2-100MO64U-30-54	60	37.8	160	117	8.8	28	18.1	107	200	70	46
	DSD2-100LO64U-30-54	76	47.2	210	155	9.8	31	19.8	109	200	88	56
	DSD2-100BO64U-30-54	90	56	265	190	9.8	31	19.8	109	200	105	66
	DSD2-100XO64U-30-54	95	53	375	300	5.1	16.3	10.1	123	200	139	73.2
4500	DSD2-100SO64U-45-54	42	39.4	105	117	6.5	14	13.5	71.7	300	52	36
	DSD2-100MO64U-45-54	60	56	160	175	7.2	15	15	72.3	300	70	46
	DSD2-100LO64U-45-54	76	68	210	220	6.3	13	12.8	75.2	300	88	56
	DSD2-100BO64U-45-54	90	84	265	285	3.8	8.1	8.5	72.9	300	105	66
6000 ⁴⁾	DSD2-100SO64U-60-54	42	54.8	105	155	4.2	6.7	9.2	54.8	400	52	36
	DSD2-100MO64U-60-54	60	71	160	220	2.8	4.5	6.5	56.6	400	70	46
	DSD2-100LO64U-60-54 ⁵⁾	76	94	210	305	1.87	3.2	5.3	54.3	366,6	88	56

¹⁾ Coil overtemperature $\Delta T < 105K$; direct flange mounting (mounting plate 450 mm x 400 mm x 30 mm)

²⁾ Rotor moment of inertia with PE brake: +17.6 kgcm²

³⁾ Weight with PE brake: +6.0 kg

⁴⁾ in combination with Heidenhain absolute signal encoders available on request

⁵⁾ The technical details refer to a nominal speed of 5,500 min⁻¹

DSD2-100..64O-.. (with fan)3 AC 400 V mains voltage for converters with unregulated supply

Rated speed	Motor type	Stand- still torque ¹⁾	Stand- still current ¹⁾	max. static torque	Max. static current	Rated out- put ¹⁾	Rated torque ¹⁾	Rated current ¹⁾	Voltage constant	Rated fre- quency	Rotor inertia (motor) ²⁾	Weight ³⁾
n _N rpm		M ₀ Nm	I ₀ A	M _{0,max} Nm	I _{0,max} A	P _N kW	M _N Nm	I _N A	k _{E/cold} V/10001/min	f _N Hz	J kgcm ²	m kg
1200	DSD2-100SO64O-12-54	55	15.3	110	35	6	48	13.3	240	80	52	41
	DSD2-100MO64O-12-54	77	21.2	160	51	8.5	67	18.6	245	80	70	51
	DSD2-100LO64O-12-54	97	26.1	210	66	11	85	23	251	80	88	61
	DSD2-100BO64O-12-54	110	30.9	260	83	12	99	27.4	250	80	105	71
	DSD2-100XO64O-12-54	130	35.6	380	143	15	115	31.3	258	80	139	77.8
2000	DSD2-100SO64O-20-54	55	24.9	110	57	9	43	19.6	148	133.3	52	41
	DSD2-100MO64O-20-54	77	33.8	160	81	13	61	26.8	154	133.3	70	51
	DSD2-100LO64O-20-54	97	42.4	210	108	16	77	33.9	155	133.3	88	61
	DSD2-100BO64O-20-54	110	49.4	260	133	19	90	39.9	156	133.3	105	71
	DSD2-100XO64O-20-54	130	53	380	215	22	105	42.8	172	133.3	139	77.8
3000	DSD2-100SO64O-30-54	55	34.8	110	80	12	37	23.9	105	200	52	41
	DSD2-100MO64O-30-54	77	48.7	160	117	17	53	33.7	107	200	70	51
	DSD2-100LO64O-30-54	97	60	210	155	21	67	42.2	109	200	88	61
	DSD2-100BO64O-30-54	110	71	260	190	25	79	49.9	109	200	105	71
	DSD2-100XO64O-30-54	130	75	380	300	29	91	53	123	200	139	77.8
4500	DSD2-100SO64O-45-54	55	51	110	117	14	29	27.6	71.7	300	52	41
	DSD2-100MO64O-45-54	77	72	160	175	19	41	39	72.3	300	70	51
	DSD2-100LO64O-45-54	97	87	210	220	25	52	47.4	75.2	300	88	61
	DSD2-100BO64O-45-54	100	95	260	285	29	61	58	72.9	300	105	71
	DSD2-100XO64O-45-54	130	93	375	375	31	66	52	98.3	300	139	77.8
6000 ⁴⁾	DSD2-100SO64O-60-54	55	67	110	155	13	21	26.6	54.8	400	52	41
	DSD2-100MO64O-60-54	77	92	160	220	18	29	36.2	56.6	400	70	51
	DSD2-100LO64O-60-54	76	95	210	305	18	29	37	54.3	400	88	61
	DSD2-100BO64O-60-54	79	95	260	365	24	38	46.7	57.3	400	105	71
	DSD2-100XO64O-60-54	130	124	375	500	9.4	16	17.8	73.7	373.3	139	77.8

¹⁾ Coil overtemperature $\Delta T < 105K$; direct flange mounting (mounting plate 450 mm x 400 mm x 30 mm)²⁾ Rotor moment of inertia with PE brake: +17.6 kgcm²³⁾ Weight with PE brake: +6.0 kg⁴⁾ in combination with Heidenhain absolute signal encoders available on request

DSD2-100..64W-.. (water cooled)

3 AC 400 V mains voltage for converters with unregulated supply

Rated speed	Motor type	Stand- still torque ¹⁾	Stand- still current ¹⁾	max. static torque	Max. static current	Rated out- put ¹⁾	Rated torque ¹⁾	Rated current ¹⁾	Voltage constant	Rated fre- quen- cy	Rotor inertia (motor) ²⁾	Weight ³⁾
n _N rpm		M ₀ Nm	I ₀ A	M _{0,max} Nm	I _{0,max} A	P _N kW	M _N Nm	I _N A	k _{E/cold} V/10001/min	f _N Hz	J kgcm ²	m kg
1200	DSD2-100SO64W-12-54	89	32.2	110	44.3	10	82	29.1	189	80	52	41
	DSD2-100MO64W-12-54	130	46.6	165	65	15	120	42.8	191	80	70	51
	DSD2-100LO64W-12-54	170	58	220	85	20	160	55	196	80	88	62
	DSD2-100BO64W-12-54	210	71	280	105	25	205	67	197	80	105	72
	DSD2-100XO64W-12-54	240	96	400	215	29	235	93	172	80	139	86.5
2000	DSD2-100SO64W-20-54	89	52	110	71	16	77	43.3	118	133.3	52	41
	DSD2-100MO64W-20-54	130	75	165	105	23	110	63	119	133.3	70	51
	DSD2-100LO64W-20-54	170	94	220	138	31	150	82	121	133.3	88	62
	DSD2-100BO64W-20-54	185	95	280	165	37	180	93	125	133.3	105	72
	DSD2-100XO64W-20-54	240	141	400	315	48	225	135	117	133.3	139	86.5
2500	DSD2-100SO64W-25-54	89	63	110	87	19	73	49.8	96.6	166.6	52	41
	DSD2-100MO64W-25-54	130	89	165	125	27	105	70	100	166.6	70	51
	DSD2-100LO64W-25-54	145	95	220	165	35	135	88	99.9	166.6	88	62
	DSD2-100BO64W-25-54	160	95	280	190	42	160	94	109	166.6	105	72
	DSD2-100XO64W-25-54	240	180	400	400	59	225	170	92.1	166.7	139	86.5
3000	DSD2-100SO64W-30-54	89	72	110	100	22	69	54	84	200	52	41
	DSD2-100MO64W-30-54	120	95	165	148	31	99	78	84.5	200	70	51
	DSD2-100LO64W-30-54	130	95	210	180	37	120	88	91.9	200	88	62
	DSD2-100BO64W-30-54	130	95	280	235	41	130	95	88.2	200	105	72
	DSD2-100XO64W-30-54	240	205	400	460	69	220	190	79.8	200	139	86.5
4500	DSD2-100SO64W-45-54	86	95	110	138	26	55	60	60.9	300	52	41
	DSD2-100MO64W-45-54	89	95	165	200	32	69	73	62.6	300	70	51
	DSD2-100LO64W-45-54	98	95	220	250	46	97	79	66.6	300	88	62
	DSD2-100XO64W-45-54	240	225	395	500	86	180	200	73.7	300	139	86.5
6000	DSD2-100SO64W-60-54	73	95	110	165	25	40	52	50.4	400	52	41
	DSD2-100MO64W-60-54	70	95	165	265	21	34	47.9	47	400	70	51
	DSD2-100XO64W-60-54	240	225	395	500	88	140	190	73.7	400	139	86.5

¹⁾ Coil overtemperature $\Delta T < 105K$; direct flange mounting (mounting plate 450 mm x 400 mm x 30 mm)

²⁾ Rotor moment of inertia with PE brake: +17.6 kgcm²

³⁾ Weight with PE brake: +6.0 kg

2.7. DSD2-132

DSD2-132..54A(R)-... (surface cooled)*

3 AC 400 V mains voltage for converters with unregulated supply

Rated speed	Motor type	Stand-still torque ¹⁾	Stand-still current ¹⁾	max. static torque	Max. static current	Rated output ¹⁾	Rated torque ¹⁾	Rated current	Voltage constant	Rated frequency	Rotor inertia (motor)	Weight ³⁾	Terminal box page 92
n_N rpm		M_0 Nm	I_0 A	$M_{0,max}$ Nm	$I_{0,max}$ A	P_N kW	M_N Nm	I_N A	$k_E/cold$ V/10001/min	f_N Hz	J kgm ²	m kg	
1000	DSD2-132KO54.-10-5	175	34,2	380	89	16	155	30.6	351	66,6	0.029	127	20
	DSD2-132MO54.-10-5	235	45,3	505	118	22	210	41,1	353	66,6	0.038	142	22
	DSD2-132LO54.-10-5	290	57	630	149	27	265	52	350	66,6	0.046	157	22
	DSD2-132BO54.-10-5	345	67	750	175	33	320	62	358	66,6	0.055	172	22
	DSD2-132XO54.-10-5	405	77	875	200	39	375	73	362	66,6	0.063	187	22
	DSD2-132YO54.-10-5	460	91	995	240	48	430	87	351	66,6	0.072	203	22
	DSD2-132YZ54.-10-5	485	97	1060	255	48	460	93	351	66,6	0.076	210	24
1500	DSD2-132KO54.-15-5	175	49,8	375	130	22	140	41,5	241	100	0.029	127	22
	DSD2-132MO54.-15-5	235	67	500	175	30	190	57	238	100	0.038	142	22
	DSD2-132LO54.-15-5	290	81	625	210	38	245	70	246	100	0.046	157	22
	DSD2-132BO54.-15-5	345	103	745	270	47	295	91	232	100	0.055	172	22
	DSD2-132XO54.-15-5	405	120	870	310	55	350	107	234	100	0.063	187	24
	DSD2-132YO54.-15-5	460	130	990	340	64	410	117	247	100	0.072	203	24
	DSD2-132YZ54.-15-5	485	142	1050	370	69	440	129	240	100	0.076	210	24
2000	DSD2-132KO54.-20-5	175	64	375	170	27	130	49,5	187	133,3	0.029	127	22
	DSD2-132MO54.-20-5	230	86	495	225	37	175	68	186	133,3	0.038	142	22
	DSD2-132LO54.-20-5	290	111	620	290	47	225	89	181	133,3	0.046	157	22
	DSD2-132BO54.-20-5	345	130	745	340	58	275	106	185	133,3	0.055	172	24
	DSD2-132XO54.-20-5	405	155	865	405	68	325	130	180	133,3	0.063	187	26
	DSD2-132YO54.-20-5	460	175	985	455	80	380	147	184	133,3	0.072	203	26
	DSD2-132YZ54.-20-5	485	195	1050	510	85	410	170	174	133,3	0.076	210	26
2500	DSD2-132KO54.-25-5	175	77	370	200	30	115	54	155	166,6	0.029	127	22
	DSD2-132MO54.-25-5	235	103	495	270	41	160	74	155	166,6	0.038	142	22
	DSD2-132LO54.-25-5	290	130	615	340	53	205	95	154	166,6	0.046	157	24
	DSD2-132BO54.-25-5	345	155	740	405	65	250	116	154	166,6	0.055	172	26
	DSD2-132XO54.-25-5	405	195	860	510	78	295	149	143	166,6	0.063	187	26
	DSD2-132YO54.-25-5	460	225	980	585	91	345	175	143	166,6	0.072	203	26
	DSD2-132YZ54.-25-5	485	225	1040	585	98	375	175	152	166,6	0.076	210	26
3000	DSD2-132KO54.-30-5	175	91	370	240	32	100	57	132	200	0.029	127	22
	DSD2-132MO54.-30-5	235	130	490	340	44	140	83	124	200	0.038	142	24
	DSD2-132LO54.-30-5	290	155	610	405	56	180	102	128	200	0.046	157	26
	DSD2-132BO54.-30-5	345	195	735	510	69	220	130	123	200	0.055	172	26
	DSD2-132XO54.-30-5	405	225	855	585	83	265	150	125	200	0.063	187	26
	DSD2-132YO54.-30-5	460	225	980	585	94	300	155	143	200	0.072	202	26
	DSD2-132YZ54.-30-5	485	225	1040	585	95	305	155	152	200	0.076	210	26

* The integrated axial fan is available on request

¹⁾ Coil temperature $\Delta T < 105K$; direct flange mounting (mounting plate 450 mm x 400 mm x 30 mm)

²⁾ Rotor moment of inertia with PE brake: +0.004kgm²

³⁾ Weight with PE brake: +13 kg

DSD2-132..54W-.. (water cooled)
3 AC 400 V mains voltage for converters with unregulated supply

Rated speed	Motor type	Stand- still torque	Stand- still current	max. static torque	Max. static current	Rated output	Rated torque	Rated current	Voltage constant	Rated fre- quency	Rotor inertia (motor)	Weight	Ter- minal box page 92
n_N		M_0	I_0	$M_{0,max}$	$I_{0,max}$	P_N	M_N	I_N	$k_E/cold$	f_N	J	m	
rpm		Nm	A	Nm	A	kW	Nm	A	$V/_{10001/min}$	Hz	kgm ²	kg	
1000	DSD2-132KO54W-10-54	240	49,9	400	98	23	220	46,2	321	66,7	0.029	95	22
	DSD2-132MO54W-10-54	330	69	525	129	32	305	65	324	66,7	0.038	115	24
	DSD2-132LO54W-10-54	420	97	640	165	41	390	88	313	66,7	0.046	135	24
	DSD2-132BO54W-10-54	515	111	770	190	50	480	104	329	66,7	0.055	155	24
	DSD2-132XO54W-10-54	615	132	890	220	60	570	125	329	66,7	0.063	170	24
	DSD2-132YO54W-10-54	715	165	1020	265	70	665	155	313	66,7	0.072	190	24
	DSD2-132YZ54W-10-54	770	175	1080	285	75	715	165	311	66,7	0.076	200	26
1500	DSD2-132KO54W-15-54	240	73	400	143	33	210	65	219	100	0.029	95	22
	DSD2-132MO54W-15-54	330	103	525	190	46	295	92	219	100	0.038	115	24
	DSD2-132LO54W-15-54	420	141	635	250	59	375	128	209	100	0.046	135	24
	DSD2-132BO54W-15-54	515	165	765	285	72	460	150	219	100	0.055	155	24
	DSD2-132XO54W-15-54	615	200	885	335	86	545	180	219	100	0.063	170	26
	DSD2-132YO54W-15-54	715	220	1010	365	100	635	205	230	100	0.072	190	26
	DSD2-132YZ54W-15-54	770	225	1070	365	105	670	205	244	100	0.076	200	26
2000	DSD2-132KO54W-20-54	240	93	395	180	42	200	79	172	133,3	0.029	95	22
	DSD2-132MO54W-20-54	330	135	520	250	59	280	116	167	133,3	0.038	115	24
	DSD2-132LO54W-20-54	420	175	635	310	75	360	150	170	133,3	0.046	135	26
	DSD2-132BO54W-20-54	515	210	755	365	93	440	185	172	133,3	0.055	155	26
	DSD2-132XO54W-20-54	615	240	875	400	108	515	210	183	133,3	0.063	170	26
	DSD2-132YO54W-20-54	715	305	995	500	128	615	270	167	133,3	0.029	190	28
	DSD2-132YZ54W-20-54	770	310	1060	500	137	655	275	177	133,3	0.038	200	28
2500	DSD2-132KO54W-25-54	240	114	395	220	49	190	92	141	166,7	0.029	95	24
	DSD2-132MO54W-25-54	330	165	520	310	70	270	137	136	166,7	0.038	115	26
	DSD2-132LO54W-25-54	420	205	630	365	90	345	175	144	166,7	0.049	135	26
	DSD2-132BO54W-25-54	515	230	755	400	108	410	195	157	166,7	0.055	153	26
	DSD2-132XO54W-25-54	615	300	870	500	130	500	255	146	166,7	0.046	170	28
	DSD2-132YO54W-25-54	715	305	995	500	144	550	260	167	166,7	0.029	190	28
	DSD2-132YZ54W-25-54	770	310	1060	500	150	570	265	177	166,7	0.038	200	28
3000	DSD2-132KO54W-30-54	240	128	395	250	55	175	98	125	200	0.029	95	24
	DSD2-132MO54W-30-54	330	195	515	365	80	255	155	115	200	0.038	115	26
	DSD2-132LO54W-30-54	420	225	625	400	101	320	185	131	200	0.046	135	26
	DSD2-132BO54W-30-54	515	290	750	500	126	400	240	240	125	0.055	153	28
4000	DSD2-132KO54W-40-54	240	170	390	335	65	155	115	94	266,7	0.029	95	26
	DSD2-132MO54W-40-54	330	215	515	400	91	215	155	104	266,7	0.038	115	26
	DSD2-132LO54W-40-54	390	310	615	600	125	300	255	87	266,7	0.046	135	28
6000	DSD2-132KO54W-60-54	240	225	385	445	64	100	108	70.5	400	0.029	95	26

¹⁾ Coil temperature $\Delta T < 105K$; direct flange mounting (mounting plate 450 mm x 400 mm x 30 mm)

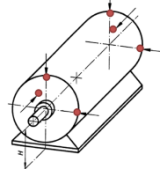
²⁾ Rotor moment of inertia with PE brake: +0.004kgm²
³⁾ Weight with PE brake: +13 kg

2.8. Vibratory load

The vibration response of the total system at the operation site, due to output elements, mounting circumstances, alignment and installation as well as by influences of external vibrations, can lead to an increase of the vibration values at the motor.

Perhaps a complete balancing of the rotor with the output element is required.

In order to achieve a proper functioning and service life, the specified vibration values compliant to DIN ISO 10816 may not be increased at the specified measuring points of the motor (refer to figure below).

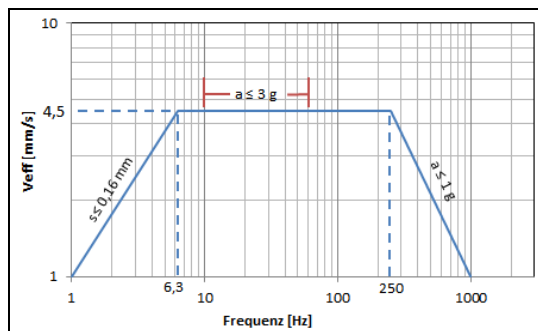


Measuring points for the vibration measuring

The specified radial and axial vibration values must be complied with, as well. They are valid for substructures that are to be designated as flexible. A flexible substructure is, if the lowest proper frequency of the total system in measuring direction is at least 25% under the basic excitation frequency of the total system (machine and base). All of the other substructures are to be designated solid. At solid substructures the manufacturer must be contacted.

Maximum radial vibration load:

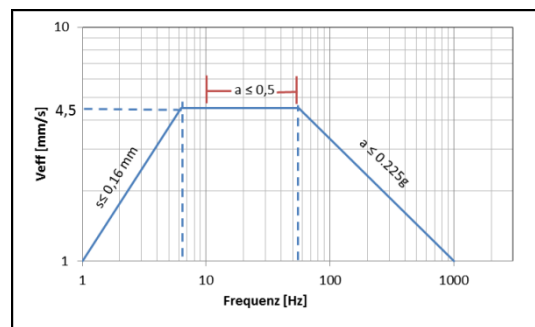
Peak vibration acceleration 1 g > 250 Hz
 Peak vibration displacement $\leq 0,16$ mm < 6,3 Hz
 Effect. vibration speed $\leq 4,5$ mm/s
 6,3 Hz – 250 Hz



Permitted radial vibratory load

Maximum axial vibration load:

Peak vibration acceleration 0,225 g > 55 Hz
 Peak vibration displacement $\leq 0,16$ mm < 6,3 Hz
 Effect. vibration speed $\leq 4,5$ mm/s
 6,3 Hz – 55 Hz



Permitted axial vibratory load

Additional resistance to vibration:

Vibration acceleration 3 g radial and 0,5g axial 10 Hz to 55 Hz

The motor additionally withstands the specified vibrations. However, the lifetime of the parts subject to wear (for example the storage) can be reduced.

Shock load:

If the vibration loads in the form of shocks are present, then measuring at the installed machine is required. On the base of this, constructive reviews and evaluation with the company Baumüller are required.

In order to evaluate the vibration speed, the measuring equipment must meet the requirements of the ISO 2954. The evaluation of the vibration acceleration is executed in the time range within the frequency range between 10 Hz and 2 kHz.

Provided that considerable vibratory excitations over 2kHz such as meshing frequencies are to be expected, the measuring range must be correspondingly adjusted. The permitted maximum values thereby do not change.

** 0.5 g Size 028-100 // 1 g Size 028-036 without brake and 132 // 1 g Size 132, 100-55 Hz

2.9. Radial force diagrams

All bearings are designed for a service life of 20.000 h L_{10h} . The load values specified below may thereby not be exceeded. The permissible radial forces F_R are valid only for the horizontal installation of the motor without additional axial forces.

Furthermore, the specified average speeds must be adhered to reach the grease consumption period of 20.000 h under the following conditions:

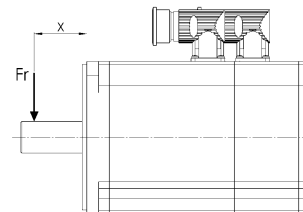
- low-vibration applications
- horizontal installation
- oscillatory bearing motion in which at least one pivot angle of 180° is performed
- Continuous bearing temperatures $<120^\circ \text{ C}$.

Axial load on the motor shaft:

No axial forces may develop when mounting clutches, pulleys, etc. on the motor shaft!

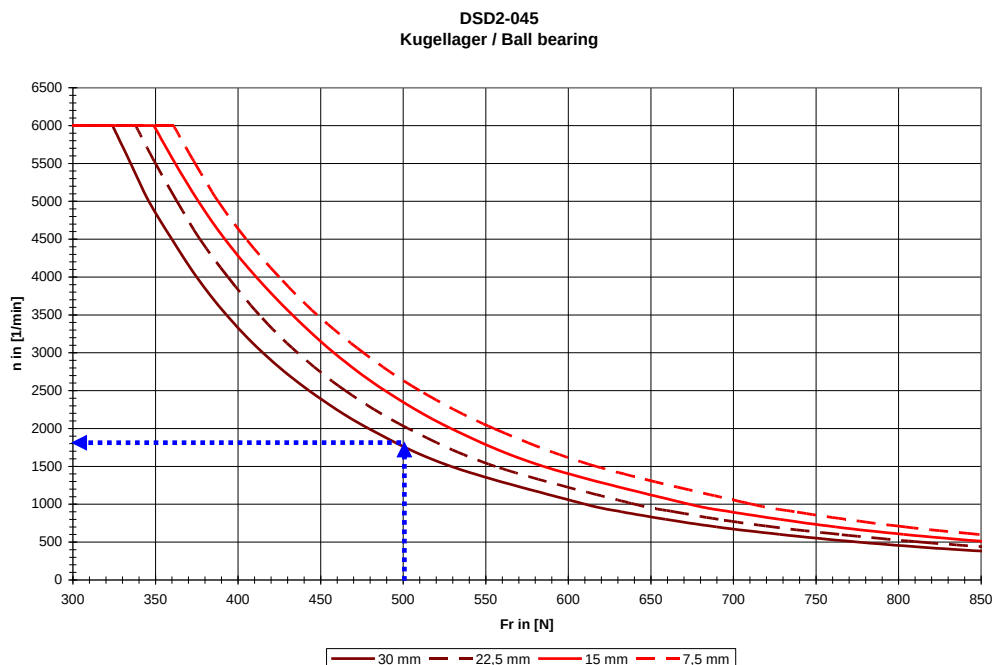
2.9.1. Sample diagram

Sample diagrams:



Driving forces $x = 30 \text{ mm}$ from the shaft shoulder

Bearing service life 20.000 h, shaft with parallel key groove

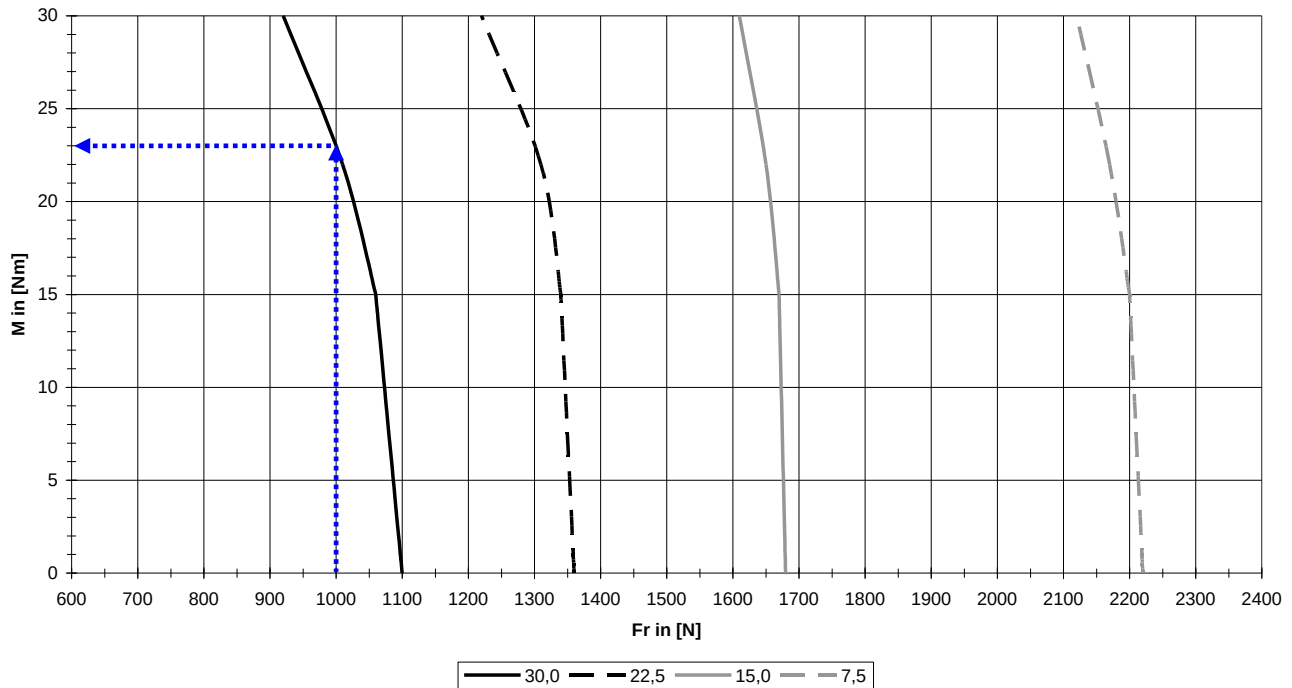


Explanation of the sample chart:

The potential maximum speed of the bearing can be calculated via radial force F_R of the application in characteristic "ball bearing".

At a radial force of 500 N with a driving force point of $x = 30 \text{ mm}$ from the shaft shoulder, a maximum speed of 1750 rpm results.

DSD2-045
glatte Welle / Shaft without key



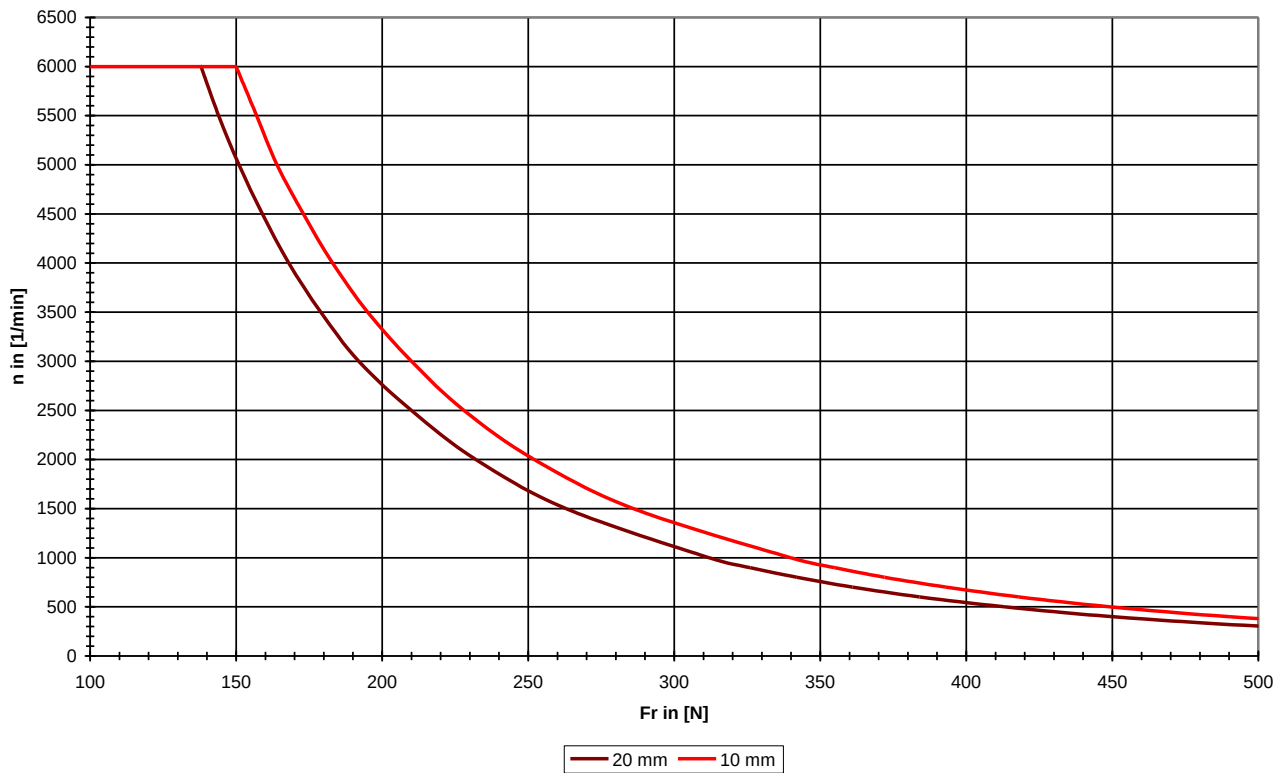
Explanation of the sample chart:

The maximum torque to be still transmitted results from the characteristic "shaft".

At a centrifugal force of 1000 N with a driving force point of $x = 30$ mm from the shaft shoulder, a torque to be still transmitted of 23 Nm results.

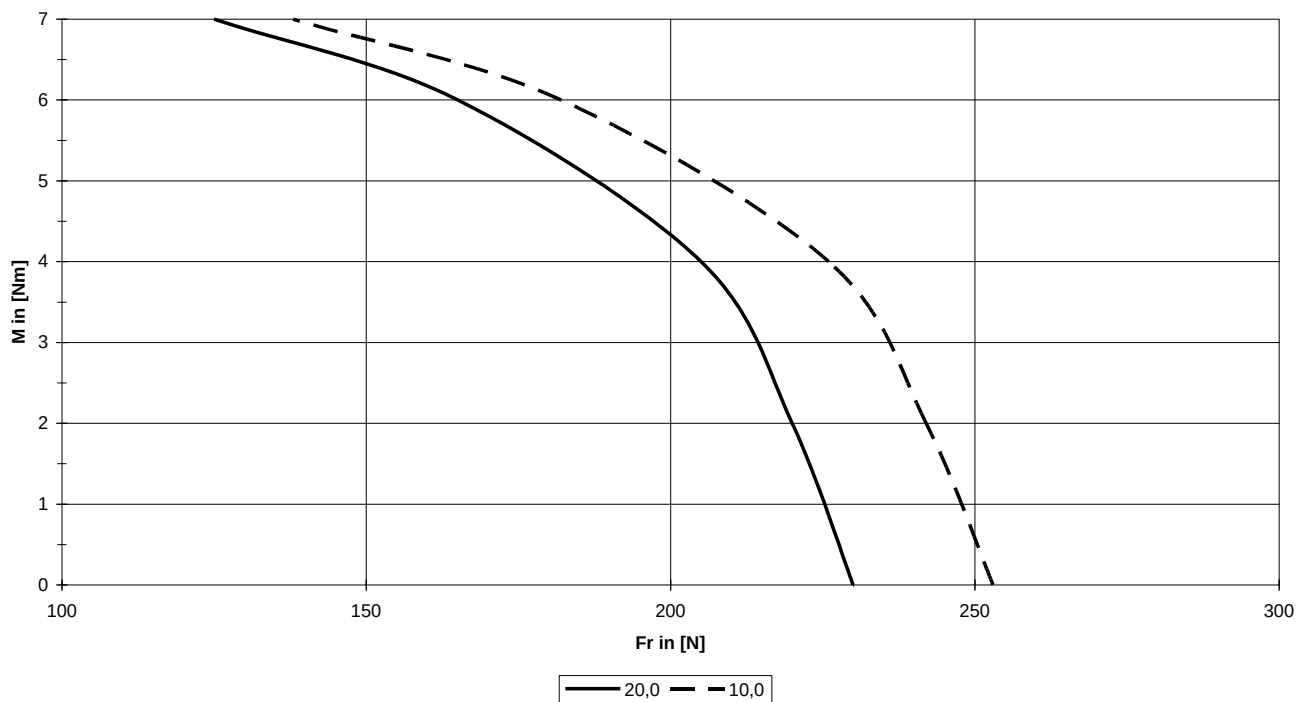
2.9.2. Diagrams DSD2-028

DSD2-028
Kugellager / Ball bearing



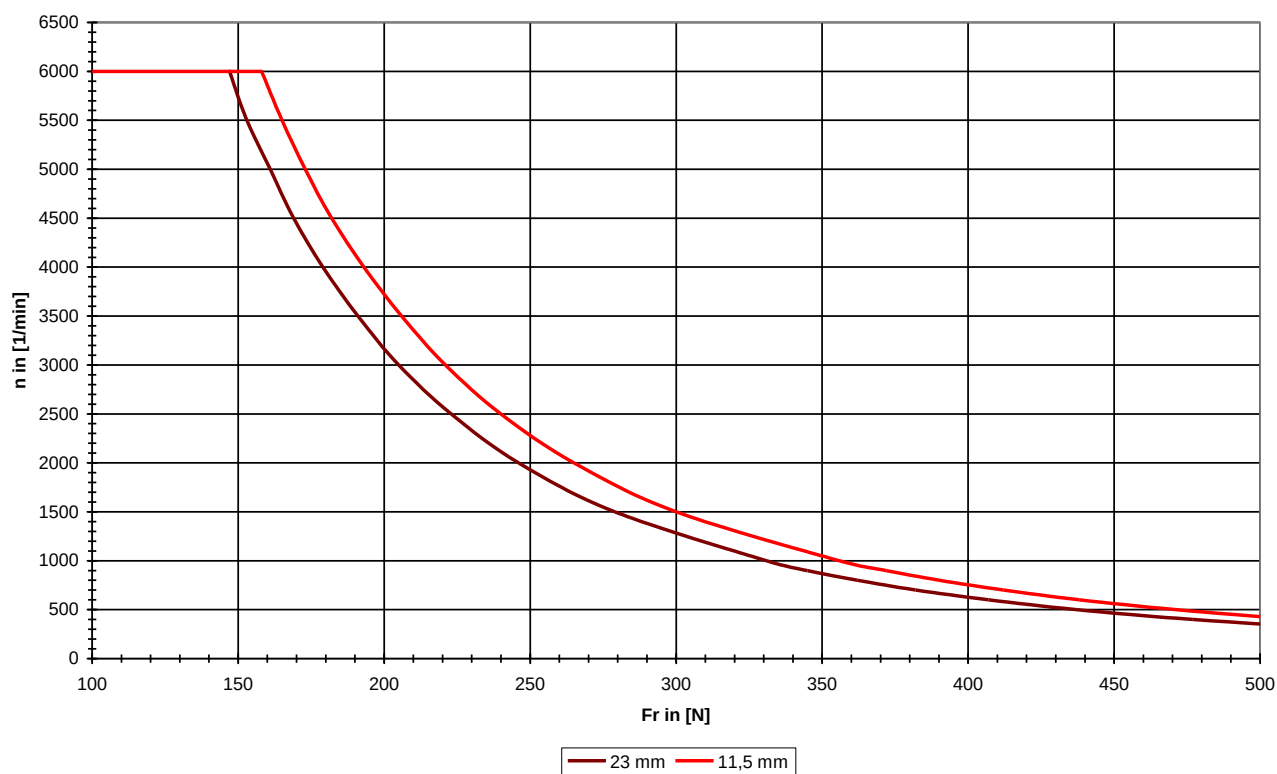
Note: 20.000h grease consumption duration at $n_{\text{effective}} \leq 6,000$ rpm

DSD2-028
glatte Welle / Shaft without key



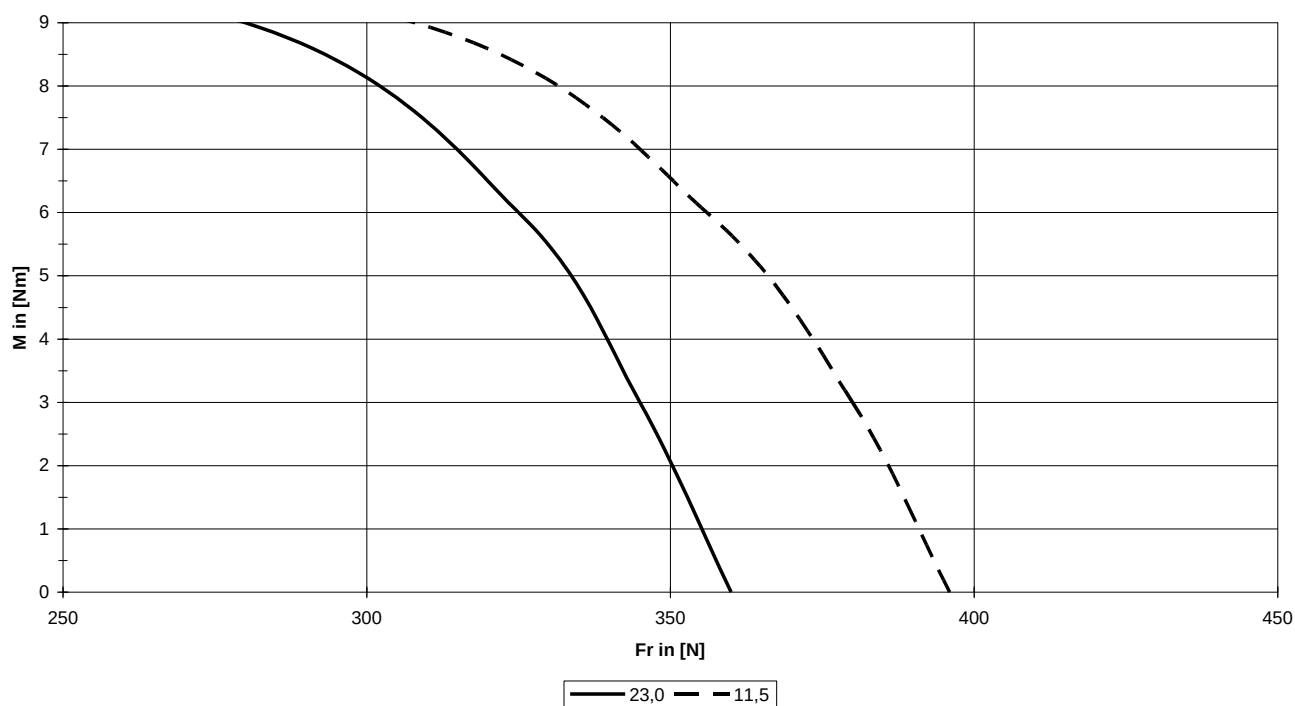
2.9.3. Diagrams DSD2-036

DSD2-036
Kugellager / Ball bearing



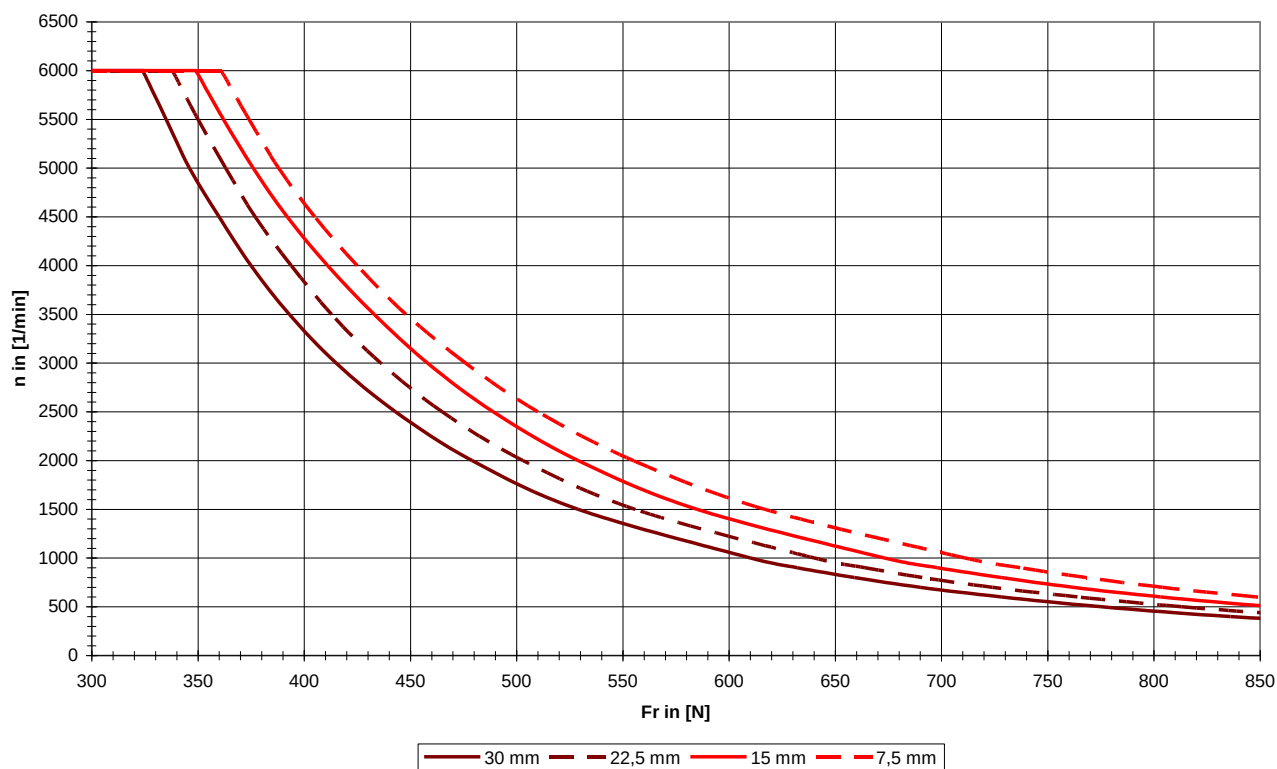
Note: 20.000h grease consumption duration at $n_{\text{effective}} \leq 6,000$ rpm

DSD2-036
glatte Welle / Shaft without key



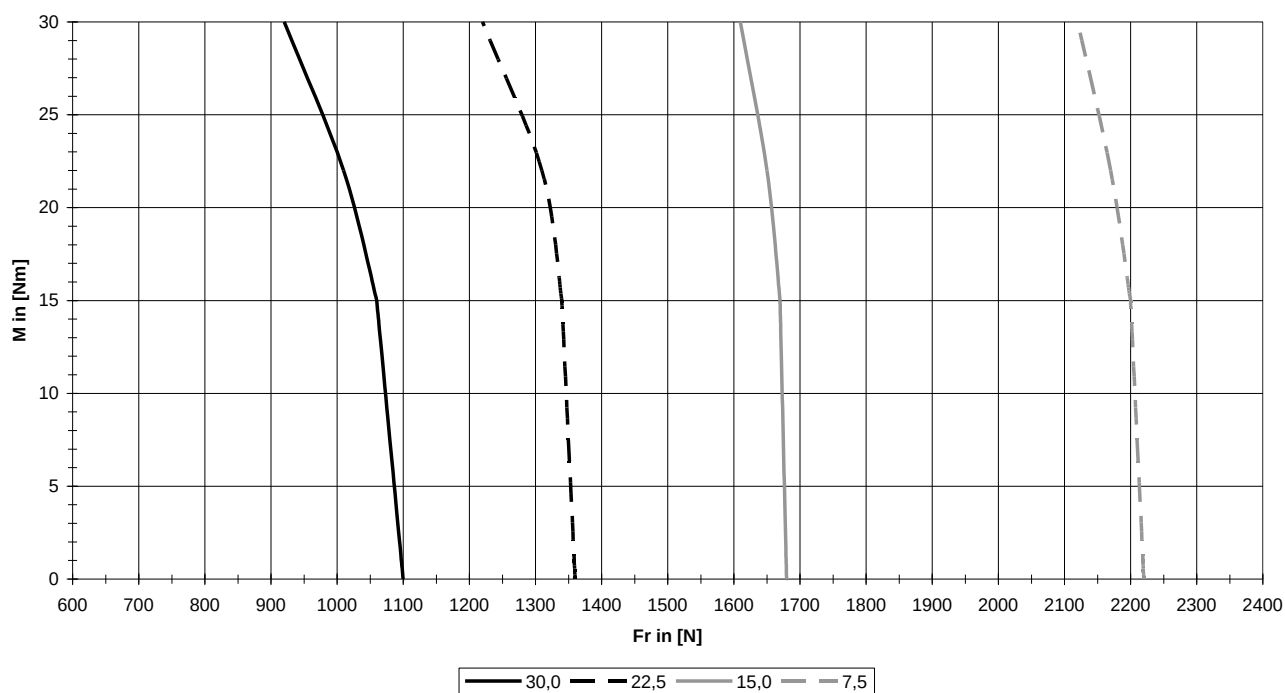
2.9.4. Diagrams DSD2-045

DSD2-045
Kugellager / Ball bearing

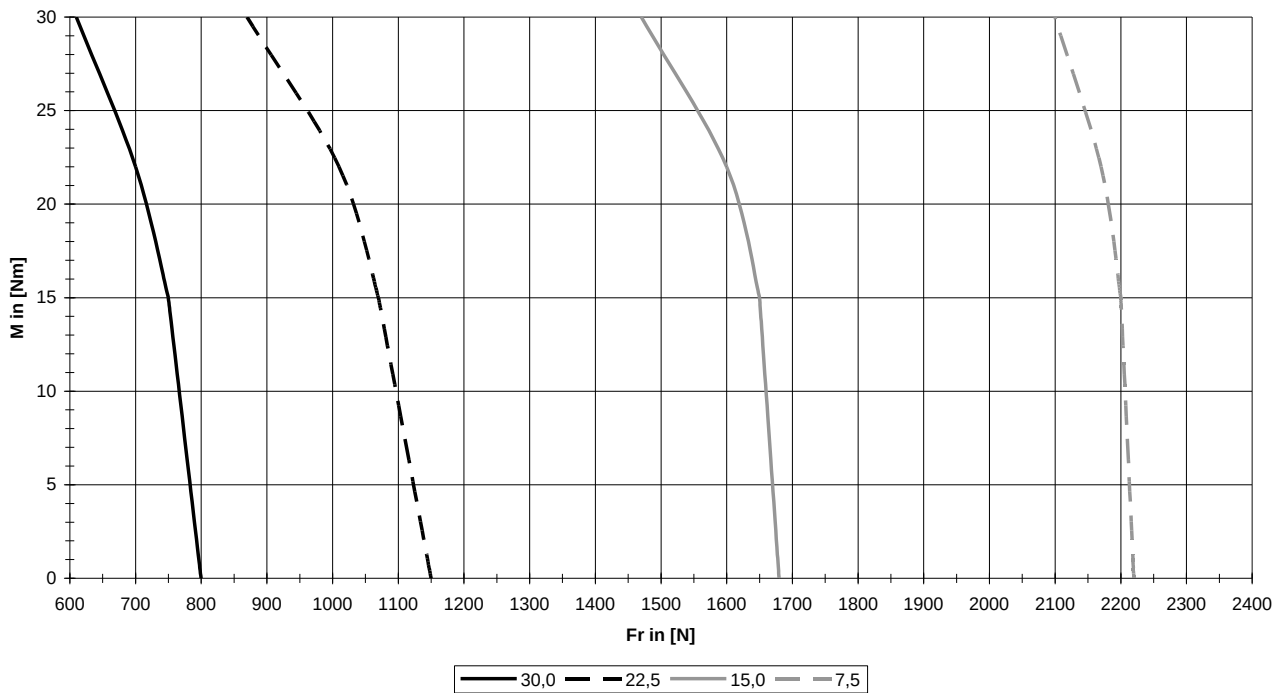


Note: 20.000h grease consumption duration at $n_{\text{effective}} \leq 6,000$ rpm

DSD2-045
glatte Welle / Shaft without key

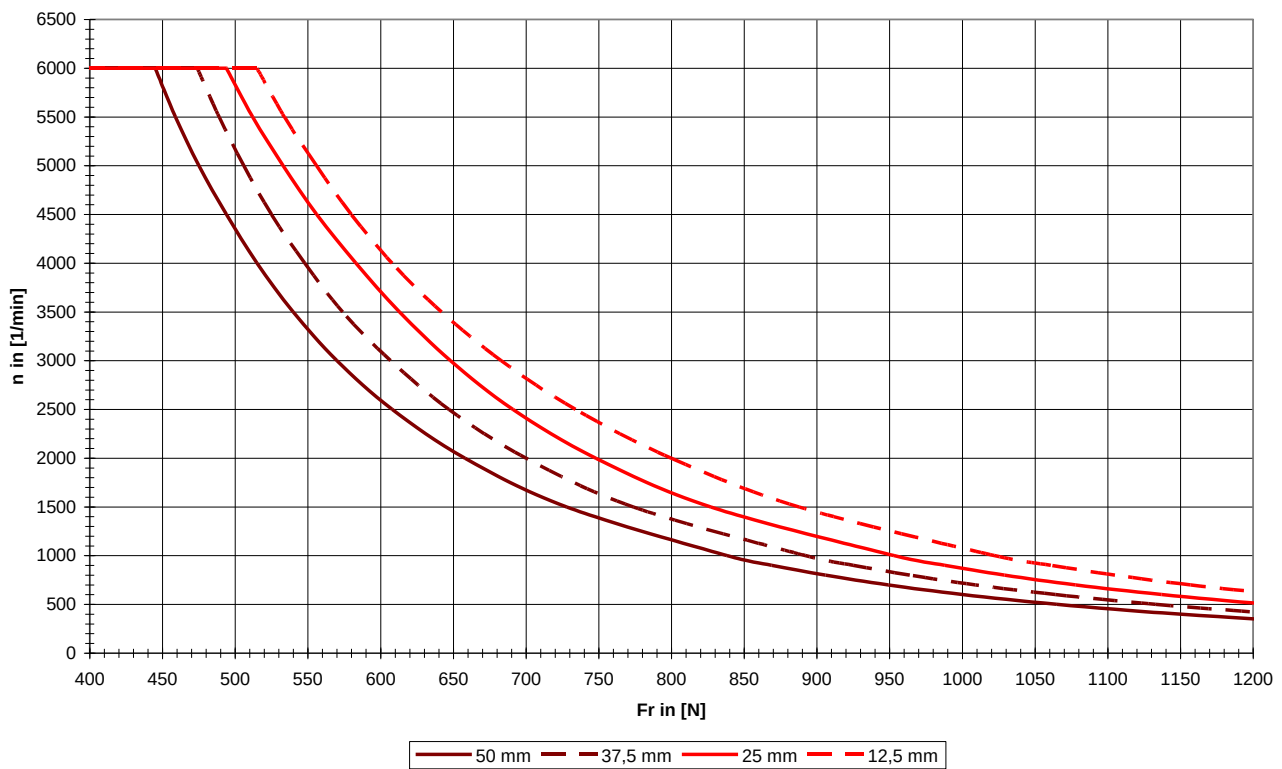


DSD2-045
Welle mit Nut / Shaft with key



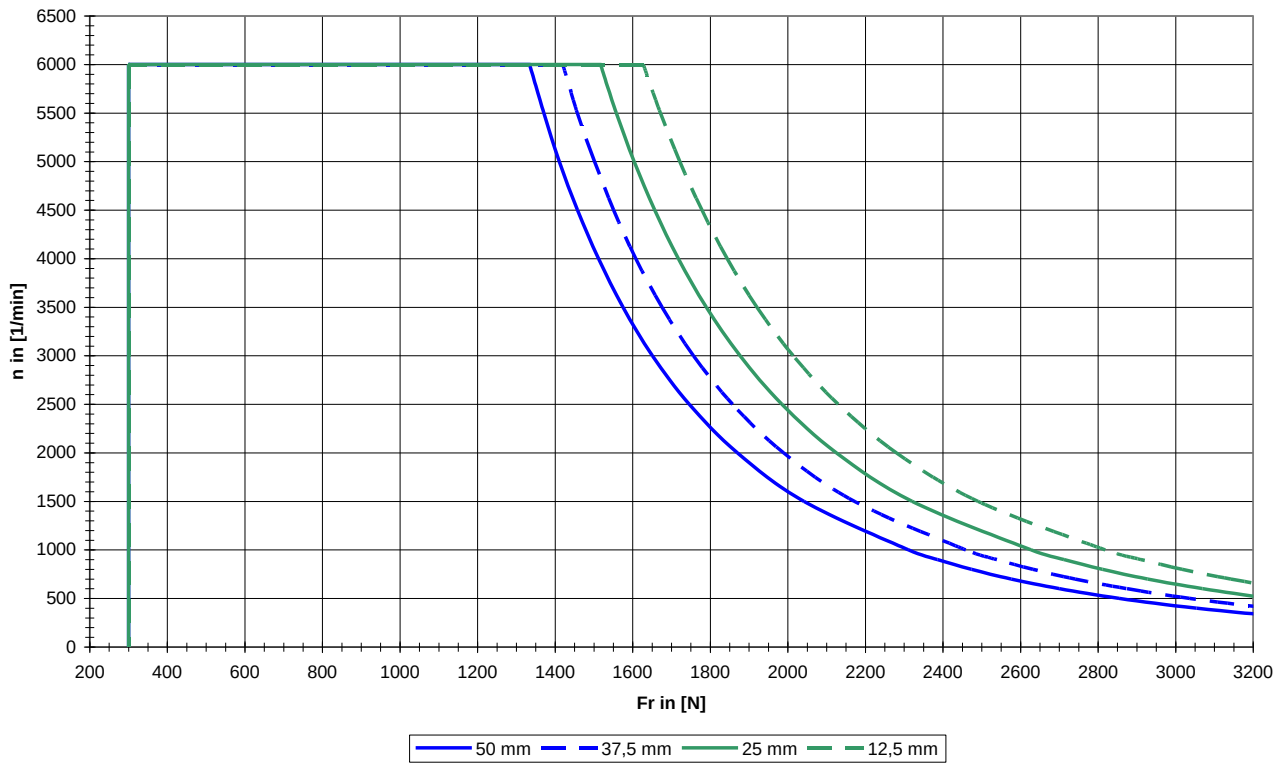
2.9.5. Diagrams DSD2-056

DSD2-056
Kugellager / Ball bearing



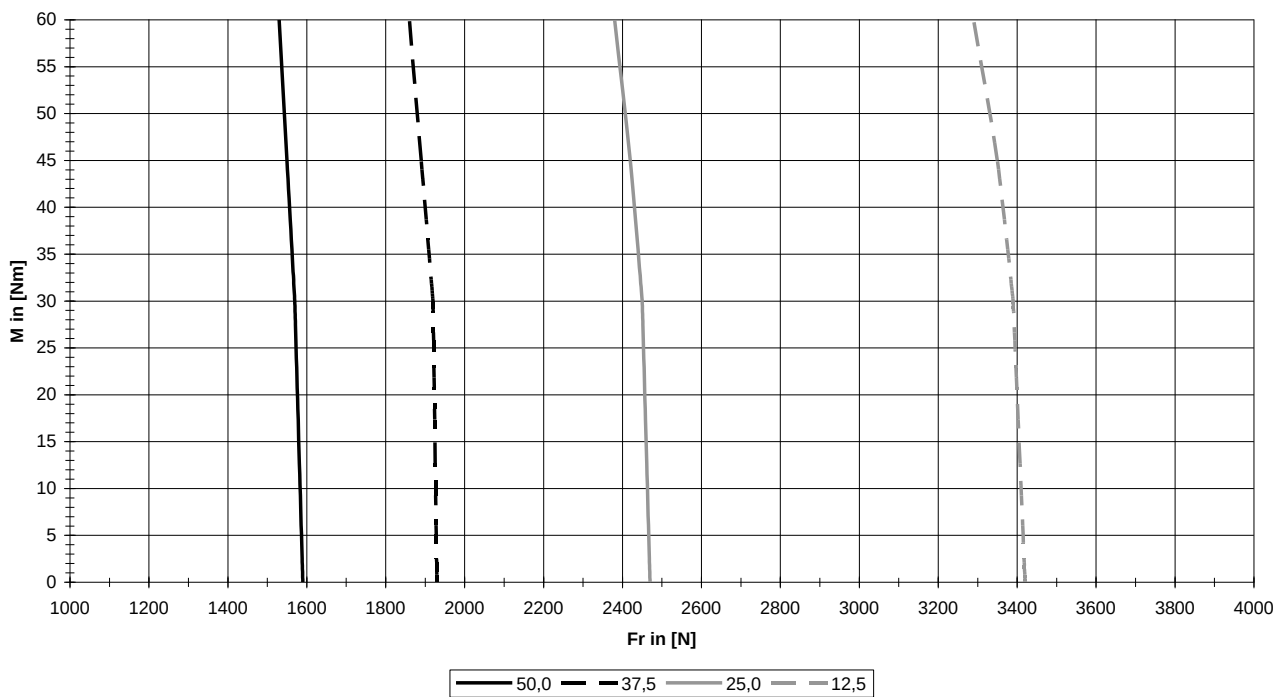
Note: 20.000h grease consumption duration at $n_{\text{effective}} \leq 6,000$ rpm

DSD2-056
Rollenlager / Roller bearing



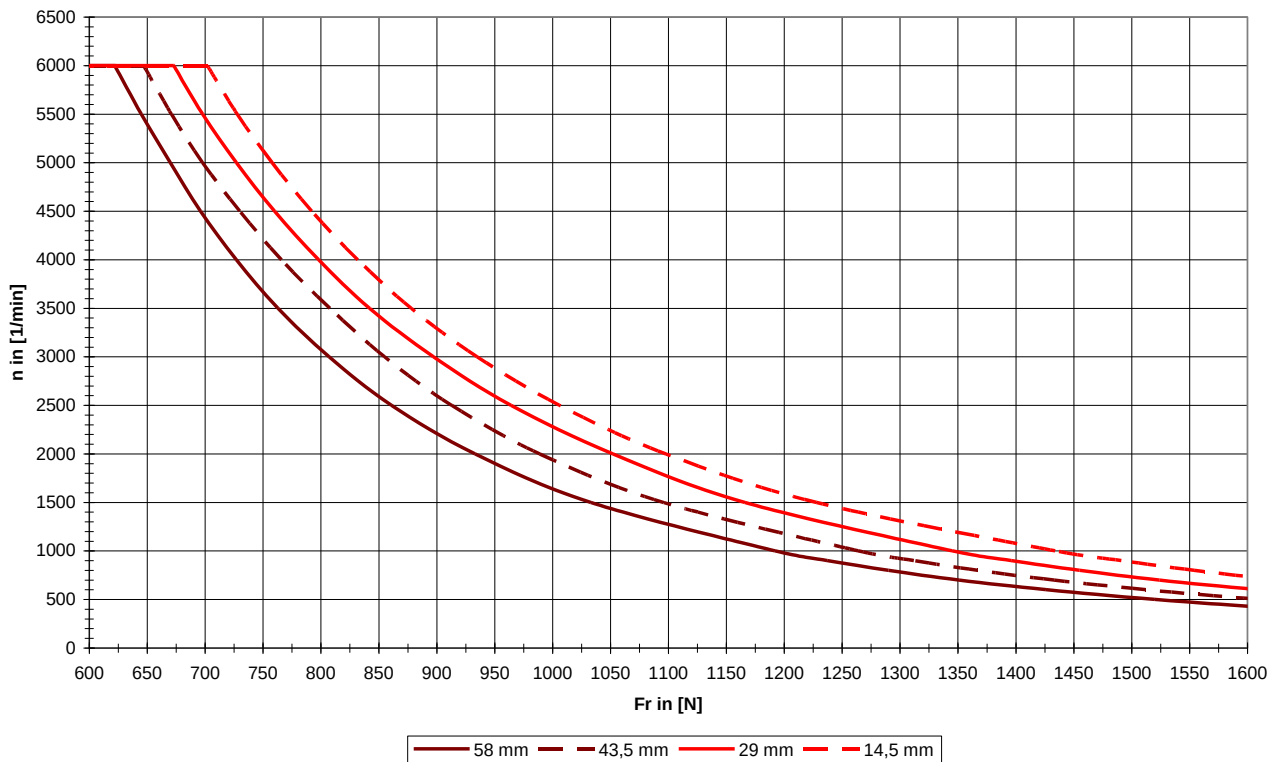
Note: 20.000h grease consumption duration at $n_{\text{effective}} \leq 2,000$ rpm

DSD2-056
Welle / Shaft



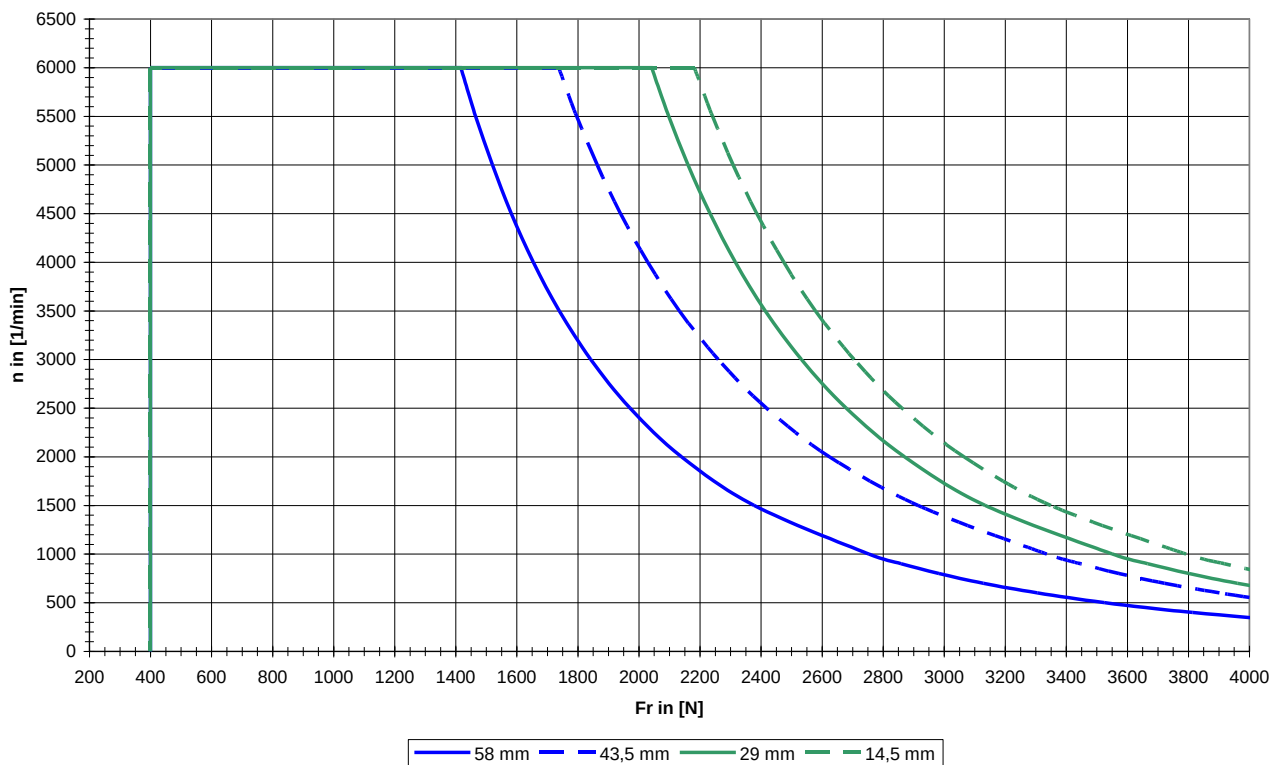
2.9.6. Diagrams DSD2-071

DSD2-071
Kugellager / Ball bearing



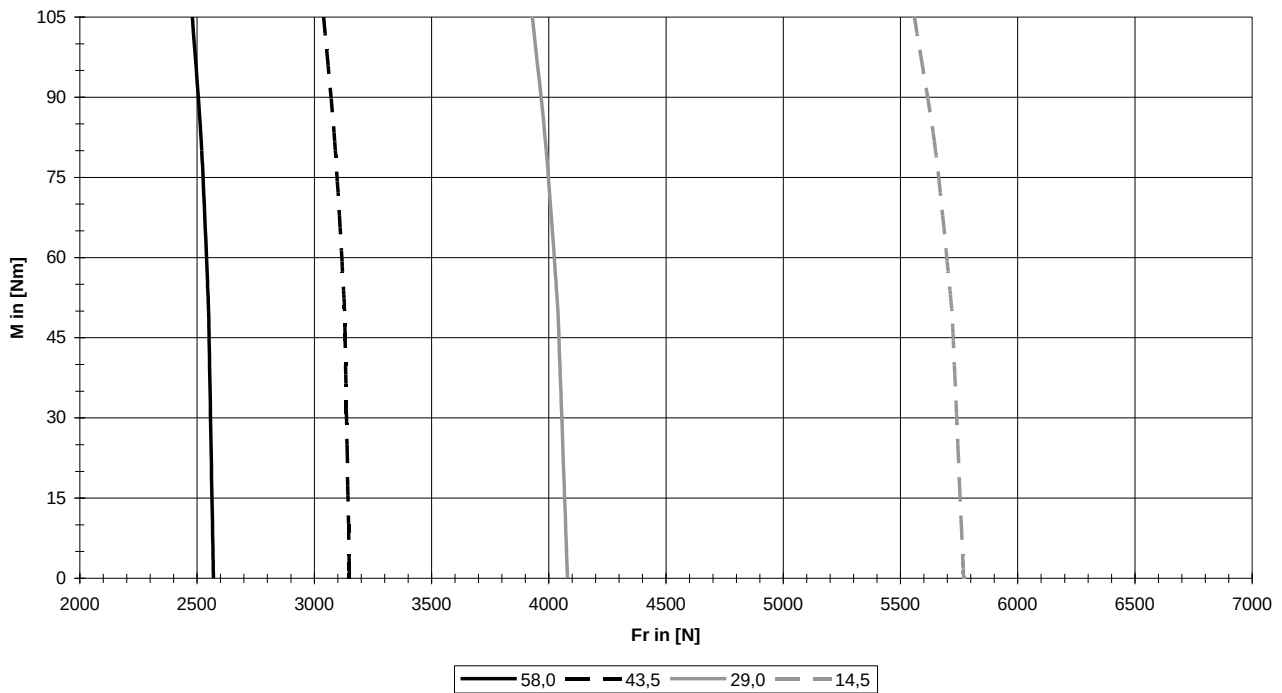
Note: 20.000h grease consumption duration at $n_{\text{effective}} \leq 6,000$ rpm

DSD2-071
Rollenlager / Roller bearing



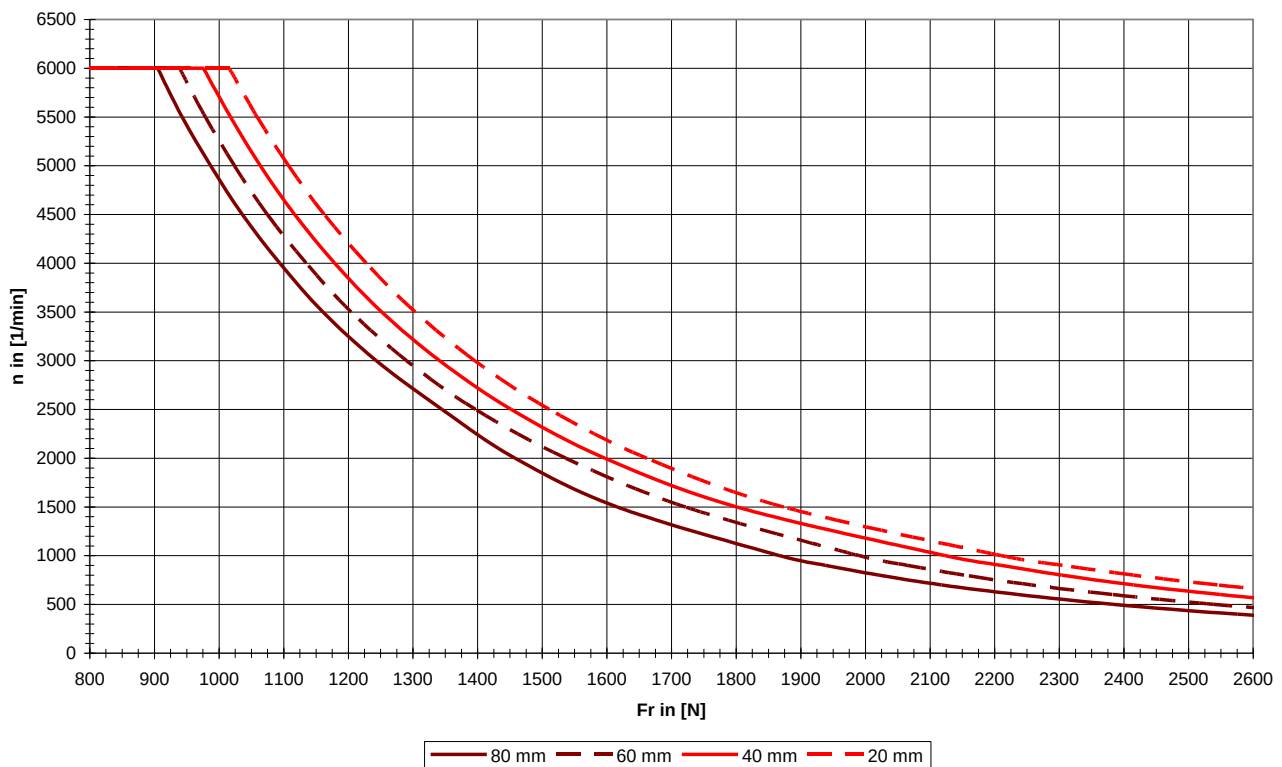
Note: 20.000h grease consumption duration at $n_{\text{effective}} \leq 1,600$ rpm

DSD2-071
Welle / Shaft



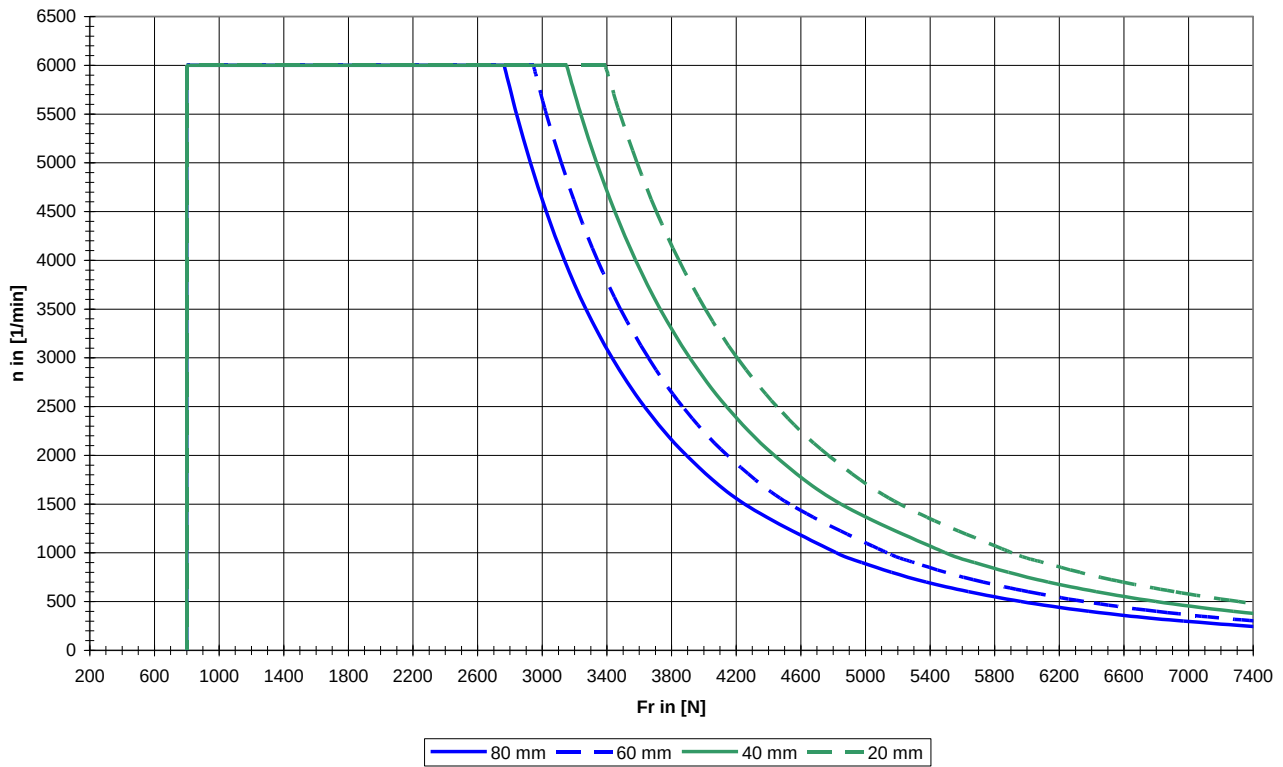
2.9.7. Diagrams DSD2-100

DSD2-100
Kugellager / Ball bearing



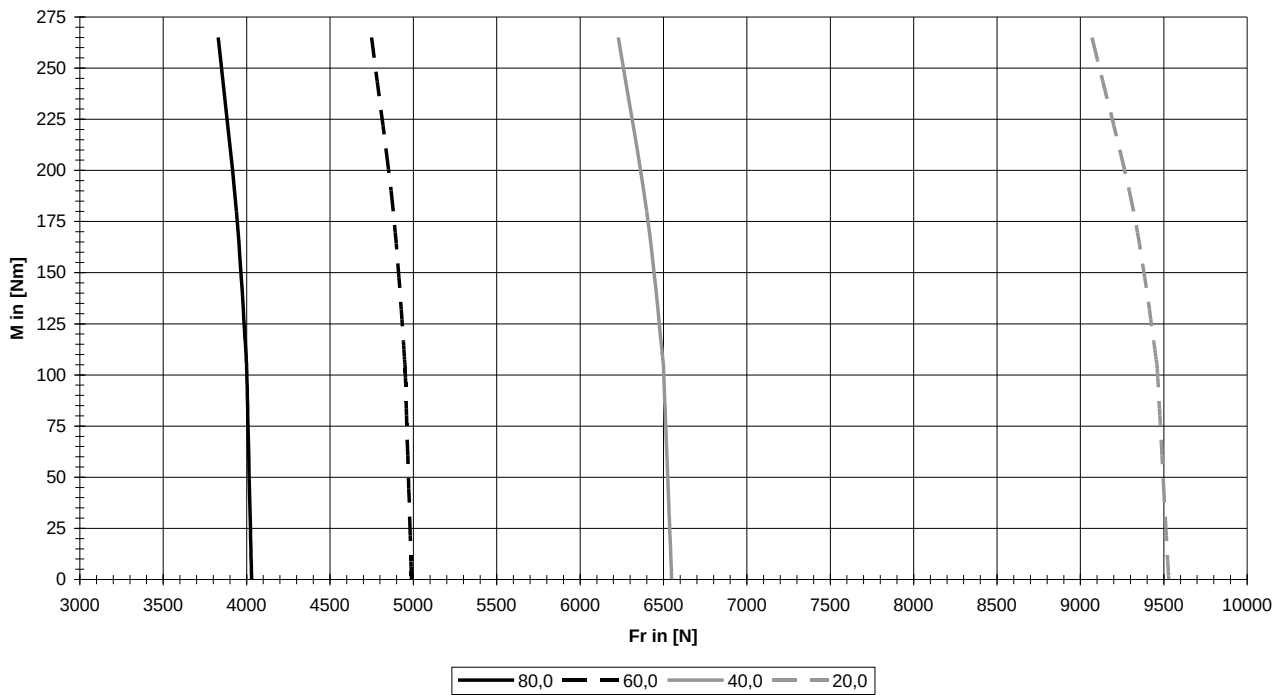
Note: 20.000h grease consumption duration at $n_{\text{effective}} \leq 4,100$ rpm

DSD2-100
Rollenlager / Roller bearing



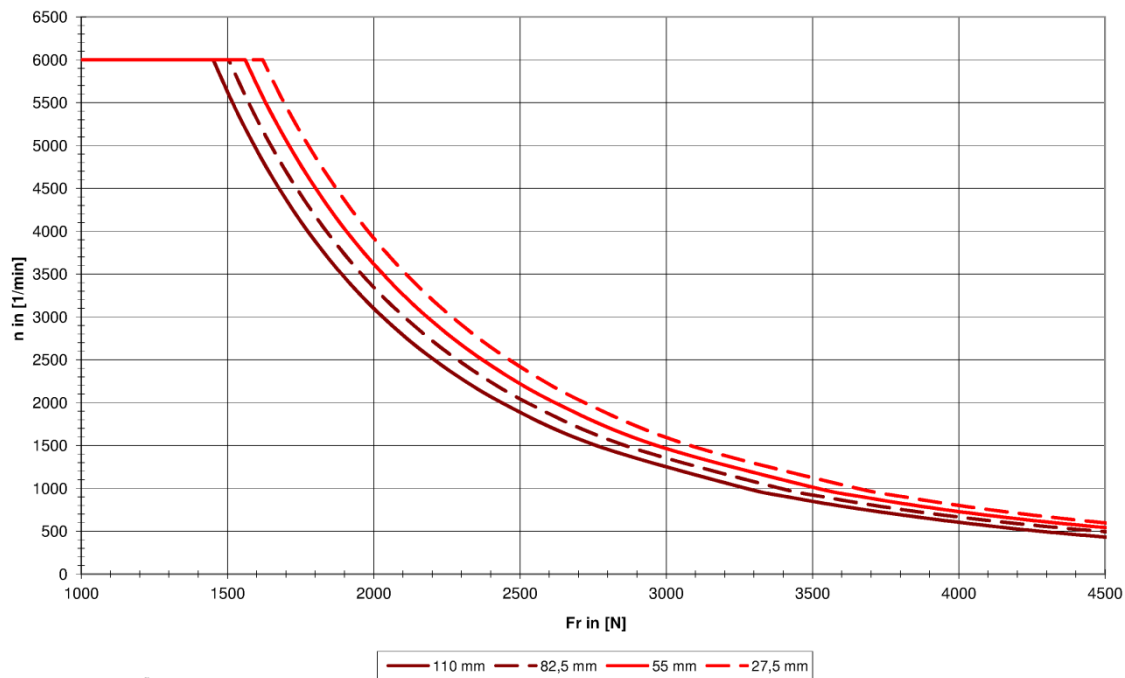
Note: 20.000h grease consumption duration at $n_{\text{effective}} \leq 1,100$ rpm

DSD2-100
Welle / Shaft



2.9.8. Diagrams DSD2-132A(R)

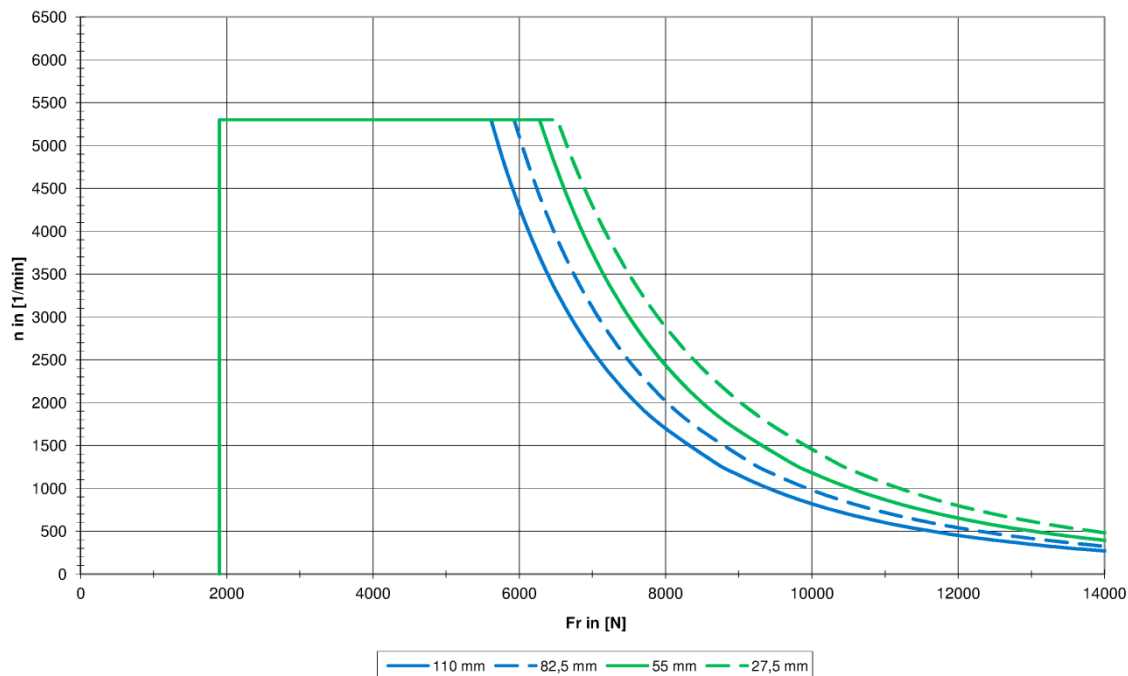
DSD2-132
Kugellager / Ball bearing



Achtung: Technische Änderungen vorbehalten !

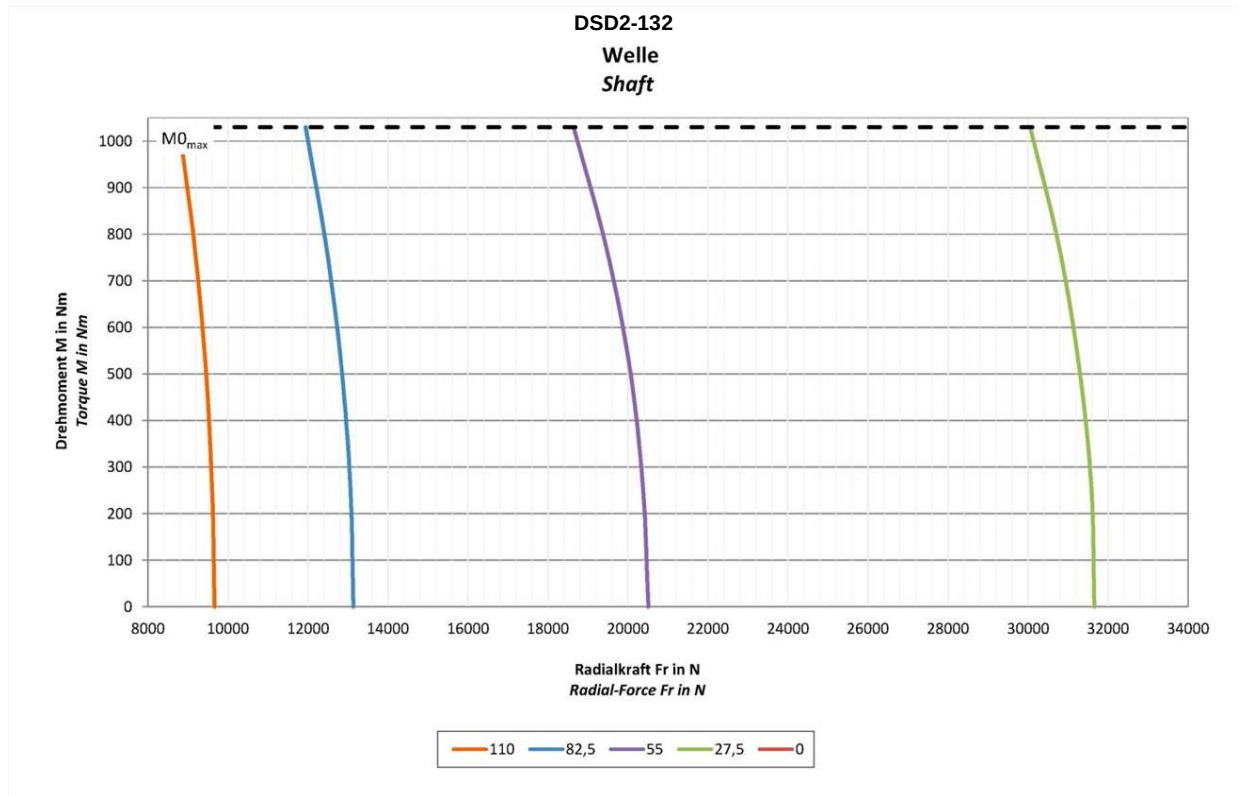
At ball bearing: Grease service life F_{h10} of 20.000 h at $n_{\text{medium}} \leq 3500 \text{ min}^{-1}$.

DSD2-132
Rollenlager / Roller bearing

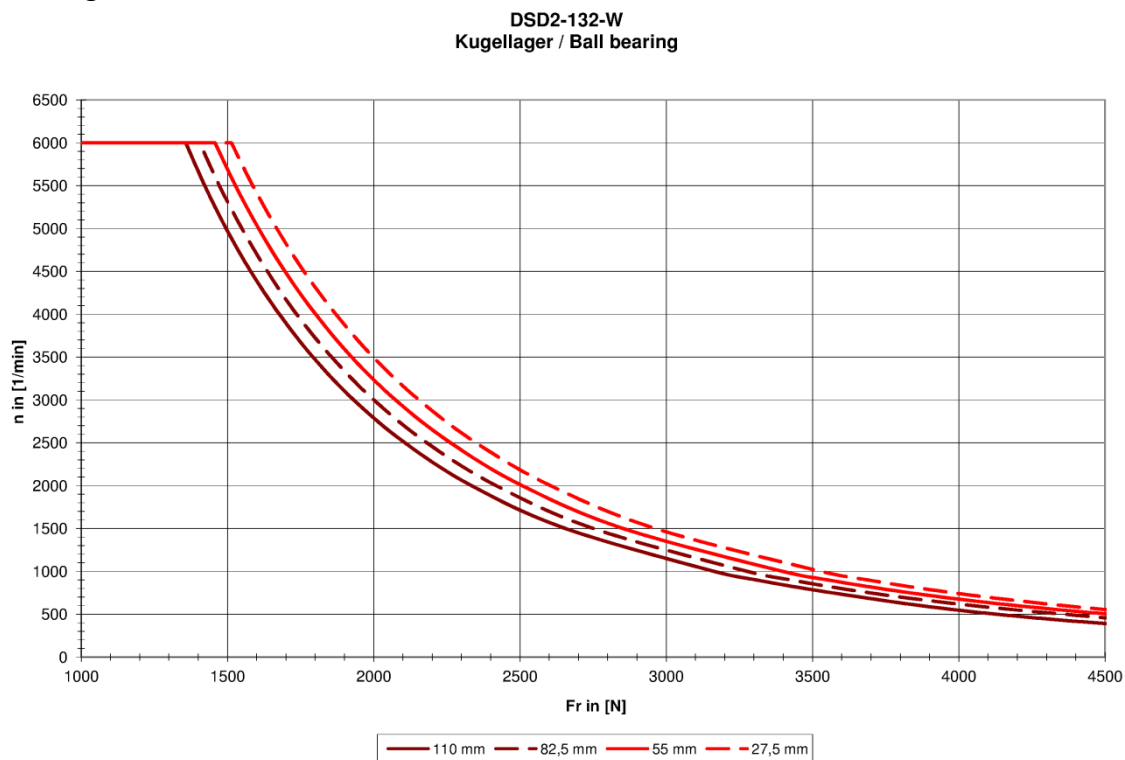


Achtung: Technische Änderungen vorbehalten !

At roller bearing: Grease service life F_{h10} of 20.000 h at $n_{\text{medium}} \leq 1000 \text{ min}^{-1}$.



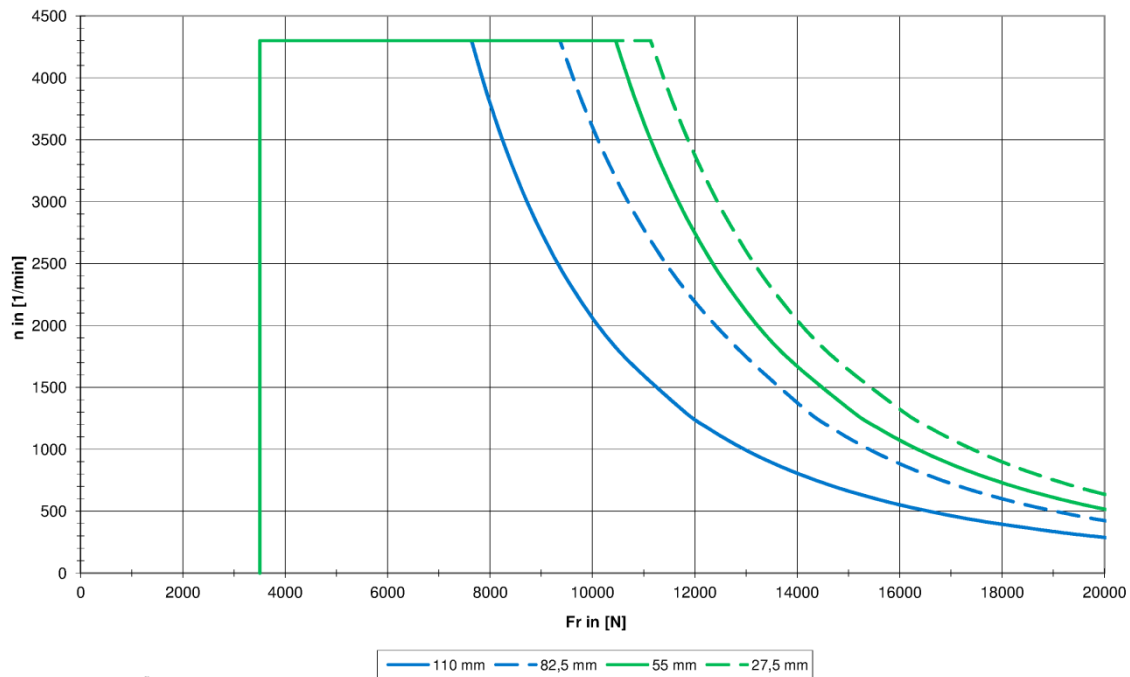
2.9.9. Diagrams DSD2-132W



Achtung: Technische Änderungen vorbehalten !

At ball bearing: Grease service life F_{h10} of 20.000 h at $n_{\text{medium}} \leq 3500 \text{ min}^{-1}$.

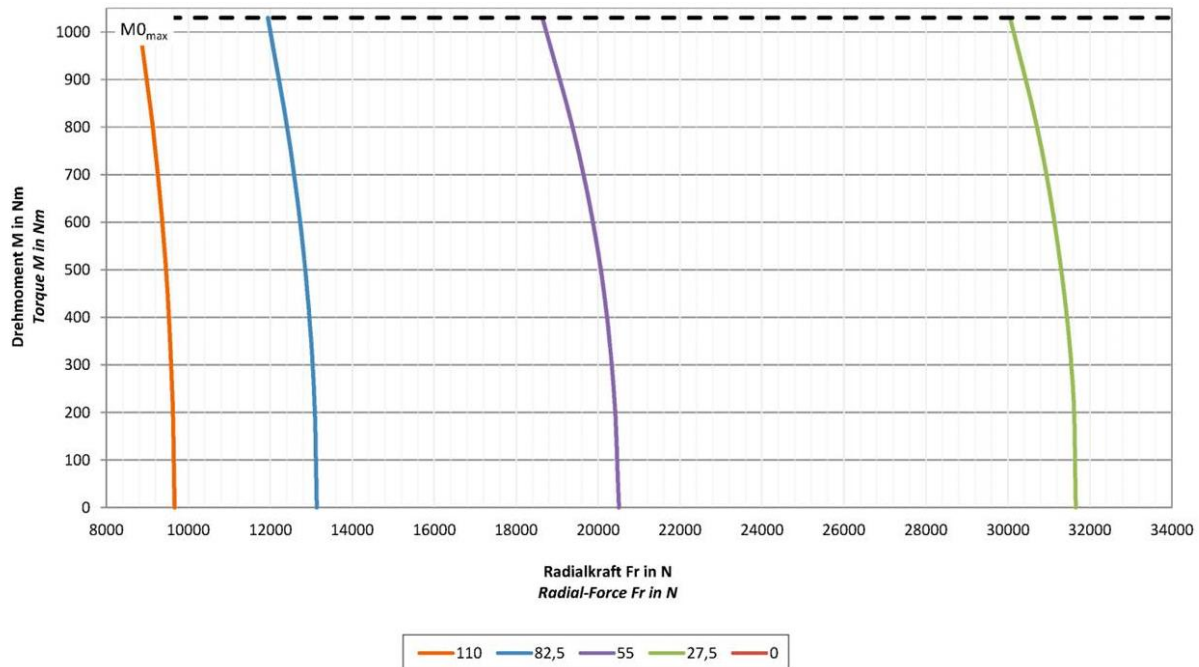
DSD2-132-W
Rollenlager / Roller bearing



Achtung: Technische Änderungen vorbehalten !

At roller bearing: Grease service life F_{h10} of 20.000 h at $n_{\text{medium}} \leq 1000 \text{ min}^{-1}$

DSD2-132
Welle
Shaft



3. Motor components (options)

3.1. Holding brake DSD2-028-100

The motors can be optionally equipped with a holding brake. The holding brake is a backlash-free permanent magnetic brake. The brakes work according to the closed current principle, i.e. the brake is applied when switched off (or at a failure of the operating voltage). The brakes are designed for an operating voltage of 24 VDC. The specifications by the brake manufacturer apply at room temperature.

The motors are available with the following holding brakes:

Motor type	DSD2-028	DSD2-036	DSD2-045	DSD2-056	DSD2-071	DSD2-100
minimal static holding torque [Nm] at 120 °C.	2	4	10	20	45	105
nominal dynamic holding torque [Nm] at 120 °C.	1,5	3,5	8	18	25	45
Maximum switching energy [J] per braking from n = 3,000 rpm	170	220	270	320	1400	2800
Connection values [V] (+6 % / -10 %)	24	24	24	24	24	24
Power consumption [W]	11	12	18	20	28	50
Moment of inertia [kgcm ²]	0.068	0.18	0.6	2,9	7,9	17,6
Switching time On [ms] Ventilation; with basic air gap	25	35	40	65	100	200
Switching time Off [ms] Braking; with basic air gap	2	2,5	20	30	40	50

All brakes are not fail safe brakes in the sense that a torque reduction cannot occur due to uninfluenceable malfunction factors. Depending on the application, the relevant accident prevention regulations, as well as basic health and safety requirements of Annex I of the Machinery Directive and the harmonized European standards must be observed.

For emergency stops or power failures, approximately 2,000 brake processes can be performed.

(Condition: maximum external inertia = motor inertia and $n_{\max}$ type-related;

max. braking / hour <20; evenly distributed).

3.2. Holding brake DSD2-132

for motor type	Brake type	Brake torque M_B for the holding brake [Nm]	Power input [W]	Current at 24 V DC [A]	max. perm. switch work W_{perm} , per switching [kJ]	Response time [s]	Drop time [s]	Torque of inertia [kgm ²]	max. perm. speed [min ⁻¹]	Weight [kg]
DSD2-132	SB 200	200	170	6,5	20	0.225	0.3	0.0040	3000	13

¹⁾ on request for radial ventilation

The following must be considered when using the holding brake:

- **Three emergency stops at maximum** (single braking) per hour that were uniformly distributed are possible
- The values for the switching times are valid for AC-sided switching, in the cold state, at the standard air gap and holding brake
- Response time – time till to the complete brake release (brake without torque)
- Drop time – time till the brake torque is reached
- All specifications apply to an installation on horizontal shaft
- If vertical running is required the supplier must be checked back
- Requirements deviating from the overview can be obtained on request.

Braking time / switch work

The brake must be checked upon its case of operation. Therefore, the switch work must be determined.

Determination of braking time t_B

$$t_B = \frac{\sum J \cdot \Delta n}{9,55 \cdot (M_B \pm M_L)} + t_0 \quad \text{in s}$$

$\sum J$ Total torque of inertia in kgm² = $J_{mot} + J_{zus}$ (referring to motor shaft)

J_{mot} Motor torque of inertia in kgm²

J_{zus} Additional torque of inertia in kgm² (referring to the motor shaft)

Δn Motor speed in min⁻¹

M_B Braking torque in Nm

M_L Load torque in Nm (positively calculated if it operates in the braking mode; negatively calculated if it acts in the accelerating mode)

t_0 Time in s from the operating instant to the completion of the braking torque (response time)

I Number of work cycles per hour

Determination of switching work $[W_R]$

$$W_R = \frac{\sum J \cdot \Delta n^2}{182,4} \cdot \frac{M_B}{(M_B \pm M_L)} \quad \text{in } \frac{\text{Joule}}{\text{Switching}}$$

Determination of switching capacity $[P_R]$

$$P_R = \frac{W_R \cdot i}{1000} \quad \text{in } \frac{\text{kJ}}{\text{h}}$$

$W_{Rperm} \leq$ Value from the table

In the most cases t_0 is negligible small. If this isn't the case and the time t_0 shall be reduced, this can be reached by interrupting the magnet circuit DC-voltage-sided.

However, this measure must be known before dimensioning the brake motor.

3.2.1. Brake supply

Standard: Normal voltage 24 V DC supply with transformer and rectifier.

Option: Normal voltage 104 and 176 V DC supply with brake supply device.

The brake supply device must be ordered separately. The brakes have micro-switches (NO contacts). The silver contacts are covered with a gold coating, so that there are two application ranges available. The gold coating can be irreversibly burnt by exceeding the maximum, load of the gold layer. In this case the contact material "gold coating" may not be used anymore.

Electrical characteristics of the switches:

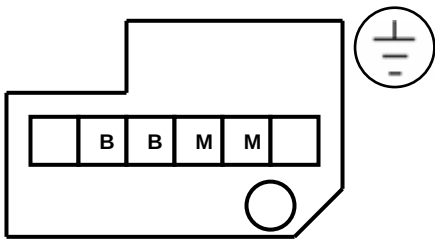
Contact material	Minimum load	Ideal range of application		Max. load
Gold coating	0 mA; 0 V to 3 million cycles	0 mA; 0 V to 3 million cycles	10 mA; 12 V to 1 million cycles	0.1 A; 12 V to 100.000 cycles
Silver	10 mA; 12 V to 3 million cycles	100 mA; 12 V to 3 million cycles	5 A; 30 V to 50.000 cycles	5 A; 30 V to 50.000 cycles

Optionally the brakes can be made with a manual lifting and locking devices.

3.2.2. Brake connection

DSD2-132 at radial forced ventilation

Terminal:

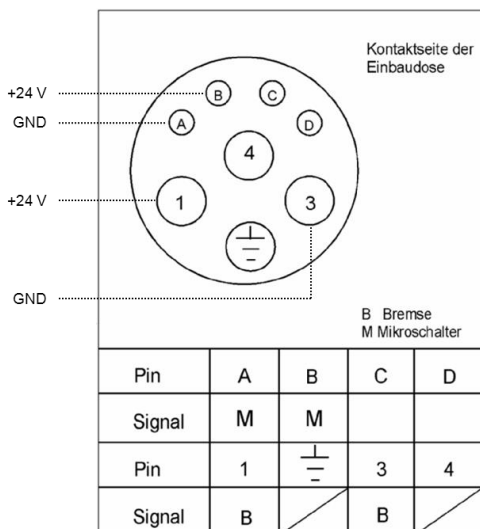


B - Brake

M – Micro switch

DSD2-132 at axial forced ventilation

Connection diagram and pin assignment:

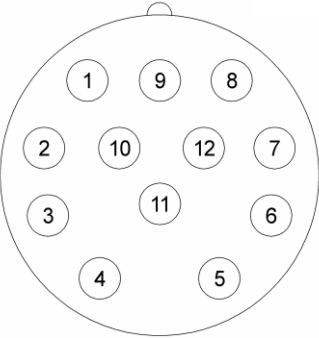


3.3. Encoder

3.3.1. Resolver

Motor size	DSD2-028-036	DSD2-045-100
Pole pair number	1	1
Transmission ratio	0.5 ± 0.05	0.5 ± 0.05
Frequency	5 kHz	5 kHz
Nominal input voltage	$7 V_{rms}$	$7 V_{rms}$
Effective input power at no-load speed	33 mW	112 mW
Current consumption at no-load speed	58 mA	70 mA
Max. output voltage at no-load speed	$3,5 V \pm 10\%$	$3,5 V \pm 10\%$
Voltage constant	61 mV/°	61 mV/°
Rotor resistance	$40 \Omega \pm 10\%$	$48 \Omega \pm 10\%$
Stator resistance	$102 \Omega \pm 10\%$	$31 \Omega \pm 15\%$
Rotor impedance at no-load speed	$75 + j 100 \Omega \pm 15\%$	$70 + j 74 \Omega \pm 15\%$
Rotor impedance with short circuit	$70 + j 85 \Omega \pm 15\%$	$62 + j 66 \Omega \pm 15\%$
Stator impedance at no-load speed with minimum coupling	$180 + j 230 \Omega \pm 15\%$	$108 + j 206 \Omega \pm 15\%$
Stator impedance with short circuit and maximum coupling	$170 + j 200 \Omega \pm 15\%$	$97 + j 183 \Omega \pm 15\%$
Phase shift	$8^\circ \pm 3^\circ$	$8^\circ \pm 3^\circ$
Zero voltage	30 mV	30 mV
Angle error in relation to $(\Delta\varphi_{max} + \Delta\varphi_{min})/2$	$\pm 10'$	$\pm 6'$
Shock according to DIN EN 60068-2-27 (11 ms)	$\leq 1000 \text{ m/s}^2$	$\leq 1000 \text{ m/s}^2$
Vibration according to DIN EN 60068-2-6	$\leq 500 \text{ m/s}^2$ (10-500 Hz)	$\leq 500 \text{ m/s}^2$ (55-2000 Hz)

Resolver connection

	Pin	Signal	Assignment option PT1000 (R1/R2) on encoder socket
	1	cos -	cos -
	2	-	-
	3	-	-
	4	-	-
	5	sin -	sin -
	6	sin +	sin +
	7	-	R2
	8	cos +	cos +
	9	-	R1
	10	ref +	ref +
	11	-	-
	12	ref -	ref -

View on the contact side of the socket

NOTE:

Use only at low demands on the true running characteristics of the motor.
The technical data is specification from the encoder manufacturer.



3.3.2. Resolver for safety-oriented applications

Motor size	DSD2-028-036	DSD2-045-132
Pole pair number	1	1
Transmission ratio	0.5 ± 0.05	0.5 ± 0.05
Frequency	5 kHz	5 kHz
Safety integrity level	SIL 3 (IEC 61508) in combination with b maXX 5000	
Performance level	PL e (EN ISO 13849) in combination with b maXX 5000	
Maximum angular acceleration	200.000 rad/s^2	100.000 rad/s^2
Effective input power at no-load speed	33 mW	112 mW
Nominal input voltage	7 V_{rms}	7 V_{rms}
Current consumption at no-load speed	58 mA	70 mA
Max. output voltage at no-load speed	$3,5 \text{ V} \pm 10\%$	$3,5 \text{ V} \pm 10\%$
Voltage constant	61 mV/°	61 mV/°
Rotor resistance	$40 \Omega \pm 10\%$	$48 \Omega \pm 10\%$
Stator resistance	$102 \Omega \pm 10\%$	$31 \Omega \pm 15\%$
Rotor impedance at no-load speed	$75 + j 100 \Omega \pm 15\%$	$70 + j 74 \Omega \pm 15\%$
Rotor impedance with short circuit	$70 + j 85 \Omega \pm 15\%$	$62 + j 66 \Omega \pm 15\%$
Stator impedance at no-load speed with minimum coupling	$180 + j 230 \Omega \pm 15\%$	$108 + j 206 \Omega \pm 15\%$
Stator impedance with short circuit and maximum coupling	$170 + j 200 \Omega \pm 15\%$	$97 + j 183 \Omega \pm 15\%$
Phase shift	$8^\circ \pm 3^\circ$	$8^\circ \pm 3^\circ$
Zero voltage	30 mV	30 mV
Angle error related to $(\Delta\varphi_{\text{max}} + \Delta\varphi_{\text{min}})/2$	$\pm 10'$	$\pm 6'$
Shock according to DIN EN 60068-2-27 (11 ms)	$\leq 1000 \text{ m/s}^2$	$\leq 1000 \text{ m/s}^2$
Vibration according to DIN EN 60068-2-6	$\leq 500 \text{ m/s}^2$ (10-500 Hz)	$\leq 500 \text{ m/s}^2$ (55-2000 Hz)

Resolver connection

	Pin	Signal	Assignment option at PT1000 (R1/R2) on encoder socket
	1	cos -	cos -
	2	-	-
	3	-	-
	4	-	-
	5	sin -	sin -
	6	sin +	sin +
	7	-	R2
	8	cos +	cos +
	9	-	R1
	10	ref +	ref +
	11	-	-
	12	ref -	ref -

View of contact side of the socket

NOTE:

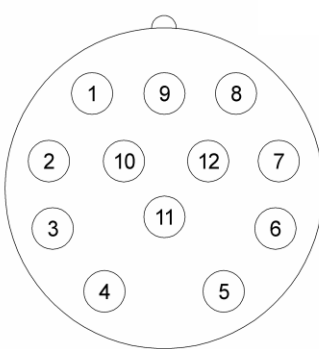
Use only, if the requirements to the smooth running characteristics of the motor are low.
The technical data were specified by the encoder manufacturer.

Information regarding configuration options of the safety encoder with the different motor versions can be abstracted from the product configurator. The axial ventilation and the brake installation with the safety-oriented resolvers at the DSD2-132 are available on request.

3.3.3. SINCOS SEK/SEL37 (SICK)

Motor size	DSD2-028-132	
	SEK37	SEL37
Number of sine-, cosine periods per revolution	16	
Measuring step at interpolation of the sine-, cosine periods such as with 12 bits	20°	
Number of absolutely completed revolutions	1	4.096
Code type for the absolute value	binary	
Error limits at the evaluation of the sine-, cosine periods; integral non-linearity	+/- 288°	
Non-linearity within a sine-, cosine period, differential non-linearity at nominal position +/- 0.1mm	+/- 144°	
Operating speed up to which the absolute position can be made	6.000 1/min	
Maximum operating speed	12.000 1/min	
Output signal	serial RS 485, asynchronous, half duplex	
Operating voltage range	7-12 V	
max. operating current without load	50 mA	
Shock according to DIN EN 60068-2-27 (10 ms)	980 m/s²	
Vibration according to DIN EN 60068-2-6 (10-2000 Hz)	490 m/s²	

SEK/SEL37 Connection

	Pin	Signal	Assignment option PT1000 (R1/R2) at encoder socket
	1	cos -	cos -
	2	+ 485	+ 485
	3	-	R1
	4	-	R2
	5	sin +	sin +
	6	sin -	sin -
	7	- 485	- 485
	8	cos +	cos +
	9	-	-
	10	GND	GND
	11	-	-
	12	+ U	+ U

View on the contact side of the socket

NOTE:

Relevant for the sizes 045-100

SEK/SEL37 – with mechanical adaption	The notch position is 0°(identical to SEK/SEL52 encoders)
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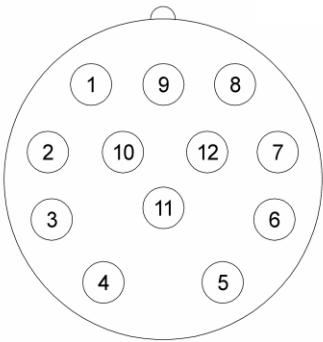
This encoder is a component susceptible to ESD.

The technical data is specification from the encoder manufacturer.

3.3.4. SINCOS SKS/SKM36 and SRS/SRM50 (SICK)

Motor size	DSD2-028-036		DSD2-045-100	
	SKS36	SKM36	SRS50	SRM50
Number of sine and cosine periods per revolution	128		1024	
Number of steps per revolution	4096		32768	
Number of absolute completed revolutions	1	4096	1	4096
Code type for the absolute value	Binary		Binary	
Output frequency of the sine and cosine signals	0-65 kHz		0-200 kHz	
Fault limits in evaluation of the sine, cosine signals; integral non-linearity in the relaxed state	+/- 80"		+/- 45"	
Nonlinearity within a sine or cosine period; differential nonlinearity	+/- 40"		+/- 7"	
Maximum speed at which the absolute position can be defined	6000 rpm		6000 rpm	
Maximum operating speed	12000 rpm		12000 rpm	
Output signals; 2 x 90° offset sinusoidal signals	1 V _{SS}		1 V _{SS}	
Output signal	Serial RS 485, asynchronous, half-duplex		Serial RS 485, asynchronous, half-duplex	
Operating voltage range	7-12 V		7-12 V	
Operating current without load	60 mA		80 mA	
Shock according to DIN EN 60068-2-27	100 g / 6 ms		100 g / 10 ms	
Vibration according to EN 60068-2-6 (10-2000 Hz)	50 g		20 g	

SKS/SKM36 and SRS/SRM50 connection

	Pin	Signal	Assignment option PT1000 (R1/R2) at encoder socket
	1	cos -	cos -
	2	+ 485	+ 485
	3	-	R1
	4	-	R2
	5	sin +	sin +
	6	sin -	sin -
	7	- 485	- 485
	8	cos +	cos +
	9	-	-
	10	GND	GND
	11	-	-
	12	+ U	+ U

View on the contact side of the socket

NOTE:

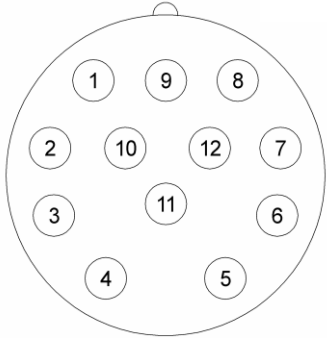
This encoder is a component susceptible to ESD.
The technical data is specification from the encoder manufacturer.

3.3.5. SINCOS SKS/SKM36-S and SRS/SRM50-S (SICK)



Motor size	DSD2-028-036		DSD2-036-132	
	SKS36-S	SKM36-S	SRS50-S	SRM50-S
Safety integrity level	SIL2 (IEC 61508), SILCL2 (IEC 62061)			
Category	3 (EN ISO 13849)			
Performance Level	PL d (EN ISO 13849)			
Maximum angular acceleration	200.000 rad/s ²			
Number of sine-, cosine periods per revolution	128		1.024	
Number of steps per revolution	4.096		32.768	
Number of the absolutely completed revolutions	1	4.096	1	4.096
Code type of the absolute value	binary		binary	
Output frequency of the sine-, cosine signals	0-65 kHz		0-200 kHz	
Error limits at the evaluation of the sine-, cosine signals, integral non-linearity	+/- 80"		+/- 45"	
Non-linearity within a sine-, cosine period, differential non-linearity	+/- 40"		+/- 7"	
Operating speed up to which the absolute position can be made	6.000 1/min		6.000 1/min	
Maximum operating speed	12.000 1/min		12.000 1/min	
Output signals; 2x90° shifted sine-shaped signals	1 V _{SS}		1 V _{SS}	
Output signal	serial RS 485, asynchronous, half duplex		serial RS 485, asynchronous, half duplex	
Operating voltage range	7-12 V		7-12 V	
Operating current without load	60 mA		80 mA	
Shock according to DIN EN 60068-2-27	980 m/s ² (6 ms)		980 m/s ² (10 ms)	
Vibration according to DIN EN 60068-2-6 (10-2000 Hz)	490 m/s ²		196 m/s ²	

SKS/SKM36-S und SRS/SRM50-S Anschluss

	Pin	Signal	Assignment option PT1000 (R1/R2) at encoder socket
	1	cos -	cos -
	2	+ 485	+ 485
	3	-	R1
	4	-	R2
	5	sin +	sin +
	6	sin -	sin -
	7	- 485	- 485
	8	cos +	cos +
	9	-	-
	10	GND	GND
	11	-	-
	12	+ U	+ U

View on the contact side of the socket

NOTE:

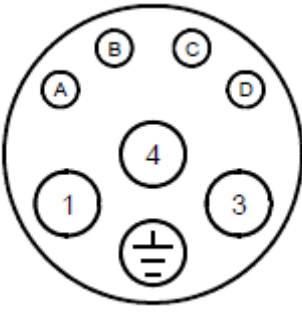
This encoder is a ESD-sensitive device. The technical data is an information of the encoder manufacturer. The configuration options of the safety encoder with different motor versions can be taken from the product configurator. The combination SRS/SRM50-S with installation gear is available on request.

The axial ventilation and the brake installation in combination with the above-mentioned encoders at the DSD2-132 can be obtained on request.

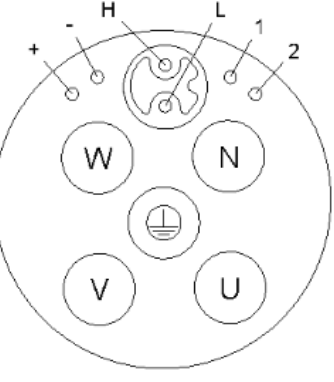
3.3.6. EES37/EEM37 Hiperface DSL® (SICK)

Motor size	DSD2-045-132	
	EES37	EEM37
Number of the absolutely completed revolutions	1 (15 bit)	4096 (15 bit)
Code type of the absolute value	Binary	
Interface signals	Digital, RS-485	
Position values/revolution	32.768	
Maximum angular acceleration	500.000 rad/s ²	
Maximum operating speed	12.000 1/min	
Voltage supply	7...12 V	
Current consumption without load	≤ 150 mA	
Shock according to DIN EN 60068-2-27 (6 ms)	980 m/s ²	
Vibration according to DIN EN 60068-2-6 (10-2000 Hz)	490 m/s ²	
Working temperature	-40°C...+115°C	

Connection EES37/EEM37 DSL Hiperface

	Pin	Signal
	A	U
	B	V
	C	W
	D	GN / GE
	E	Blue
	F	White
	G	Inner shield encoderr
	H	+U / DSL+
	L	GND / DSL-

View on the contact side of the socket size 1

	Pin	Signal
	U	U
	V	V
	W	W
	N	/
	⊕	GN/GE
	+	blue
	-	white
	1	Inner shield
	2	/
	H	+U / DSL+
	L	GND / DSL-

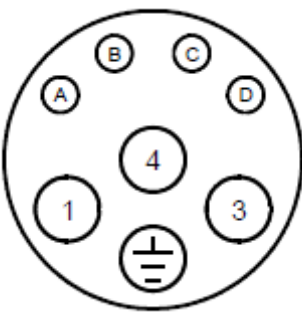
View on the contact side of the socket size 1.5

The configuration options of the Hiperface DSL encoder with different motor versions can be taken from the product configurator. The encoder can be used up to a cable length of 60 m.

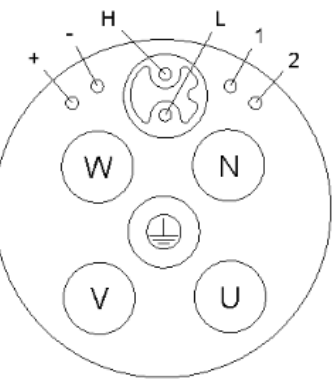
3.3.7. EKS36/EKM36 Hiperface DSL® (SICK)

Motor size	DSD2-028-036	
	EKS36	EKM36
Number of the absolutely completed revolutions	1 (18 bit)	4096 (18 bit)
Code type for the absolute value	Binary	
Interface signals	Digital, RS-485	
Position values/revolution	262.144	262.144
Maximum operating speed	12.000 1/min	9.000 1/min
Power supply	7...12 V	
Current consumption without load	≤ 150 mA	≤ 150 mA
Shock according to DIN EN 60068-2-27 (6 ms)	980 m/s ²	
Vibration according to DIN EN 60068-2-6 (10-2000 Hz)	490 m/s ²	
Working temperature	-20°C...+115°C	

Connection EKS36/EKM36 DSL Hiperface

	Pin	Signal
	A	U
	B	V
	C	W
	D	GN / GE
	E	blue
	F	white
	G	Inner shield encoder
	H	+U / DSL+
	L	GND / DSL-

View on the contact side of the socket size 1

	Pin	Signal
	U	U
	V	V
	W	W
	N	/
	⊕	GN / GE
	+	blue
	-	white
	1	Inner shield encoder
	2	/
	H	+U / DSL+
	L	GND / DSL-

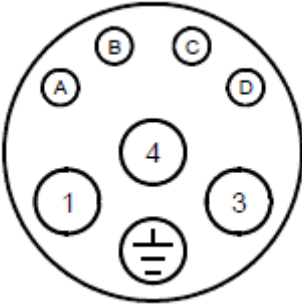
View of the contact side of the socket size 1,5

The configuration options for the Hiperface DSL encoder with different motor versions can be found in the product configurator. The encoders can be used up to a cable length of 60 m.

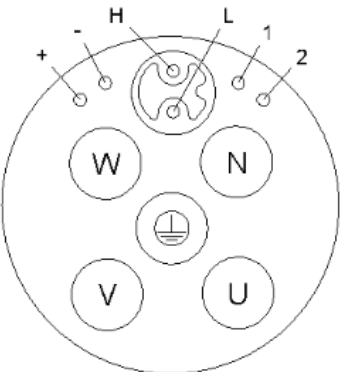
3.3.8. EFS50/EFM50 Hiperface DSL® (SICK)

Motor size	DSD2-045-132	
	EFS50	EFM50
Number of absolutely completed revolutions	1 (21 bit)	4096 (21 bit)
Code type of the absolute value	Binary	
Interface signals	Digital, RS-485	
Positioning values/revolution	2.097.152	
Maximum angular acceleration	200.000 rad/s ²	
Maximum operating speed	12.000 1/min	9.000 1/min
Voltage supply	7...12 V	
Current consumption without load	≤ 150 mA	
Shock according to DIN EN 60068-2-27 (6 ms)	980 m/s ²	
Vibration according to DIN EN 60068-2-6 (10-2000 Hz)	196 m/s ²	
Working temperature	-20°C...+120°C	

Connection EFS50/EFM50 DSL Hiperface

	Pin	Signal
	A	U
	B	V
	C	W
	D	GN / GE
	E	blue
	F	white
	G	Inner shield encoder
	H	+U / DSL+
	L	GND / DSL-

View on the contact side of the socket size 1

	Pin	Signal
	U	U
	V	V
	W	W
	N	/
	⊕	GN/GE
	+	Blue u
	-	white
	1	Inner shield
	2	/
	H	+U / DSL+
	L	GND / DSL-

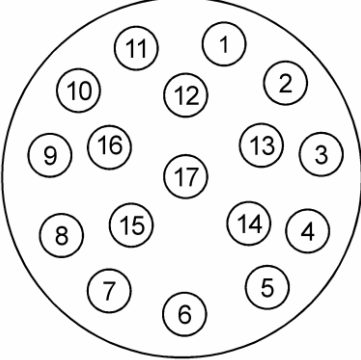
View of the contact side of the socket size 1.5

The configuration options of the Hiperface DSL encoder with different engine versions can be found in the product configurator. The encoders can be used up to a cable length of 60 m.

3.3.9. ECN1313/EQN1325 (Heidenhain)

Motor size	DSD2-045-132	
	ECN 1313	EQN 1325
Number of sine and cosine periods per revolution	2.048	
System accuracy	± 20"	
Number of absolute completed revolutions	1	4.096 (12 bit)
Code type for the absolute value	EnDat 2.1	
Sampling limit frequency or limit frequency	0-200 kHz	
Position values/revolution	8.192 (13 bit)	
Maximum speed at which the absolute position can be defined	12.000 1/min	
Maximum operating speed (1/min)	12.000 1/min	
Voltage supply	3,6-14 V	
Current consumption without load	≤ 160 mA	≤ 200 mA
Shock 6ms as per DIN EN 60068-2-27 (6 ms)	≤ 2.000 m/s ²	
Vibration 55-2000Hz as per DIN EN 60068-2-6 (55-2000 Hz)	≤ 300 m/s ²	

ECN1313/EQN1325 Anschluss

	Pin	Signal	Assignment option PT1000 (R1/R2) at encoder socket
	1	U _p	U _p
	2	-	-
	3	-	-
	4	0V	0V
	5	-	R1
	6	-	R2
	7	U _p	U _p
	8	Clock	Clock
	9	Clock inv.	Clock inv.
	10	0V	0V
	11	-	-
	12	B +	B +
	13	B -	B -
	14	Data	Data
	15	A +	A +
	16	A -	A -
	17	Data inv.	Data inv.

View on the contact side of the socket

NOTE:

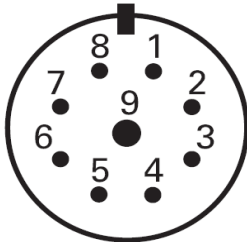
This encoder is a component susceptible to ESD.

The technical data is specification from the encoder manufacturer.

3.3.10. ECI1319/EQI1331 (Heidenhain)

Motor size	DSD2-056-132	
	ECI 1319	EQI 1331
Number of lines	-	
System accuracy	$\pm 65''$	
Number of absolutely completed revolutions	1	4096 (12 bit)
Code type for the absolute value	EnDat 2.2	
Position values/revolution	524 288 (19 bit)	
Maximum speed at which the absolute position can be defined	15000 1/min	
Maximum operating speed	15000 1/min	12000 1/min
Power supply	3,6... 14 V	
Current consumption without load	95 mA	95 mA
Shock 6ms according to DIN EN 60068-2-27 (6ms)	$\leq 2000 \text{ m/s}^2$	
Vibration 55-2000Hz according to DIN EN 60068-2-6 (55-2000 Hz)	$\leq 400 \text{ m/s}^2$	

ECN1319/EQN1331 connection

	Pin	Signal
	1	Clock
	2	Clock inv.
	3	U_p
	4	0V
	5	Data
	6	Data inv.
	7	Sensor U_p
	8	Sensor 0V
	9	-

View on the contact side of the socket

NOTE:

This encoder is a component susceptible to ESD.

The technical data is specification from the encoder manufacturer.



3.3.11. ECI1319/EQI1331-S (Heidenhain)

Motor size	DSD2-056-132	
	ECI 1319-S	EQI 1331-S
Number of lines	-	
System accuracy	$\pm 65''$	
Number of absolutely completed revolutions	1	4096 (12 bit)
Code type for the absolute value	EnDat 2.2	
Position values/revolution	524 288 (19 bit)	
Maximum speed at which the absolute position can be defined	15000 1/min	
Maximum operating speed	15000 1/min	12000 1/min
Power supply	3,6... 14 V	
Current consumption without load	95 mA	95 mA
Shock 6ms according to DIN EN 60068-2-27 (6ms)	$\leq 2000 \text{ m/s}^2$	
Vibration 55-2000Hz according to DIN EN 60068-2-6 (55-2000 Hz)	$\leq 400 \text{ m/s}^2$	

ECN1319/EQN1331-S connection

	Pin	Signal
	1	Clock
	2	Clock inv.
	3	U_p
	4	0V
	5	Data
	6	Data inv.
	7	Sensor U_p
	8	Sensor 0V
	9	-

View on the contact side of the socket

NOTE:

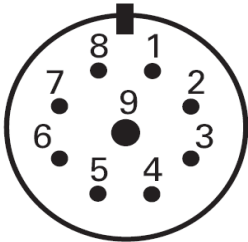
This encoder is a component susceptible to ESD.

The technical data is specification from the encoder manufacturer.

3.3.12. ECN1325/EQN1337 (Heidenhain)

Motor size	DSD2-045-132	
	ECN 1325	EQN 1337
Number of lines	2048	
System accuracy	± 20“	
Number of absolute completed revolutions	1	4096 (12 bit)
Code type for the absolute value	EnDat 2.2	
Position values/revolution	33554432 (25 bit)	
Maximum speed at which the absolute position can be defined	12000 1/min	
Maximum operating speed	12000 1/min	
Power supply	3,6...14 V	
Current consumption without load	≤ 160 mA	≤ 200 mA
Shock 6ms according to DIN EN 60068-2-27 (6ms)	≤ 2000 m/s²	
Vibration 55-2000Hz according to DIN EN 60068-2-6 (55-2000 Hz)	≤ 300 m/s²	

ECN1325/EQN1337 connection

	Pin	Signal
	1	Clock
	2	Clock inv.
	3	U _p
	4	0V
	5	Data
	6	Data inv.
	7	Sensor U _p
	8	Sensor 0V
	9	-

View on the contact side of the socket

NOTE:
This encoder is a component susceptible to ESD.
The technical data is specification from the encoder manufacturer.



3.3.13. ECN1325/EQN1337-S (Heidenhain)

Motor size	DSD2-045-132	
	ECN 1325-S	EQN 1337-S
Safety integrity level	SIL 2 nach EN 61508	
Category	3 (EN ISO 13849)	
Performance Level	PL d (EN ISO 13849)	
Maximum angular acceleration	50.000 rad/s ²	
Pulses per revolution	2.048	
System accuracy	± 20"	
Number of absolute completed revolutions	1	Number of absolute completed revolutions
Code type for the absolute value	EnDat 2.2	
Position values / revolution	33.554.432 (25 bit)	
Maximum speed at which the absolute position can be defined	12.000 1/min	
Maximum operating speed	12.000 1/min	
Power supply	3,6...14 V	
Current consumption without load	≤ 160mA	Current consumption without load
Shock according to DIN EN 60068-2-27 (6 ms)	≤ 2000 m/s ²	
Vibration according to DIN EN 60068-2-6 (55-2000 Hz)	≤ 300 m/s ²	

ECN1325/EQN1337-S Connection

	Pin	Signal
	1	Clock
	2	Clock inv.
	3	U _p
	4	0V
	5	Data
	6	Data inv.
	7	Sensor U _p
	8	Sensor 0V
	9	-

View of the contact side of the socket

NOTE:

This encoder is a component susceptible to ESD.
The technical data is the specifications of the encoder manufacturer.

The configuration options for the safety encoders with different engine versions can be found in the product configurator.

The combination ECN1325/EQN1337-S with add-on gearbox is available by request. These encoders are not available for uncooled motors with $n_{\text{medium}} \geq 4500 \text{ min}^{-1}$.

The axial ventilation and the braking installation are available in combination with the above mentioned encoders at the DSD2-132 on request.

3.4. Encoder cables for b maXX 4000

General Information

A prefabricated encoder cable is used for all encoder systems. The connection at the motor end consists of a 12-pole circular signal connector on resolvers and Hyperface® – encoders, a 17-pole circular signal connector on ECN1313/EQN1325 and a 9-pole circular signal connector on ECN1325/EQN1337. The connection at the controller side consists of a 15-pole D-Sub connector. Alternatively, the signal connector on the motor side is available for Speedtec versions with trailing cables.

The dragable cable is suitable for mobile applications such as drag chains, for example. Unlike non-dragable cables made from PVC, the cable sheath is made from durable PU (suitable for environments where acids and bases are present).

3.4.1. Technical data

Technical description - non-dragable for resolver/ SinCos Hyperface®-interface / SinCos - and TTL - incremental encoder

- LiYCY, 5x (2x0.14mm²) + 2 x 0.5mm² copper strand, twisted pairs
- PVC sheath, grey; inscription with Baumüller logo, black
- 1st side: 12-pole circular signal plug connector with 12 socket contacts
- 2nd side: 15-pole Sub-D plug connector with pin contacts and locking screws 4-40UNC
- Outer diameter 9.0 mm (+/- 0.3mm)
- Bending radius: $r \geq 60$ mm (fixed routing), $r \geq 135$ mm (flexible use)
- Nominal voltage: 250V_{AC}

Technical description - dragable for resolver/ SinCos Hyperface®-interface / SinCos - and TTL - incremental encoder

- Li12YC11Y, 5x (2x0.14mm²) + 2 x 0.5mm² copper strand, twisted pairs
- PU sheath, black; inscription with Baumüller logo, white
- 1st side: 12-pole circular signal plug connector with 12 socket contacts
- 2nd side: 15-pole Sub-D plug connector with pin contacts and locking screws 4-40UNC
- Outer diameter 9.0 mm (+/- 0.3mm)
- Bending radius: $r \geq 70$ mm (fixed routing), $r \geq 100$ mm (flexible use)
- Nominal voltage: 300V_{AC}

Technical description - non-dragable for EnDat® 2.1-interface

- LiYCY, 5x (2x0.14mm²) + 2 x 0.5mm² copper strand, twisted pairs
- PVC sheath, grey; inscription with Baumüller logo, black
- 1st side: 17-pole circular signal plug connector with 17 socket contacts
- 2nd side: 15-pole Sub-D plug connector with pin contacts and locking screws 4-40UNC
- Outer diameter 9.0 mm (+/- 0.3mm)
- Bending radius: $r \geq 60$ mm (fixed routing), $r \geq 135$ mm (flexible use)
- Nominal voltage: 250V_{AC}

Technical description - dragable for EnDat® 2.1-interface

- Li12YC11Y, 5x (2x0.14mm²) + 2 x 0.5mm² copper strand, twisted pairs
- PU sheath, black; inscription with Baumüller logo, white
- 1st side: 17-pole circular signal plug connector with 17 socket contacts
- 2nd side: 15-pole Sub-D plug connector with pin contacts and locking screws 4-40UNC
- Outer diameter 9.0 mm (+/- 0.3mm)
- Bending radius: $r \geq 70$ mm (fixed routing), $r \geq 100$ mm (flexible use)
- Nominal voltage: 300V_{AC}

Technical description - dragable for EnDat® 2.2-interface

- PUR sheath, 1x(4x0.14mm²) + (4x0.34mm²)
- 1 twisted foursome 0.14mm², 4 wires 0.34mm², copper, tin-plated
- Total shield CuSn, inscription Heidenhain
- 1st side: 9-pole circular signal plug connector with 8 socket contacts
- 2nd side: 15-pole Sub-D plug connector with pin contacts and locking screws 4-40UNC
- Outer diameter 6.0 mm
- Bending radius: $r \geq 20$ mm (fixed routing), $r \geq 75$ mm (flexible use)
- Dielectric strength wire/wire and wire/shield: 0.5kV at 50Hz, 1 minute

3.4.2. Application references
Operating temperature of encoder cable resolver/ SinCos Hiperface®-interface / SinCos - and TTL - incremental encoder / EnDat® 2.1

	Dragable	Not dragable
Limit temperature	on the surface	on the surface
Static use/minimal movement	- 40 °C to + 80 °C	- 30 °C to + 80 °C
Permanent movement	- 30 °C to + 80 °C	- 5 °C to + 70 °C

Operating temperature of encoder cable EnDat® 2.2

	Dragable
Limit temperature	on the surface
Static use/minimal movement	- 40 °C to + 80 °C
Permanent movement	- 10 °C to + 80 °C

Routing of cable on motor

The cables must not touch the surface of the motor.

3.4.3. Order information for encoder cables

Encoder cables for resolver/ SinCos Hiperface®-interface / SinCos - and TTL - incremental encoder - prefabricated cables with connector

Not dragable, prefabricated Cable 5 x (2x0.14mm ²) + 2 x 0.5 mm ² with plug connector		Dragable, prefabricated Cable 5 x (2x0.14mm ²) + 2 x 0.5 mm ² with plug connector		
Länge [m]	Part No.	Länge [m]	Part No.	Part No. (Speedtec)
1	243601	3	246658	448944
2	211338	4	243379	448945
3	219333	5	239540	448948
4	231166	6	242954	448946
5	209879	8	239541	448949
6	220197	10	239542	448956
7	216455	15	239543	448962
8	220429	20	239544	448967
10	210052	25	239545	448970
15	215716	30	239546	448971
20	218568	35	239547	448973
25	218569	40	240520	448976
30	217094	45	240521	448978
35	216444	50	240522	448980
40	217095	55	244033	448981
45	217567	60	245484	448982
50	217568			
55	217569			
60	217570			
70	232088			

Encoder cables for EnDat® 2.1- prefabricated cables with plug connector

Not dragable, prefabricated Cable 5 x (2x0.14mm ²) + 2 x 0.5 mm ² with plug connector		Dragable, prefabricated Cable 5 x (2x0.14mm ²) + 2 x 0.5 mm ² with plug connector		
Länge [m]	Part No.	Länge [m]	Part No.	Part No. (Speedtec)
2	383152	2	393889	448816
3	383923	3	369864	448817
5	393885	5	394014	448818
7	389445	7	389807	448819
8	380138	8	393890	448820
9	389446	9	389808	448821
10	393886	10	393891	448822
15	388505	15	393892	448823
20	388418	17	371494	448824
25	393887	20	393893	448825
30	393888	25	393894	448826
35	387958	30	380358	448827
40	382006	35	391216	448828
50	388419	40	382005	448830
70	384473	50	378022	448832
90	387391			

Encoder cables for EnDat® 2.2 - prefabricated cables with plug connector

Dragable, prefabricated

cable 1x4x0.14 + 4x0.34 PUR Ø 6mm with plug connector

Länge [m]	Part No.	Part No. (Speedtec)
2	434056	459031
3	434057	459032
5	434058	459033
10	434059	459035
15	434060	459036
20	434061	459037
25	434062	459038
50	434063	459042

3.5. Encoder cables for b maXX 5000

A prefabricated encoder cable is used for all encoder systems. The connection at the motor end consists of a 12-pole circular signal connector on resolvers and Hyperface® encoder, a 17-pole circular signal connector on ECN1313/EQN1325. The connection at the controller side consists of a 26-pole D-Sub connector. Alternatively, the signal connector on the motor side is available in a Speedtec version.

3.5.1. Technical data

Technical description - dragable for resolver

- Li9YC, 1 x (2 x 0.25) + Li9Y, 2 x (2x0.25) + Li9YC11Y, 1 x (2 x 0.34), copper strand, twisted pairs
- PUR sheat, green; inscription with Baumüller Nürnberg and encoder cable Resolver
- 1st side: 12-pole circular signal plug connector with 12 socket contacts
- 2nd side: 26-pole Sub-D plug connector with pin contacts and locking screws 4-40UNC
- Outer diameter 7.3 mm (+/- 0.3mm)
- Bending radius: $r \geq 4 \times D$ (fixed routing), $r \geq 10 \times D$ (flexible use)

Technical description - dragable for SinCos Hyperface®-interface und SinCos - and TTL - incremental encoder

- Li9YC, 3 x (2 x 0.25) , + Li9Y, 3 x (2 x 0.25) + Li9YC11Y, 1 x (2x0.34), copper strand, twisted pairs
- PUR sheat, green; inscription with Baumüller Nürnberg and encoder cable Hyperface or Incremental
- 1st side: 12-pole circular signal plug connector with 12 socket contacts
- 2nd side: 26-pole Sub-D plug connector with pin contacts and locking screws 4-40UNC
- Outer diameter 9.6 mm (+/- 0.3mm)
- Bending radius: $r \geq 4 \times D$ (fixed routing), $r \geq 10 \times D$ (flexible use)

Technical description – dragable for EnDat® 2.1-interface

- Li9YC, 3 x (2 x 0.25) , + Li9Y, 3 x (2 x 0.25) + Li9YC11Y, 1 x (2x0.34), copper strand, twisted pairs
- PUR sheat, green; inscription with Baumüller Nürnberg and encoder cable Endat 2.1
- 1st side: 17-pole circular signal plug connector with 17 socket contacts
- 2nd side: 26-pole Sub-D plug connector with pin contacts and locking screws 4-40UNC
- Outer diameter 9.6 mm (+/- 0.3mm)
- Bending radius: $r \geq 4 \times D$ (fixed routing), $r \geq 10 \times D$ (flexible use)

Technical description – Dragable hybrid cable for Hiperface DSL®

- Hybrid cable
- Shielding braid Copper wires, tinned
- PUR sheath, orange, flame-retardant, self-extinguishing
- 1st side: metal round plug Speedtec M23 hybrid socket 8-pole for wire with 4G1.5 and 4G2.5
metal round plug Speedtec M40 hybrid socket 5+4-pole + 2 -pole insulating body for wire with 4G2.5, 4G4 and 4G6
- 2nd side: Metal 45°-D-Sub plug, 26-pole with electronics

3.5.2. Application references**Operating temperature of encoder cable resolver/ SinCos Hiperface®-interface / SinCos - and TTL - incremental encoder / EnDat® 2.1**

Limit temperature	on the surface
Static use/minimal movement	- 40 °C to + 80 °C
Permanent movement	- 20 °C to + 60 °C

Routing of cable on motor

The cables must not touch the surface of the motor.

3.5.3. Order information for encoder cables**Encoder cable - prefabricated with plug****For Resolver**

Length [m]	Item Number	Item Number. (Speedtec)
1	429914	448746
2	429915	448747
3	429916	448748
5	429917	448749
7	429918	448750
10	429919	448751
15	429920	448752
20	429921	448753
25	429922	448754
30	429923	448755
35	429924	448756
40	429925	448757
50	429926	448758
75	429927	448759

For SinCos Hiperface® - interface

Length [m]	Item Number	Item Number (Speedtec)
1	429958	448761
2	429959	448762
3	429960	448763
5	429961	448764
7	429962	448765
10	429963	448766
15	429964	448767
20	429965	448768
25	429966	448769
30	429967	448770
35	429968	448772
40	429969	448773
50	429970	448774
75	429971	448775

For SinCos - and TTL - incremental encoder

Length [m]	Item Number	Item Number (Speedtec)
1	430015	448777
2	430016	448778
3	430017	448779
5	430018	448780
7	430019	448781
10	430020	448782

For SinCos EnDat® 2.1 - interface

Length [m]	Item Number	Item Number (Speedtec)
1	429986	448796
2	429987	448797
3	429988	448798
5	429989	448799
7	429990	448800
10	429991	448801

15	430021	448783
20	430022	448784
25	430023	448785
30	430024	448786
35	430025	448787
40	430026	448788
50	430027	448789
75	430028	448790

15	429992	448802
20	429993	448803
25	429994	448804
30	429995	448805
35	429996	448806
40	429997	448807
50	429998	448808
75	429999	448809

For Hiperface DSL® Hybrid cables size 1

Length [m]	Rated current 15A 4G1.5+(2x0.75)+ (2x22AWG)	Rated current 20A 4G2.5+(2x1.0)+ (2x22AWG)
	Part No.	Part No.
3	464201	464217
5	464202	464218
7	464203	464219
10	464204	464220
15	464205	464221
20	464206	464222
25	464207	464223
30	464208	464224
35	464209	464225
40	464210	464226
50	464211	464227
60	464212	464228

For Hiperface DSL® Hybrid cables size 1.5

Length [m]	Rated current 21A 4G2.5+(2x1.0)+(2x22AWG)	Rated current 28A 4G4.0+(2x1.0)+(2x22AWG)	Rated current 36A 4G6.0+(2x1.0)+(2x22AWG)
	Part No.	Part No.	Part No.
3	464235	464278	464294
5	464236	464279	464295
7	464237	464280	464296
10	464238	464281	464297
15	464239	464282	464298
20	464240	464283	464299
25	464241	464284	464300
30	464242	464285	464301
35	464243	464286	464302
40	464244	464287	464303
50	464245	464288	464304
60	464246	464289	464305

3.6. Motor cables

The motor cables are highly flexible trailing cables with overall shielding. They comply with VDE, UL and CSA regulations. The control cables are integrated as star quads. The brake control and the temperature sensor are connected via the main connector. The cables are particularly suited for the optimum use of cable racks thanks to their small cross-section, low weight, and non-impeding surface. As a result, they can be used efficiently in trailing chains. The overall shielding with an optical coverage of more than 85% makes the cable non-critical from an EMC perspective.

3.6.1. Technical data

- Sheath resistance to media such as coolants and machine and gearbox oils
- Abrasion resistance thanks to a special surface in cable racks and trailing chains
- Highly flexible trailing cable, minimum bending radius for flexible use: $12 \times D$
- Non-blocking sheath surface with satin finish
- Shield made of tinned copper braid with optical coverage of $\geq 85\%$
- Core insulation made from TPE or polyester, sheath material: Halogen-free PUR
- Cable is CFC and silicone-free
- Behavior in the event of fire: Fire-inhibiting, halogen-free
- Cable color RAL 1028, melon yellow
- Label features Baumüller logo and VDE, UL and CSA marks

Rated voltage

- U_0/U 600/1.000 V (power cores)
- U 24 V DC (control cores)

Core labeling

- Power cores U, VV, WWW
- Colored control cable pairs as star quads in red, white, black, yellow

Assignment of pairs: (note the polarity)

- Red – black (brake)
- white – yellow (temperature)

3.6.2. Main connection via connector

Note:

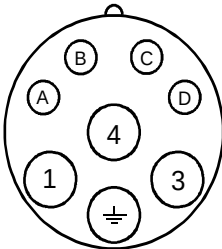
The connector size is determined by the standstill current I_0 of the motor used.

Motors with a standstill current of ≤ 20 A feature a size 1 main connector.

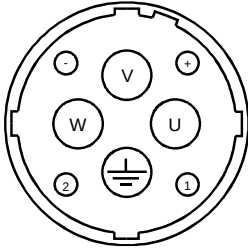
For standstill currents of $20 \text{ A} < I_0 \leq 36 \text{ A}$, a size 1.5 main connector is used.

A terminal box must be used at a $I_0 > 36 \text{ A}$.

Poles of the female main connectors:

		Pin	Signal	Color/labeling
Size 1 $I_0 \leq 20 \text{ A}$		1	Phase U	U
			PE	Green/yellow
		3	Phase V	V V
		4	Phase W	W W W
		A	B+	Red
		B	B-	Black
		C	K-	White
		D	K+	Yellow

View of contact side of female connector

		Pin	Signal	Color/labeling
Size 1.5 $I_0 \leq 36 \text{ A}$		U V W ⏏ + - 1 2	Phase U Phase V Phase W PE B+ B- K- K+	U V V W W W green / yellow Red Black White Yellow

View of contact side of female connector

Cable cross-section ²⁾	Rated current [A] ^{1) 2)}	Connector 540 V Size ²⁾	Cable diameter ²⁾ [mm]
4x1.5 mm ² + 4x0.75 mm ²	15	1	11.7-12.3
4x2.5 mm ² + 4x0.75 mm ²	20	1	12.7-14.6
4x4 mm ² + 4x0.75 mm ²	28	1.5	14.2-15.4
4x6 mm ² + 4x0.75 mm ²	36	1.5	16.6-17.9
4x10 mm ² + 4x0.75 mm ²	50	1.5	20.5-21.5
4x16 mm ² + 4x0.75 mm ²	66	-	23.0-25.8
4x25 mm ² + 2x(2x1.5 mm ²)	84	-	26.3-29.7
4x35 mm ² + 2x(2x1.5 mm ²)	104	-	30.8-32.5

¹⁾ Current carrying capacity acc. to Table 5, laying type C or E VDE 0113/EN 60204 Part 1 issue 1997)
Ambient temperature 40 °C

²⁾ Deviating regulations apply for  approved motors.

3.6.3. Application notes

Operating temperature

The cables can be operated within a temperature range of between -20 °C and +80 °C.

Cable laying at the motor

The cables must not touch the motor surface.

Smallest permissible bending radii

12x outer cable diameter.

3.6.4. Ordering data for main connection cables

Rated current: 15 ACable 4 x 1.5 mm² + 4 x 0.75 mm²

With connector size 1

Length in m	Item Number	Item Number Speedtec
5	324781	445872
7	324782	445887
10	324783	445889
15	324784	447675
20	324785	447676
25	324786	447677
30	324787	447678
35	324788	447679
40	324789	447680
50	324790	447681
75	324791	447682
100	324792	447683

Rated current: 28 ACable 4 x 4 mm² + 4x 0.75 mm²

With connector size 1.5

Length in m	Item Number	Item Number Speedtec
5	326589	448063
7	326591	448064
10	326592	448065
15	326593	448066
20	326594	448067
25	326596	448069
30	326597	448070
35	326598	448071
40	326599	448072

Rated current: 20 ACable 4 x 2.5 mm² + 4x 0.75 mm²

With connector size 1

Length in m	Item Number	Item Number Speedtec
5	414840	447684
7	380967	447687
10	413410	447688
15	414841	447692
20	414842	447698
25	414843	447852
30	414846	447853
35	414848	447854
40	414849	447855
50	414850	447856
75	414851	447857
100	414852	447858

Rated current: 36 ACable 4 x 6 mm² + 4x 0.75 mm²

With connector size 1.5

Length in m	Item Number	Item Number Speedtec
5	326600	448080
7	326601	448118
10	326602	448119
15	326603	448120
20	326604	448121
25	326605	448122
30	326606	448123
35	326607	448124
40	326608	448125

Rated current: 21 ACable 4 x 2.5 mm² + 4x 0.75 mm²

With connector size 1.5

Length in m	Article number	Item Number Speedtec
5	326577	447686
7	326578	447689
10	326579	447690
15	326580	447691
20	326581	447693
25	326582	447694
30	326583	447695
35	326584	447696
40	326585	447697
50	326586	447699
75	326587	448060
100	326588	448061

Rated current: 50 ACable 4 x 10 mm² + 4x 0.75 mm²

With connector size 1.55

Length in m	Article number	Item Number Speedtec
5	326609	448129
7	326610	448131
10	326611	448132
15	326612	448133
20	326613	448134
25	326614	448135
30	326615	448136
35	326616	448137
40	326617	448138

Mating Plug

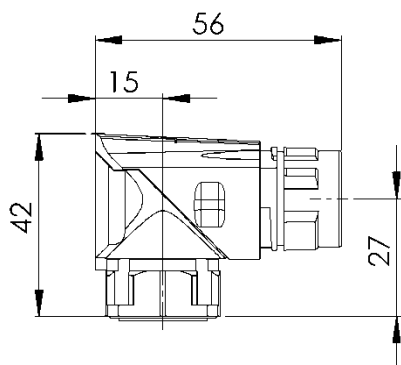
Type		Art. Nr.	Art. Nr. Speedtec
Gr. 1	4x1,5mm ² o. x2,5mm ²	261740	445486
Gr. 1	4x2,5mm ² bis 4x4mm ²		500064
Gr. 1,5	4x2,5mm ² bis 4x6mm ²	326574	445487
Gr. 1,5	4x10mm ²	326569	445488

Deviating regulations apply for  approved motors.

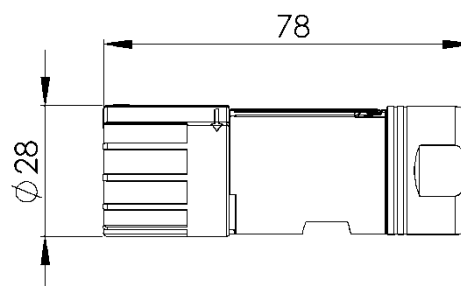
3.7. Dimensional drawings for equipment socket and plug

3.7.1. Main connection:

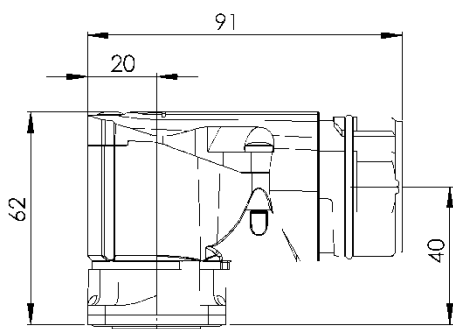
Speedtec - rotary angle socket
(Size 1 for Current I_0 up to 20 A)
or hybrid socket DSL Hiperface



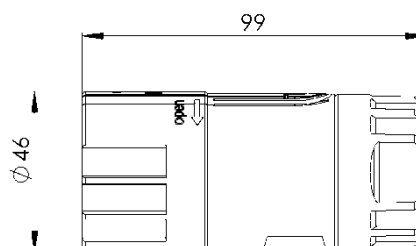
Speedtec - mating connector
(Size 1 for Current I_0 up to 20 A)



Speedtec - rotary angle socket
(Size 1.5 for Current I_0 up to 36 A)
or hybrid socket DSL Hiperface

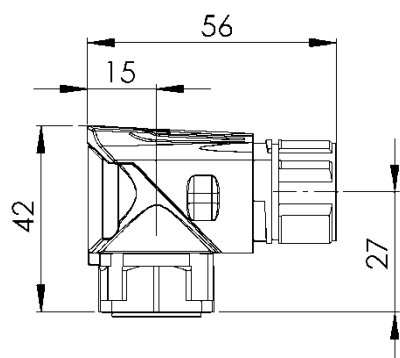


Speedtec - mating connector
(Size 1.5 for Current I_0 up to 36 A)

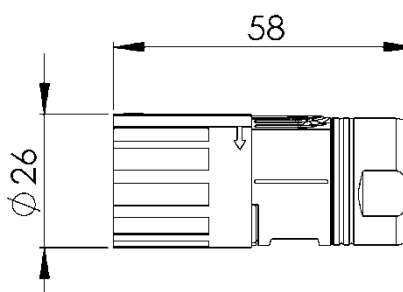


3.7.2. Encoder connection

Speedtec - rotary angle socket

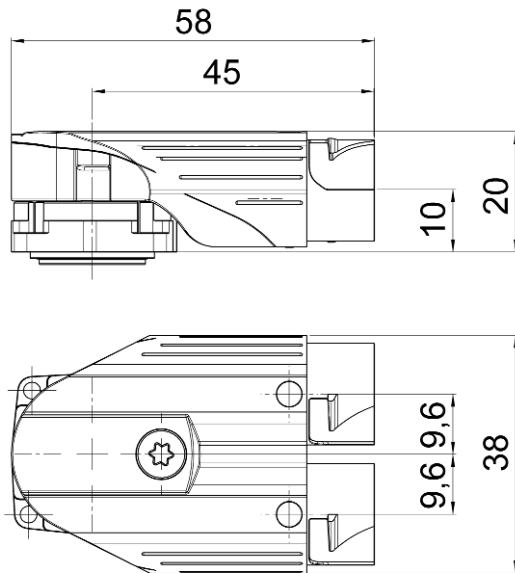


Speedtec - mating connector

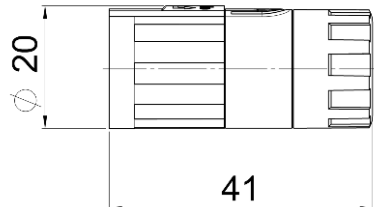


3.7.3. Combination angle socket

rotating combination angle socket
(current I_0 up to 14 A)



Opposing plug socket combination angle socket
(current I_0 up to 14 A)



NOTE:

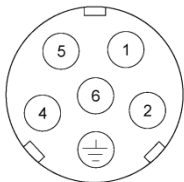
The connection allocations of the combination angle socket or opposing plug socket are not identical with the sensor and main connection allocations described in this documentation. The connection allocation of the combination angle socket is available on request.

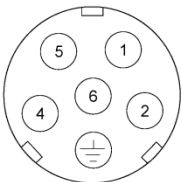
3.8. Fan

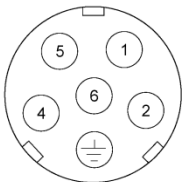
3.8.1. DSD2-056-100

	DSD2-056..100			
Rated voltage [V]	24 V DC	115 V AC	230 V AC	
Rated frequency [Hz]	-	60	50	60
Rated current [A]	0.52	0.47	0.22	0.20
Rated speed [rpm]	2758	2394	2385	2099
Power rating [W]	12.4	35	32	30
Connection	6-pole plug			
Protection type	IP65			

Fan connection 24 V DC / 115 V AC / 230 V AC

Kontaktseite der Einbaudose Contact side mounting box 				
Pin	1	2	4	5
Signal	/	/	L+ (+24V)	L- (0V)
Pin	6			
Signal	/			
Anschlußschema / Connection diagram: Steckeranschluß DC-Lüfter 24V DC-Fan-Motor 24V				KA 2038
				19.07.12
Baumüller Nürnberg GmbH				

Kontaktseite der Einbaudose Contact side mounting box 				
Pin	1	2	4	5
Signal	/	N	/	/
Pin	6			
Signal	U			
Anschlußschema / Connection diagram: Steckeranschluß Lüfter 115V Fan-Motor 115V				KA 2039
				19.07.12
Baumüller Nürnberg GmbH				

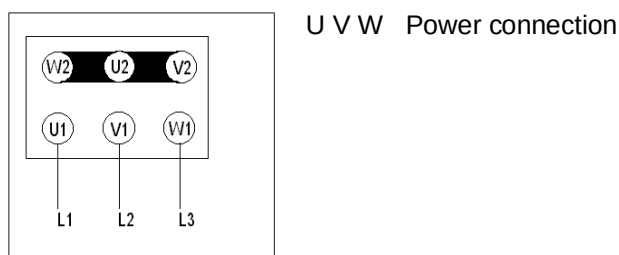
Kontaktseite der Einbaudose Contact side mounting box 				
Pin	1	2	4	5
Signal	U	N	/	/
Pin	6			
Signal	/			
Anschlußschema / Connection diagram: Steckeranschluß Lüfter 230V Fan-Motor 230V				KA 2028_a AM 30764 19.07.12
Baumüller Nürnberg GmbH				

View on the contact side of the socket

3.8.2. DSD2-132

Fan connection at a standardized blower motor via the terminal box

Connection diagram



Standardized blower motor for radial ventilation

200/400V // 230/460V 50 // 60Hz IE3

Size	Power [kW]	Rated current [A]	Fan	Rated input power [kW]	Volume flow rate [m³/min]	Stat. pressure [Pa]	Speed [1/min]	Spez. ratio
132	0.25 // 0.25	1.3 / 1.65 // 1.1 / 0.55	FCA 63B-2	0.36	11.5	599	2825	1

The rated currents are maximum values.

Standardized blower motor for axial ventilation

Δ/Y 200-265V / 345-460V 50 // 60Hz

Size	Power [kW]	Rated current [A]	Fan	Rated input power [kW]	Volume flow rate [m³/min]	Stat. pressure [Pa]	Speed [1/min]	Spez. ratio
132	0.08 // 0.12	0.4 / 0.23 // 0.61 / 0.35	FCA 56A-2	0.16	5,15	514	2800	1

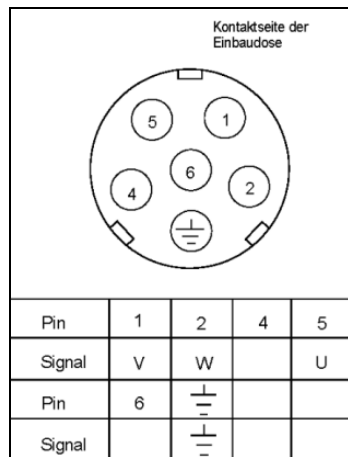
Δ/Y 265-345V / 460-600V 50 // 60Hz

Size	Power [kW]	Rated current [A]	Fan	Rated input power [kW]	Volume flow rate [m³/min]	Stat. pressure [Pa]	Speed [1/min]	Spez. ratio
132	0.08 // 0.12	0.3/ 0.18 // 0.46 / 0.26	FCA 56A-2	0.16	5,15	514	2800	1

The rated currents are maximum values.

Fan connector at axial integrated blower

Connection diagram



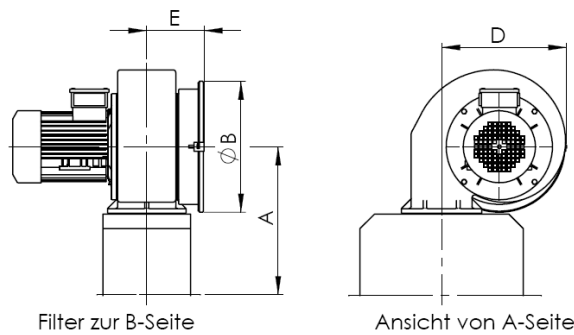
Y 400V // 460V 50 // 60Hz

Size	Power [kW]	Rated current [A]
132	0.47 // 0.54	0.7 // 0.75

The rated currents are maximum values.

3.9. Filter

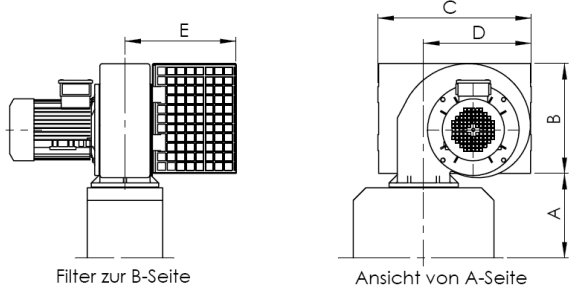
Flat filter for DSD2-132..R



Dimensions at the fan on the top or at the side

Motor size	Fan type	A	Ø B	D	E
132	BFB 519	264	210	214	105

Rectangle filter for DSD2-132..R



Dimensions at the fan on the top or at the side

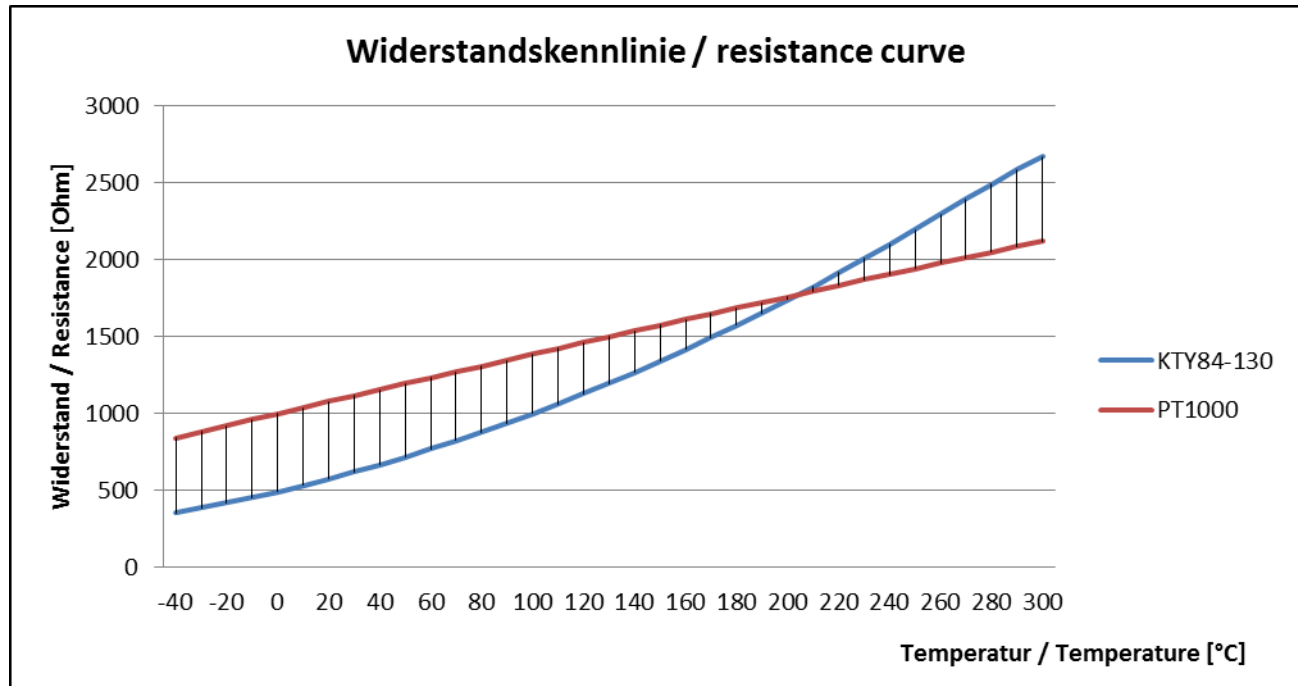
Motor size	Fan type	A	B	C	D	E
132	BFB 519	175	207	306	213	163

Note:

At IP54 motors the filter mat rough is preferred.

3.10. Temperature sensor

The connection of the temperature sensor of the DSD2 series is implemented via the main connection. An option is the connection at the DSD2-028-100 via the encoder socket. The particular version must be marked in the ordering code.



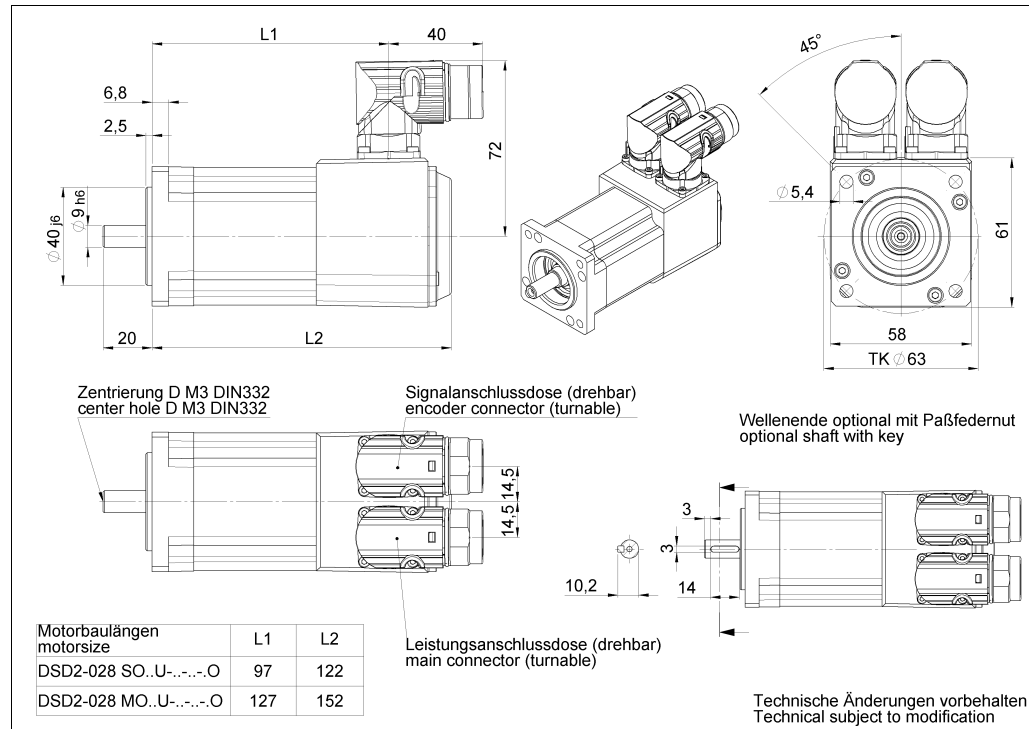
The temperature sensor PT1000 continuously monitors the motor temperature. If the sensor is supplied with a measured current of 2 mA the above shown resistance curve results.

4. Dimension drawings

4.1. Dimension drawings DSD2-028

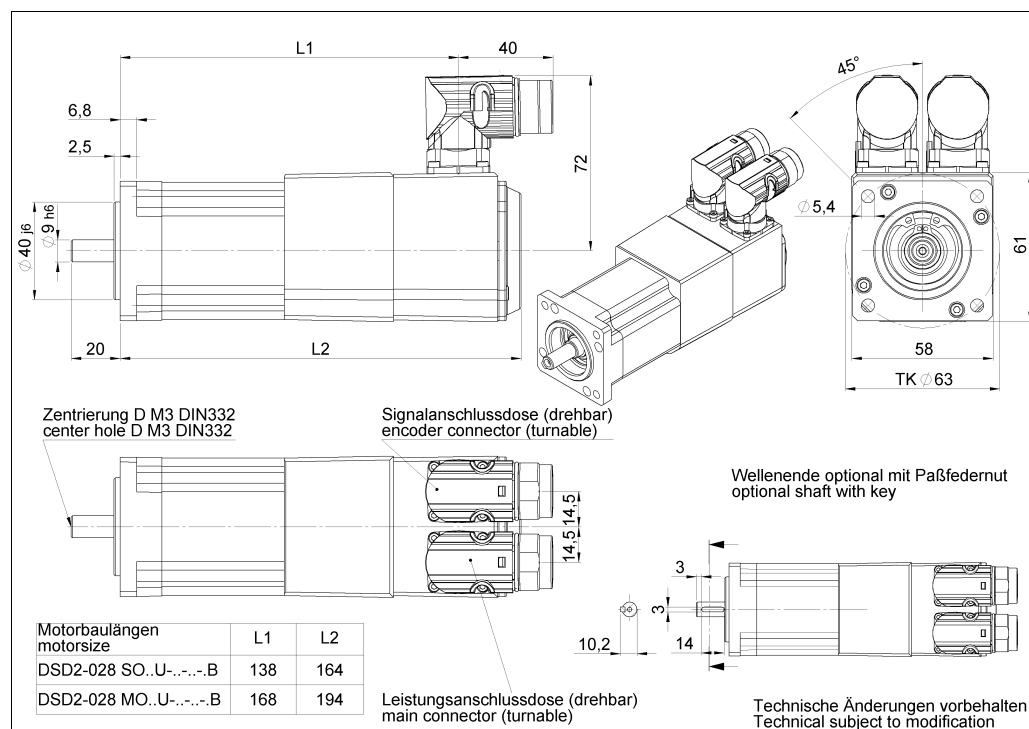
Dimension drawing DSD2-028....U-...-O.-SPP-...-O-000

Version IM B5



Dimension drawing DSD2-028....U-...-B.-SPP-...-O-000

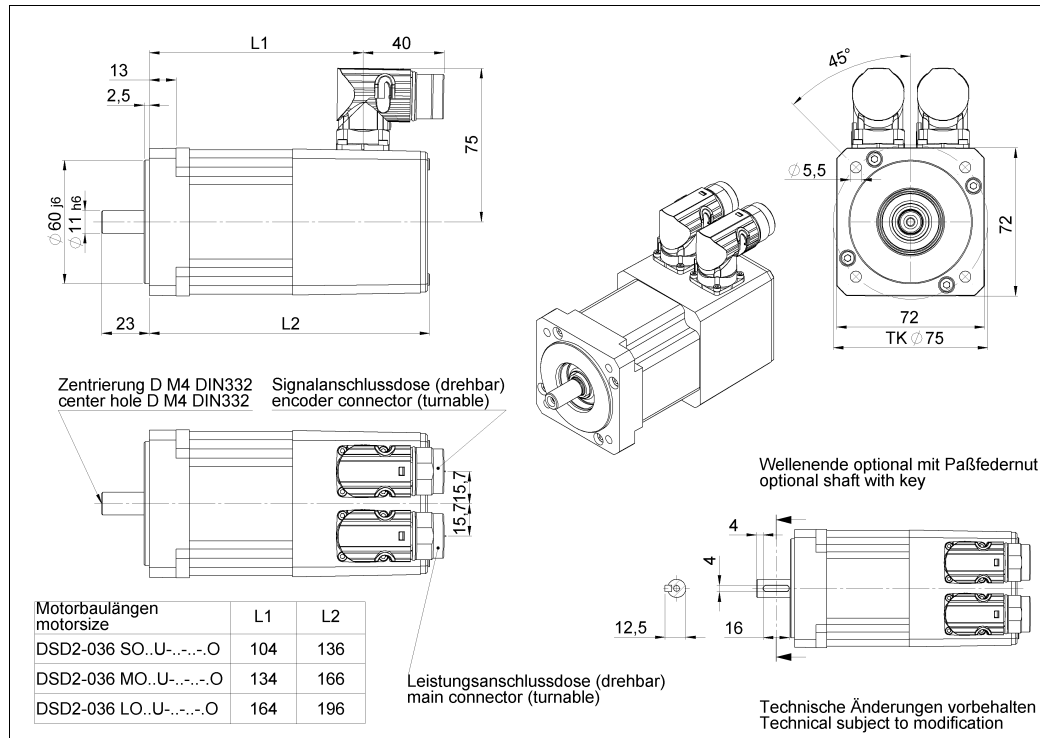
Version IM B5



4.2. Dimension drawings DSD2-036

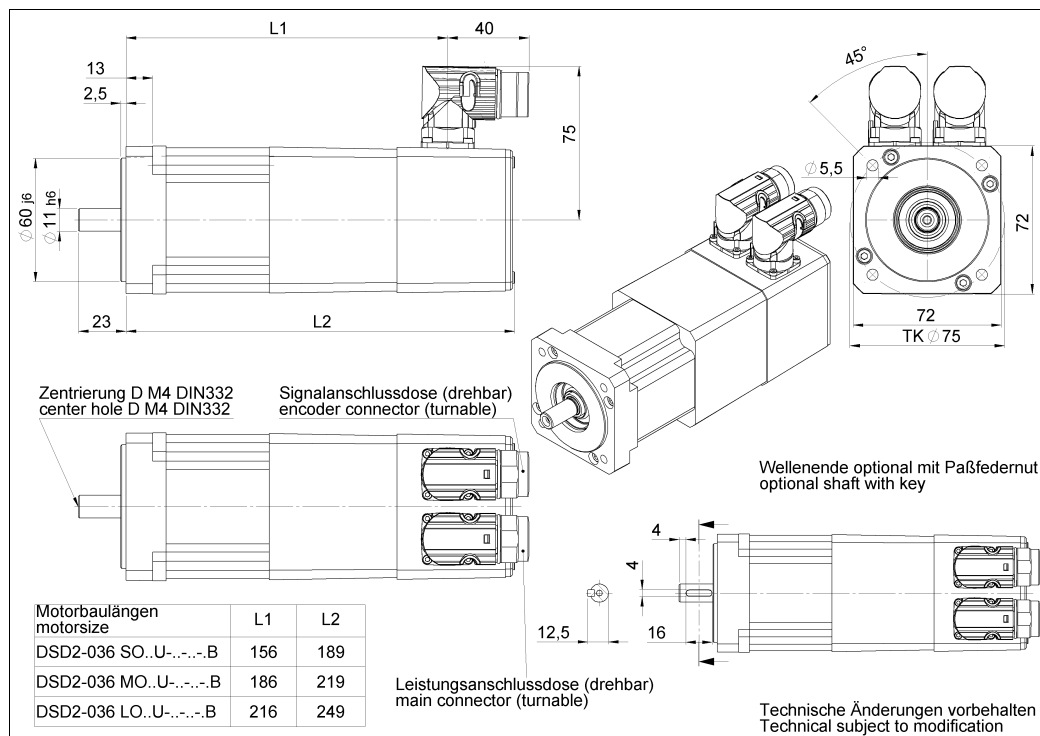
Dimension drawing DSD2-036...U-...-O.-SPP-...-O-000

Version IM B5



Dimension drawing DSD2-036...U-...-B.-SPP-...-O-000

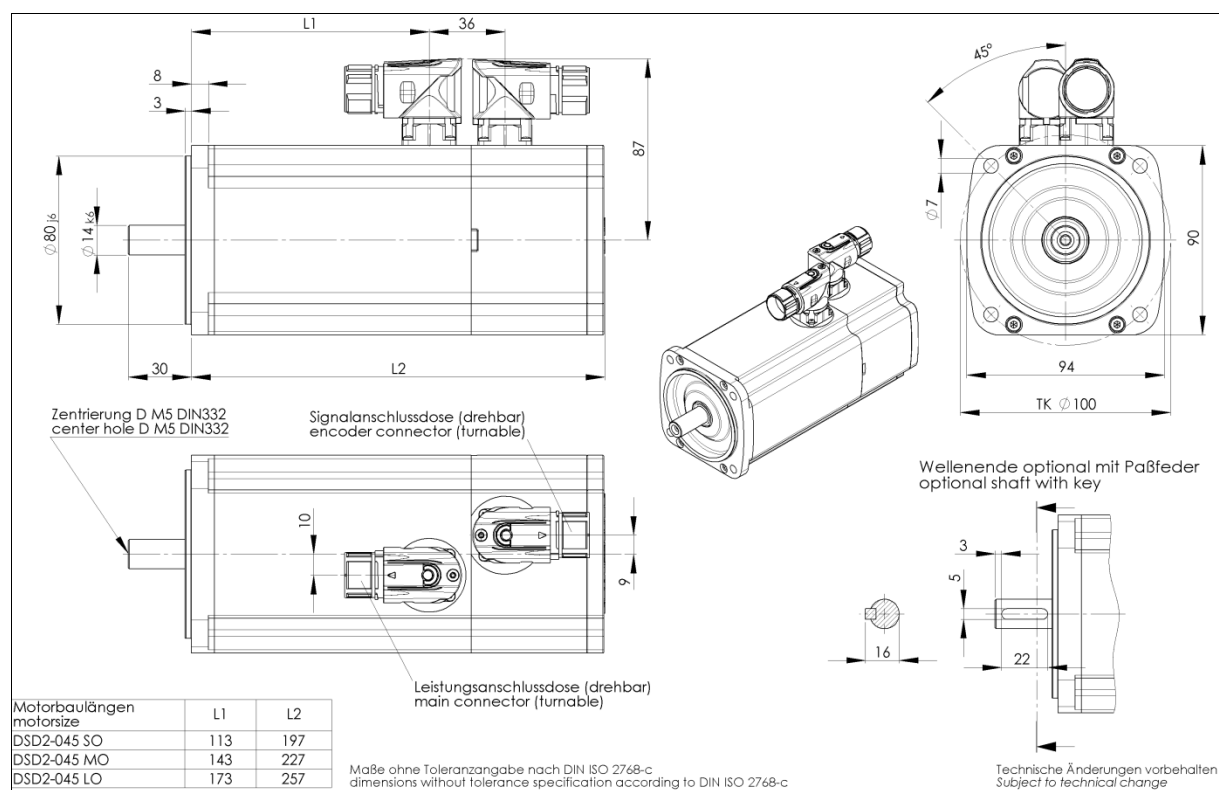
Version IM B5



4.3. Dimension drawings DSD2-045

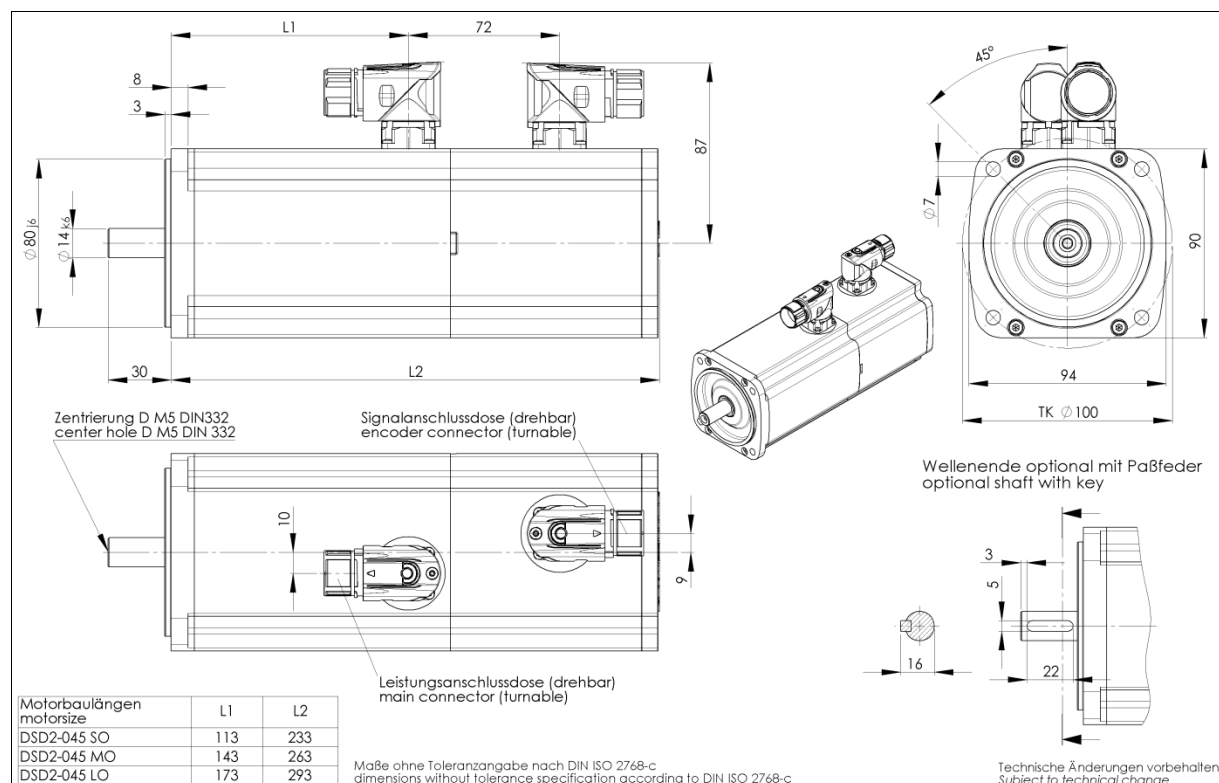
Dimension drawing DSD2-045....U-...-O.-SPP-...-O-000

Version IM B5



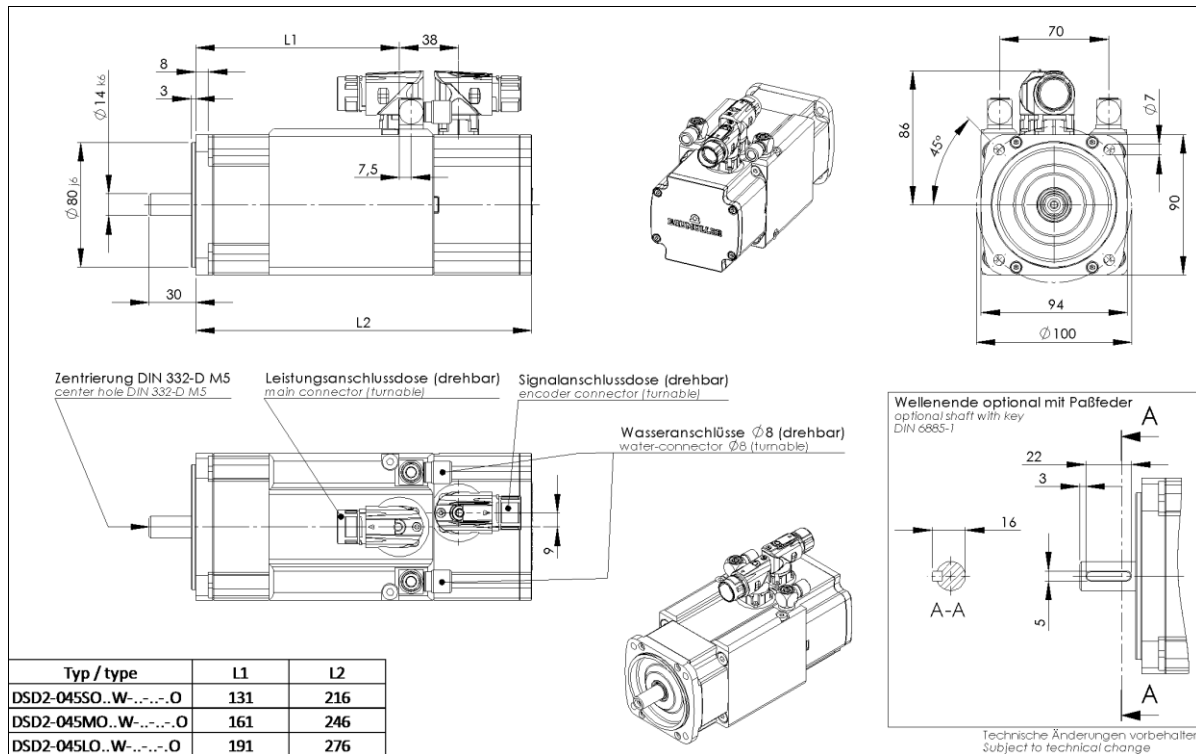
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Version IM B5



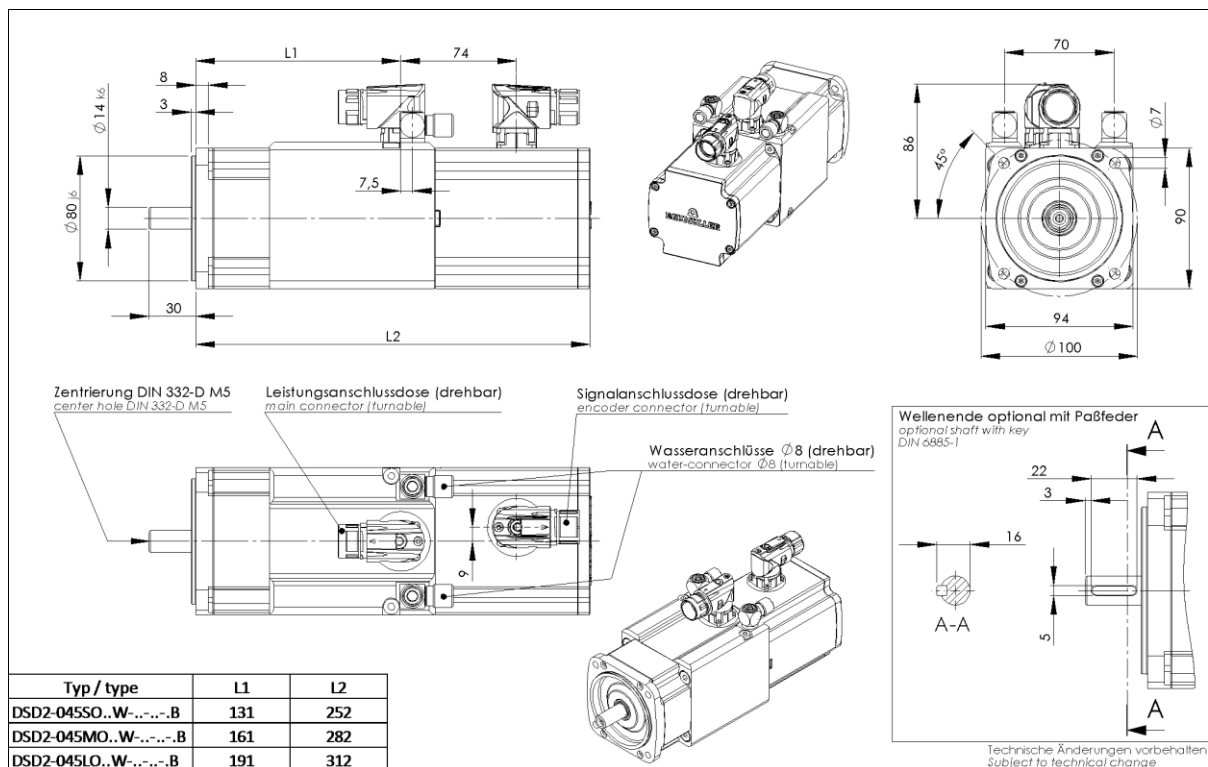
Dimension drawing DSD2-045...W-...-O-PP-...-O-000

Version IM B5



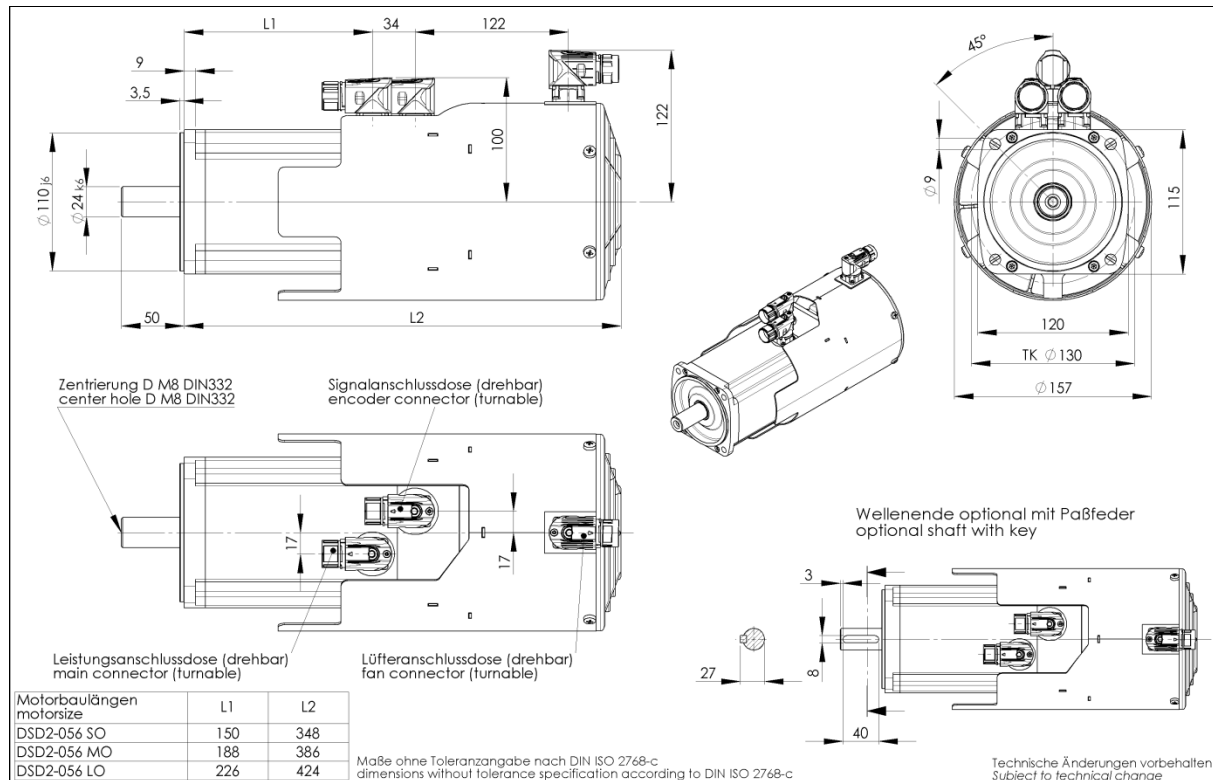
Dimension drawing DSD2-045...W-...-B-PP-...-O-000

Version IM B5



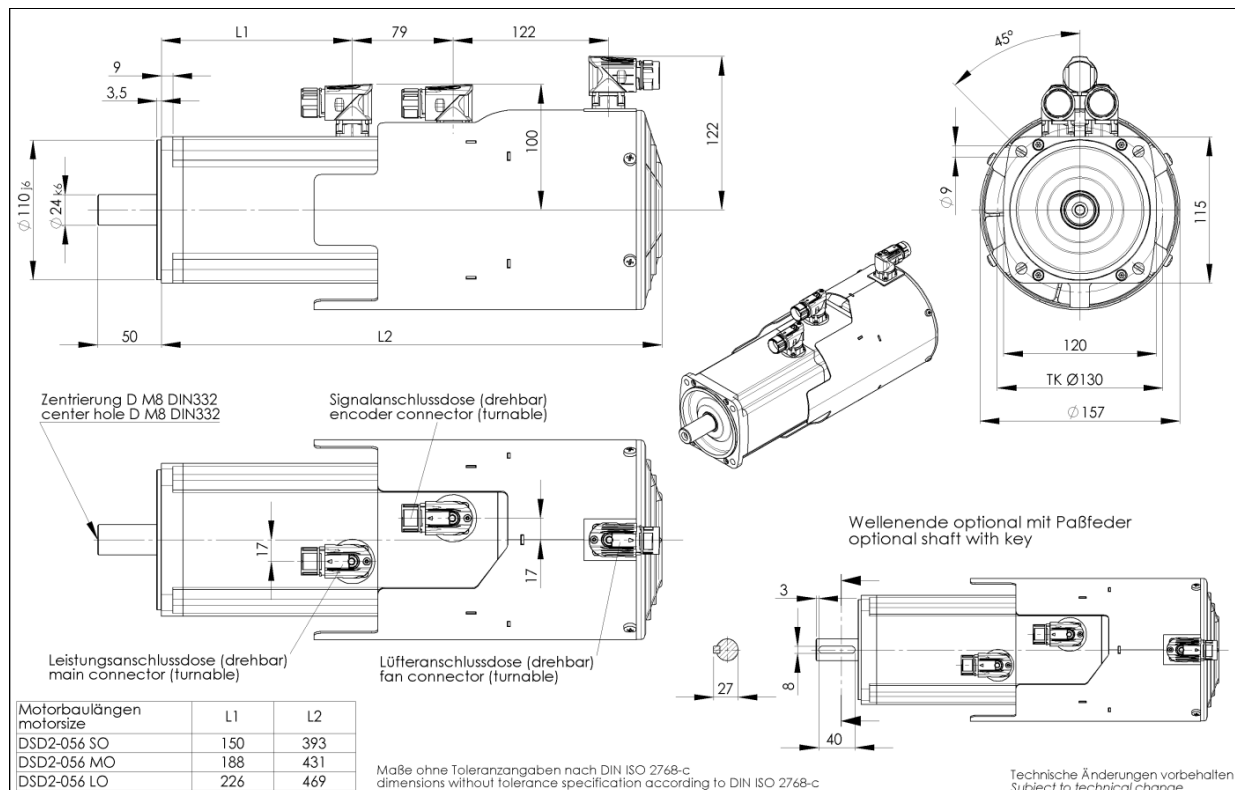
Dimension drawing DSD2-056....O-...-O.-SPP-...-O-000

Version IM B5

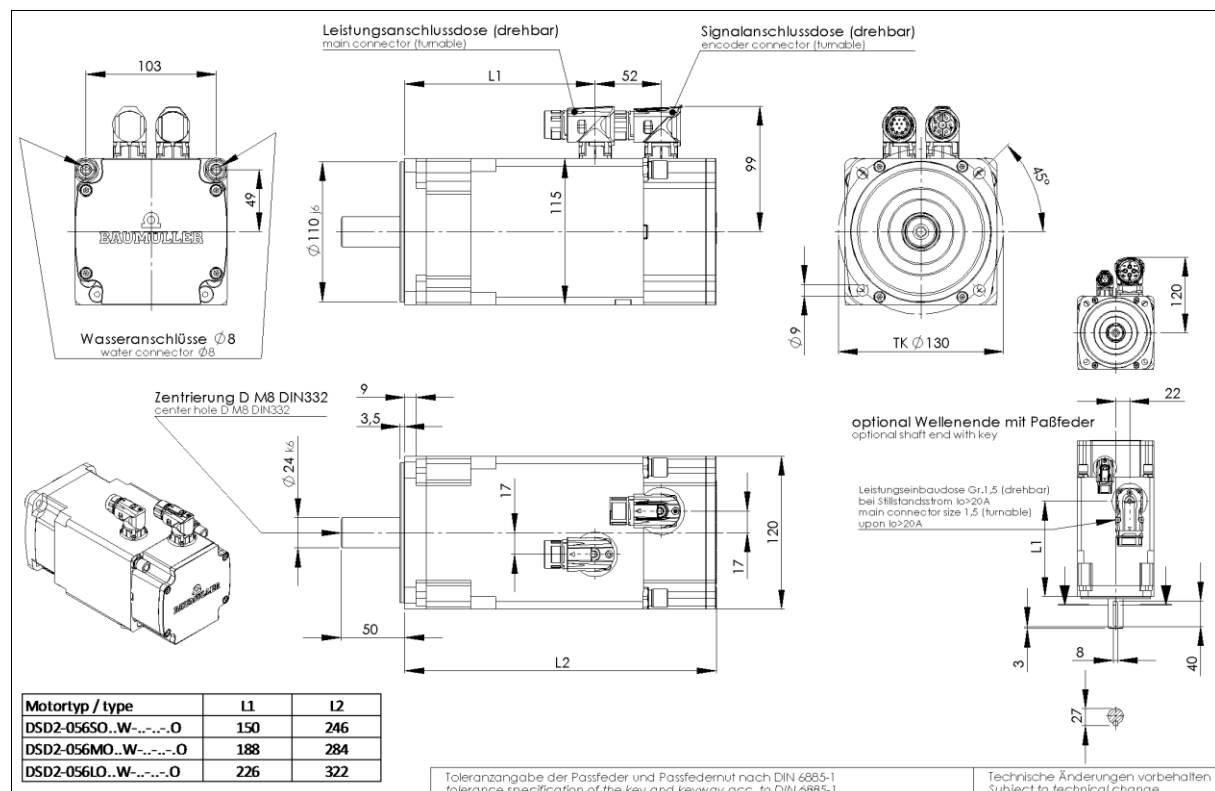


Dimension drawing DSD2-056....O-...-B.-SPP-...-O-000

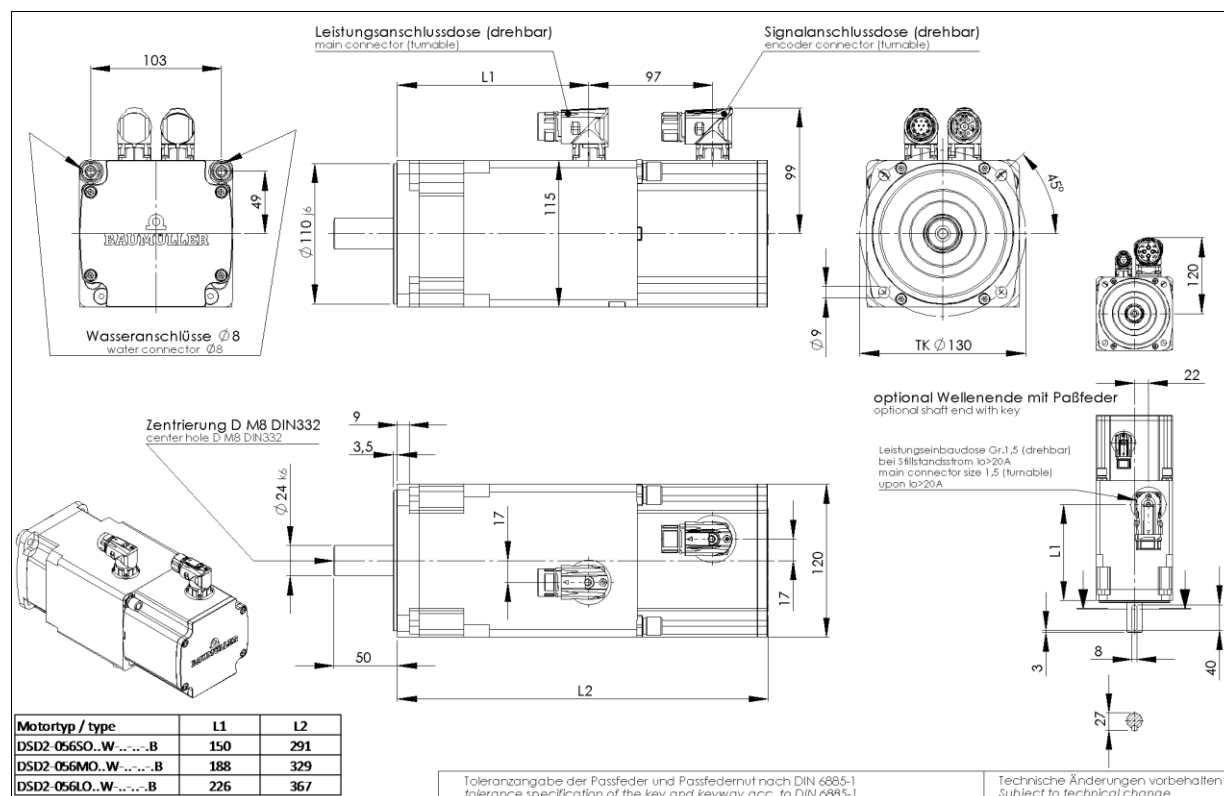
Version IM B5



Dimension drawing DSD2-056...W-...-O-PP-...-O-000 Version IM B5

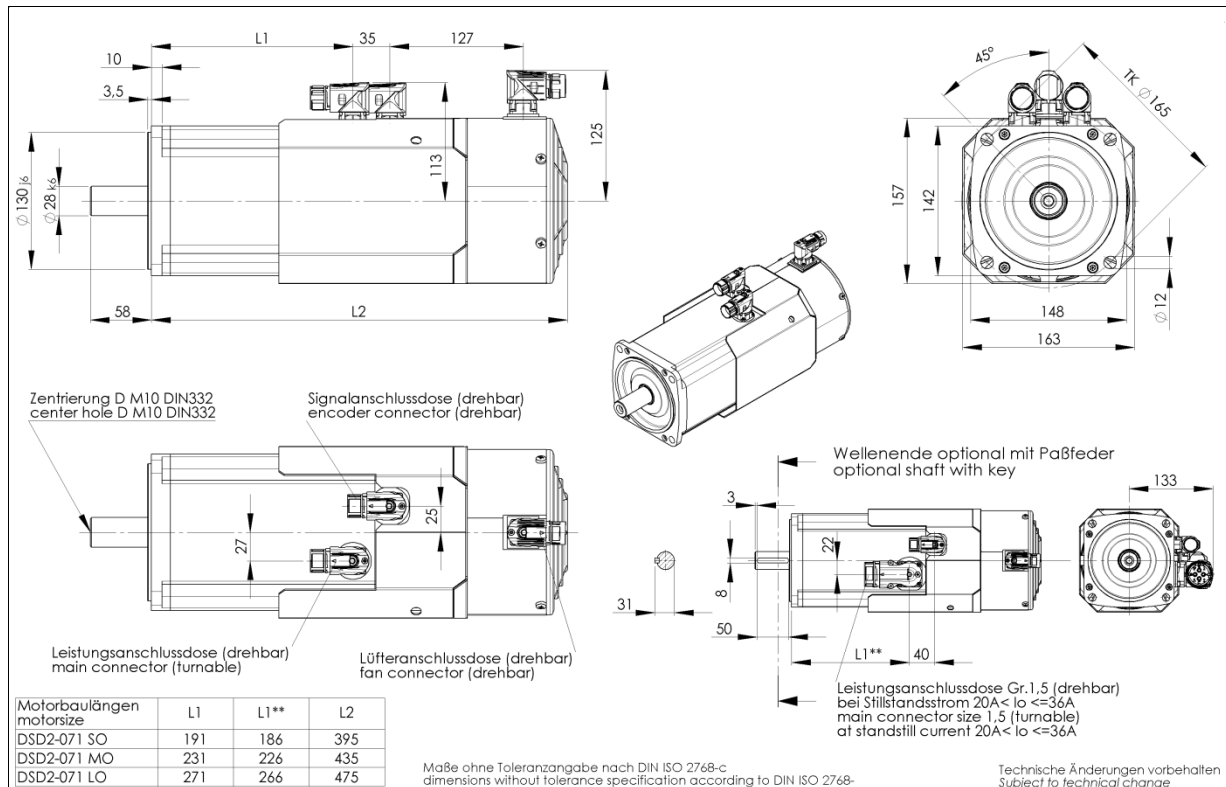


Dimension drawing DSD2-056...W-...-B-PP-...-O-000 Version IM B5



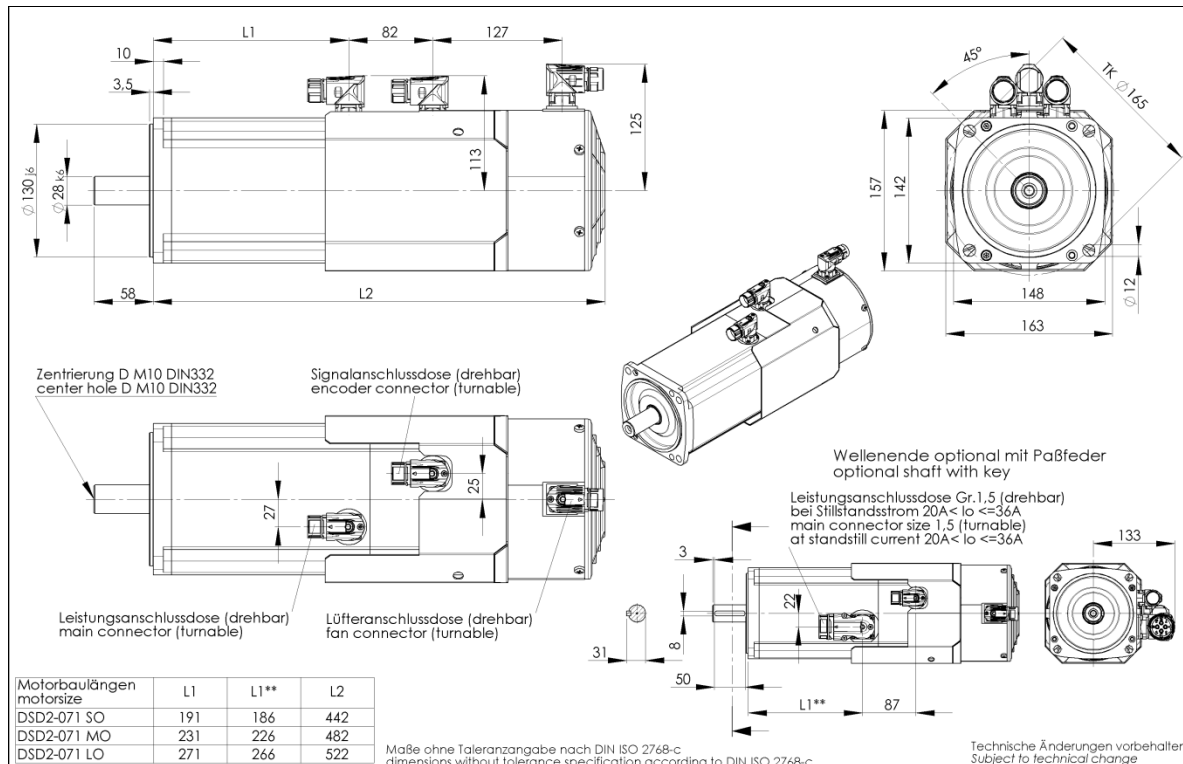
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Version IM B5



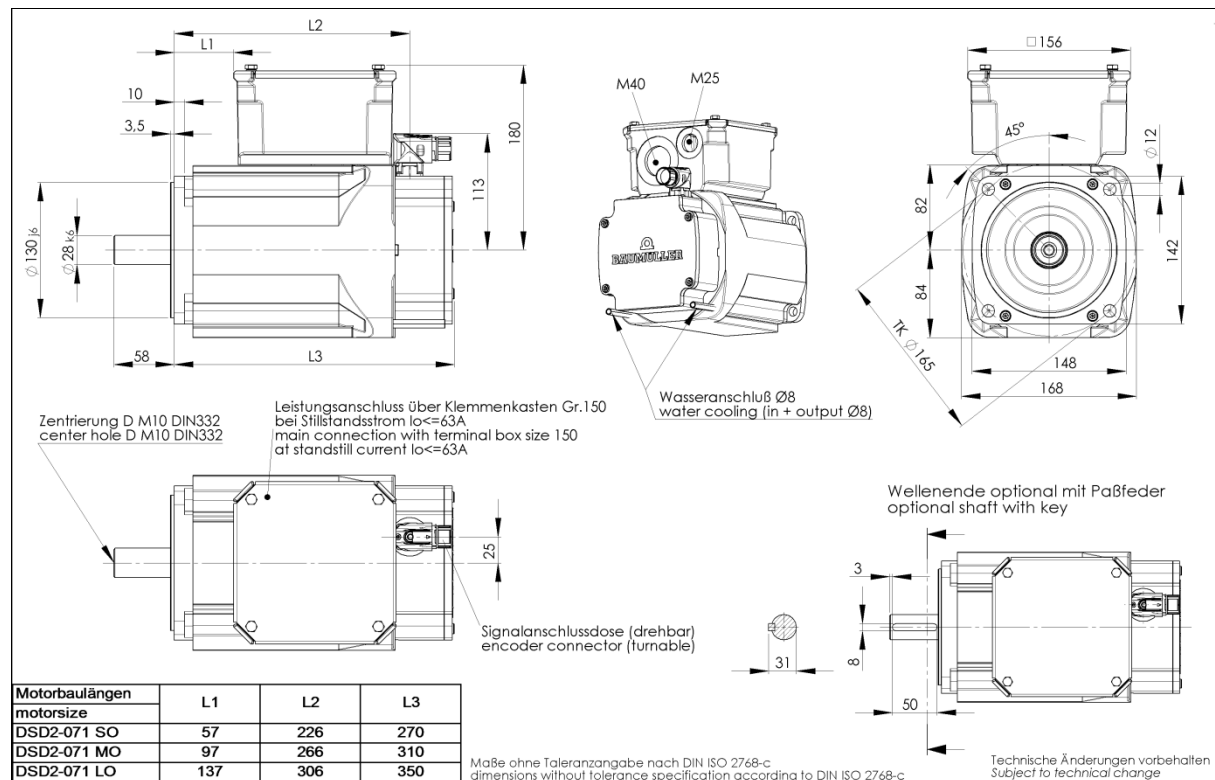
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Version IM B5



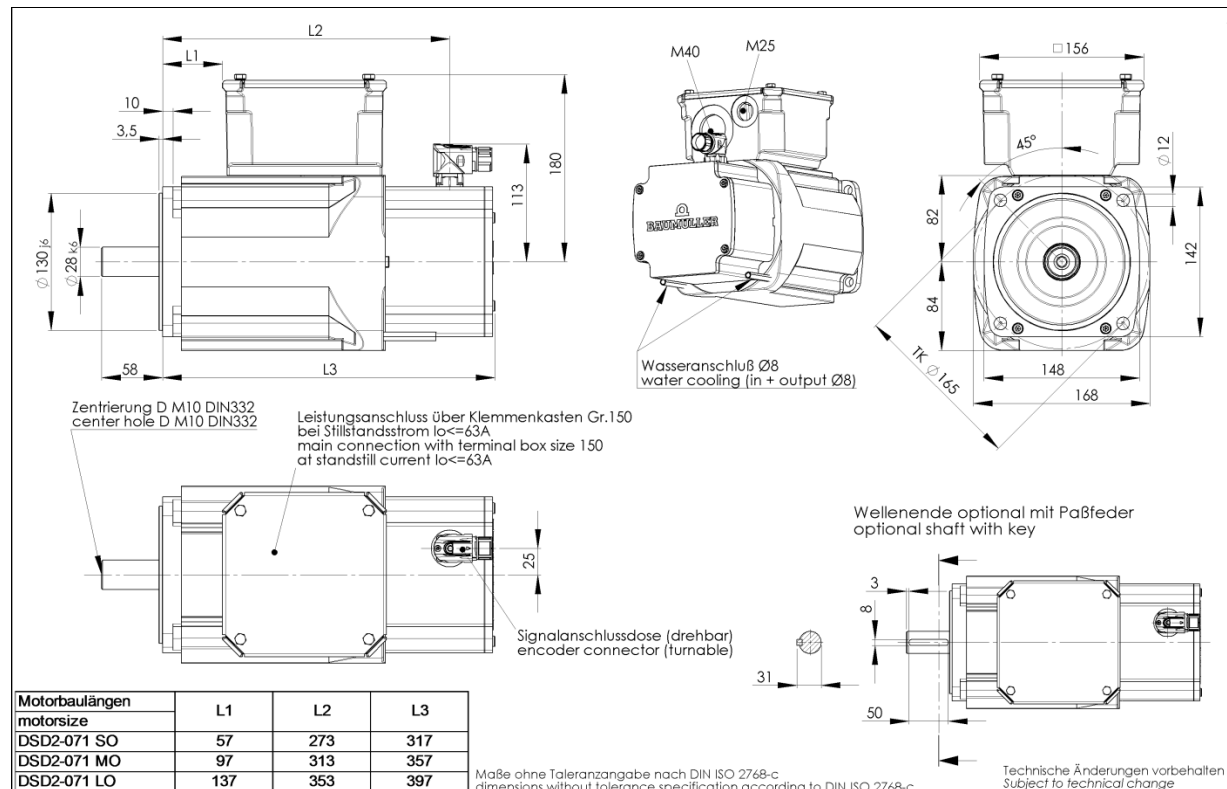
Dimension drawing DSD2-071....W-....O.-KNP-....O-000

Version IM B5



Dimension drawing DSD2-071....W-....B.-KNP-....O-000

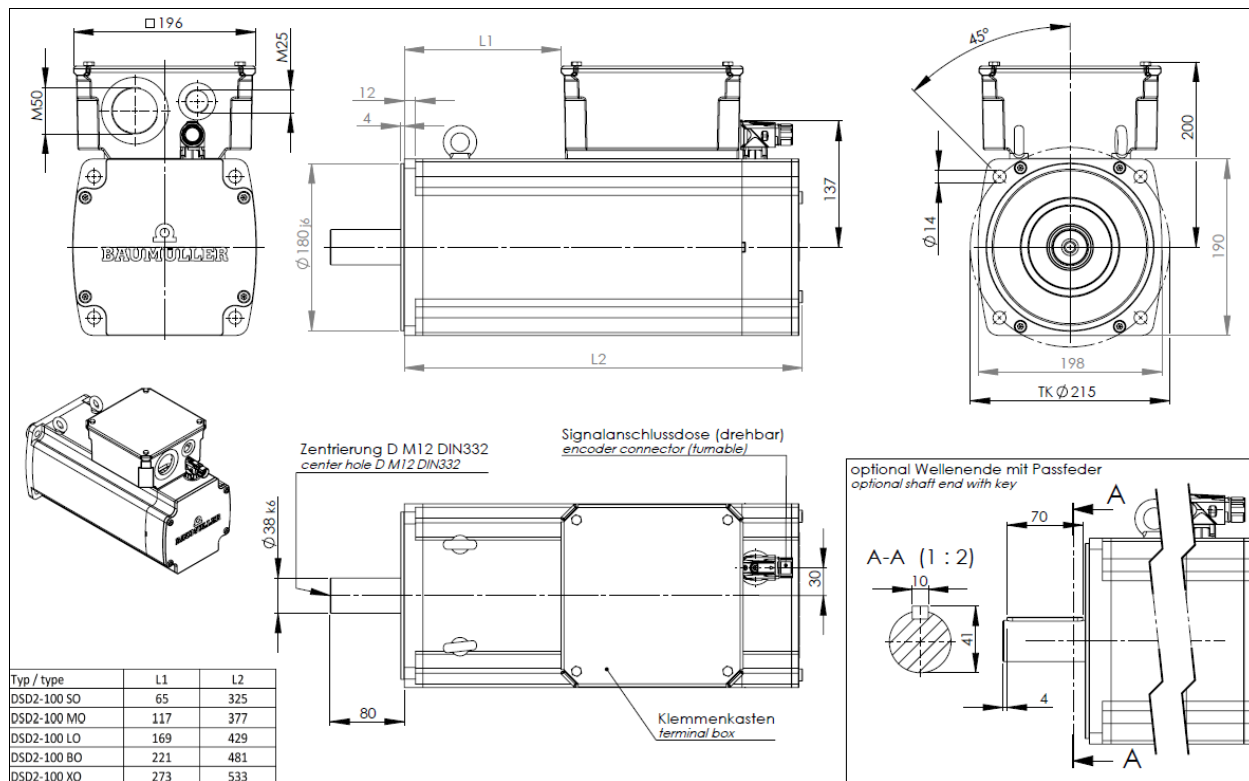
Version IM B5



4.6. Dimension drawings DSD2-100

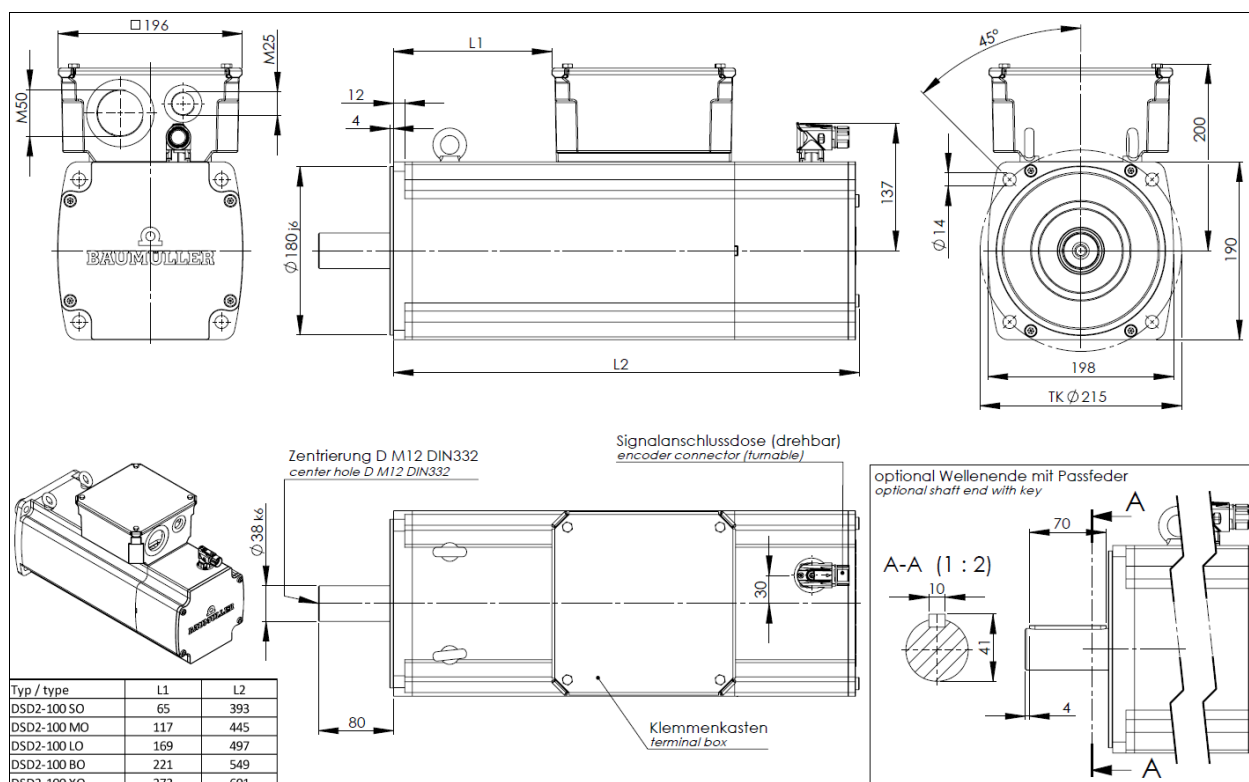
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Version IM B5

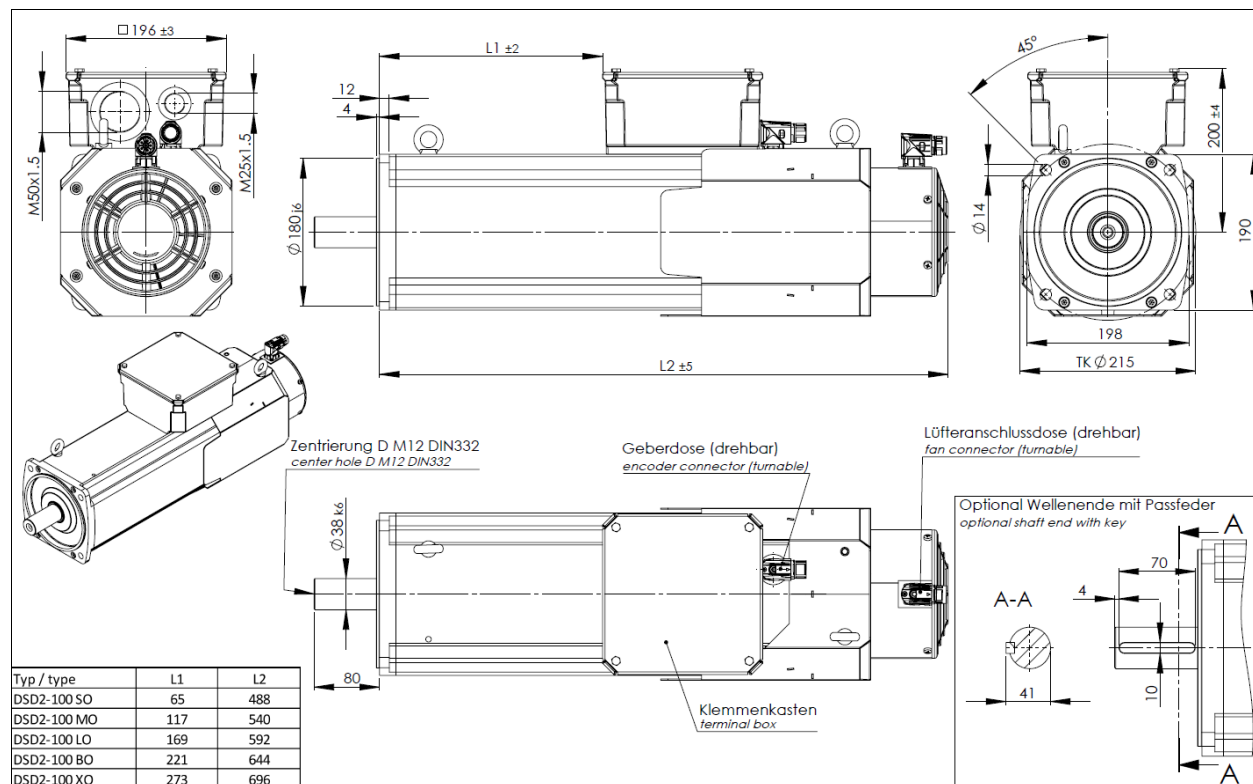


Dimension drawing DSD2-100....U-...-B-PP-...-O-000

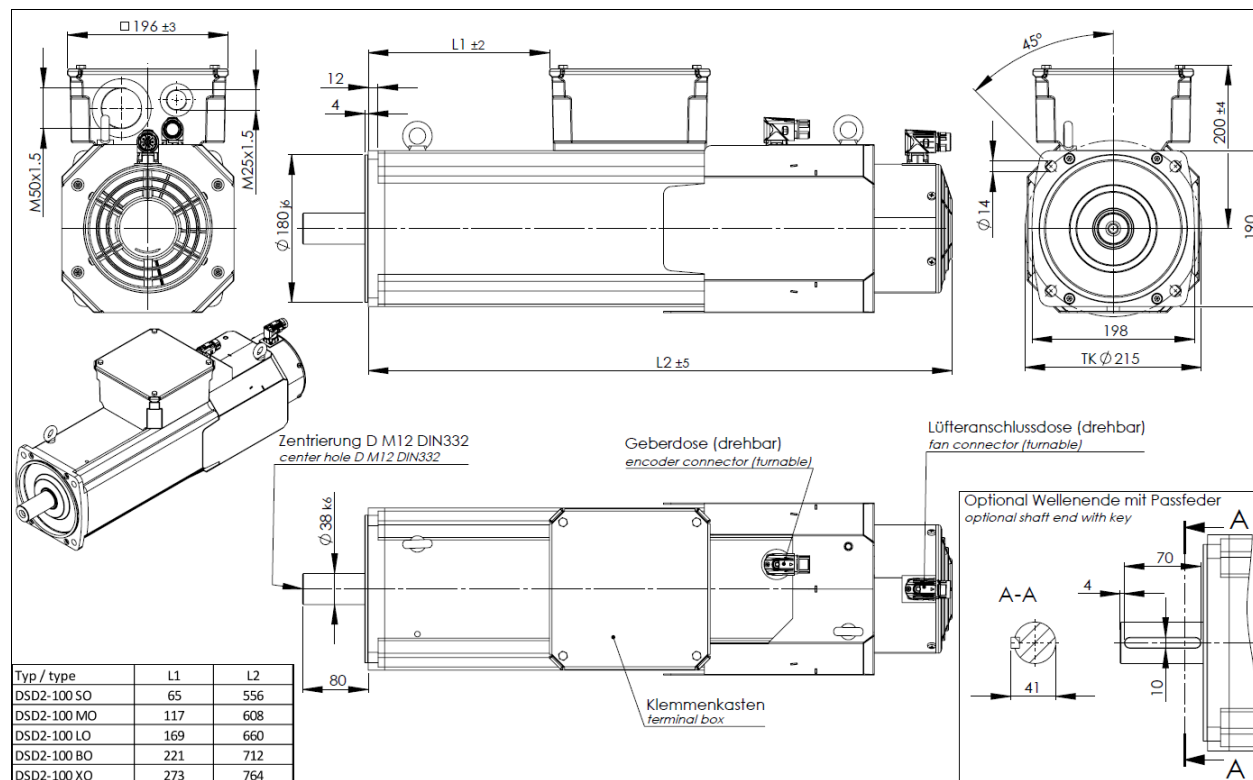
Version IM B5



Dimension drawing DSD2-100....O-...-O-PP-...-O-000
Version IM B5

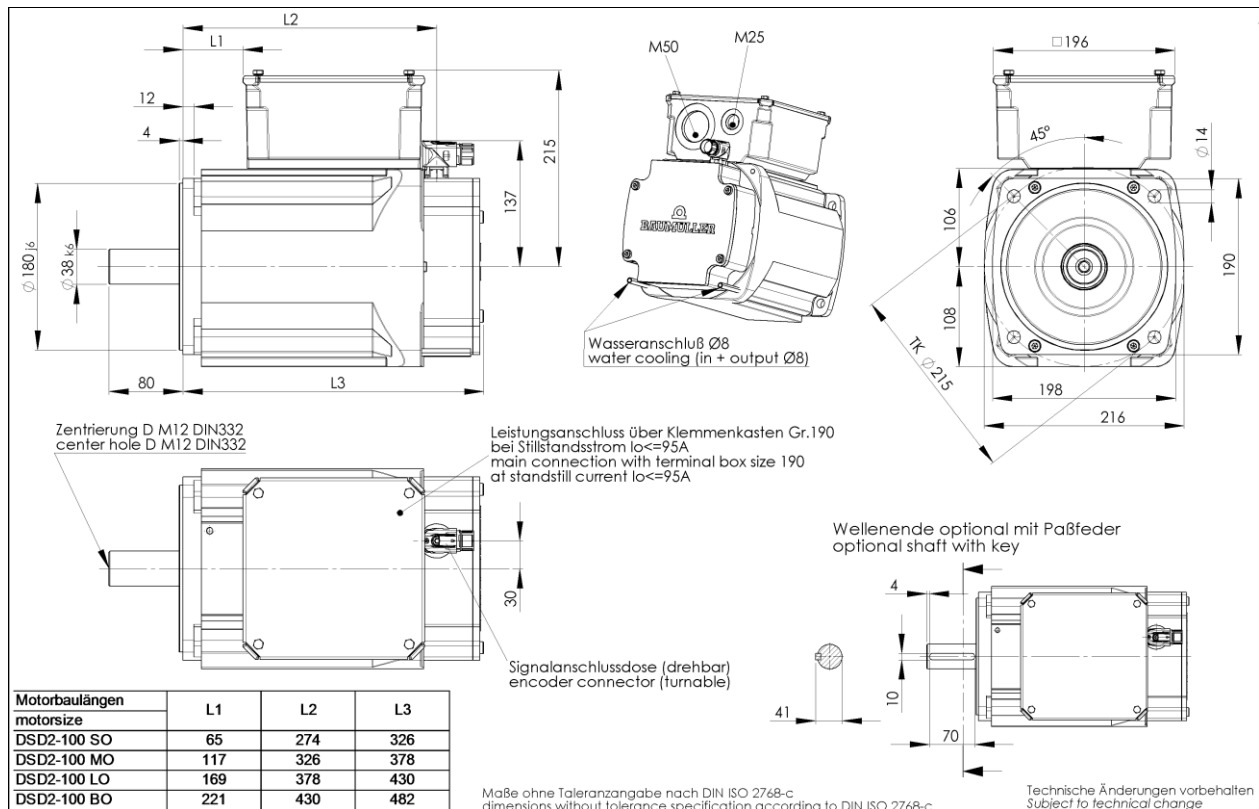


Dimension drawing DSD2-100....O-...-B-PP-...-O-000
Version IM B5



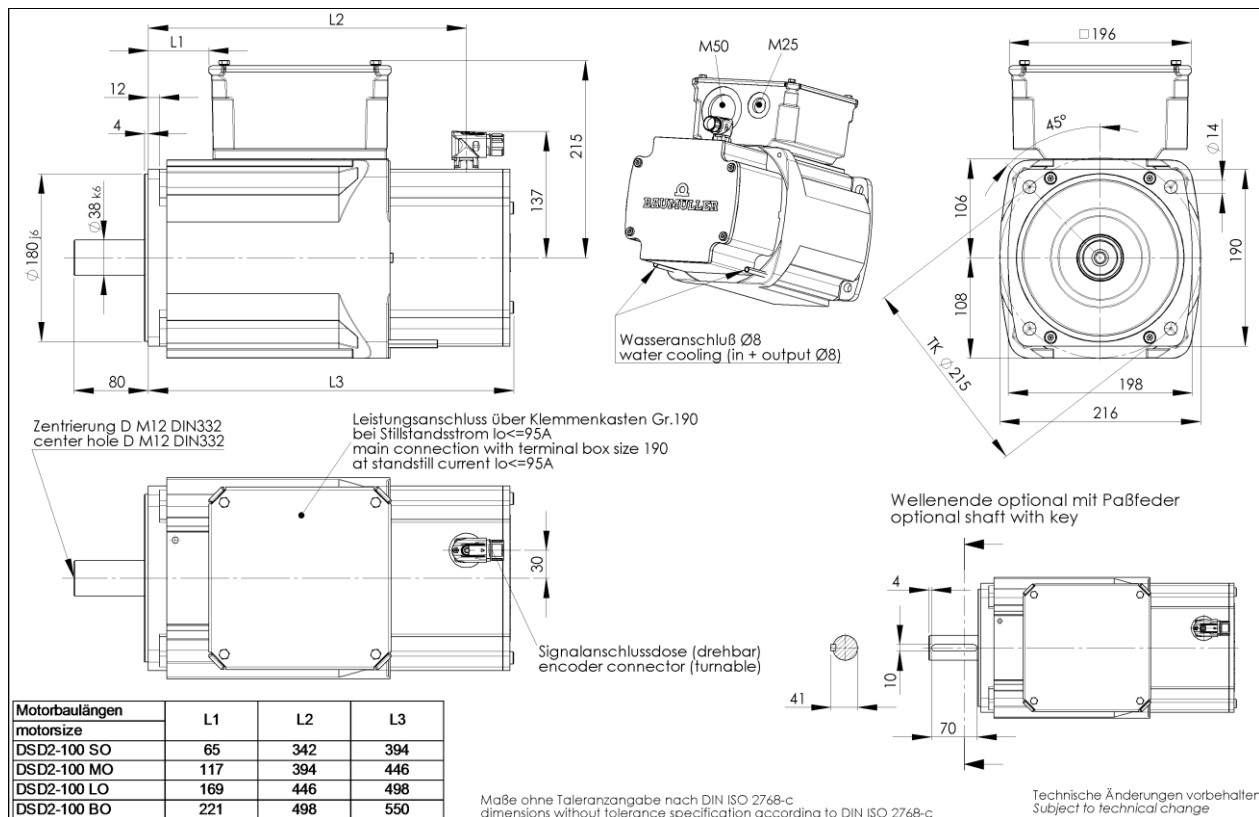
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Version IM B5

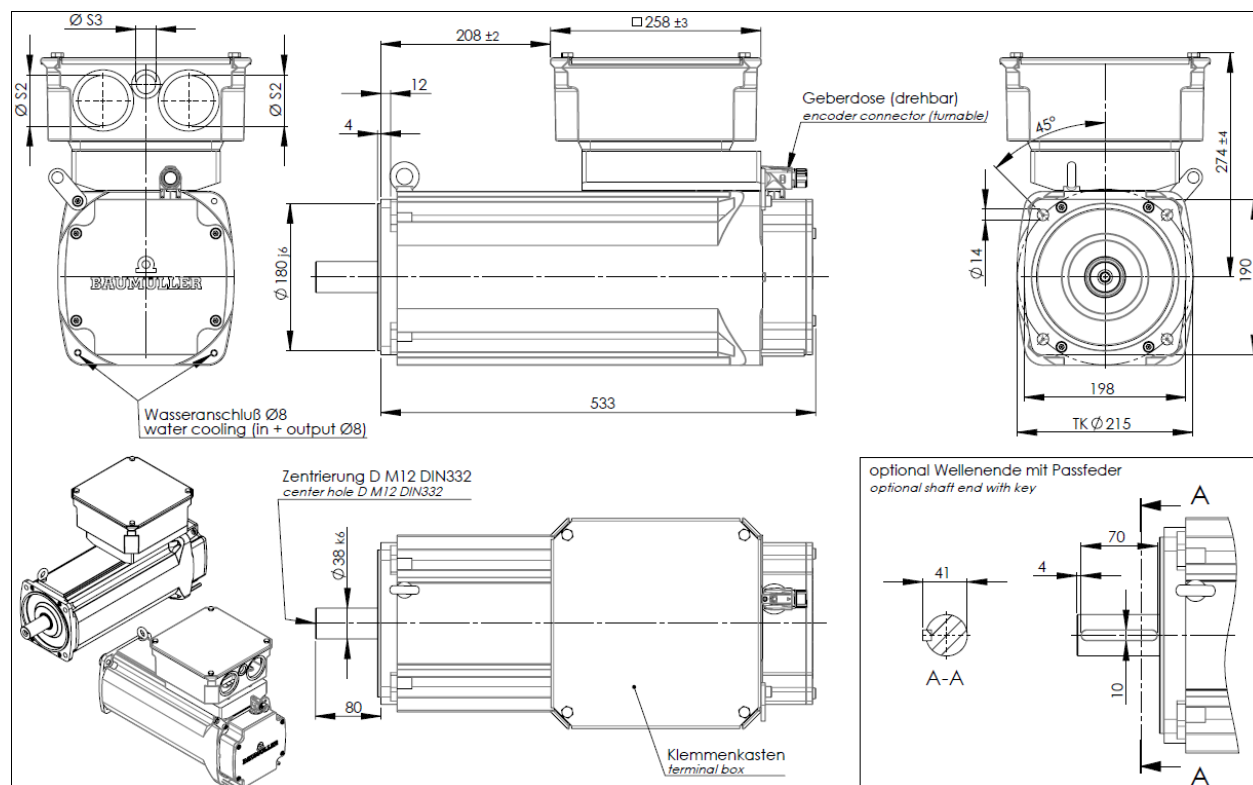


Dimension drawing DSD2-100...W-...-B..NP-...-O-000

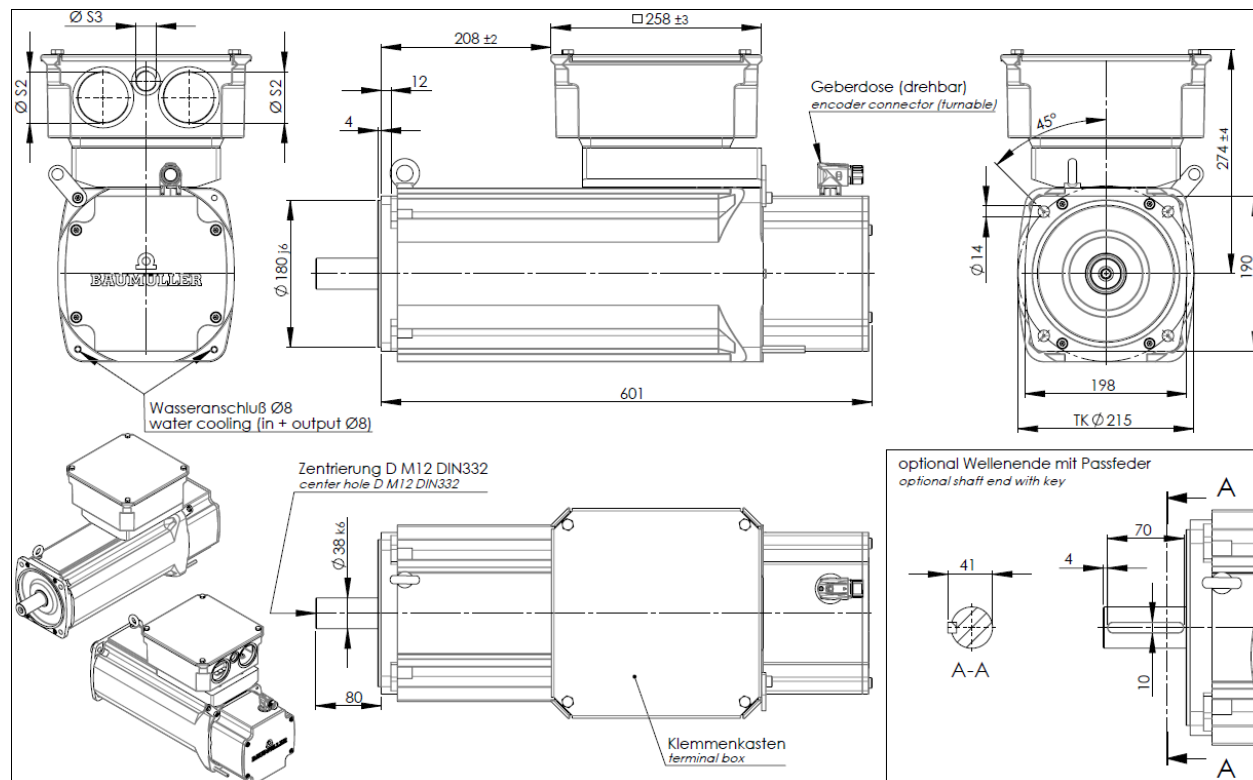
Version IM B5



Dimension drawing DSD2-100XO..W-...-O..NP-...-O-000

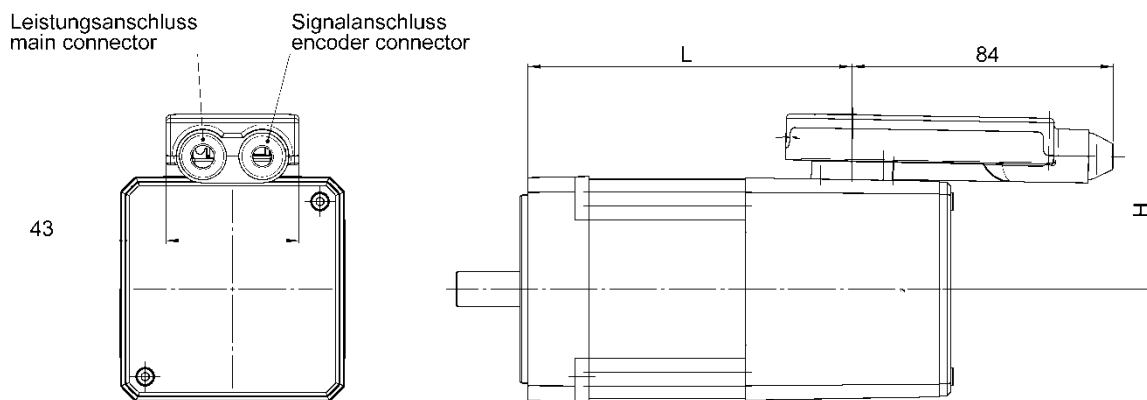


Dimension drawing DSD2-100XO..W-...-B..NP-...-O-000



4.8. Terminal box dimensions

4.8.1. Terminal box dimensions DSD2-028-036



Motor size	L	H
DSD2-028SO	97	56
DSD2-028MO	127	56

DSD2-036SO	104	58
DSD2-036MO	134	58
DSD2-036LO	164	58

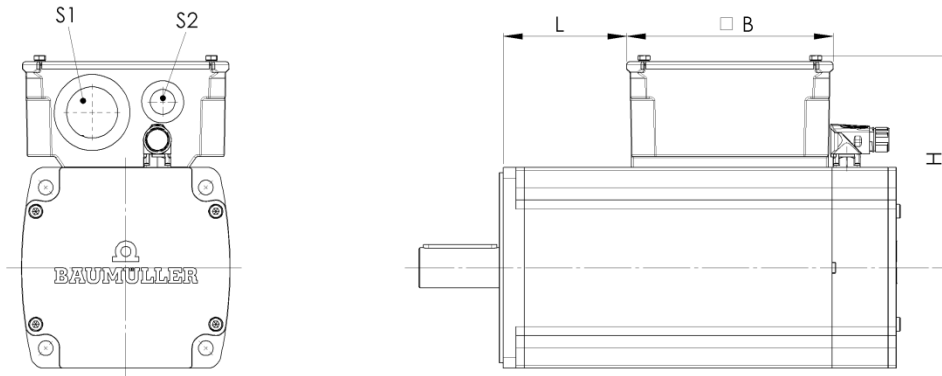
The terminal box can be mounted turned in steps of 90°. The discharge direction of the main and sensor connections is to be marked in the order index.

NOTE:

The connection allocation of the terminal box is not identical with the allocations of the sensor and main connection allocations described in this documentation. The connection allocation of the terminal box is available on request.

4.8.2. Terminal box dimensions DSD2-071-100

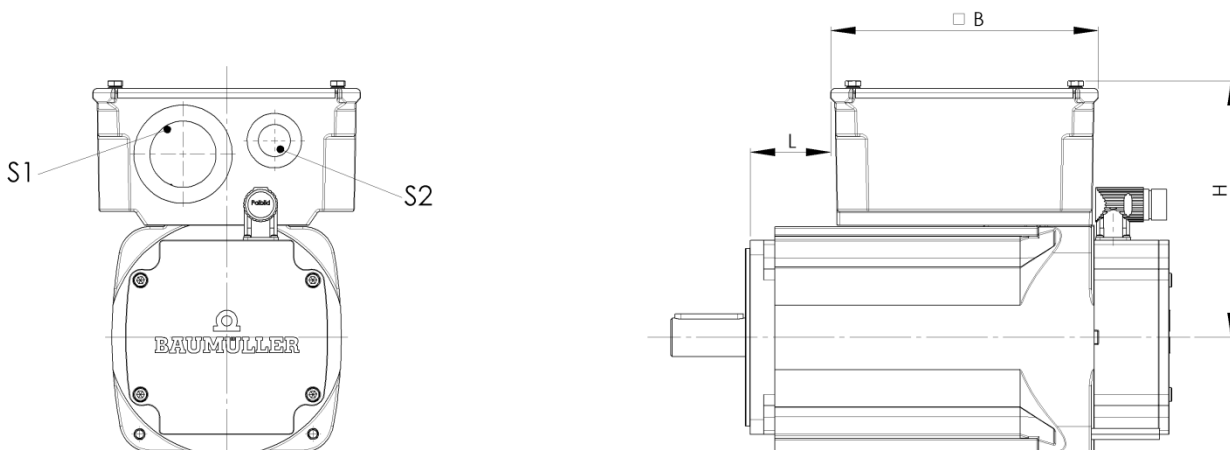
A terminal box must be used if $I_0 > 36$ A. The outlet direction of the main connection is at the B side.



Motor Length	L	B	H	S1	S2	I_0
DSD2-071SO	57	156	170	M40	M25	≤ 63 A
DSD2-071MO	97	156	170	M40	M25	≤ 63 A
DSD2-071LO	137	156	170	M40	M25	≤ 63 A

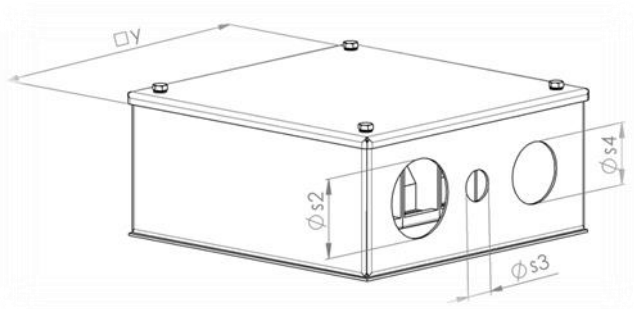
DSD2-100SO	65	196	205	M50	M25	≤ 95 A
DSD2-100MO	117	196	205	M50	M25	≤ 95 A
DSD2-100LO	169	196	205	M50	M25	≤ 95 A
DSD2-100BO	221	196	205	M50	M25	≤ 95 A
DSD2-100XO	208	258	264	2x M63	M25	> 95 A

In a water cooled DSD2-71 with an $I_0 > 63$ A the following terminal box has to be used. The discharge direction of the main connection is the N-side.



Motor Length	L	B	H	S1	S2	I_0
DSD2-071MO	59	196	190	M50	M25	≤ 95 A
DSD2-071LO	99	196	190	M50	M25	≤ 95 A

4.8.3. Terminal box dimensions DSD2-132



Nr.	Cable entries		Terminal	Thread	
	□ y [mm]	g1 [mm] compare dimensional drawings in chapter 44		s2 [mm]	s3 [mm]
20	258	250	3 x M8	2 x M25x1,5	1 x M25x1,5
22	258	250	3 x M8	2 x M40x1,5	1 x M25x1,5
24	258	250	3 x M8	2 x M63x1,5	1 x M25x1,5
26	258	250	3 x M10	2 x M63x1,5	1 x M25x1,5
28	318	305	3 x M12	2 x M75x1,5	1 x M25x1,5

4.8.4. Terminal designation

DSD2-071-100

	Connection diagram U V W Power connection R1/ R2 Temperature sensor (PT1000) BD+ / BD- Brake SL Protective conductor
--	---

DSD2-132

	Terminal designation Connection diagram U V W Power connection R1/ R2 Temperature sensor (PT 100) R2-1/ R2-2 Temperature sensor reserve (PT1000)
--	--

4.8.5. Position of the terminal boxes and the outlet direction of the main connection

The terminal box is on the B-side. The following positions of the terminal box are possible at the DSD2-132:

- Terminal box at the top
- Terminal box on the left (with viewing direction A-side onto the shaft end)
- Terminal box on the right (with viewing direction A-side onto the shaft end)

Dependent on the position of the terminal box at the DSD2-132 the following outlet directions of the main connection can be configured (also see product configurator).

Cooling type	Position terminal box	Output direction main connection				
		top	bottom	left ³⁾	right ³⁾	B-Seite
Radially mounted external fan ¹⁾	top	-	-	☒	☒	☒ ²⁾
	left ³⁾	☒	☒	-	-	☒
	right ³⁾	☒	☒	-	-	☒
Water cooling	top	-	-	☒	☒ ²⁾	☒
	left ³⁾	-	-	-	-	-
	right ³⁾	-	-	-	-	-
Axially mounted external fan	top	-	-	☒	☒ ²⁾	-
	left ³⁾	☒	☒	-	-	-
	right ³⁾	☒	☒	-	-	-

¹⁾ At motors with a radial blower the position of the terminal box is dependent of the position of the cooling (e.g. position of the terminal box ("on the top" isn't possible if the position of the cooling is „on the top“).

²⁾ Preferred option

³⁾ Viewing direction A-side onto the shaft end

At the DSD2-071-100 the position of the terminal box is on the top and the outlet direction of the main connection is on the B-side.

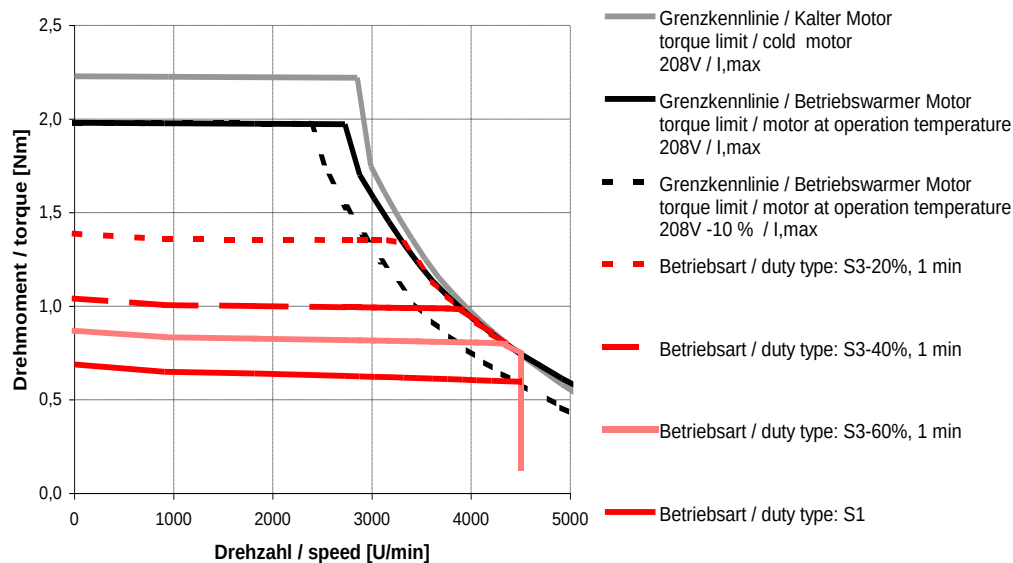
5. Motor characteristic curves

Definition

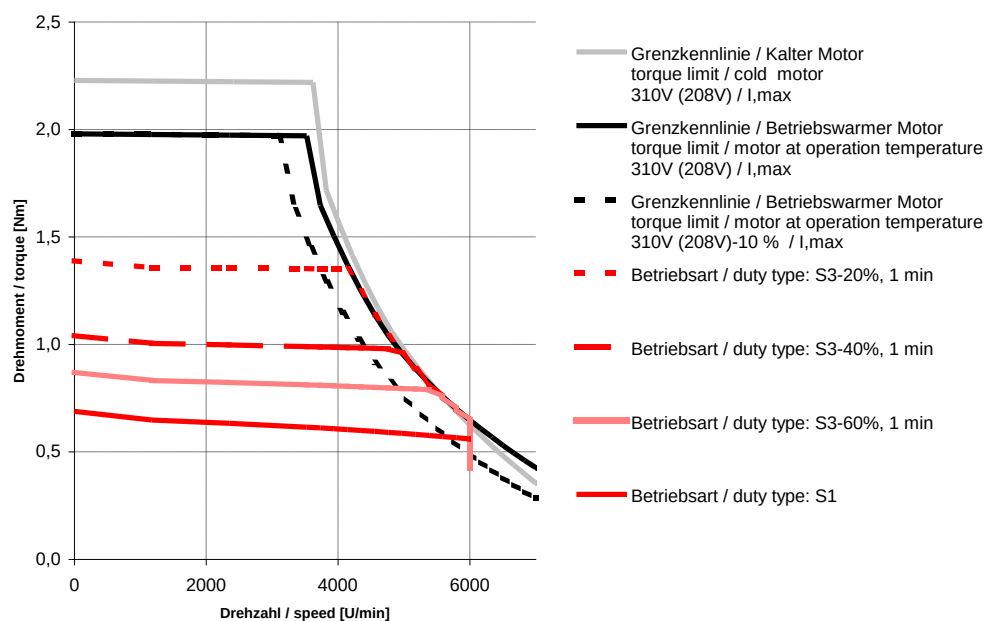
Cold motor	Environmental temperature (0°C to 40°C)
Motor at operating temperature	Continuous operation (S1) with nominal data of the motor or cyclical operation with corresponding effective output --> Environmental temperature + Delta temperature rise (105K)

5.1. Motor characteristic curve DSD2-028

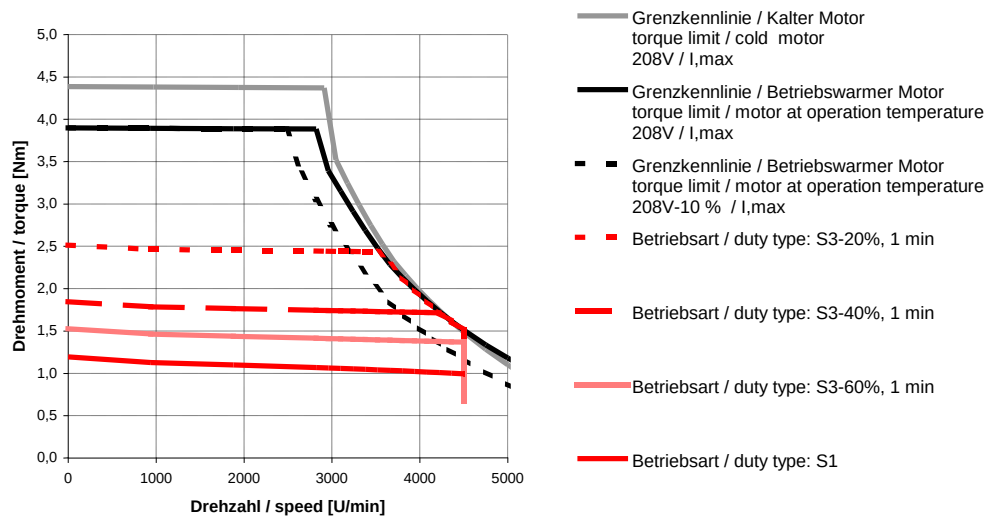
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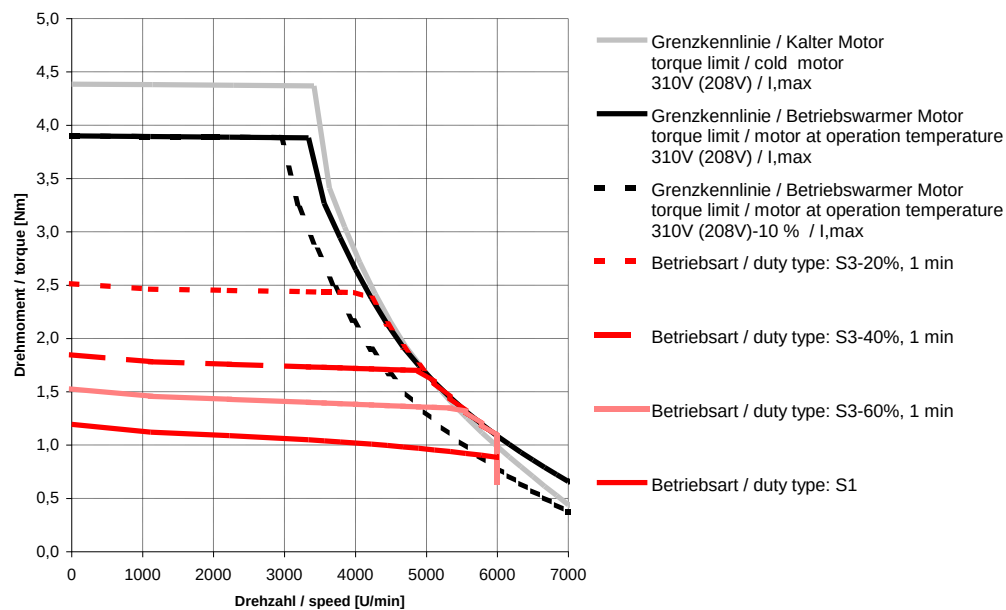
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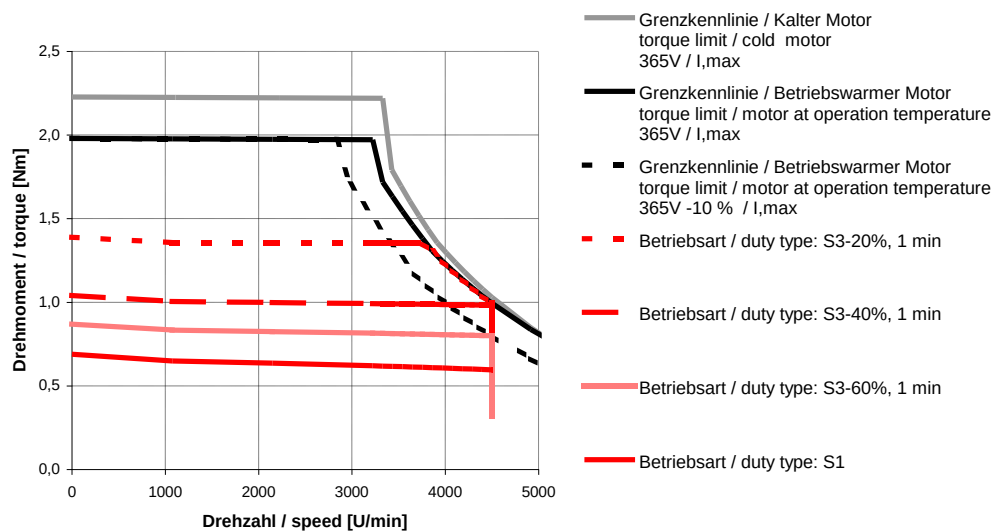
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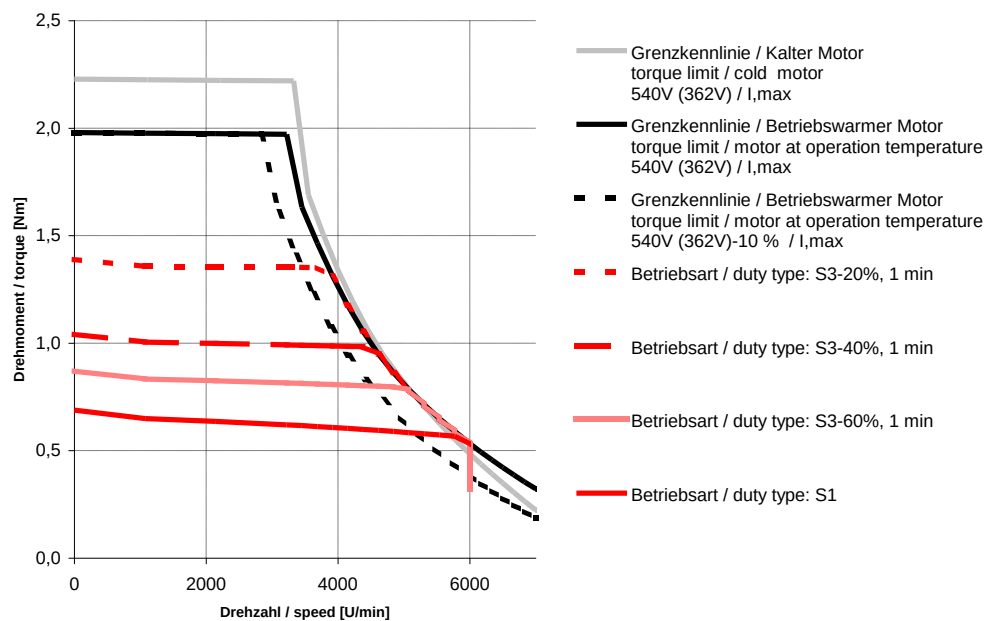
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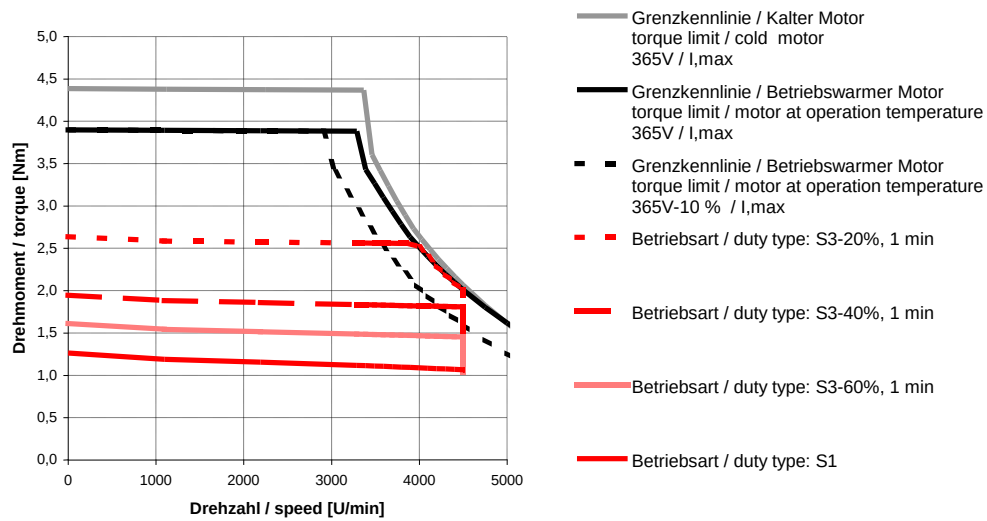
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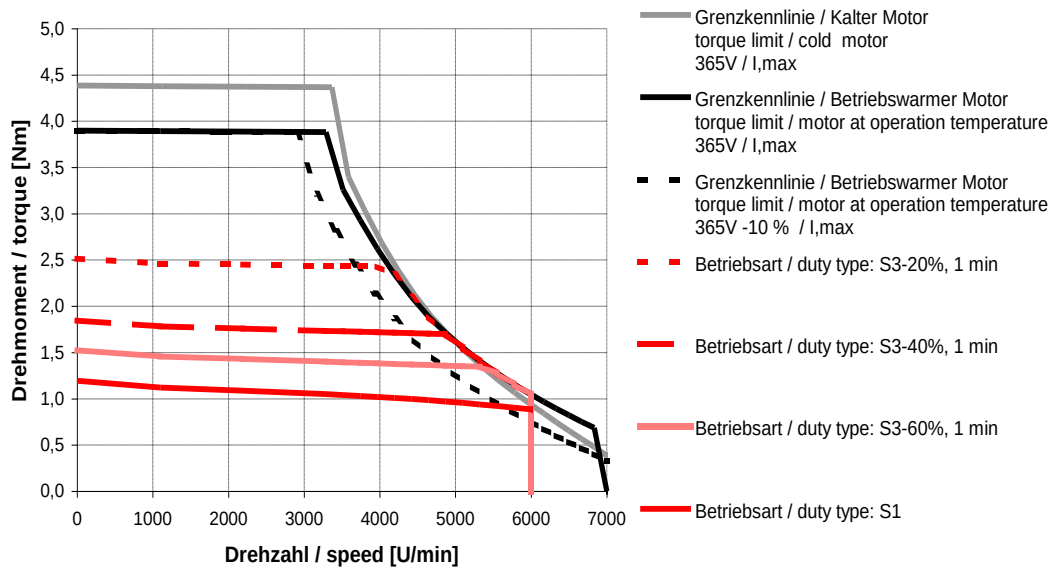
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DSD2-028MO44U-45-54

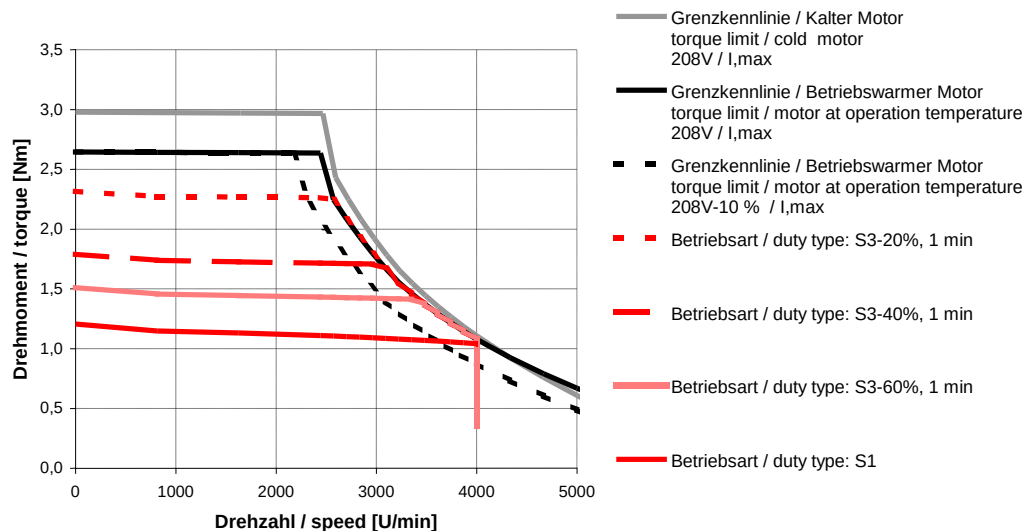


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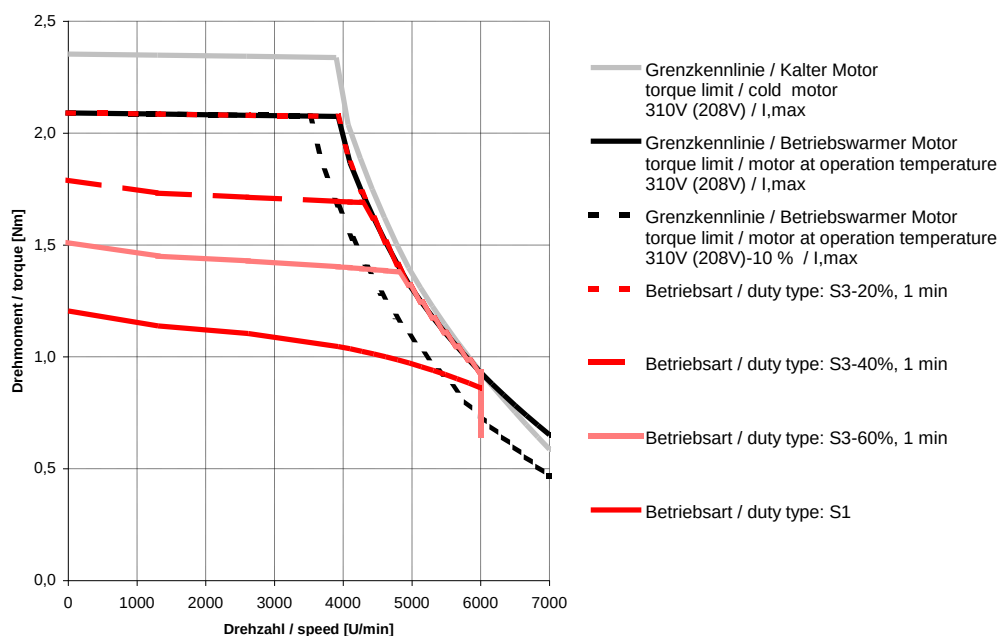


5.2. Motor characteristic curve DSD2-036

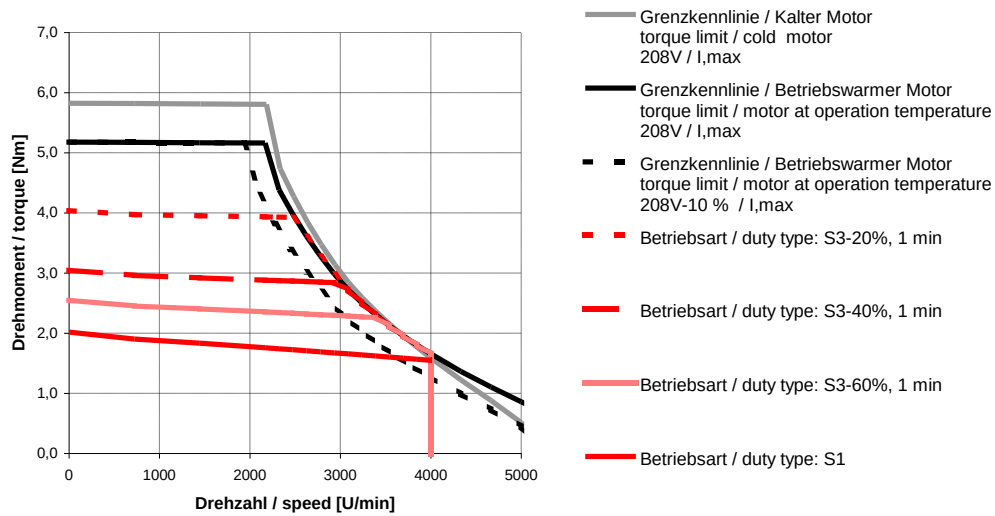
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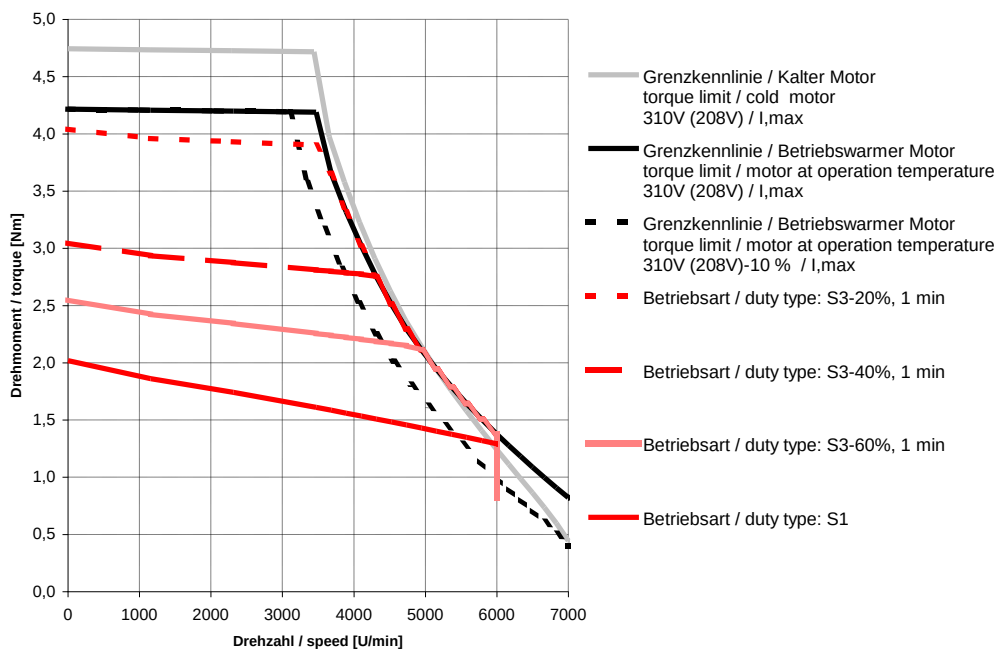
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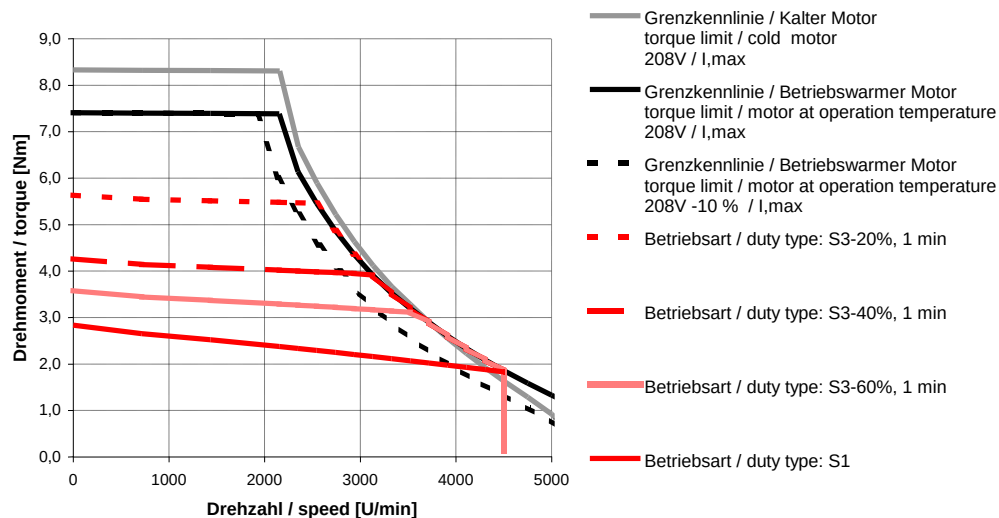
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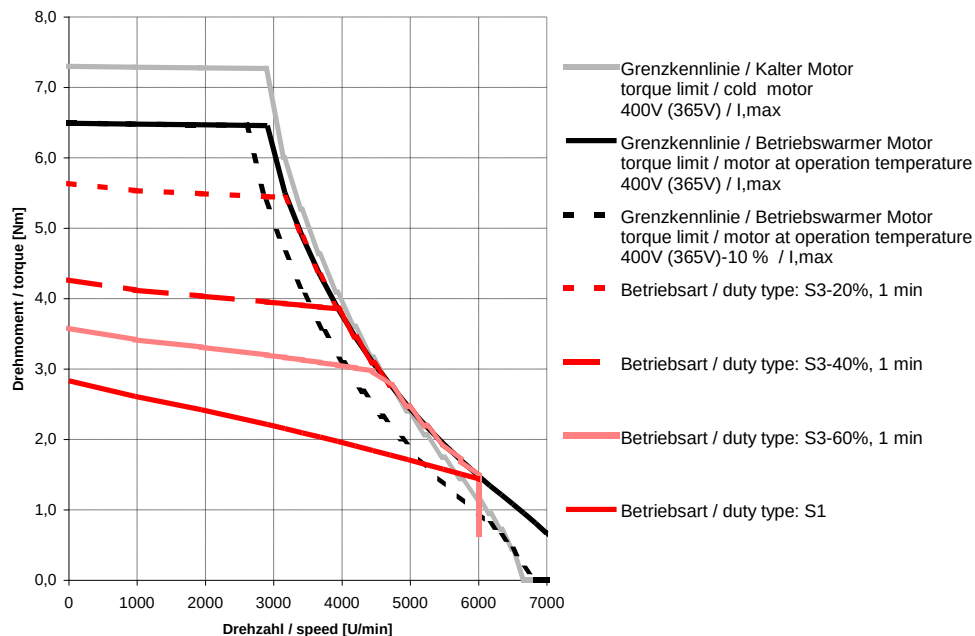
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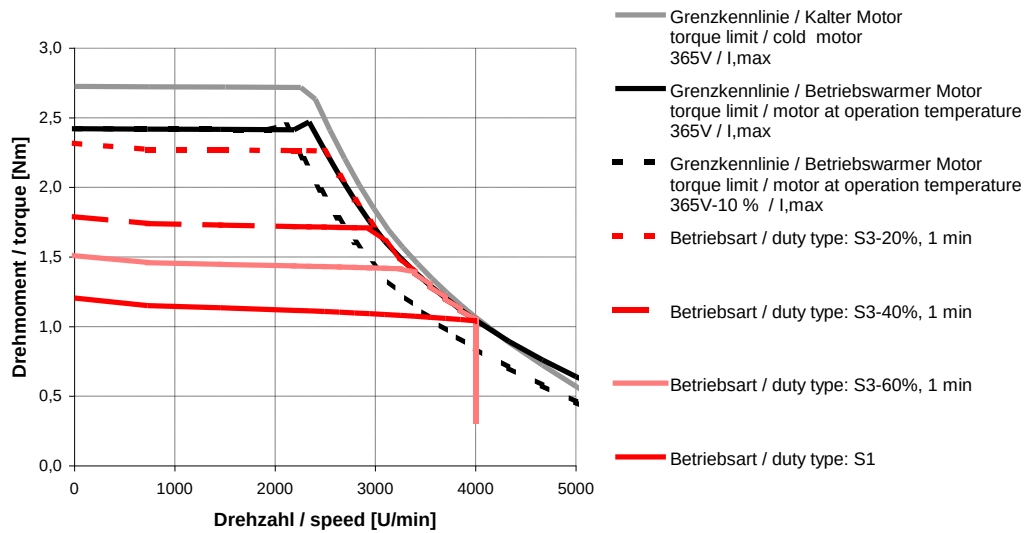
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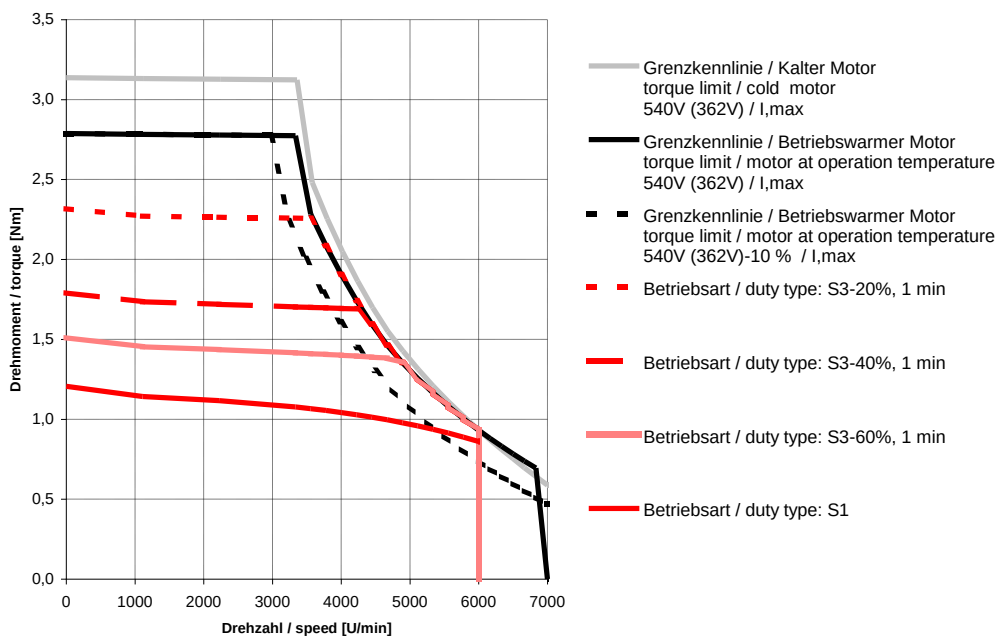
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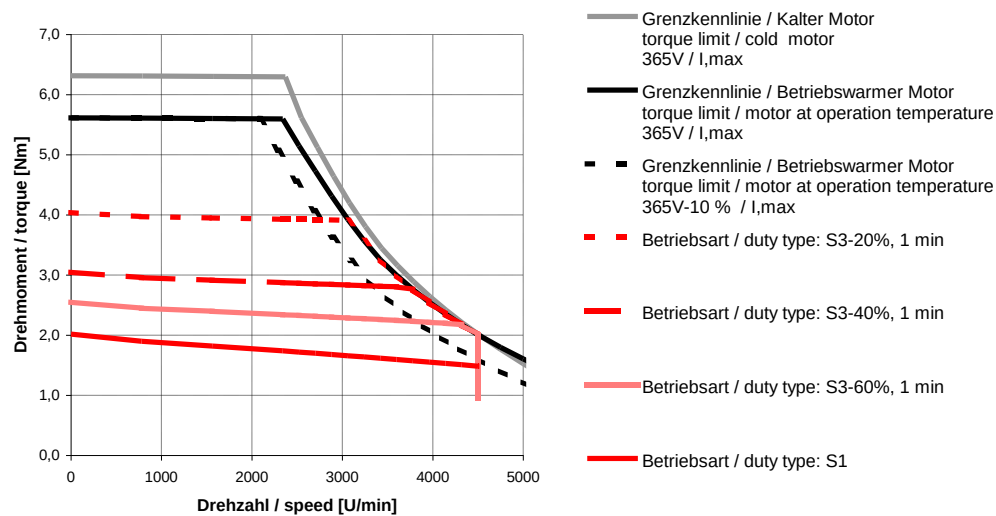
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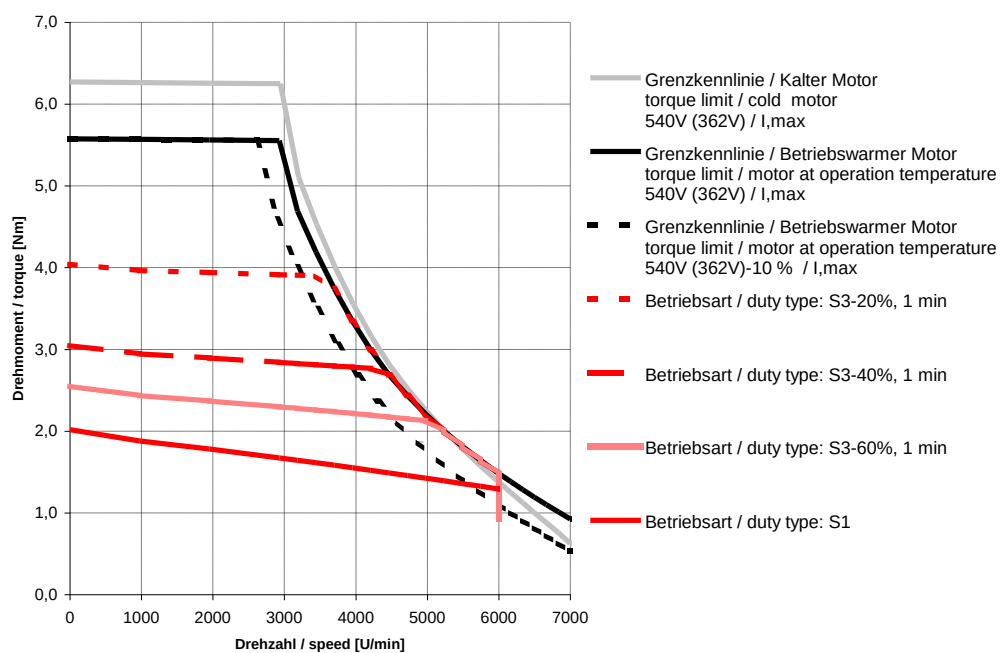
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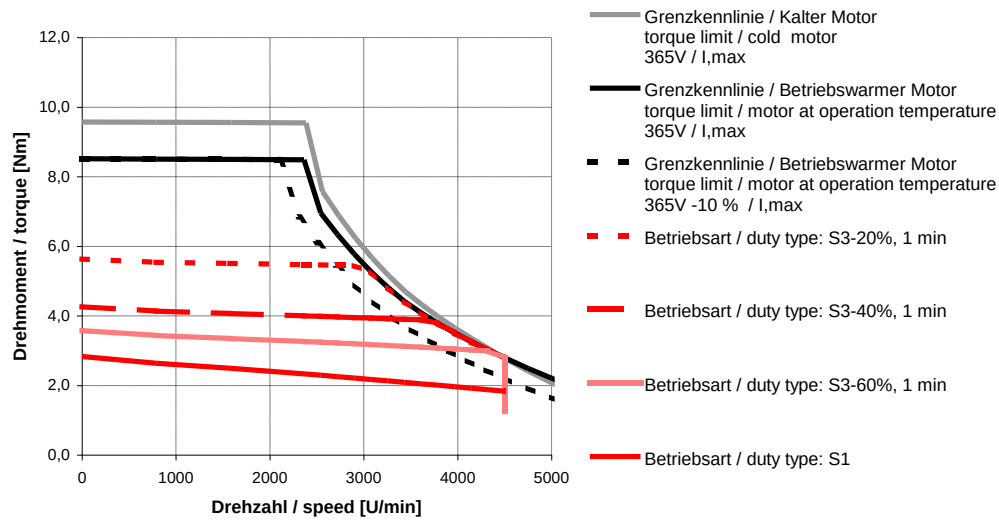
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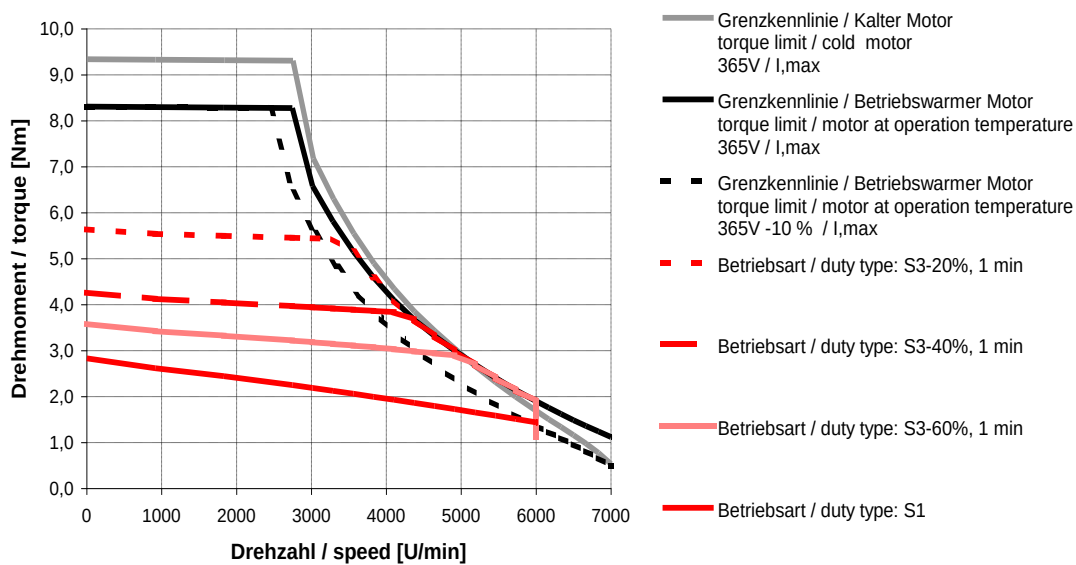
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DSD2-036LO44U-45-54



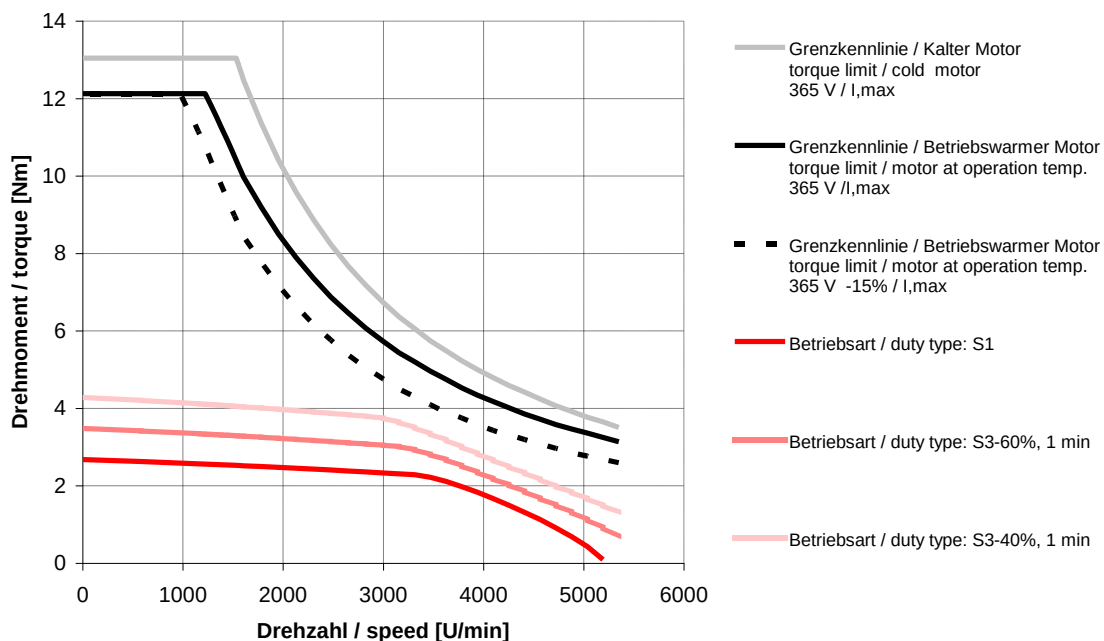
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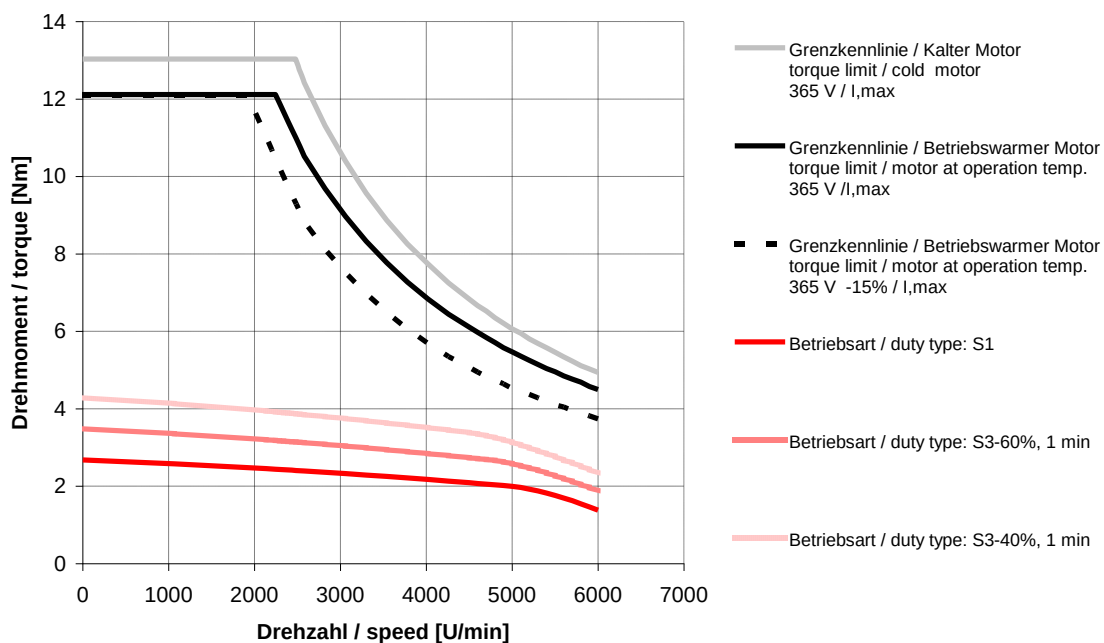
5.3. Motor characteristic curve DSD2-045

5.3.1. DSD2-045..64U..(without fan)

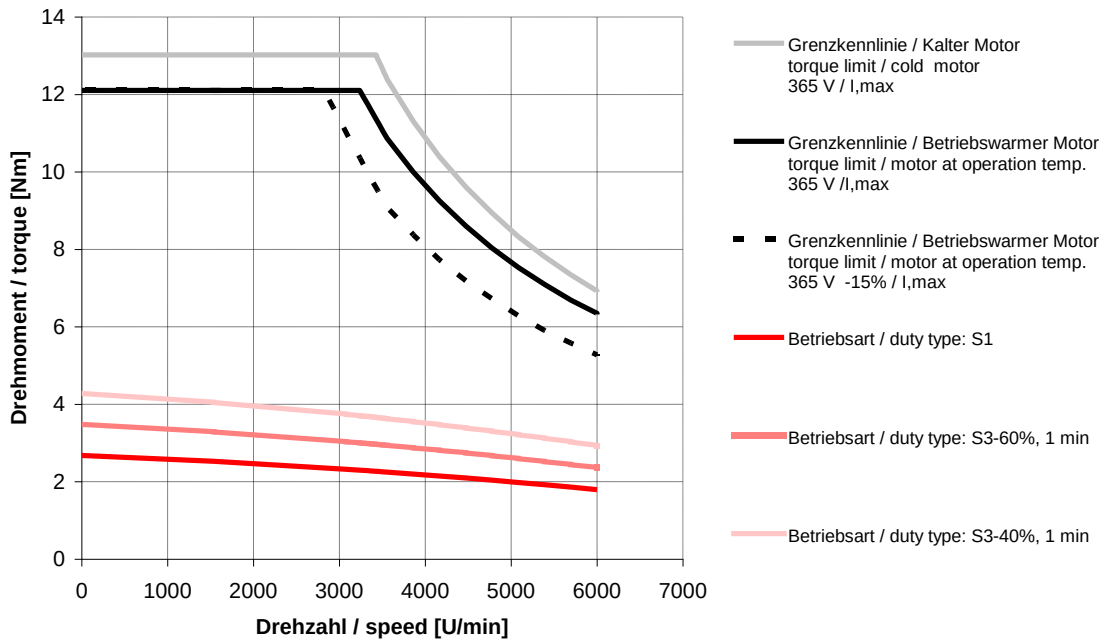
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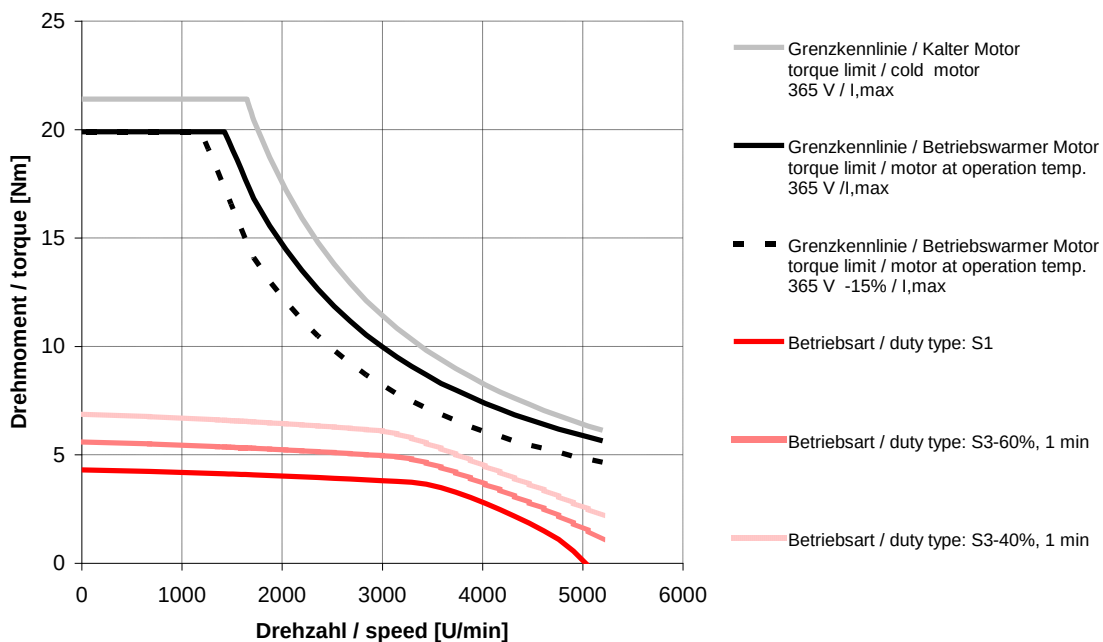
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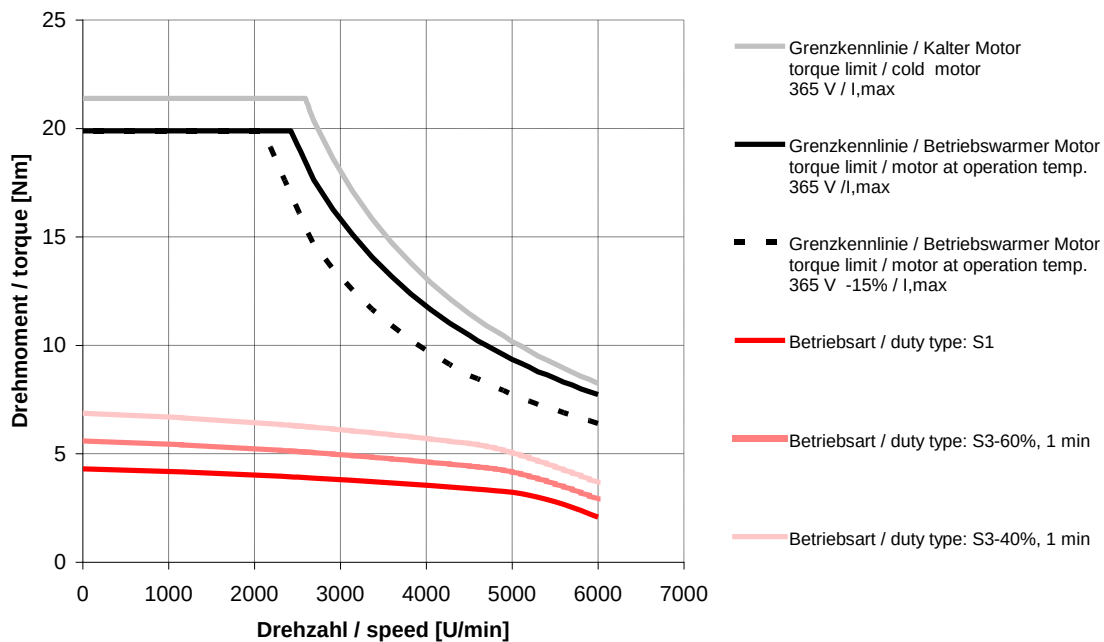
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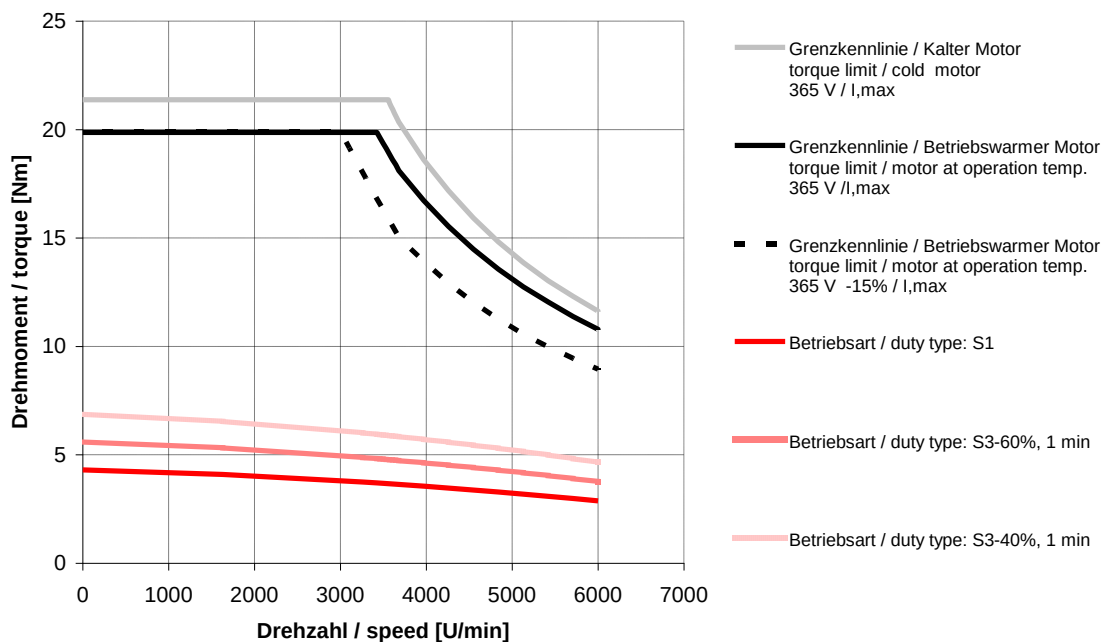
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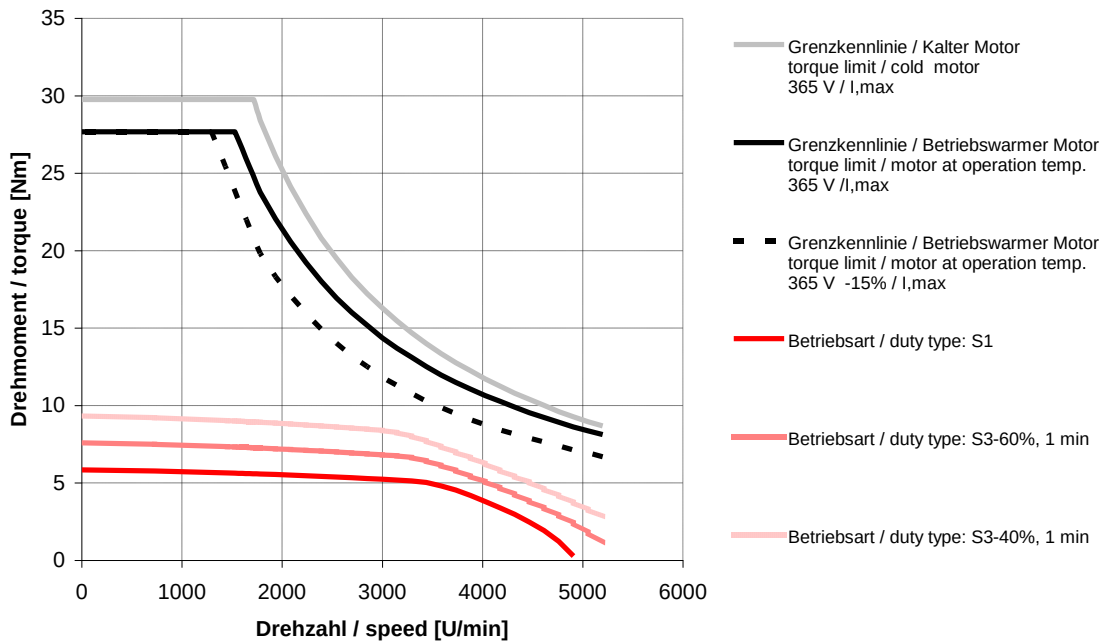
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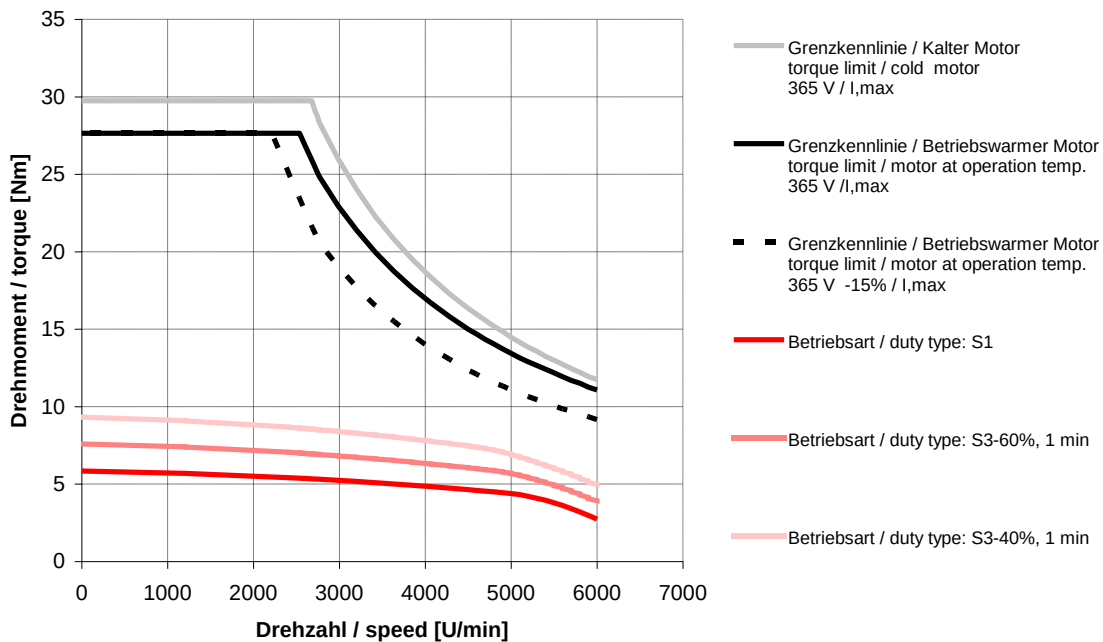
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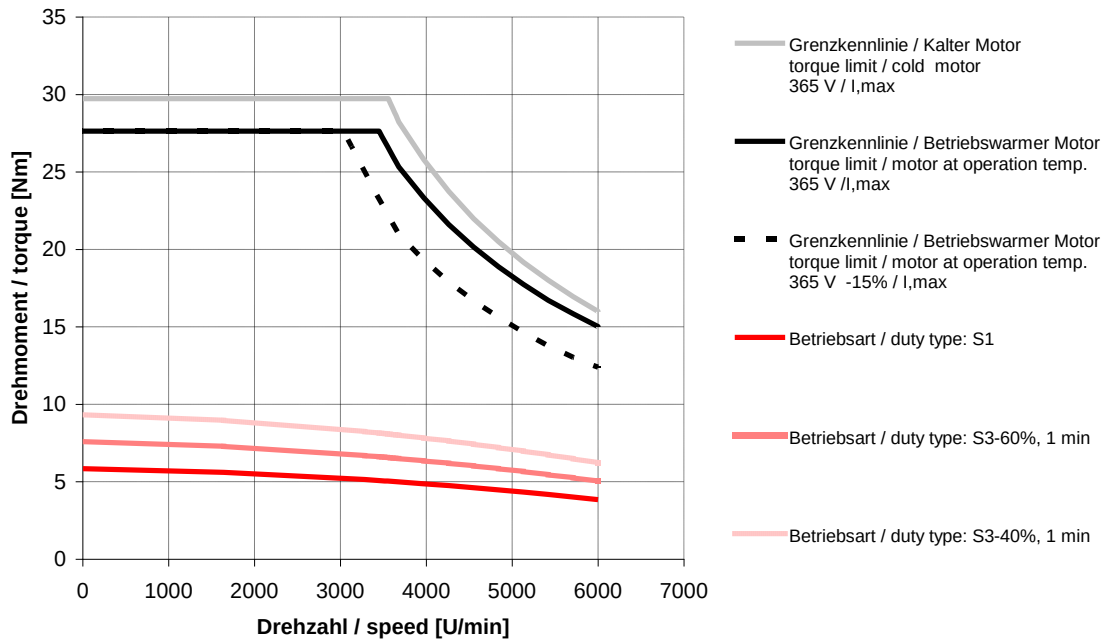
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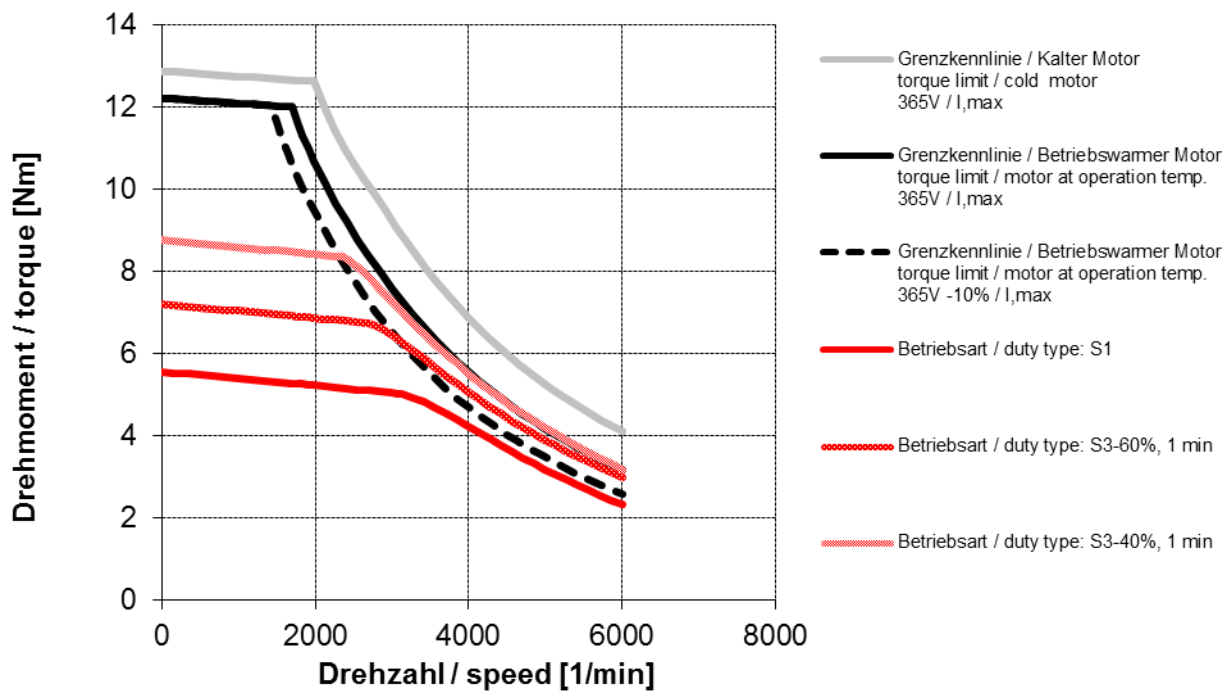


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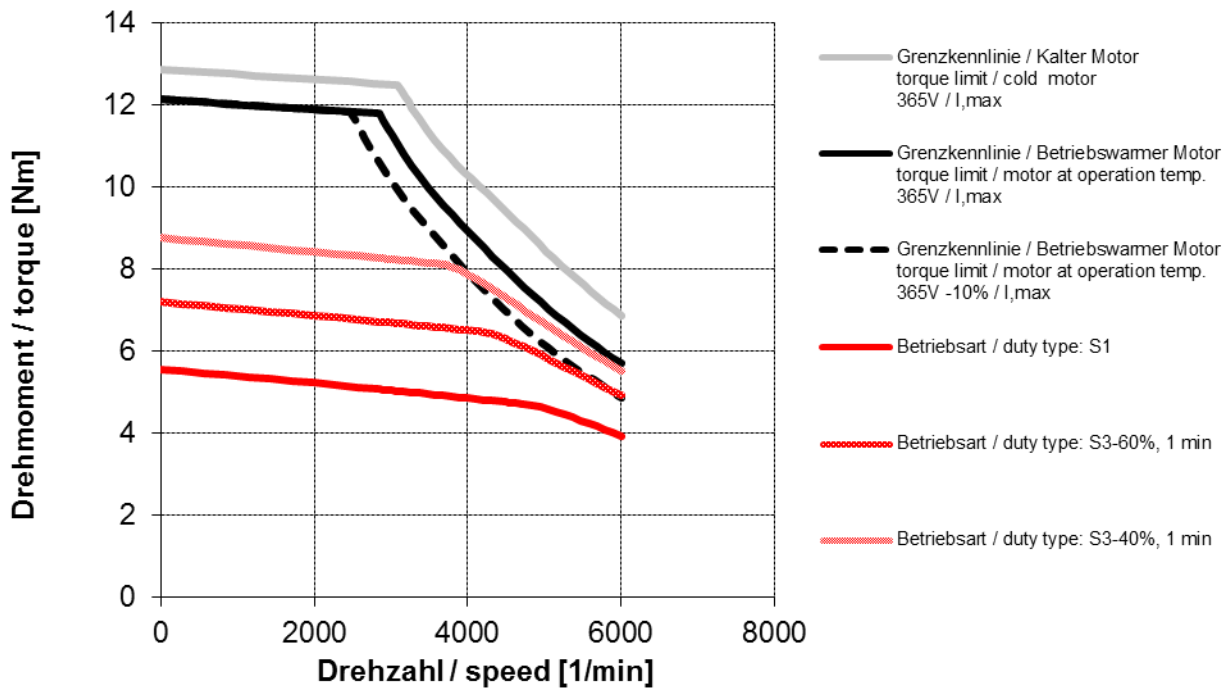


5.3.2. DSD2-045..64W-.. (water cooled)

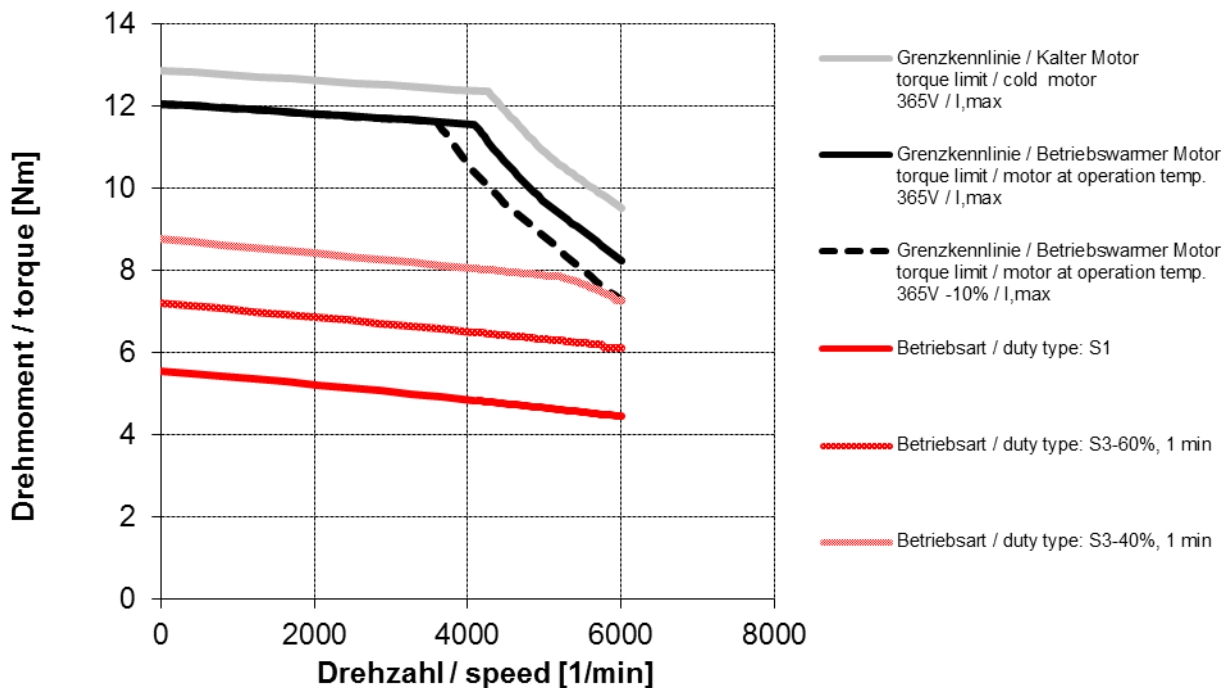
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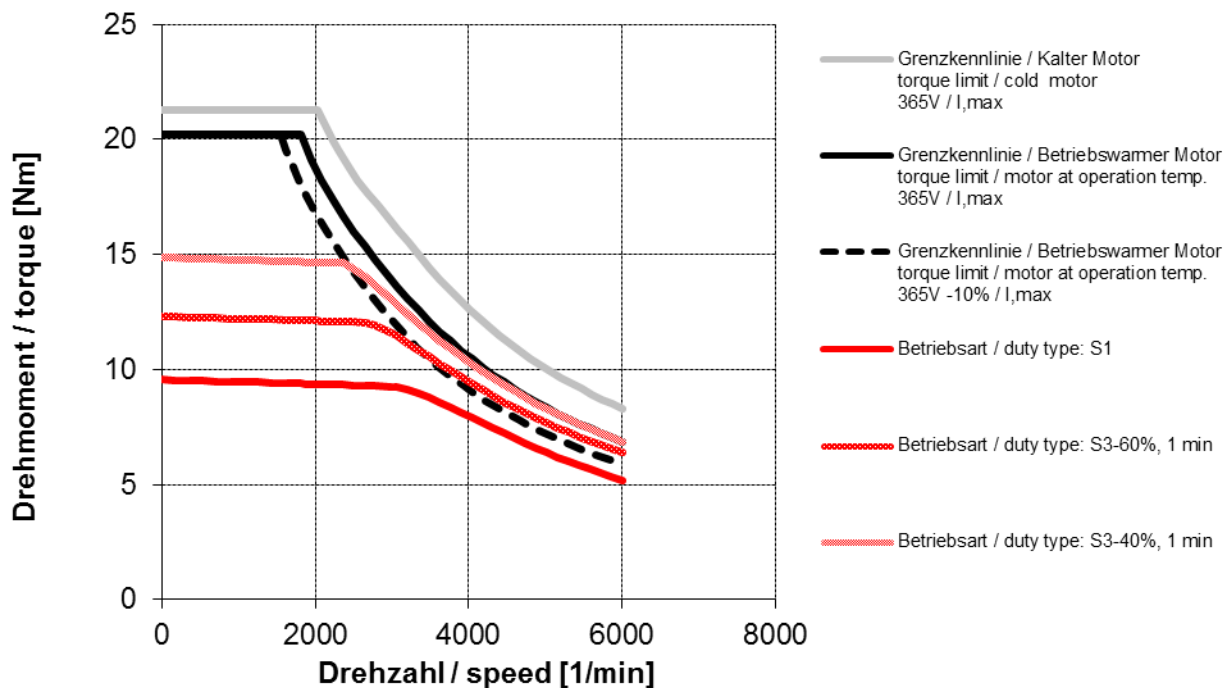
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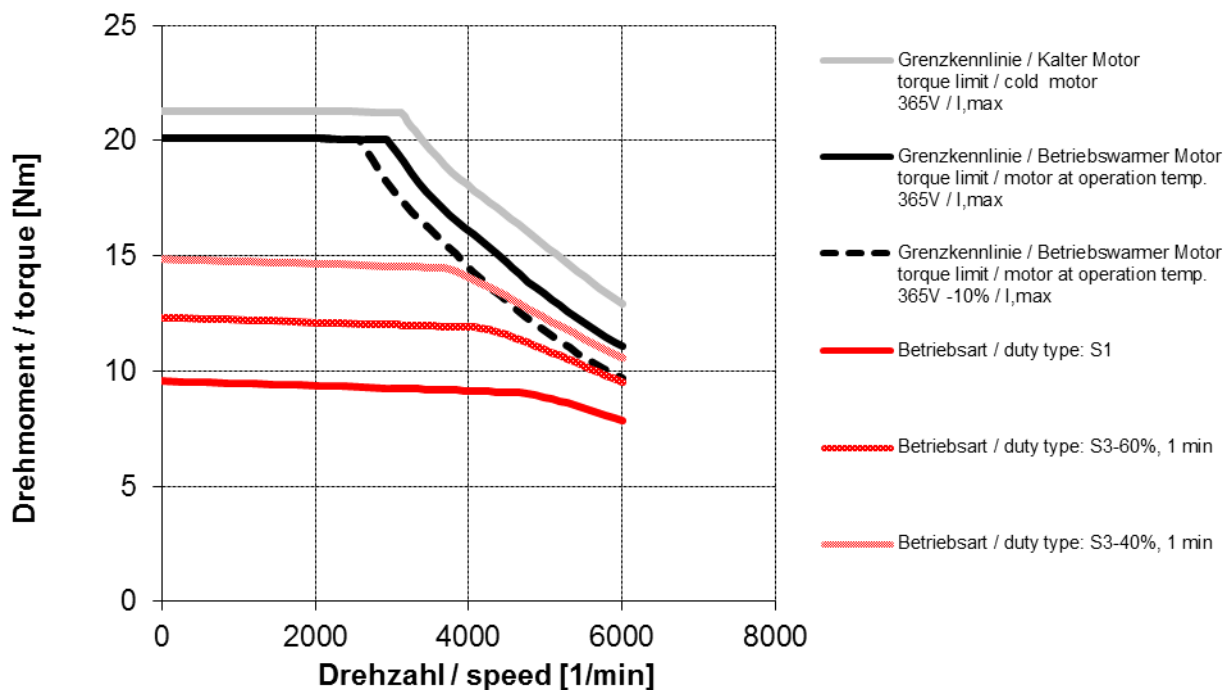
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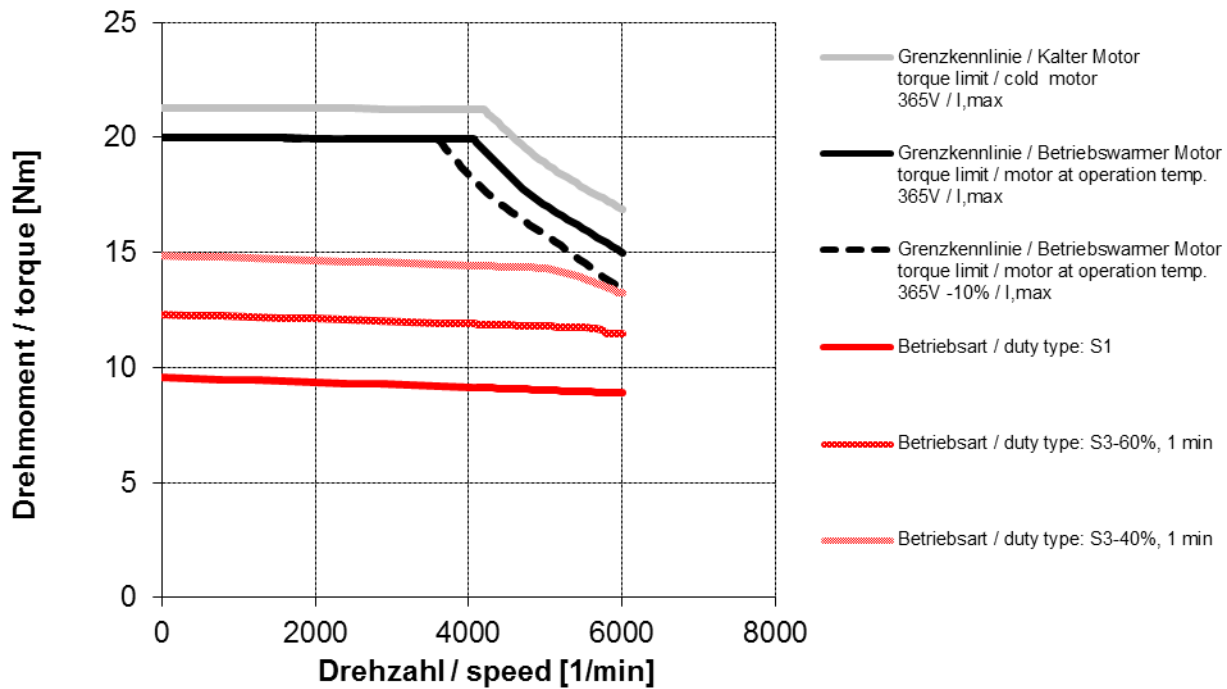
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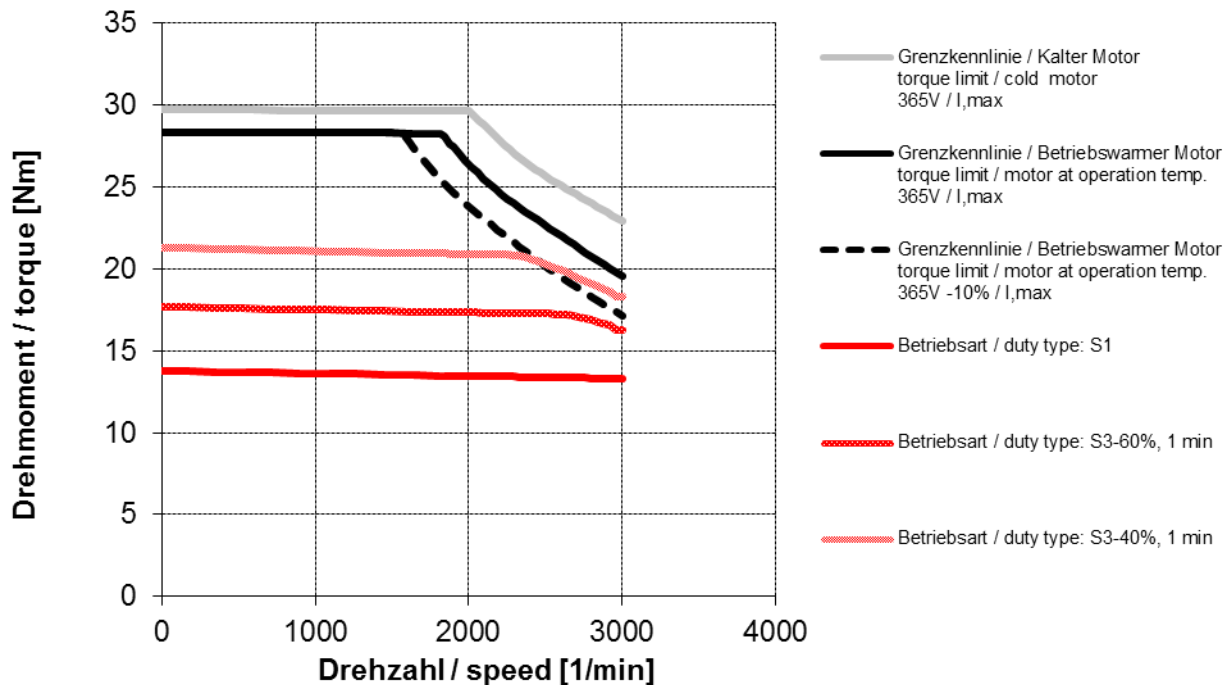
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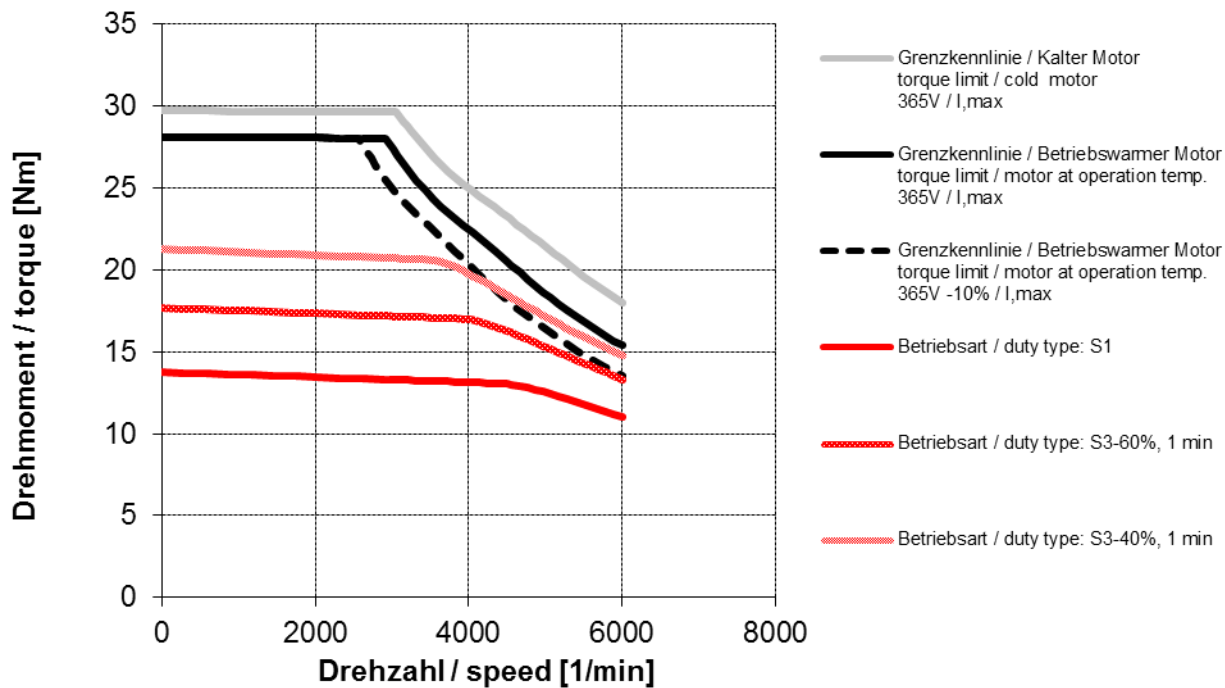
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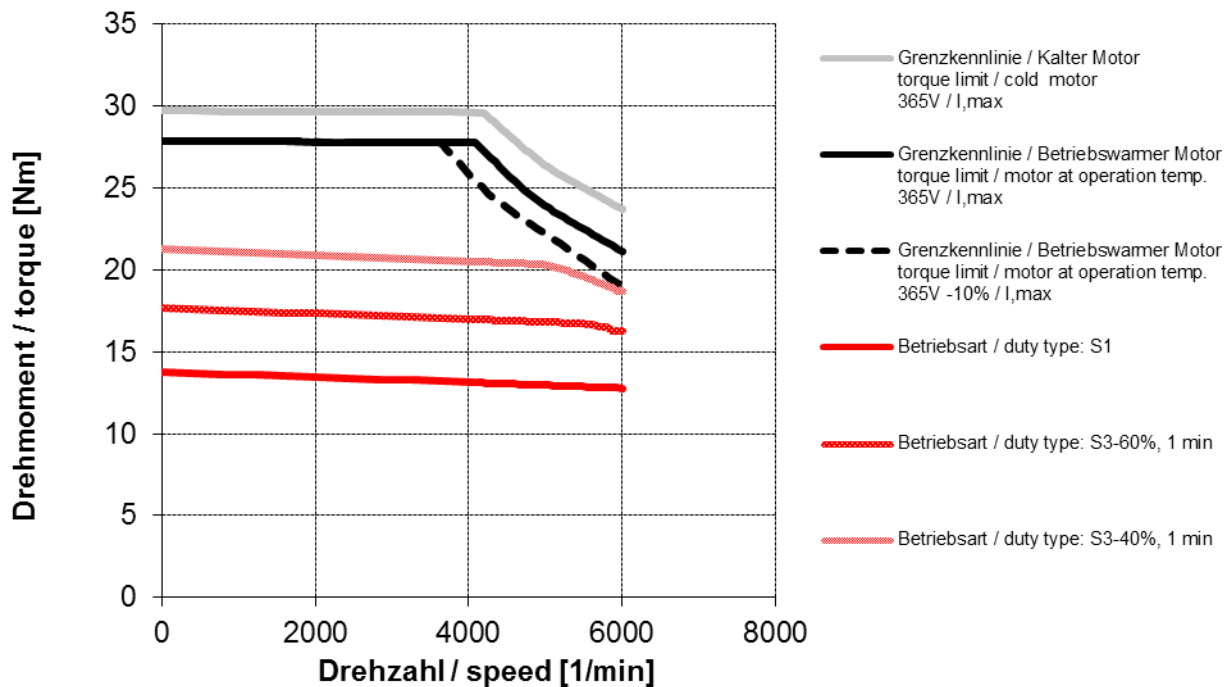
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DSD2-045LO64W-45-54



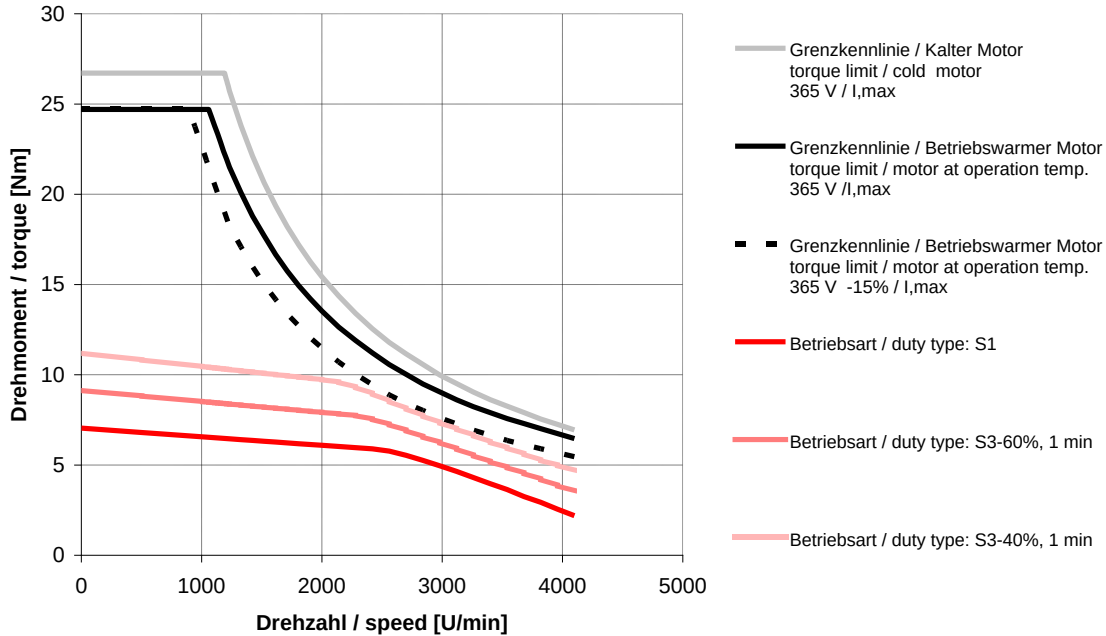
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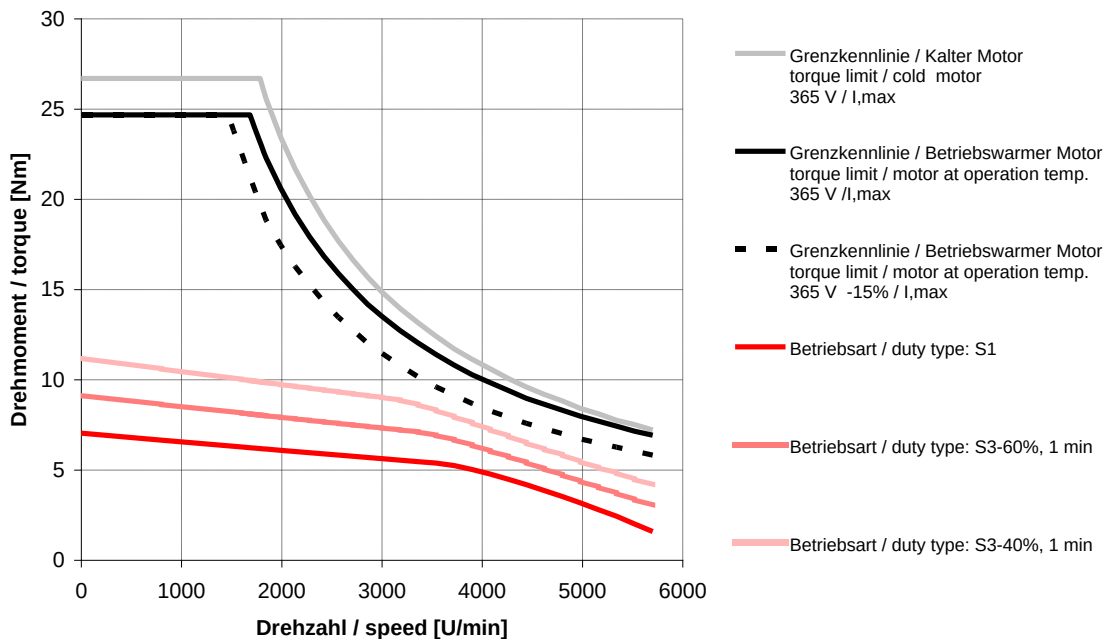
5.4. Motor characteristic curve DSD2-056

5.4.1. DSD2-056..64U-.. (without fan)

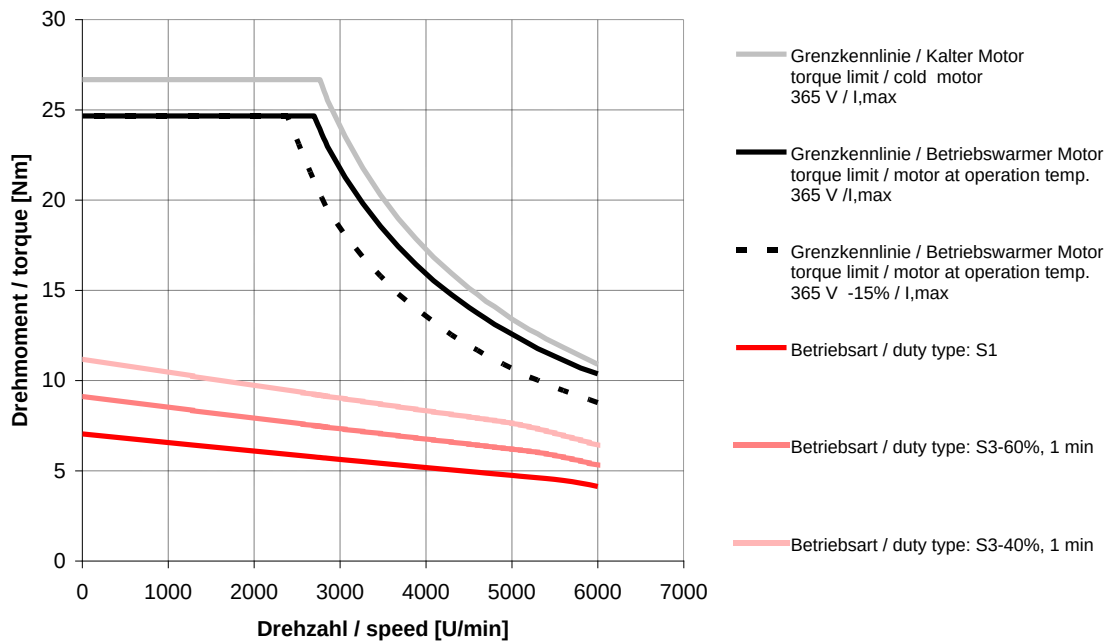
DSD2-056SO64U-20-54



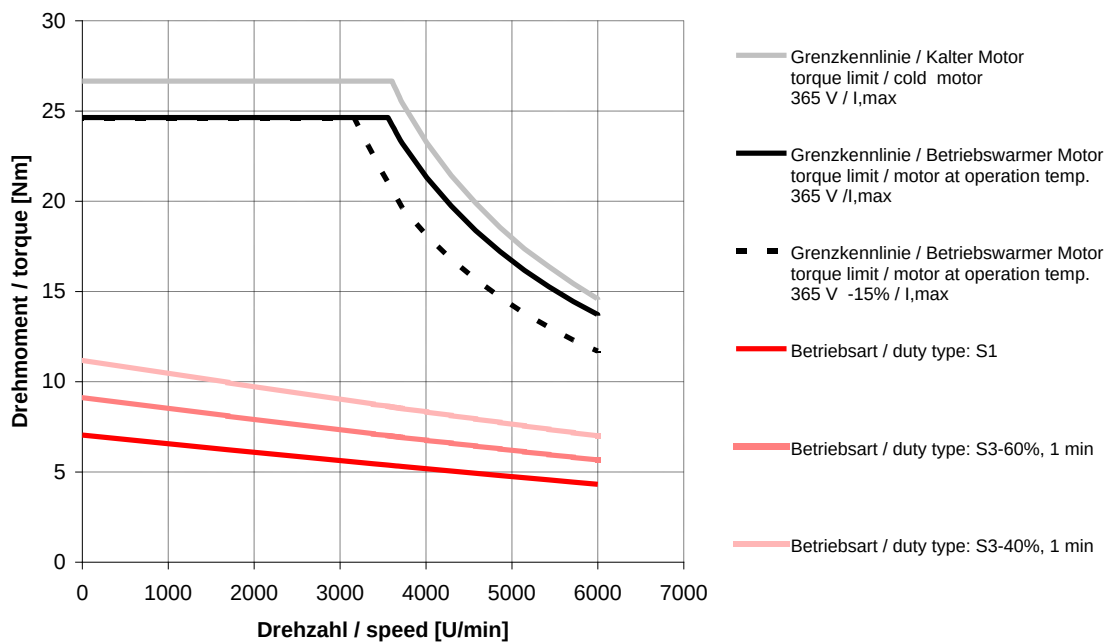
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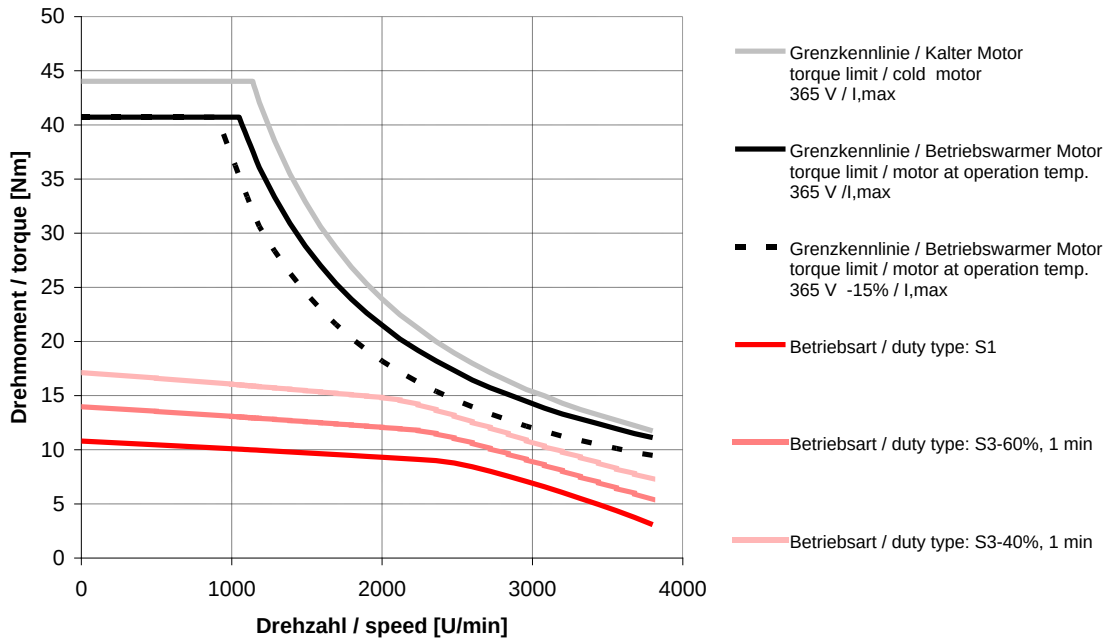
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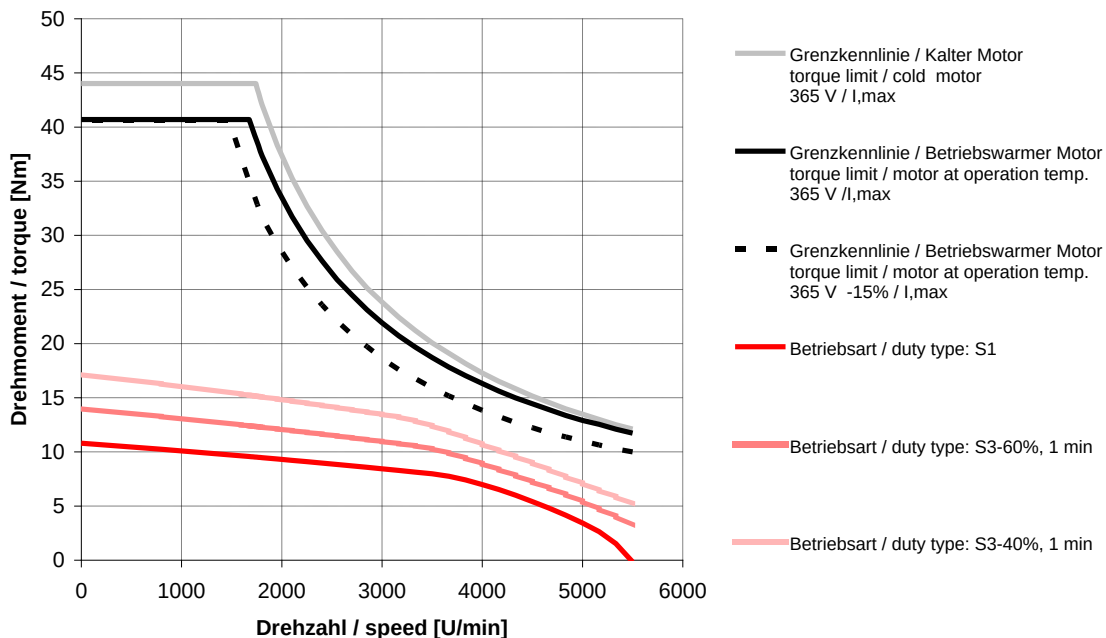
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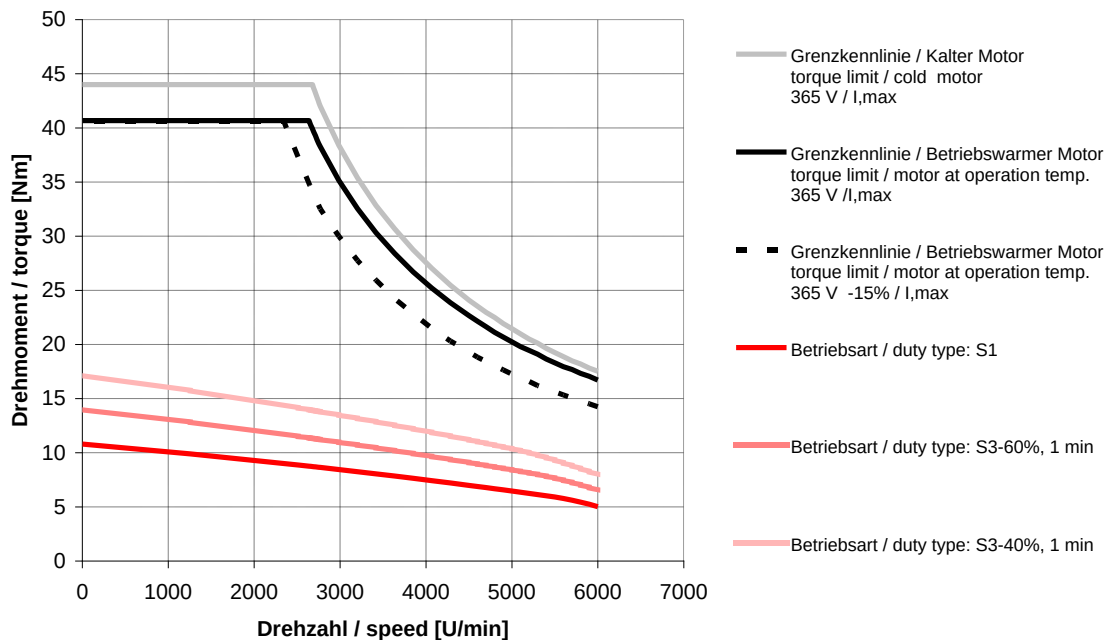
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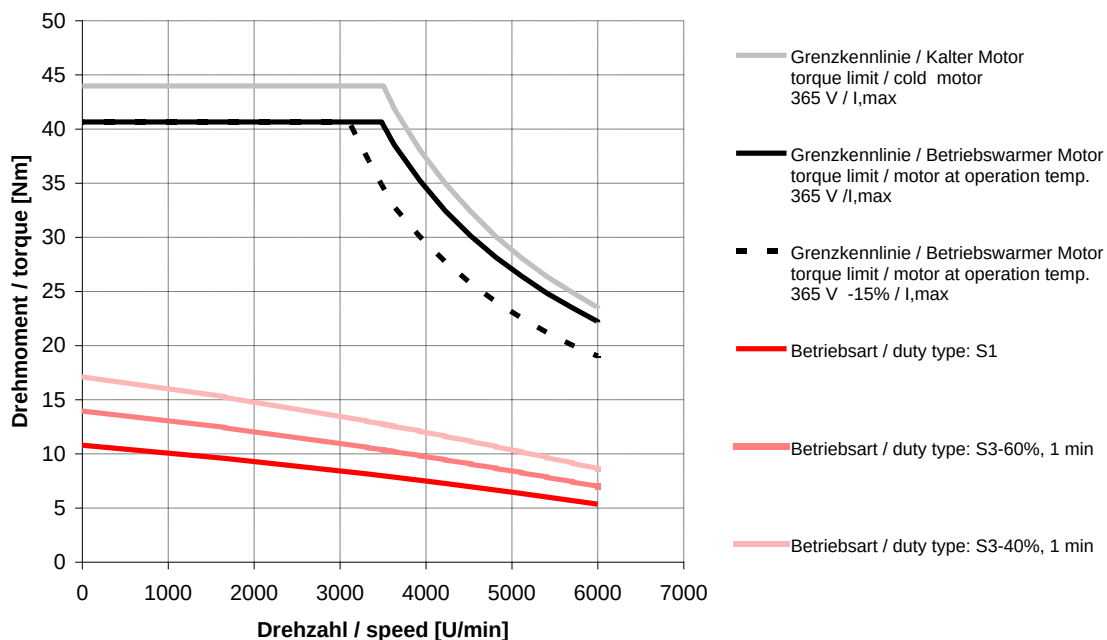
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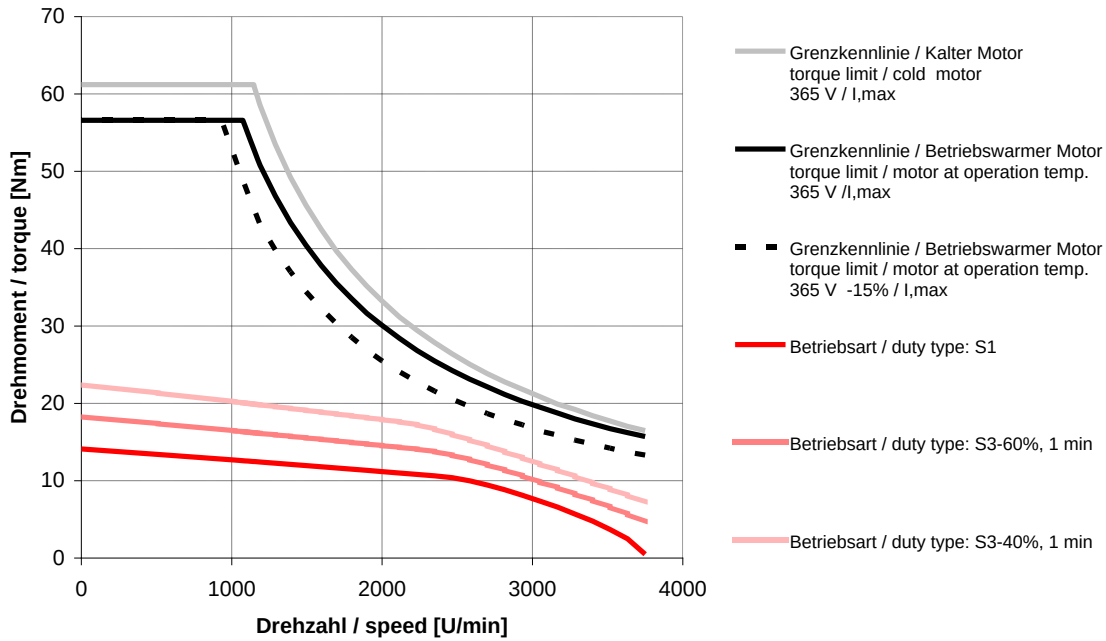
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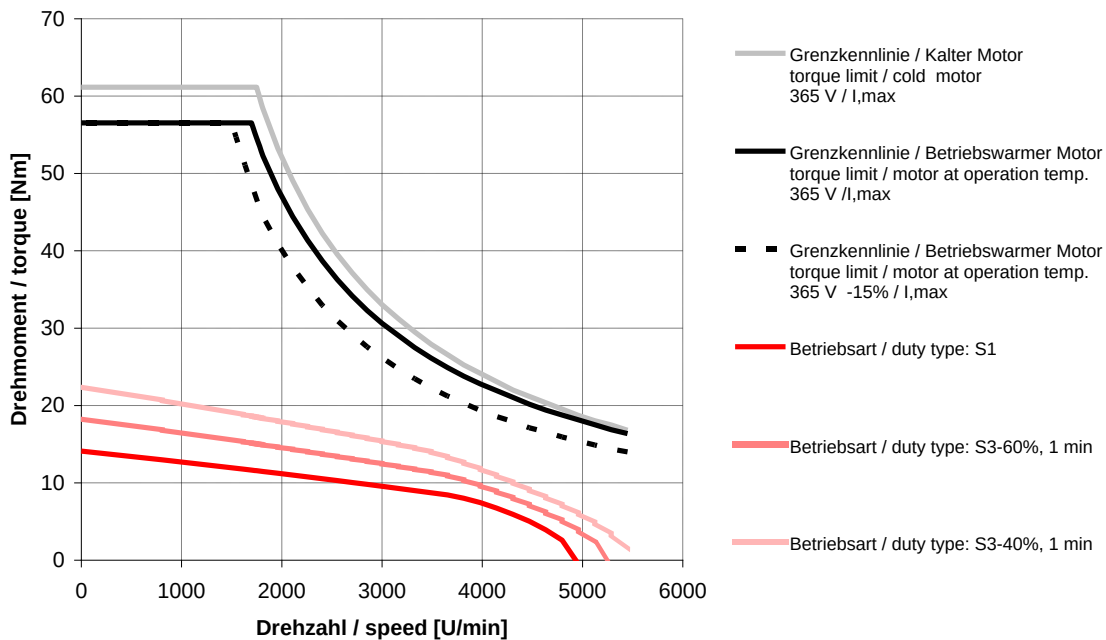
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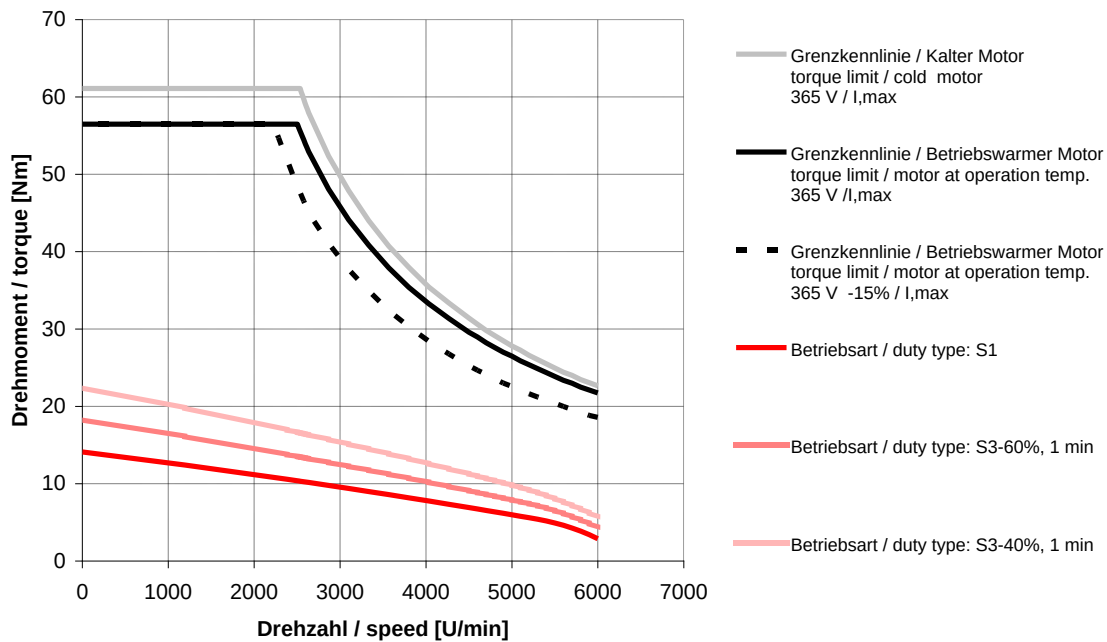
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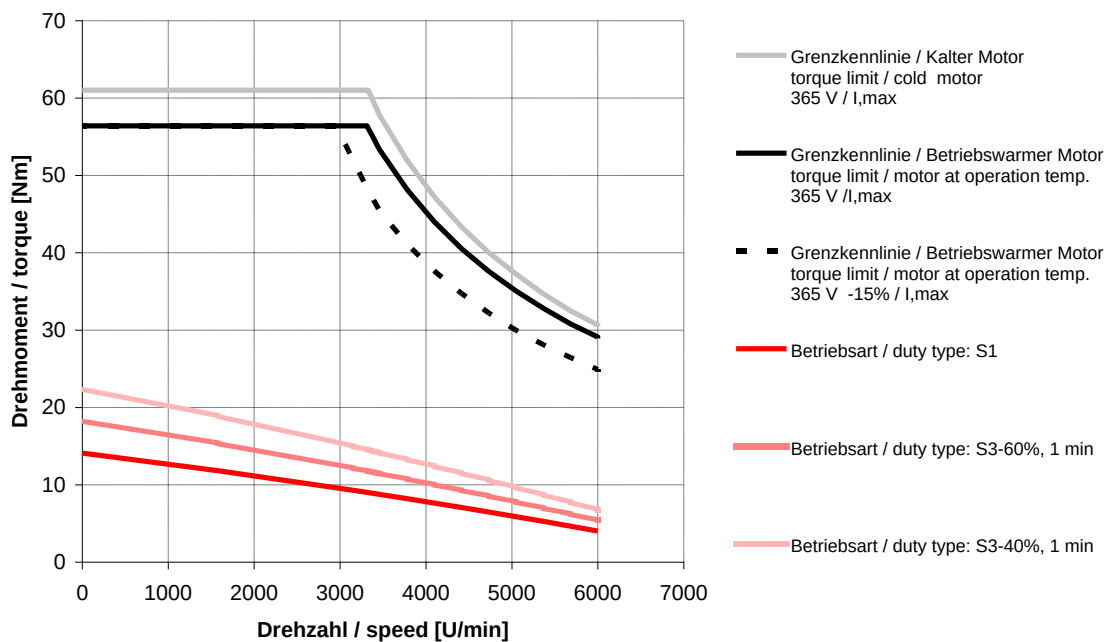
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DSD2-056LO64U-45-54

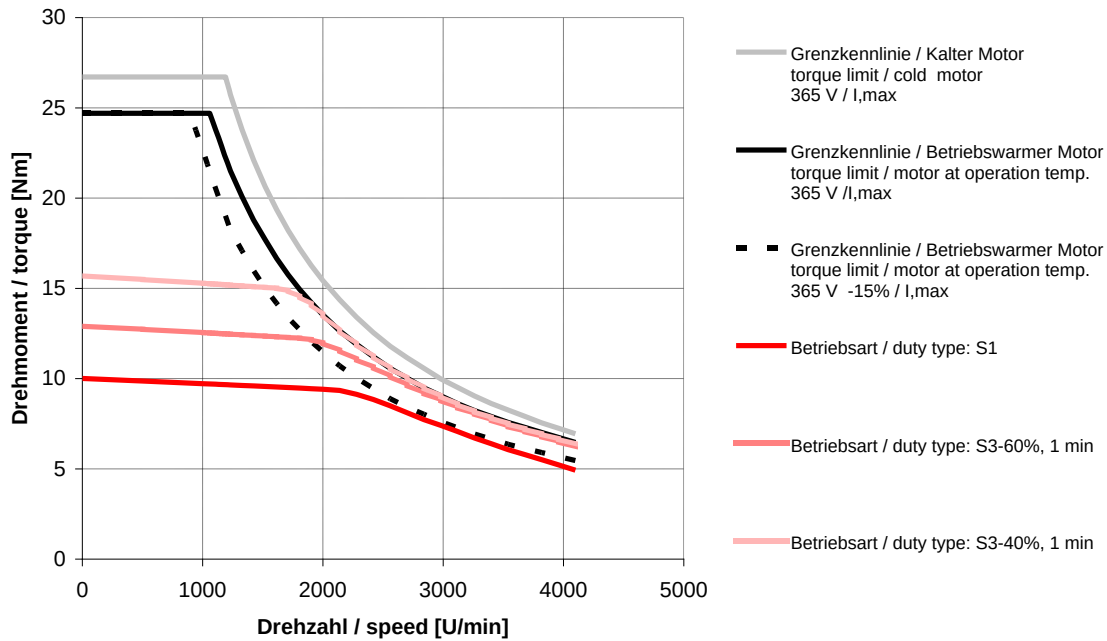


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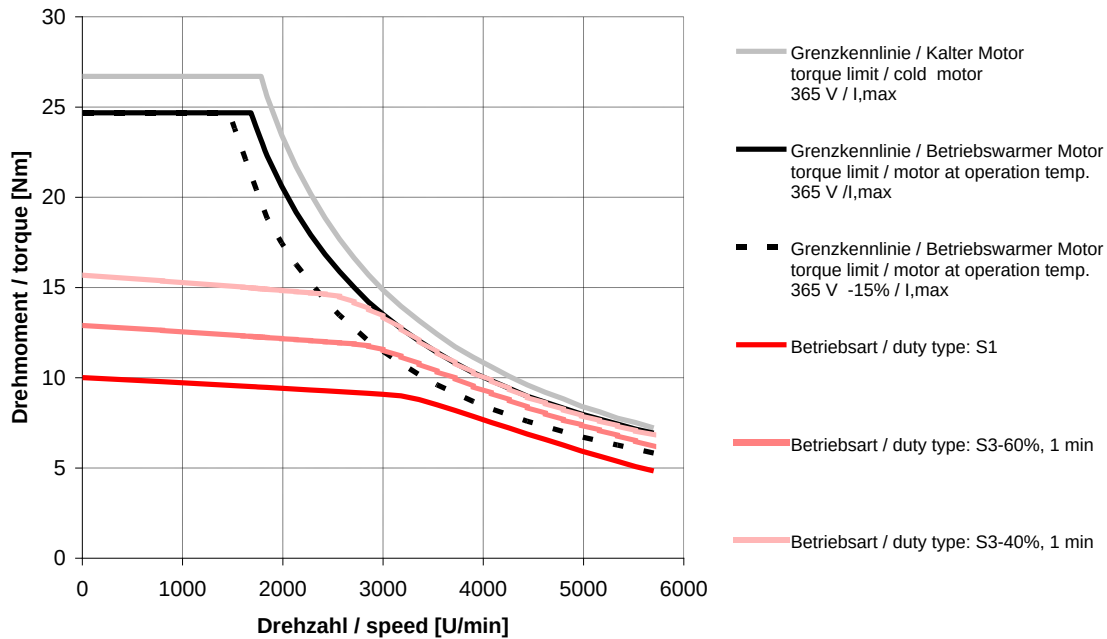


5.4.2. DSD2-056..640-.. (with fan)

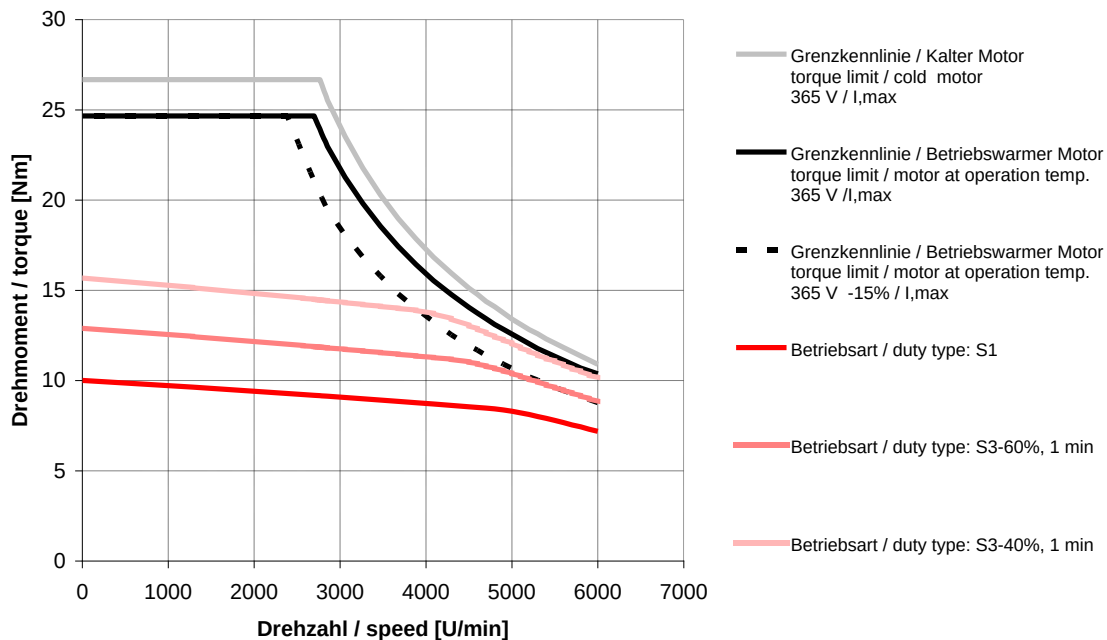
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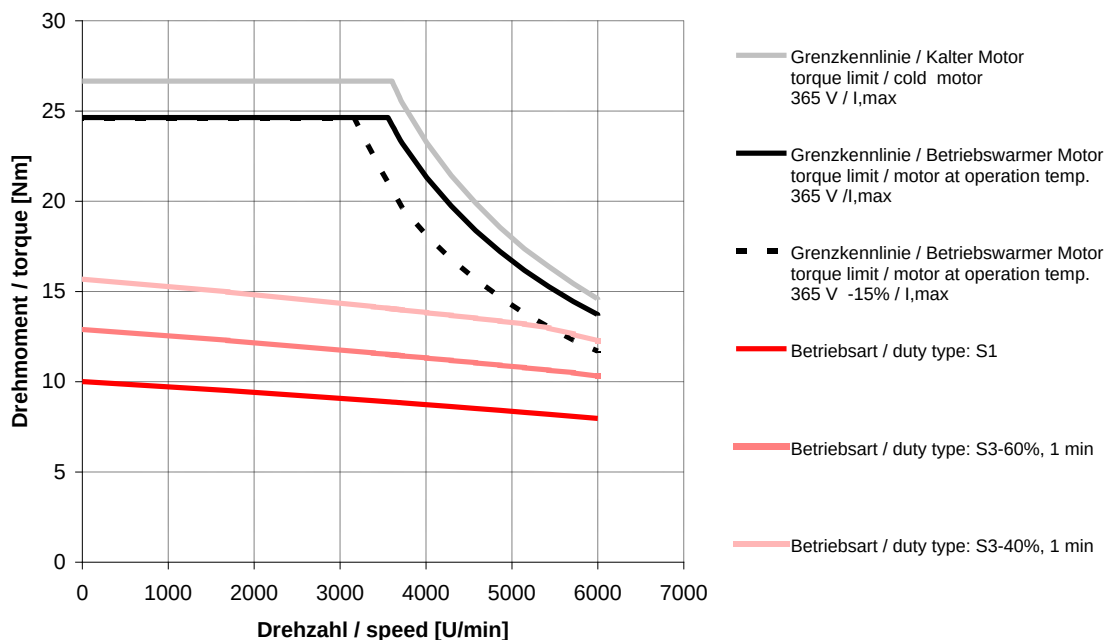
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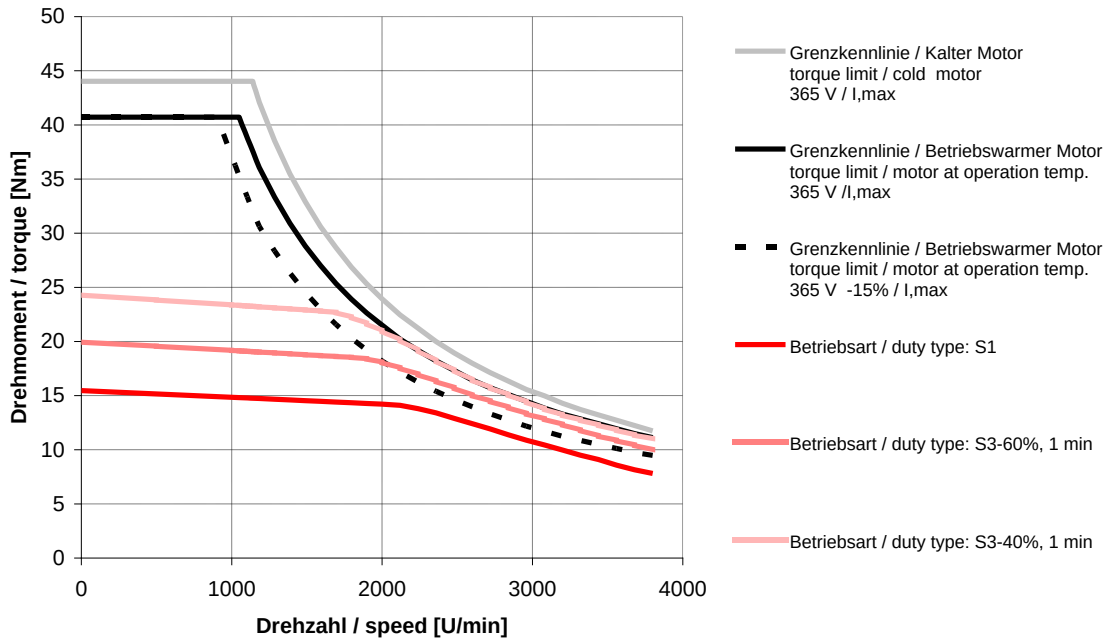
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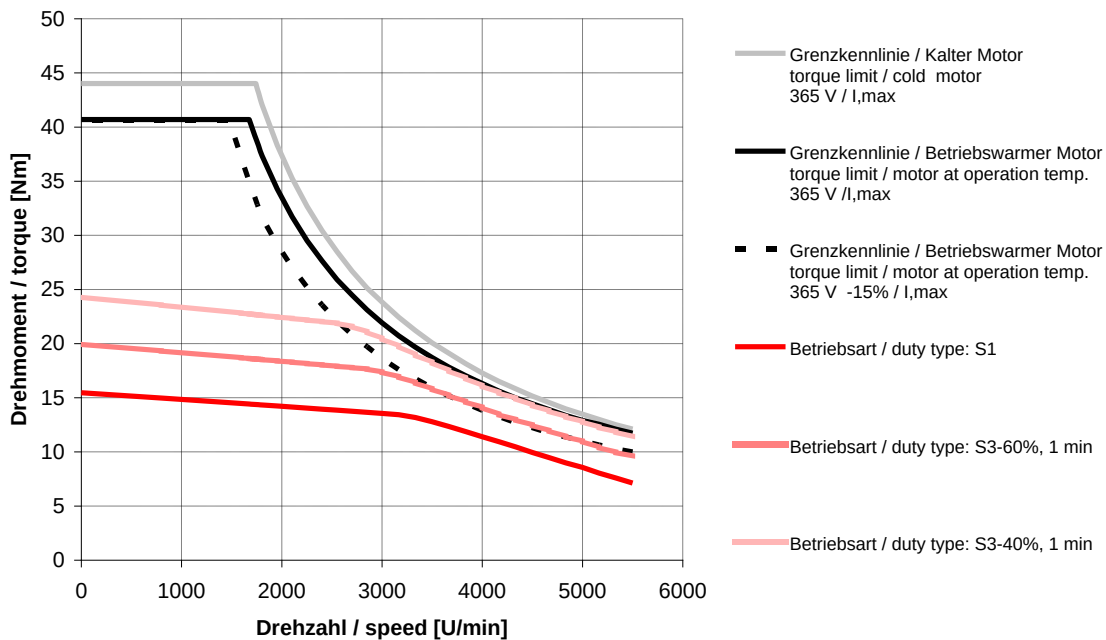
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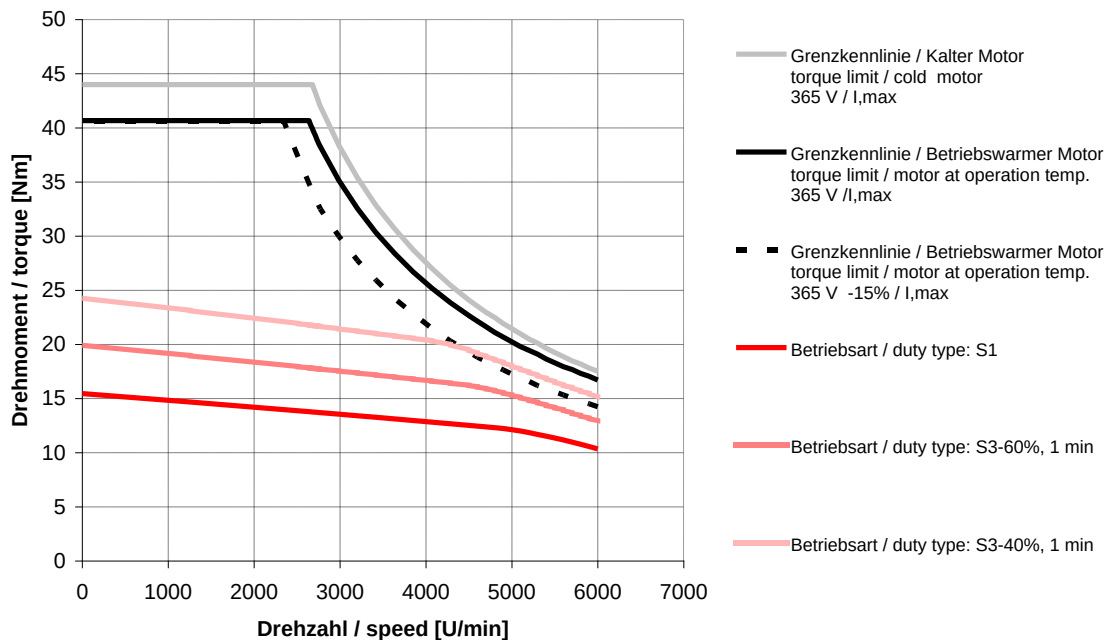
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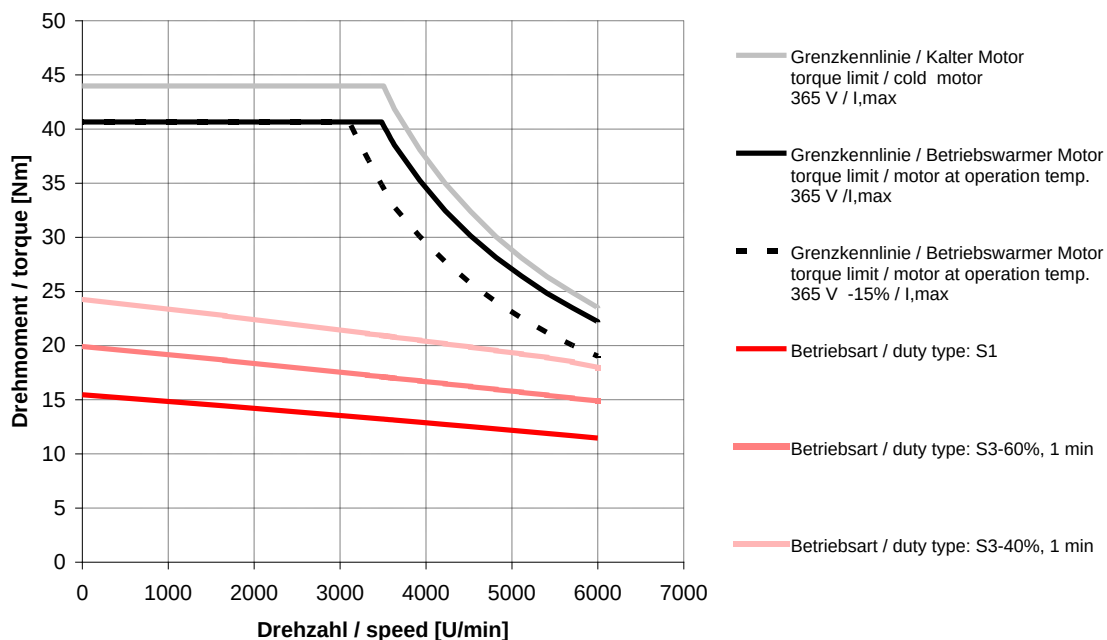
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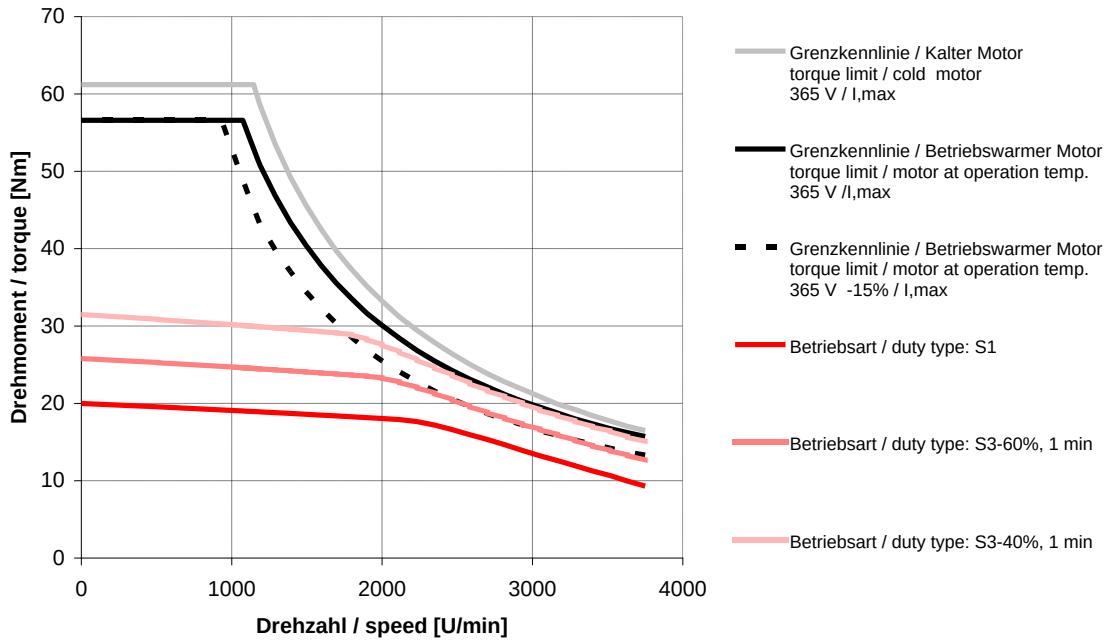
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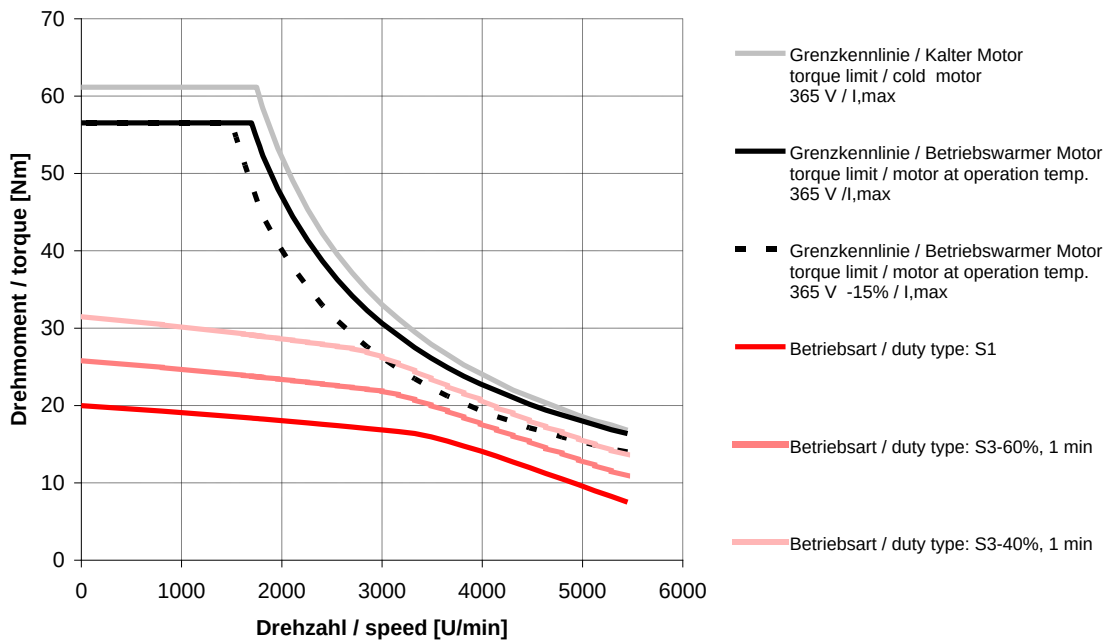
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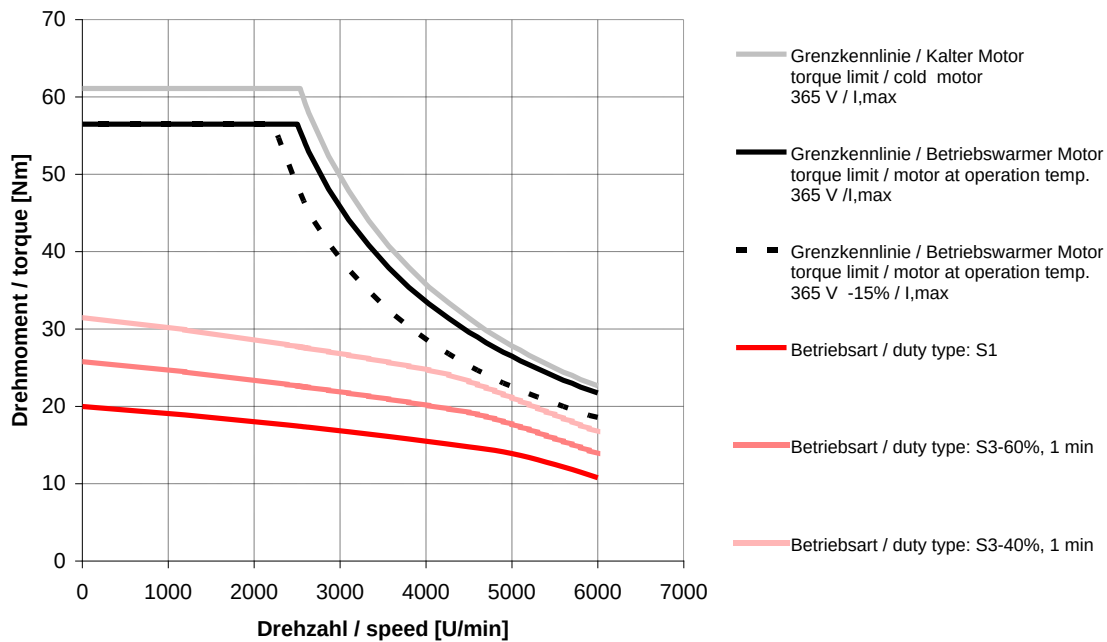
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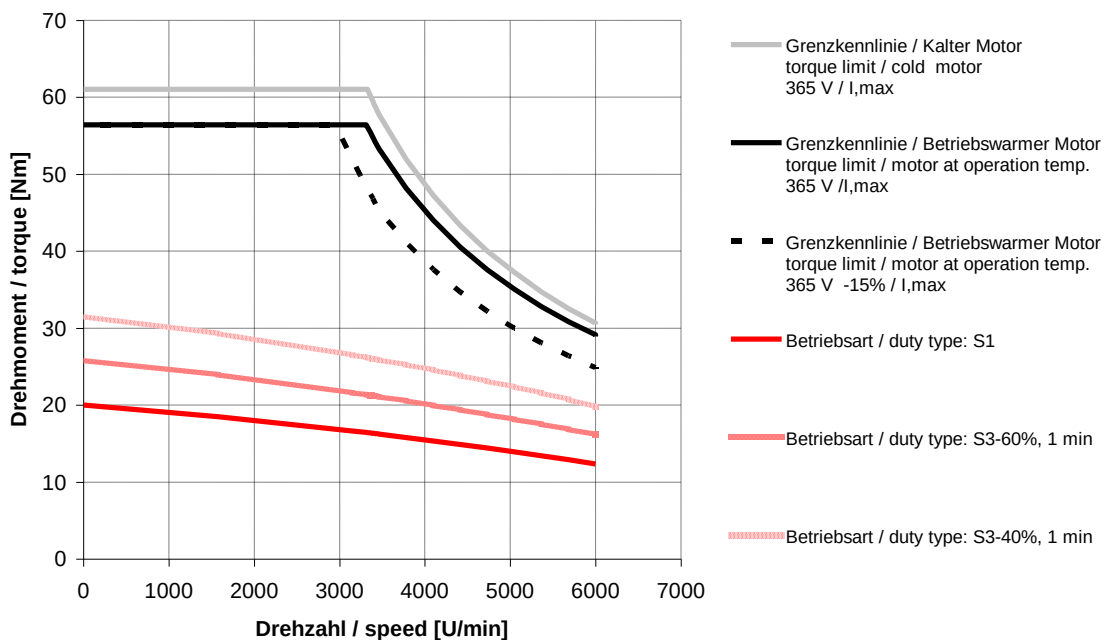
DSD2-056LO64O-30-54



DSD2-056LO64O-45-54

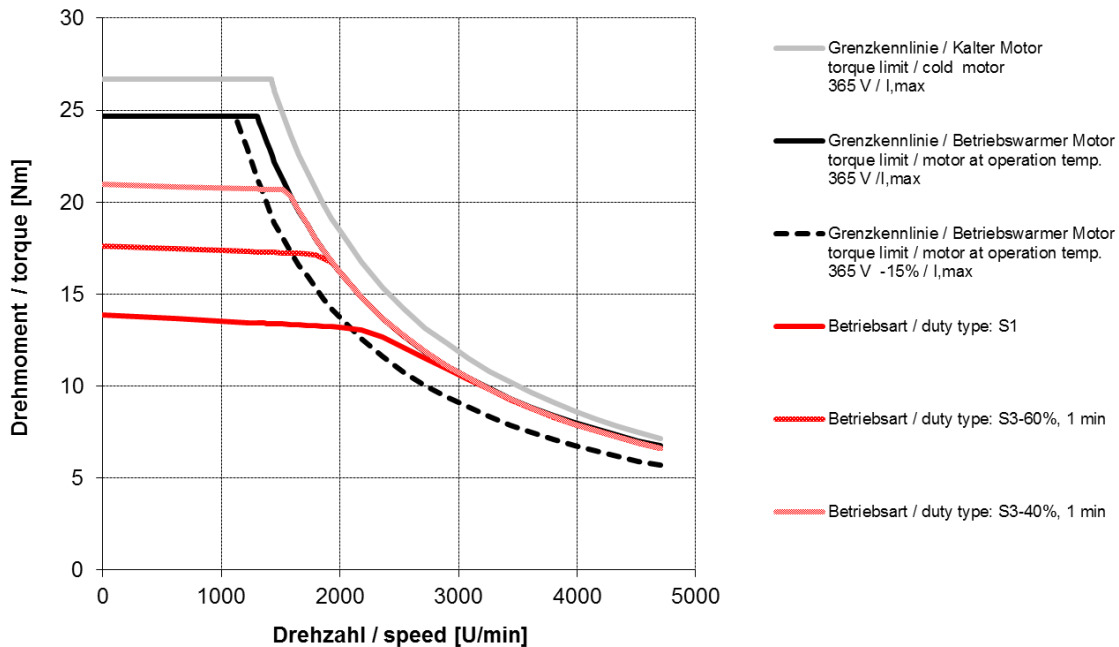


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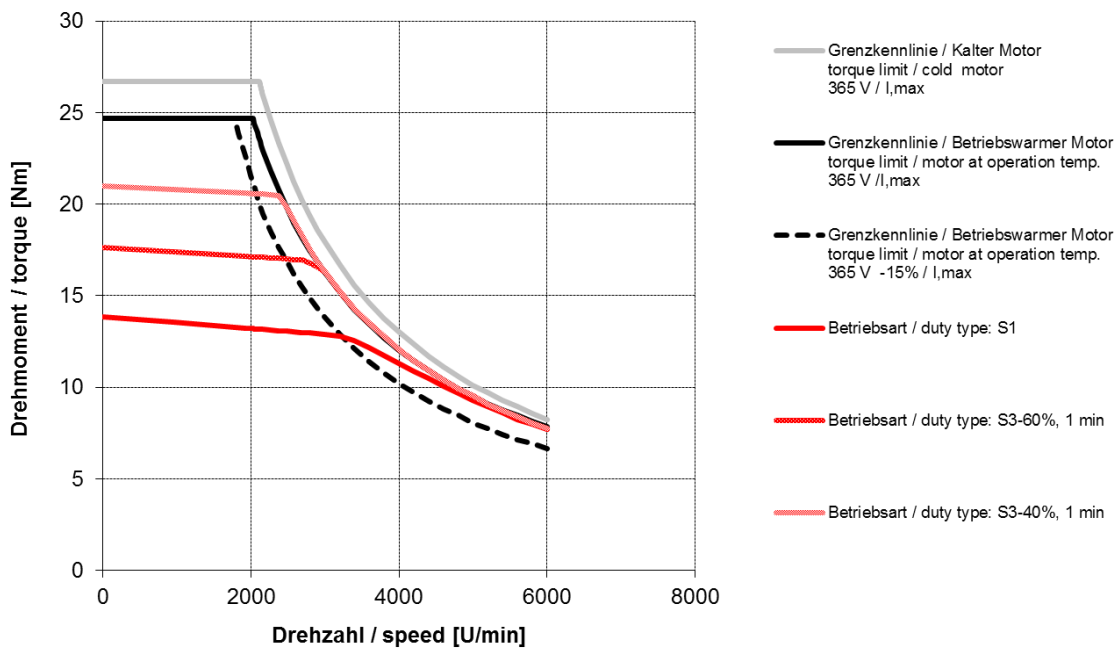


5.4.3. DSD2-056..64W-.. (water cooled)

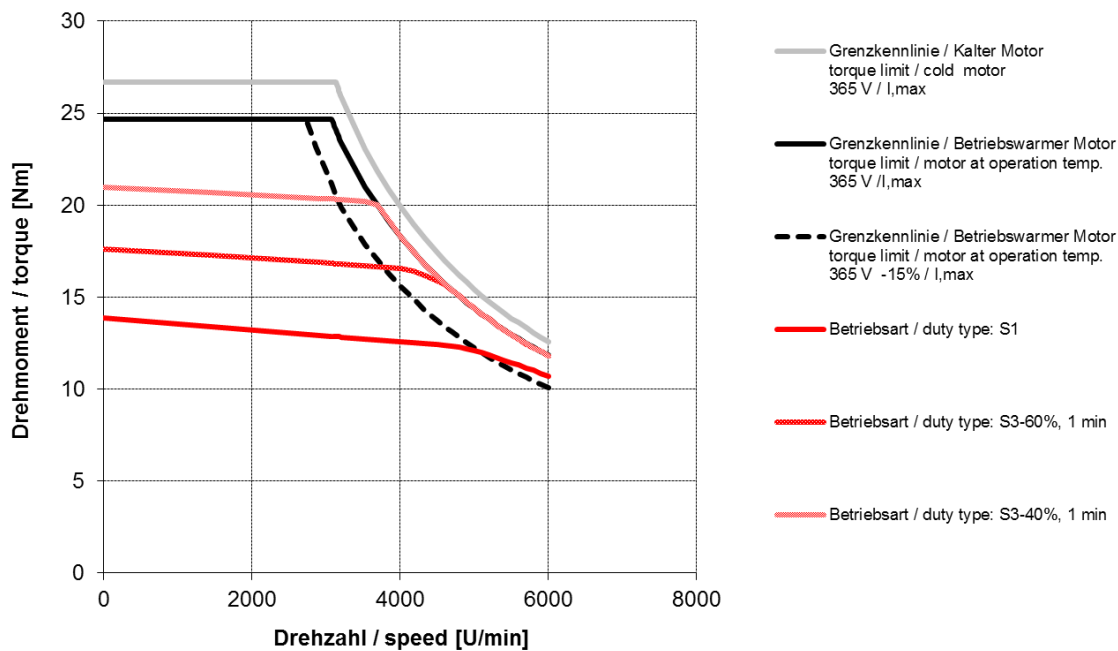
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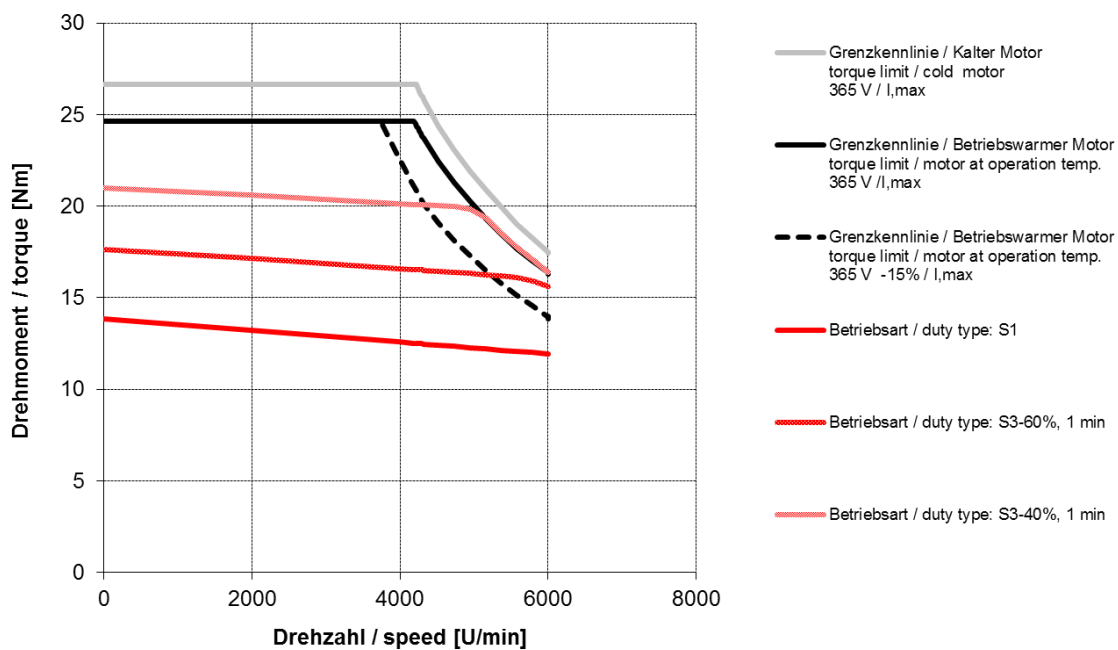
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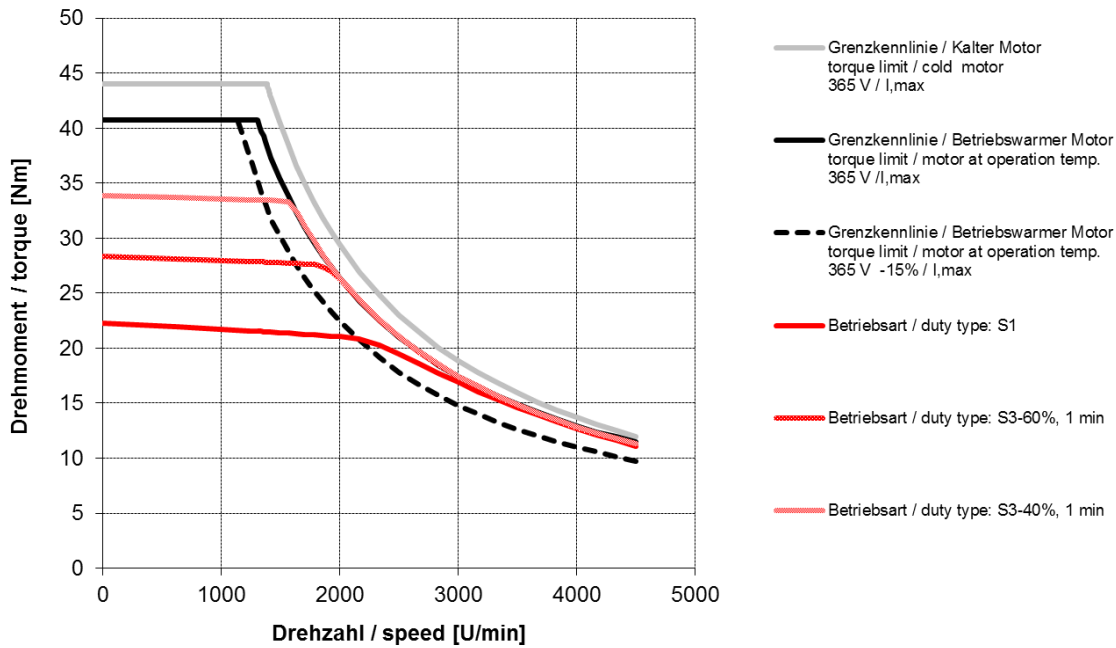
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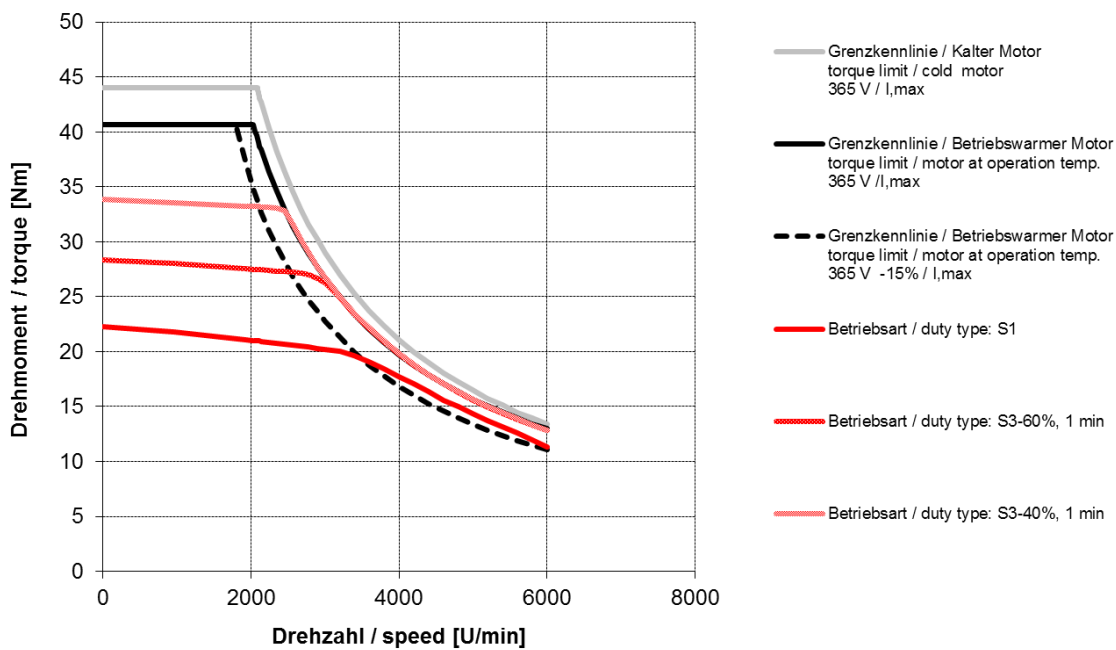
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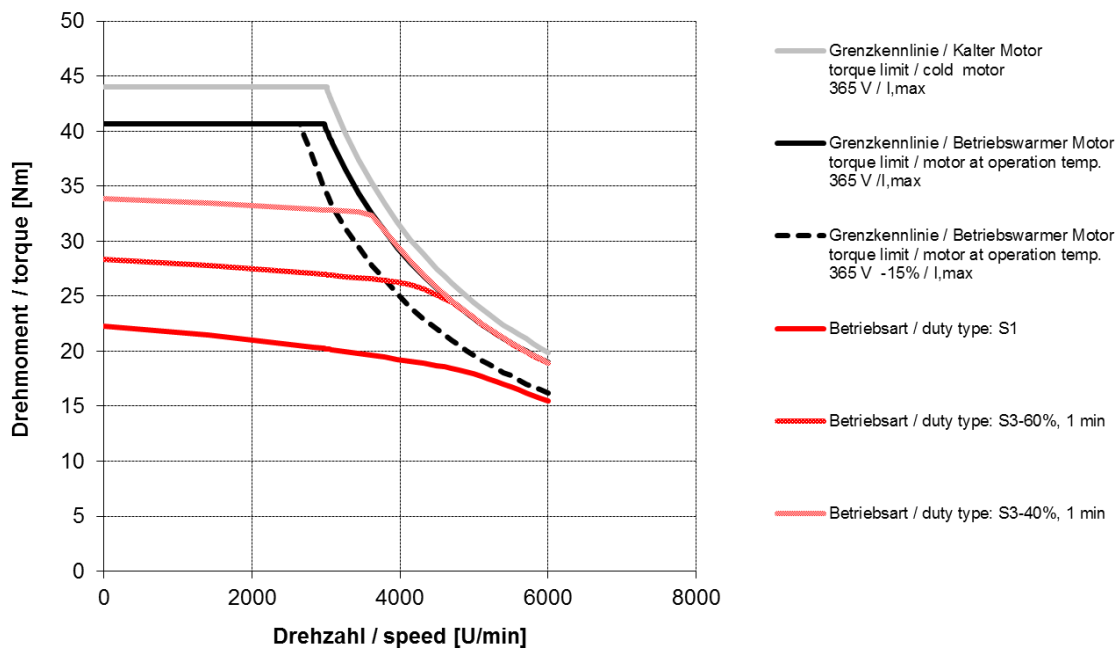
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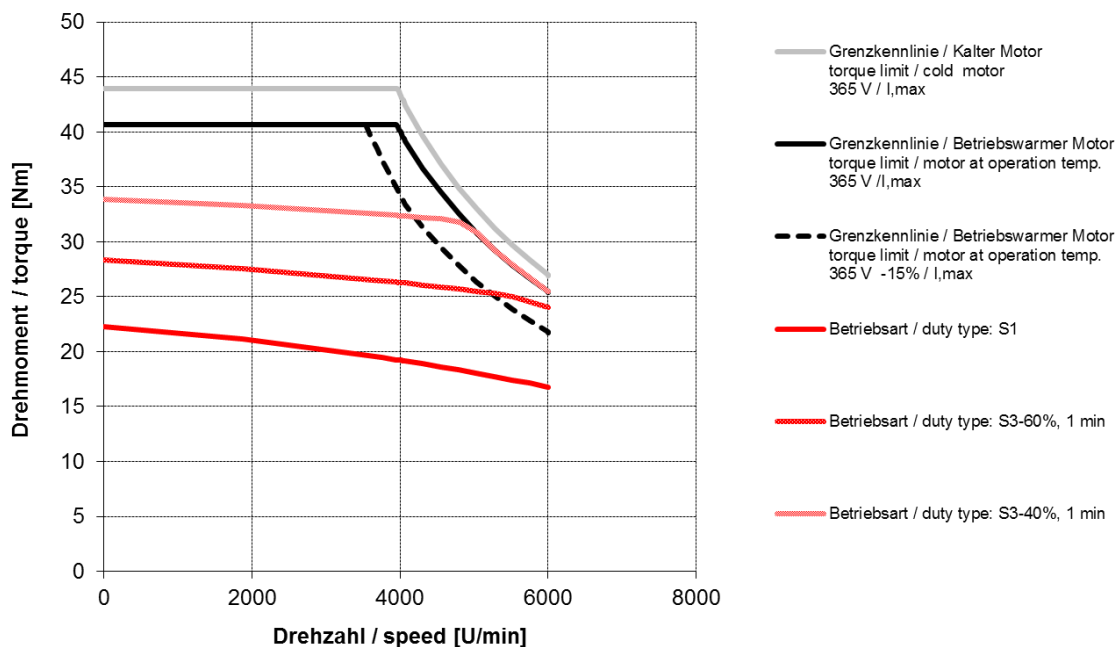
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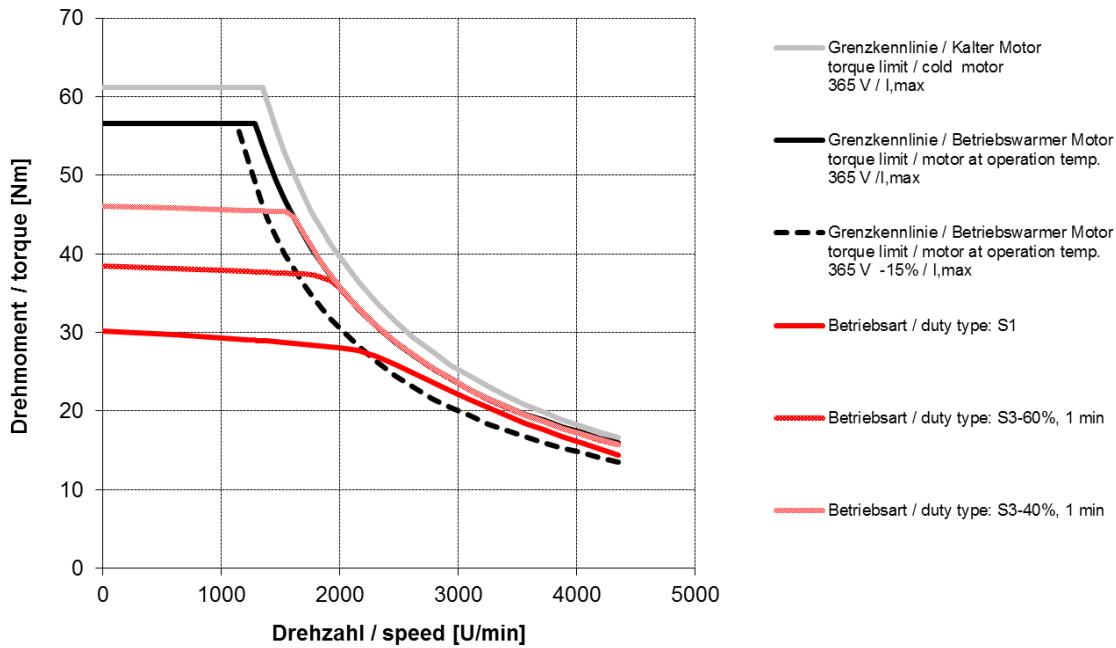
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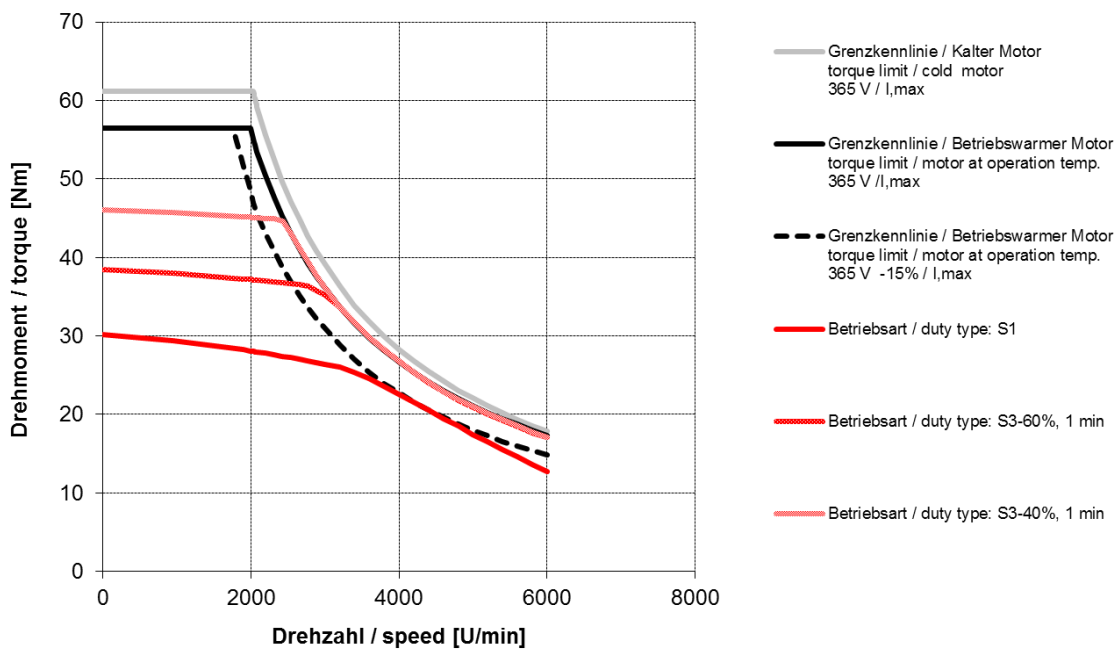
DSD2-056MO64W-60-54



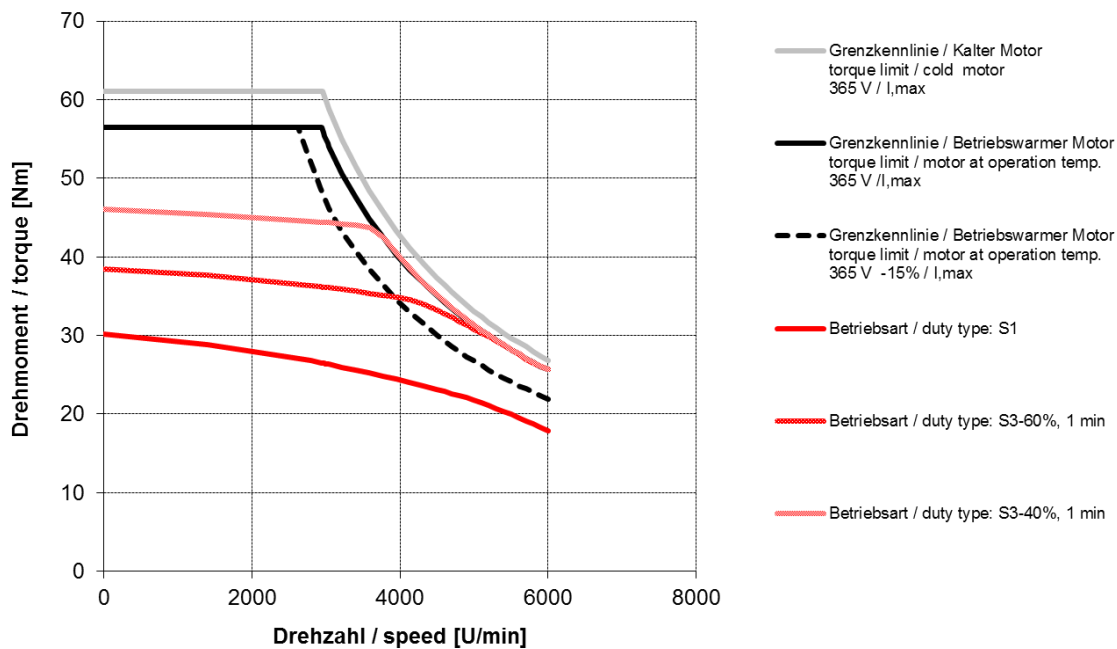
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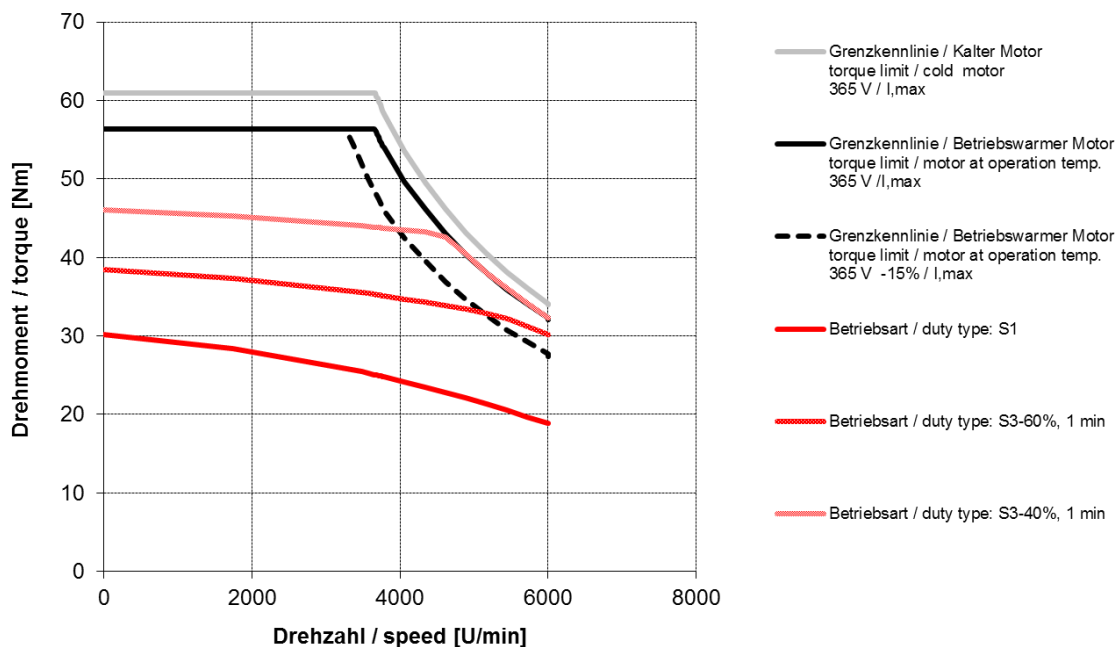
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DSD2-056LO64W-45-54



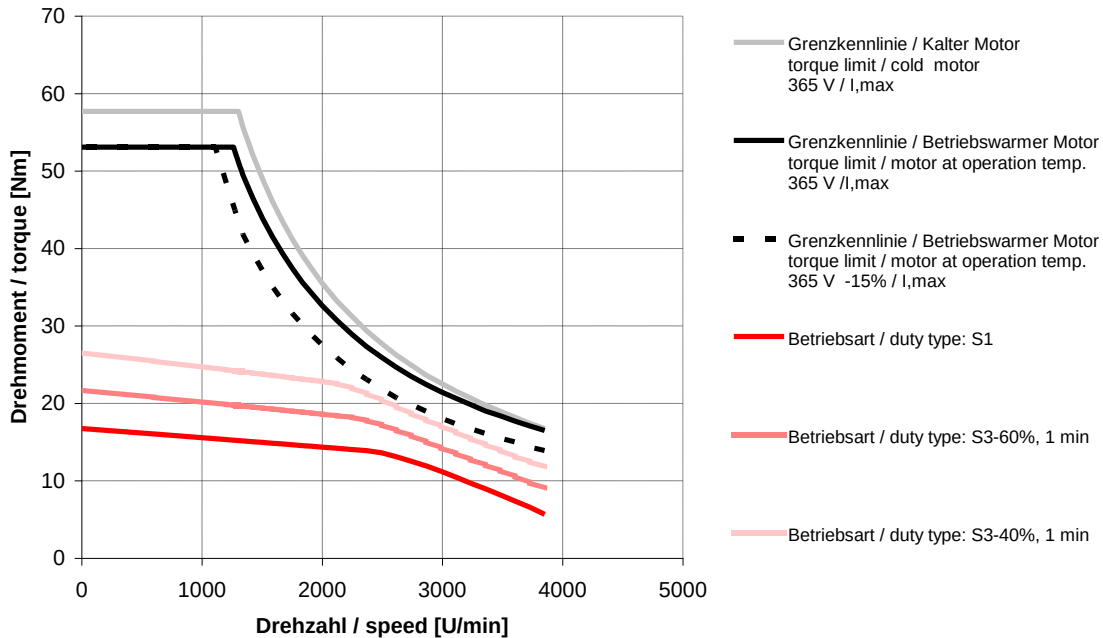
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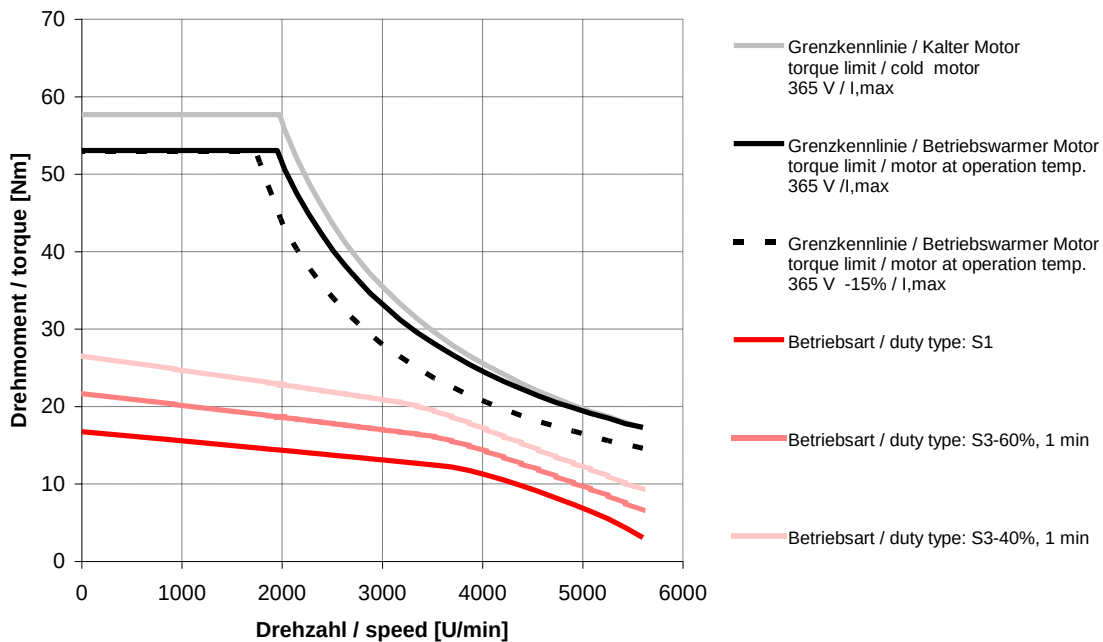
5.5. Motor characteristic curve DSD2-071

5.5.1. DSD2-071..64U-.. (without fan)

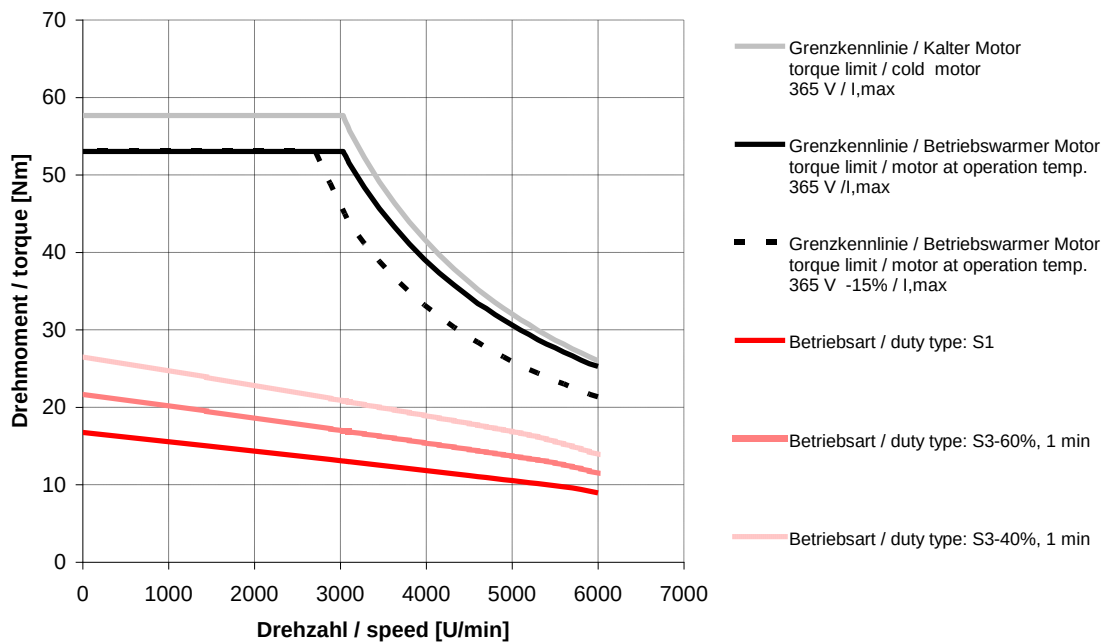
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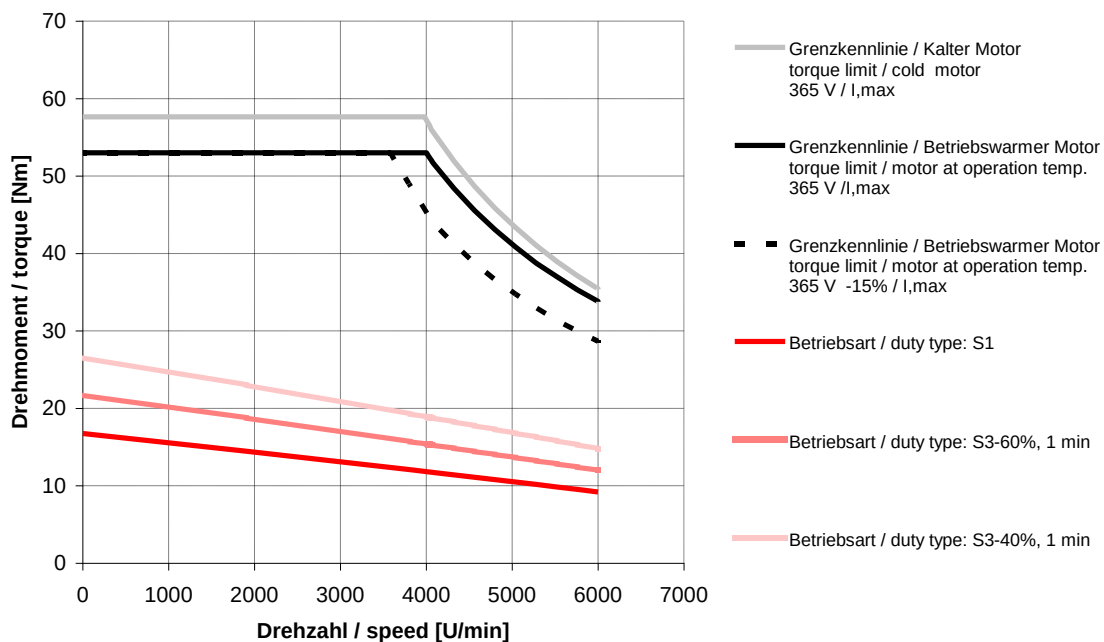
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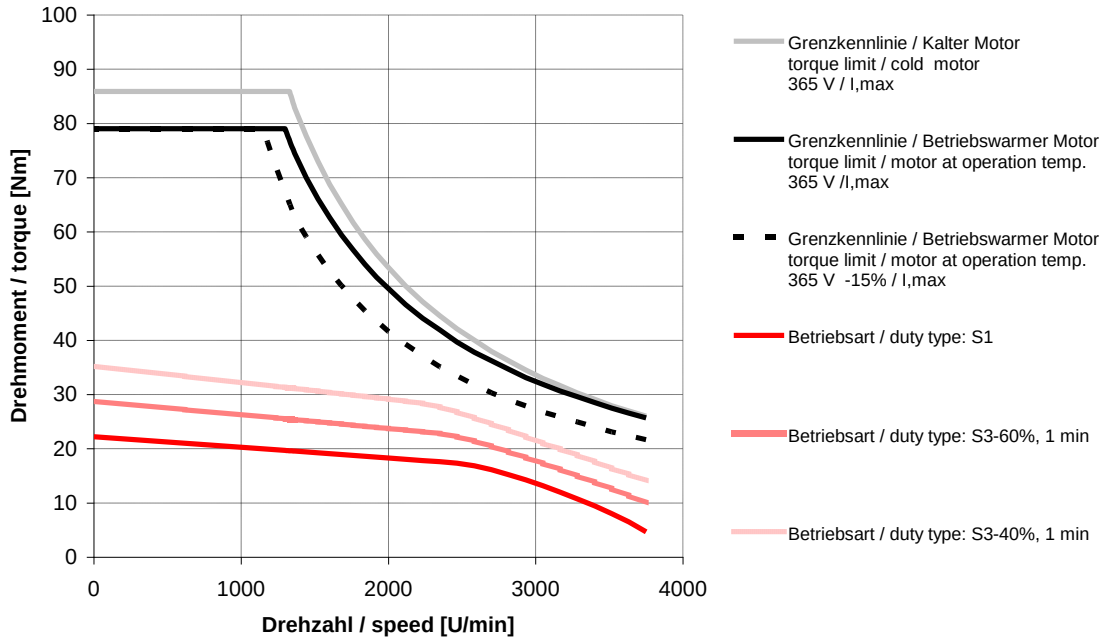
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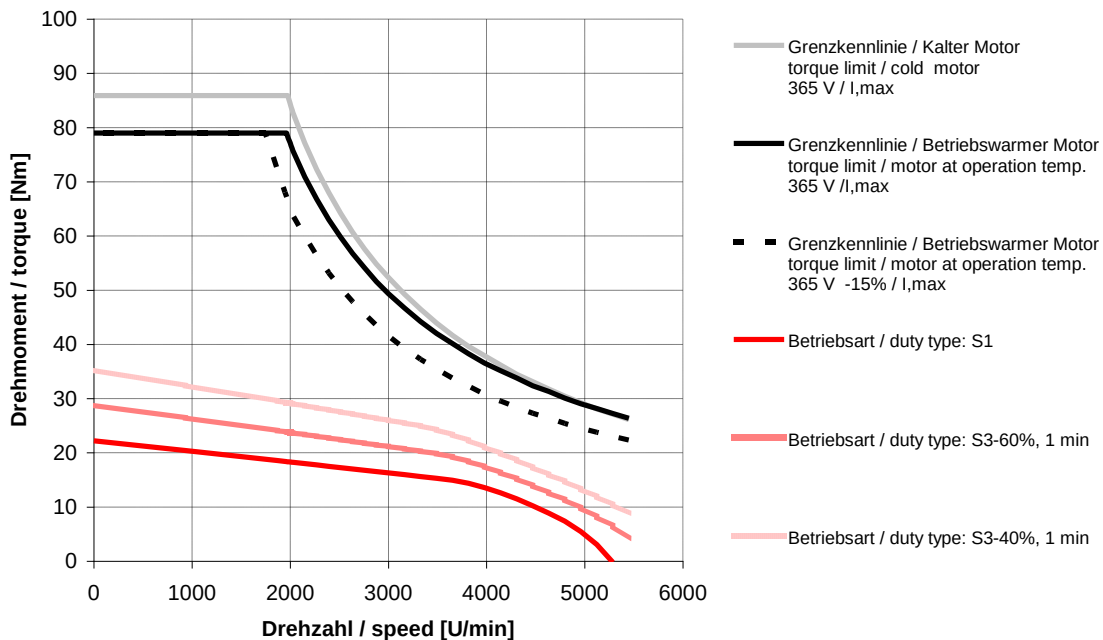
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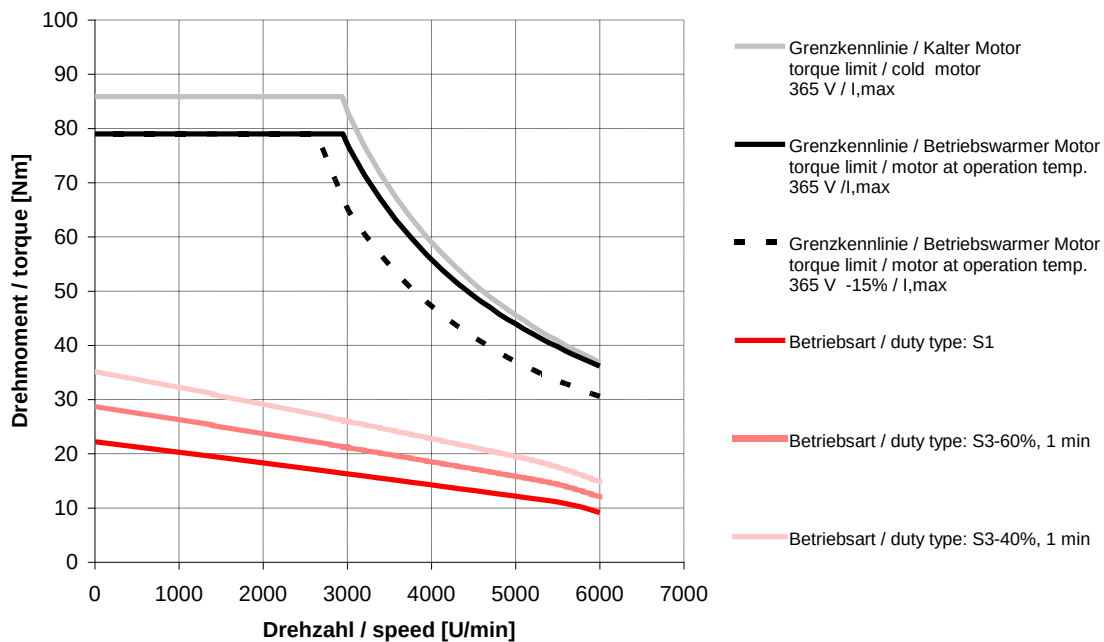
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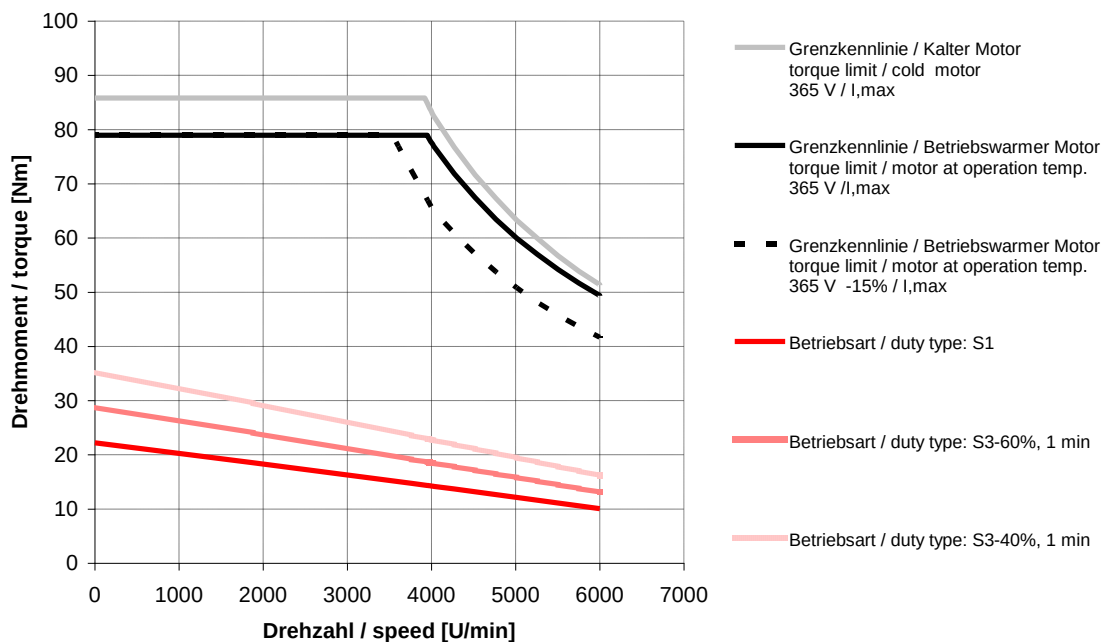
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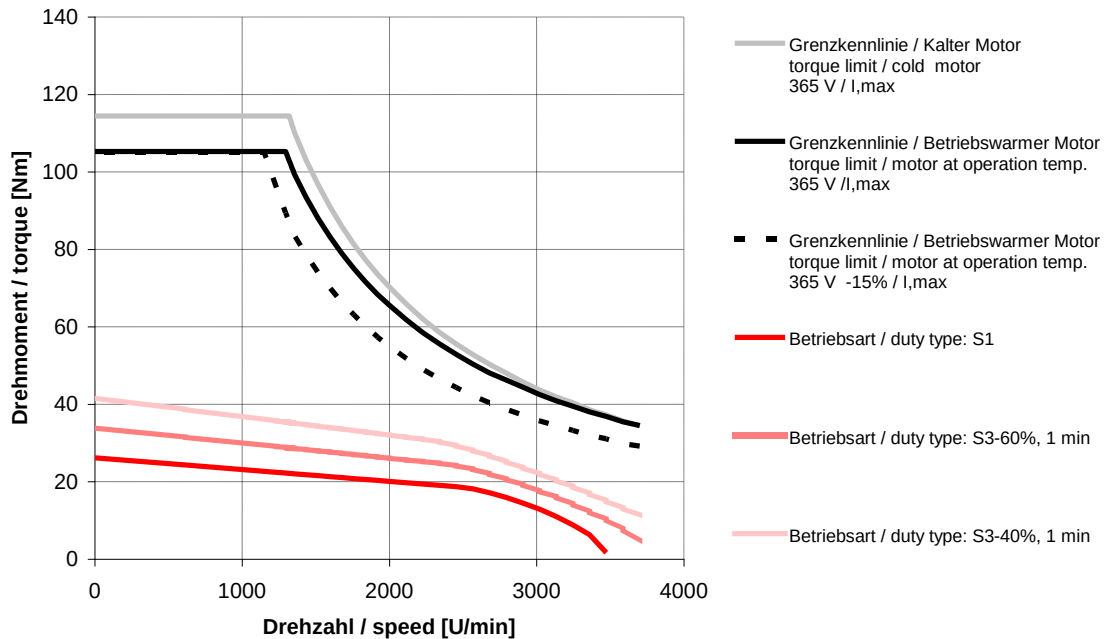
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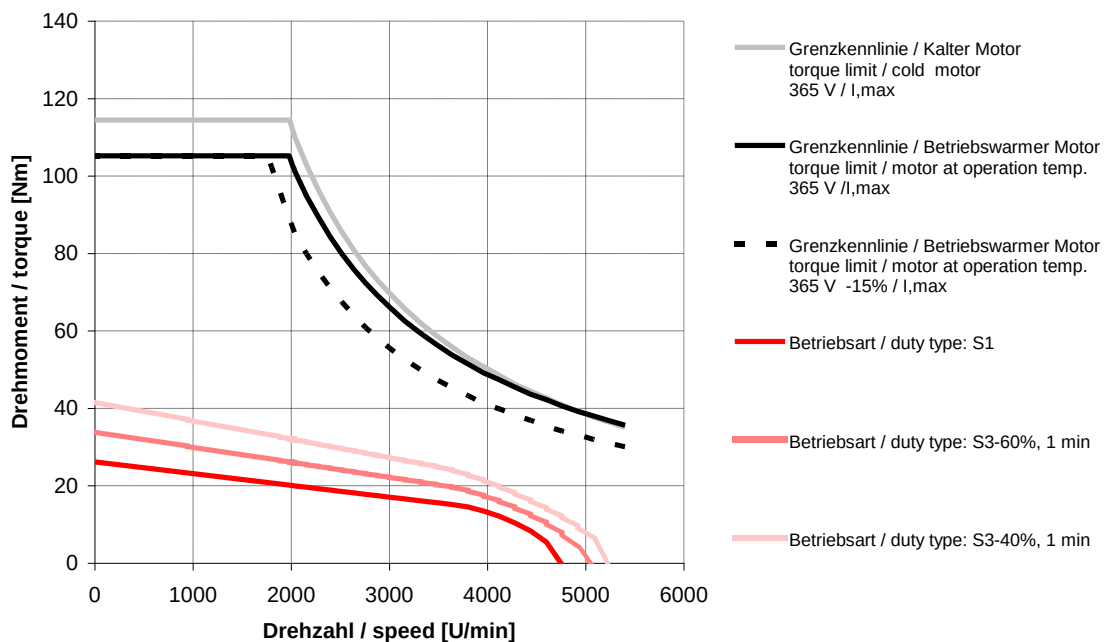
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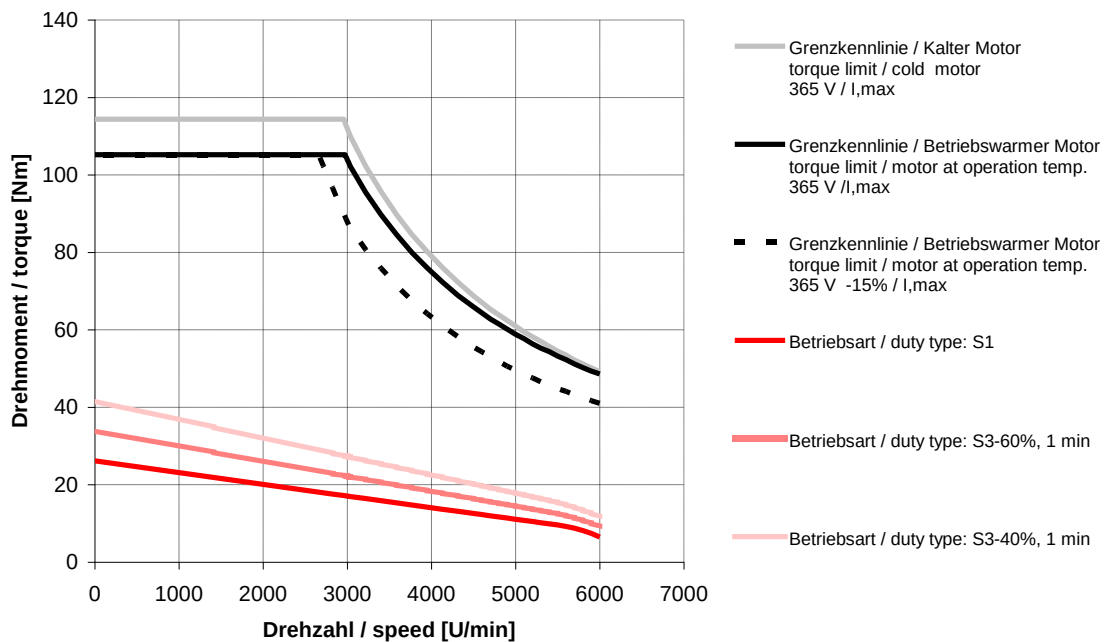
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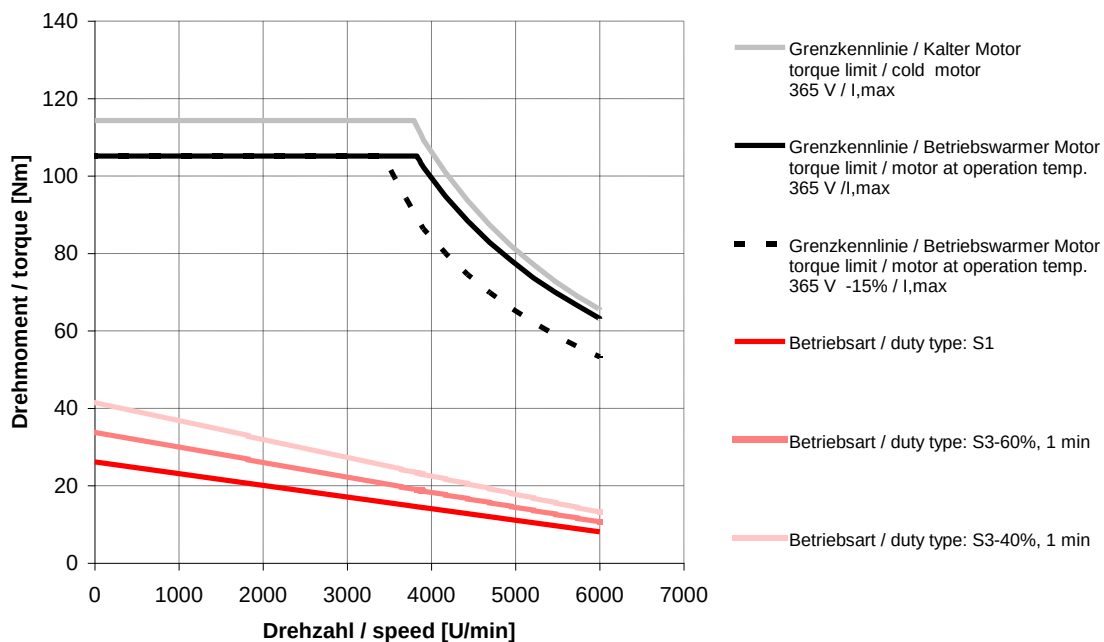
DSD2-071LO64U-30-54



DSD2-071LO64U-45-54

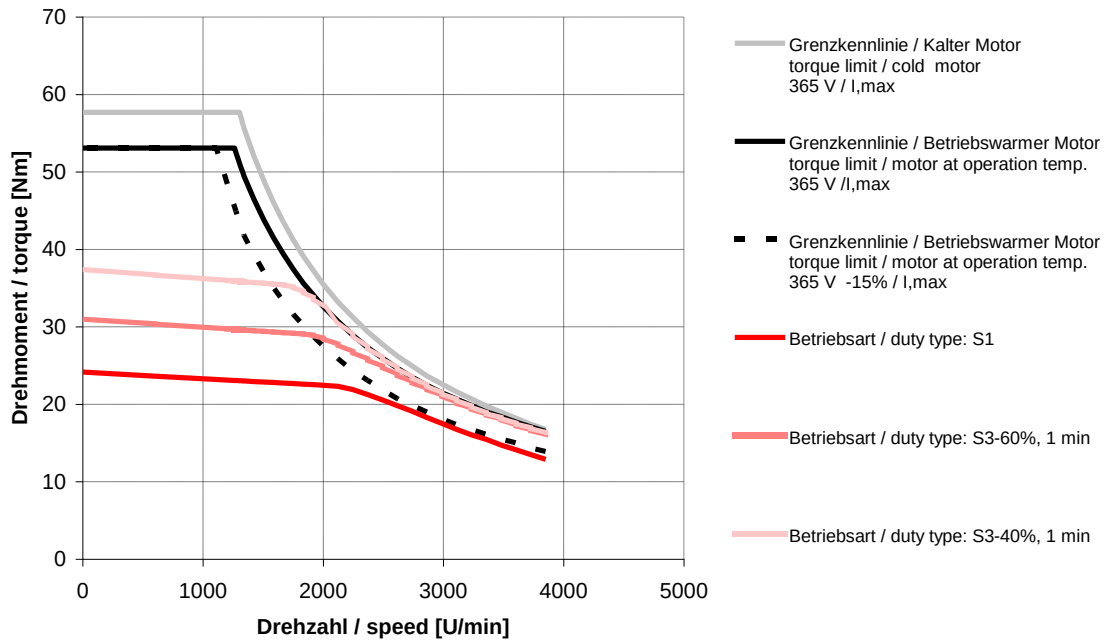


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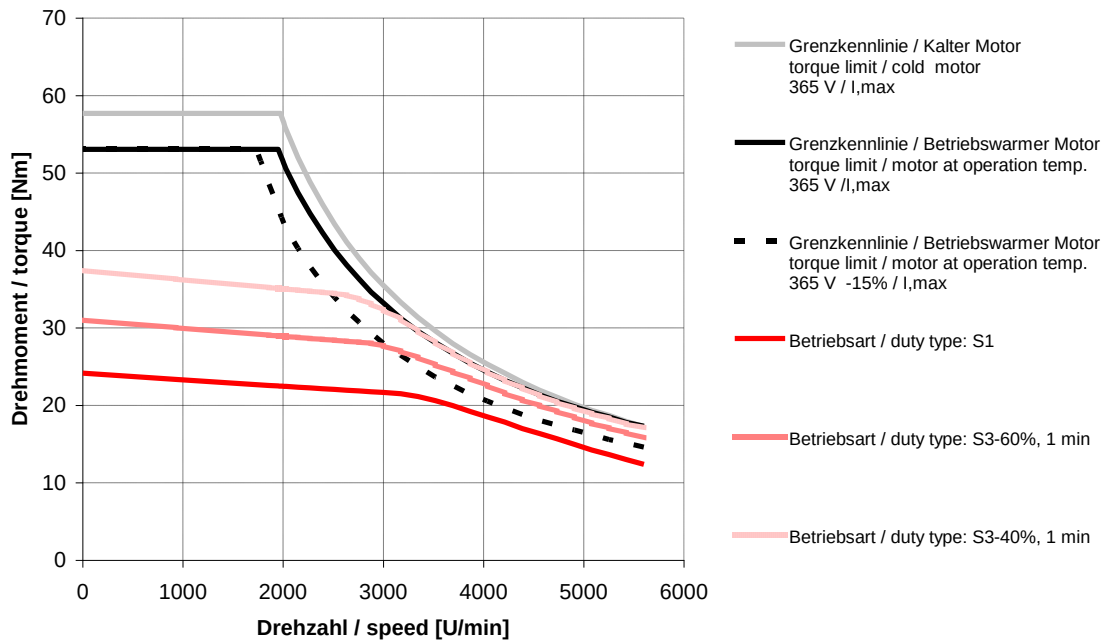


5.5.2. DSD2-071..640-.. (with fan)

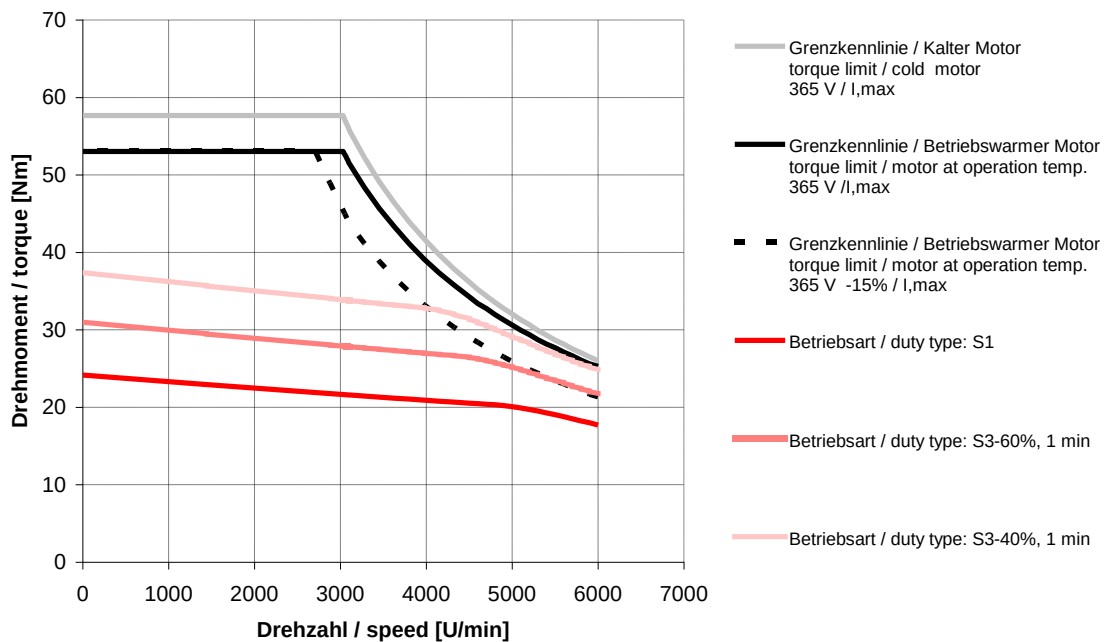
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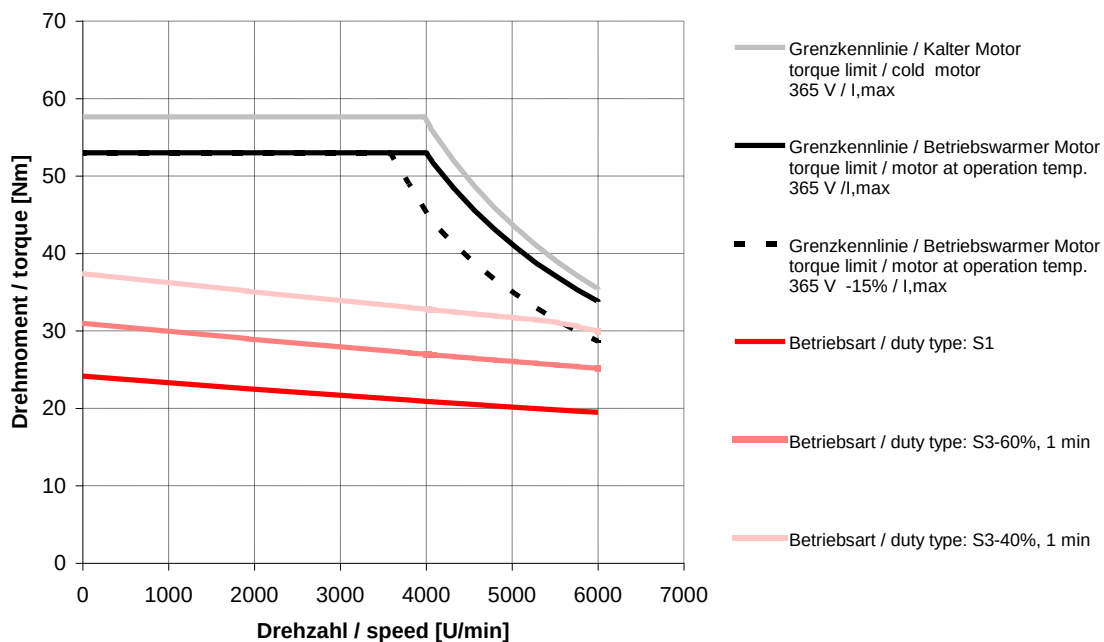
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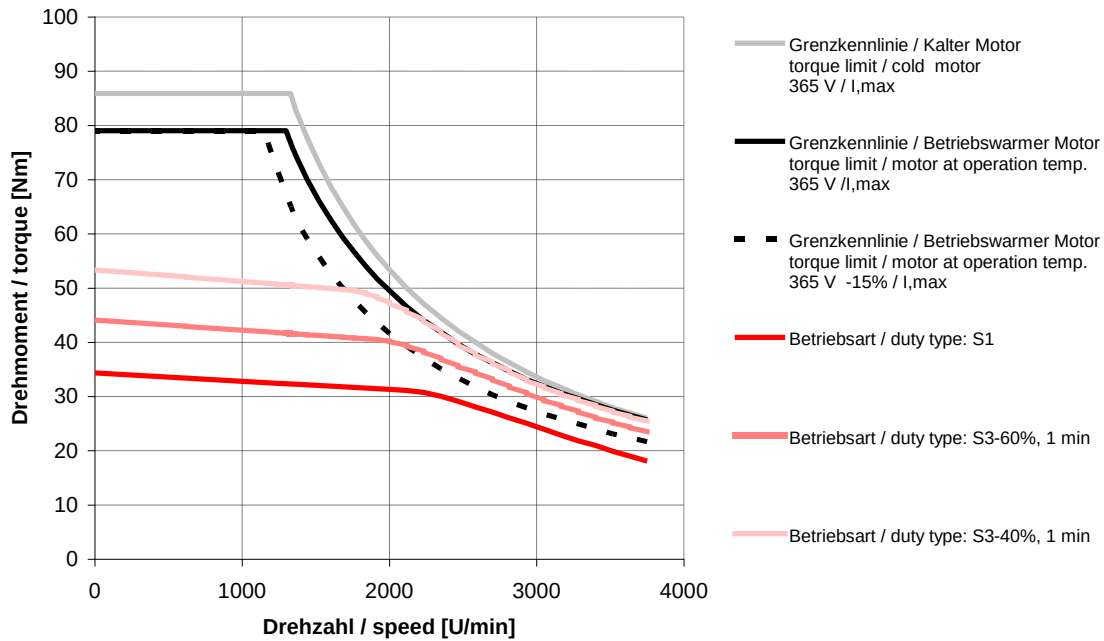
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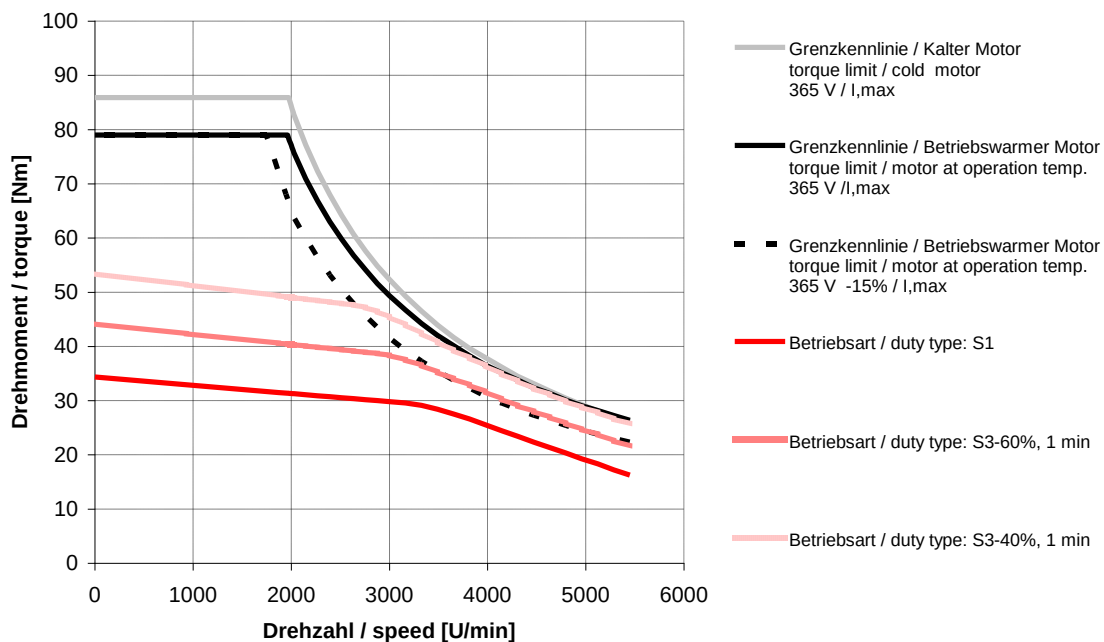
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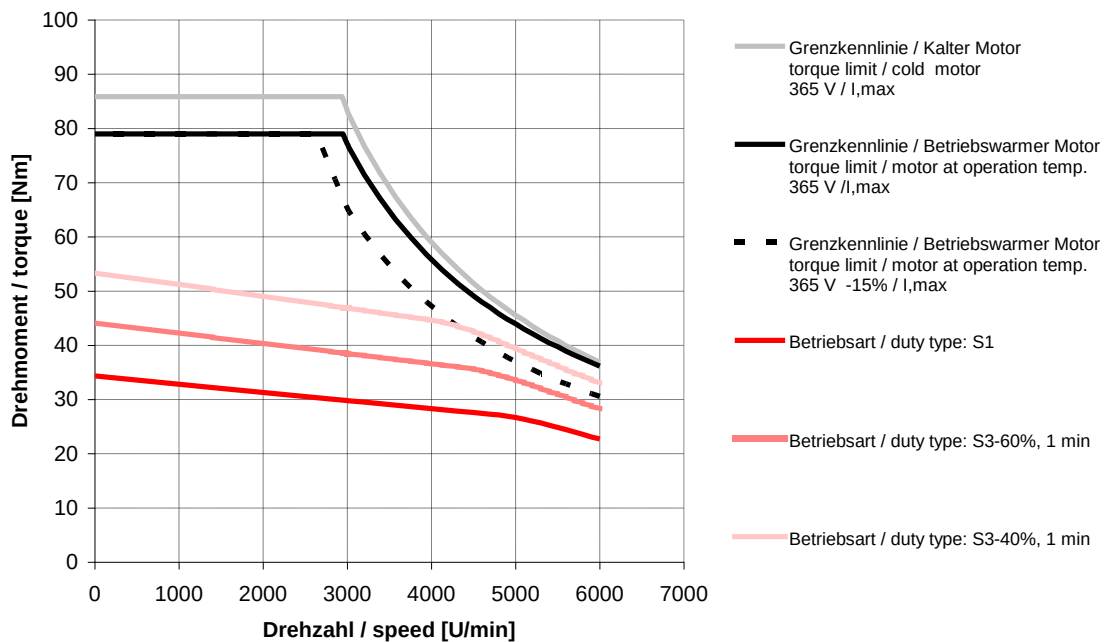
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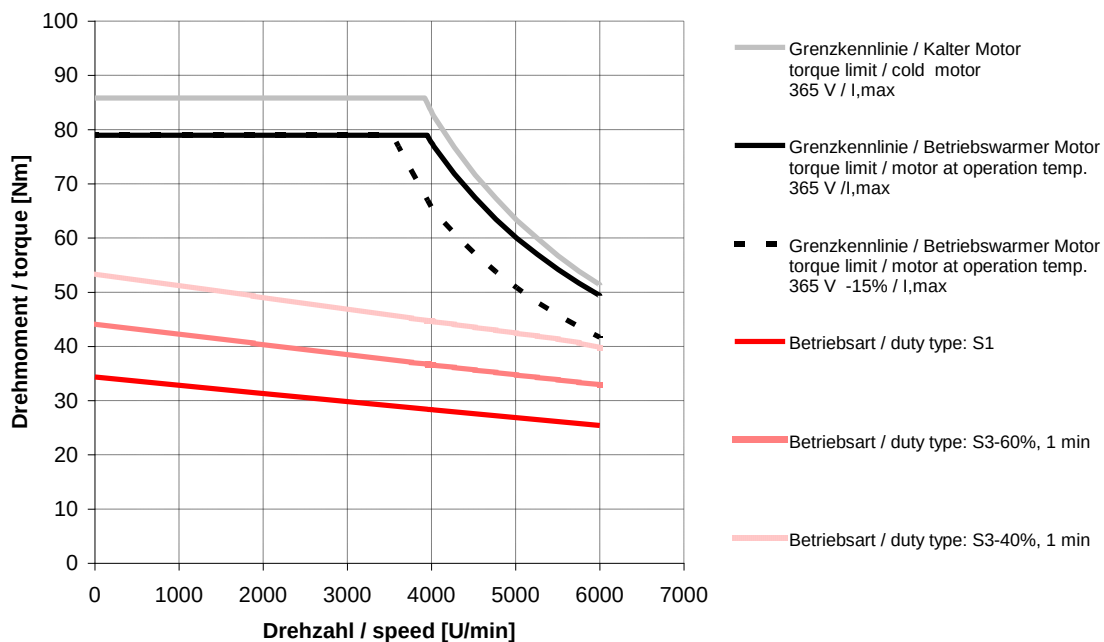
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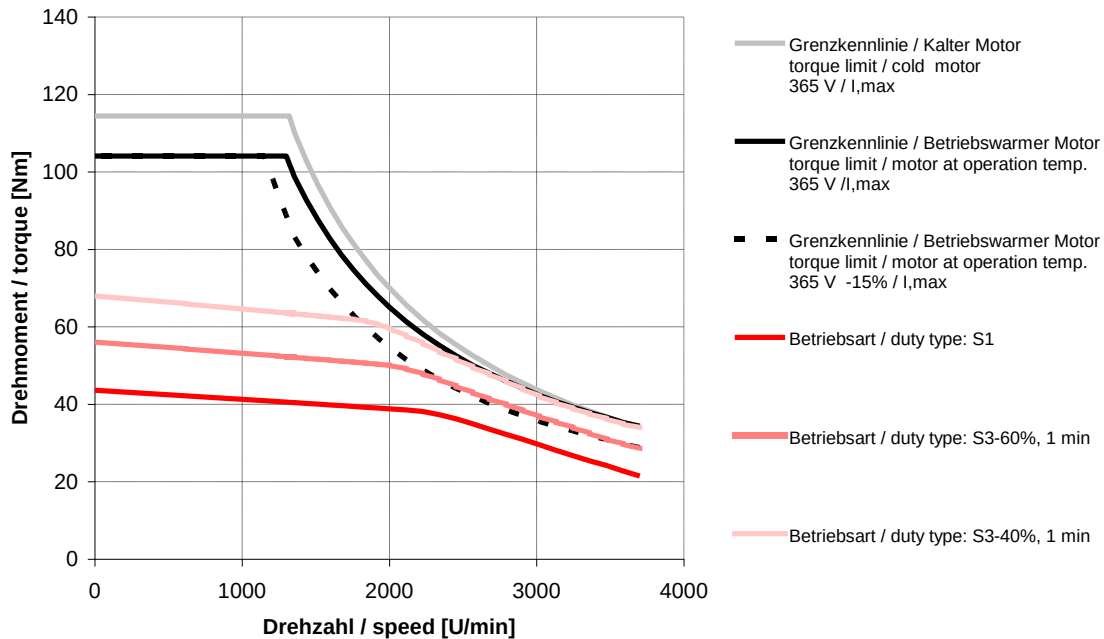
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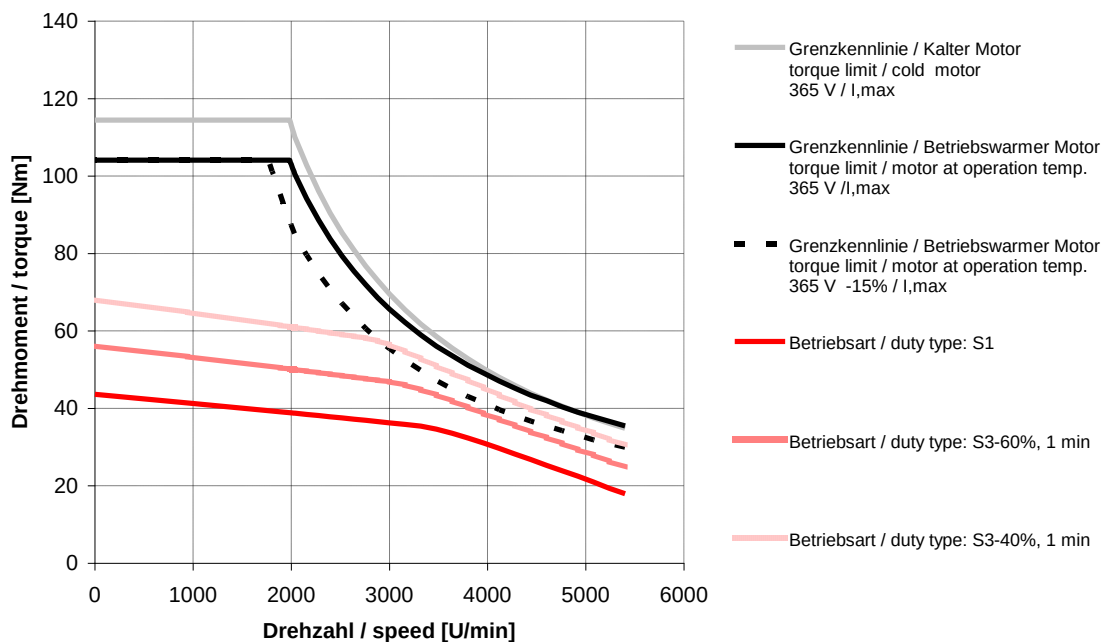
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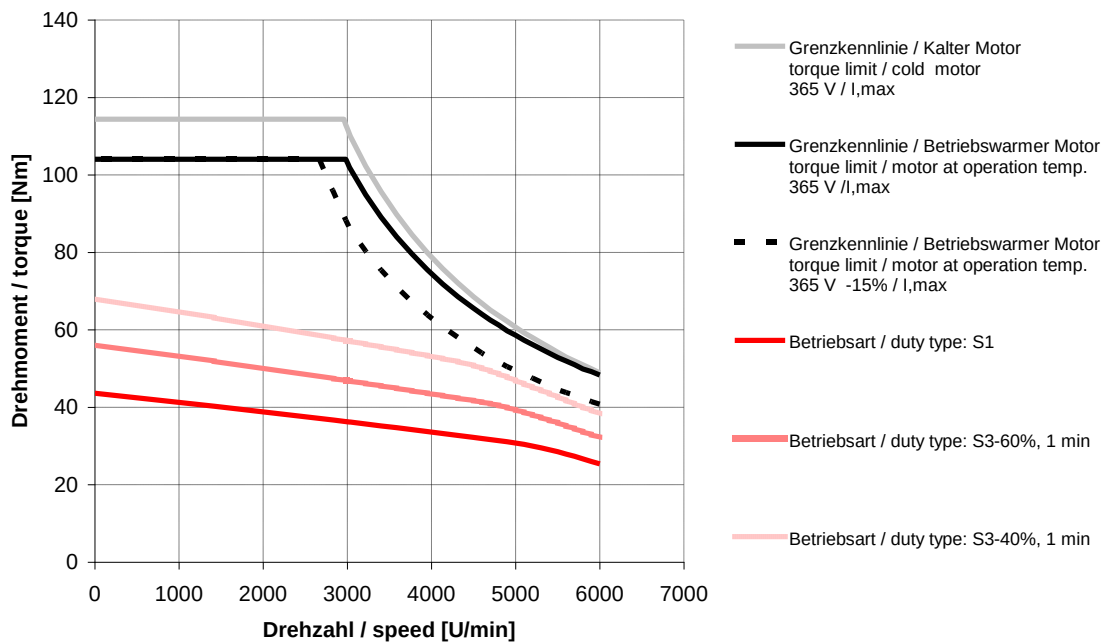
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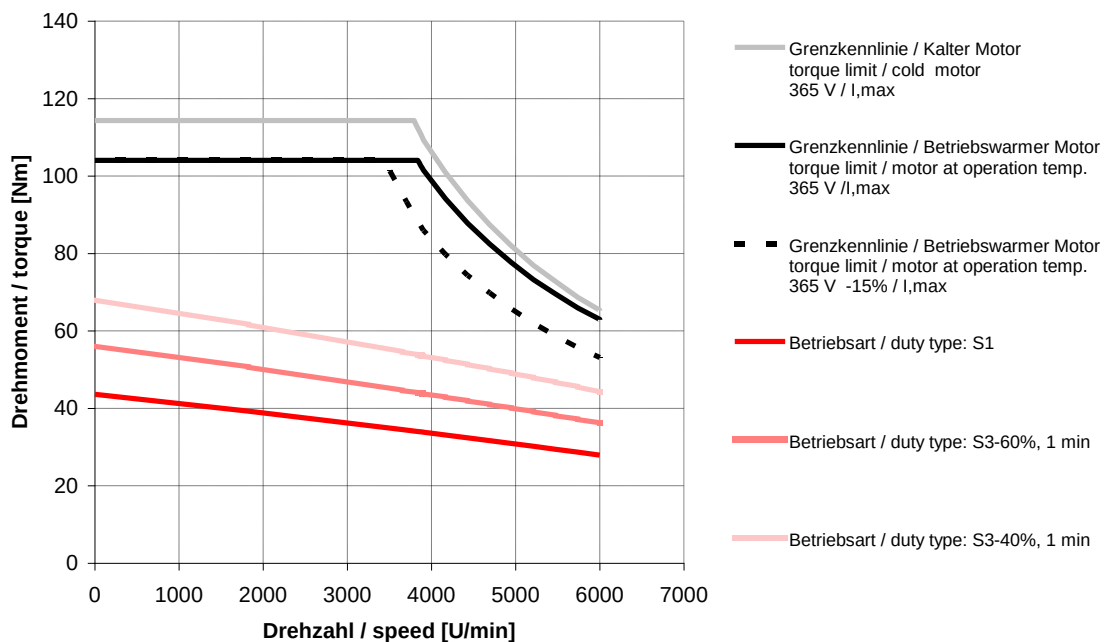
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DSD2-071LO64O-45-54

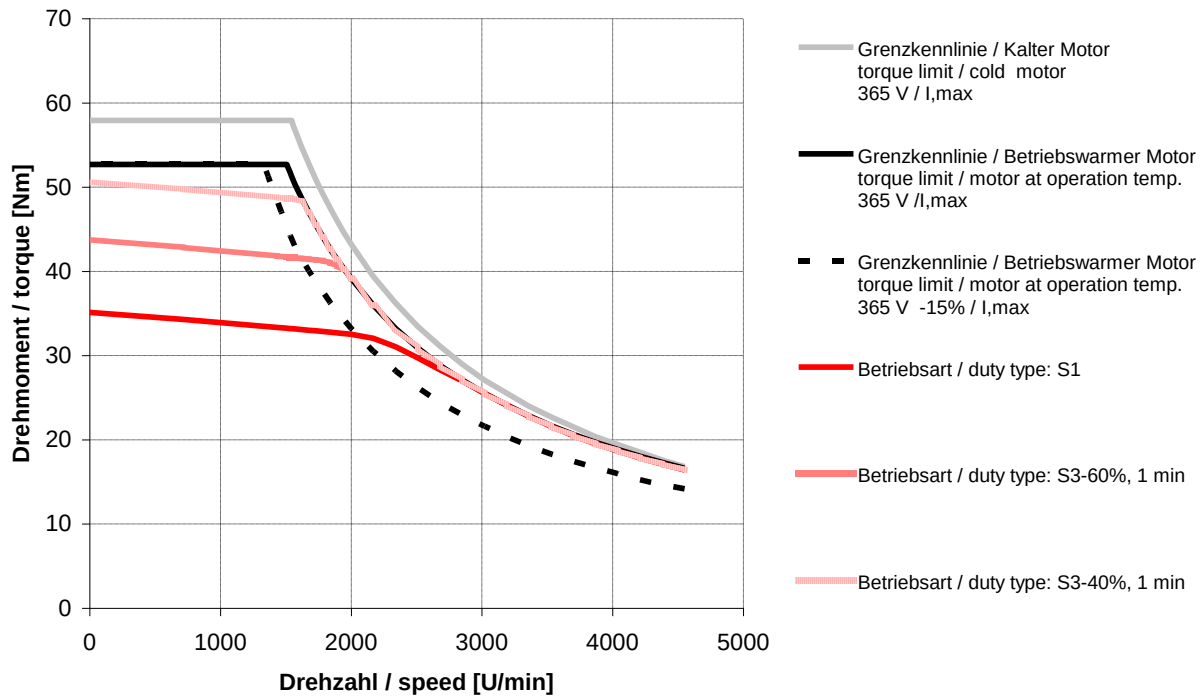


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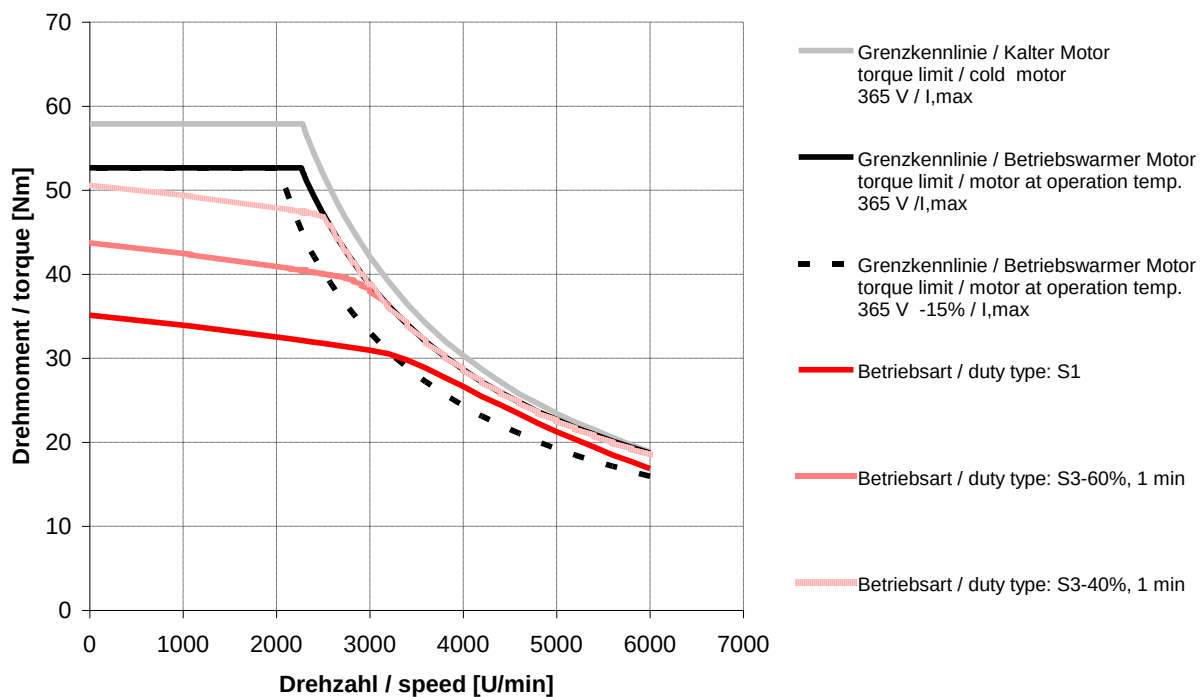


5.5.3. DSD2-071..64W-.. (water cooled)

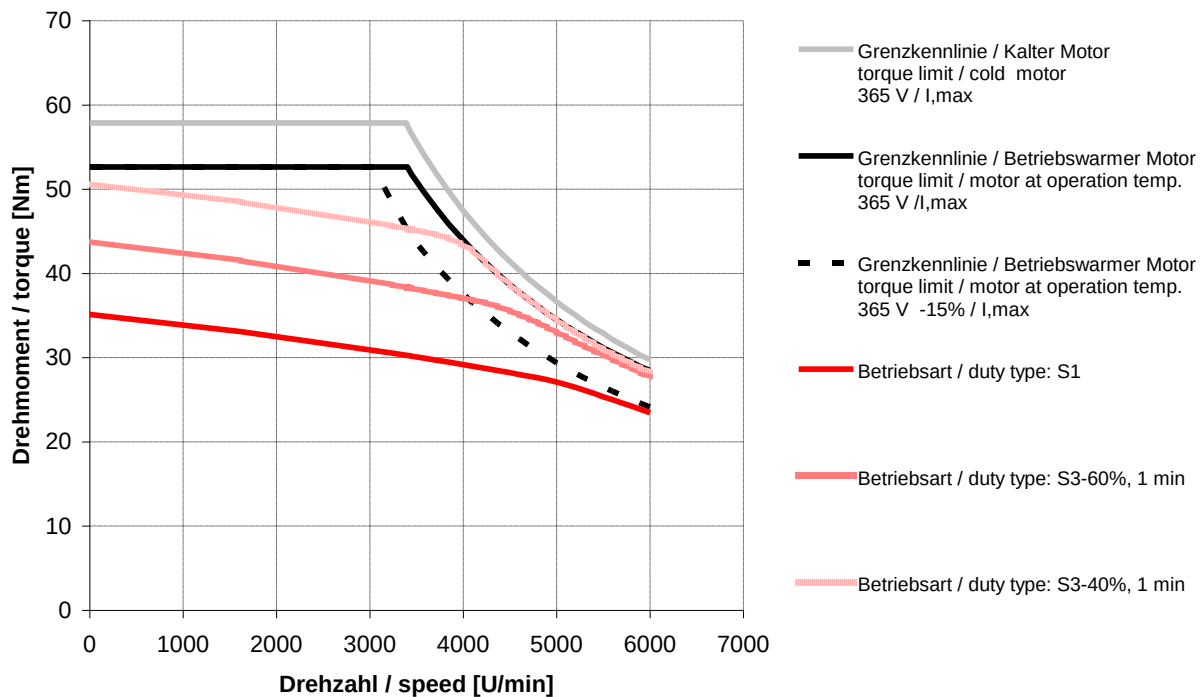
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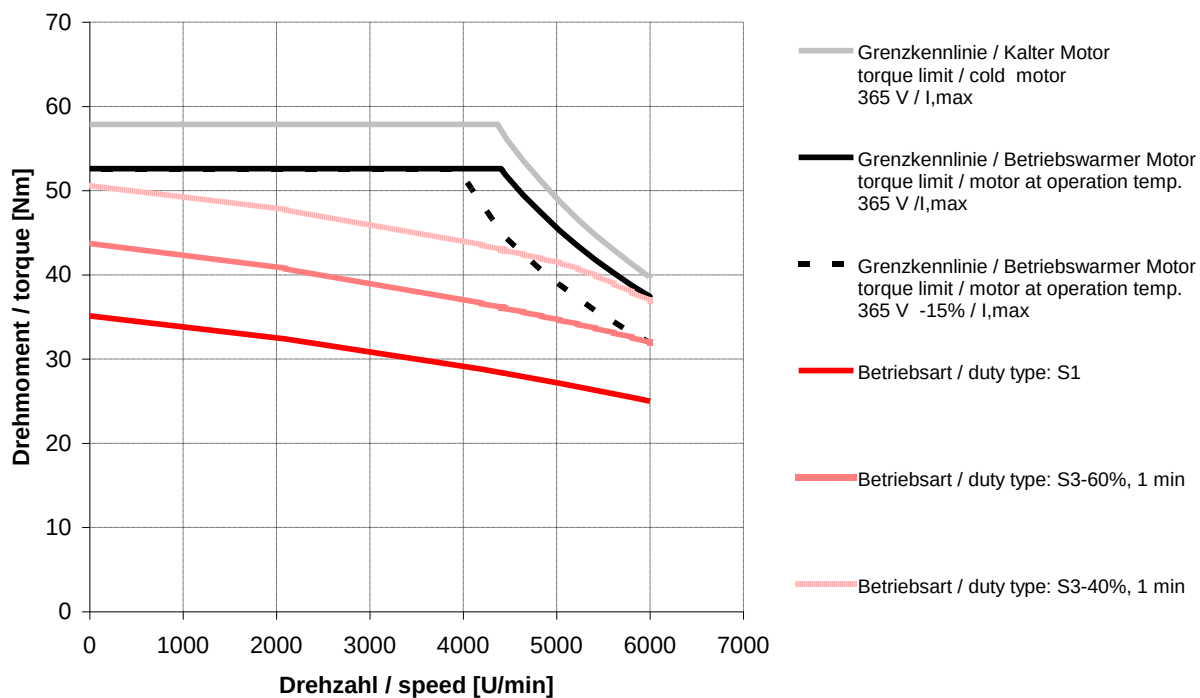
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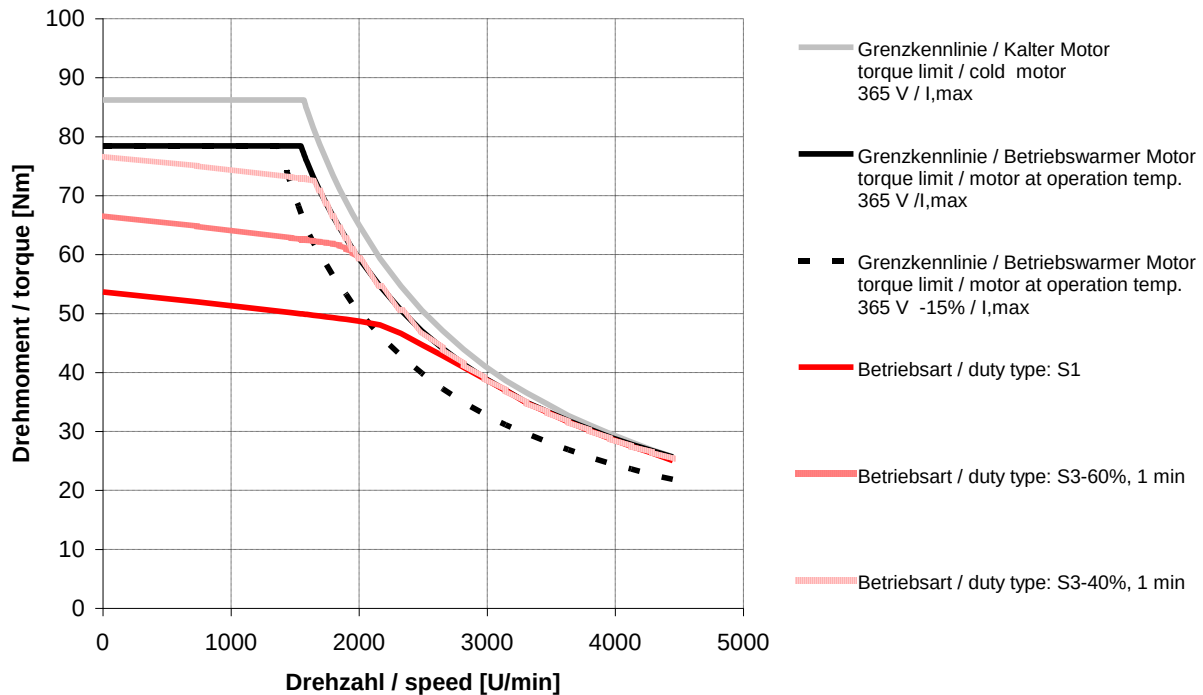
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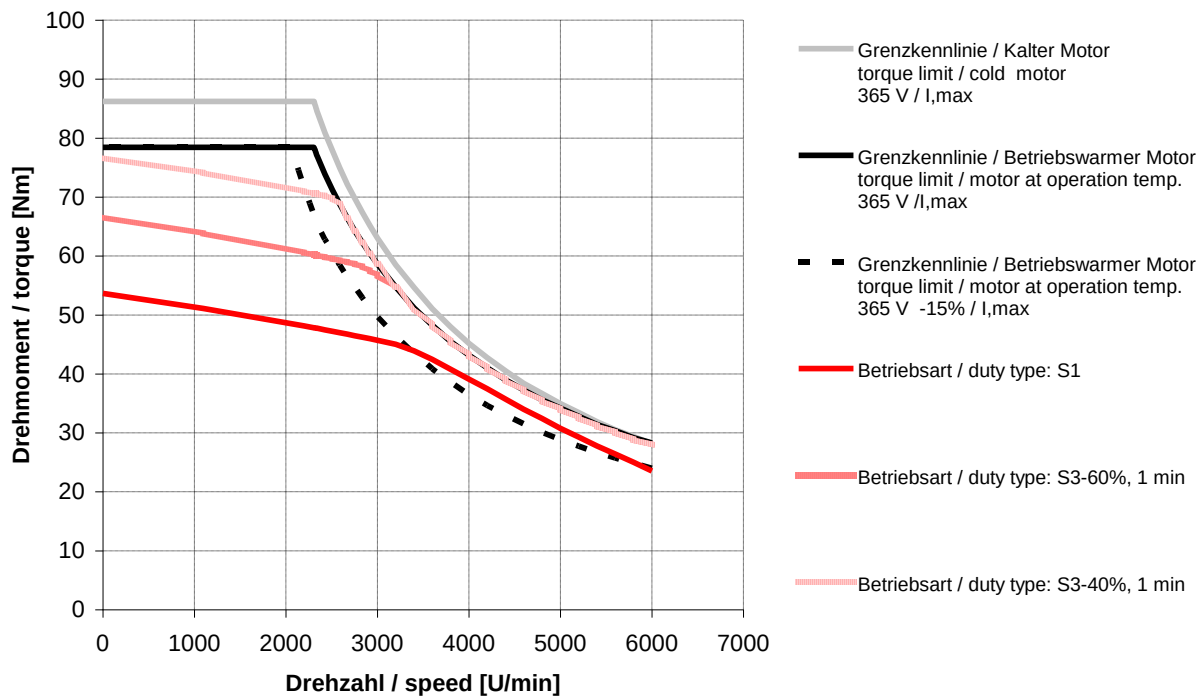
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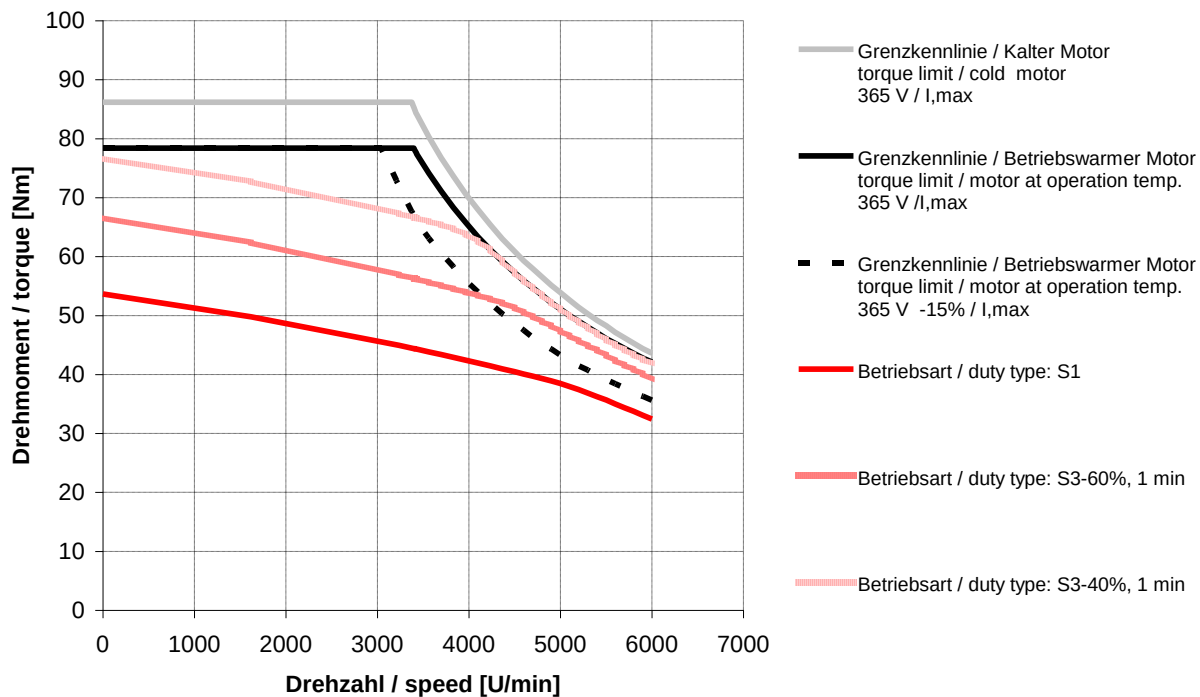
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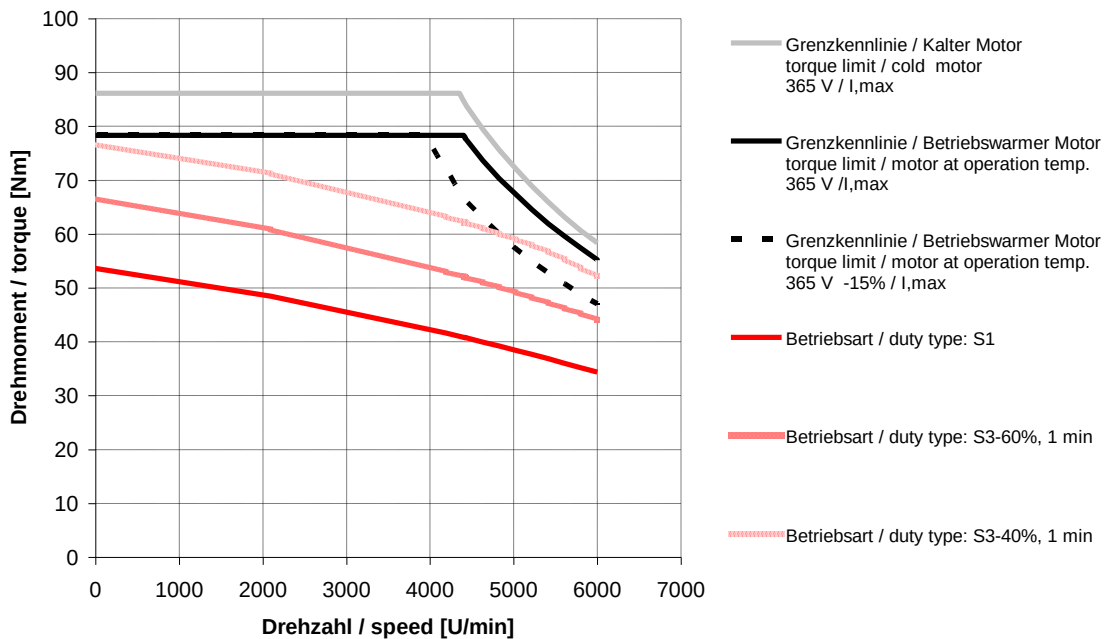
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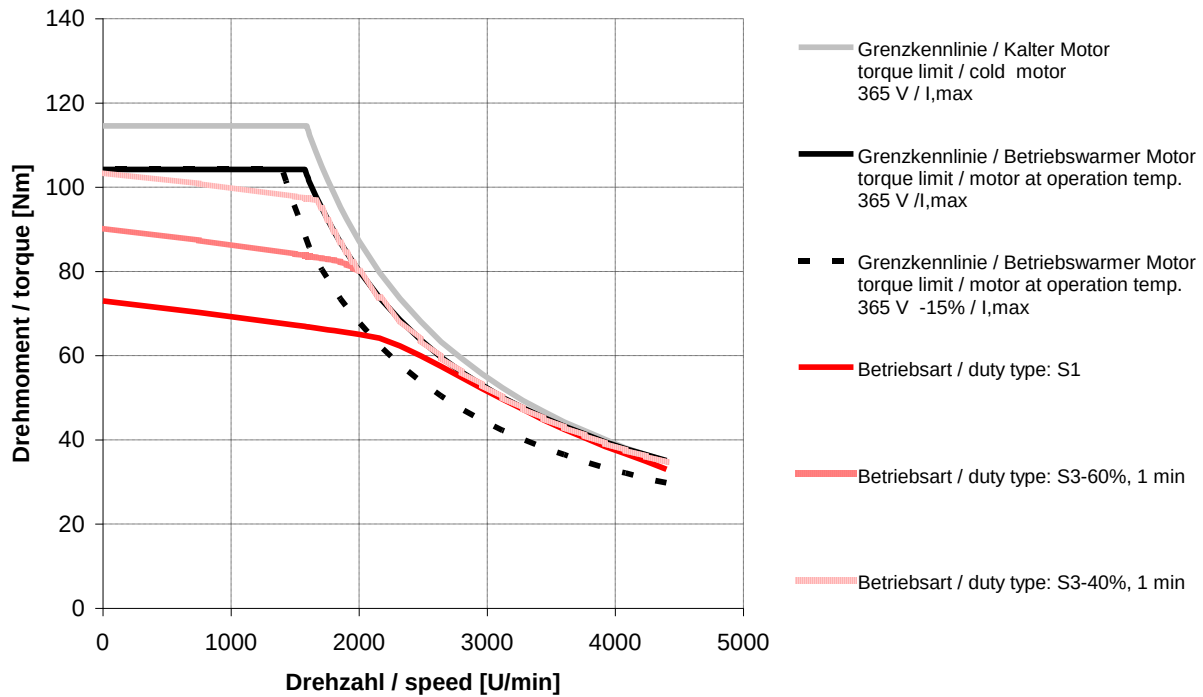
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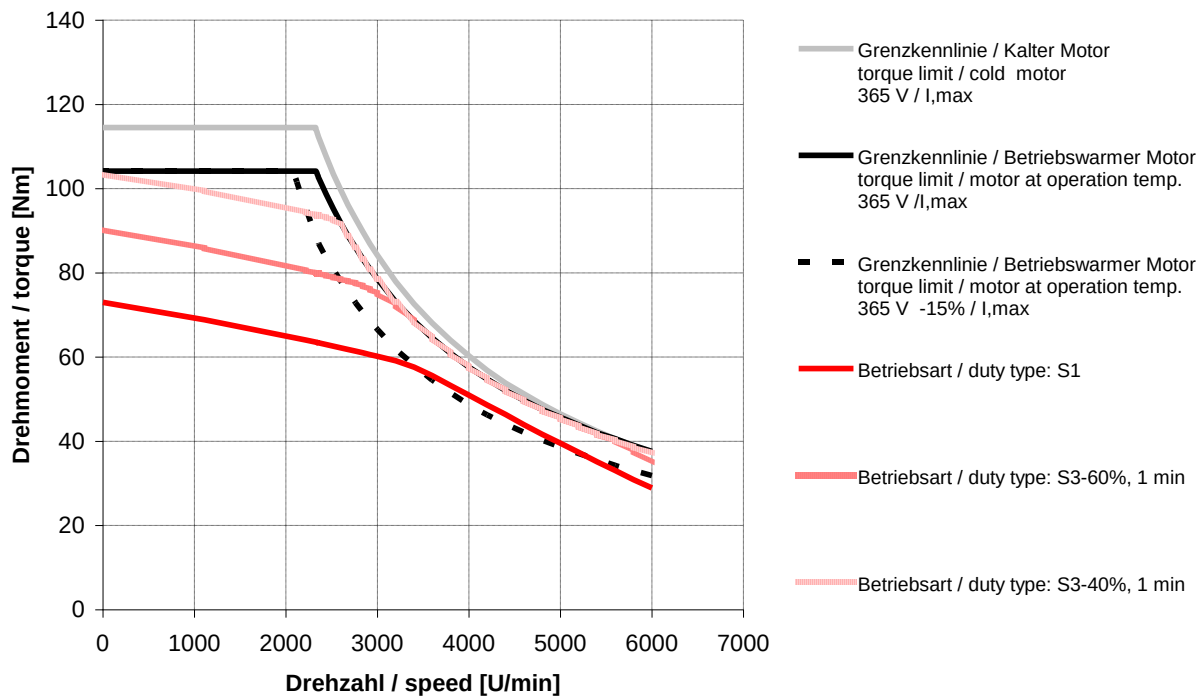
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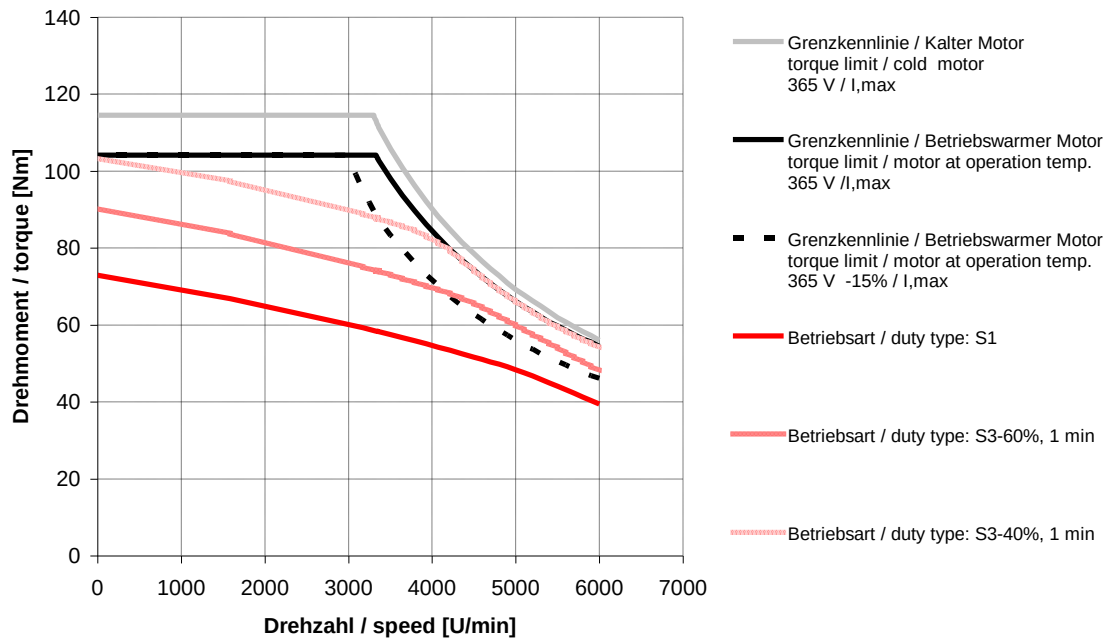
DSD2-071LO64W-20-54



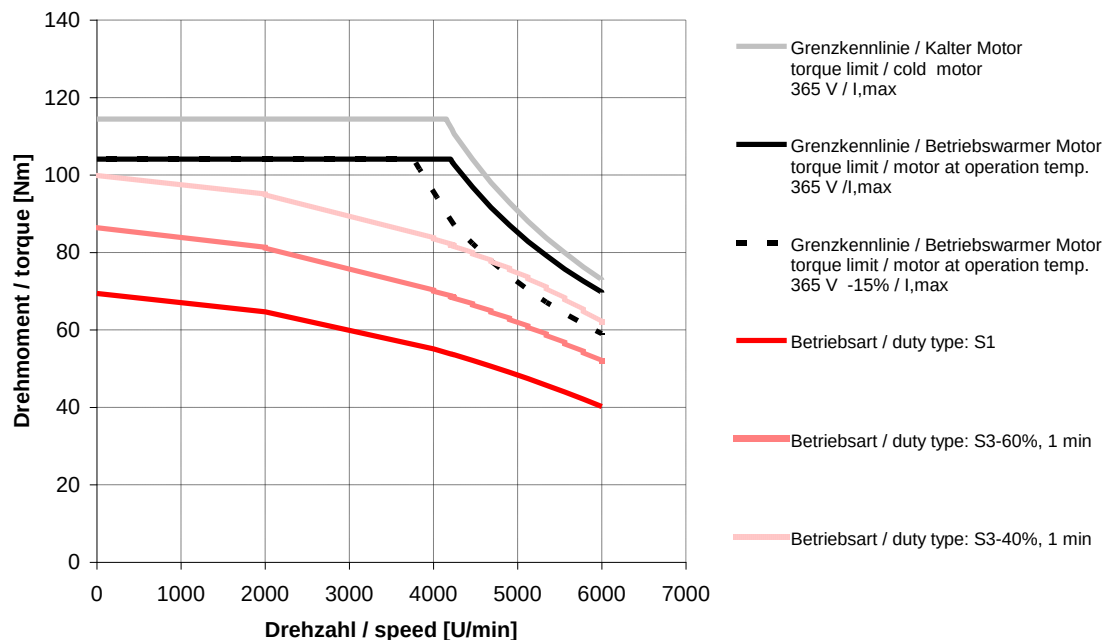
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DSD2-071LO64W-45-54



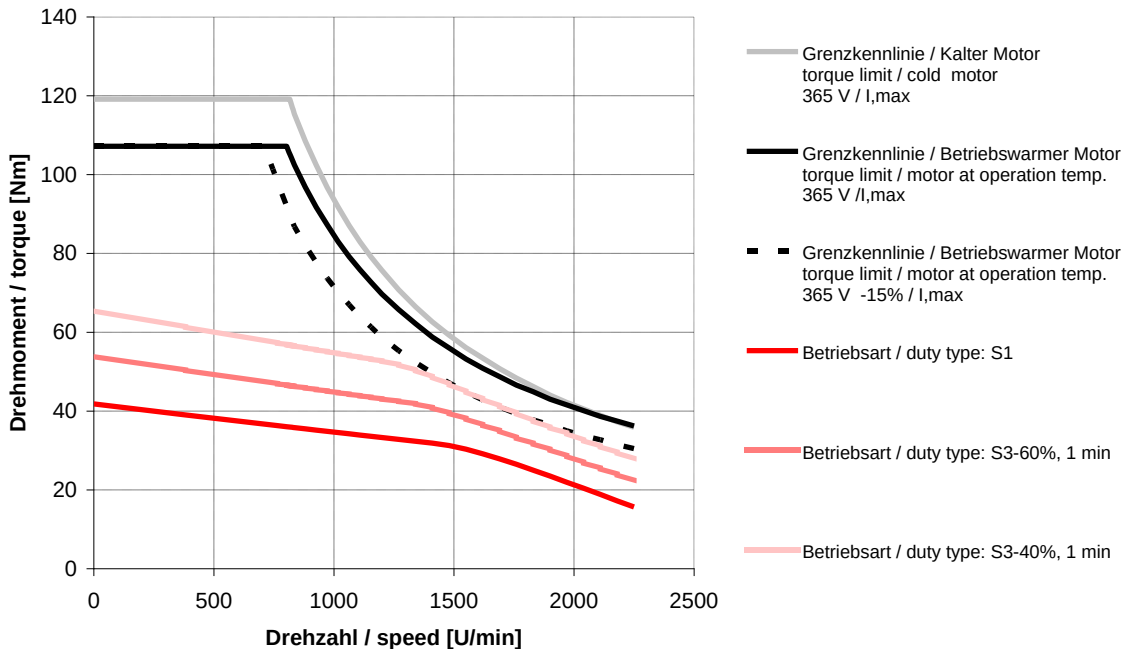
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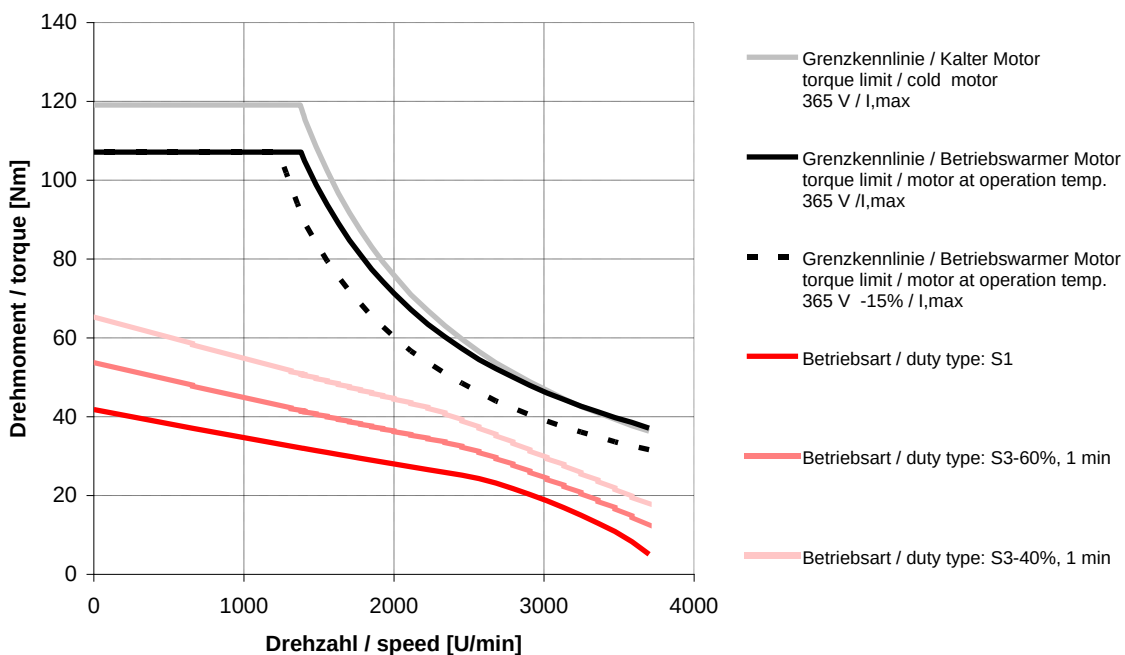
5.6. Motor characteristic curve DSD2-100

5.6.1. DSD2-100..64U-.. (without fan)

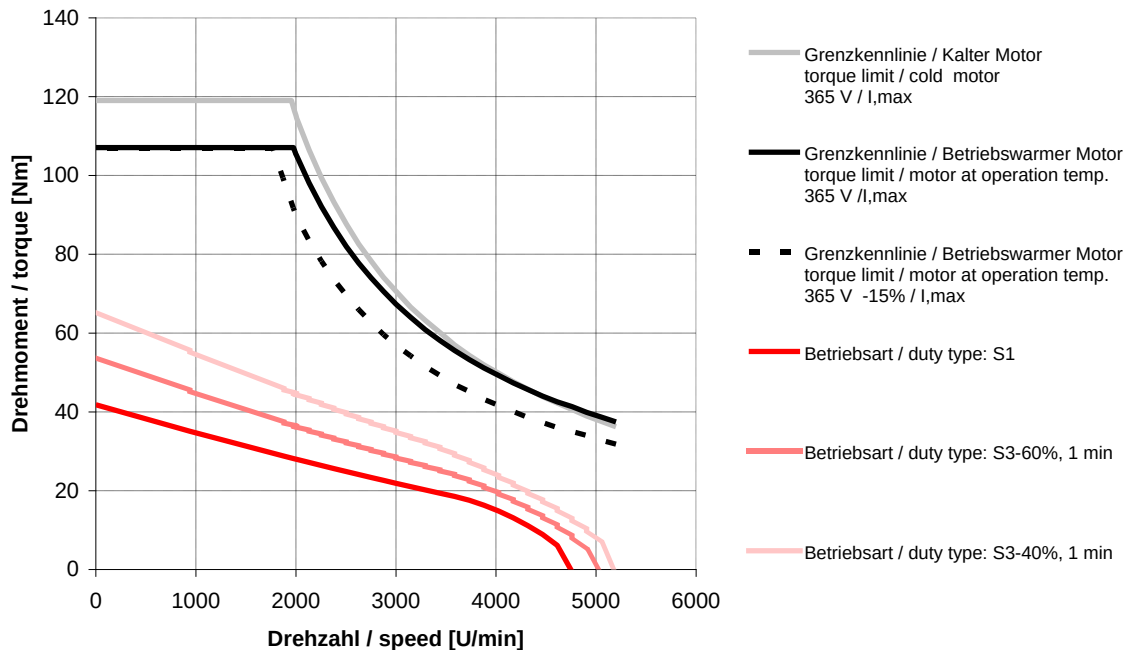
DSD2-100SO64U-12-54



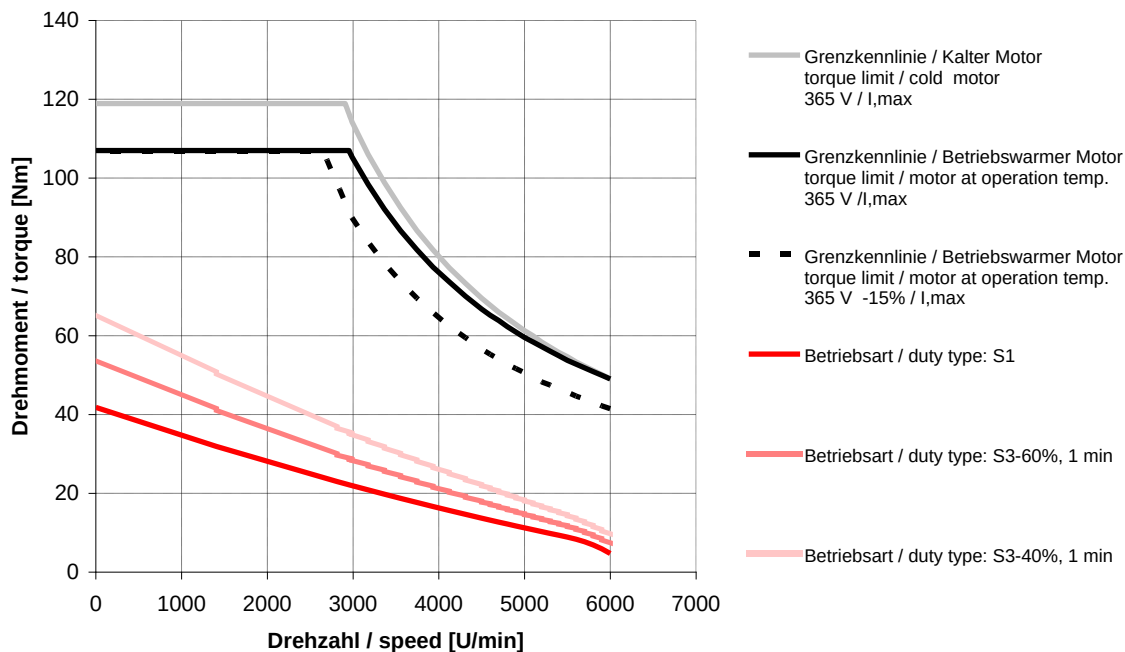
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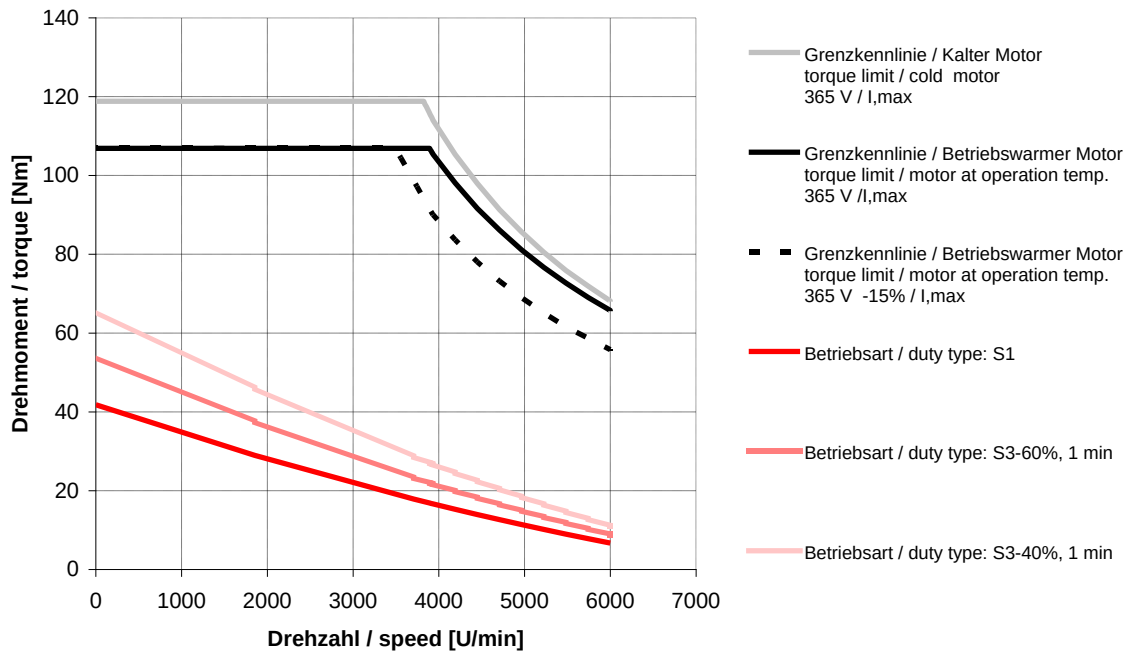
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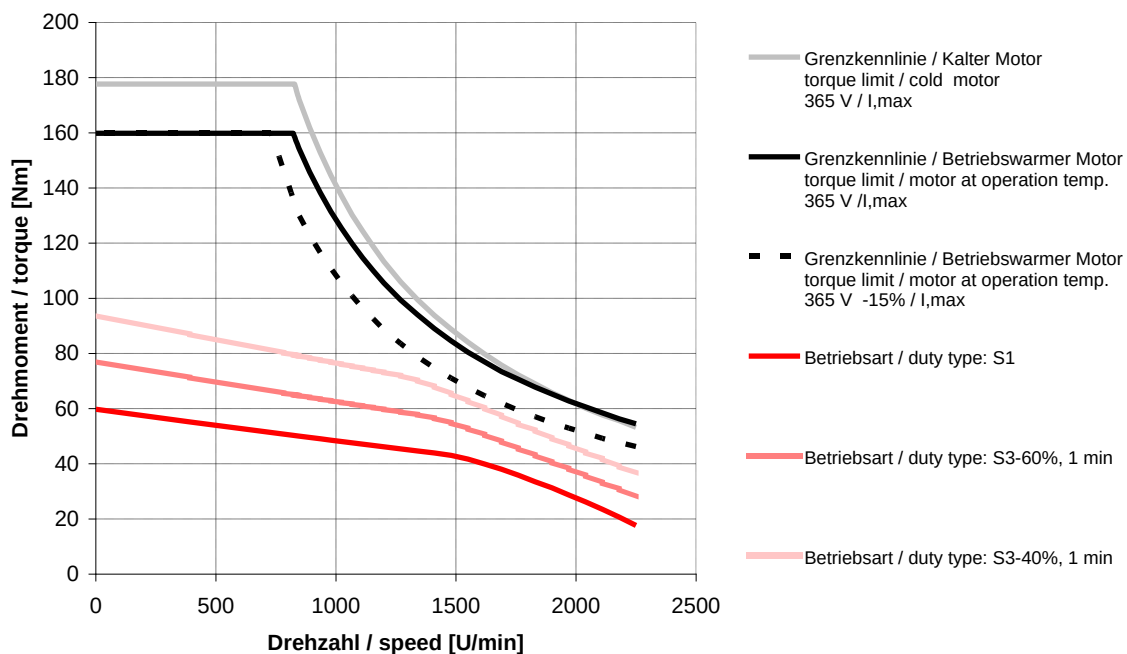
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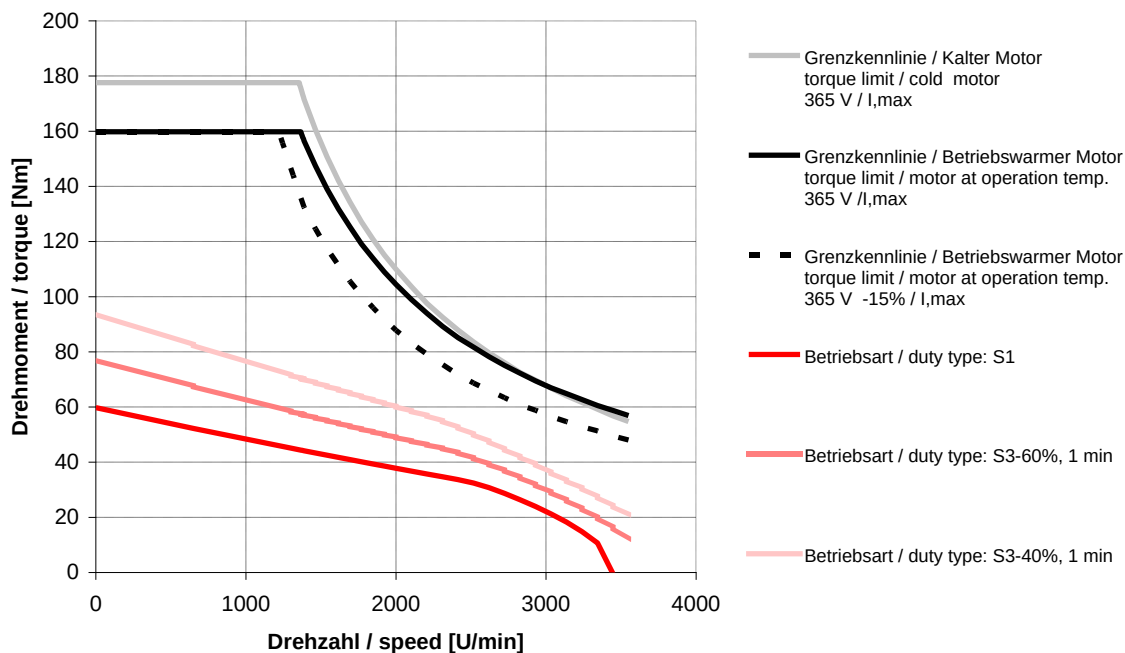
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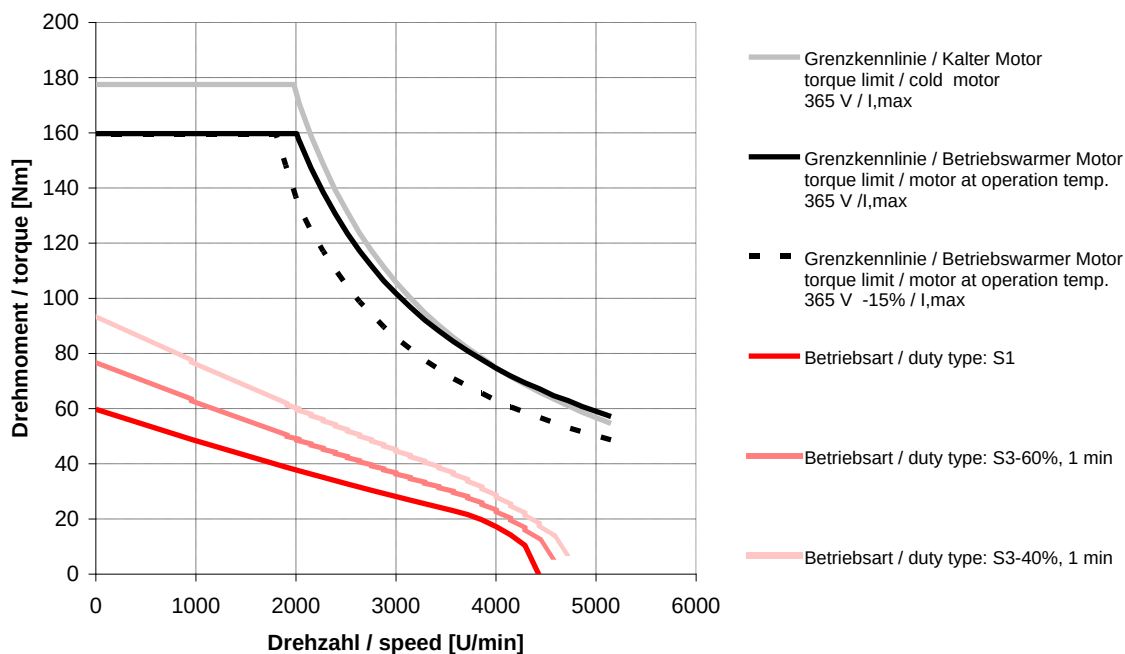
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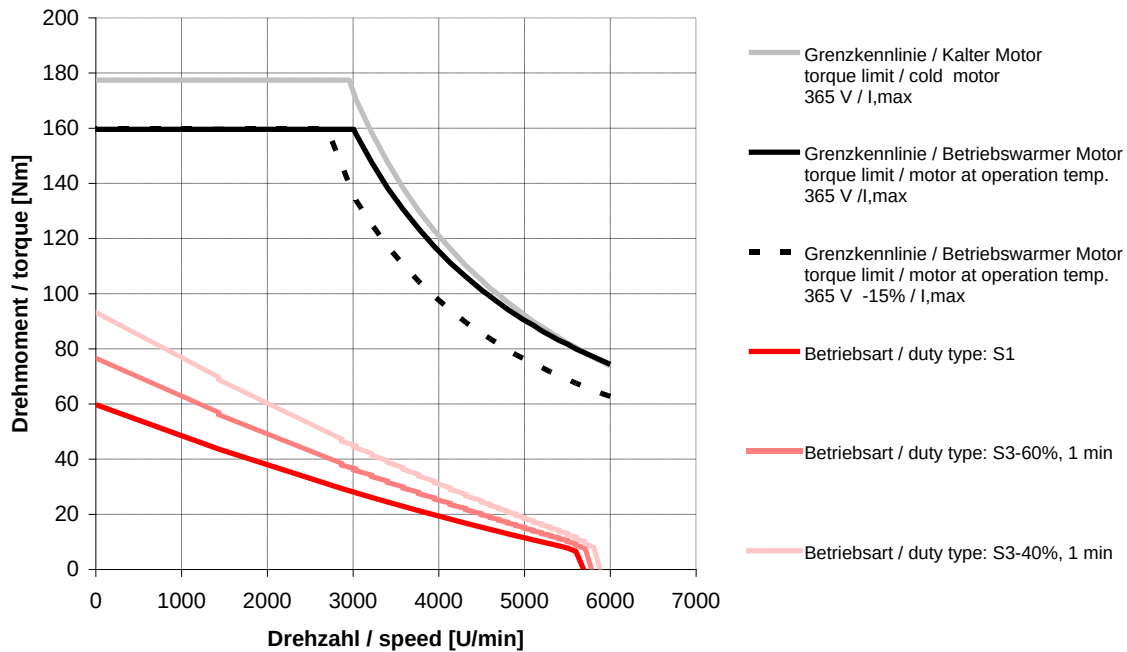
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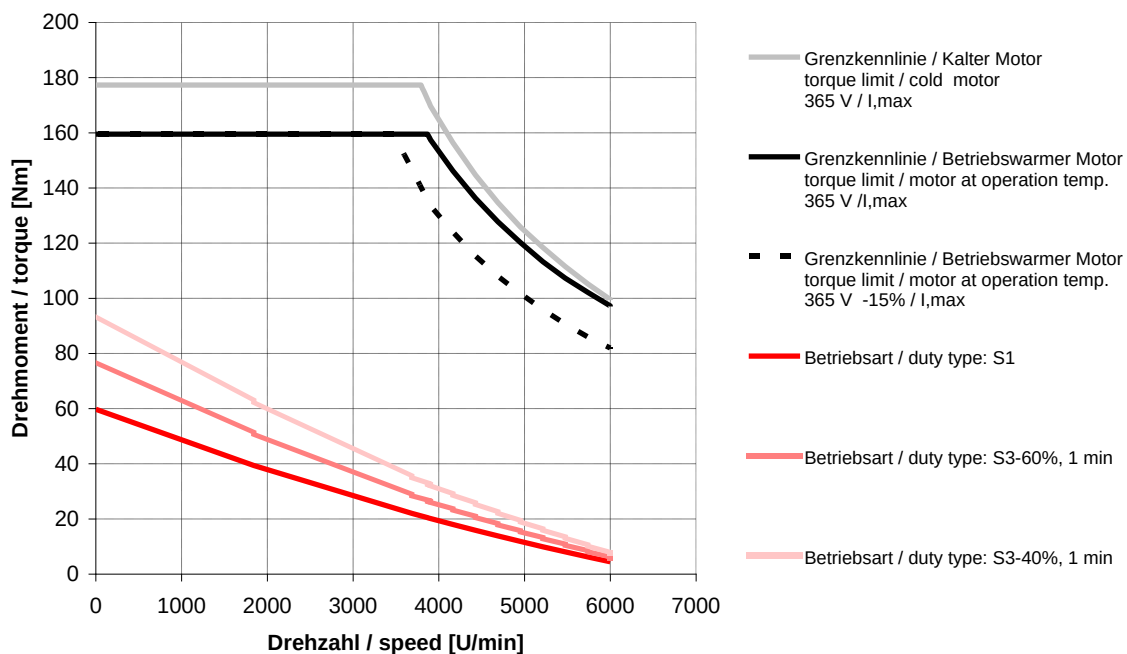
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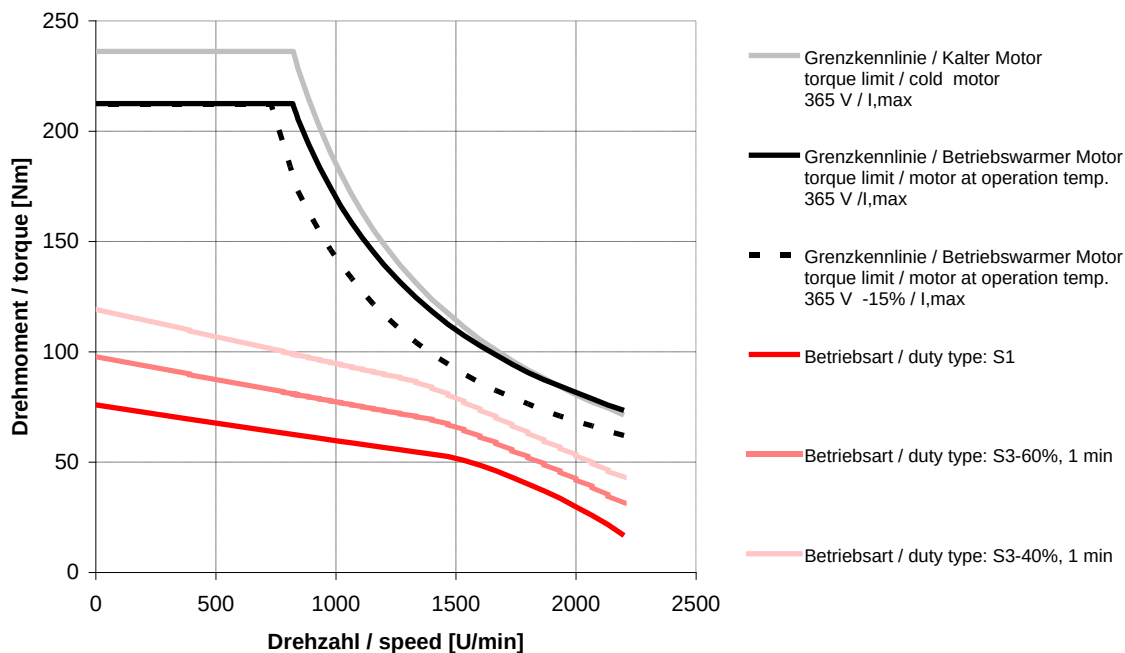
DSD2-100MO64U-45-54



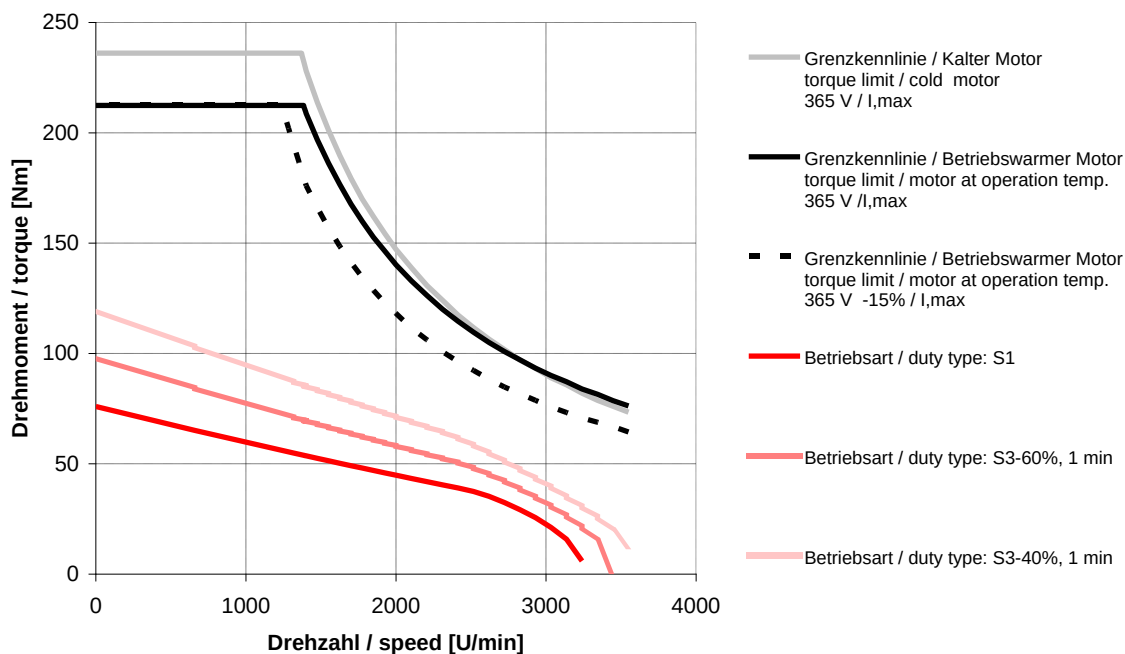
DSD2-100MO64U-60-54



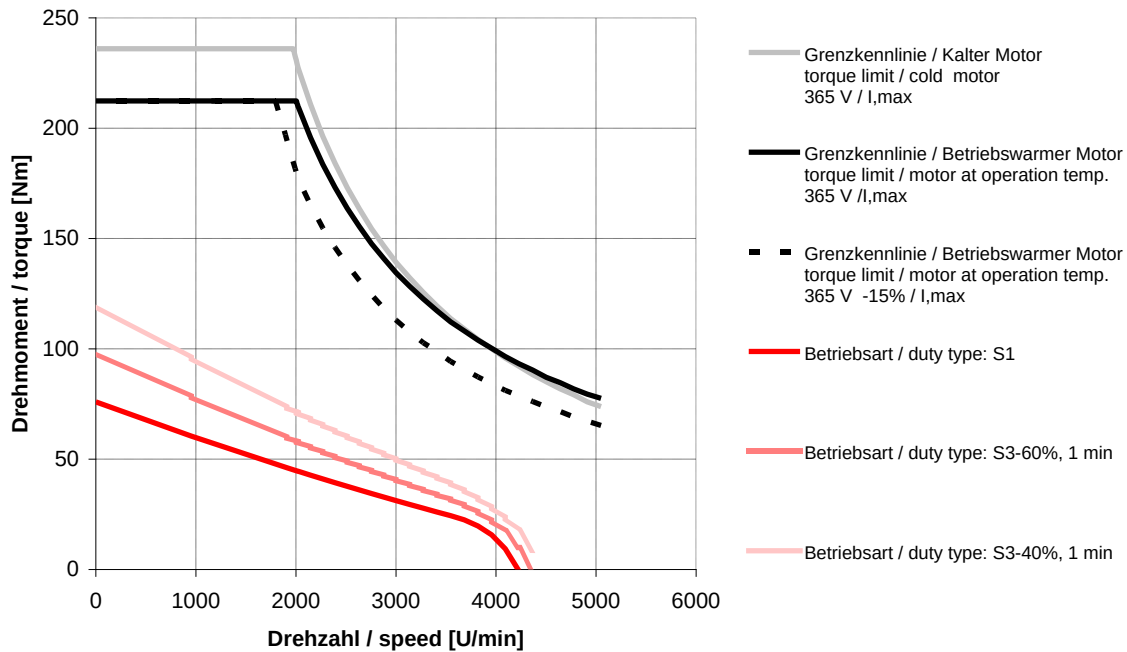
DSD2-100LO64U-12-54



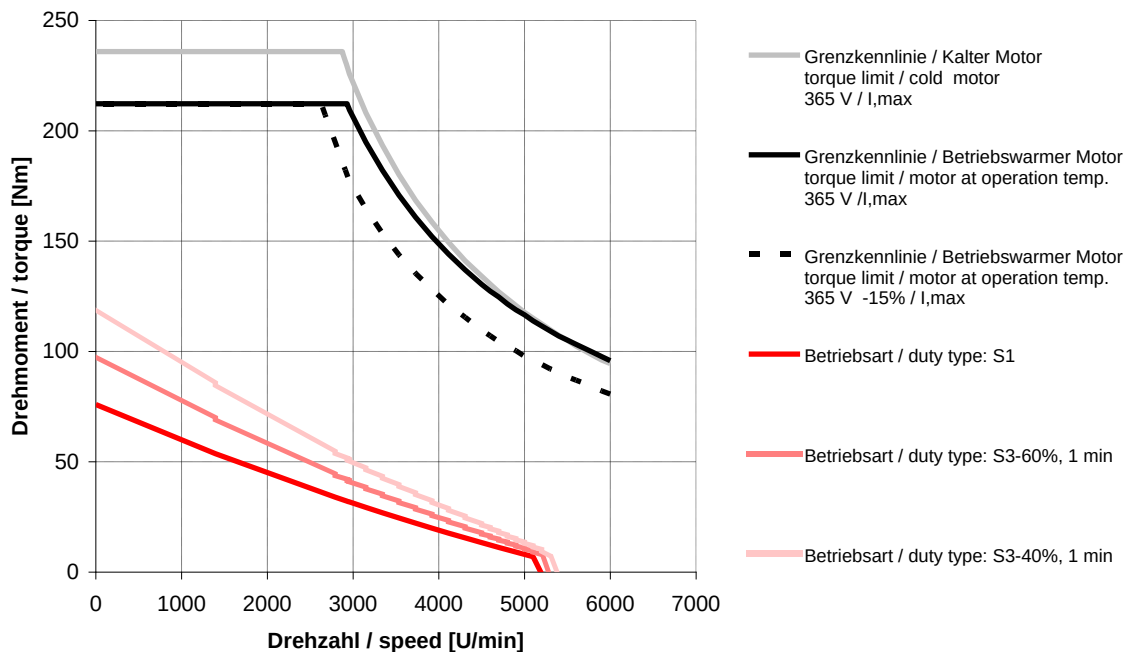
DSD2-100LO64U-20-54



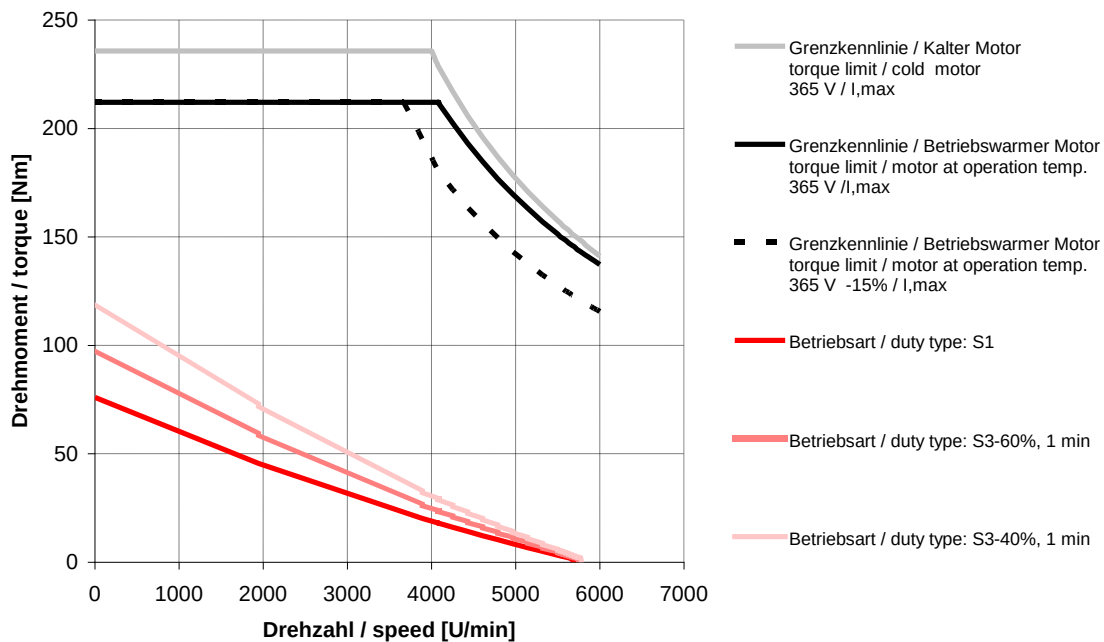
DSD2-100LO64U-30-54



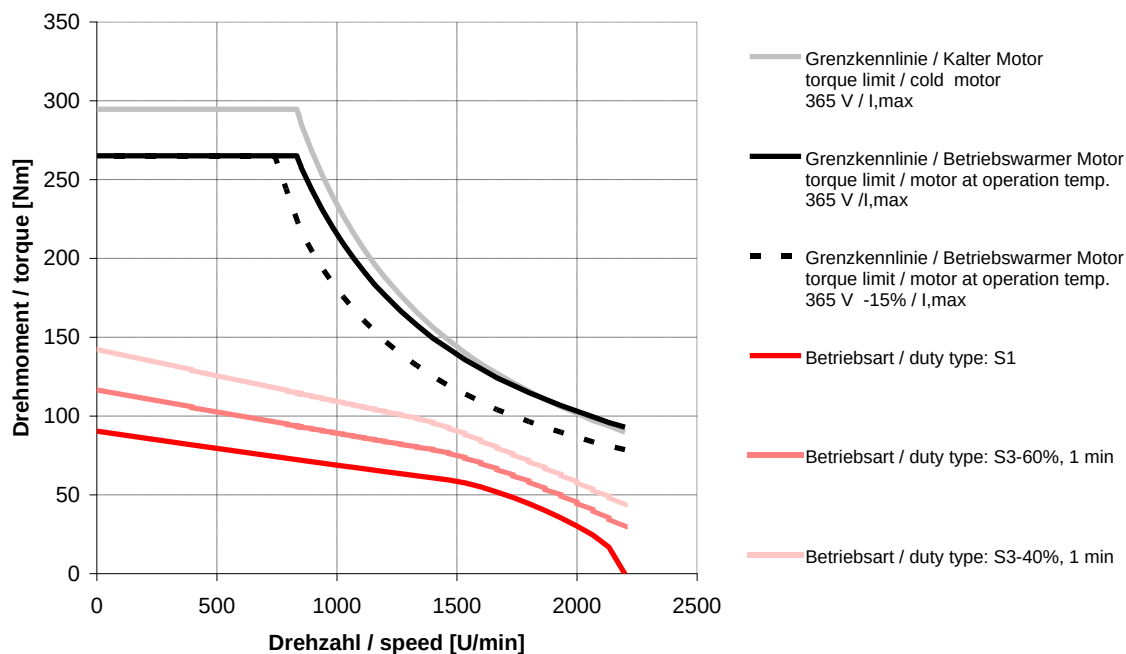
DSD2-100LO64U-45-54



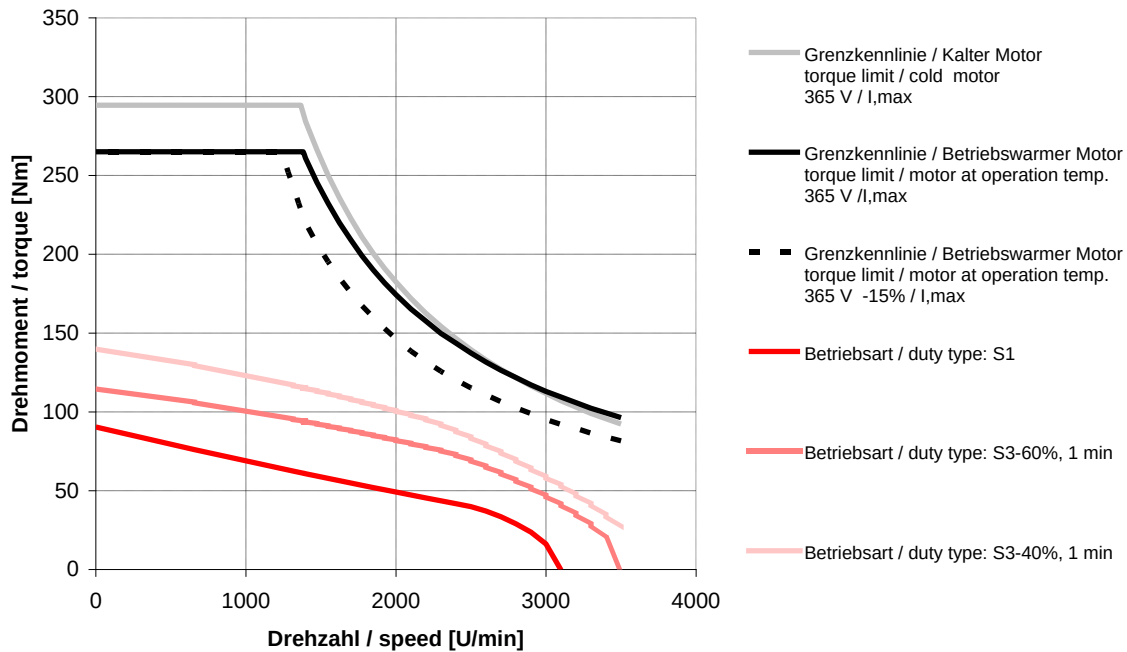
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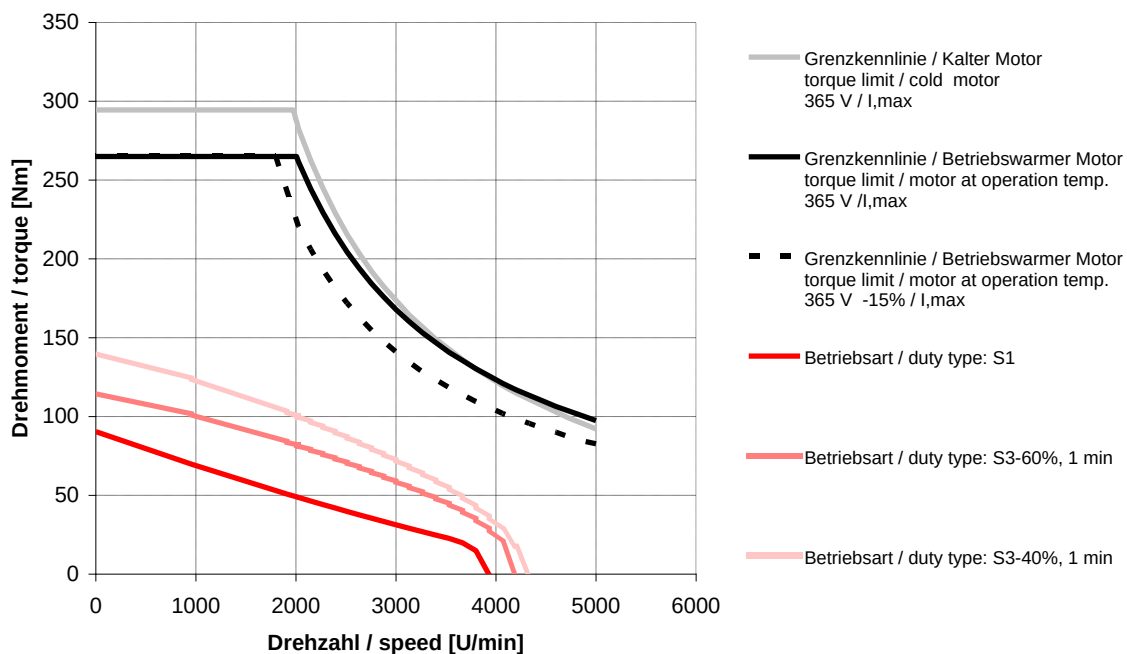
DSD2-100BO64U-12-54



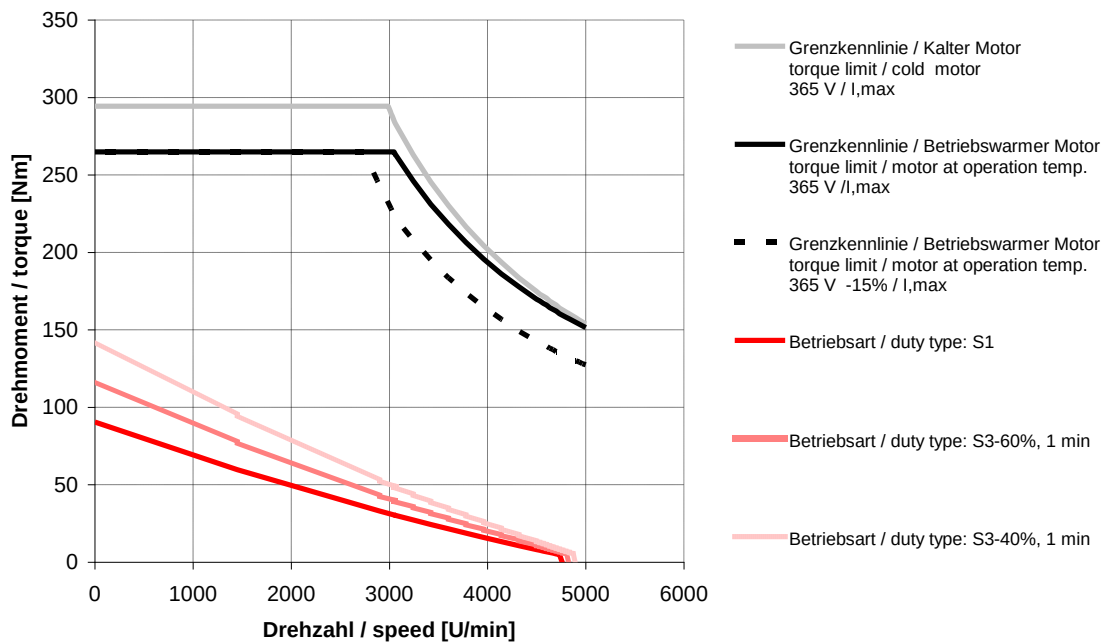
DSD2-100BO64U-20-54



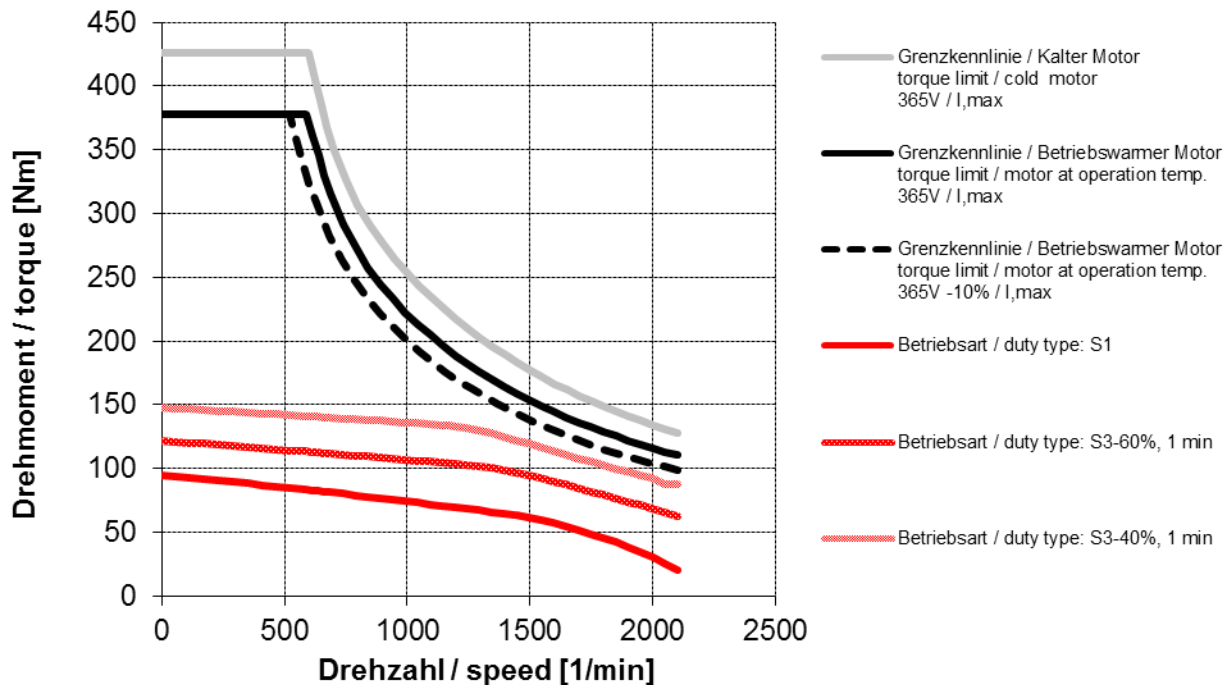
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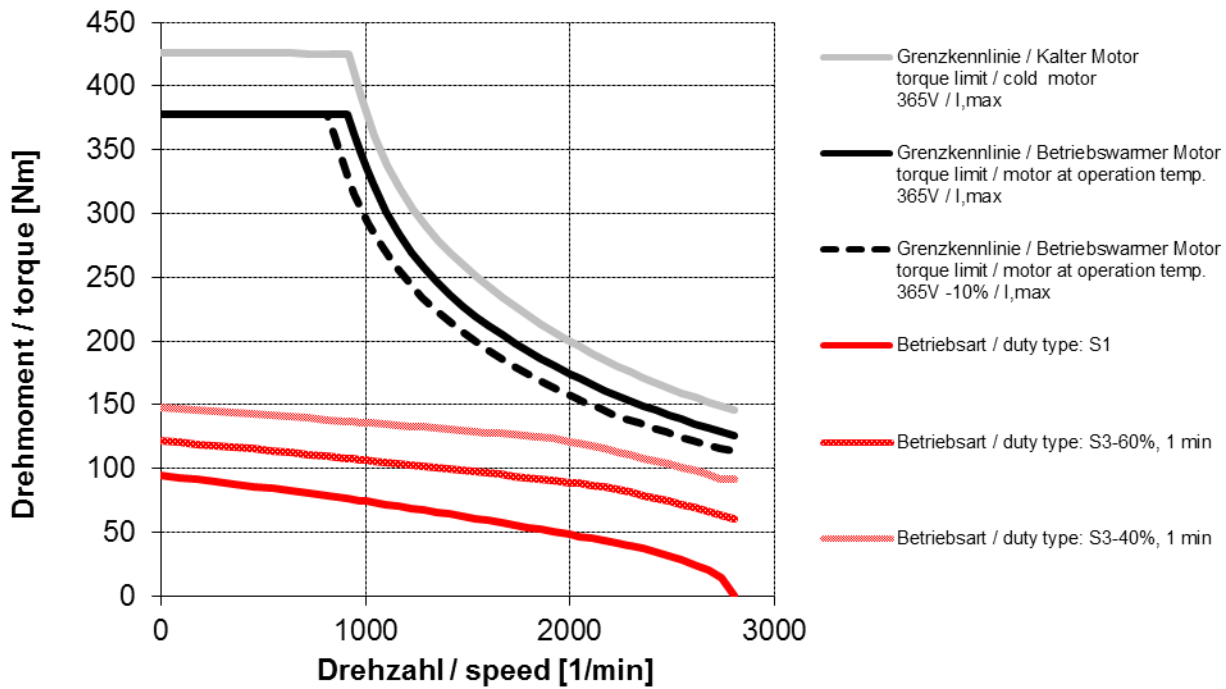
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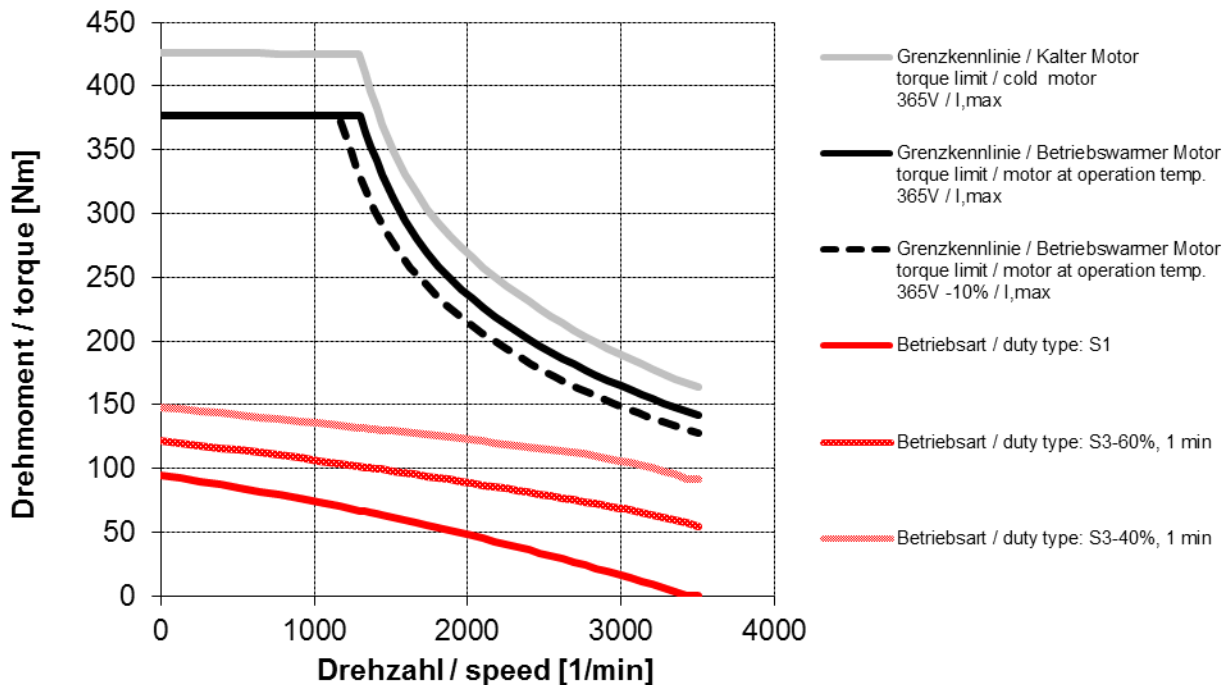
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DSD2-100XO64U-20-54

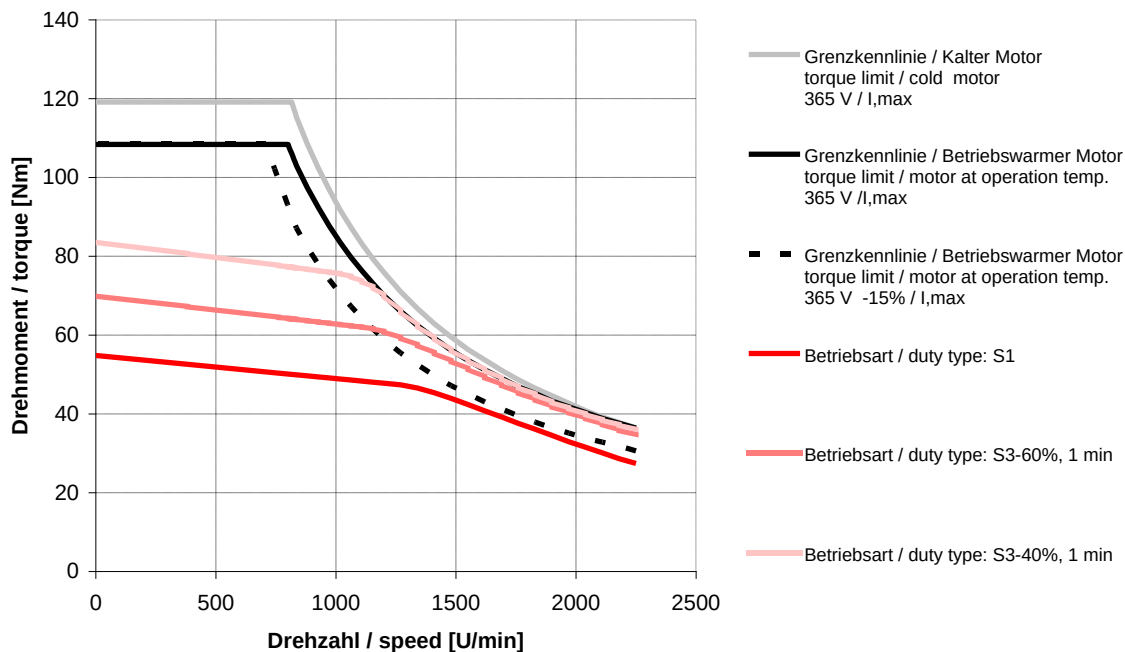


DSD2-100XO64U-30-54

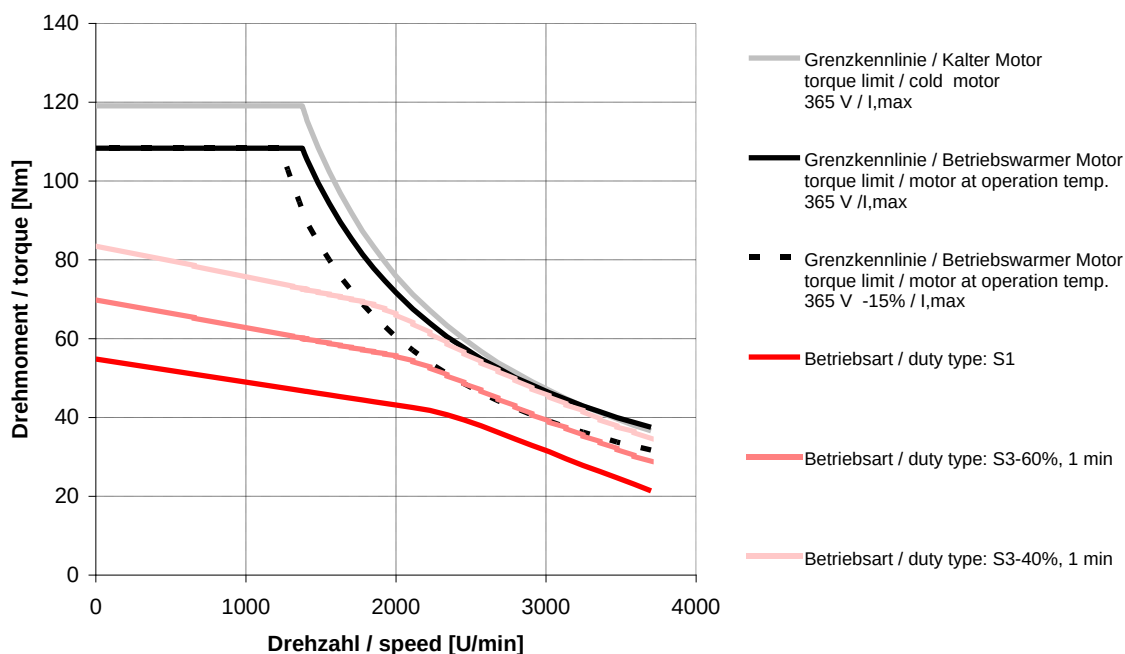


5.6.2. DSD2-100..640-.. (with fan)

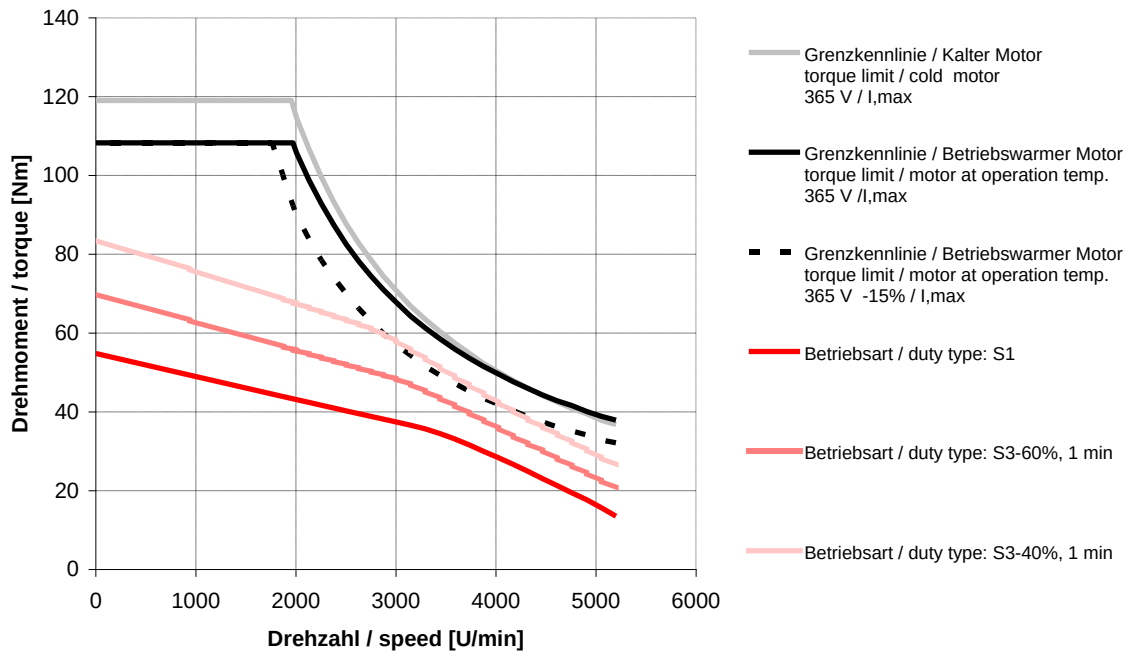
DSD2-100SO640-12-54



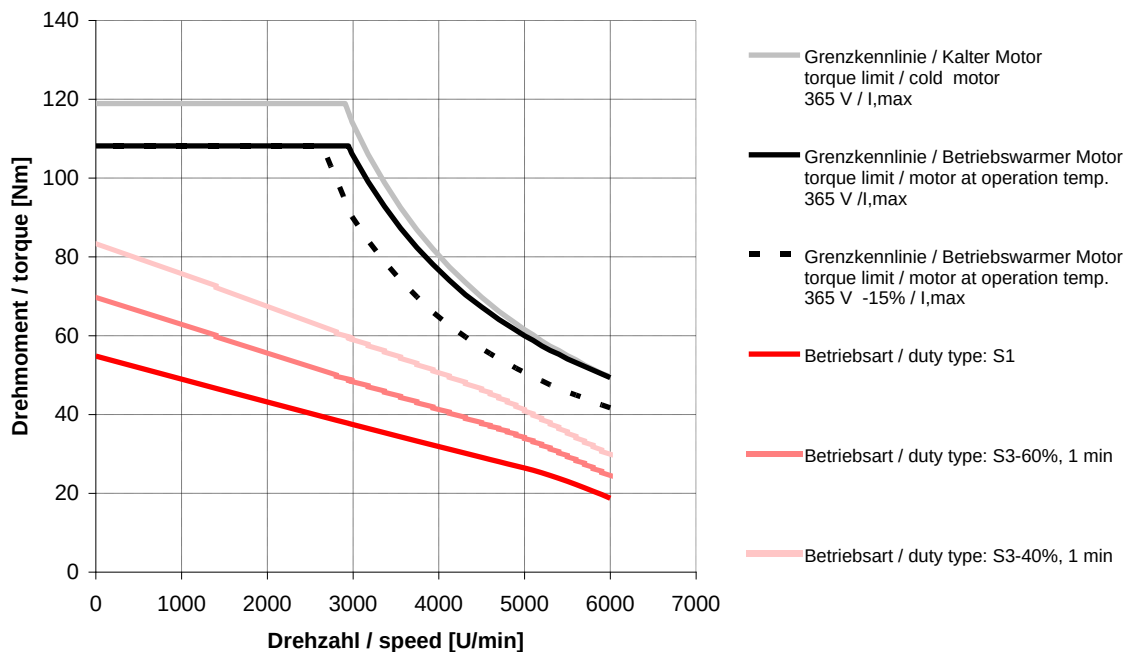
DSD2-100SO640-20-54



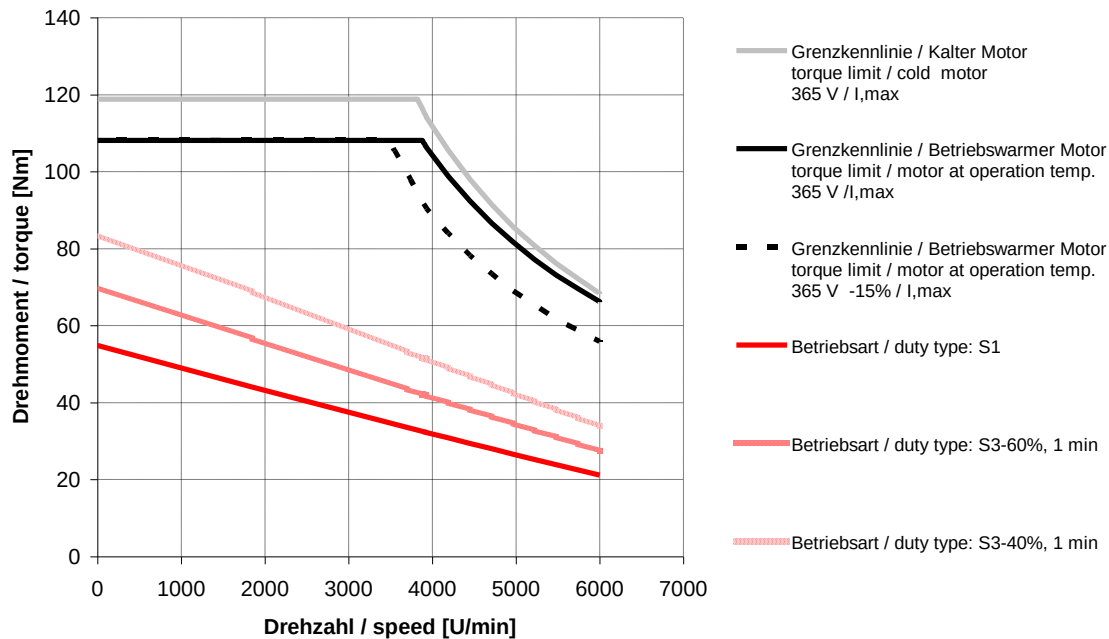
DSD2-100SO640-30-54



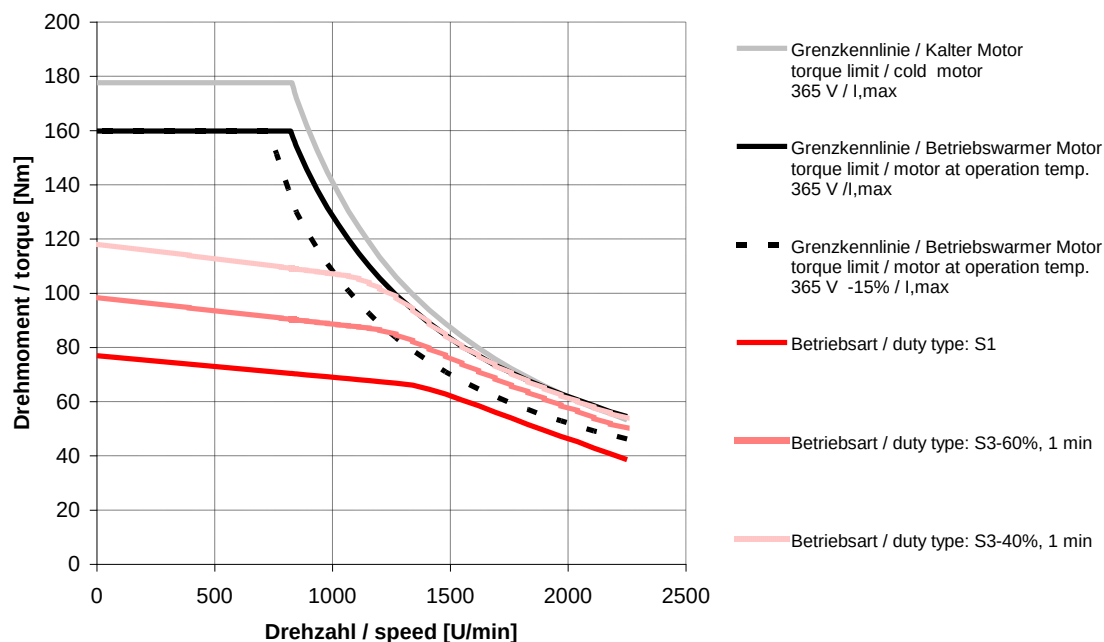
DSD2-100SO640-45-54



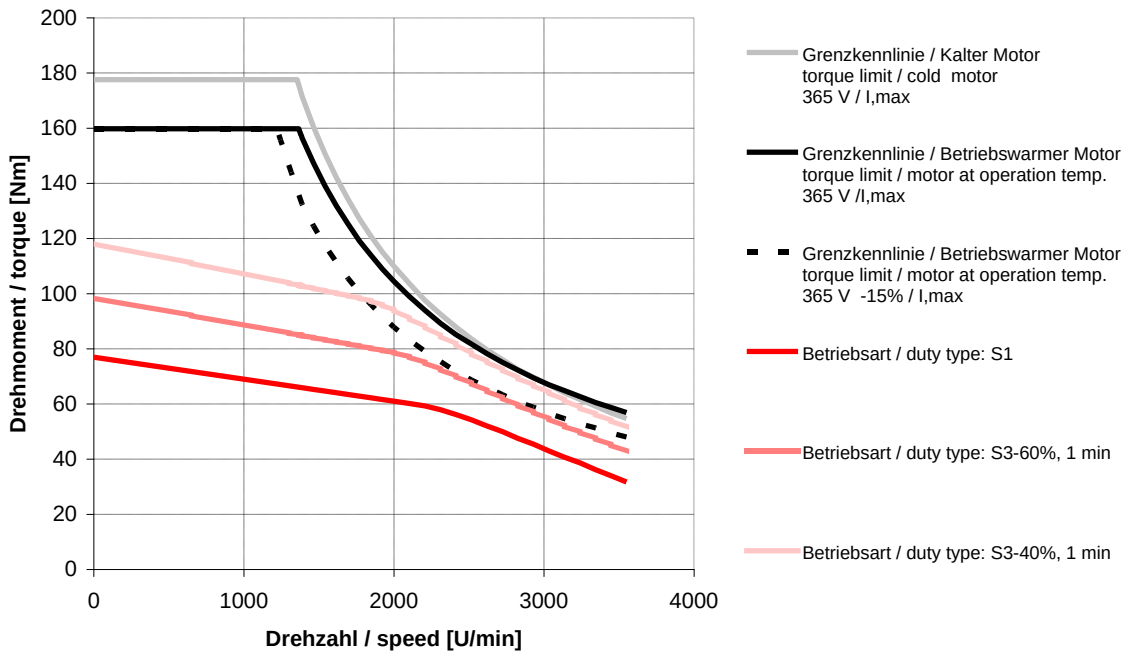
DSD2-100SO64O-60-54



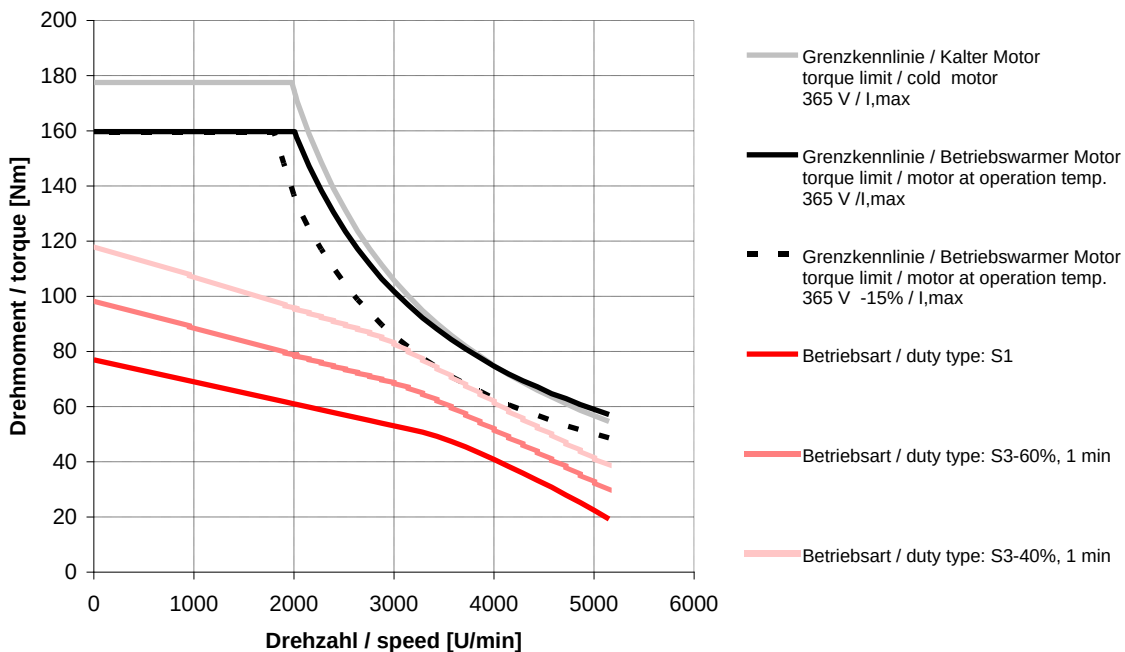
DSD2-100MO64O-12-54



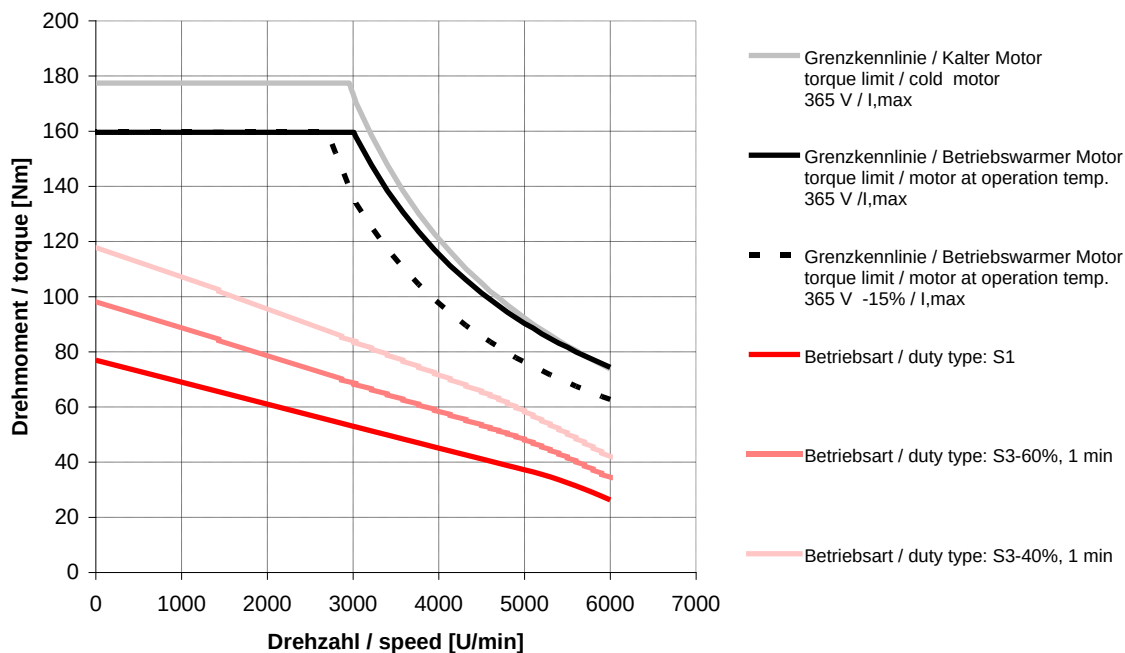
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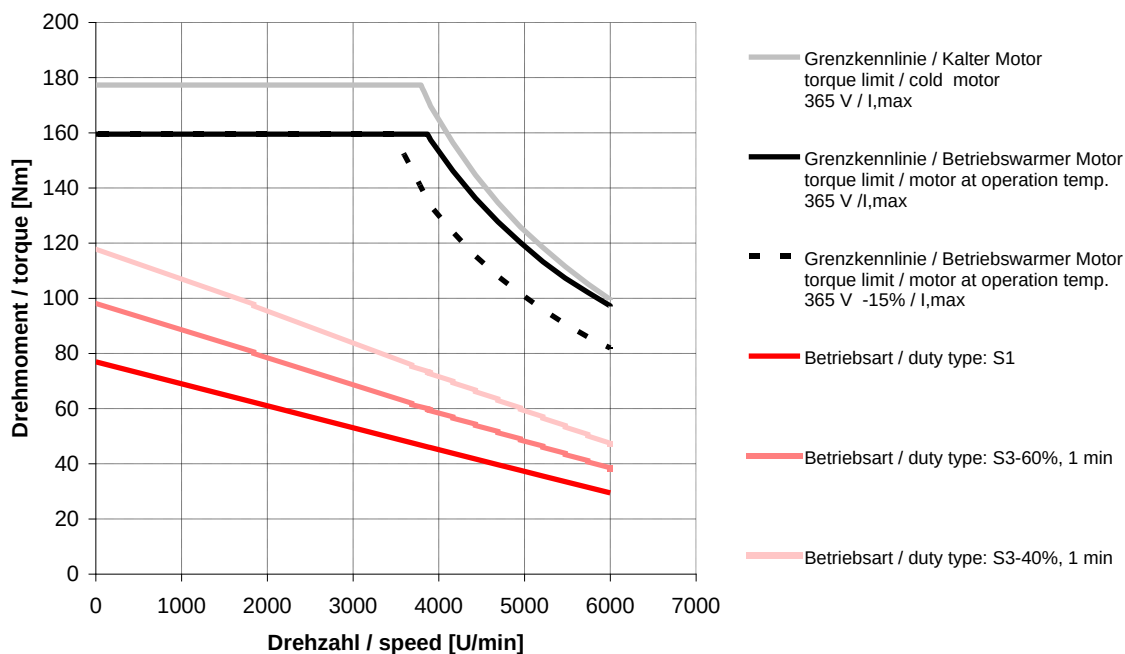
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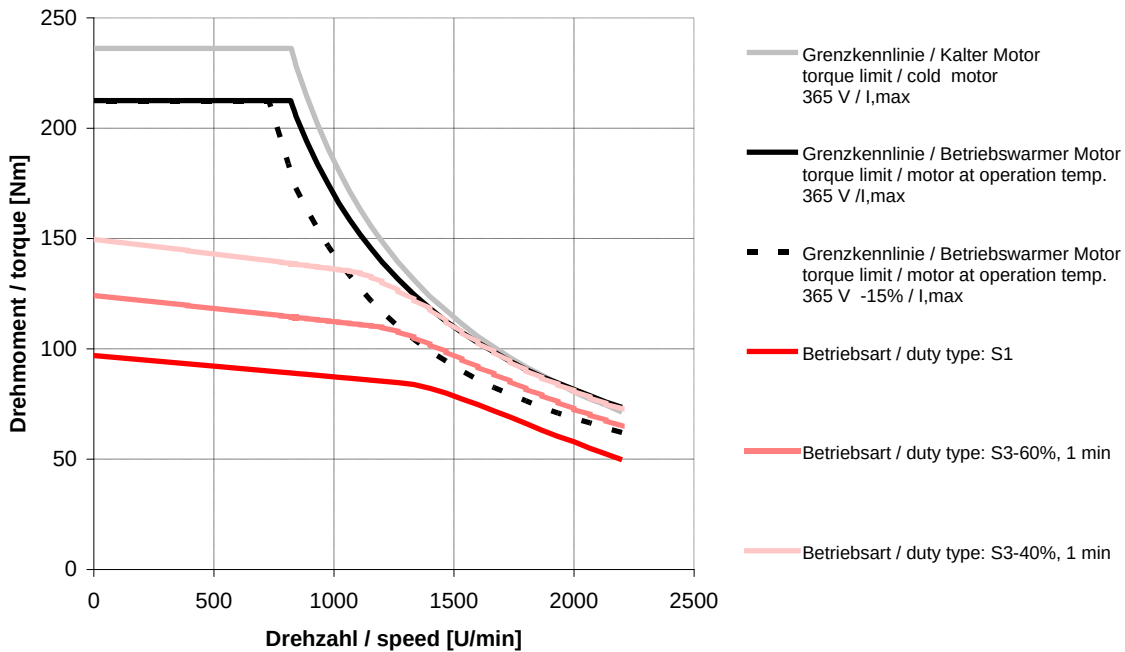
DSD2-100MO64O-45-54



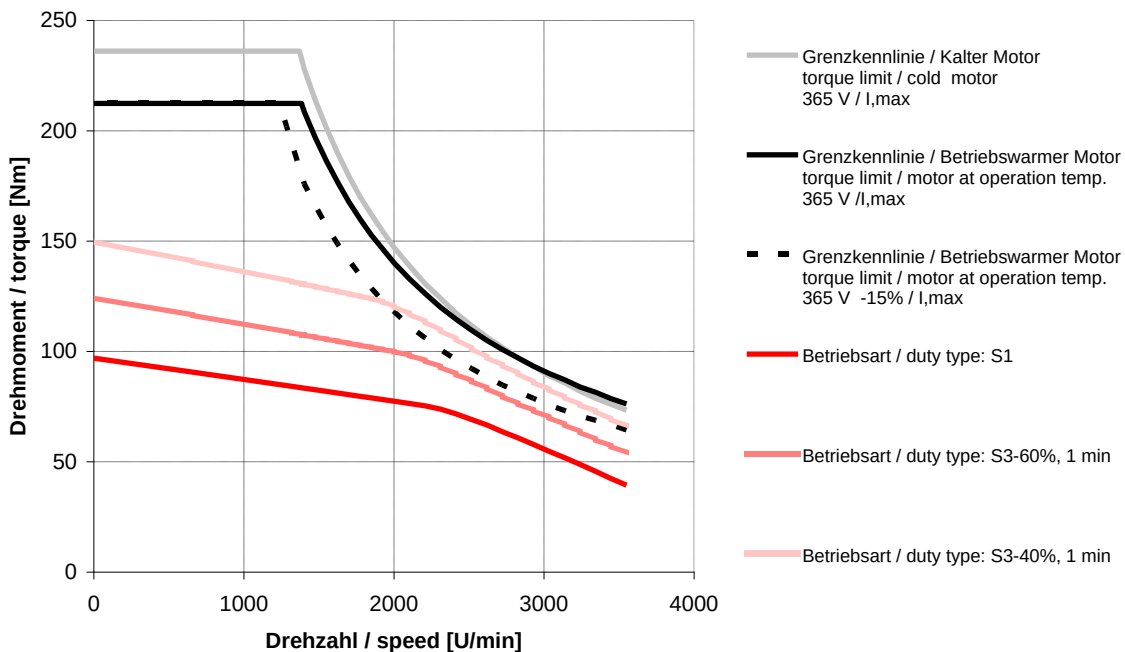
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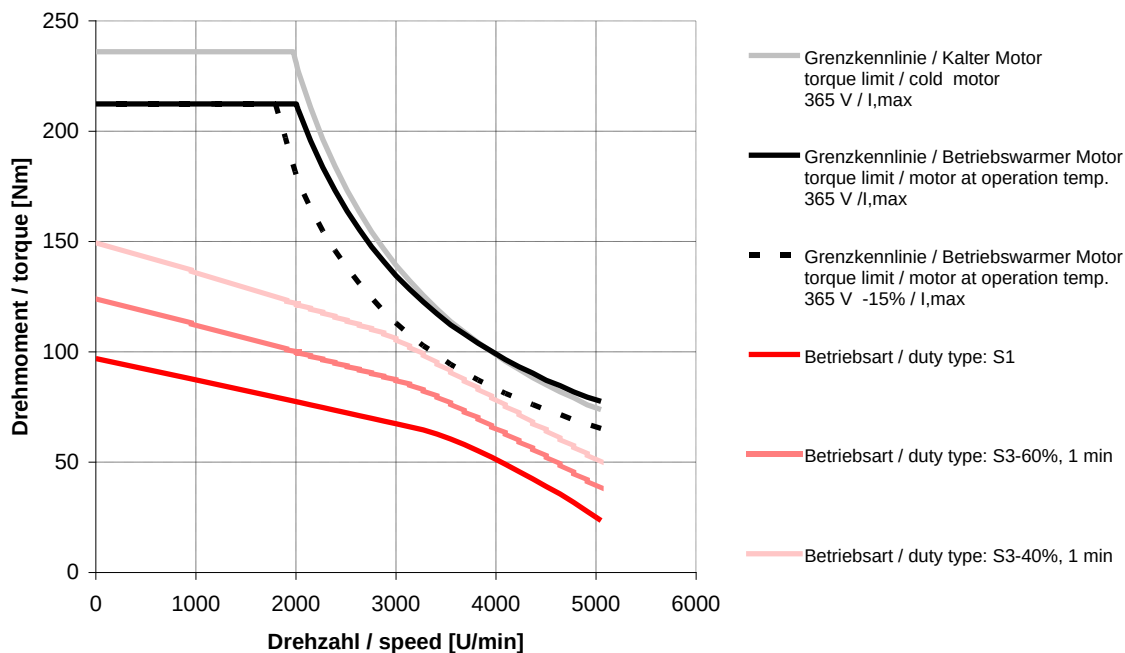
DSD2-100LO64O-12-54



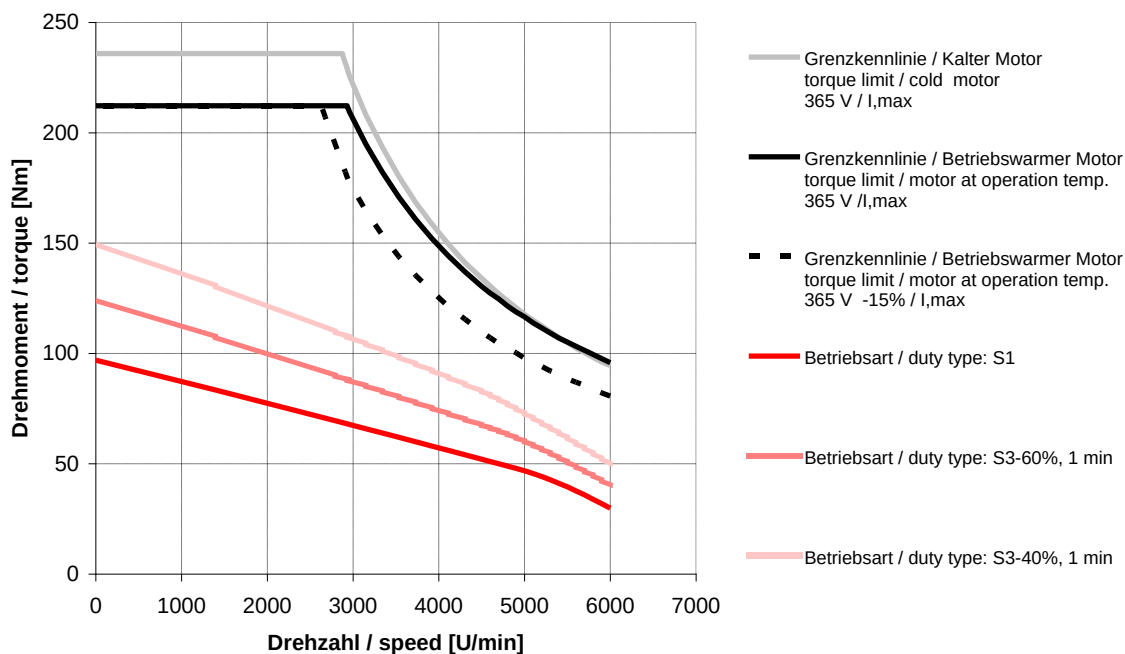
DSD2-100LO64O-20-54



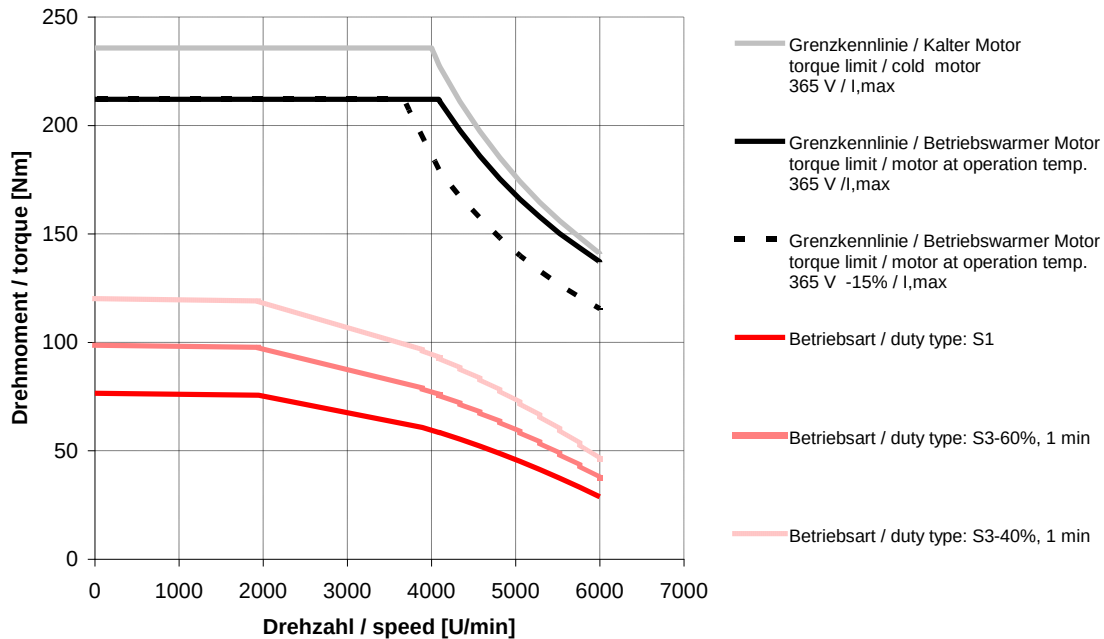
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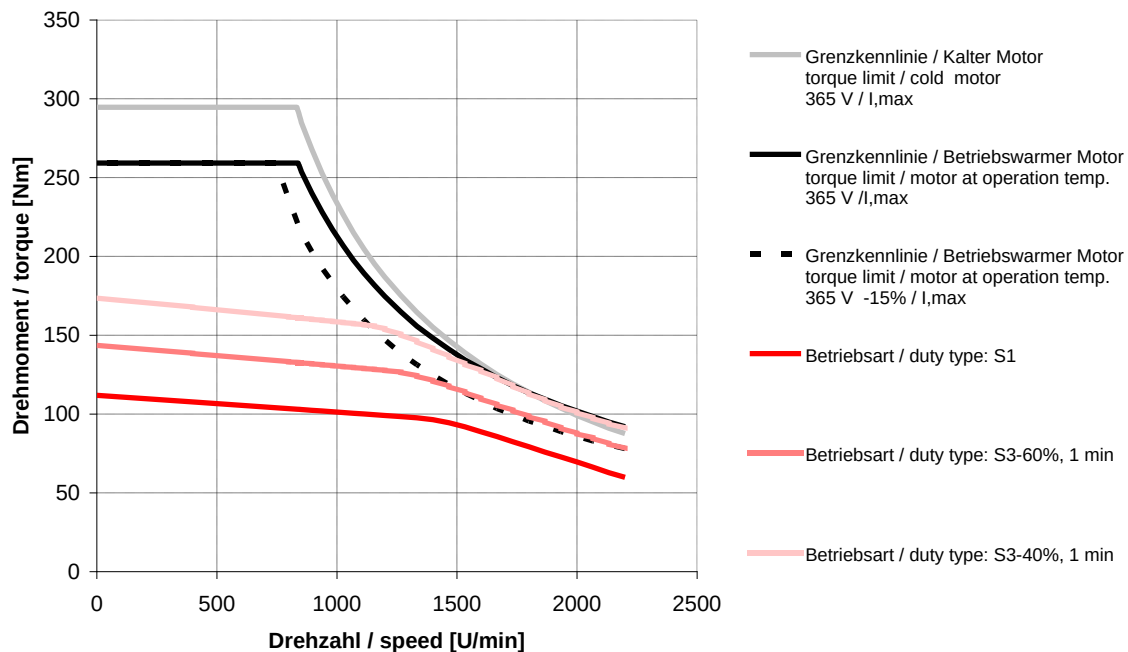
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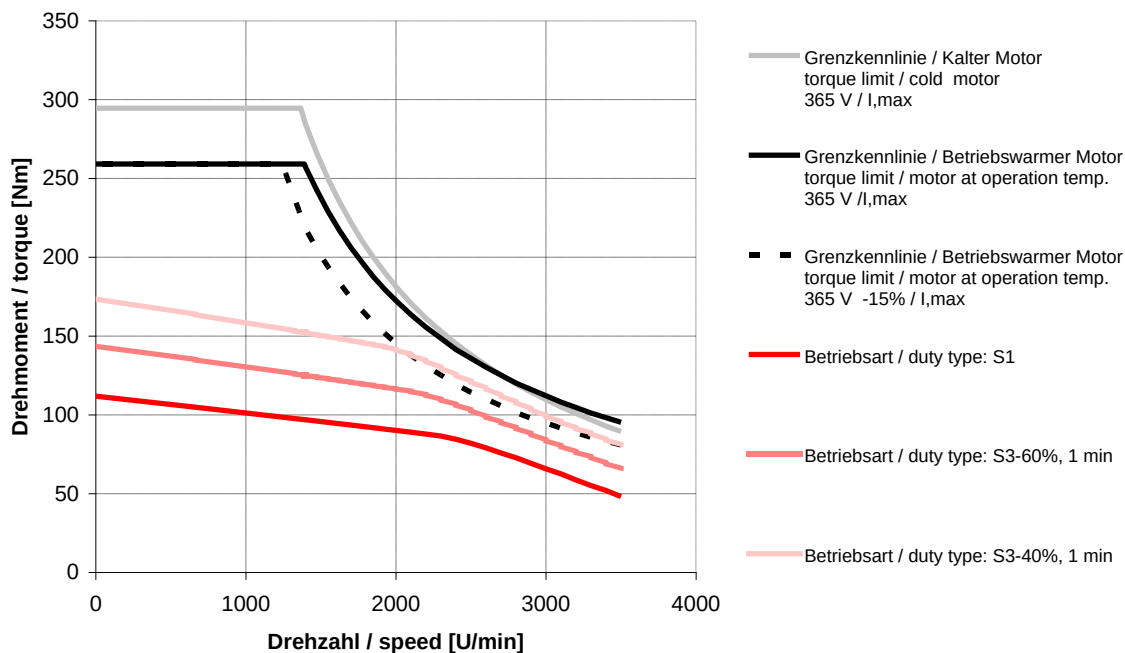
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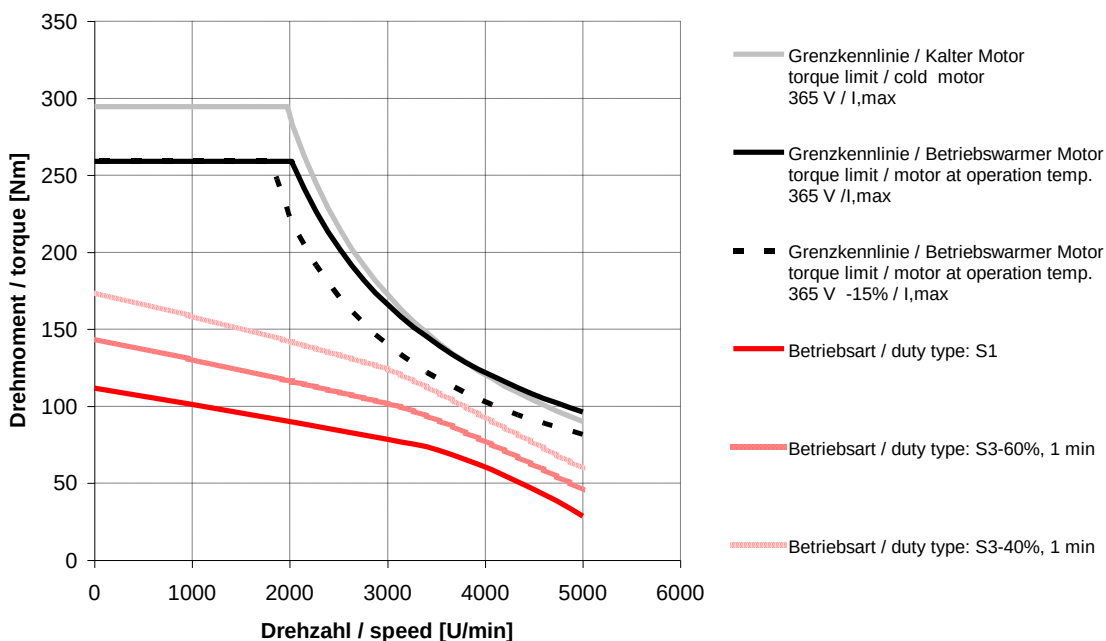
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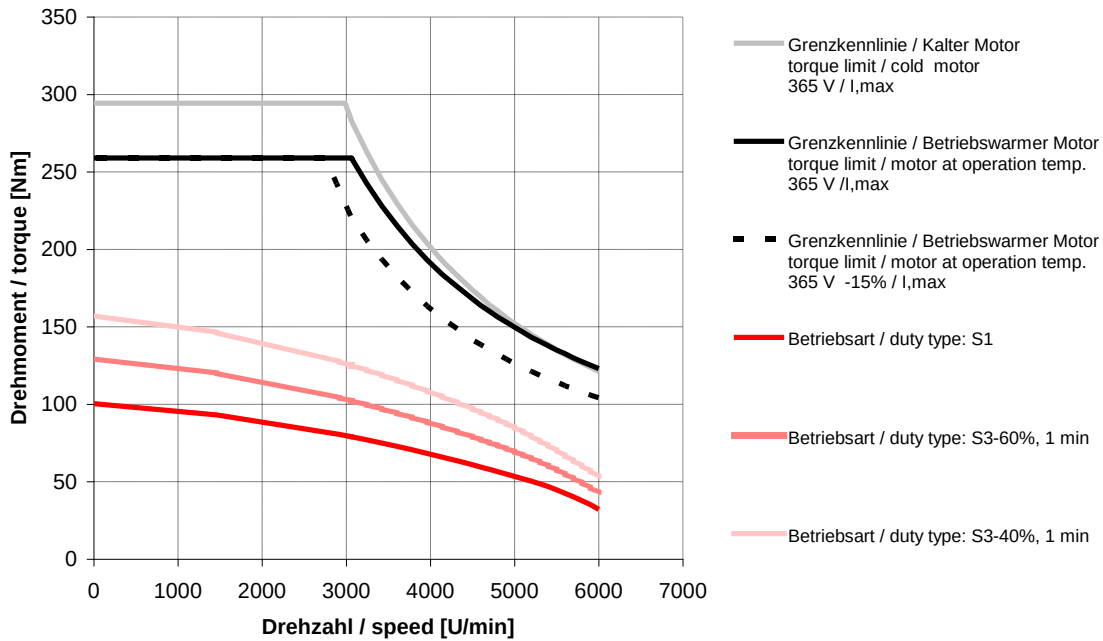
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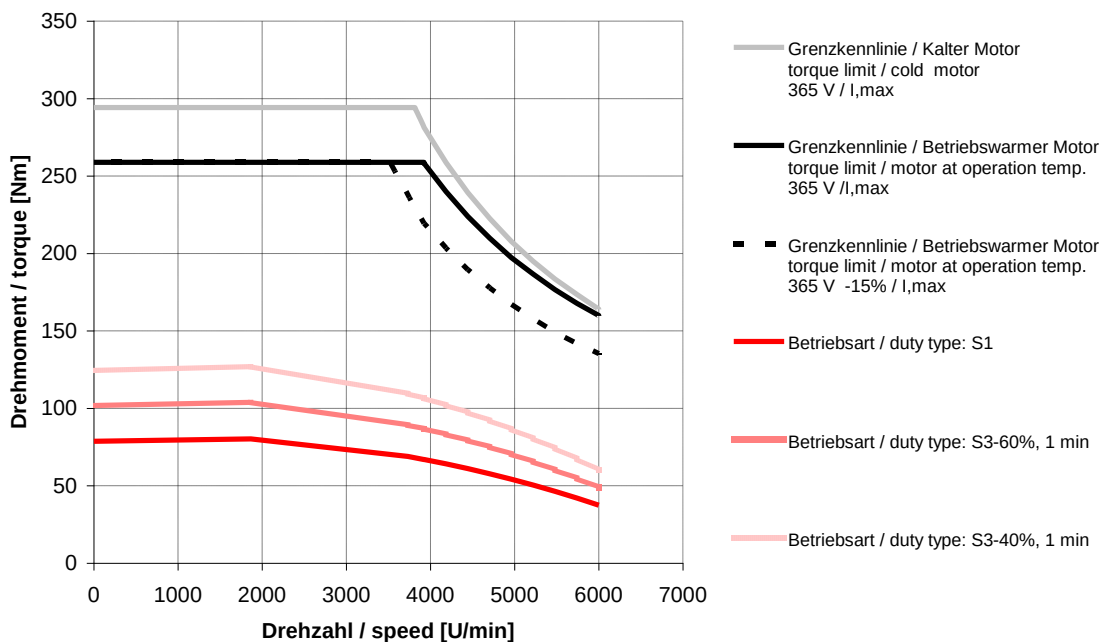
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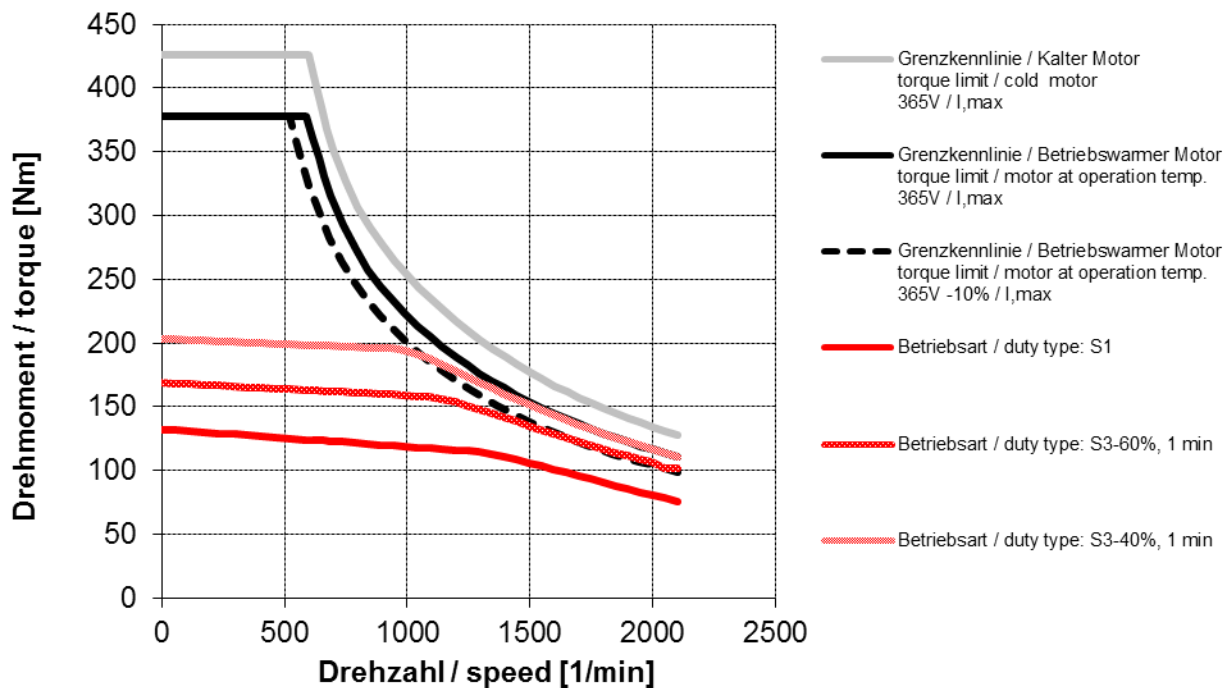
DSD2-100BO64O-45-54



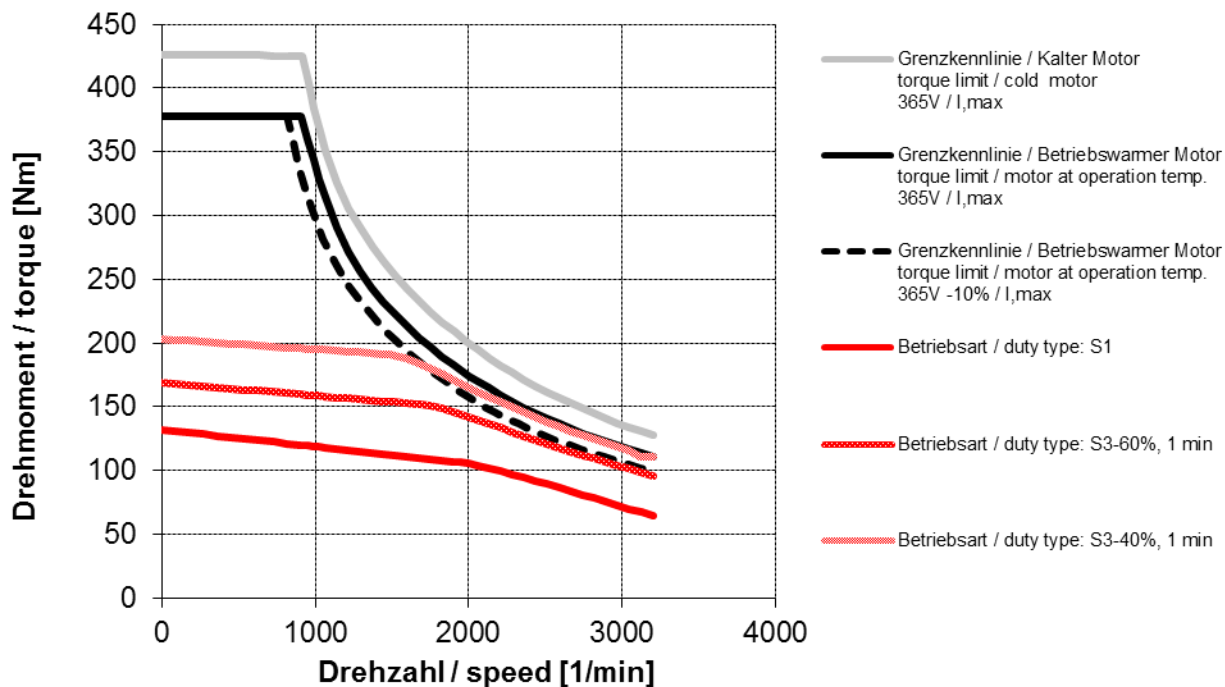
DSD2-100BO64O-60-54



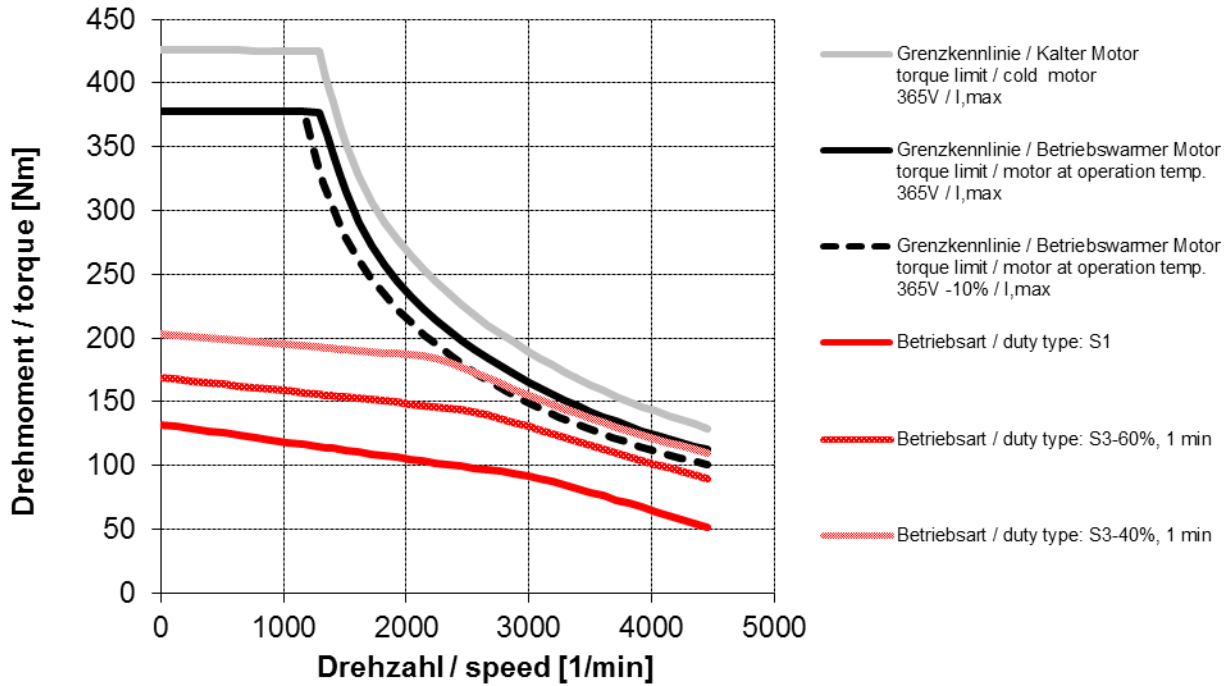
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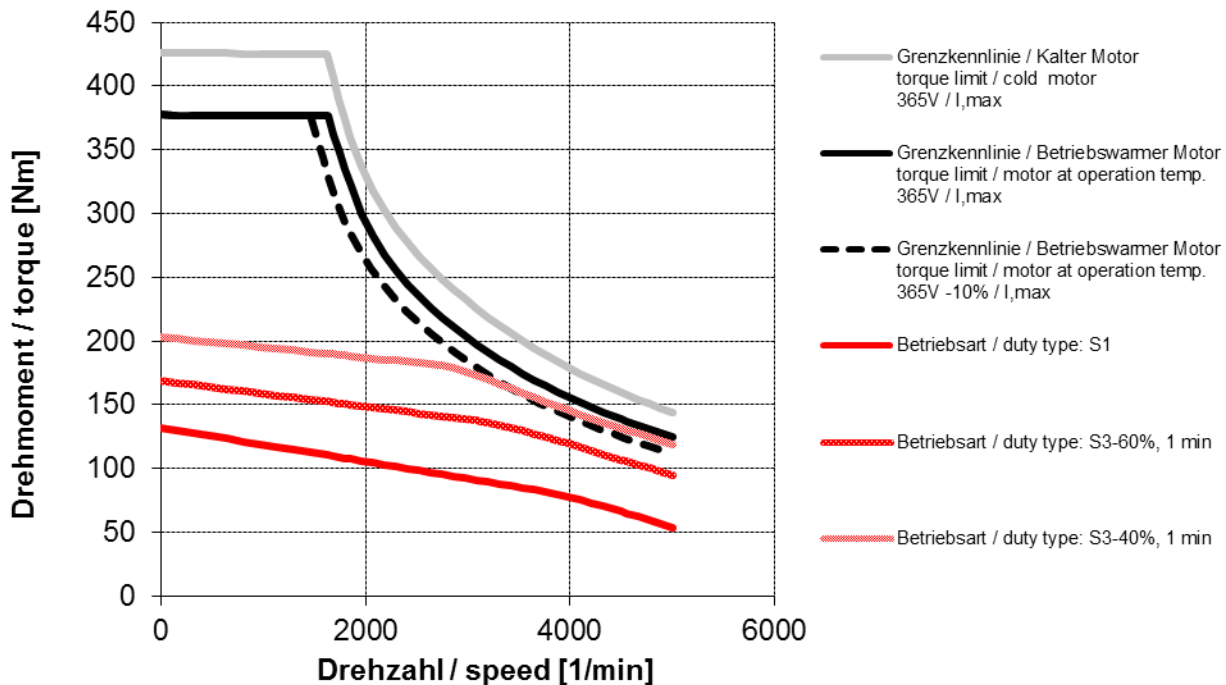
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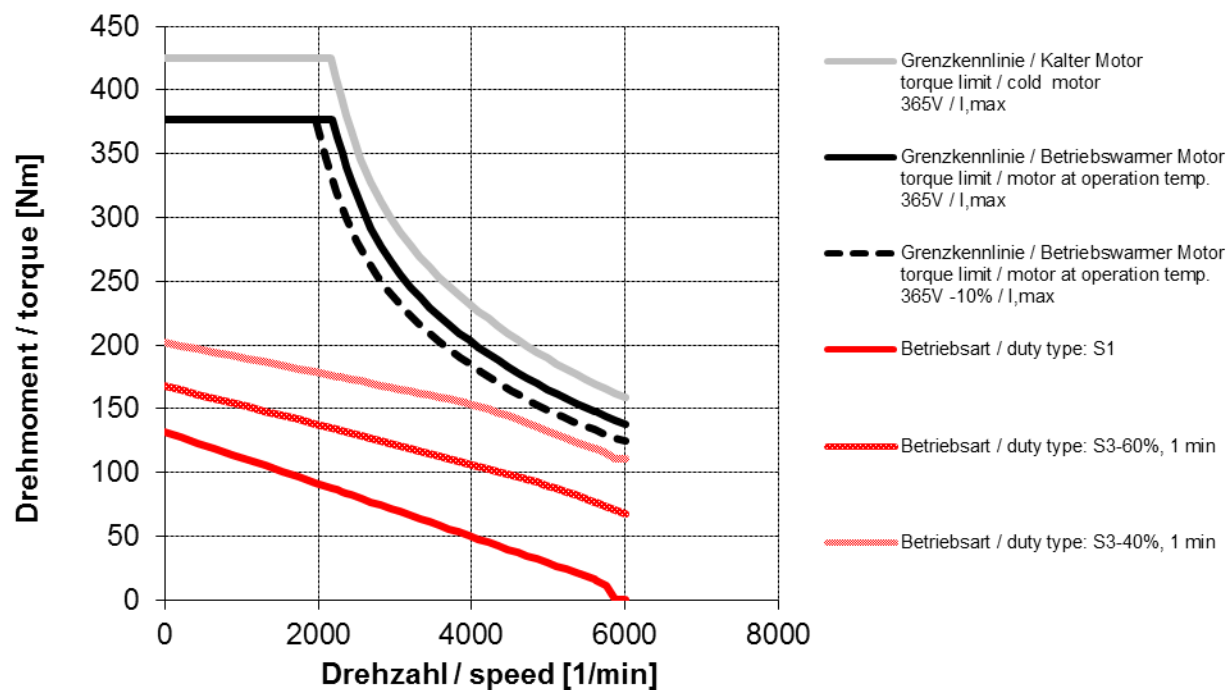
DSD2-100XO64O-30-54



DSD2-100XO64O-45-54

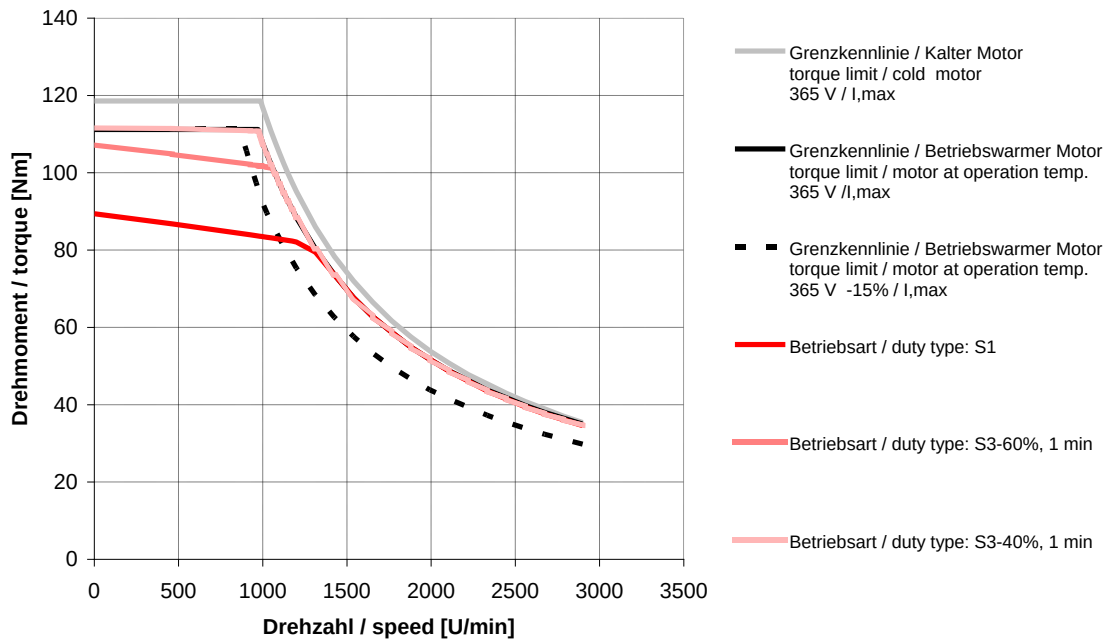


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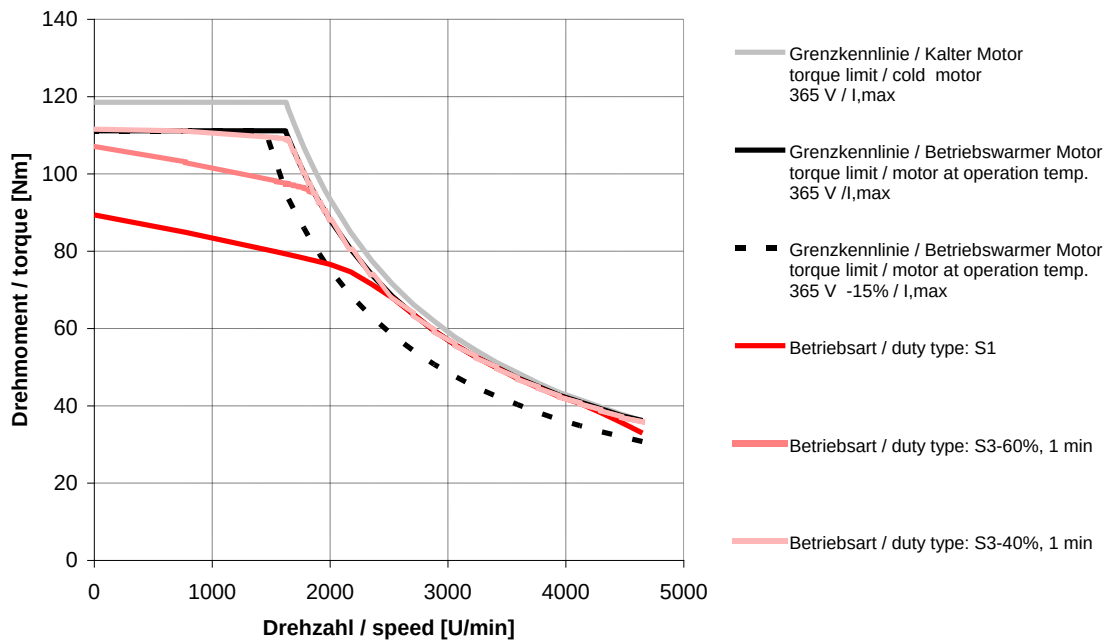


5.6.3. DSD2-100..64W-.. (water cooled)

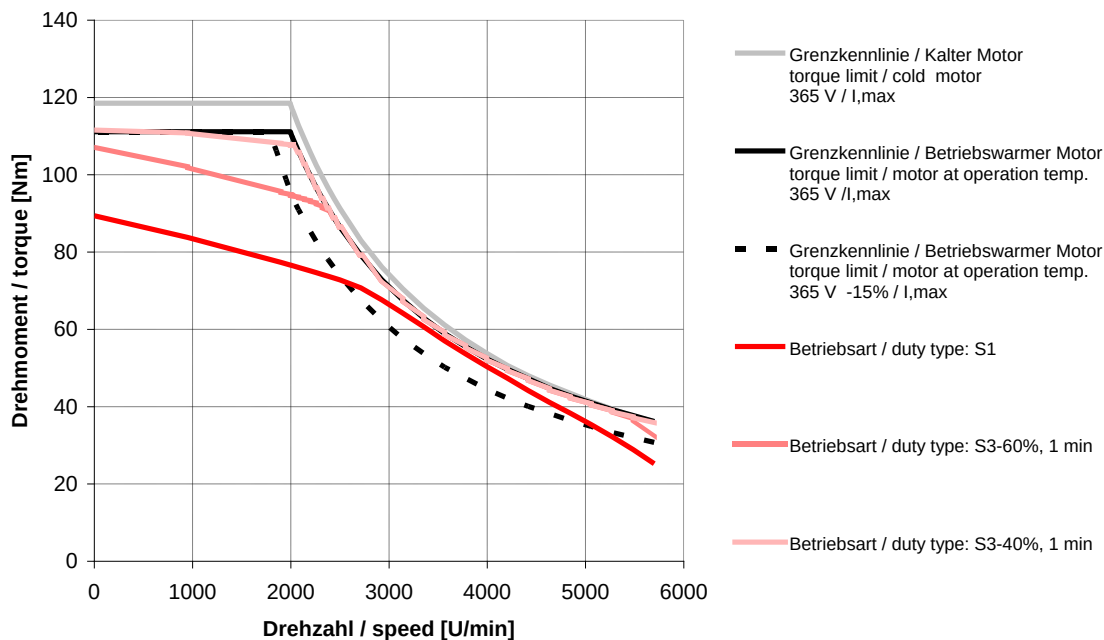
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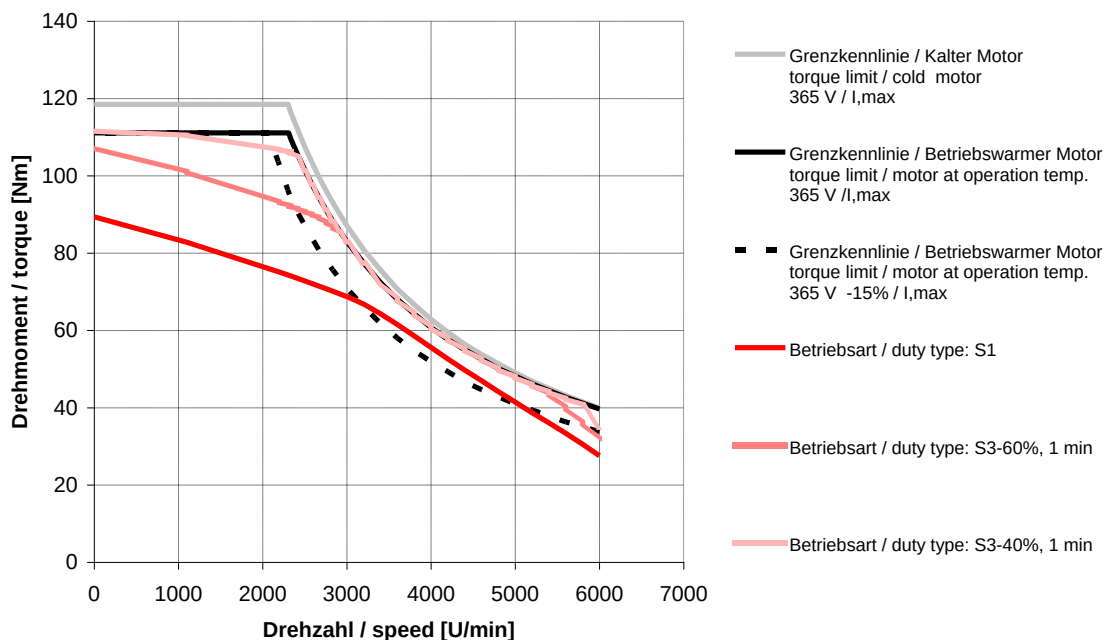
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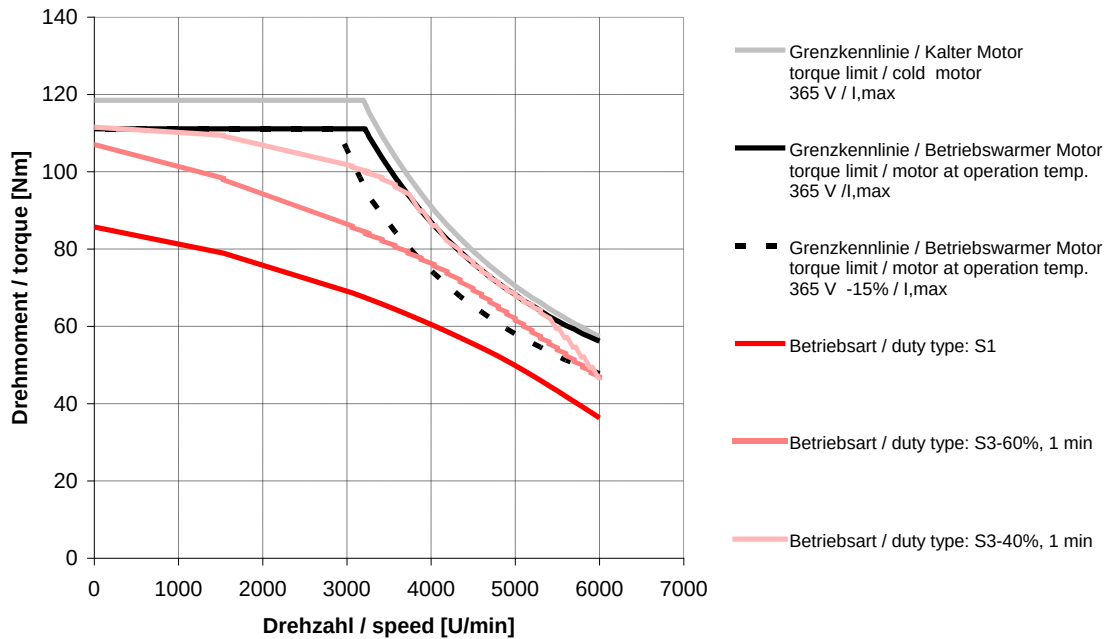
DSD2-100SO64W-25-54



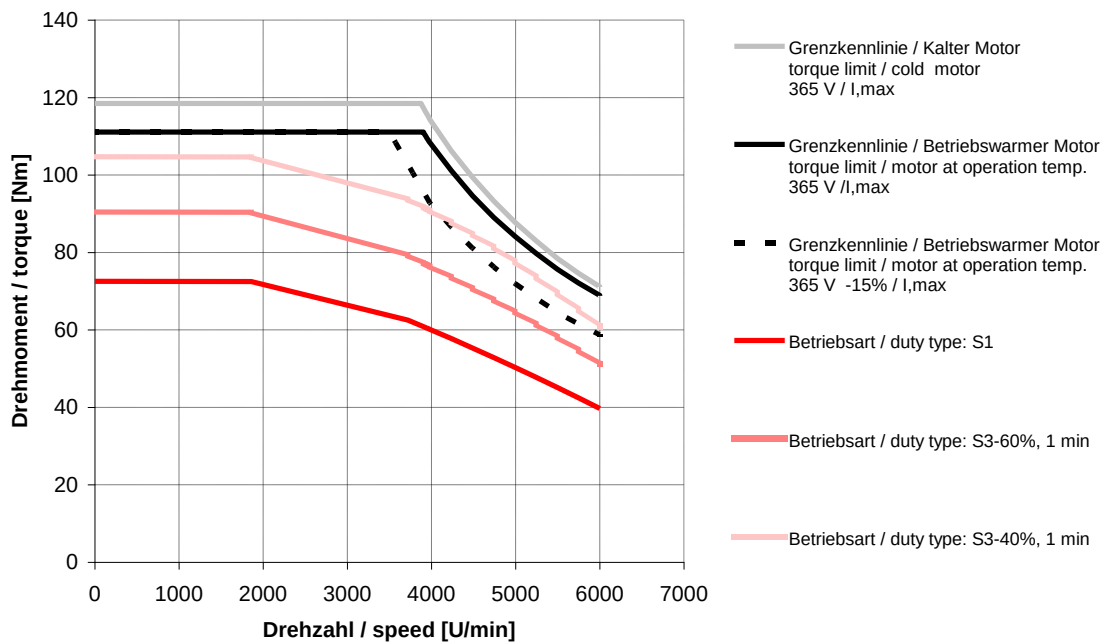
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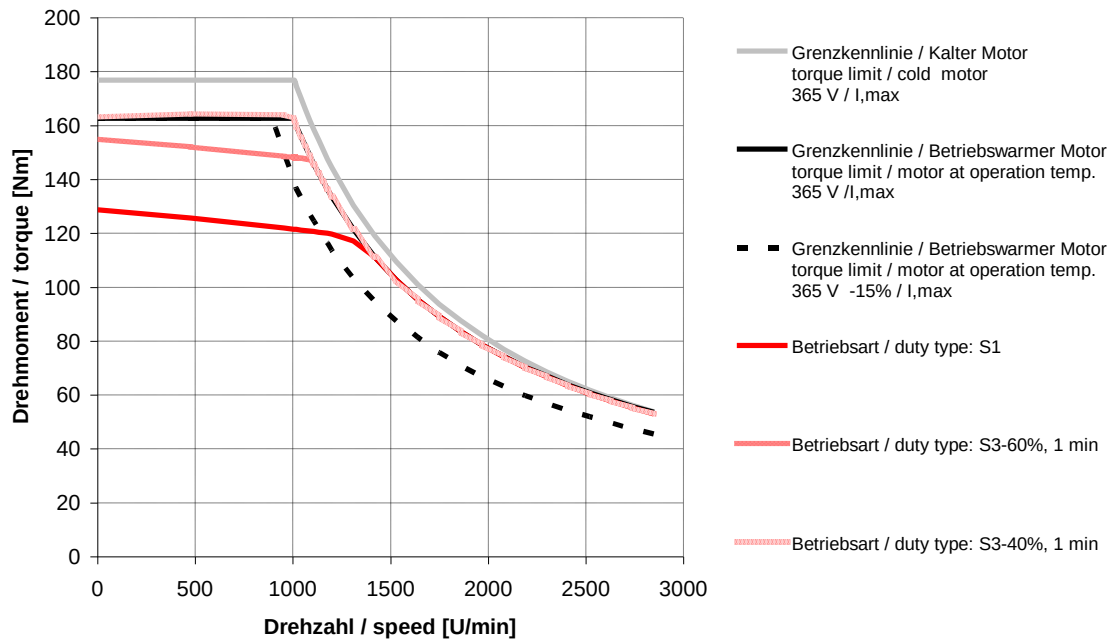
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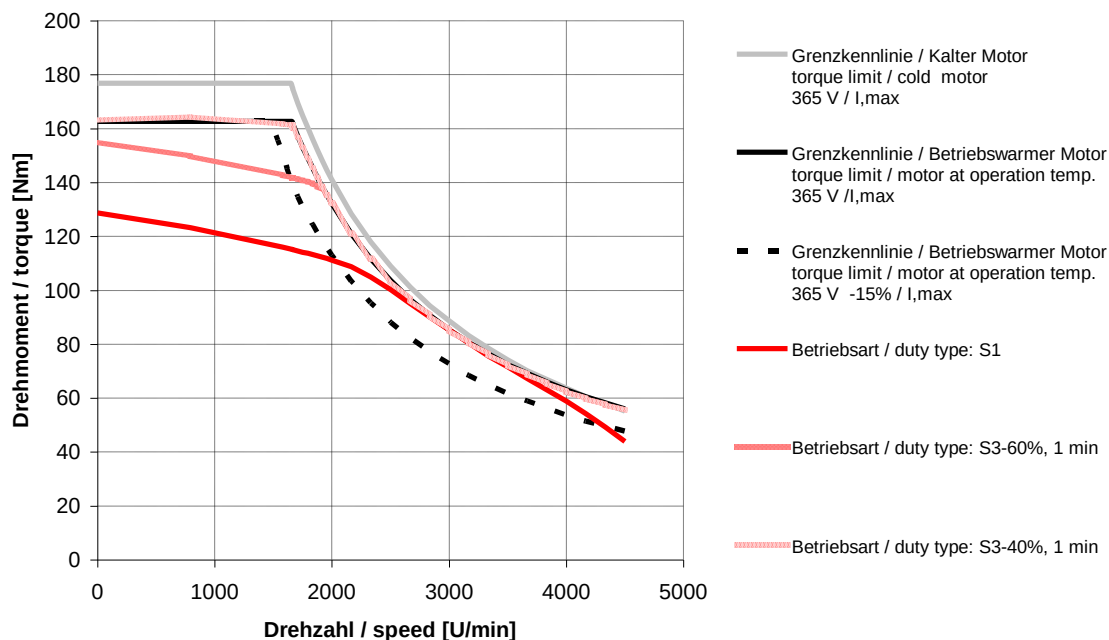
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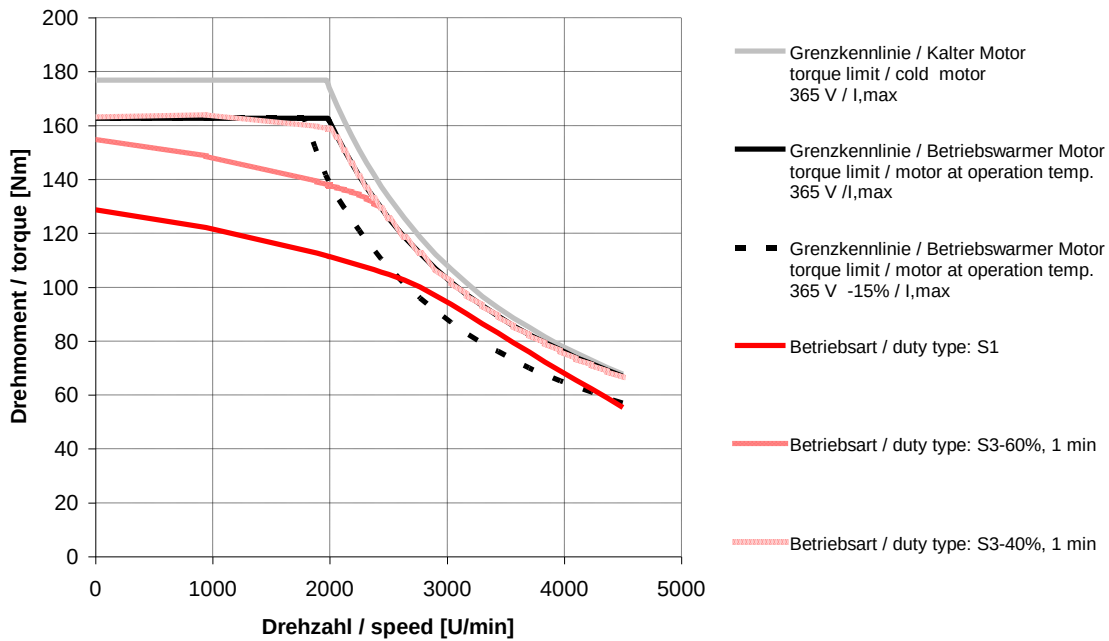
DSD2-100MO64W-12-54



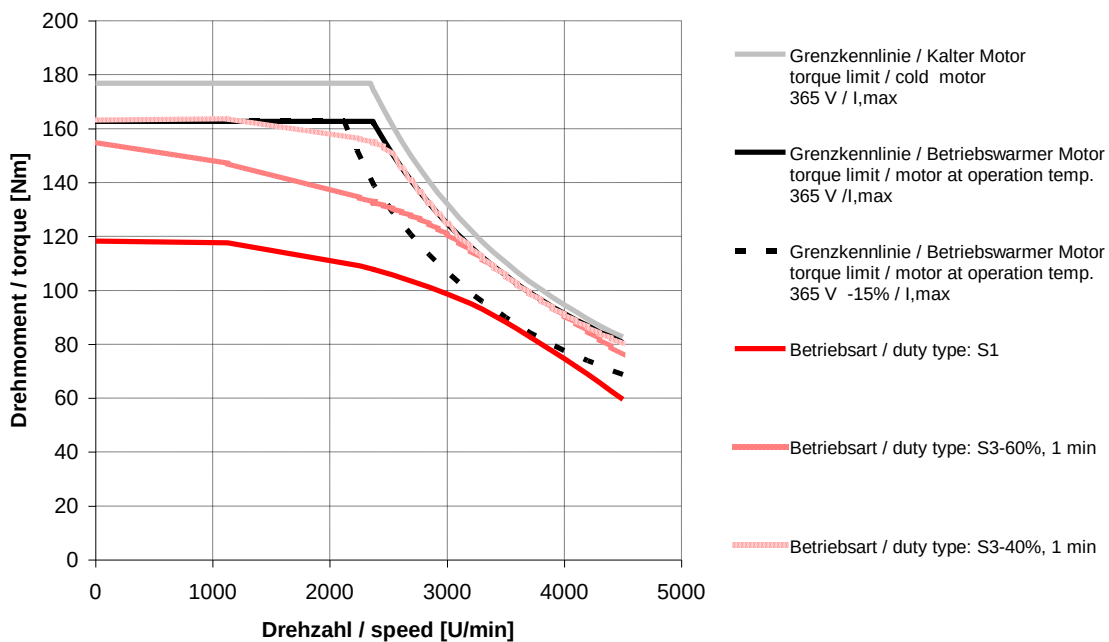
DSD2-100MO64W-20-54



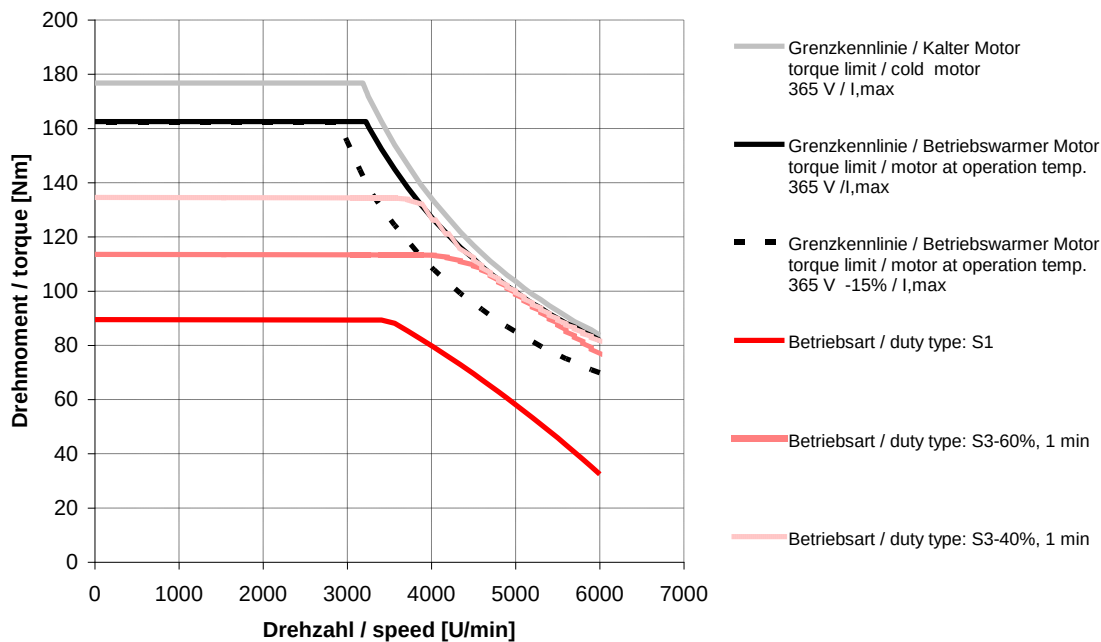
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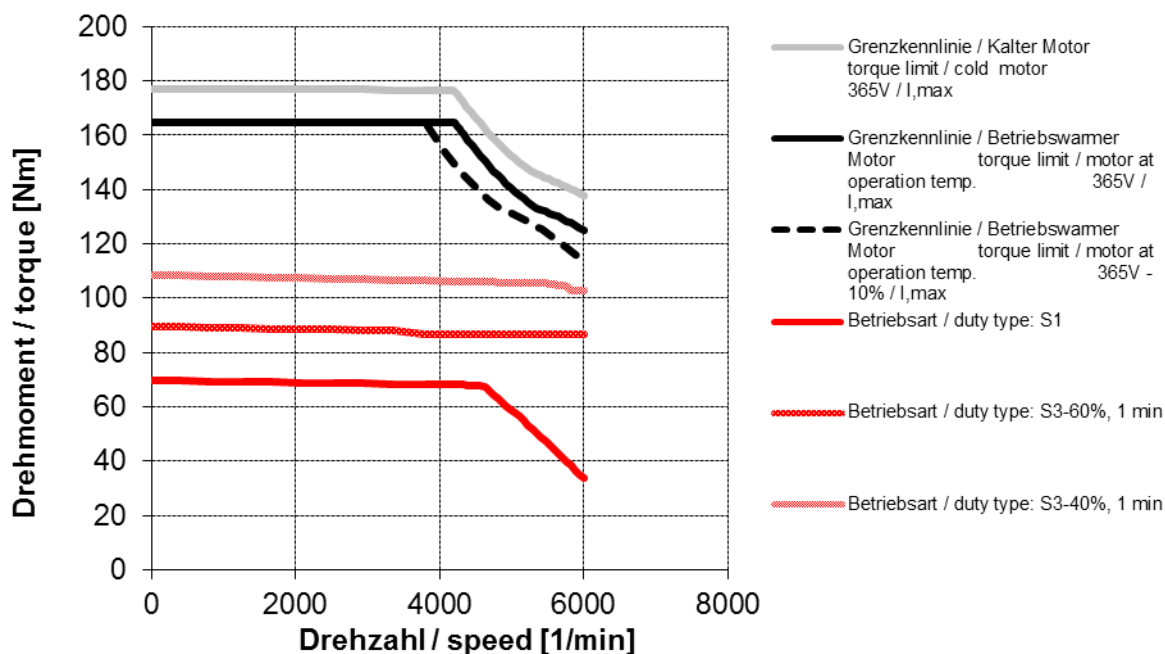
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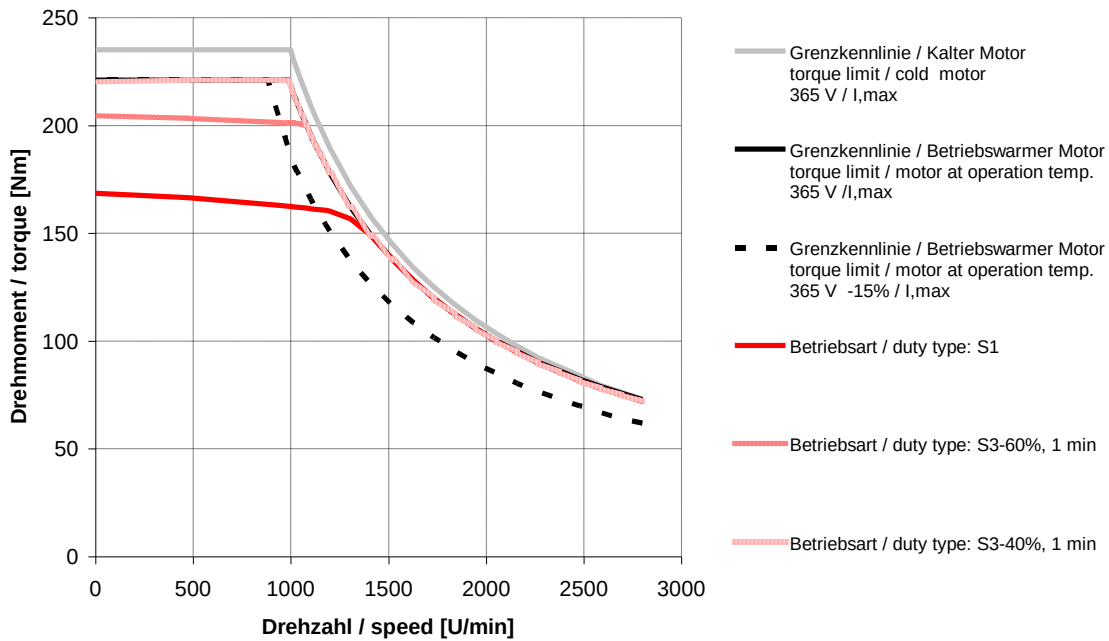
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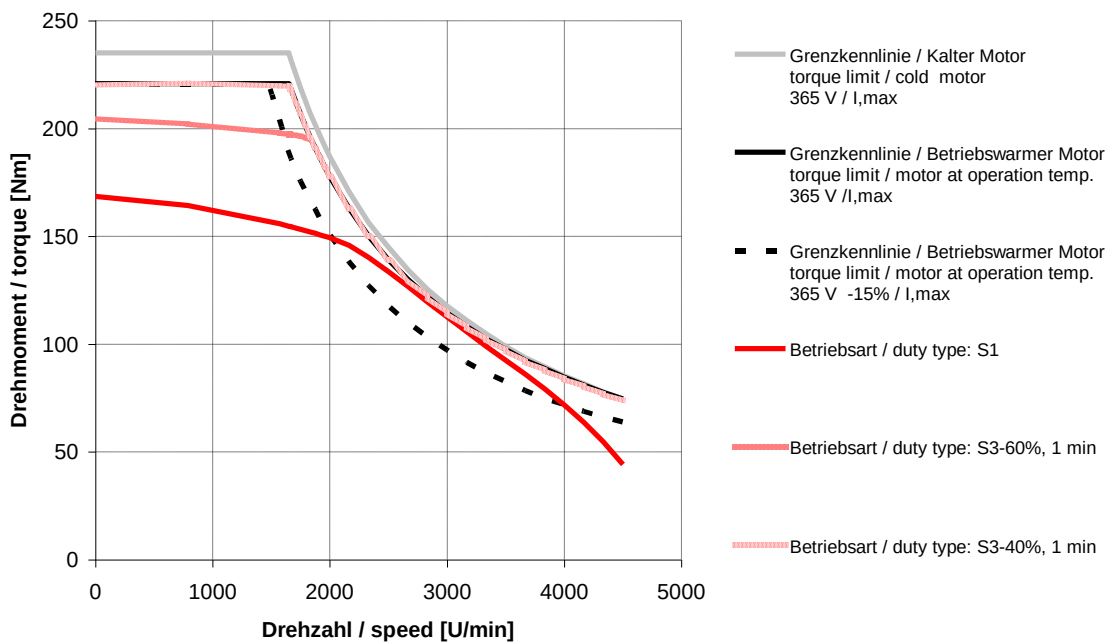
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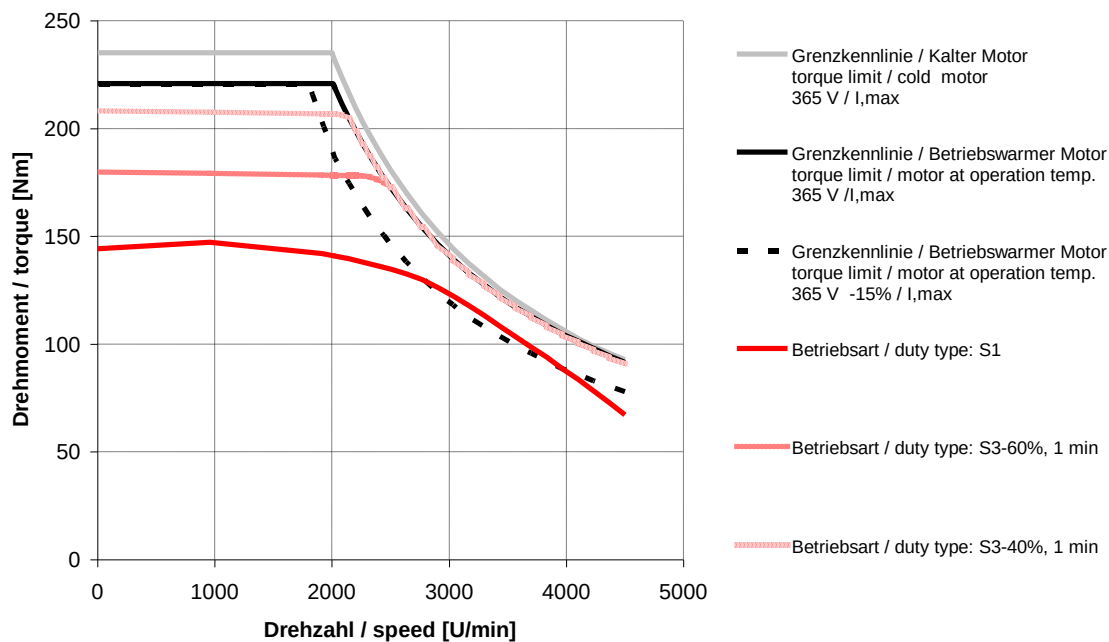
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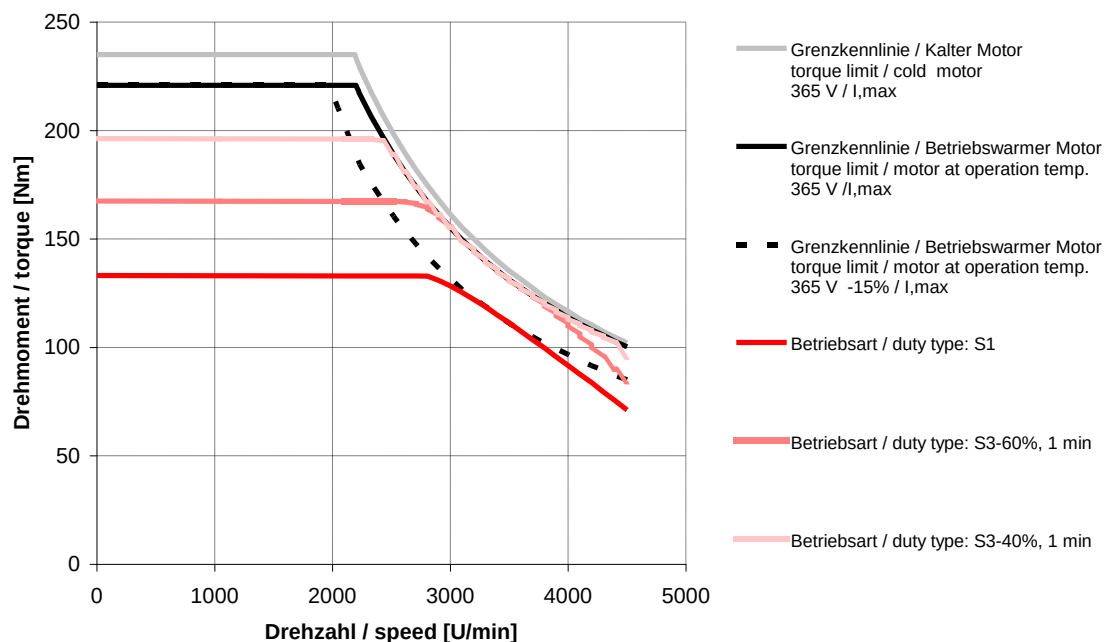
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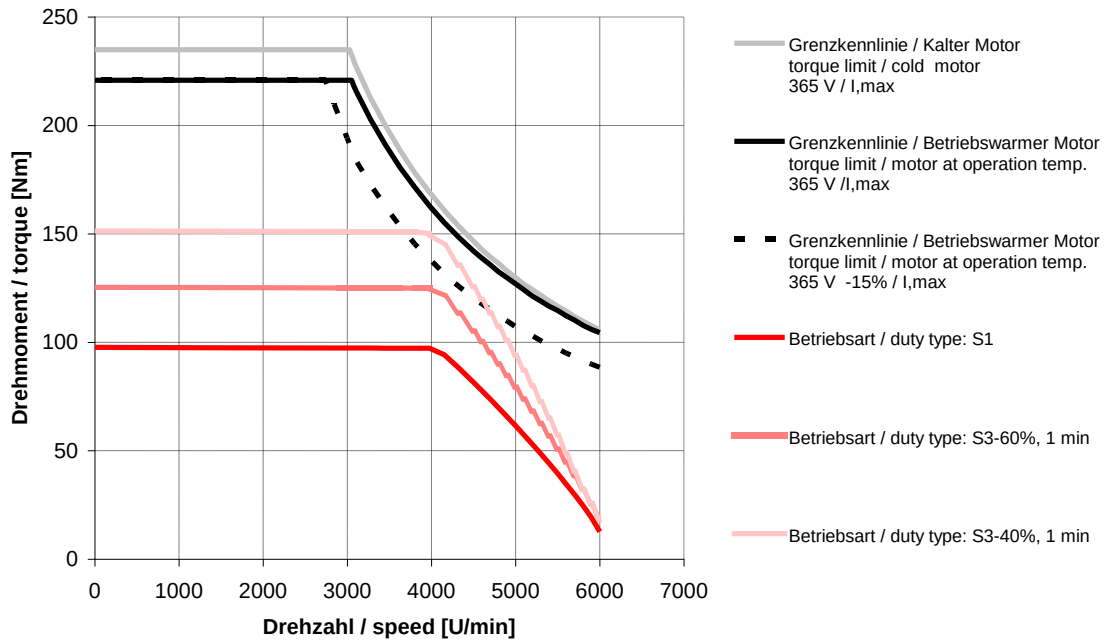
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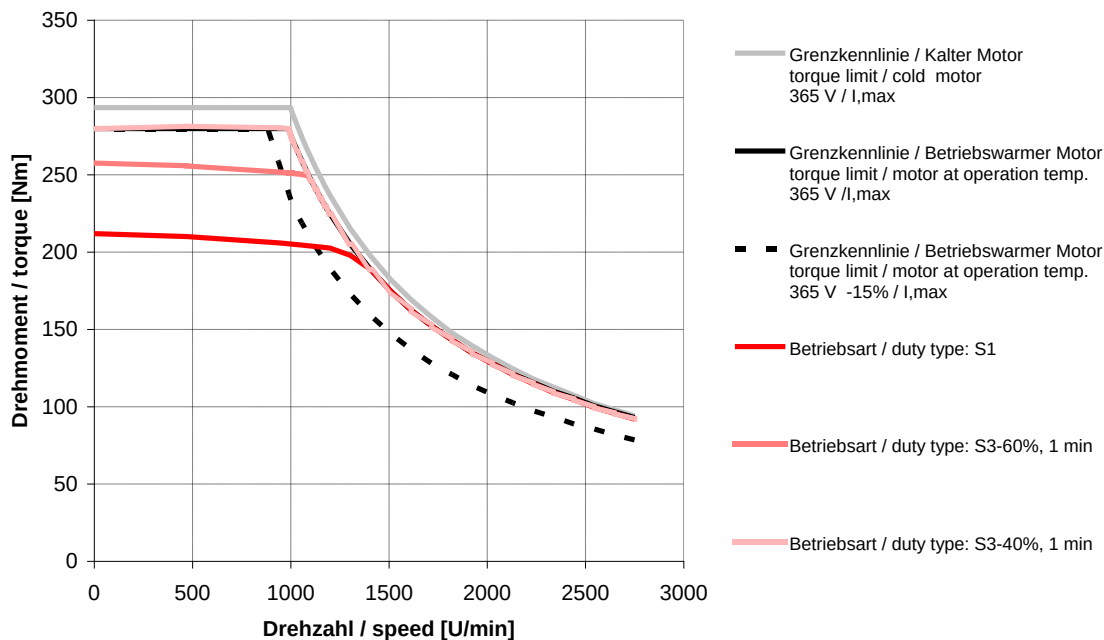
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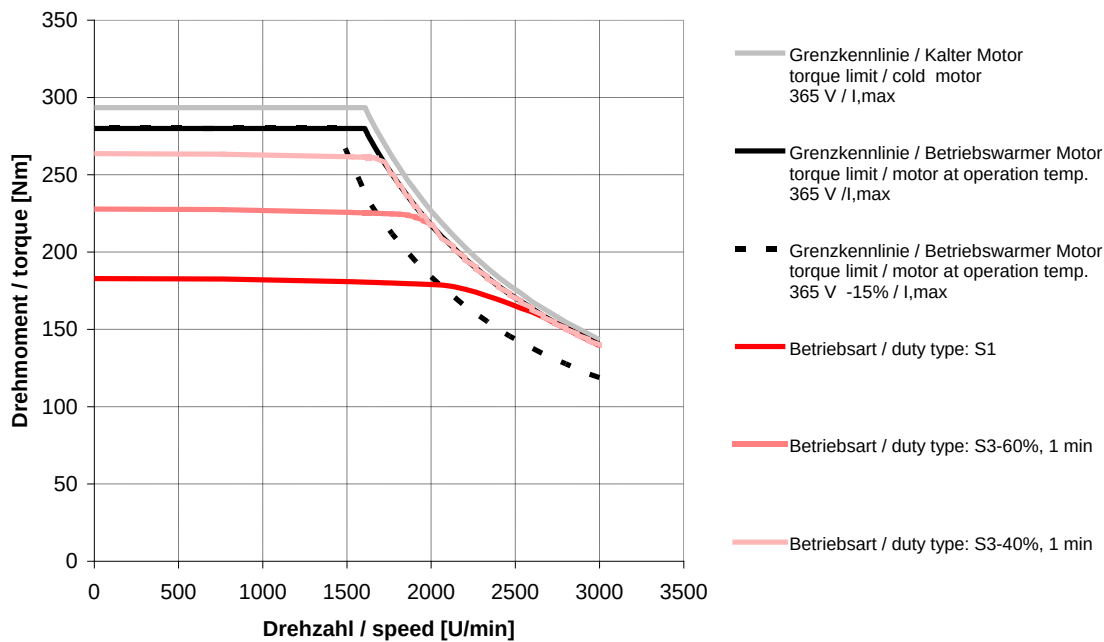
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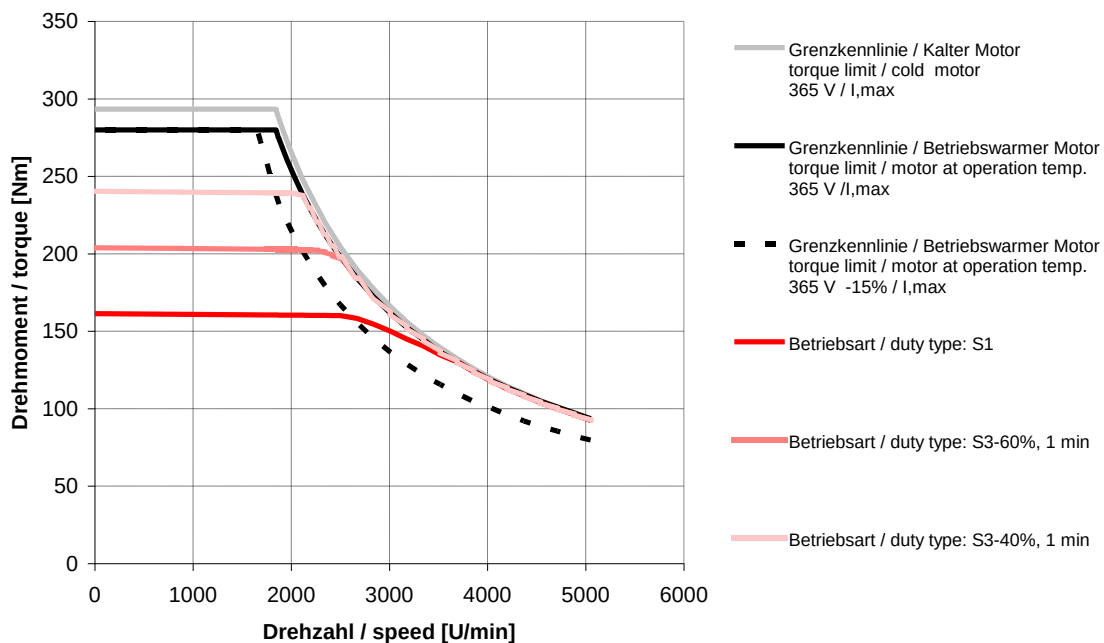
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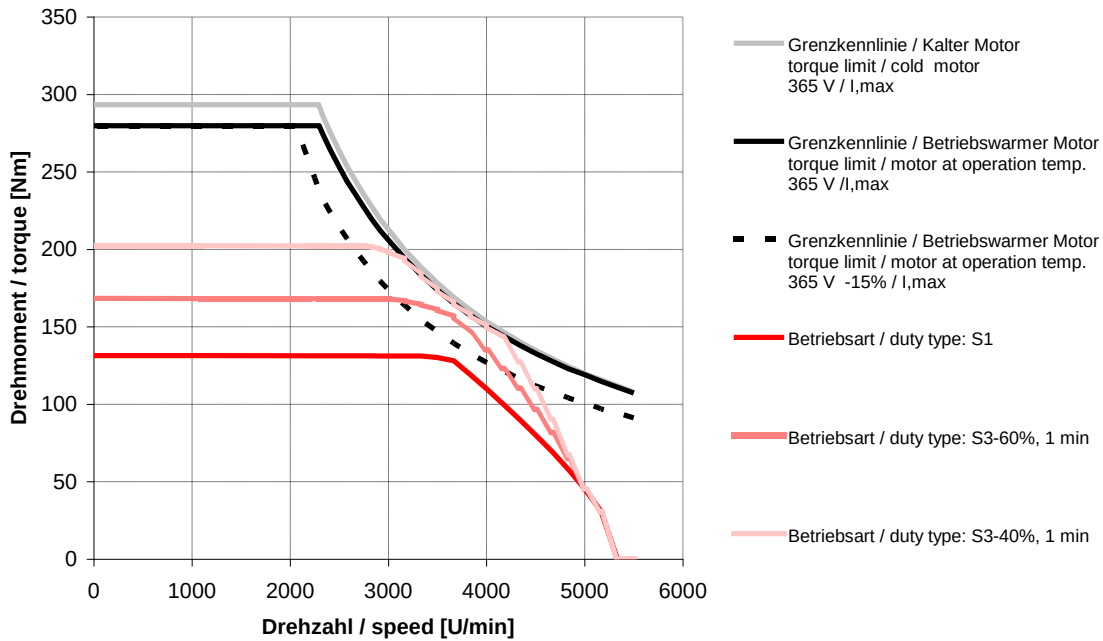
DSD2-100BO64W-20-54



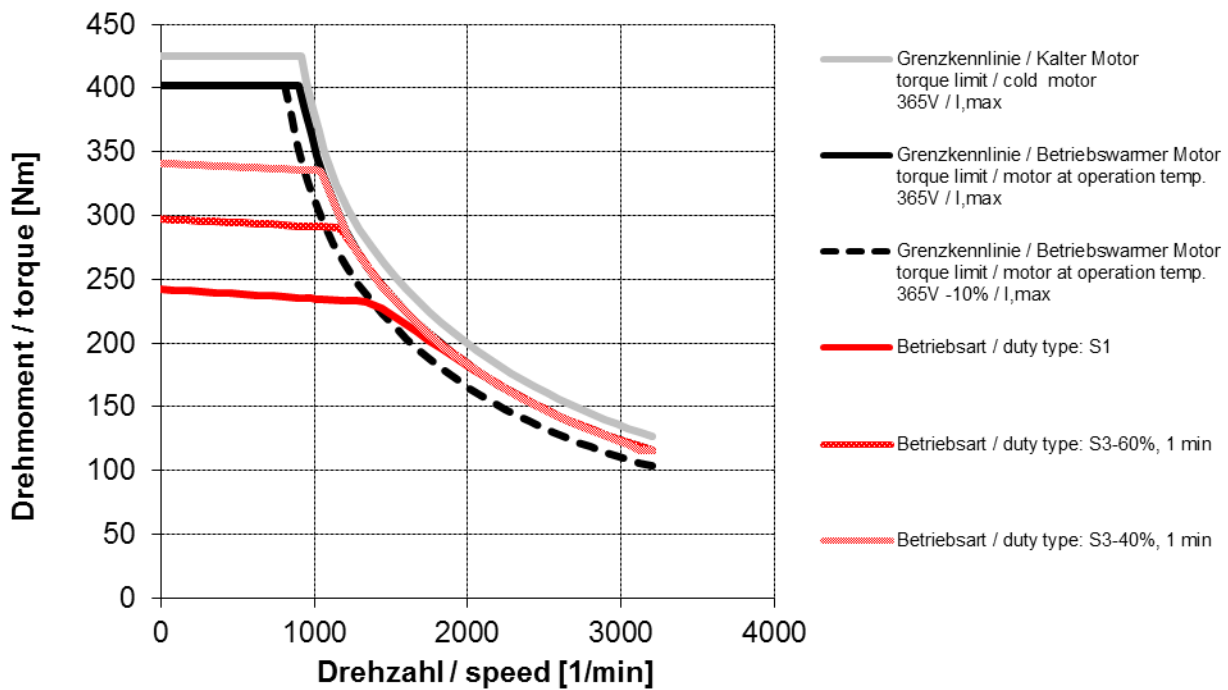
DSD2-100BO64W-25-54



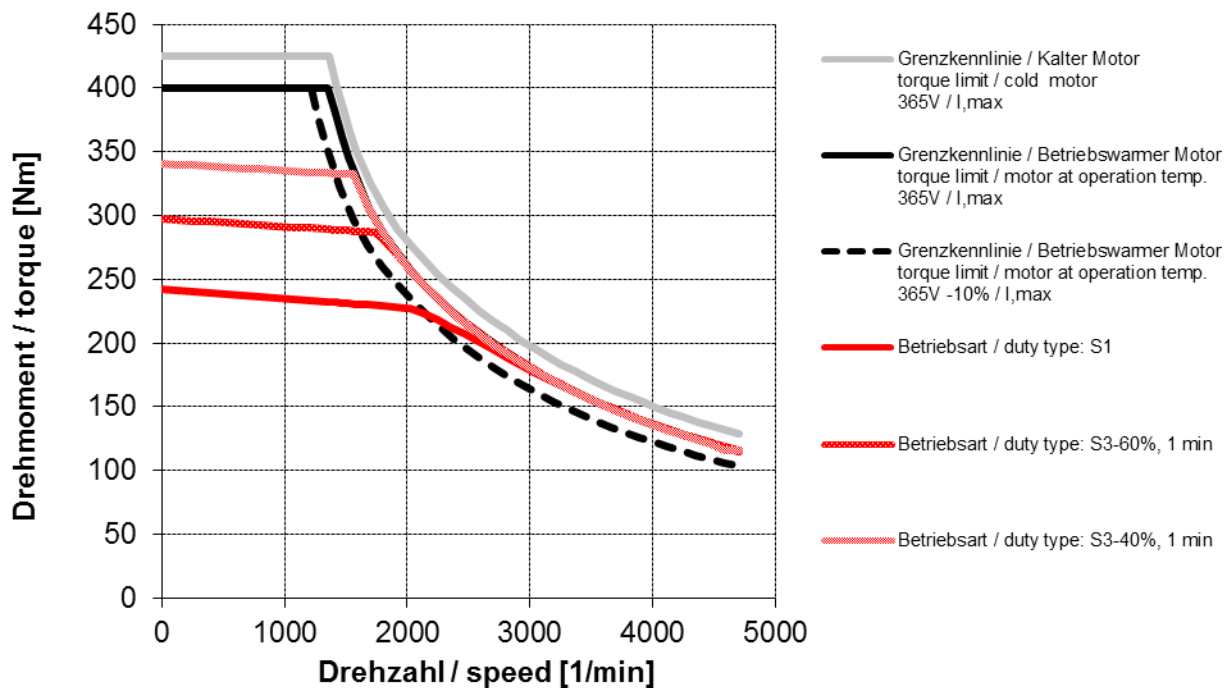
DSD2-100BO64W-30-54



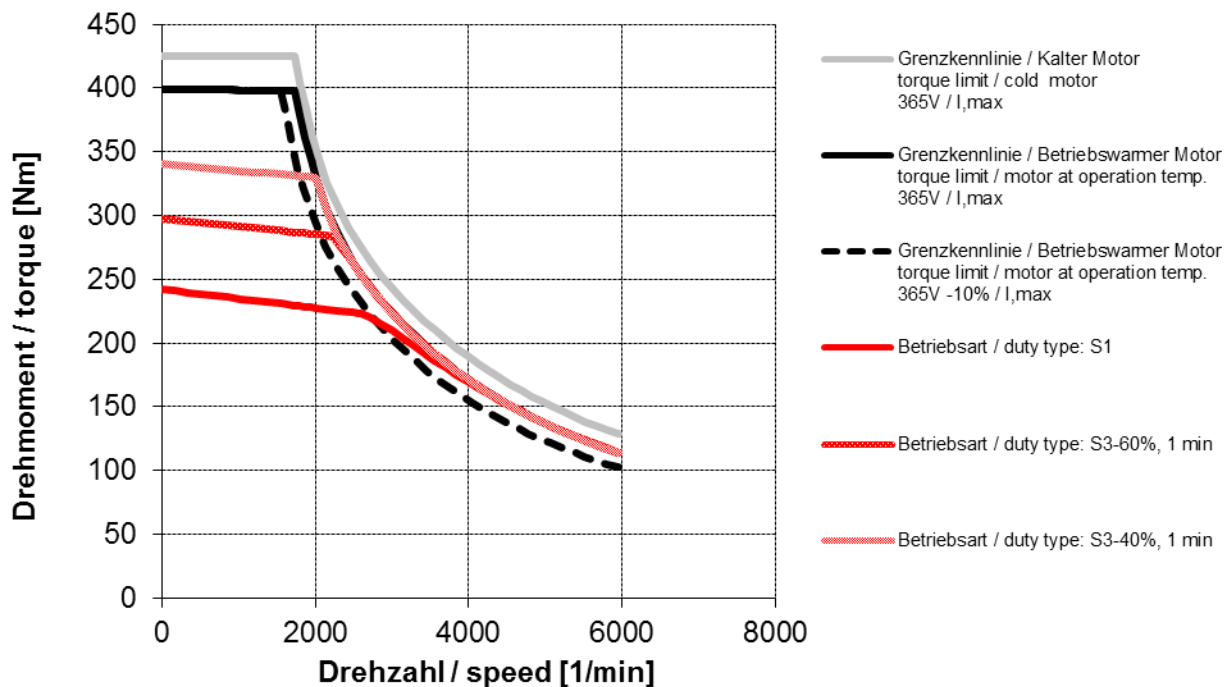
DSD2-100XO64W-12-54



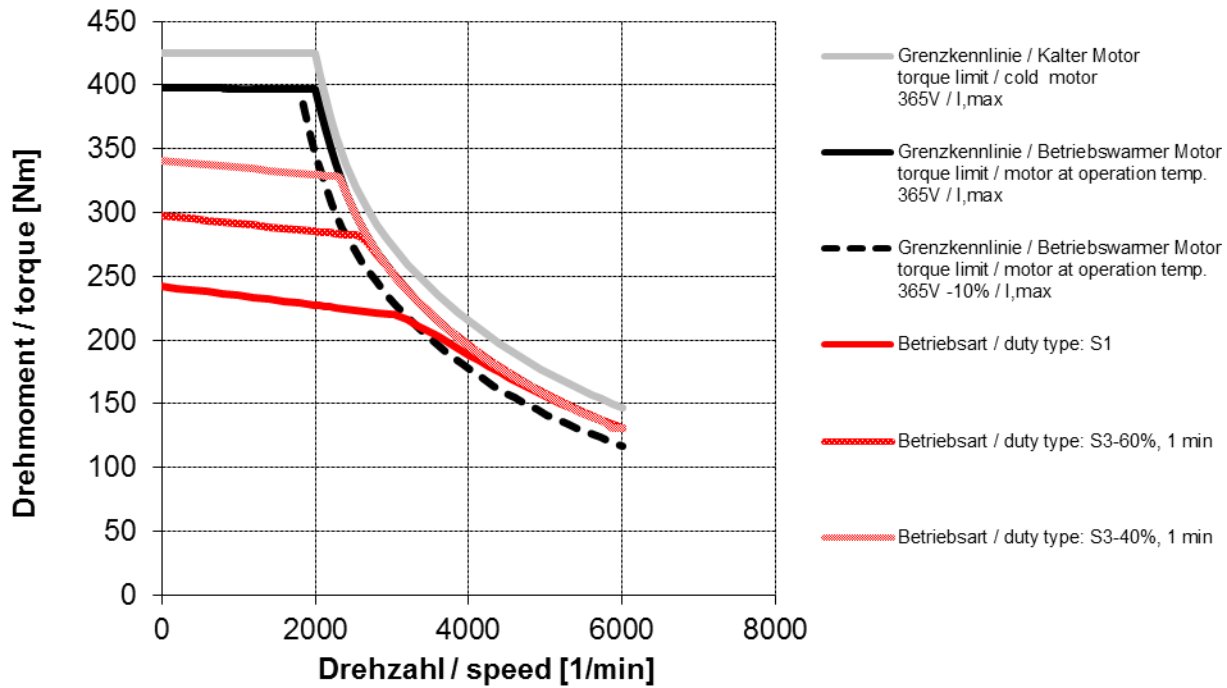
DSD2-100XO64W-20-54



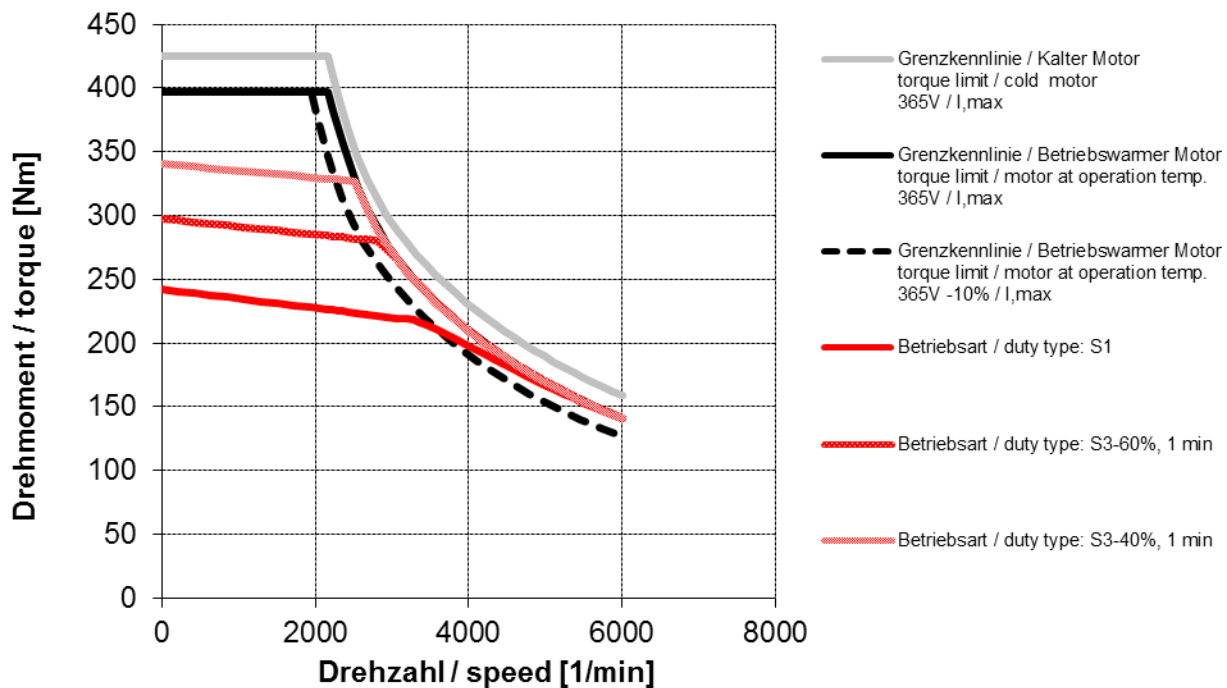
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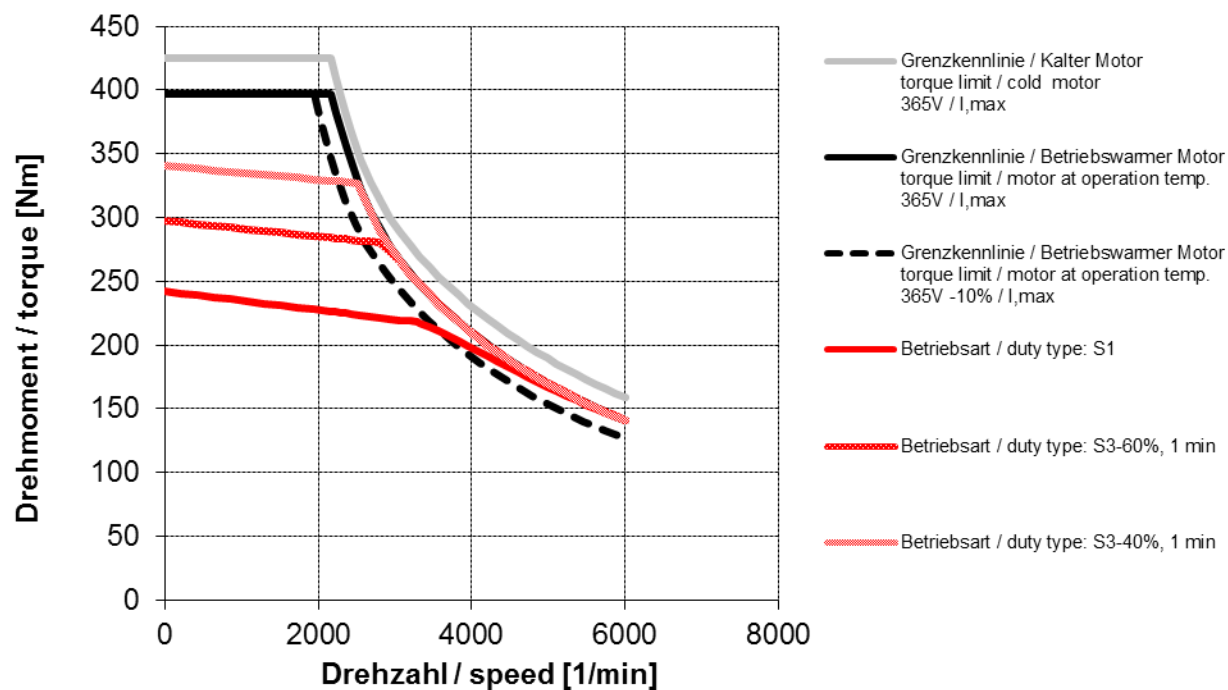
DSD2-100XO64W-30-54



DSD2-100XO64W-45-54



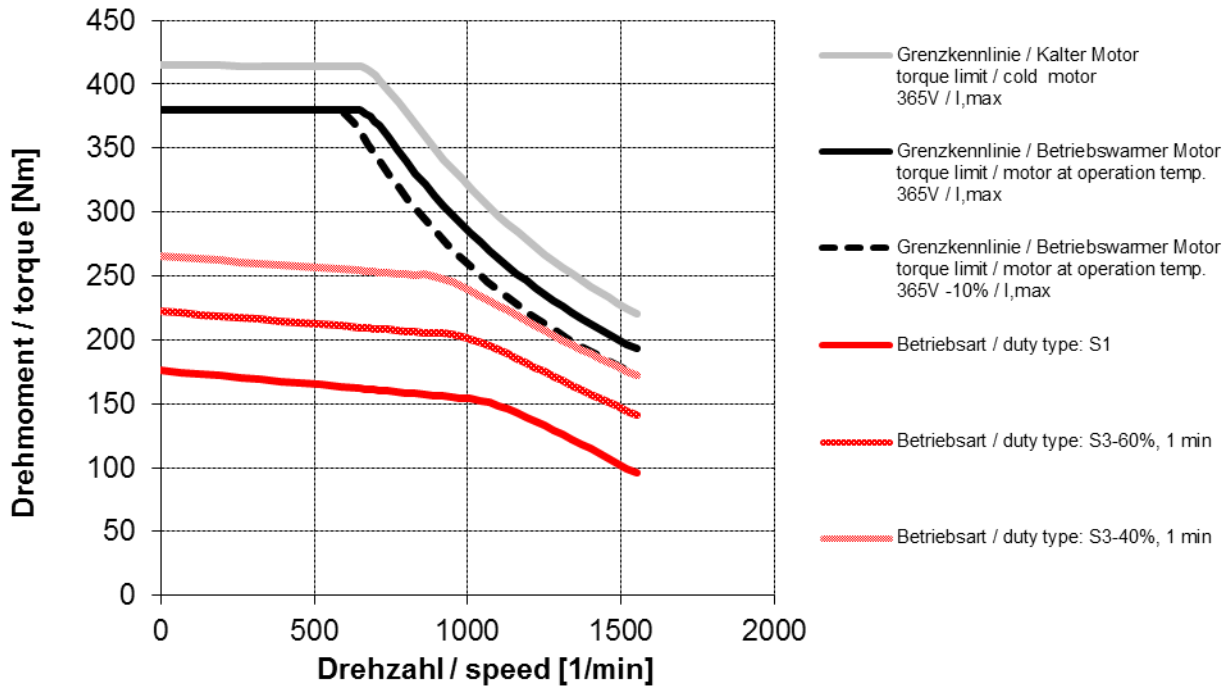
DSD2-100XO64W-60-54



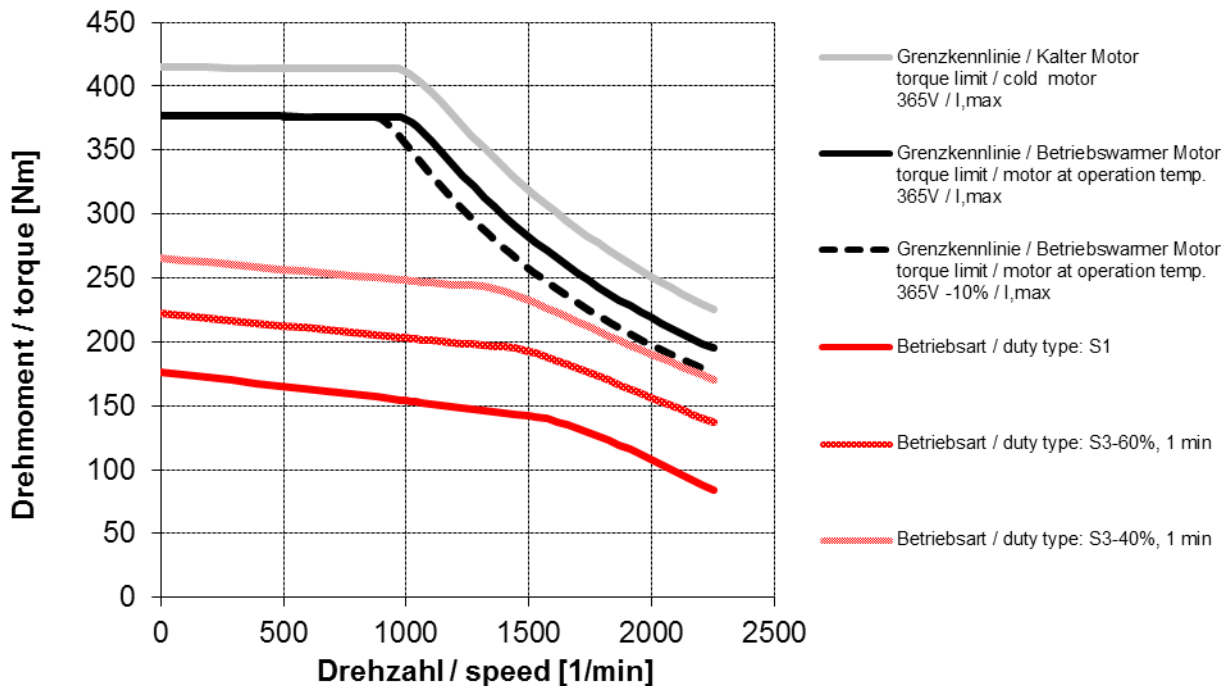
5.7. Motor characteristic curve DSD2-132

5.7.1. DSD2-132..54A(R)-..(IP54 surface cooled)

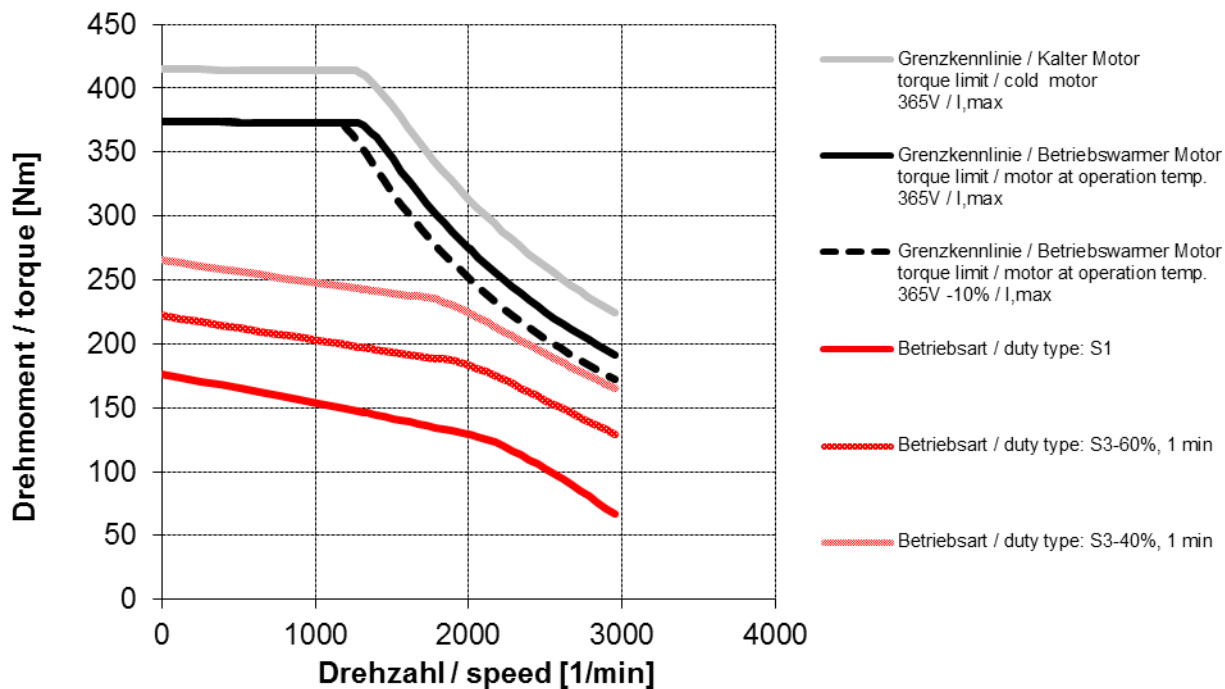
DSD2-132KO54A(R)-10-5



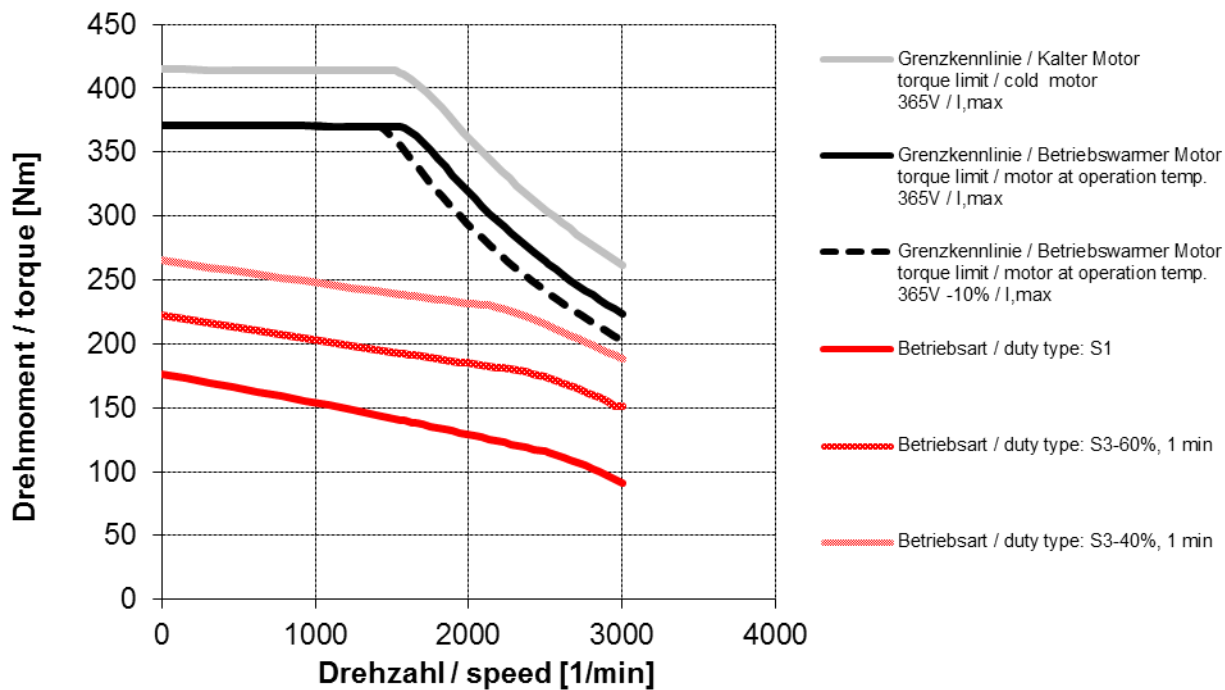
DSD2-132KO54A(R)-15-5



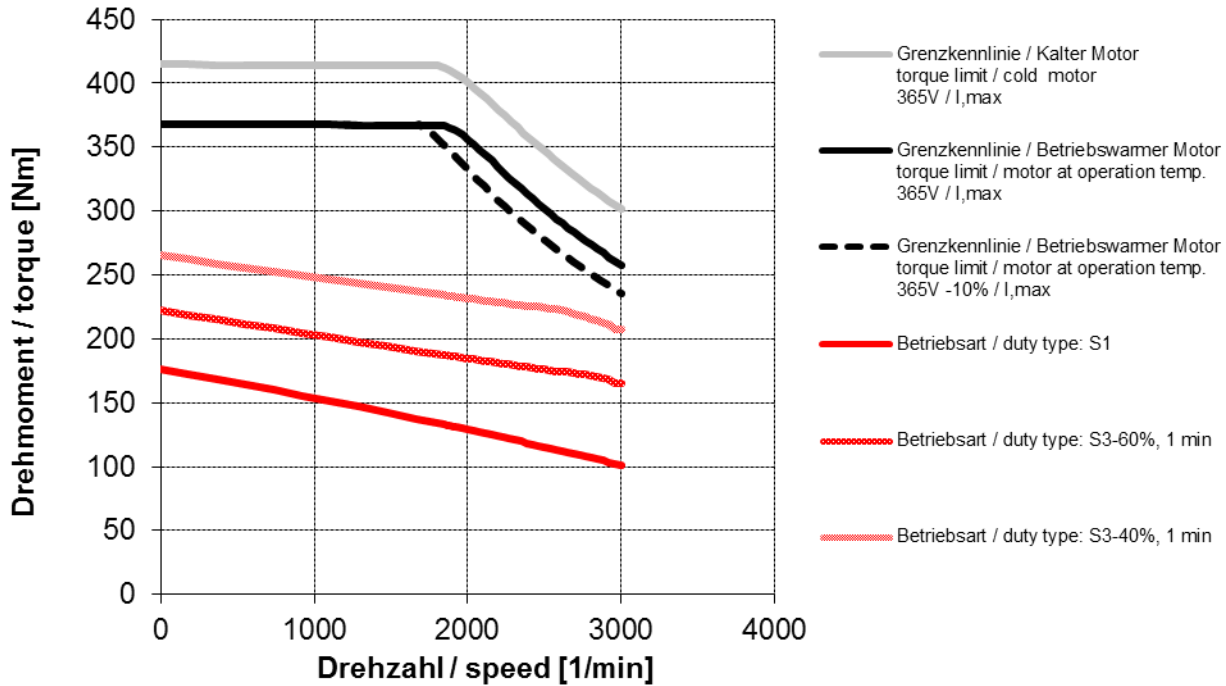
DSD2-132K054A(R)-20-5



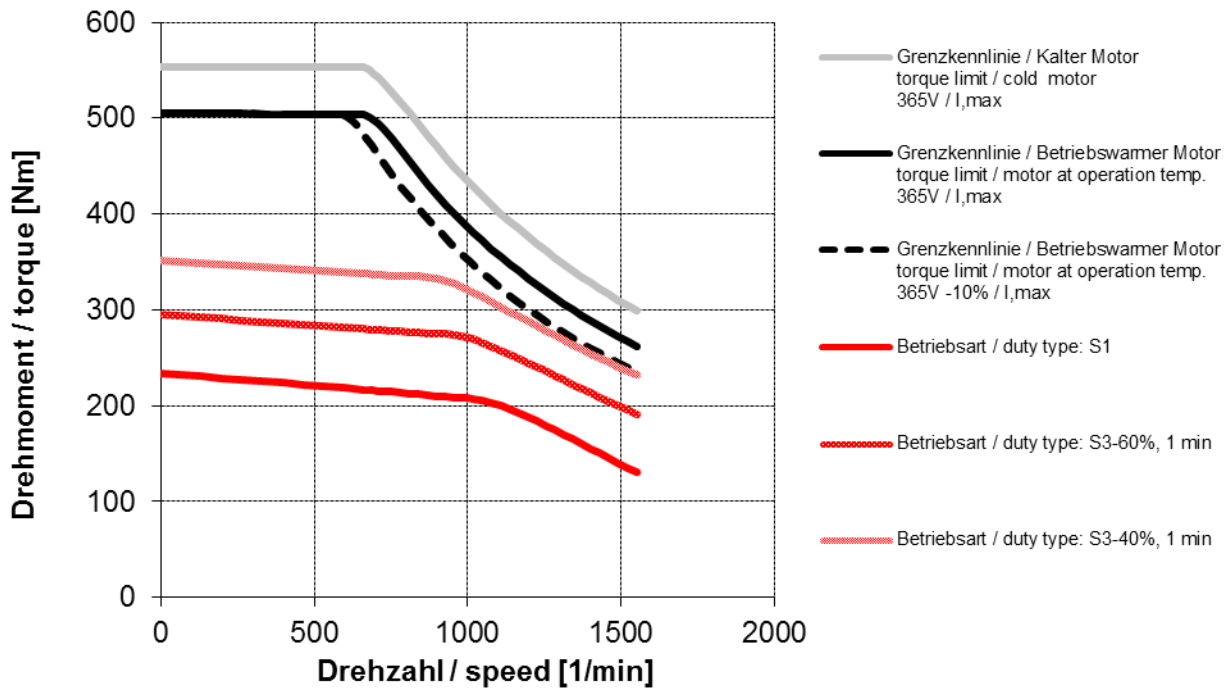
DSD2-132K054A(R)-25-5



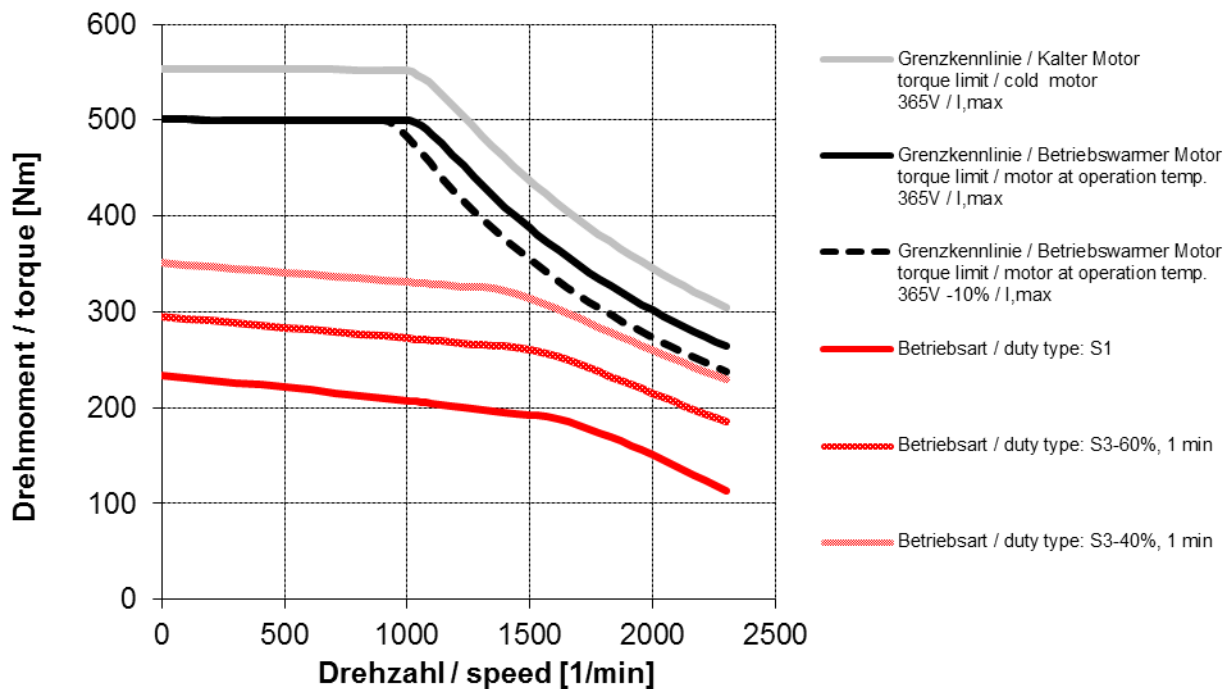
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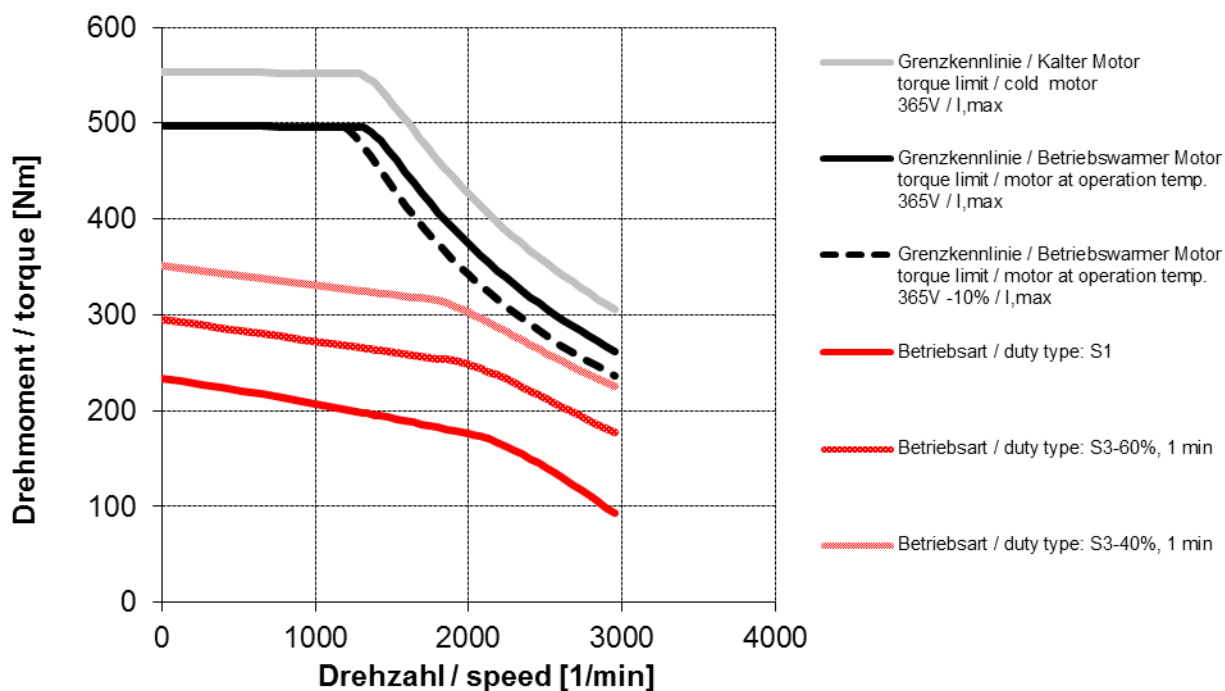
DSD2-132MO54A(R)-10-5



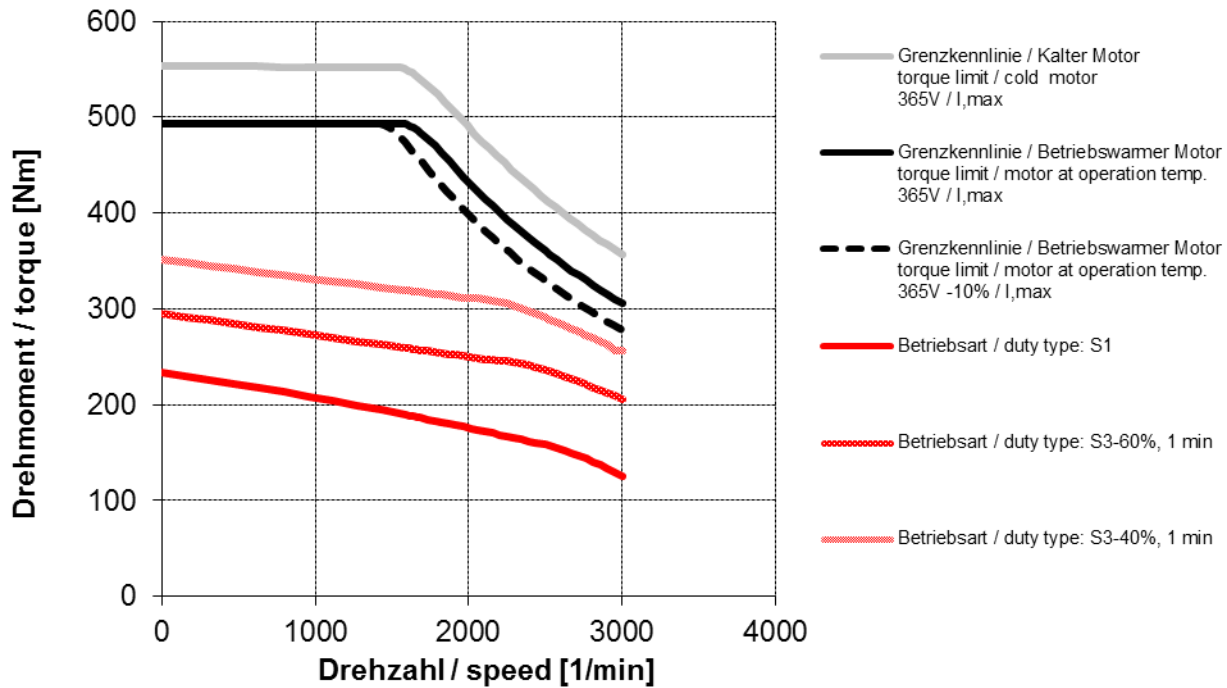
DSD2-132MO54A(R)-15-5



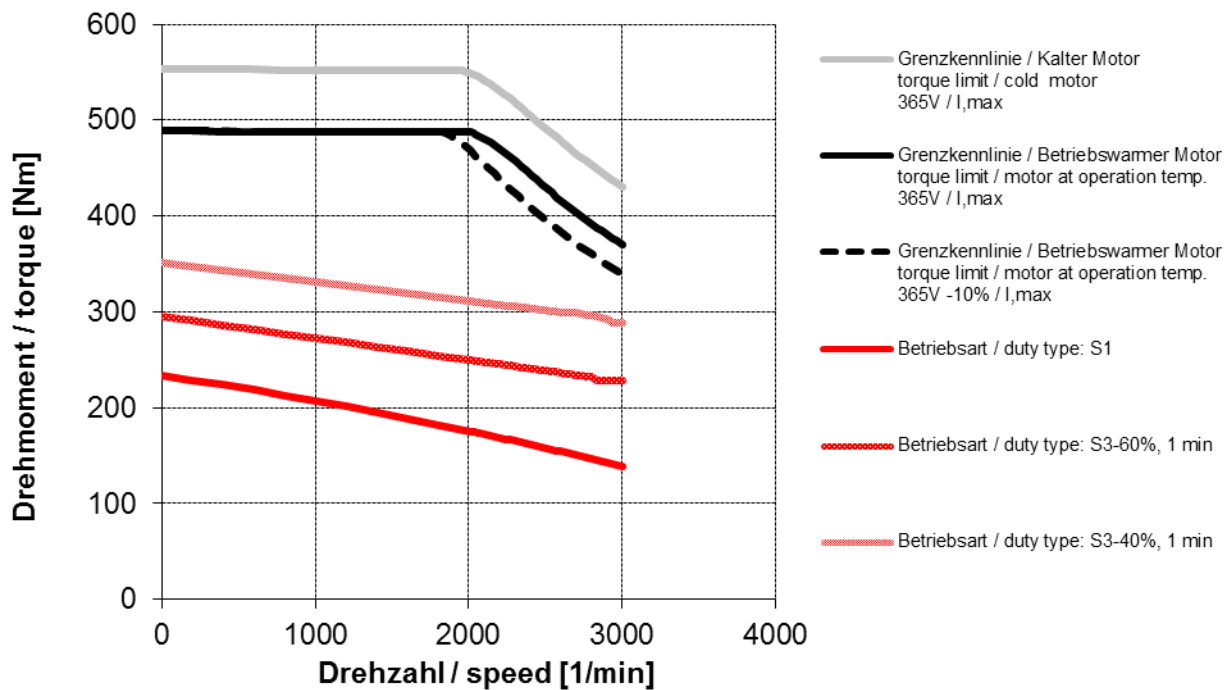
DSD2-132MO54A(R)-20-5



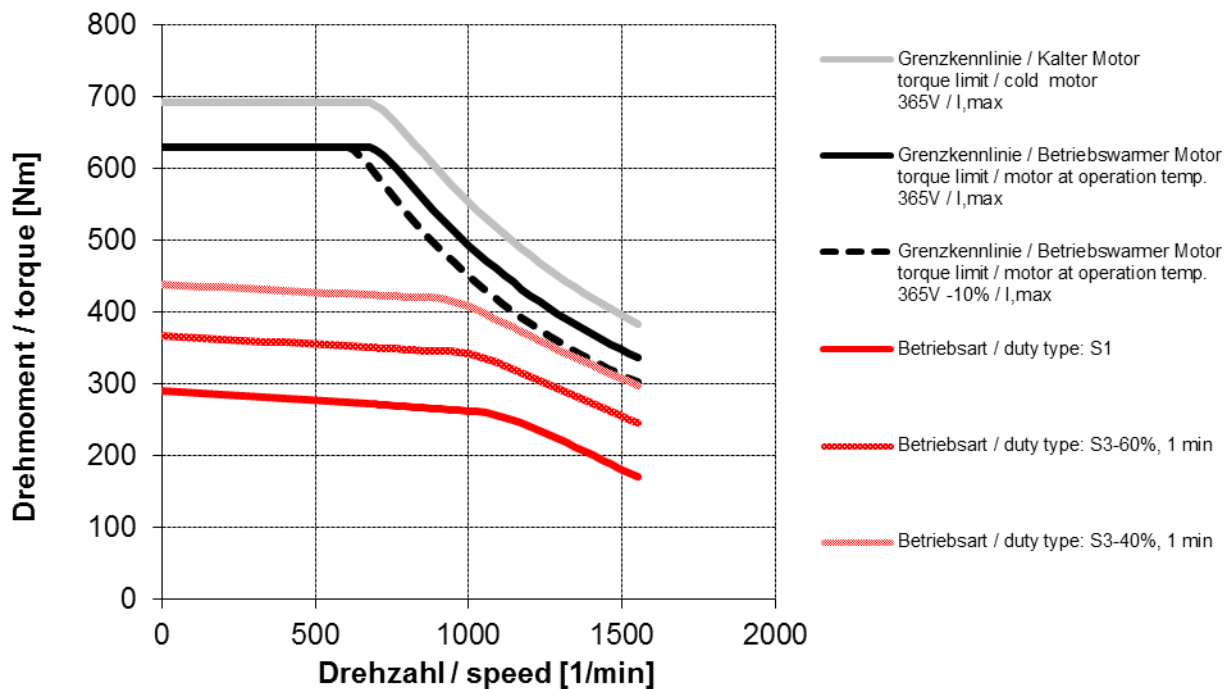
DSD2-132MO54A(R)-25-5



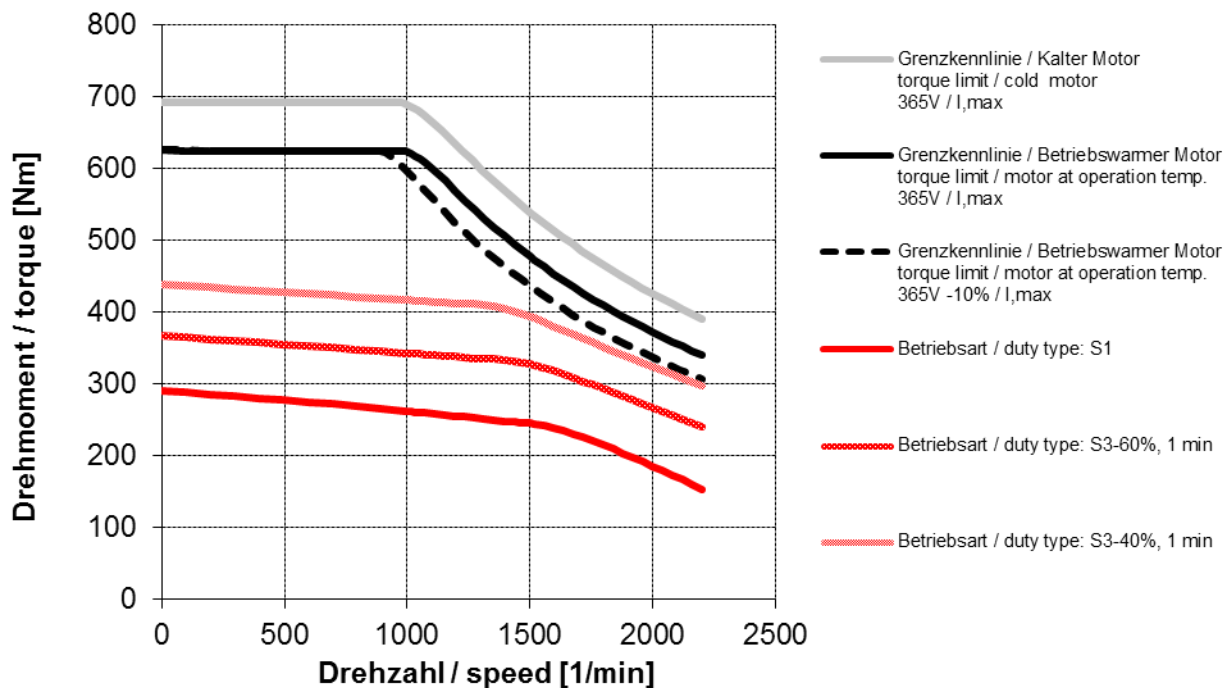
DSD2-132MO54A(R)-30-5



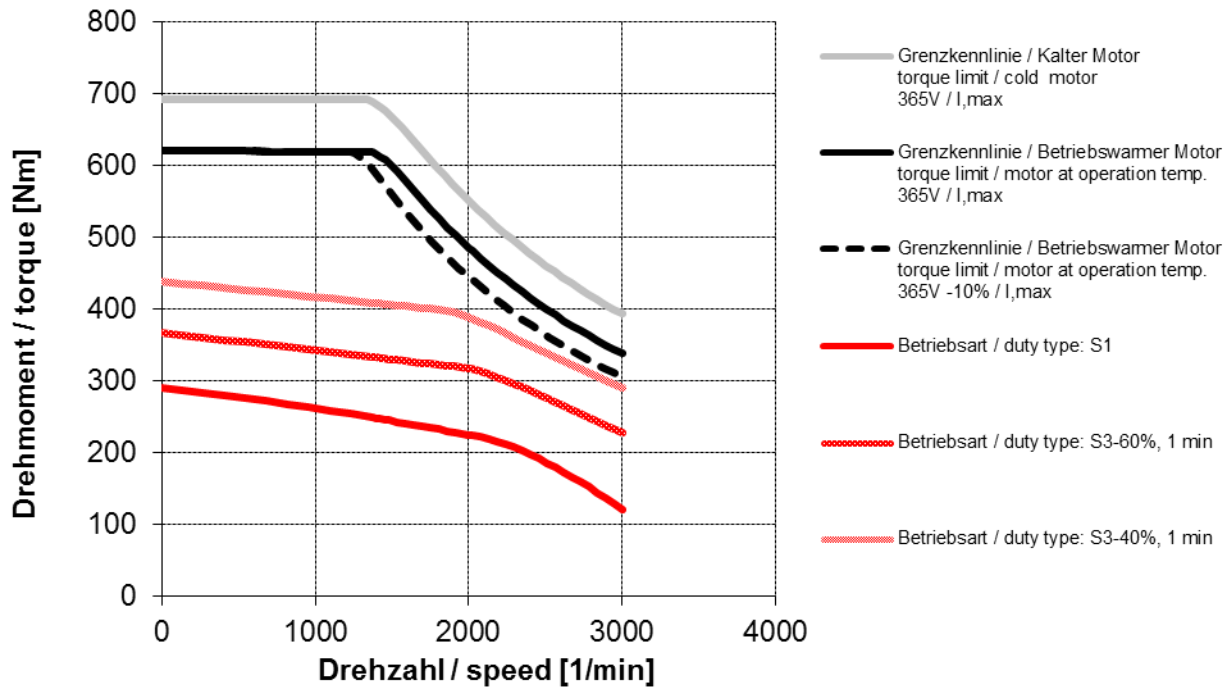
DSD2-132LO54A(R)-10-5



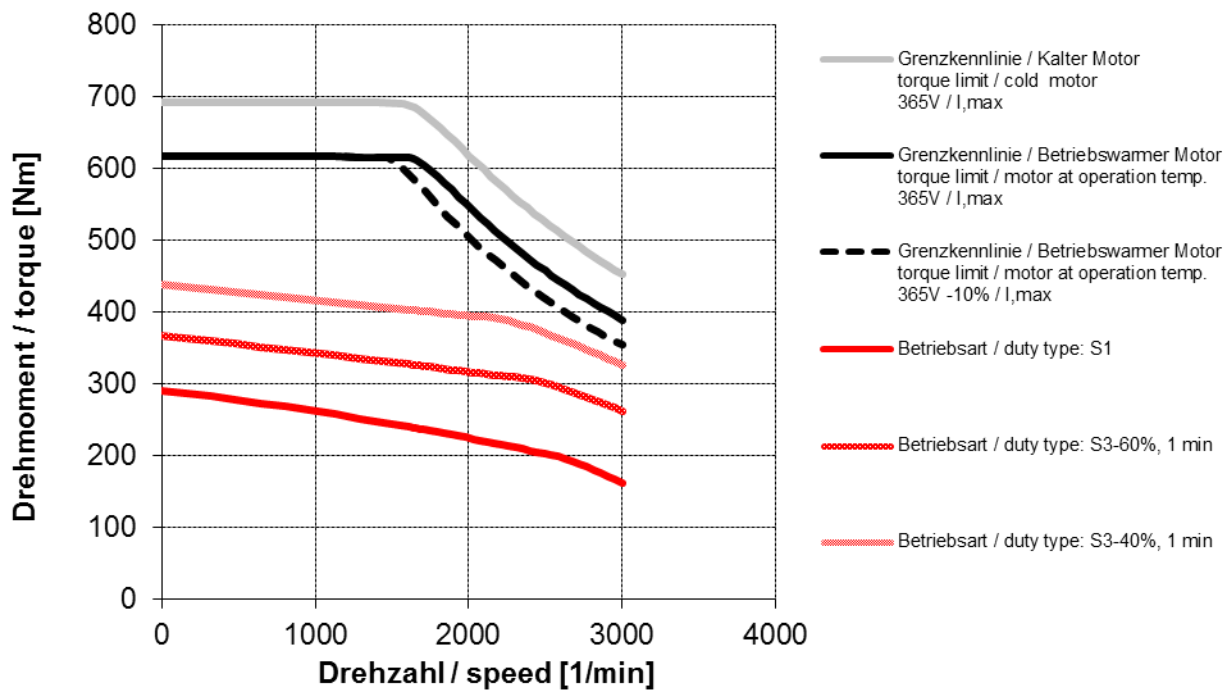
DSD2-132LO54A(R)-15-5



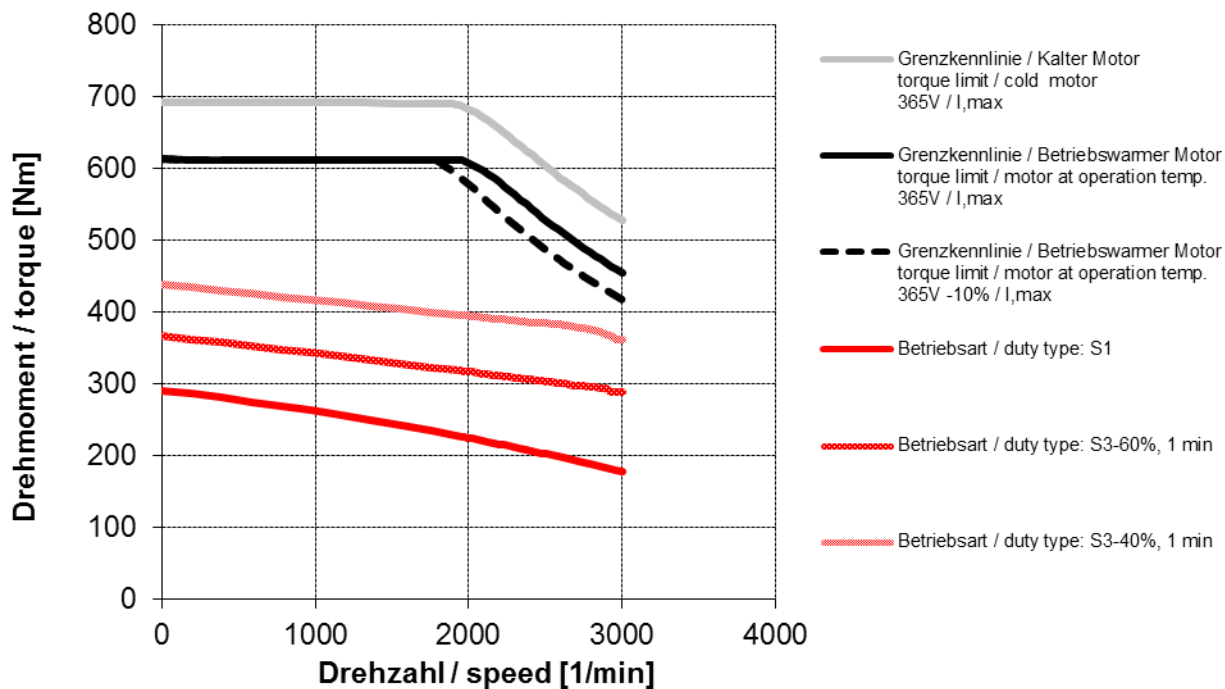
DSD2-132LO54A(R)-20-5



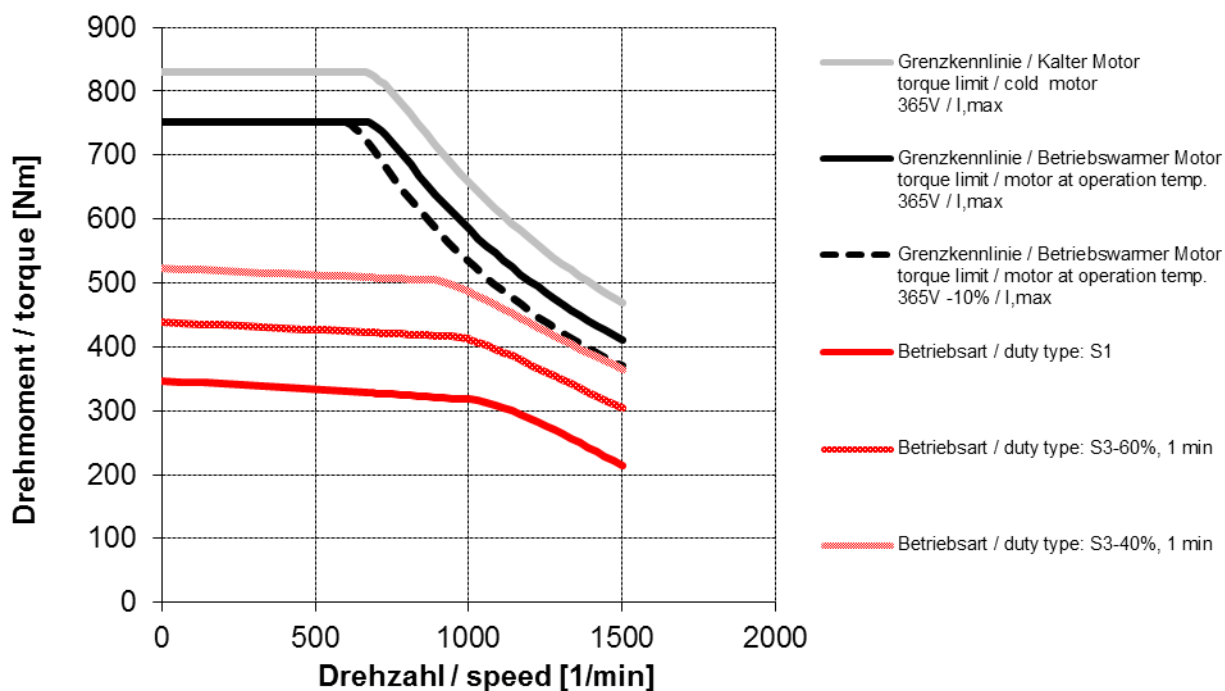
DSD2-132LO54A(R)-25-5



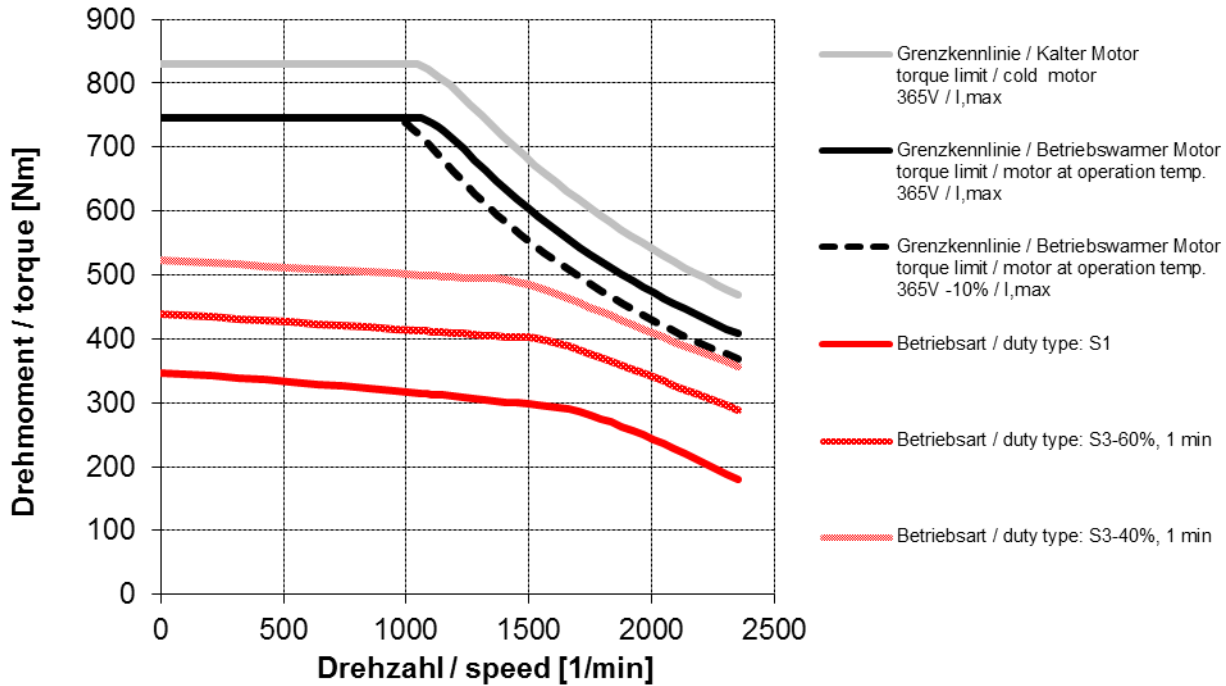
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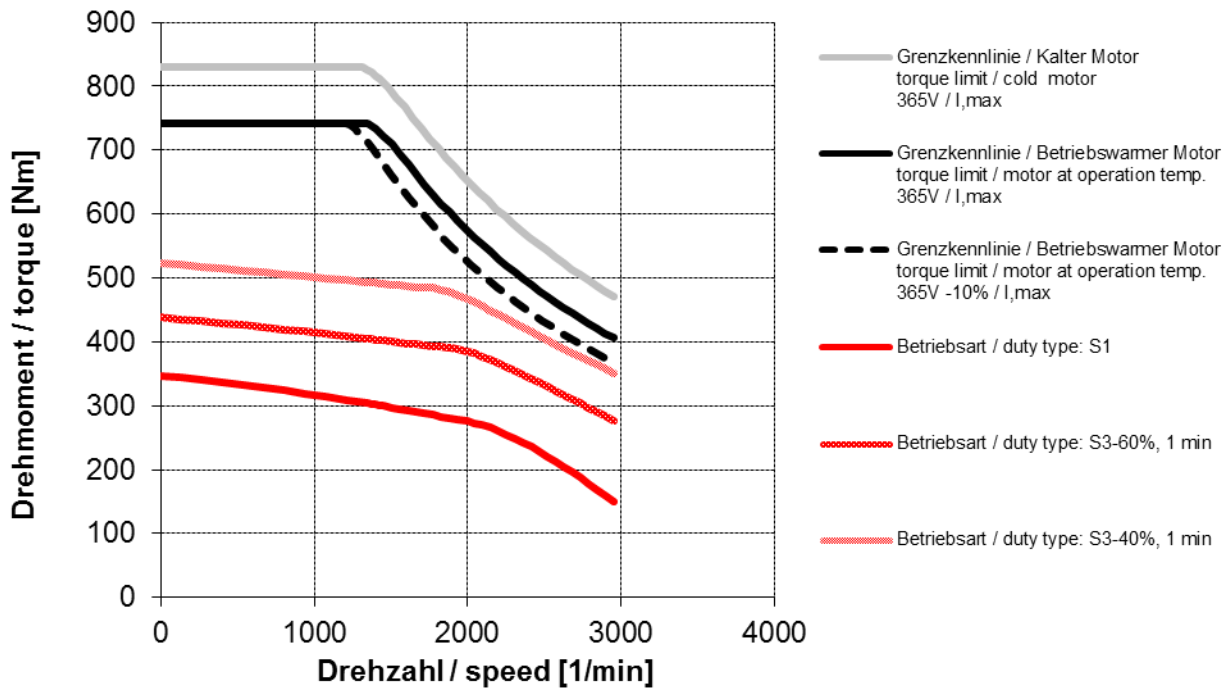
DSD2-132BO54A(R)-10-5



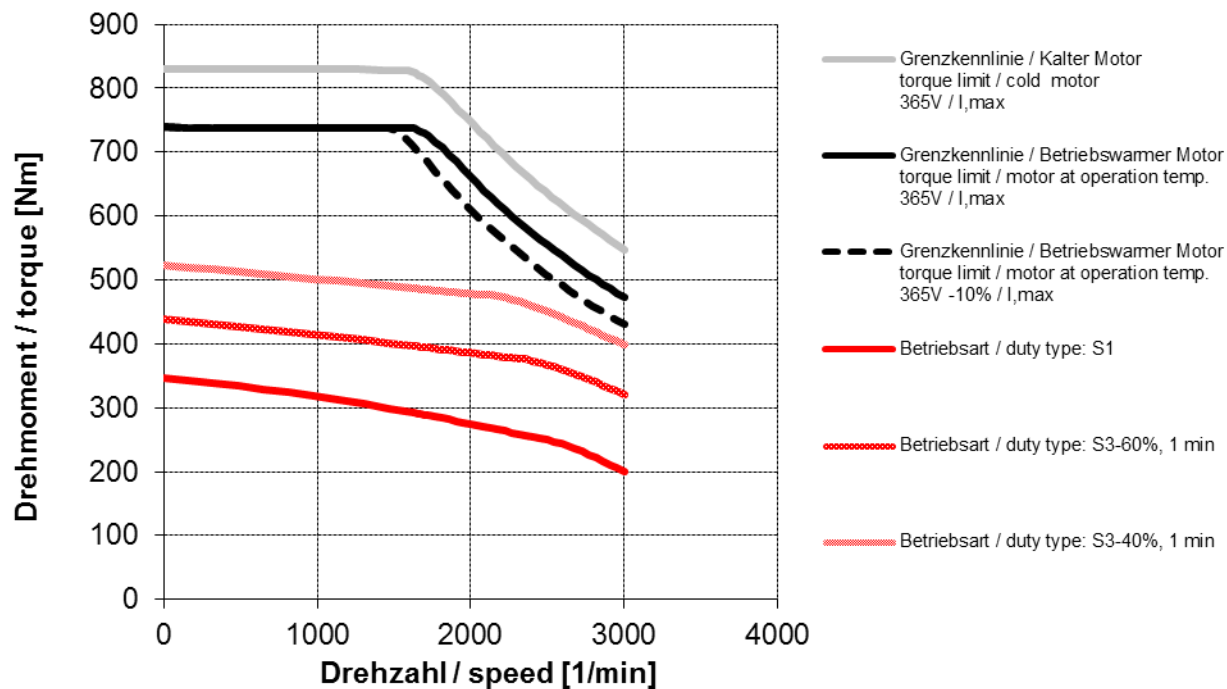
DSD2-132BO54A(R)-15-5



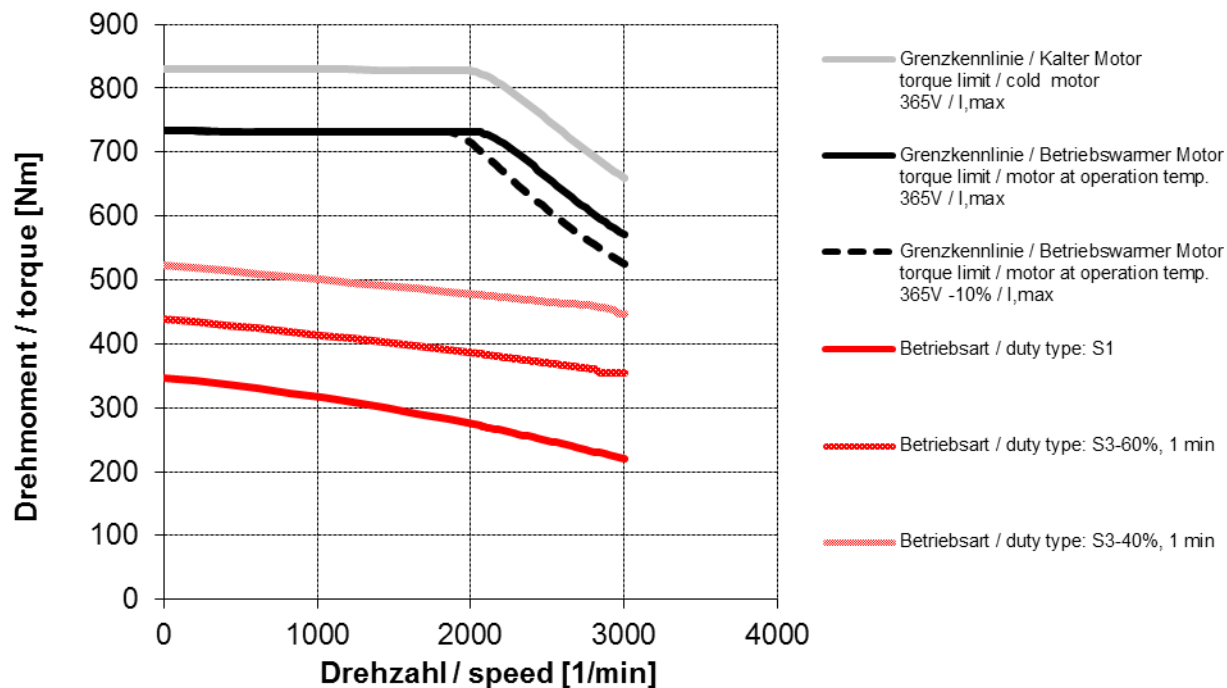
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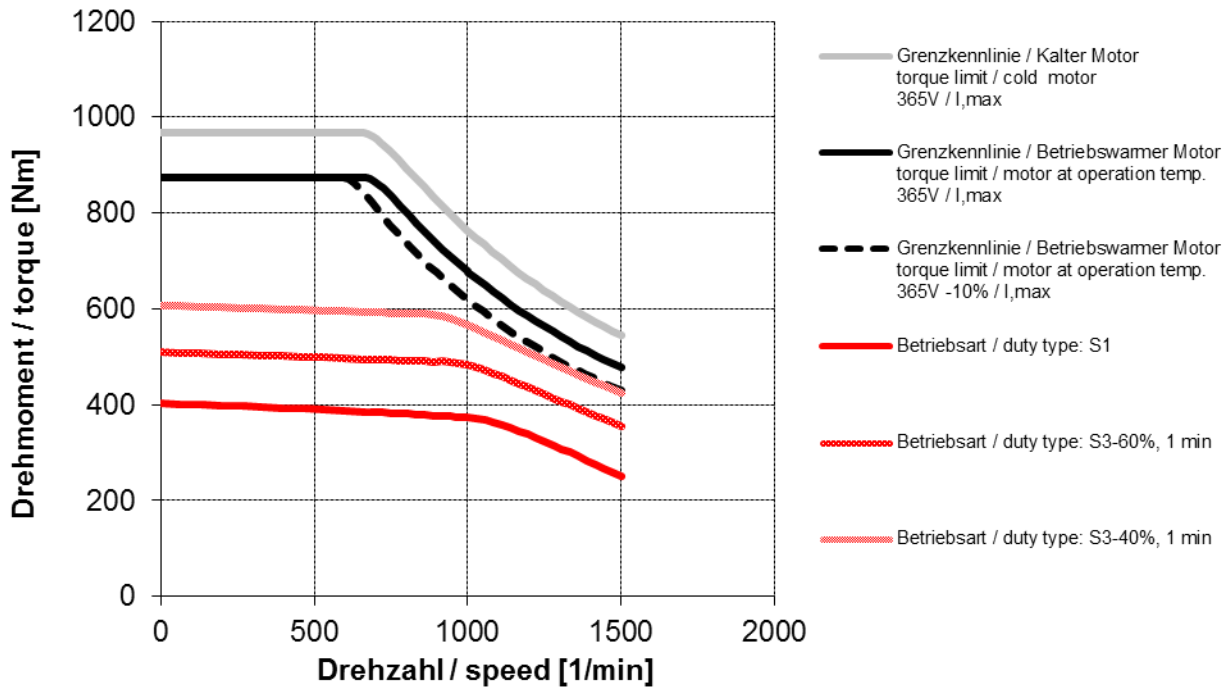
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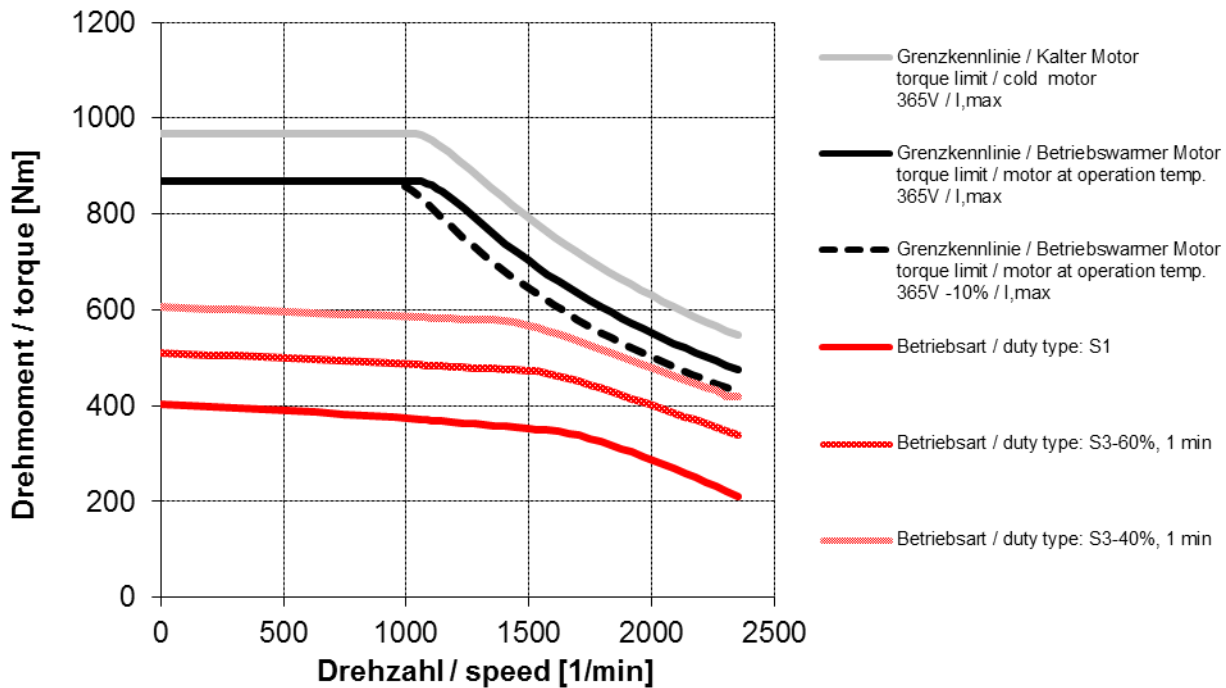
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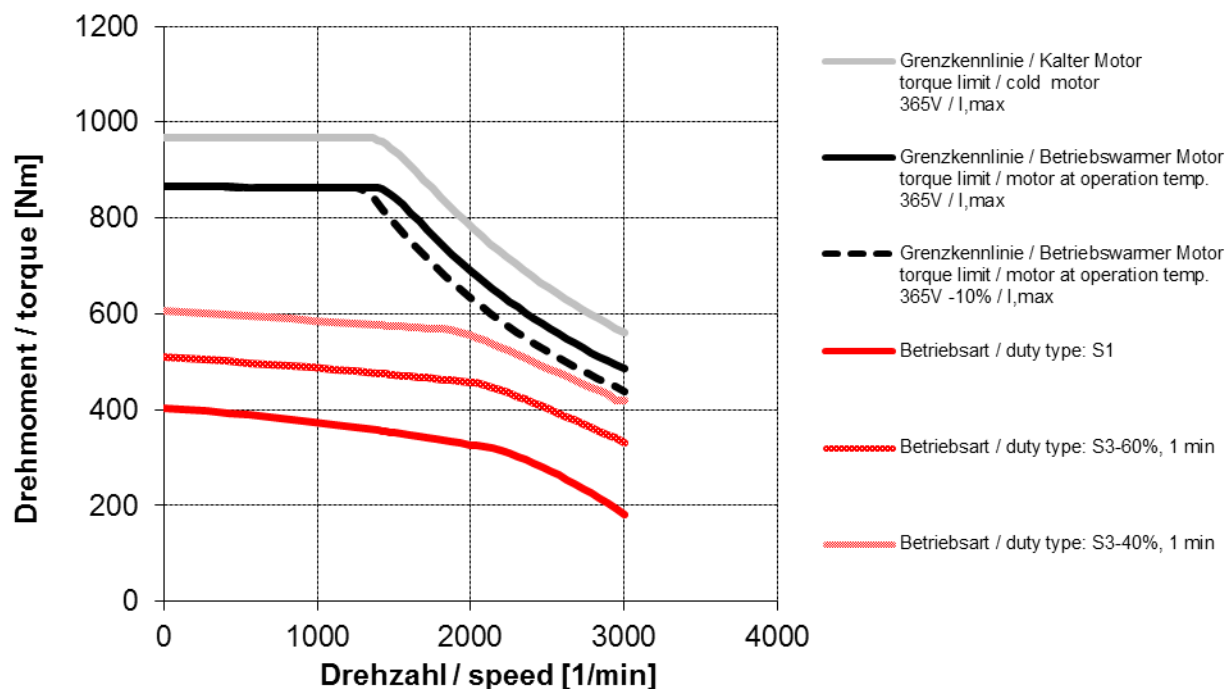
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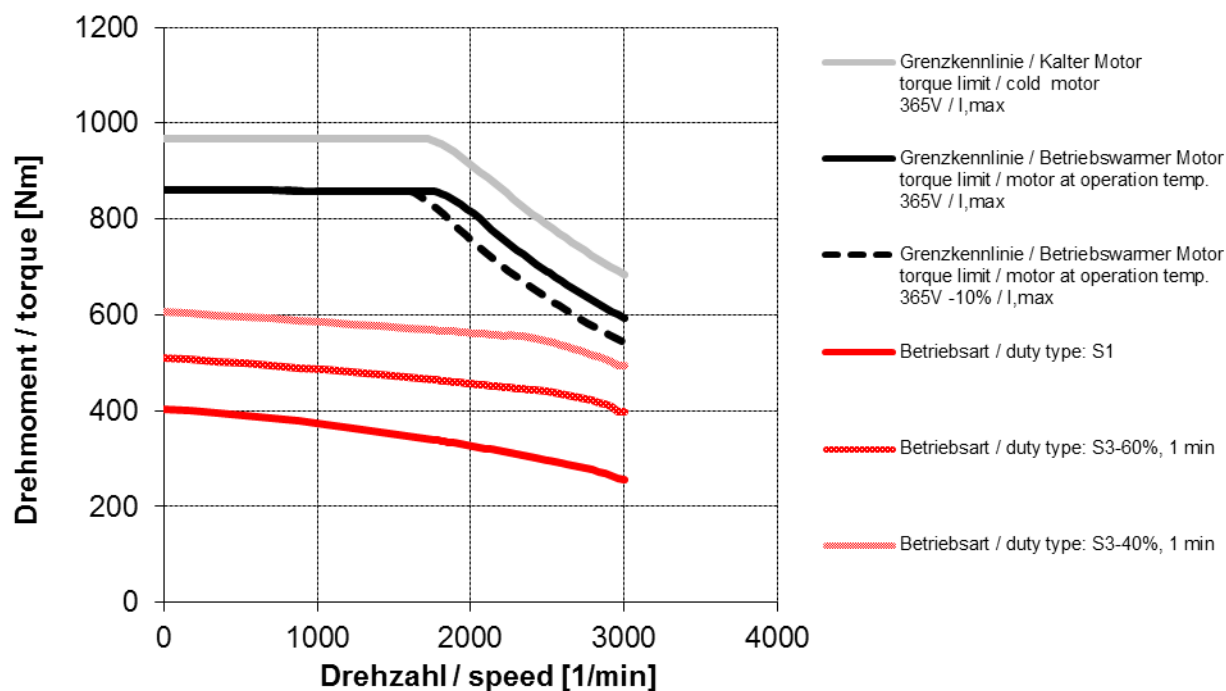
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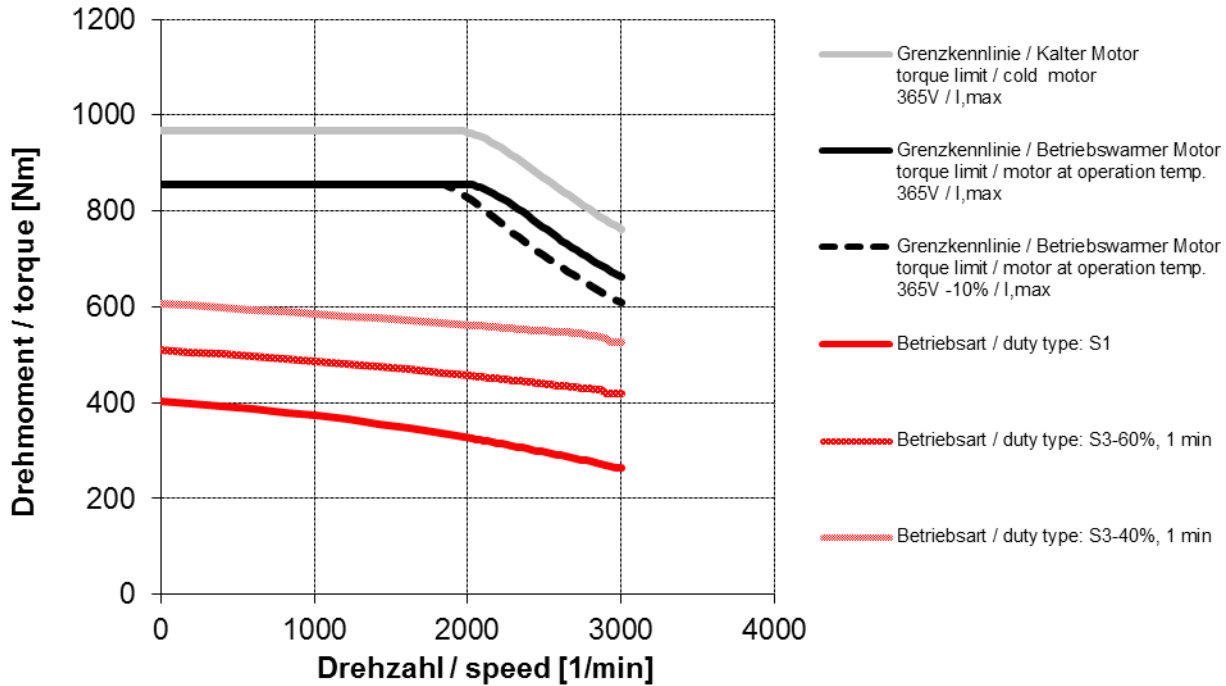
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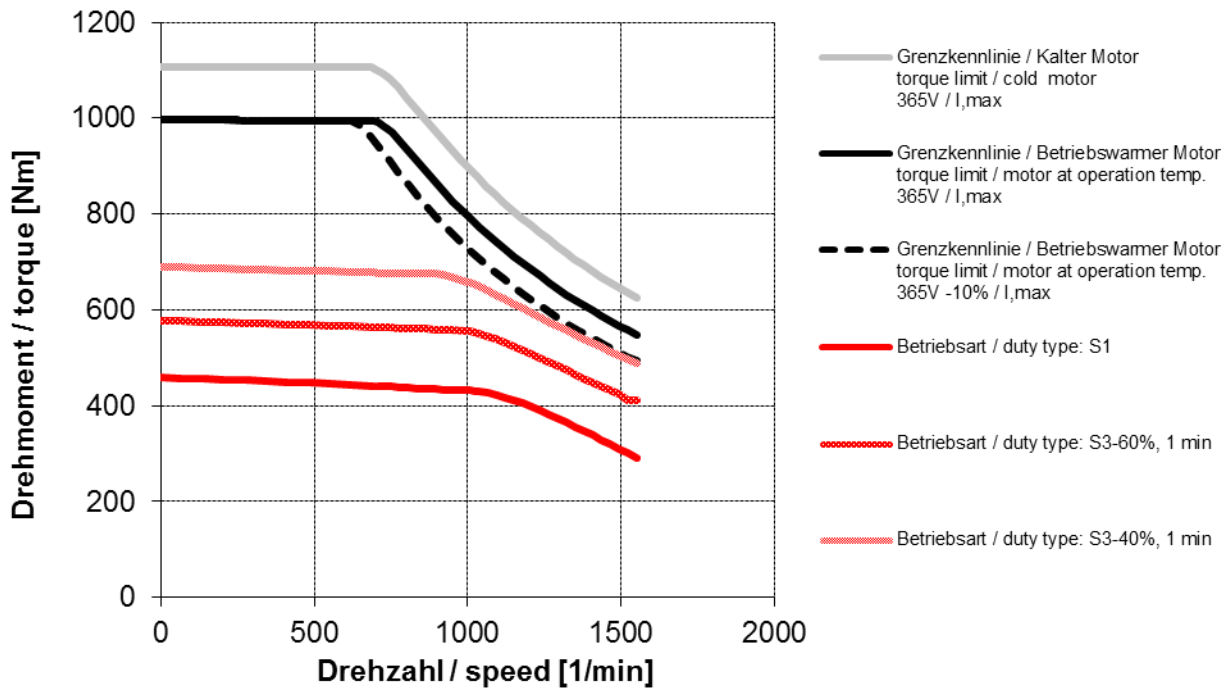
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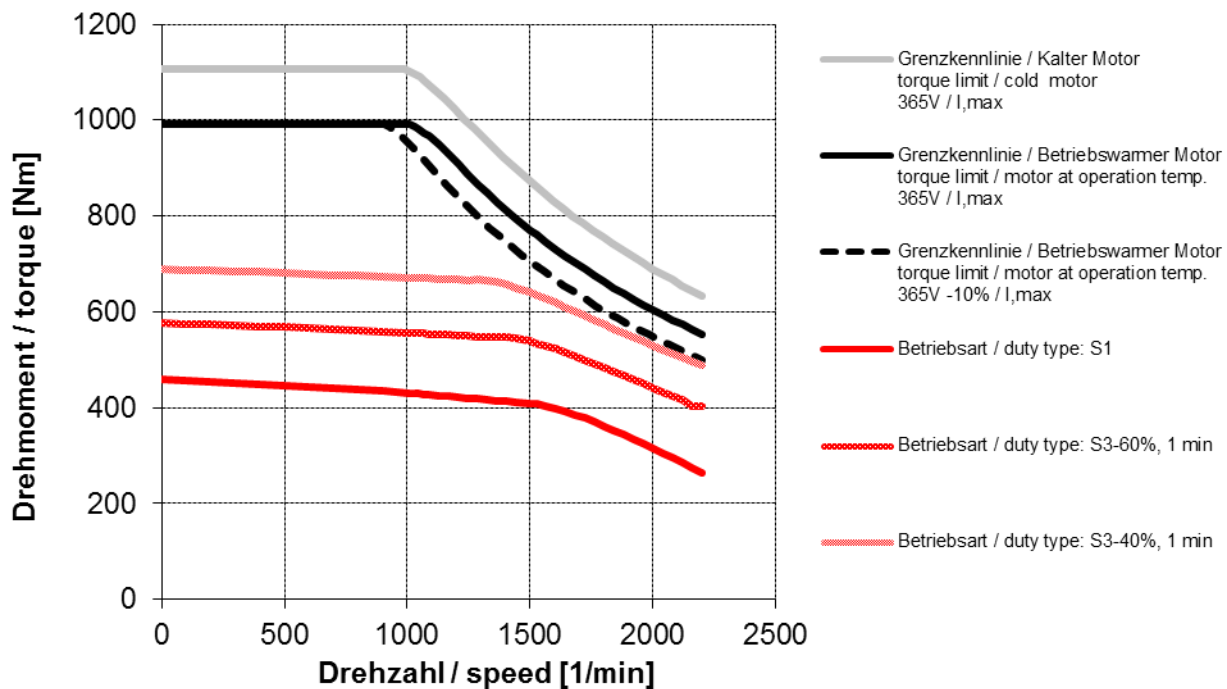
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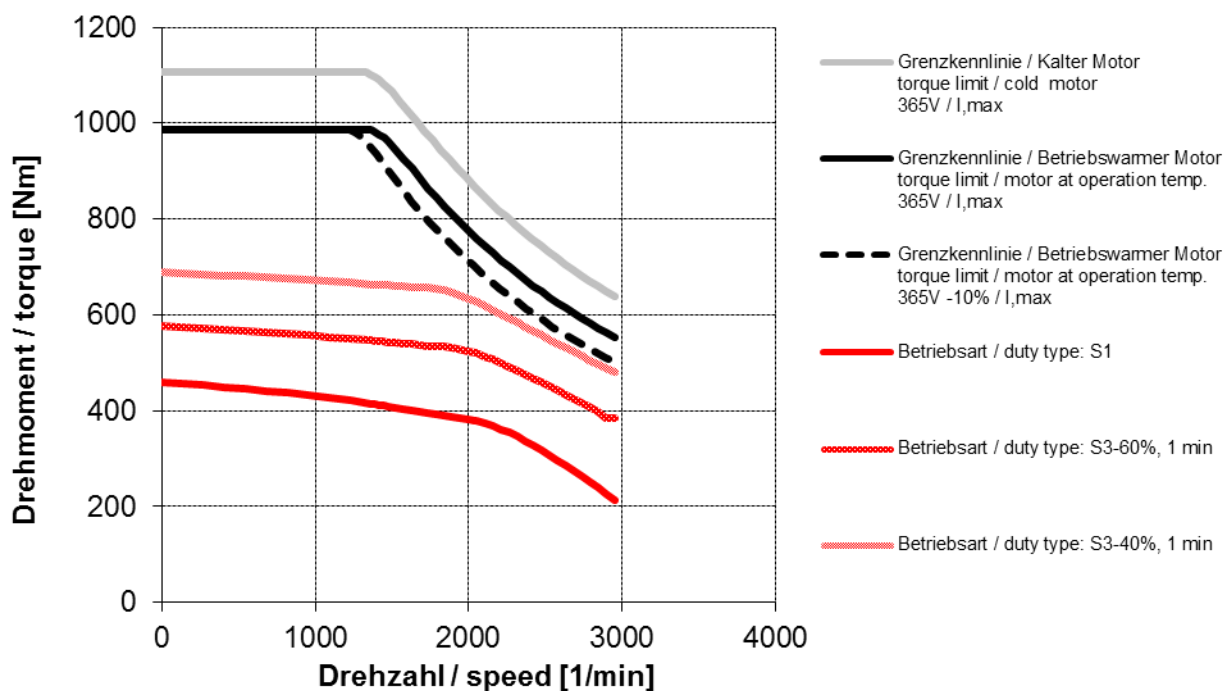
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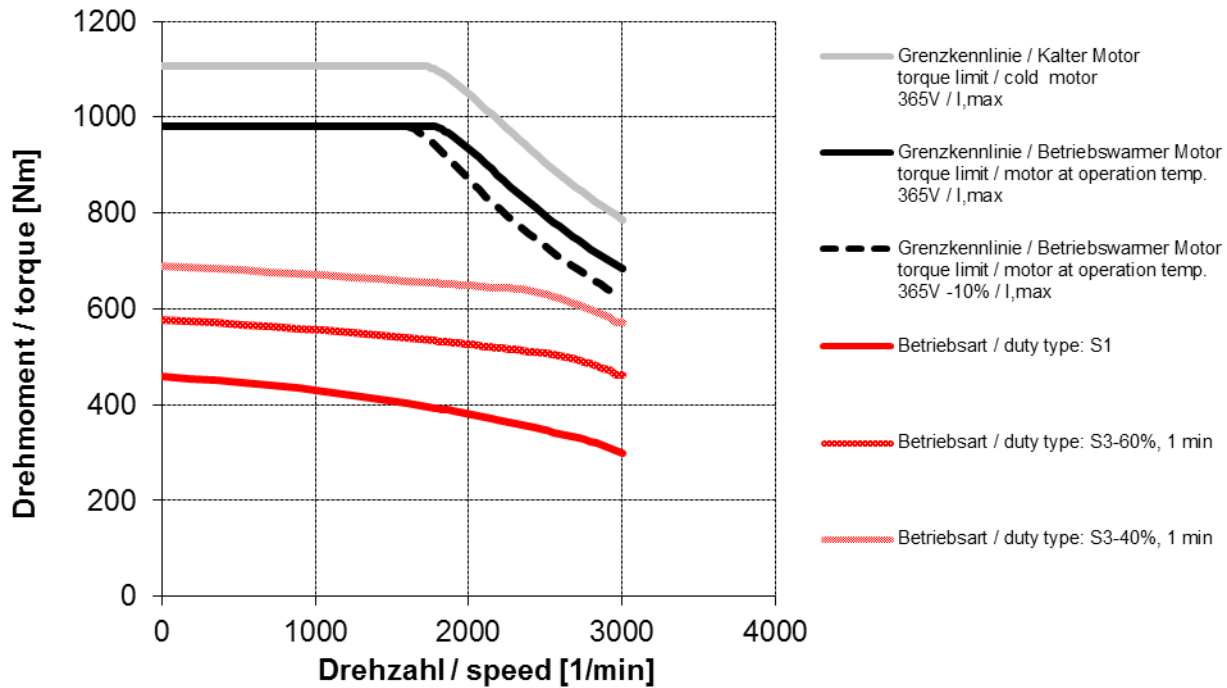
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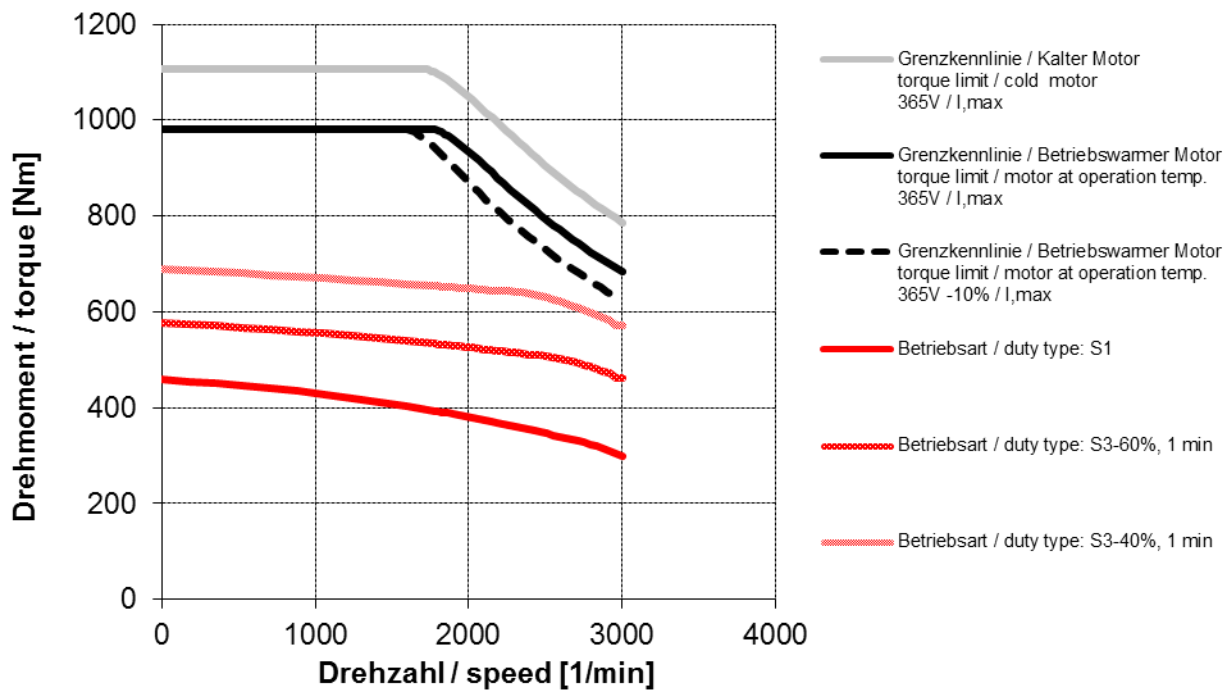
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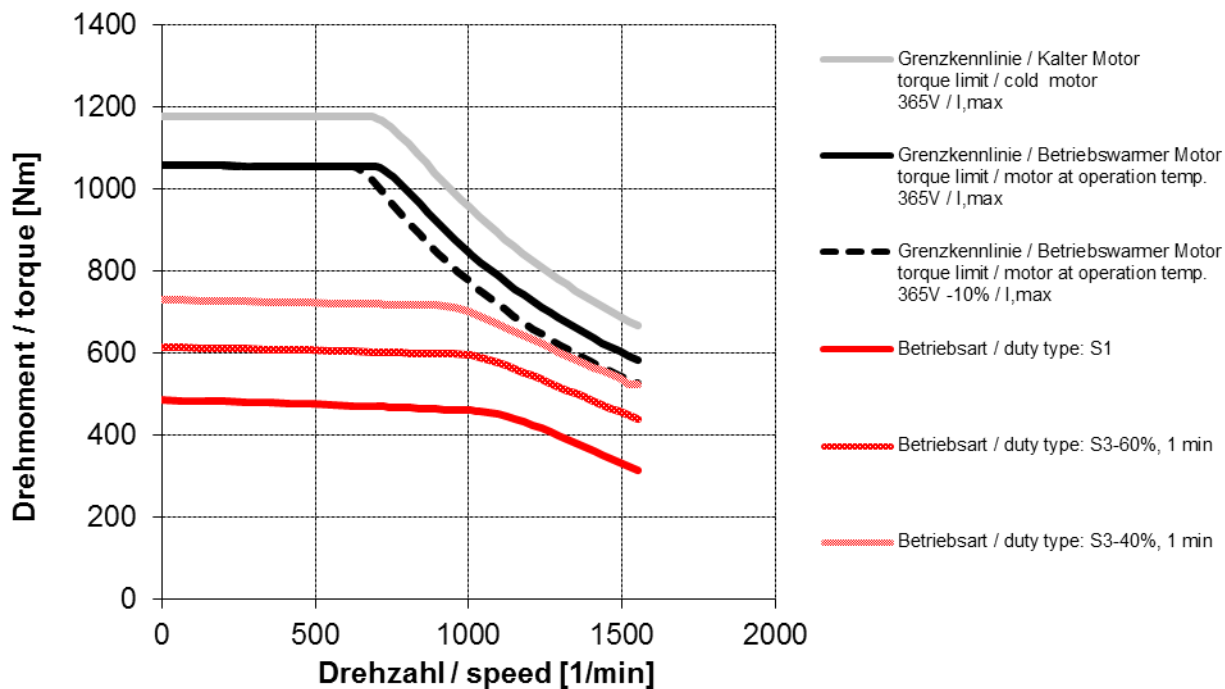
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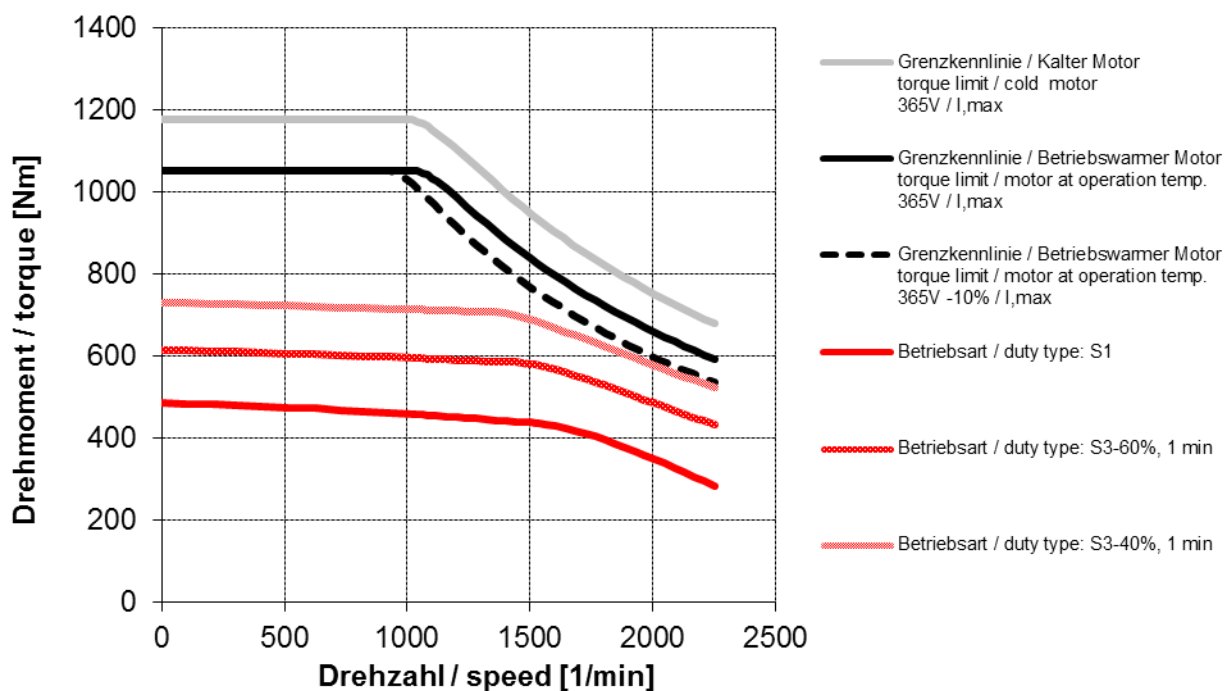
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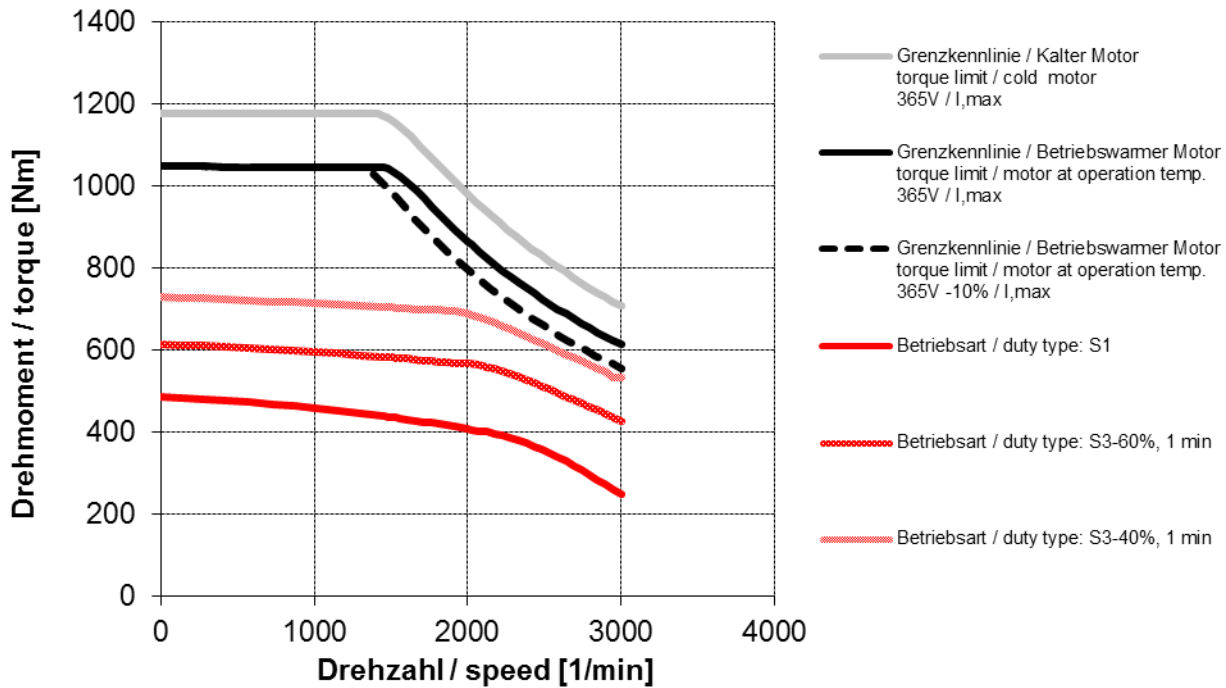
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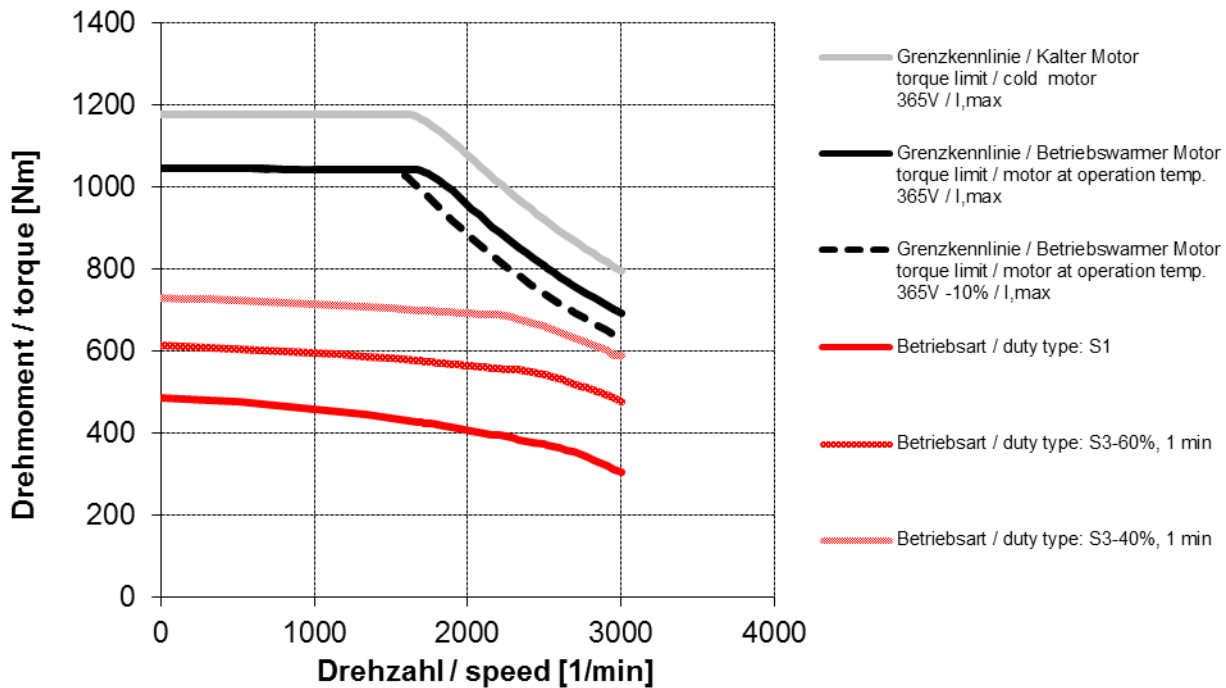
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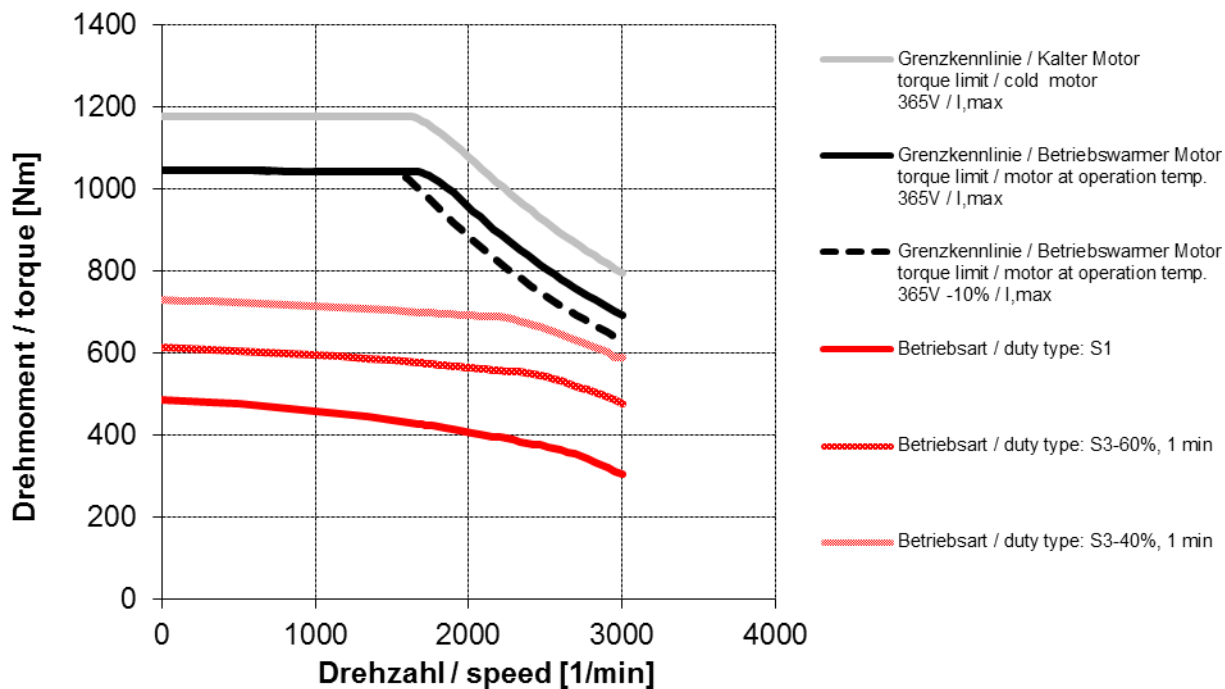
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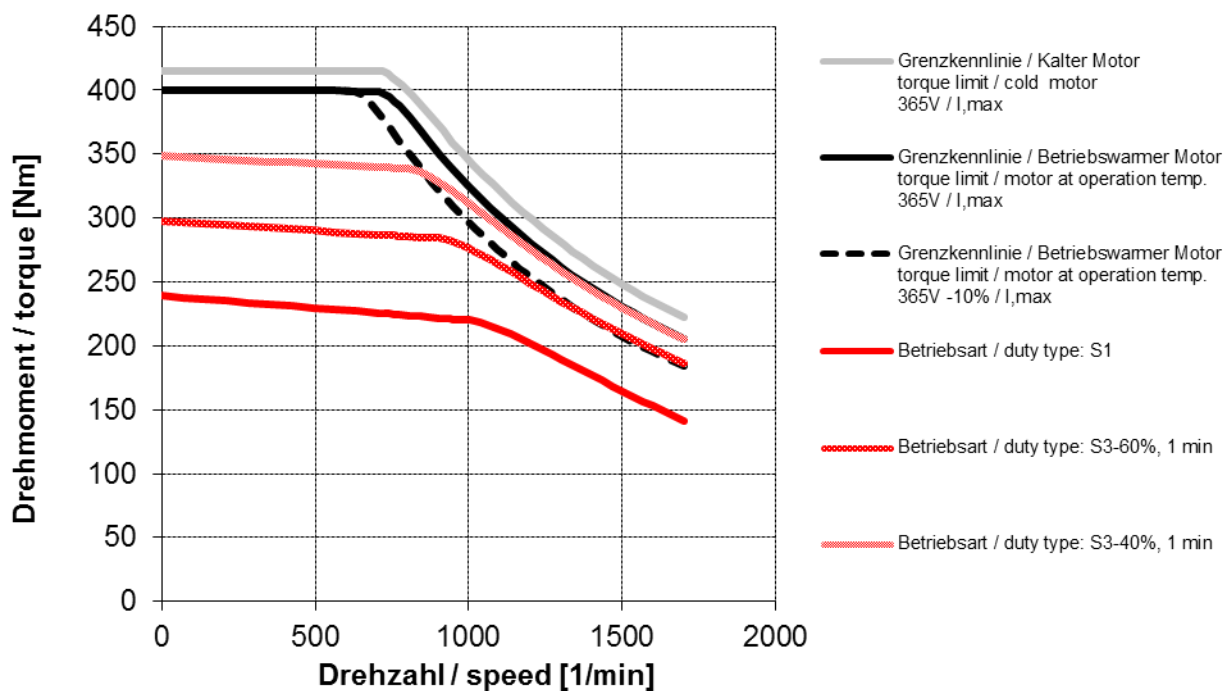


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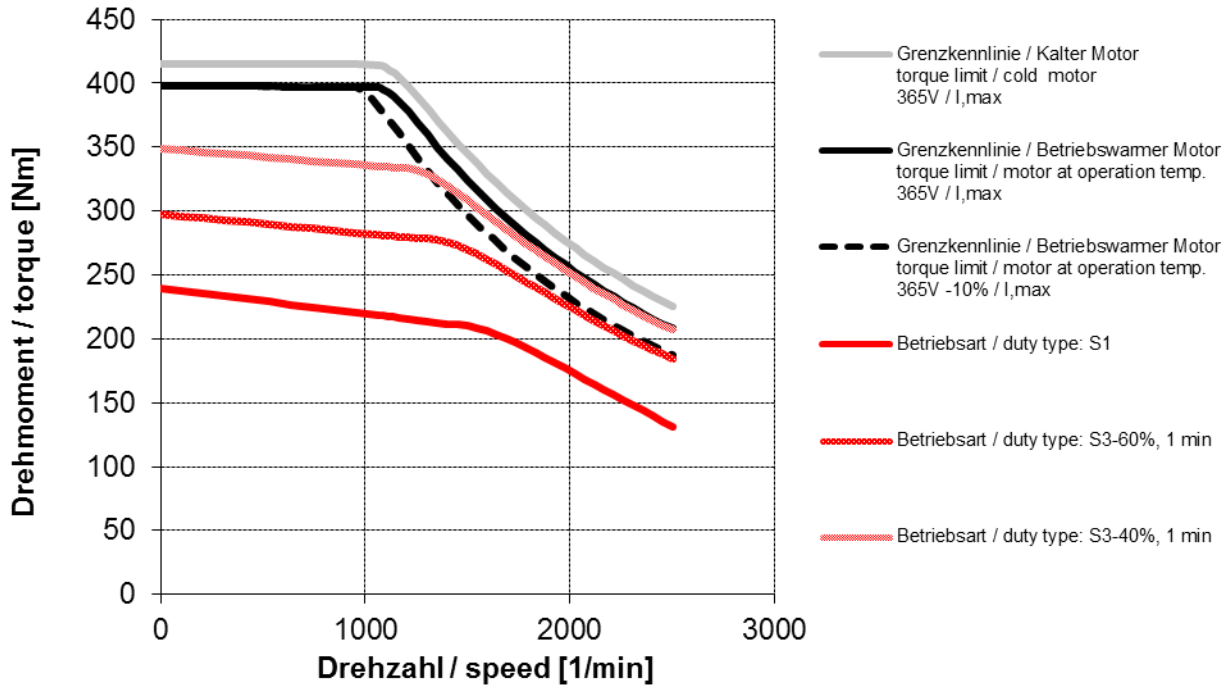


5.7.2. DSD2-132..54W-.. (IP54 water cooled)

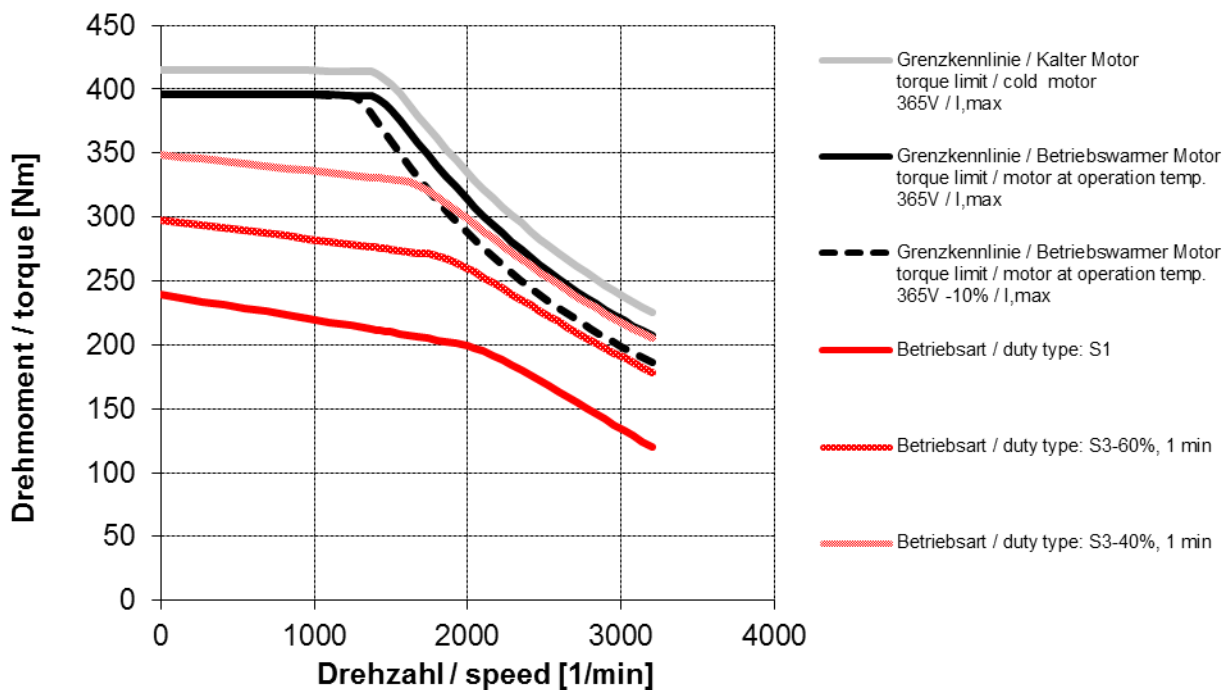
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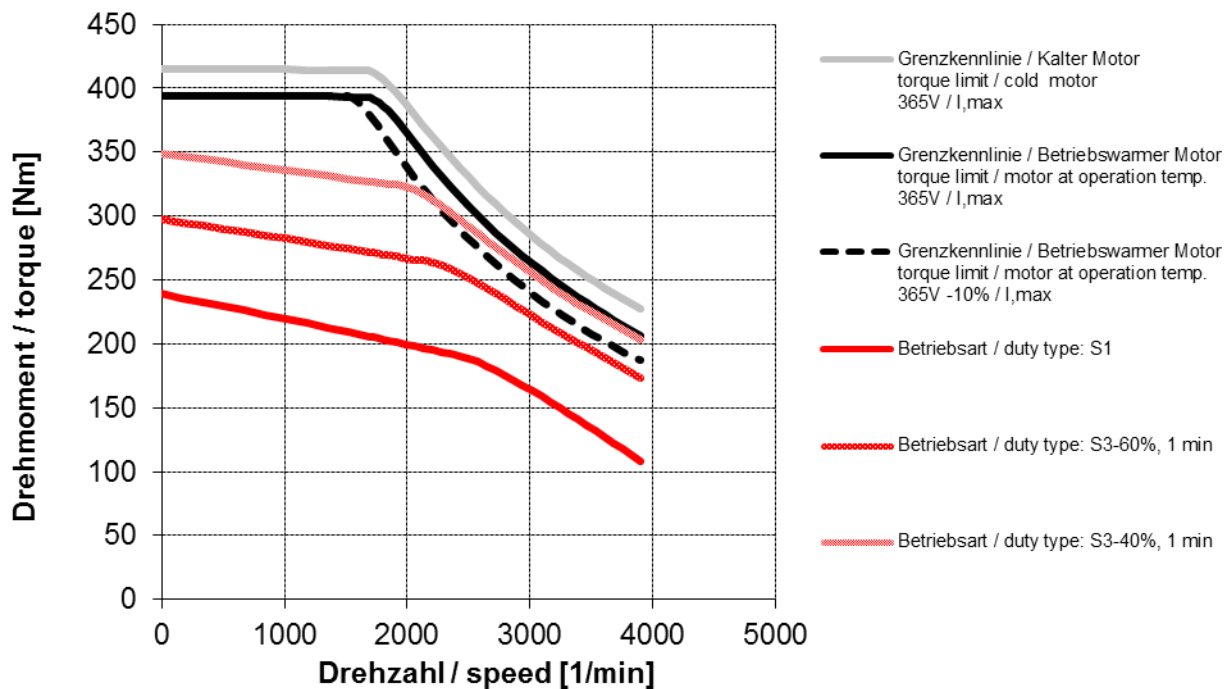
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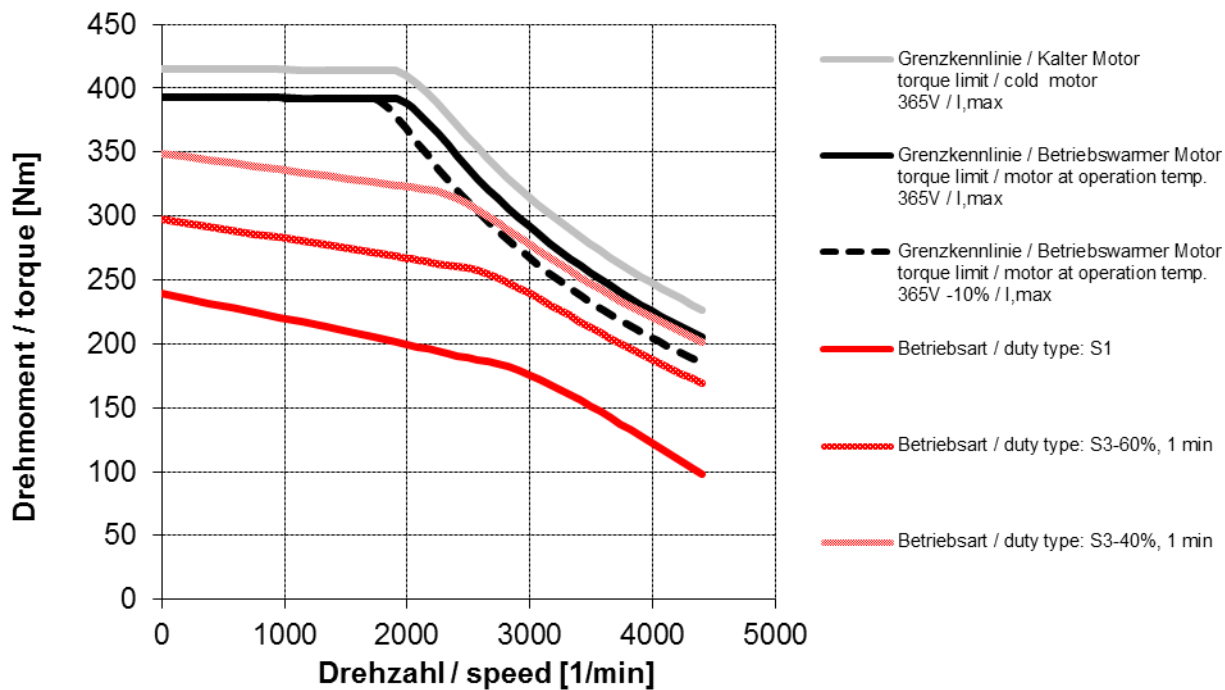
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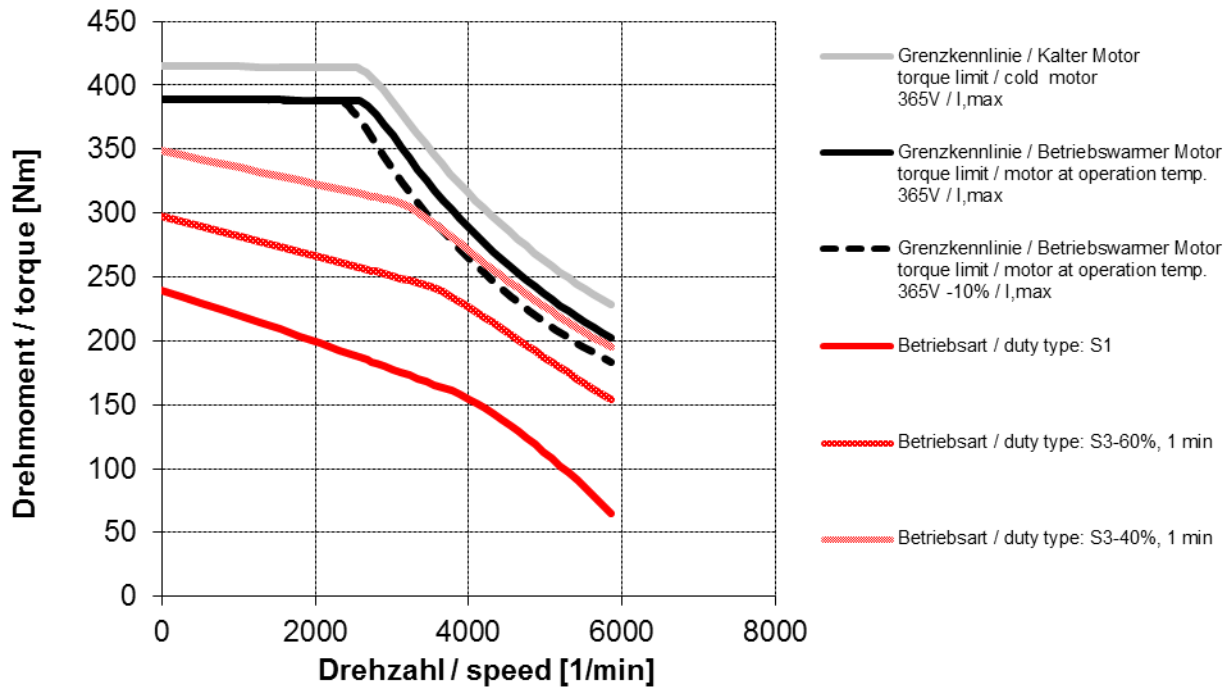
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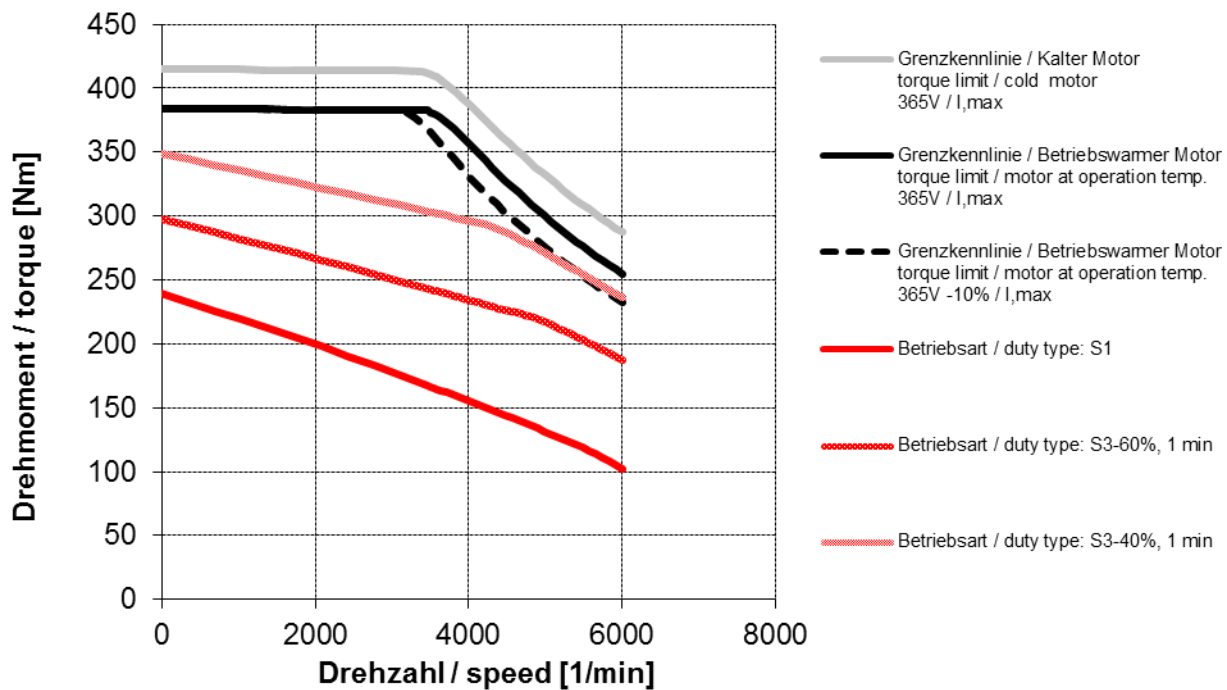
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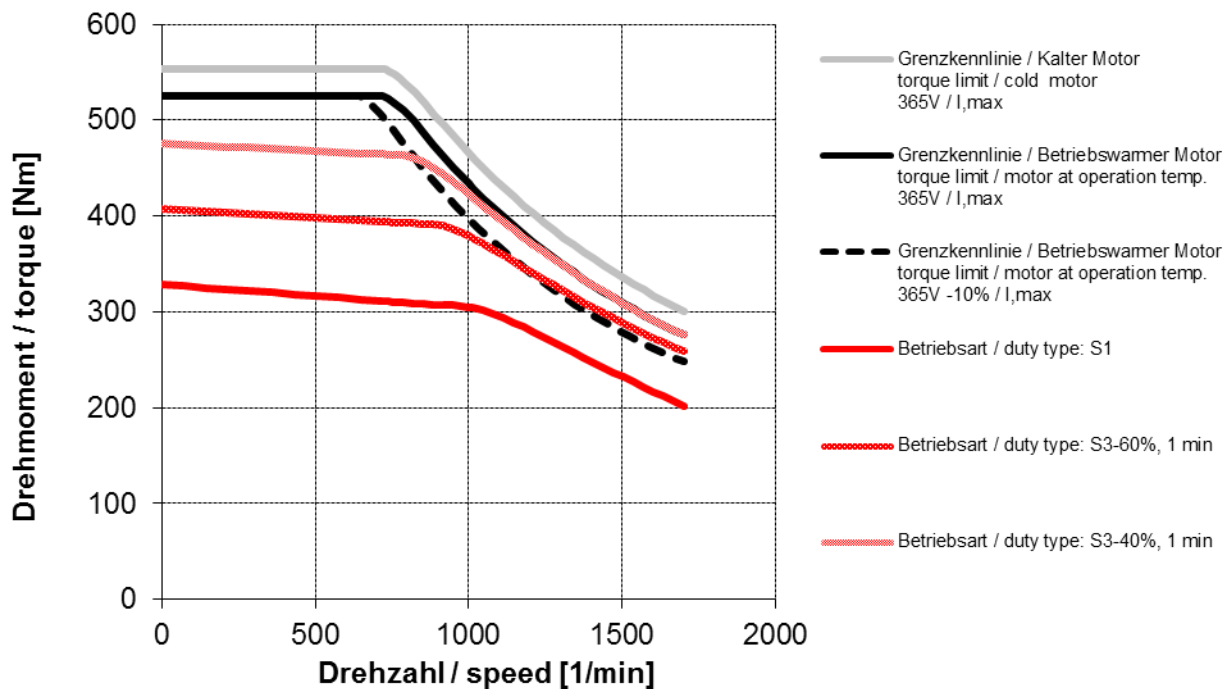
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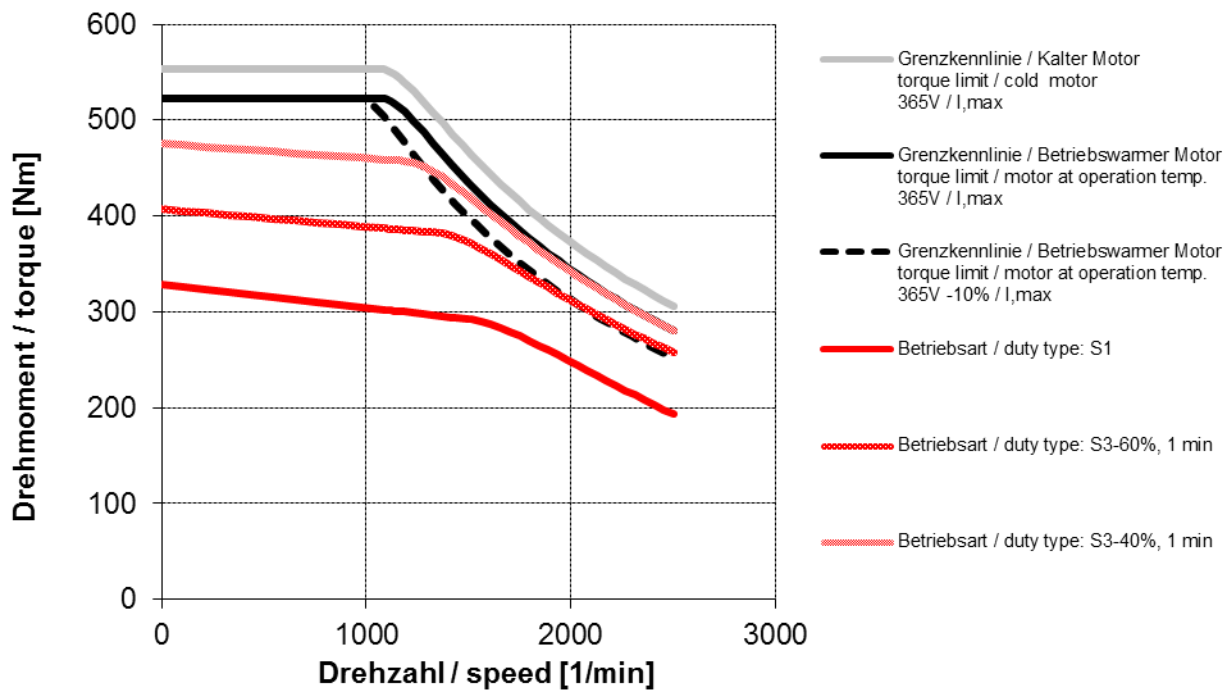
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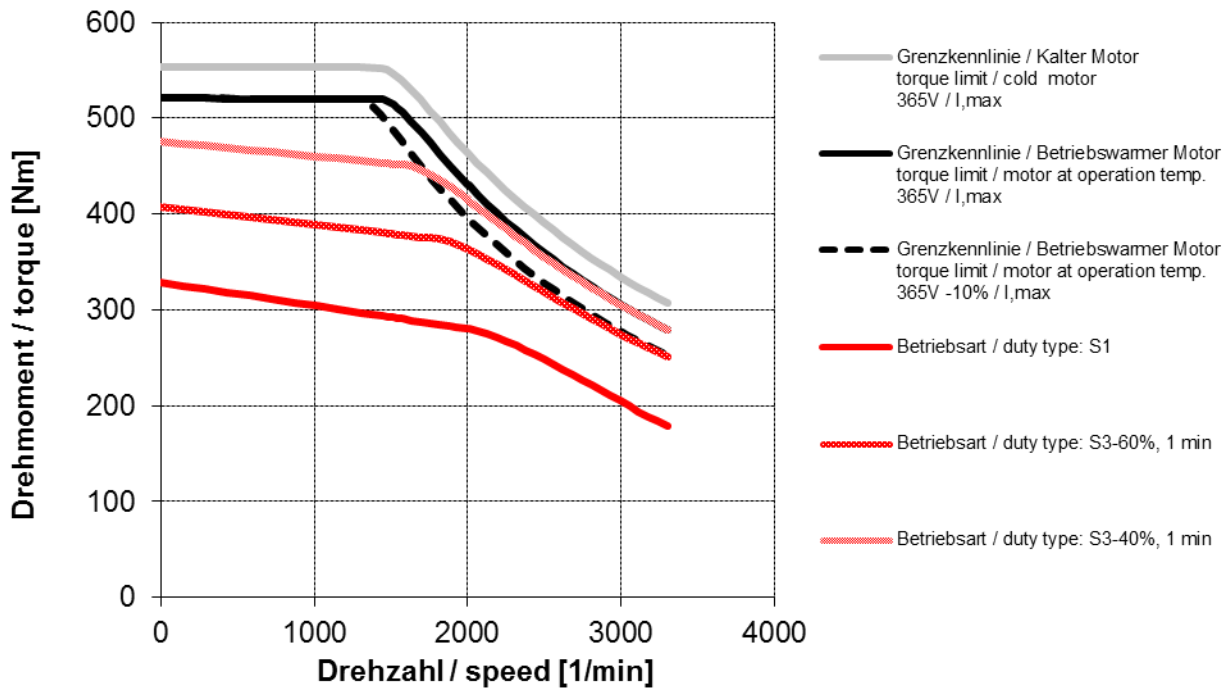
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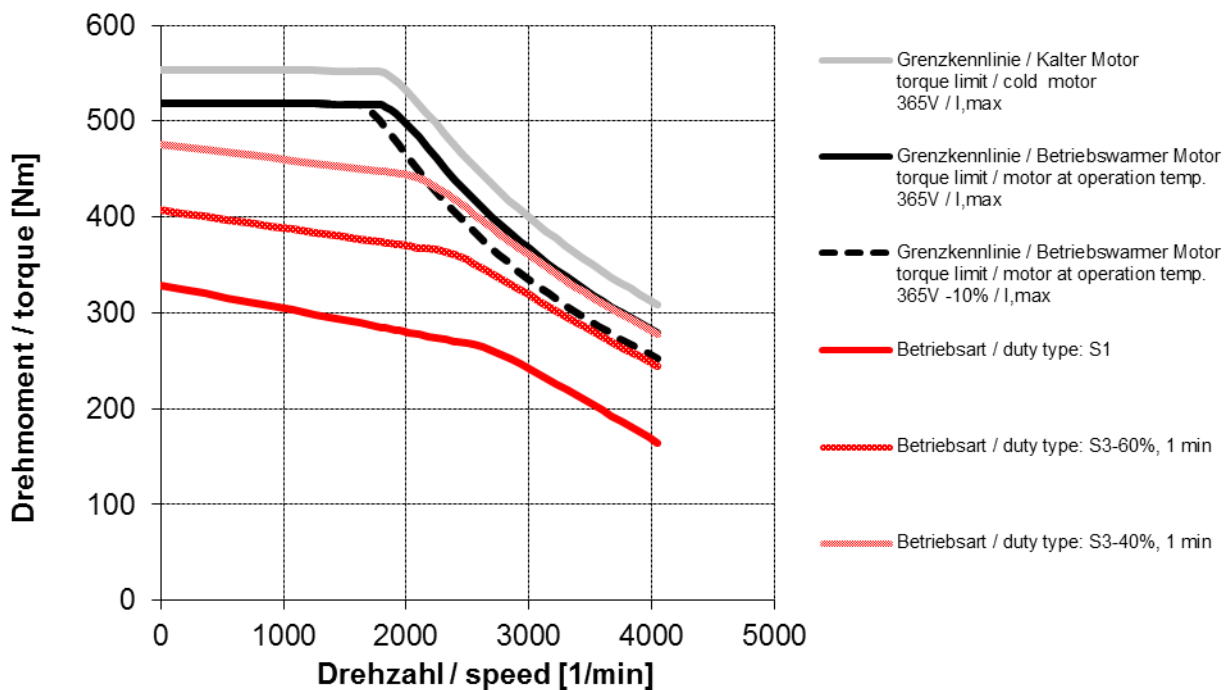
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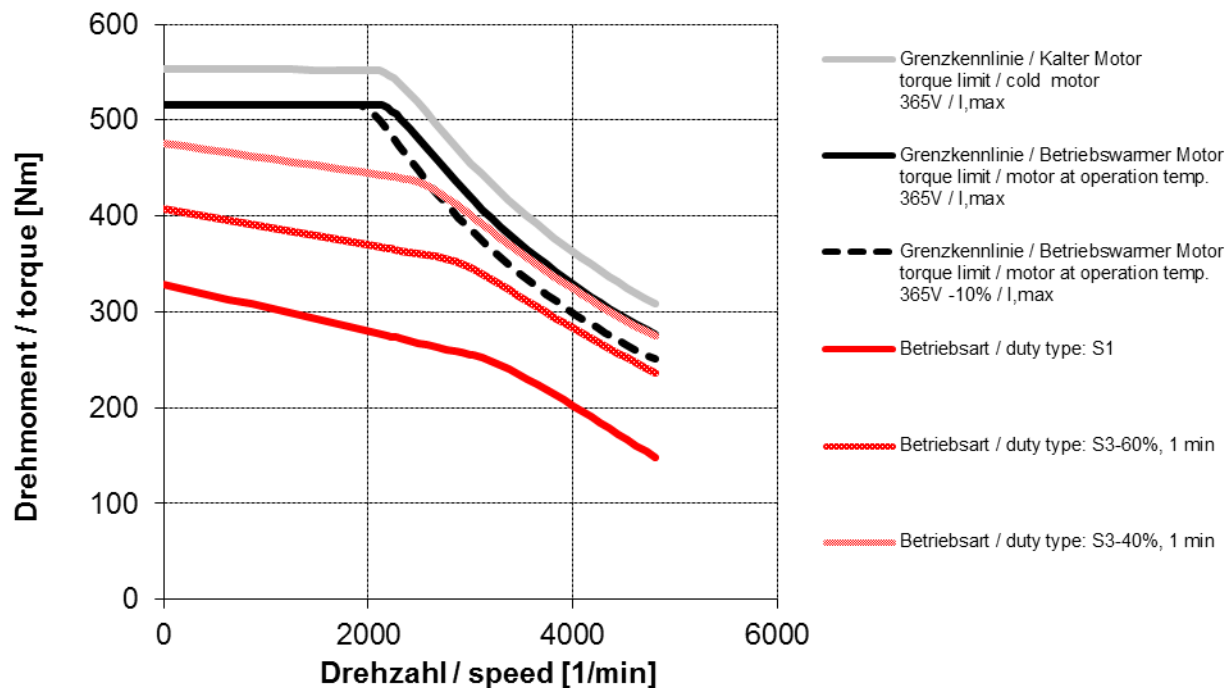
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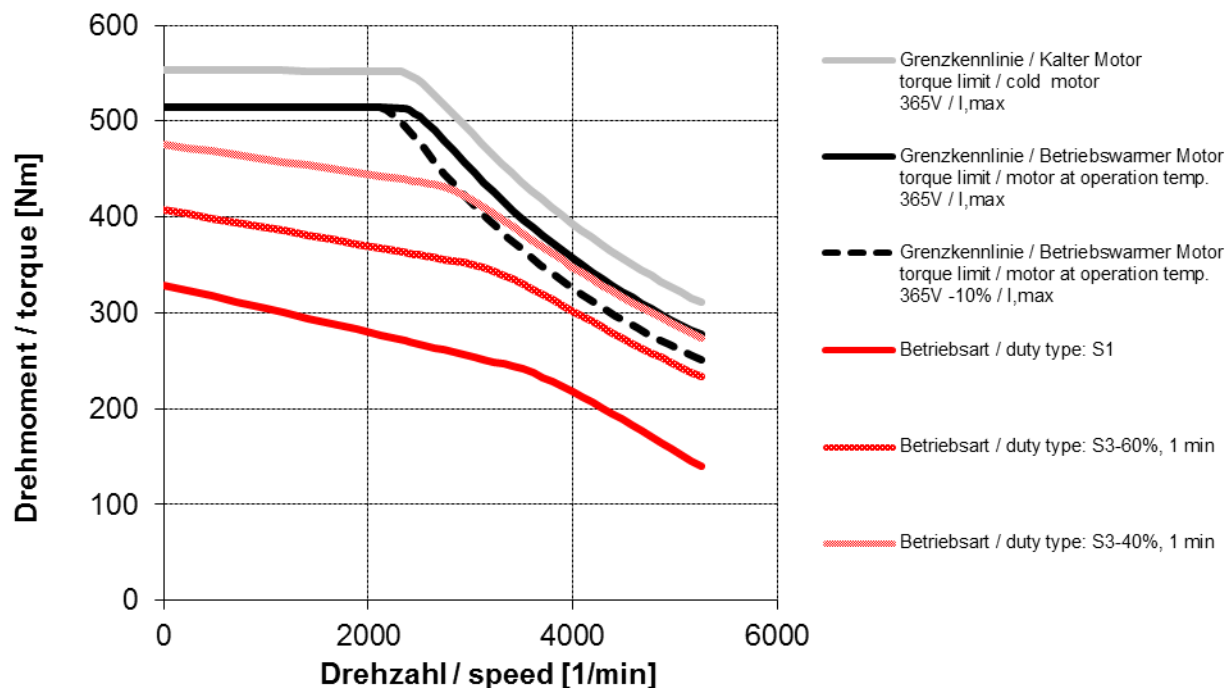
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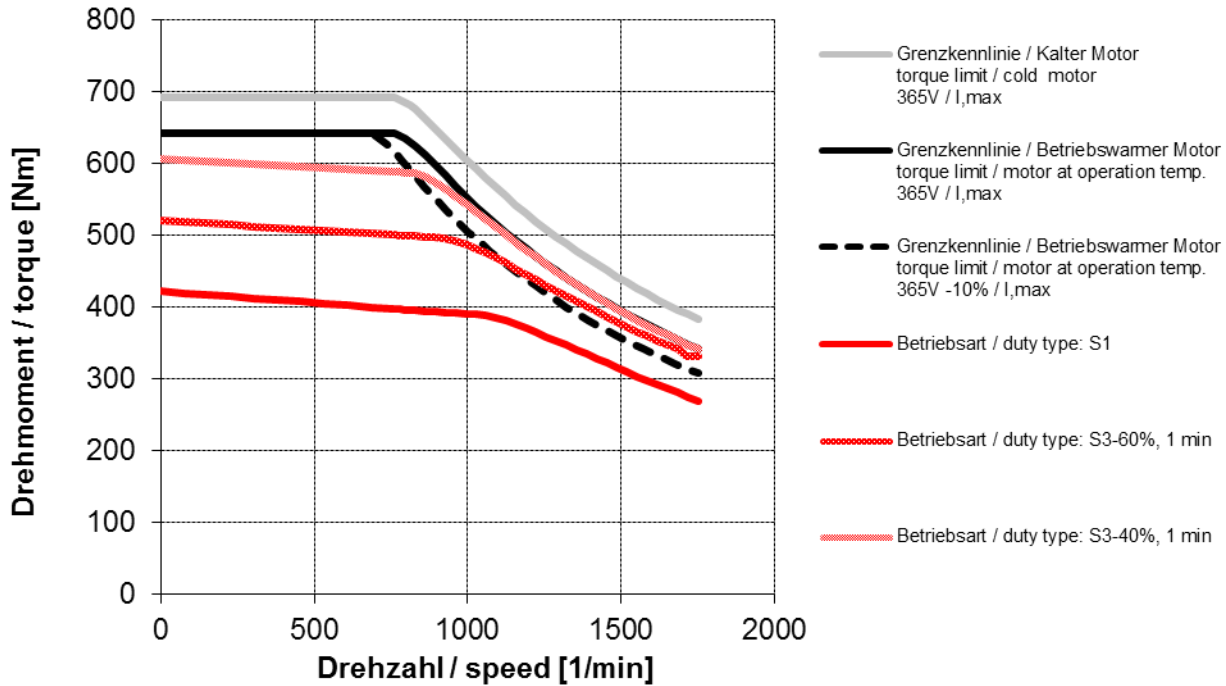
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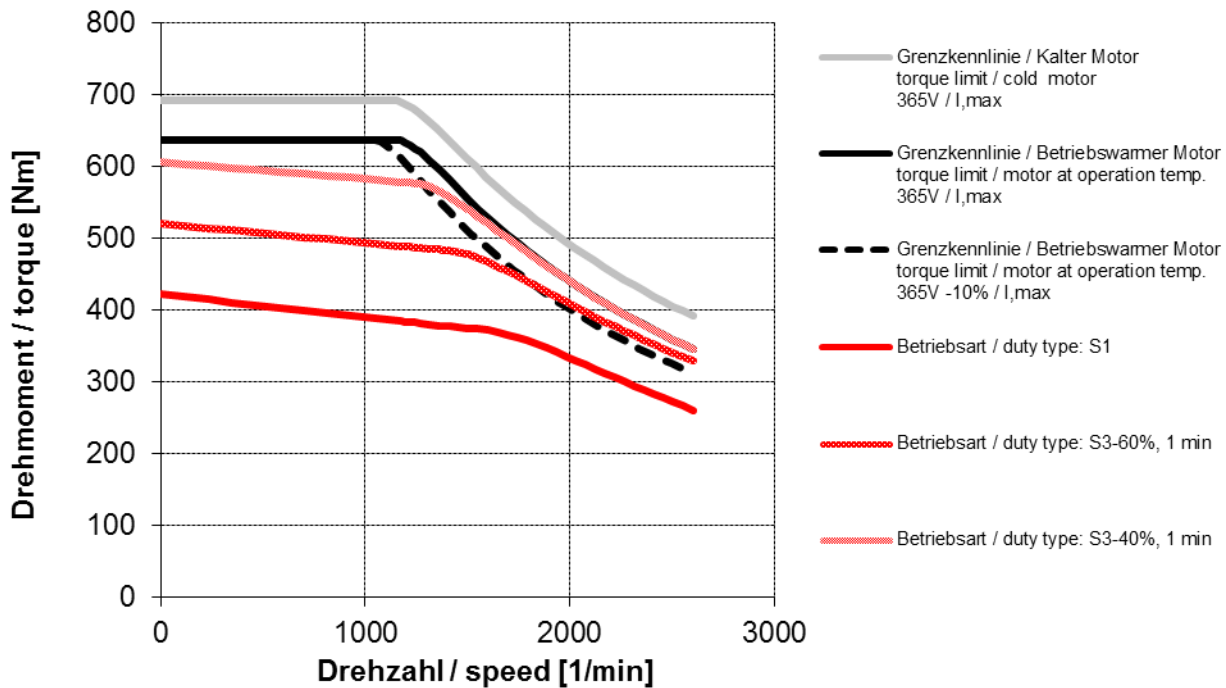
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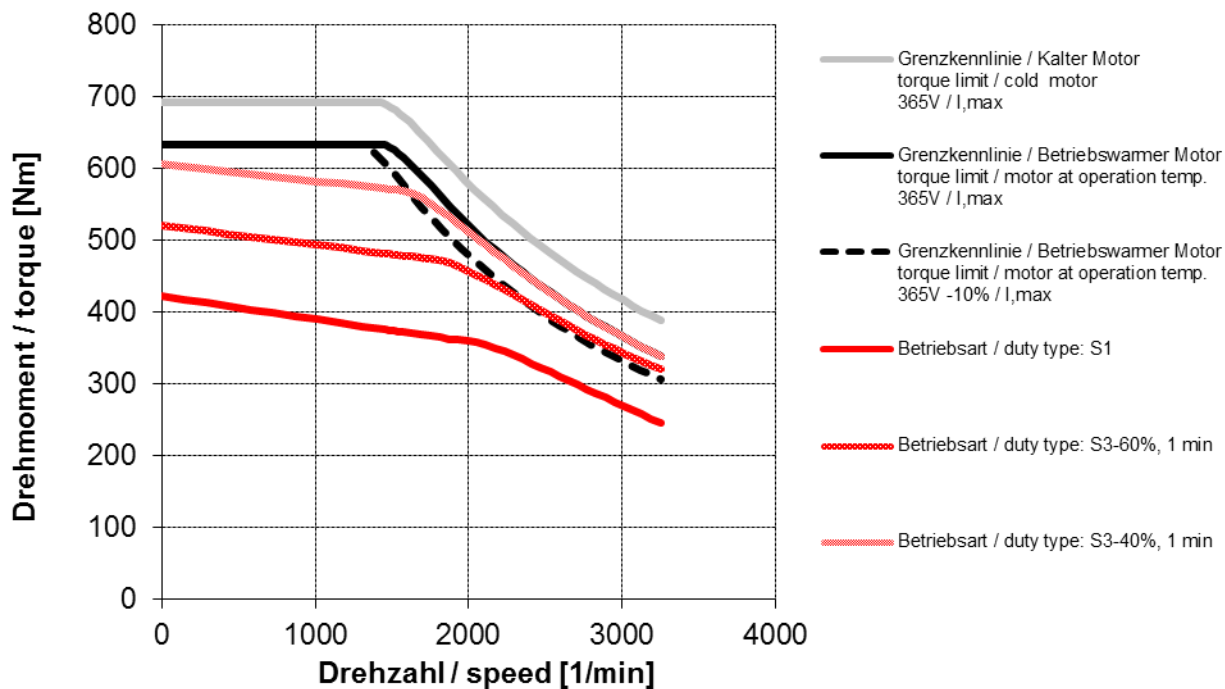
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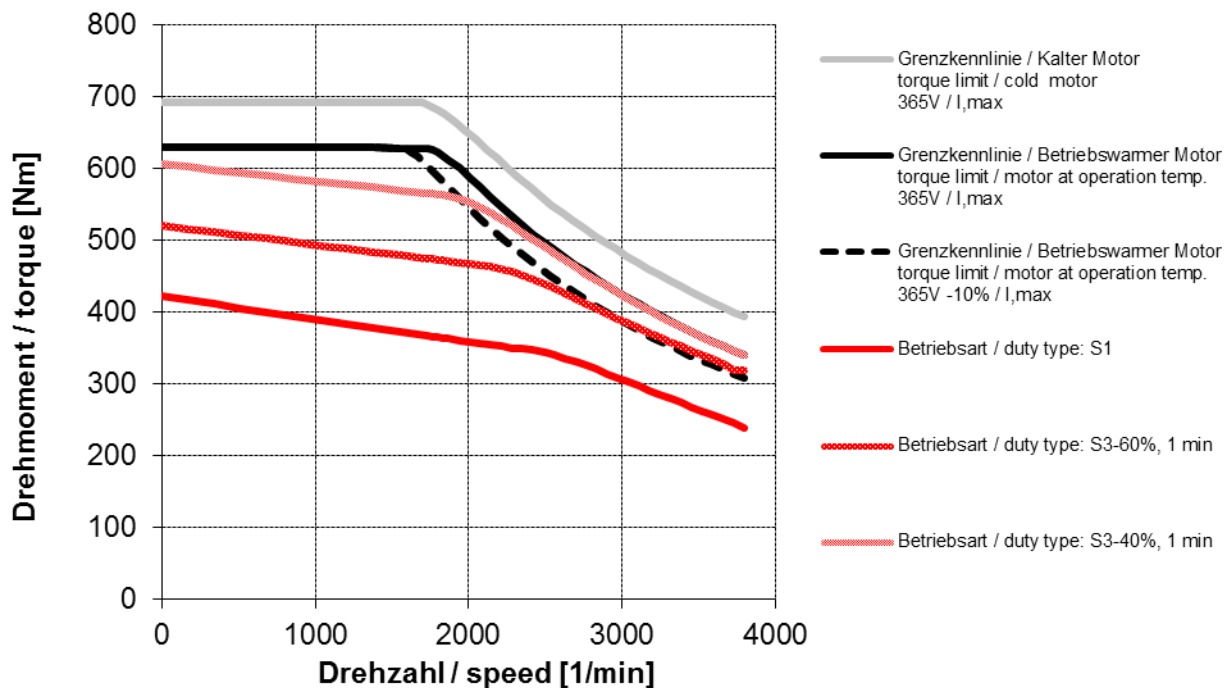
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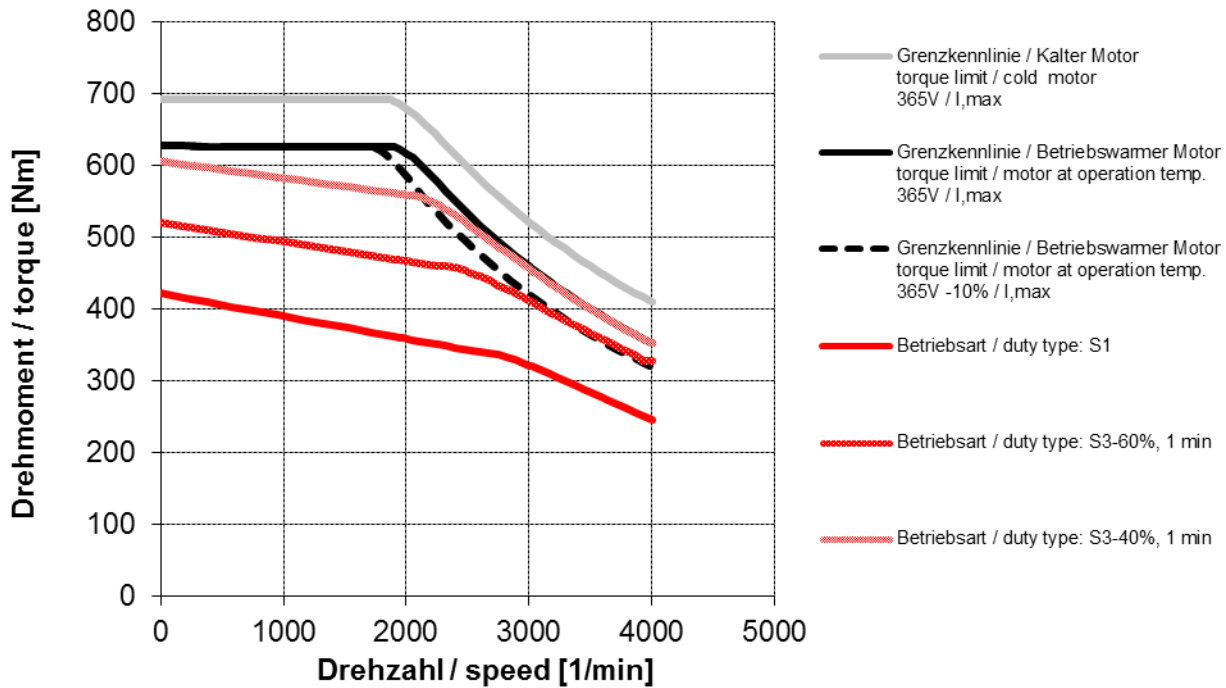
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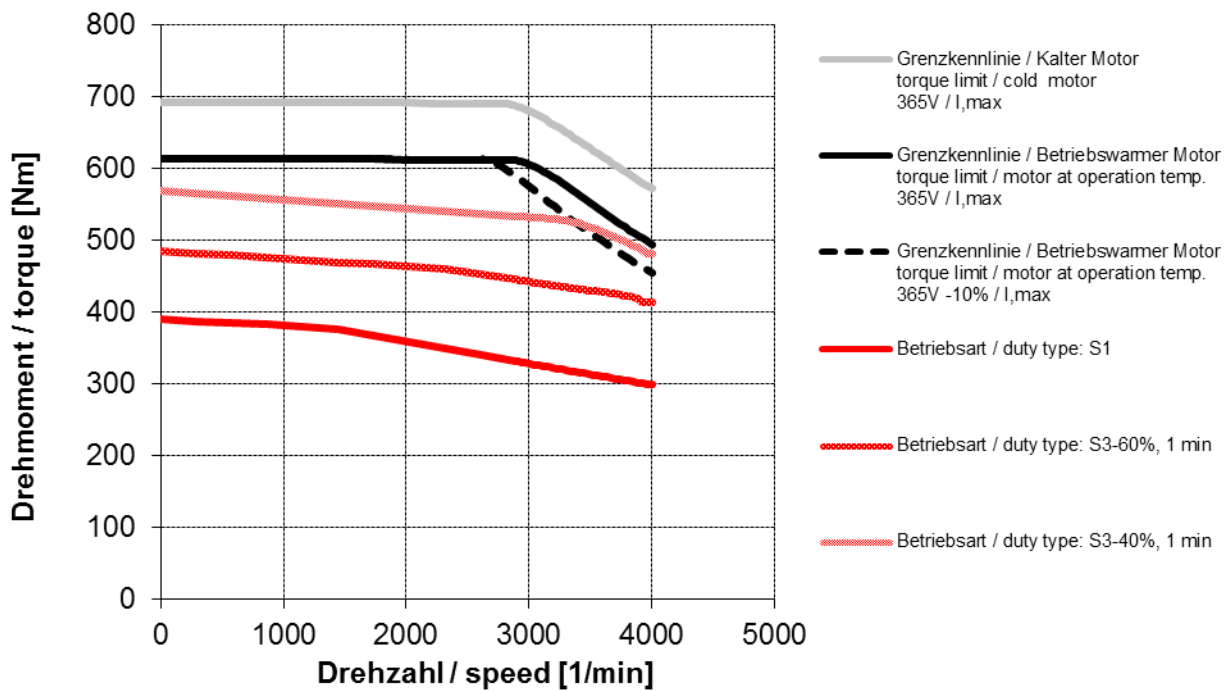
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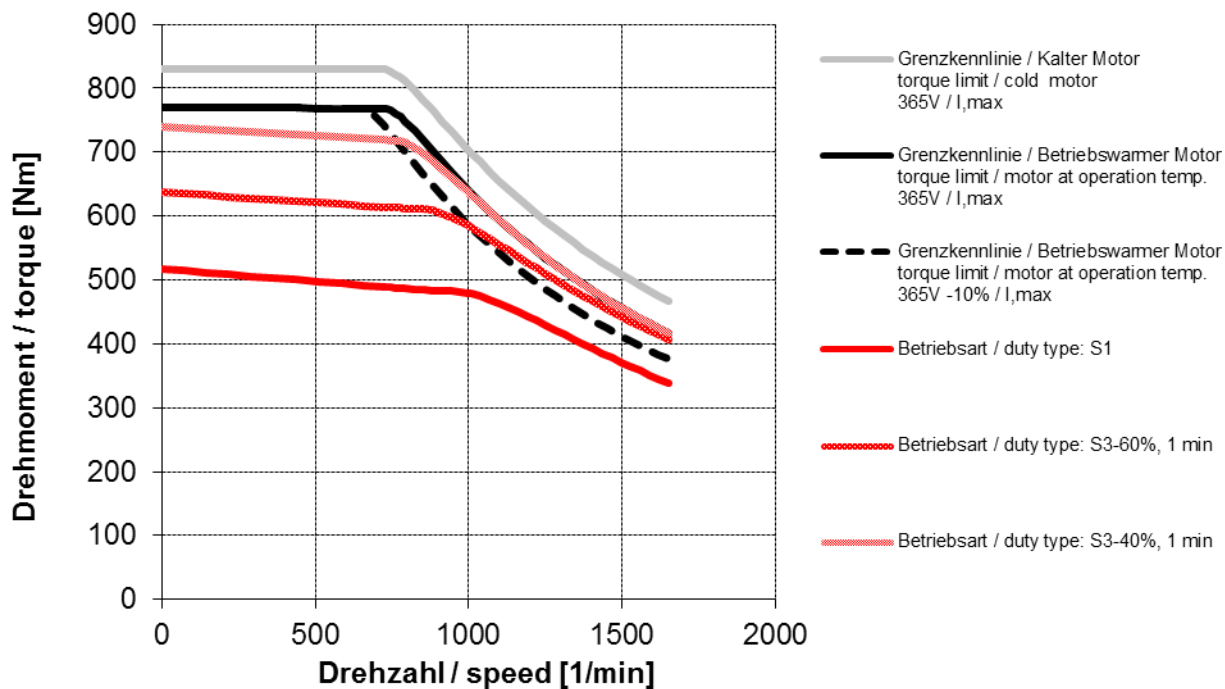
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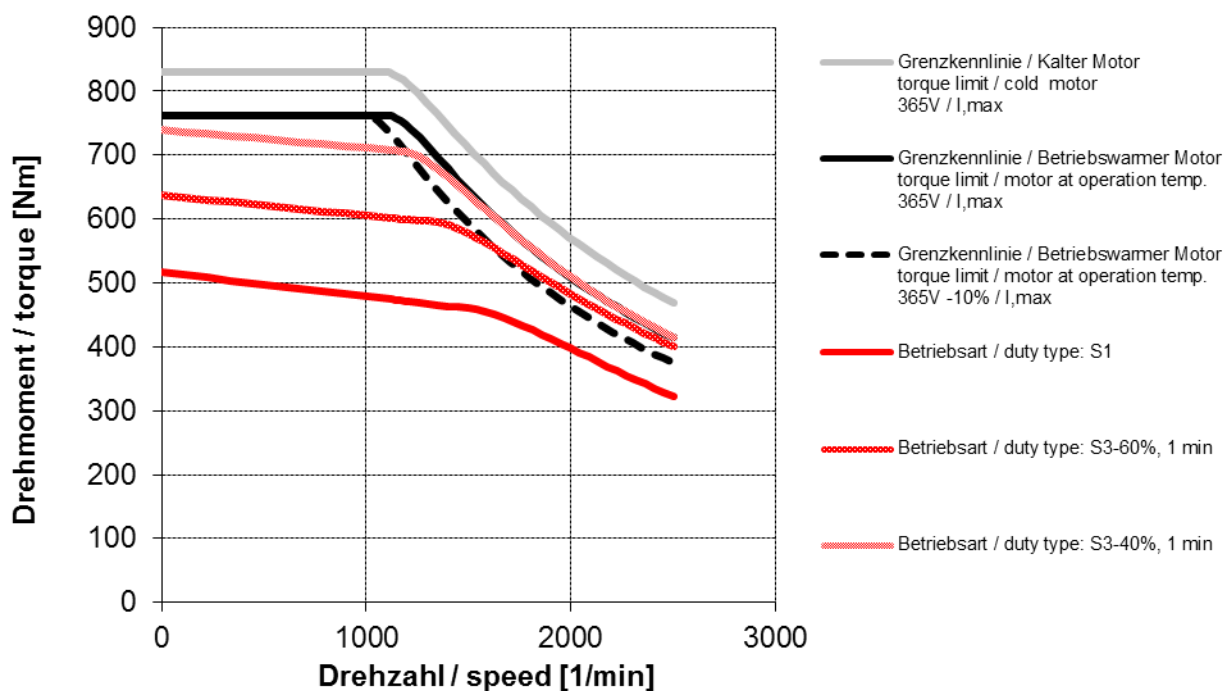
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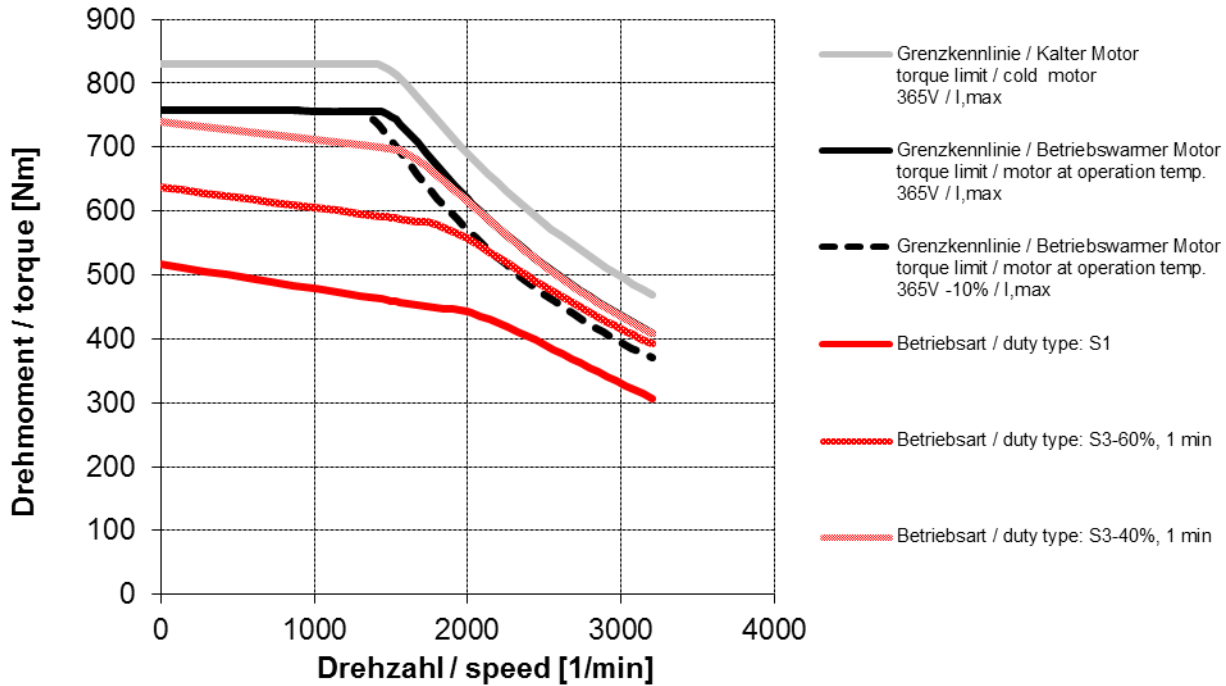
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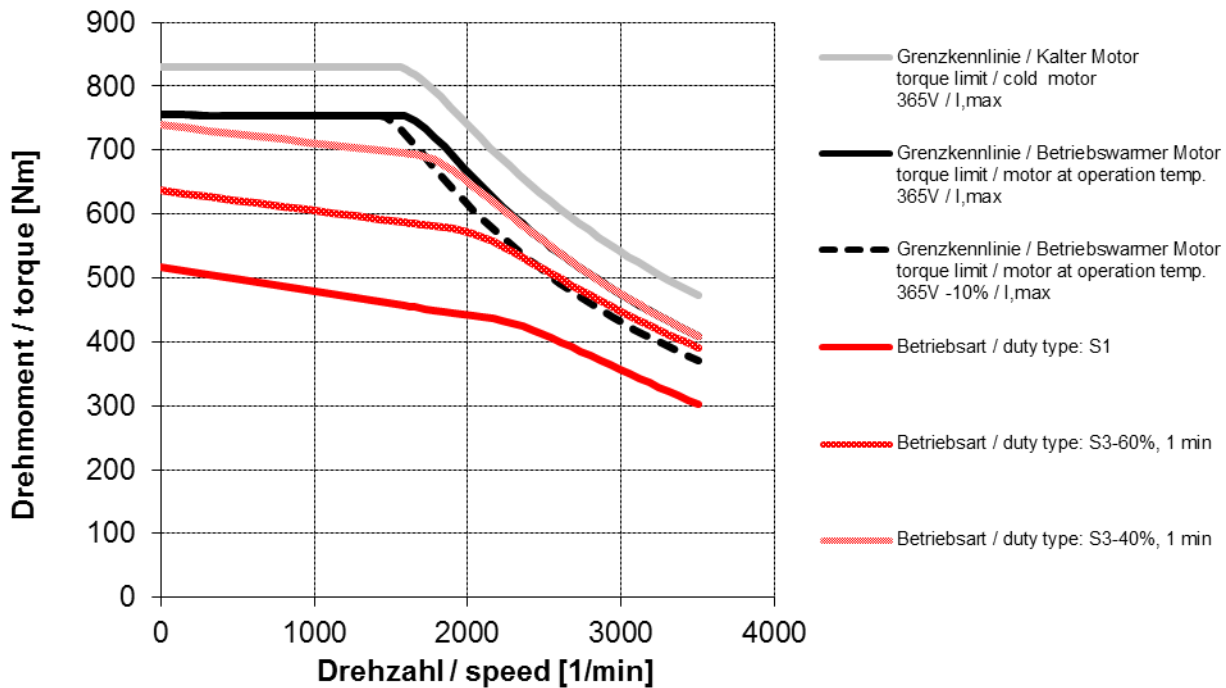
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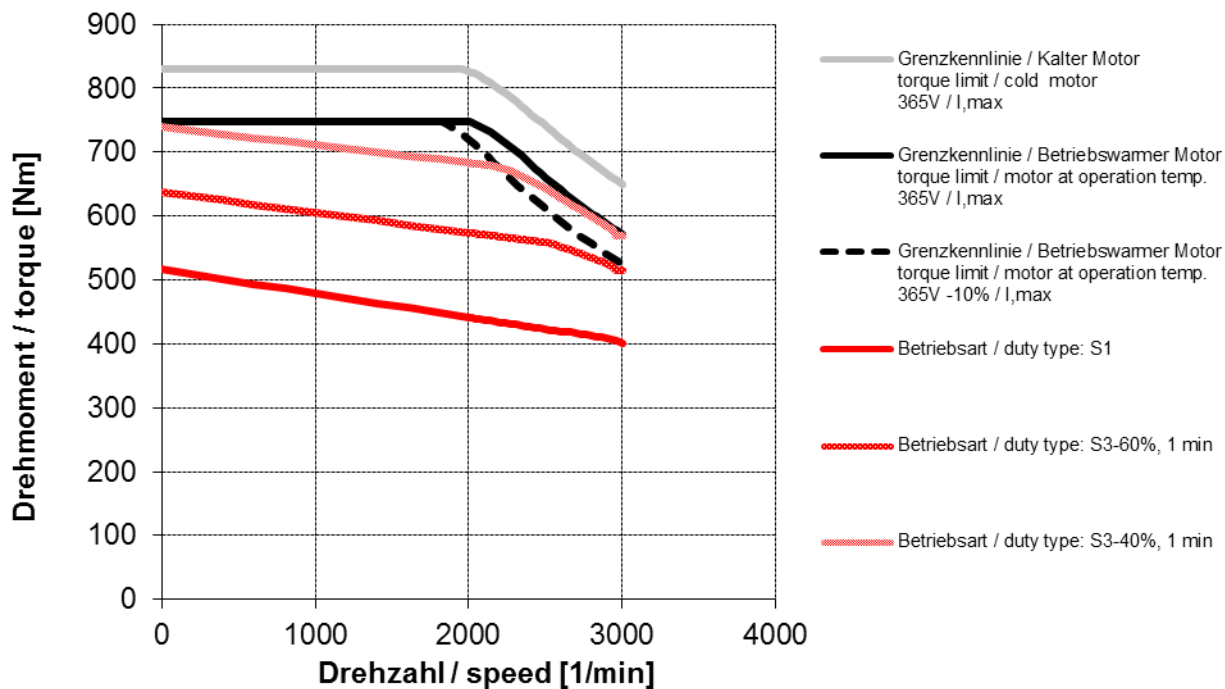
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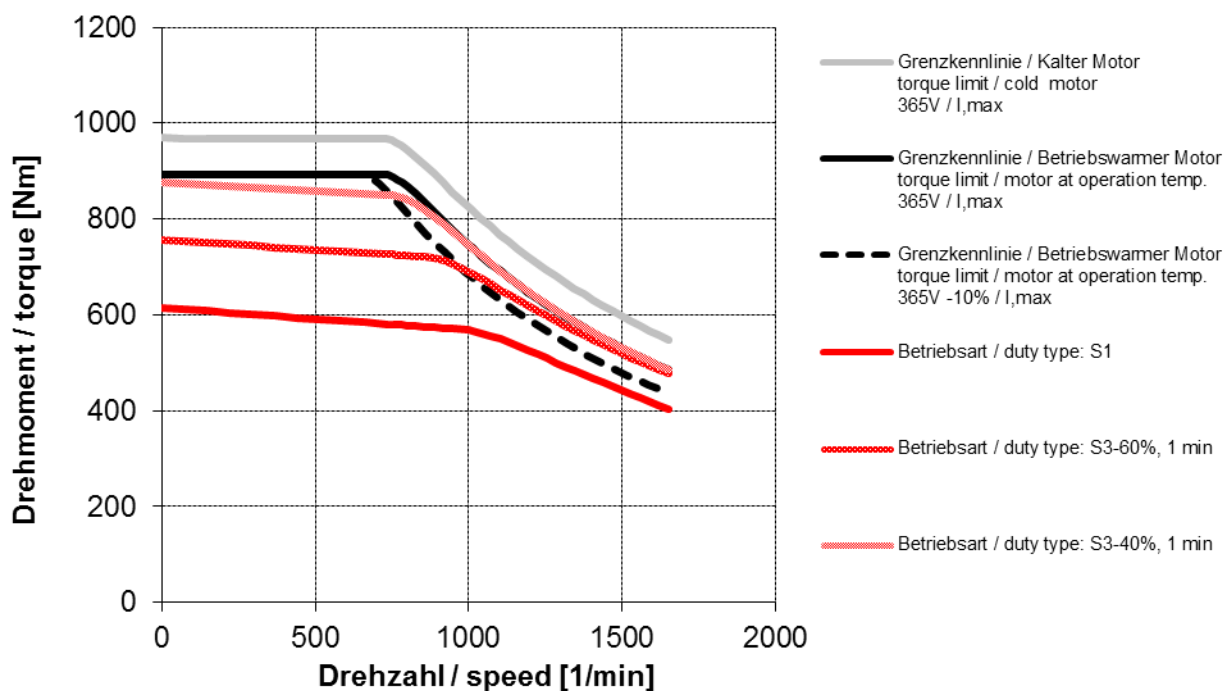
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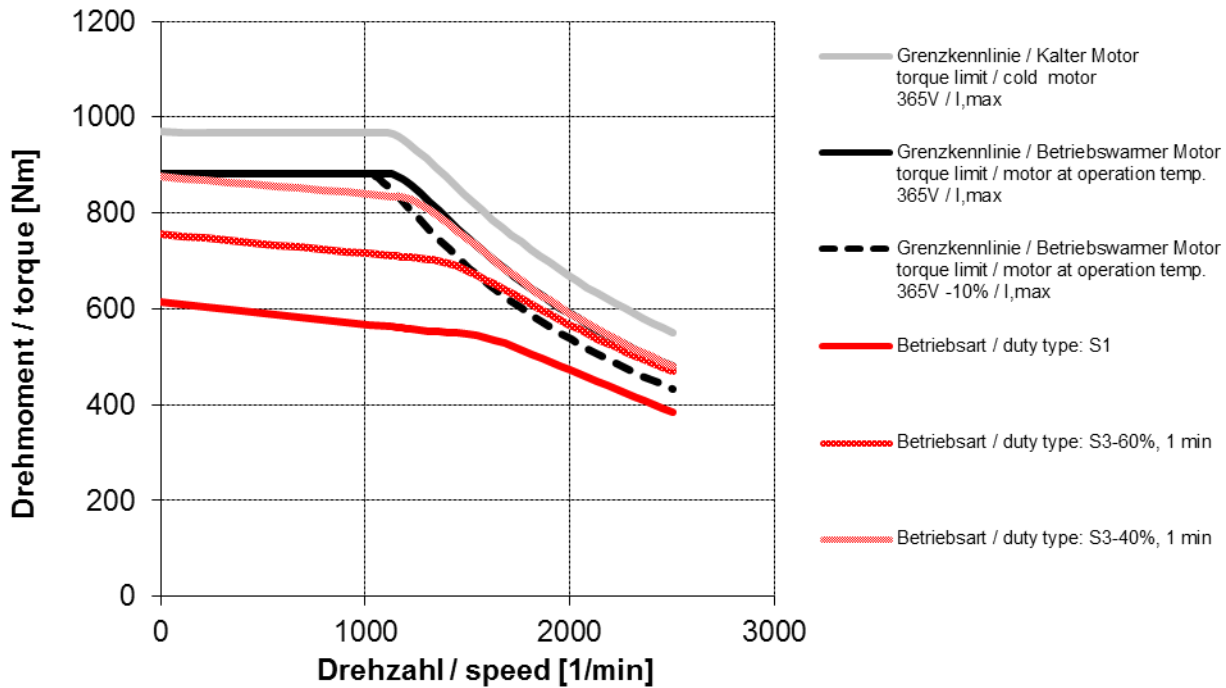
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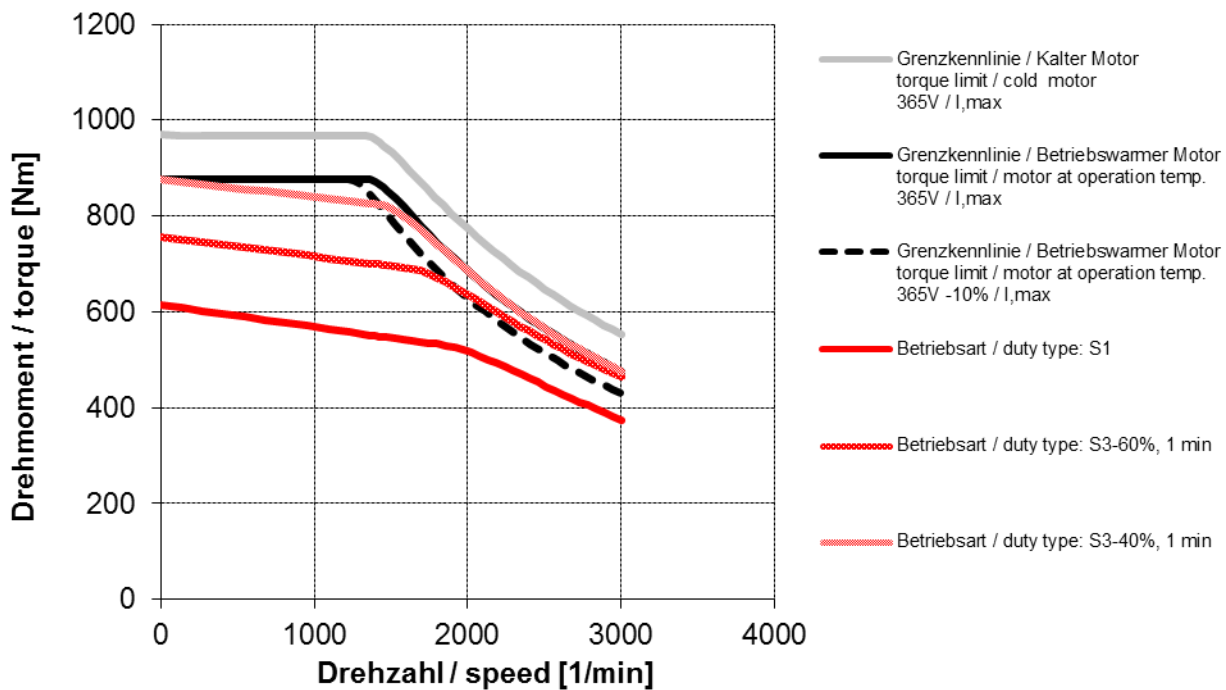
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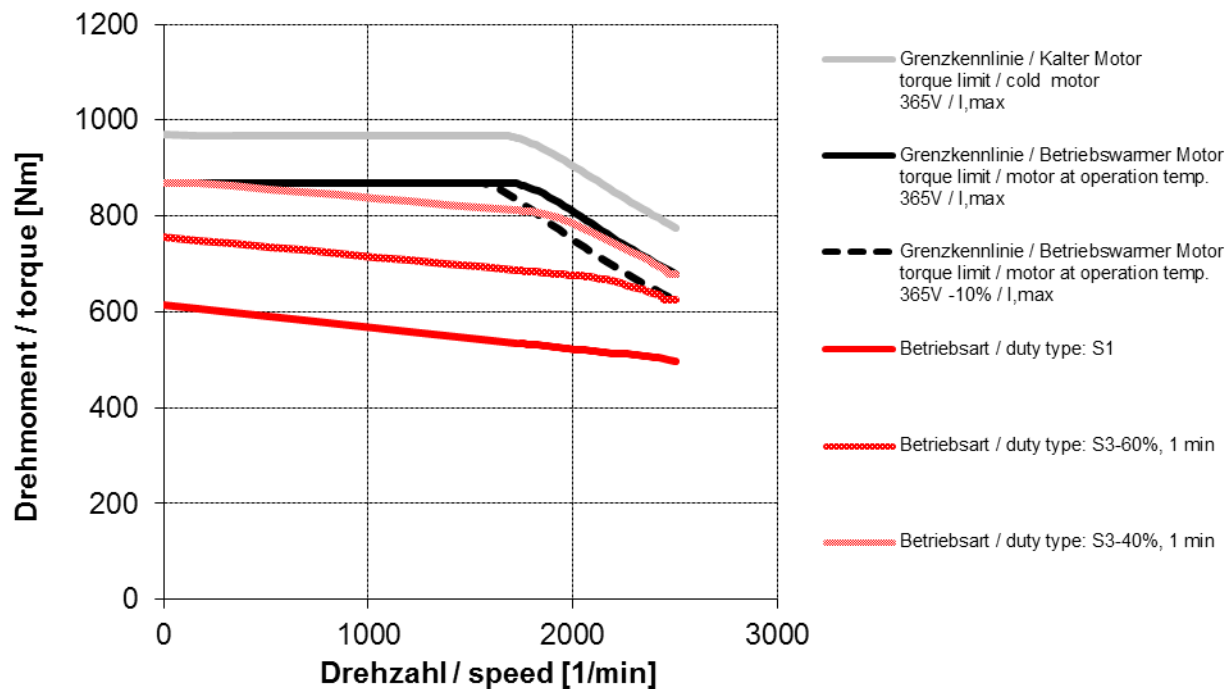
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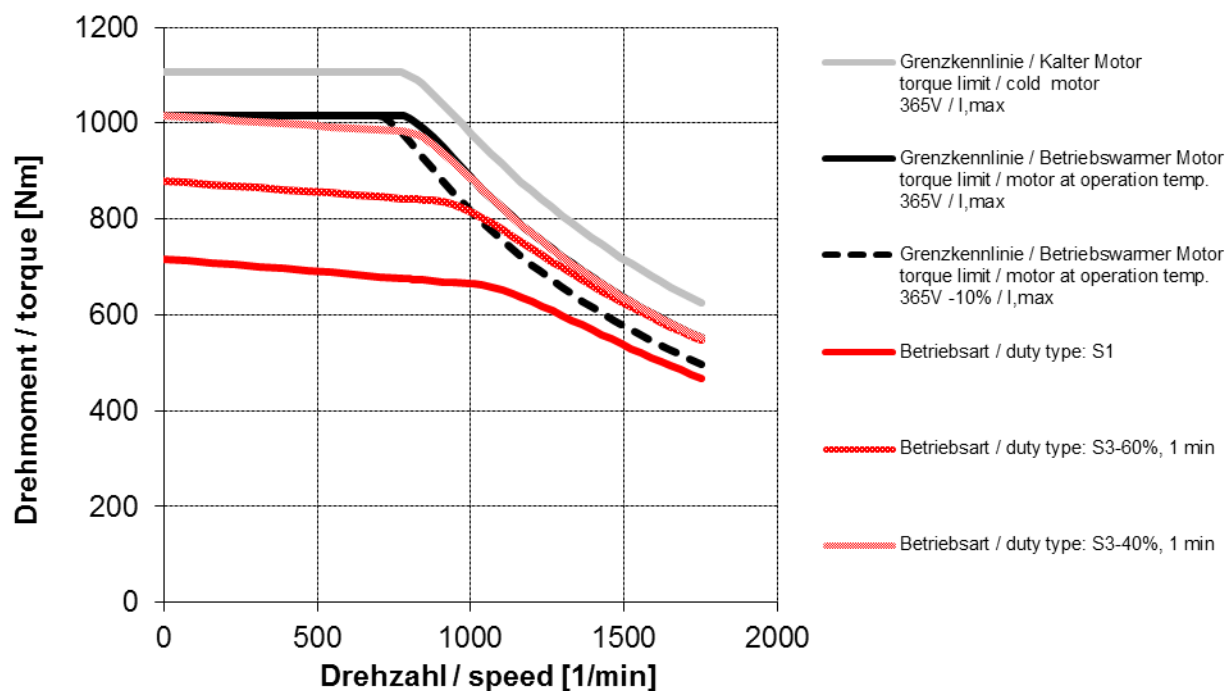
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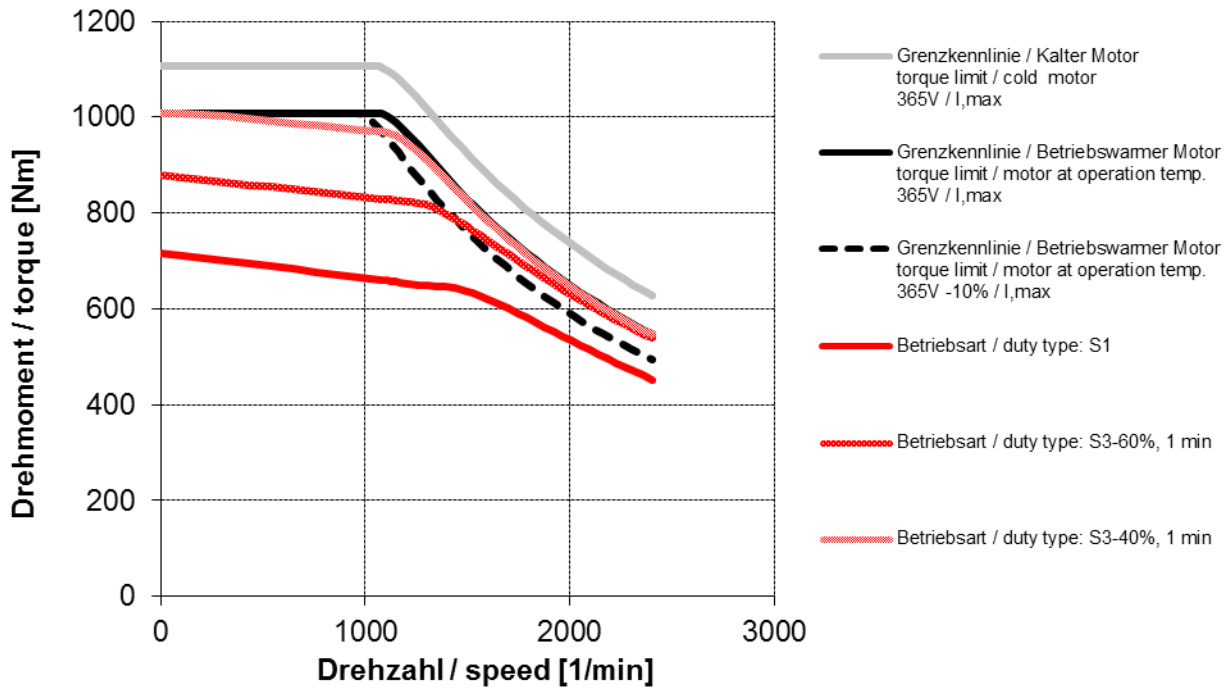
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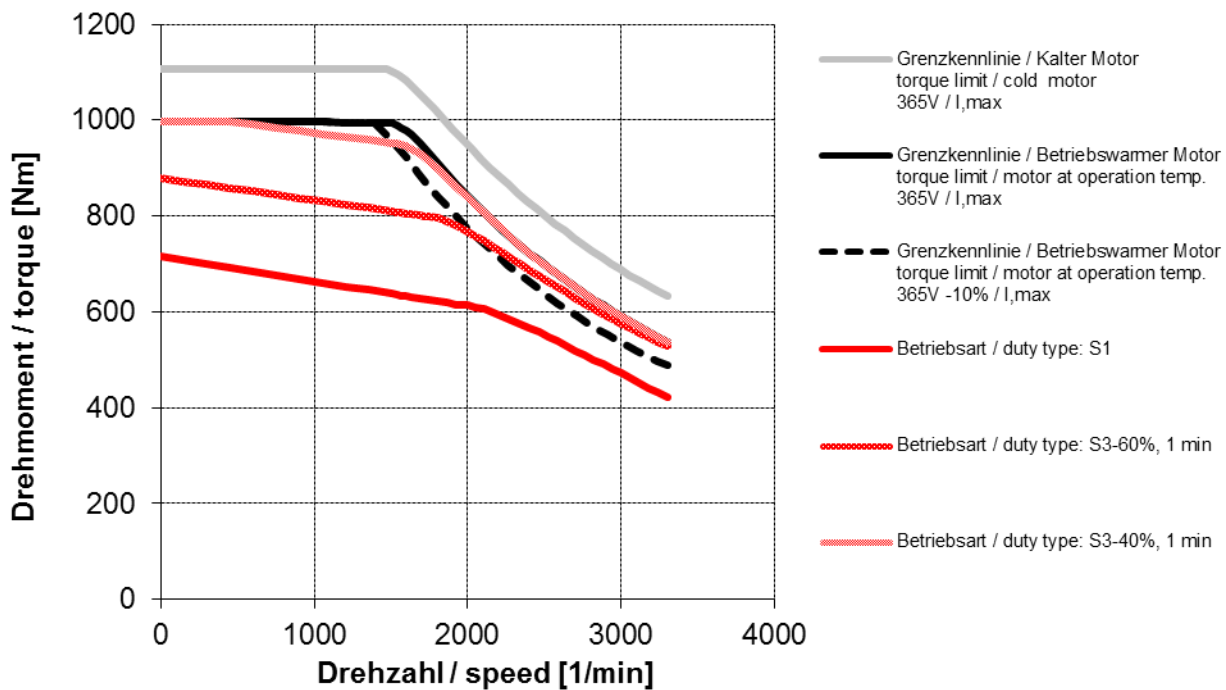
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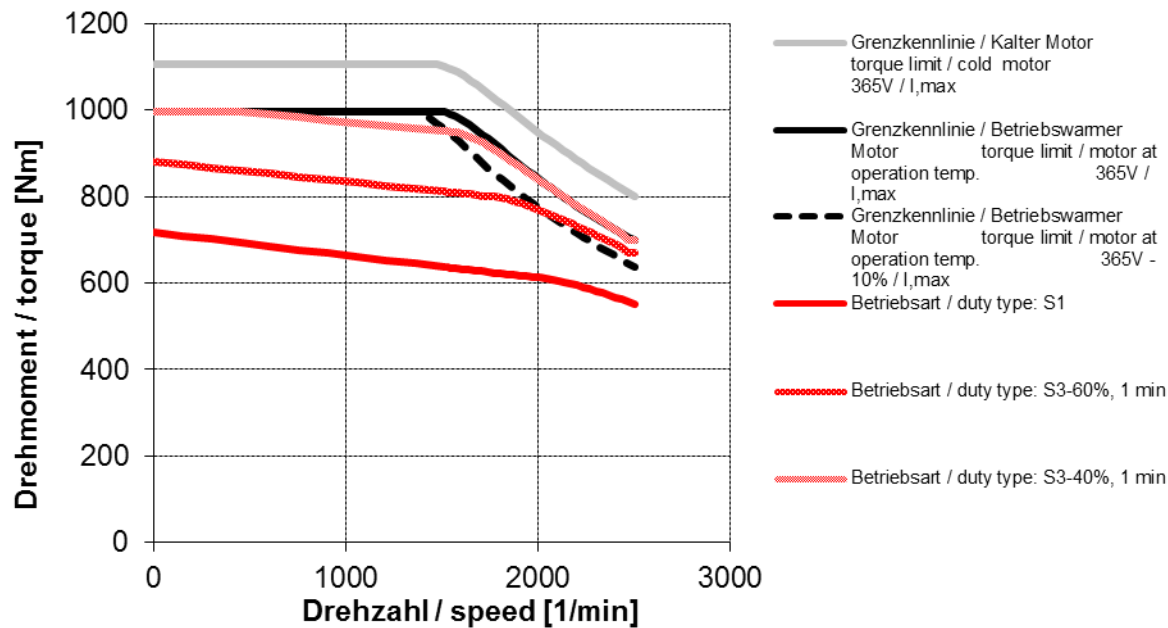
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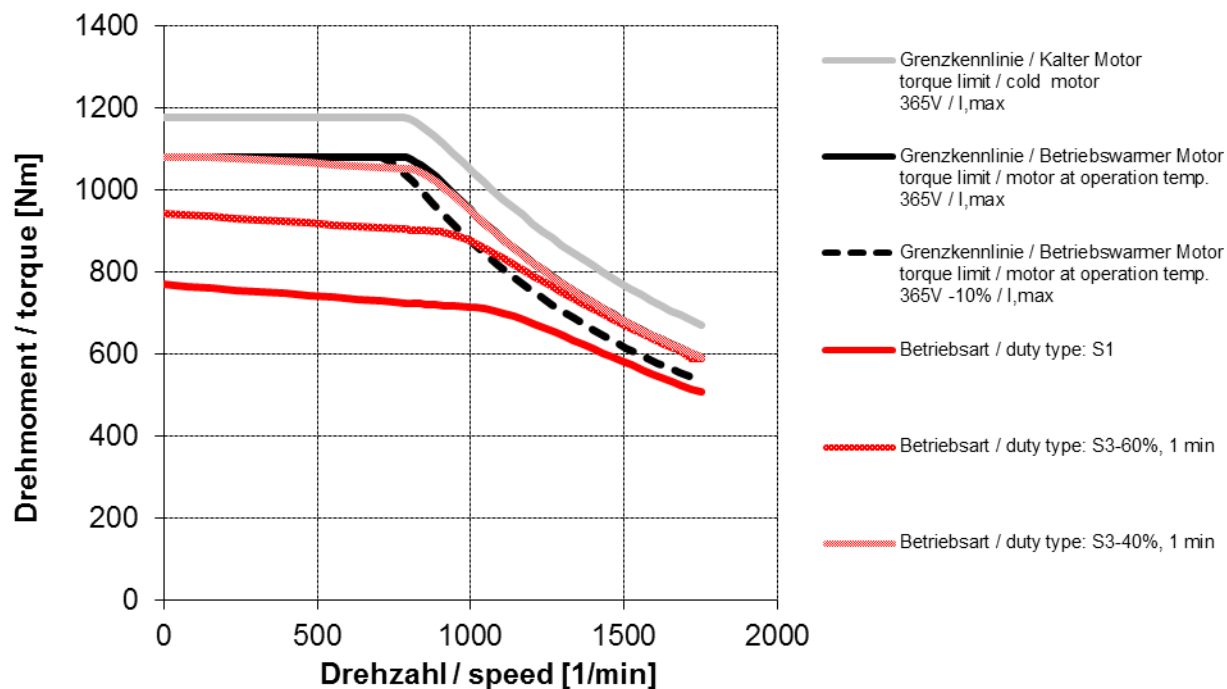
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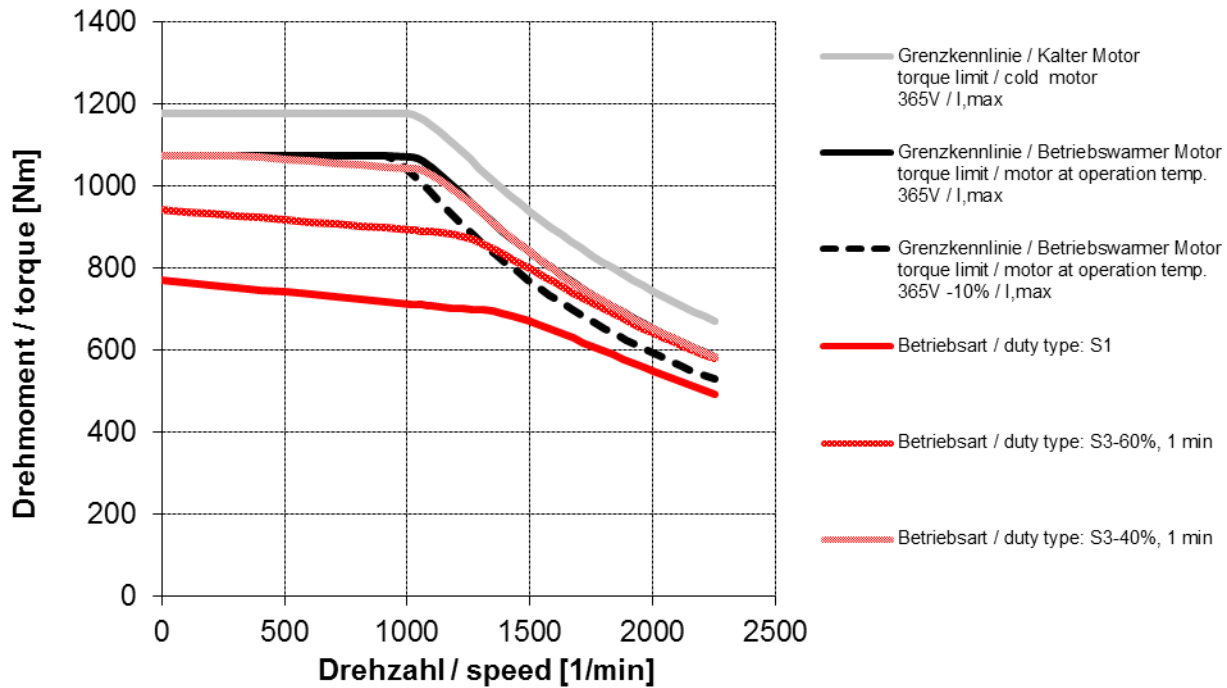
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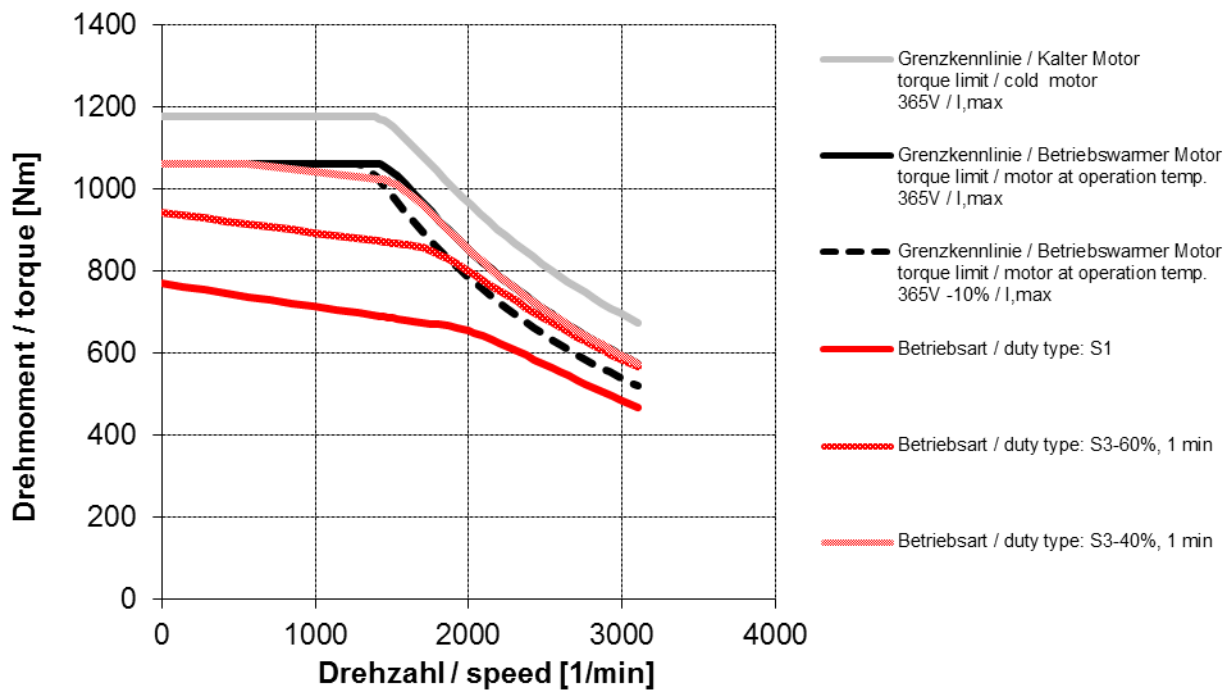
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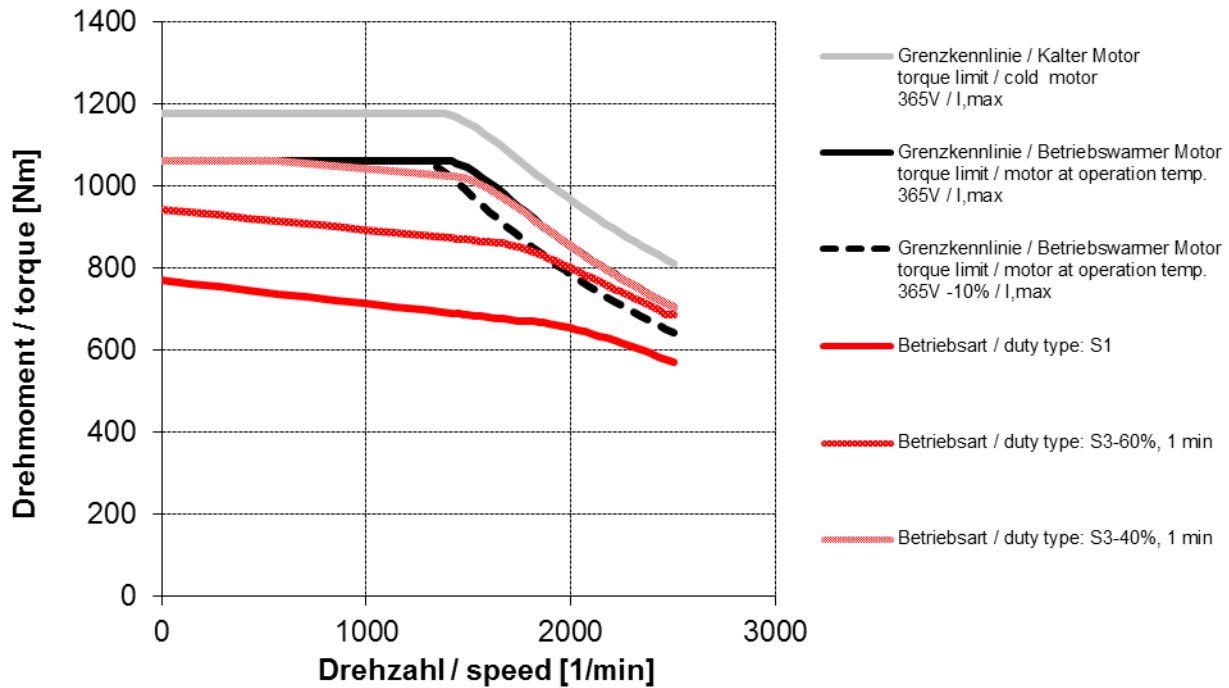
DSD2-132YZ54W-15-5



DSD2-132YZ54W-20-5



DSD2-132YZ54W-25-5



6. Commissioning and maintenance instructions

For information on commissioning the motors, please request a copy of our commissioning and maintenance instructions.

Motor	Commissioning and maintenance instructions
DSD2-028...036	No. 00721
DSD2-045...132	No. 00699

7. Declaration of Conformity

7.1. EU – Declaration of Conformity



EU-Konformitätserklärung gemäß

- Richtlinie 2014/35/EU
(Niederspannungsrichtlinie)

Richtlinie 2014/30/EU
(EMV-Richtlinie)

Hersteller

Baumüller Nürnberg GmbH
Ostendstr. 80 - 90
90482 Nürnberg
Deutschland
Tel. +49 9 11 54 32 - 0
Fax: +49 9 11 54 32 - 1 30
E-Mail: mail@baumueller.de
Internet: www.baumueeller.de

Hiermit erklären wir, dass die nachfolgend genannten Produkte aufgrund ihrer Konzeption, Konstruktion und Bauart in der von uns in Verkehr gebrachten Ausführung den Anforderungen der oben genannten Richtlinien einschließlich der zum Zeitpunkt der Erklärung geltenden Änderungen entsprechen.

Hinweise:

1. Bei Umbau oder Änderungen am Produkt verliert diese Erklärung mit sofortiger Wirkung ihre Gültigkeit.
2. Diese Erklärung bescheinigt die Übereinstimmung mit der genannten Richtlinie, stellt aber keine Zusicherung von darüber hinaus gehenden Produkteigenschaften dar.
3. Diese Konformitätserklärung wird unter der alleinigen Verantwortung des Herstellers ausgestellt.

Angewandte harmonisierte Normen:

- EN 60034-1:2010 + Cor.:2010
Drehende elektrische Maschinen – Teil 1:
Bemessung und Betriebsverhalten
- EN 60034-5:2001 + A1:2007
Drehende elektrische Maschinen – Teil 5:
Schutzarten aufgrund der Gesamtkonstruktion von
drehenden elektrischen Maschinen (IP-Code) –
Einteilung
- EN 60034-6:1993
Drehende elektrische Maschinen – Teil 6:
Einteilung der Kühlverfahren (IC-Code)

(Wird fortgesetzt auf der nächsten Seite ...)



EU-Declaration of Conformity according

- Directive 2014/35/EU
(Low-voltage-directive)

Directive 2014/30/EU
(EMC-directive)

Manufacturer

Baumüller Nürnberg GmbH
Ostendstr. 80 - 90
90482 Nürnberg
Germany
Tel. +49 9 11 54 32 - 0
Fax: +49 9 11 54 32 - 1 30
E-Mail: mail@baumueller.de
Internet: www.baumueeller.de

We declare, that the products referred to in the following conform in their concept, construction and design as launched by us to the above mentioned directives and their respective changes which were valid at the point of declaration.

Notes:

1. By modifying or alternating the device(s) this declaration immediately becomes invalid.
2. This declaration confirms the compliance with the directive listed, but it is no covenant of any further product properties.
3. This declaration of conformity is issued under the sole responsibility of the manufacturer.

Applied harmonised standards:

- EN 60034-1:2010 + Cor.:2010
Rotating electrical machines – Part 1:
Rating and performance
- EN 60034-5:2001 + A1:2007
Rotating electrical machines – Part 5:
Degree of protection provided by the integral design
of rotating electrical machines (IP-Code) –
Classification
- EN 60034-6:1993
Rotating electrical machines – Part 6:
Methods of cooling (IC-Code)

(To be continued on the next page ...)

(... Fortsetzung von der vorherigen Seite)

- EN 60034-9:2005 + A1:2007
Drehende elektrische Maschinen – Teil 9:
Geräuschgrenzwerte
- EN IEC 60034-14:2018
Drehende elektrische Maschinen – Teil 14:
Mechanische Schwingungen von bestimmten
Maschinen mit einer Achshöhe von 56 mm und höher –
Messung, Bewertung und Grenzwerte der
Schwingstärke
- EN 61800-5-1:2007 + A1:2017
Elektrische Leistungsantriebssysteme mit einstellbarer
Drehzahl – Teil 5-1:
Anforderungen an die Sicherheit – Elektrische,
thermische und energetische Anforderungen
- EN 60204-1:2018
Sicherheit von Maschinen - Elektrische Ausrüstung von
Maschinen - Teil 1:
Allgemeine Anforderungen

(... continued from the previous page)

- EN 60034-9:2005 + A1:2007
Rotating electrical machines – Part 9:
Noise limits
- EN IEC 60034-14:2018
Rotating electrical machines – Part 14:
Mechanical vibration of certain machines with shaft
heights 56 mm and higher – Measurement,
evaluation and limits of vibration severity
- EN 61800-5-1:2007 + A1:2017
Adjustable speed electrical power drive systems –
Part 5-1:
Safety requirements – Electrical, thermal and energy
- EN 60204-1:2018
Safety of machinery - Electrical equipment of
machines - Part 1:
General requirements

Markenname: Baumüller Produktbezeichnung: Drehstrommotor	Brand Name: Baumüller Product Name: AC motor
Produkt:	Product:

DSD2	0 2 8	A A 0 0 A	0 0	0 0	0 0 0	0 0 0	0	0 0	0	0 0 0
	0 3 6	bis/ bis/ bis/ bis/ bis/	bis/ bis/	bis/ bis/	bis/ bis/ bis/	bis/ bis/ bis/	bis/	bis/ bis/	bis/	bis/ bis/ bis/
	0 4 5	to to to to to	to to	to to	to to to	to to to	to	to to	to	to to to
	0 5 6	Z Z 9 9 Z	9 9	9 9	9 9 9	9 9 9	9	9 9	9	9 9 9
	0 7 1				oder/ oder/ oder/	oder/ oder/ oder/	oder/	oder/ oder/	oder/	oder/ oder/ oder/
	1 0 0				or or or	or or or	or	or or	or	or or or
					a a a	a a a	a	a a a	a	a a a
					bis/ bis/ bis/	bis/ bis/ bis/	bis/	bis/ bis/	bis/	bis/ bis/ bis/
					to to to	to to to	to	to to	to	to to to
					Z Z Z	Z Z Z	Z	Z Z	Z	Z Z Z

DSD2	1 3 2	A A 0 0 A	0 0	0	0 0 0	0	0 0 0	0 0 0	0 0 0	0 0 0
		bis/ bis/ bis/ bis/ bis/	bis/ bis/	bis/	bis/ bis/ bis/	bis/	bis/ bis/ bis/	bis/ bis/ bis/	bis/ bis/ bis/	bis/ bis/ bis/
		to to to to to	to to	to	to to to	to	to to to	to to to	to to to	to to to
		Z Z 9 9 Z	9 9	9	9 9 9	9	9 9 9	9 9 9	9 9 9	9 9 9
					oder/ oder/ oder/	oder/	oder/ oder/ oder/	oder/ oder/ oder/	oder/ oder/ oder/	oder/ oder/ oder/
					or or or	or	or or or	or or or	or or or	or or or
					a a a	a	a a a	a a a	a a a	a a a
					bis/ bis/ bis/	bis/	bis/ bis/ bis/	bis/ bis/ bis/	bis/ bis/ bis/	bis/ bis/ bis/
					to to to	to	to to to	to to to	to to to	to to to
					Z Z Z	Z	Z Z Z	Z Z Z	Z Z Z	Z Z Z

Nürnberg, 07.11.2022

Dr.-Ing. Michael Wengler

Geschäftsführer
COO Technology/Production

Nuremberg, 7-Nov-2022

I.V. Michael Veeh

Entwicklungsleiter Motoren
Manager R&D Motors

7.2. UKCA Declaration of Conformity



UKCA-Declaration of Conformity according

- Electrical Equipment Regulation 2016 (Statutory Instrument 2016/1101)
- Electromagnetic Compatibility Regulation 2016 (Statutory Instrument 2016/1091)

Manufacturer

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United Kingdom
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Fax: +44 161 432 8493
E-Mail: mail@baumueller.co.uk
Internet: www.baumueller-services.com/uk

We declare, that the products referred to in the following conform in their concept, construction and design as launched by us to the above mentioned directives and their respective changes which were valid at the point of declaration.

Notes:

1. By modifying or alternating the device(s) this declaration immediately becomes invalid.
2. This declaration confirms the compliance with the directive listed, but it is no covenant of any further product properties.
3. This declaration of conformity is issued under the sole responsibility of the manufacturer.
4. responsibility of the manufacturer This motor series isn't in scope of guideline 2005/32/EG

Applied harmonised standards:

- **BS EN 60034-1:2010**
Rotating electrical machines – Part 1:
Rating and performance
- **BS EN 60034-5:2020**
Rotating electrical machines – Part 5:
Degrees of protection provided by the integral design of rotating electrical machines (IP code). Classification
- **BS EN 60034-6:1994**
Rotating electrical machines – Part 6:
Methods of cooling (IC-Code)
- **BS EN 60034-9:2005**
Rotating electrical machines – Part 9:
Noise limits
- **BS EN IEC 60034-14:2018**
Rotating electrical machines – Part 14:
Mechanical vibration of certain machines with shaft heights 56 mm and higher. Measurement, evaluation and limits of vibration severity. The following applies to roller bearing motors : Based on EN 60034-14 or requirements according to customer agreement.

(To be continued on the next page ...)

FM_0009, Version 3.0

Seite 1 von 2

(... continued from the previous page)

- **BS EN 61800-5-1:2007 + A11:2021**
Adjustable speed electrical power drive systems – Part 5-1:
Safety requirements – Electrical, thermal and energy
- **BS EN 60204-1:2018**
Safety of machinery - Electrical equipment of machines - Part 1:
General requirements

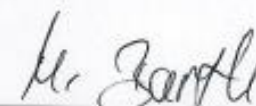
Brand Name: Baumüller
Product Name: AC motor

Produkt / Product (x): optionaler Buchstabe / optional character (x, y): alternative Buchstaben oder Zahlen / alternative characters	Jahr der erstmaligen CE-Kennzeichnung / Year of first CE marking
DS 3 Phase AC Servomotoren DS (x)(x)-xxx-x-x-x GN(A,F)xxxx(S,M,L)N DSC1-XXXXXXXX-XX-XX-XXX-XXX-X-XX-X-XXX DSH1-XXXXXXXX-XX-XX-XXX-XXX-X-XX-X-XXX DSP1-XXXXXXXX-XX-XX-XXX-XXX-X-XX-X-XXX HYG1-XXXXXXXX-XX-XX-XXX-XXX-X-XX-X-XXX DS2--XXXXXXXX-XX-XX-XXX-XXX-X-XX-X-XXX DSD2-XXXXXXXX-XX-XX-XXX-XXX-X-XX-X-XXX DST2-XXXXXXXX-XX-XX-XXX-XXX-X-XX-X-XXX DA1- XXXXXXXX-XX-XX-XXX-XXX-X-XX-X-XXX	2022

Nürnberg, 05.04.2022



Dr.-Ing. Michael Wengler
 Director



ppa. Matthias Barth
 Manager R&D

HOUSE OF AUTOMATION



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